

PRICING AND
REVENUE OPTIMIZATION

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PRICING AND
REVENUE OPTIMIZATION

Second Edition

ROBERT L. PHILLIPS

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PREFACE TO THE SECOND EDITION

This book originally grew out of courses in pricing and revenue optimization that I developed with Michael Harrison at Stanford University and taught at Columbia Business School and at the Stanford Graduate School of Business in the early 1990s. Over the last 20 years, the number of courses on the topic has increased steadily both in business schools and in management science and operations research programs. Some of the challenges involved in developing and teaching a course on pricing and revenue optimization have been treated in articles by Peter Bell (2004) and me (Phillips 2003).

The primary audience for this book is students at the master's or undergraduate level, as well as managers in revenue management, pricing, and related areas. The book assumes some familiarity with probabilistic modeling and optimization theory and comfort with basic calculus. Sections that are more specialized or require more quantitative sophistication (or at least more patience), or that are specialized to a particular interest and may not be of general interest, have been marked with an asterisk (*) and can be skipped without loss of continuity. In pricing and revenue optimization, as in other applications of analytics, what is theoretically elegant is often not practical, and what is practical is often not theoretically elegant. When in doubt, I have erred on the side of presenting the practical. I have also simplified models at times, and I generally present heuristic arguments rather than formal proofs with the aim of providing insight. For those who want to dive deeper into the theory and see proofs of key results, I heartily recommend Kalyan Talluri and Garrett van Ryzin's *The Theory and Practice of Revenue Management* (2004b) and Guillermo Gallego and Huseyin Topaloglu's *Revenue Management and Pricing Analytics* (2019).

This second edition has been substantially rewritten. This reflects, in part, the evolution that the field has undergone in the 15 years since the first edition was published. What was fresh and exciting then (particularly classic revenue management applications at the airlines and related industries) is now standard practice, and what is fresh and exciting now (for example, "learning and earning" algorithms for online pricing) did not exist 15 years ago. This edition reflects these developments, with new sections on dynamic pricing, price-response estimation, reinforcement learning, and other more recent topics, and with

somewhat less emphasis on classic revenue management applications in airlines, hotels, and the like. The goal continues to be to provide the reader with a broad overview of the field with a focus on applications. References are provided in “Further Reading” sections at the end of each chapter for those who want to dig deeper into particular topics.

For clarity, I call anyone who is setting a price a *seller* and anyone who is considering a purchase a *customer*. This seems less clumsy than the more accurate term *prospective buyer*. I use the term *consumer* in a broader sense to refer to individuals as economic decision makers without reference to a specific purchase opportunity. Also, for clarity, I use male pronouns for sellers and female pronouns for customers.

Among those who read drafts of this book and generously provided comments and suggestions, pride of place belongs to Michael Harrison, who taught the pricing and revenue optimization course with me at the Stanford Business School. Not only did Mike read the first draft of the first edition and provide many helpful comments, but our discussions helped me focus and organize my own thinking. I am also thankful to Brenda Barnes, whose careful reading and thoughtful comments on several chapters resulted in substantial improvements. The late Ken McLeod of Stanford University Press provided encouragement and inspiration in the early days of writing the book. He is very much missed. I also thank Dean Boyd, Bill Carroll, Yosun Denizeri, Michael Eldredge, Mehran Farahmand, Scott Friend, Steve Haas, Jake Krakauer, Ahmet Kuyumcu, Bob Oliver, Rama Ramakrishnan, Carol Redfield, Alex Romanenko, and Nicola Secomandi, who all contributed comments and suggestions that improved the book. Thanks also go to my students at Columbia and Stanford, who caught many typos. Thanks go to the Columbia University Business School, the Stanford University Business School, Manugistics, and Nomis Solutions, all of whom provided office space and support at various times throughout the writing of the book.

The book has also benefited from my extensive interactions and discussions with colleagues over the years, including Bill Brunger, Simon Caufield, Glenn Colville, Guillermo Gallego, Tom Grossman, Lloyd Hansen, Peter Grønlund, Garud Iyengar, Anton Kleywegt, Steve Kou, Warren Lieberman, Ray Lyons, Costis Maglaras, Özalp Özer, Özgür Özluk, Jörn Peter Petersen, Özge Şahin, Kalyan Talluri, Garrett van Ryzin, Van Veen, Loren Williams, Graham Young, and Jon Zimmerman, among many others. Special thanks go to Christian Albright, Serhan Duran, Jiehong Kong, Warren Lieberman, Joern Meissner, and Nicola Secomandi for catching errors in previous printings and providing useful suggestions.

Since the first edition, I have benefited enormously from spending time in the Decision Risk and Operations Division of the Columbia Business School and from numerous discussions and collaborations with colleagues there, including Omar Besbes, Daniel Guetta, Gabriel Weintraub, and Assaf Zeevi. I am also grateful to Uber and Amazon for the time I spent in data science and pricing research groups at both companies—in both cases, I was fortunate to work with immensely talented colleagues who helped me grapple with the particular pricing challenges faced by modern e-commerce companies. Guillermo Gallego and Markus Husemann-Kopetzky provided comments on individual chapters that led to significant improvements. Additional thanks go to Fabio Bortone, Carolyn Lacy, Cheyenne Morgan, Frank Quilty, Kit Weathers, and Gretchen Yanda for highly focused discussions

on specific topics. Finally, thanks go to my editor, Steve Catalano; my copy editor, Anita Hueftle; and the team at Stanford University Press for their support and assistance in creating a much improved second edition.

Very special thanks go to my parents for their love and support.

Needless to say, any errors in the book are neither my fault nor the fault of any of the others mentioned here; they are, as always, due to malign outside influences.

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1

BACKGROUND

What is a cynic? . . . A man who knows the price of everything and the value of nothing.

—Oscar Wilde, *Lady Windermere's Fan*

This is a book about pricing—specifically, how companies should set and adjust their prices to maximize profitability. It takes the view that pricing decisions are commonplace, that they are often complex, and that they are critical determinants of profitability and revenue. Despite this, pricing decisions are often badly managed—or simply unmanaged. While most companies have fairly good prices in place most of the time, few have the processes and capabilities needed to ensure that they have the right prices in place for all their products, for all their customers, through all their channels, all the time. This is the goal of pricing and revenue optimization.

Pricing and revenue optimization is a tactical function. It recognizes that in many cases, prices need to change rapidly and often and provides guidance on how they should change. This makes it distinct from *strategic pricing*, where the goal is to establish a general price position in a market. While strategic pricing is concerned with how a product should in general be priced relative to the market, pricing and revenue optimization is concerned with determining the prices that will be in place tomorrow and next week. Strategic pricing sets the constraints within which pricing and revenue optimization operates. For example, a luxury brand may decide that it will not discount its products more than a certain percentage in order to maintain its luxury image. This strategic pricing decision could be implemented as a maximum-discount constraint within a pricing and revenue management system.

One of the distinguishing characteristics of pricing and revenue optimization is its use of analytical techniques derived from management science, statistics, and machine learning. The use of these techniques to set prices in a complex, dynamic environment is about 35 years old. One of the first applications of pricing optimization was revenue management, which was first used by the passenger airlines in the 1980s. Since then, the rapid development of e-commerce, the availability of customer data through customer relationship management (CRM) systems, and the increasing dynamism of online commerce have led to the adoption of pricing optimization in many other industries, including automotive,

retail, telecommunications, financial services, and manufacturing. A multitude of software vendors provide “price optimization” or “demand management” or “revenue management” solutions focused on one or more industries. Pricing and revenue optimization has become a core competency within a wide range of industries including airlines, hotels, and online and offline retail, and it has been widely adopted across a range of other industries. Pricing and revenue optimization has also been a fruitful area of academic research.

In this chapter, we begin by looking at some historical context for pricing and revenue optimization. In particular, I provide some perspective on how pricing went from being an obscure and mysterious art within many companies to becoming the subject of intense scrutiny and analysis. This chapter argues that the pricing problem has become increasingly difficult and dynamic and is likely to become even more so in the future. It argues further that improving pricing is often one of the highest-return investments available to a company. Hopefully this will whet the reader’s appetite for the more quantitative material to come in the following chapters.

1.1 HISTORICAL BACKGROUND AND CONTEXT

For most of history, philosophers took it for granted that goods had an intrinsic value in the same sense that they had an intrinsic color or weight. A fair price reflected that intrinsic value. Charging a price too much in excess of the intrinsic value was condemned as a sign of avarice and often prohibited by law. Prices were set by custom, by law, or by imperial fiat with an emphasis on both fairness and stability: Plato advocated severe penalties for any merchant who asked for more than one price for a good on the same day, and he believed that prices should be set by law at a rate that provided merchants a reasonable profit.¹ The general belief that prices should be fair and not set by the merchant to maximize profit survived the collapse of the Roman Empire and the rise of Christianity—many sermons inveighed against the sin of charging unfair prices to gain excessive profits.²

The problem of pricing did not exist on a large scale until modern market economies began to emerge in the West in the seventeenth and eighteenth centuries. With the emergence of market economies and a growing middle class with more disposable income, prices were allowed to move more freely—untethered to the traditional concept of value. Speculative bubbles such as tulipomania in the Dutch Republic in the 1630s, in which the prices of some varieties of tulips rose more than a hundredfold in 18 months before collapsing in 1637, and the South Sea bubble in England in 1720, in which the prices of shares in the South Sea Company soared before the company collapsed amid general scandal, fed a sense of anxiety and the belief that prices could somehow lose touch with reality. Furthermore, for the first time, large numbers of people could amass fortunes—and lose them—by buying and selling goods on the market. Questions naturally arose: What were prices, exactly? Where did they come from? What determined the right price? When was a price fair? When should the government intervene in pricing? The modern field of economics arose, at least in part, in response to these questions.

Possibly the greatest insight of classical economics—usually credited to Adam Smith—was that the price of a good at any time in an unregulated economy is not based on any

intrinsic value but rather on the interplay of supply and demand. This was a major intellectual breakthrough—on par in its own way with the Newtonian theory of the clockwork universe and Darwin’s theory of evolution. In essence, the price of a good or service was determined by the interaction of people willing to sell the good with the willingness of others to buy the good. That is all there is to it—intrinsic value, cost, and labor content do not enter directly into the equation. Of course, these and many other factors enter *indirectly* into pricing—sellers would not last long selling goods below cost, and a buyer will react to a price based on the value she places on the item—but these are not primary. There are many reasons why sellers sell below cost when they are in possession of a cartload of vegetables that are on the verge of going rotten—the classic sell-it-or-smell-it situation. Just so, the value that buyers place on different goods changes with their changing situation and the dictates of fashion. According to modern economics there is no normative right price for a good or service against which the price can be compared—rather, there are only the actual prices out in the marketplace, floating freely, based on the willingness of sellers to sell and buyers to buy.

While neoclassical economics solved the problem of the origin of price, it raised as many questions as it answered. In particular, if prices are not tied to fundamental values—if they have no anchor—why do they show any stability at all? Under normal circumstances, prices for most goods are reasonably stable most of the time. If prices are based only on the whims of buyers and sellers, why is the price of bread not subject to wild swings like the Dutch tulip market in 1637? Why does milk not cost five times as much in Chicago as it does in New York? How can manufacturers and merchants plan at all and make reasonable profits in order to stay in business? How can an economy based on free-floating prices work at all? And, assuming that such an economy could work, how could it possibly work better than a centralized economy where planners strive to allocate resources across the entire economy?

One of the great achievements of twentieth-century economics was to show mathematically how a largely unregulated economy could function: that an economy consisting of individuals who supply their labor in return for wages and maximize the utility of their earnings as they buy goods from firms who seek to maximize profitability can be remarkably stable and efficient. Under certain assumptions, this type of decentralized economy can be shown to be at least as efficient as any centrally planned economy at producing and allocating goods that satisfy consumers. Furthermore, prices in such an economy would generally be stable and reasonably predictable. The price for a product would equal the long-run marginal production cost of that product plus the return on invested capital necessary to produce it. A seller offering an item for less than it cost would go bankrupt. If someone tried selling for more, other sellers would undercut his price, consumers would flee to the lower-priced sellers, and the high-price seller would be forced to lower his price or go bankrupt for lack of business. As this happens simultaneously across the economy, prices equilibrate and change only as a result of exogenous factors or changes in resource availability, taxation, monetary policy, or consumer tastes.

This view of the world is based in part on the assumption that most markets are perfectly competitive, where the concept of perfect competition can be summarized as follows.

A market structure is perfectly competitive if the following conditions hold: There are many firms, each with an insubstantial share of the market. These firms produce a homogenous product using identical production processes and possess perfect information. It is also the case that there is free entry to the industry; that is, new firms can and will enter the industry if they observe that greater-than-normal profits are being earned. The effect of this free entry is to push the demand curve facing each firm downwards until each firm earns only normal profits, at which point there is no further incentive for new entrants to come into the industry. Moreover, since each firm produces a homogenous product, it cannot raise its price without losing all of its market to its competitors. . . . Thus, firms are price takers and can sell as much as they are capable of producing at the prevailing market price. (Pearce 1992, 327–328)

There are no pricing decisions in perfectly competitive markets—prices are determined by the iron law of the market. If one merchant were offering a good for a lower price than another, neoclassical economics assumes that either customers would entirely abandon the higher-price merchant and swamp the lower-price merchant or an arbitrageur would arise who would buy all the goods from the lower-price merchant and sell them at the higher price. In either case, a single market price would prevail. Furthermore, if prices were so high in a sector that sellers enjoyed higher profits than the rest of the economy, more sellers would enter that sector, lowering the average price until the return on capital dropped to the market level. In this situation, there are no pricing decisions at all: Prices are set by the market, as stock prices are set by the New York Stock Exchange or NASDAQ. The price of Microsoft stock is not set by a pricing manager but by the interplay of supply and demand—in fact, most financial instruments such as stocks and bonds satisfy the economic definition of a commodity. Certain other highly fungible goods—grain, crude oil, and some bulk chemicals—also come very close to being commodities. In these markets, there is simply no need for pricing and revenue optimization: the market sets the price.

As any shopper can tell you, most of the real world is much messier—prices vary all over the place, sometimes in ways that seem irrational. Buyers often behave erratically, sellers do not always seek to maximize short-run profit, neither buyers nor sellers are possessed of perfect information, and opportunities for arbitrage are not immediately seized. Table 1.1 shows prices for a half gallon of whole milk at different markets in a 16-block area of the Upper West Side of Manhattan on a single day in November 2019. For a half gallon of non-organic milk, prices ranged from a low of \$2.69 to a high of \$4.00—a variation of \$1.31, or 49%. Furthermore, the price varied by \$1.10 even for two stores on the same block (Duane Reade and Amsterdam Gourmet). How could this be? Why would anybody buy milk at a high price when they could walk less than a block and save \$1.10? Why do arbitrageurs not buy all the milk at the lower price and sell it at the higher?³

The price variation shown in Table 1.1 will hardly come as a shock to most people—after all, both businesses and consumers know that it pays to shop around because suppliers of the same (or similar) products often charge different prices. Furthermore, there are other ways than walking to the next store to pay a lower price for exactly the same product: wait until it goes on sale, travel to a retail outlet, clip a coupon, buy in bulk, buy online, or

TABLE 1.1
Retail prices for a half gallon of whole milk on the Upper West Side of Manhattan, November 2019

Location	Store	Nonorganic price (\$)	Organic price (\$)
74th and Broadway	Fairway	3.89	6.99
79th and Amsterdam	Amsterdam Gourmet	4.00	—
79th and Amsterdam	Duane Reade	2.79	—
77th and Broadway	CVS	2.99	5.49
74th and Columbus	Pioneer	2.69	—
73rd and Columbus	C. P. Yang	—	7.25
74th and Amsterdam	301 Amsterdam	—	7.45
75th and Broadway	Citarella	2.99	5.99
71st and Columbus	JNH Produce	—	7.25
Average		3.23	6.74
Standard deviation		0.52	0.73

try to negotiate a lower price. In fact, not only do prices vary between sellers, but a single seller will sometimes sell the same product to different customers for different prices!

This provides the setting for pricing and revenue optimization—sellers have scope to adjust prices, and the right price to charge at a given time is not usually obvious. In this situation, pricing becomes a decision rather than a fait accompli. If, furthermore, a seller has many prices to set and these prices are subject to frequent change, then the use of a computerized system to manage pricing is almost always required. Pricing and revenue optimization can be viewed as an approach that uses mathematics to set and update multiple prices in a dynamic environment. The approaches used in pricing and revenue optimization are drawn from management science, operations research, marketing science, statistics, economics, and, increasingly, machine learning.

A further impetus to the growth of pricing and revenue optimization is that companies increasingly need to make pricing decisions more and more rapidly to respond to competitive actions, market changes, or their own inventory positions. Sellers often no longer have the luxury to perform market analyses or extended spreadsheet studies every time a pricing change needs to be considered. There is a premium on speed. While there has been a general acceleration of all aspects of business, the impact on pricing and revenue optimization has been particularly notable. This acceleration—and the corresponding interest in developing tools to enable better pricing and revenue optimization (PRO) decisions—has been driven by four trends.

- The success of revenue management in the airline industry provides an example of how pricing and revenue optimization can increase profitability in a real-time pricing environment.
- Increasingly inexpensive data storage and computation makes it feasible to run sophisticated algorithms to calculate and update prices quickly.
- The rise of e-commerce necessitates the ability to manage and update prices in a fast-moving, highly transparent, online environment for many companies that had not previously faced such a challenge.

- The success of machine learning provides new ways to use data to support effective decision making.

Because of their importance to the development of pricing and revenue optimization, we spend a little time considering each of these trends.

1.1.1 The Success of Revenue Management

In 1985, American Airlines was threatened on its core routes by the low-fare carrier PeopleExpress. In response, American developed a revenue management program based on differentiating prices between leisure and business travelers. A key element of this program was a yield management (now more commonly known as *revenue management*) system that used optimization algorithms to determine the right number of seats to protect for later-booking full-fare passengers on each flight while still accepting early-booking low-fare passengers. This approach was a resounding success for American, resulting ultimately in the demise of PeopleExpress.

We delve more deeply into the American Airlines/PeopleExpress story in Chapter 8. For now, the importance of the story is in the publicity it garnered. American Airlines featured its revenue management capabilities in its annual report, emphasizing that it was an application of advanced mathematics. The team that developed the system won a prestigious prize for the best application of management science in 1991. American Airlines' revenue management system was widely touted as an important strategic application of mathematics (Anderson, Bell, and Kaiser 2003), and the tale of American using its superior capabilities to defeat PeopleExpress was the centerpiece of a popular business book (Cross 1997).

Not surprisingly this publicity resulted in widespread interest. Companies began to investigate the prospects of improving the profitability of their pricing decisions. Ford Motor Company was inspired by the success of revenue management at the airlines to institute its own successful program (Leibs 2000). Vendors arose selling revenue management software systems, and consultants offered to help companies set up their own programs. Over the next decade, revenue management spread well beyond the passenger airlines to fields such as hotels and rental cars that had similar problems of managing constrained and perishable capacity.

Under its strictest definition, revenue management has a fairly narrow field of application. In particular, revenue management is applicable when the following conditions are met.

- Capacity is limited and immediately perishable. Most obviously, an empty seat on a departing aircraft or an empty hotel room cannot be stored to satisfy future demand.
- Customers book capacity ahead of time. Advance bookings are common in industries with constrained and perishable capacity, since customers need a way to ensure ahead of time that capacity will be available when they need to consume it. This gives airlines the opportunity to track demand for future flights and adjust prices accordingly to balance supply and demand.

- Prices are changed by opening and closing predefined booking classes. This is a by-product of the design of the computerized distribution systems that the airlines developed. These systems allow airlines to establish a set of prices (fare classes) for each flight and then open or close those fare classes as they wish. This is somewhat different from the pricing issue in most industries, which is not “What fare classes should we open and close?” but “What price should we offer now for each of our products to each market segment through each channel?” The difference is subtle and may not even be visible to consumers, but it leads to major differences in system design and implementation.

Many companies are understandably wary about adopting revenue management programs, protesting that “we are not an airline.” In general, this is the right view—the algorithms behind airline revenue management do not transfer directly to most other industries. However, the experience of the airlines contains several important lessons.

- Pricing and revenue optimization can deliver more than just short-term profitability benefits. Revenue management enabled American Airlines to meet the challenge posed by PeopleExpress. It also meant the difference between survival and bankruptcy for National Rent-a-Car. In 1992, National was losing \$1 million per month and was on the verge of being liquidated by its then-owner, General Motors. At that point, National had been through two rounds of downsizing, and corporate management felt there were no more significant savings that could be achieved on the cost side. As a last-ditch effort, National decided to work on the revenue side. They worked with the revenue management company Aeronomics to develop a system that forecasted supply and demand for each car type and rental length at all 170 corporate locations and adjusted fares to balance supply and demand. The results were immediate.

National initiated a comprehensive revenue management program whose core is a suite of analytic models developed to manage capacity, pricing, and reservation. As it improved management of these functions, National dramatically increased its revenue. The initial implementation in July 1993 produced immediate results and returned National Car Rental to profitability. (Geraghty and Johnson 1997, 107)

- E-commerce both necessitates and enables pricing and revenue optimization. The airlines pioneered electronic distribution—their computerized distribution systems, SABRE and Galileo, were the internet before the internet. These systems allowed immediate receipt and processing of customer booking requests. They also enabled airlines to change prices and availability and have the updated information instantaneously transmitted worldwide. In effect, the airlines were wrestling with the complexities of e-commerce well before the arrival of the internet. As the internet becomes an ever more important channel for sales, the need to continually monitor demand and update prices will only increase.
- Effective segmentation is critical. The key to the success of revenue management in the airline industry is the ability of the airlines to segment customers between early-booking leisure passengers and late-booking business passengers. Note that, for the

airlines, this segmentation was achieved not by direct discrimination—that is, trying to charge a different fare based on demographics, age, or other customer characteristics—but via product differentiation, creating different products that appealed to different segments. Segmenting customers based on their willingness to pay and finding ways to charge different prices to different segments is a critical piece of pricing and revenue optimization—one that we address in detail in Chapter 6.

At heart, airline revenue management systems are highly sophisticated opportunity cost calculators. They forecast the future opportunities to sell a seat and seek to ensure that the seat is not sold for less than the expected value of those future opportunities. Most industries do not face capacity constraints as stark as those faced by the airlines. Manufacturers typically have the opportunity to adjust production levels or store either finished or partially finished goods. Retailers can adjust their stocks in response to changes in demand. However, this does not mean that calculating opportunity cost is irrelevant in these industries. On the contrary, in many industries facing inventory or capacity constraints, opportunity cost can be the critical link between supply chain management and pricing and revenue optimization.

1.1.2 The Evolving Computational Environment

A major motivation for the development of more sophisticated pricing and revenue optimization systems has been the explosion of data available to companies coupled with the continued precipitous decline in the cost of both computation and data storage. Four developments were particularly important in this regard.

- Between 1985—the time that airlines began implementing revenue management systems—and 2019, both the cost of computation and the cost of storage dropped by a factor of about 10,000,000.⁴ This incredible rate of technological progress has had enormous implications not just for pricing and revenue management but for society as a whole.
- Starting in the early 1990s, companies began concerted efforts to organize scattered customer data into a usable format (customer relationship management) as well as to break down corporate data silos to make information available across the organization (enterprise resource planning). The idea behind enterprise resource planning (ERP) is to provide a corporate information backbone that supports all business users with consistent and timely data from a single source. Ideally information will then flow freely among business processes with data islands and fiefdoms eliminated. While ERP efforts have not been universally successful, and many organizations still suffer from data fiefdoms, substantial progress has been made by many organizations toward the goal of retaining all pertinent data and having them readily available to all parts of the organization when they are needed.
- The move from on-premises computing to cloud computing has made information technology more flexible and reduced costs even further. When computing was primarily on-premises, information technology (IT) departments needed to make

vast investments in computers and the technology and personnel to maintain them. With cloud computing (also called computing as a service), companies pay to obtain IT services from a third party that purchases and maintains the needed hardware. Cloud computing uses economies of scale to reduce the cost of computing services below what many companies could achieve in-house. Furthermore, cloud computing converts IT costs from a combination of fixed and variable costs to variable cost only. This provides flexibility for companies to adjust their purchases of computational services to their changing demands without the need to maintain extensive internal IT departments.

- The development of agile methodologies for software development along with improved supporting software tools and languages has made development and testing of pricing approaches much easier at many organizations than in the past. Historically, implementing a new approach to pricing or pricing analytics usually required a major investment in software development and implementation often extending over a year or more. This meant that new approaches had to be rigorously tested and a strong business case established before a company would be willing to invest millions of dollars and years of development time in a new approach. Agile development and supporting tools have made it much easier to implement new ideas at less cost and in a much shorter time frame. Furthermore, it is now often easier for companies to revert to a previous approach if a new approach does not work as planned.

It is worth emphasizing that these trends have affected different companies to vastly different degrees. Long-established companies are often saddled with legacy systems that are hard to replace or modify and that can inhibit adoption of new approaches. The airlines represent a classic example—many of their IT systems are built around the concept of a fare class. Each fare class can have a different fare and airlines can open or close fare classes in order to set different prices according to algorithms described in Chapter 9. However, this is not true dynamic pricing because prices are limited to the preset fares. The concept of a fare class is embedded so deeply into many airline systems from reservations to distribution to accounting that it can be prohibitively difficult for any single airline to move to a pure dynamic pricing system even if it wanted to do so.⁵ Similarly, television networks and their customers, advertising agencies, have developed extensive systems to support a system of up-front pricing and spot pricing that dates from the early days of the commercial television industry. While this system seems outdated, it persists in part because both ad agencies and networks have heavily invested in software systems based on the current market structure.⁶

By contrast, many more recently founded tech companies have implemented more modern, flexible information technology infrastructure that allows them to move more quickly in introducing new pricing approaches. For example, ride-sharing apps such as Uber and Lyft are not restricted to a predefined set of fares and can adjust prices much more flexibly and dynamically than the airlines. The same is true of many online retailers such as Alibaba, Amazon, and Walmart.com. There is irony in this situation—the global distribution infrastructure that the airlines spent billions of dollars to implement in the

1960s and 1970s was revolutionary for the time, but in some ways this once pathbreaking infrastructure is now a barrier to innovation.

1.1.3 The Rise of E-commerce

By the late 1990s, the internet was widely predicted to be a revolutionary and transformative technology that would change everything. The fact that the internet drives a greater need for pricing and revenue optimization is contrary to some early expectations. A number of analysts predicted that internet commerce would inevitably drive prices down to the lowest common denominator: The internet would bring about the world of perfect competition, in which sellers lose control of prices. This was part of the vision behind Bill Gates's concept of "friction-free capitalism" (Gates 1996, 180). However, reality has turned out to be quite different. Studies consistently show that most online buyers actually do little comparison shopping—for example, a McKinsey study showed that 89% of online book purchasers buy from the first site they visit, as do 81% of music buyers. As a result, online prices often vary considerably, even for identical items. Table 1.2 shows the base price and shipped price for 13 online vendors as well as a list and an in-store price for Stephen King's novel *The Institute*; the prices were gathered using an online shopping application.⁷ Note that the delivered price varies by almost \$19.00 across the vendors, with none of the 13 vendors offering the book at the same delivered price. Note also that the two largest online booksellers—Amazon and Barnes and Noble—have neither the highest nor the lowest price, and their prices differ substantially from each other. At least in this case, the internet seems to be perpetuating rather than eliminating price variability!

The resemblance between the distribution of online book prices in Table 1.2 and the distribution of milk prices in Table 1.1 is not coincidental. It shows that, just like any other channel, the internet supports price differentials. For any seller offering a large catalog of products (such as an online bookseller), the price of each item needs to be set intelligently based on cost, inventory, current competitive prices, and other information. The pricing problems facing an online bookseller are similar to those facing retailers: Could I increase my profitability by raising my price? By lowering my price? How should my price be updated as inventory changes—should it be as competitive prices change? As demand changes? What is the right relationship between my base price and the total delivered price? Multiply these questions by a catalog of hundreds of thousands or even millions of items and a need to update frequently, and the full magnitude of the pricing problem faced by online merchants becomes clearer.

While one group of analysts was predicting that the internet would eliminate price differentials and drive prices inevitably toward the lowest common denominator, another group of analysts was predicting exactly the opposite—that the internet would enable one-to-one pricing crafted to the individual propensities of each buyer. According to this school of thought, e-commerce would become a market-of-one environment, in which prices would be calculated on the fly to maximize the profitability of each transaction. A customer entering a website would immediately be identified, and, based on her past buying patterns, a pricing engine would calculate personalized prices reflecting an estimate of her willingness to pay. While various online sellers have been rumored at various times to be practic-

TABLE 1.2
Online book prices for the hardback edition of The Institute, by Stephen King, November 1, 2019

Vendor	Price (\$)	Shipping cost (\$)	Total (\$)
Amazon Marketplace	11.79	4.20	15.99
The Book Depository	21.21	—	21.21
AbeBooks.com	21.44	—	21.44
Alibris	17.65	3.99	21.64
Shophity	21.99	—	21.99
Biblio.com	18.55	3.50	22.05
Amazon.com	16.19	5.99*	22.18
Superbookdeals	19.49	3.99	23.48
Barnes and Noble	21.00	3.99	24.99
Booksrun	25.90	—	25.90
Bigger Books	27.09	3.99	31.08
Ecampus.com	27.65	3.99	31.64
Chegg	28.99	5.99	34.98
Average	21.46	3.05	24.51

NOTE: All prices for new purchase.

* Shipping is free for Amazon Prime members.

ing individualized pricing—for example, the Amazon DVD example in Chapter 14—in reality pure market-of-one pricing is not widely practiced, and there are several reasons to believe it will never become standard practice. For one thing, there is strong buyer resistance to pricing discrimination that is perceived as unfair or arbitrary. People seem no more inclined to accept online price discrimination than they are willing to accept variable pricing at the time of checkout based on a clerk's estimate of their willingness to pay. (Chapter 14 digs more deeply into customer reaction to pricing that seems unfair.) The transparency of the internet means that whatever online pricing system a company adopts, the details will soon be widely known by customers. Price differentiation by online sellers is always rapidly discovered. As a result, market-of-one pricing has not taken over the internet, and when price differentiation is employed, it is usually within careful guardrails.

Given that the internet has neither driven prices to their lowest common denominator nor led to real-time personalized pricing, can we conclude that it has no impact on pricing? Not at all. On the contrary, e-commerce has been a major motivator for companies to improve their pricing capabilities. Four specific characteristics of internet commerce increase the urgency of pricing and revenue optimization.

- The internet increases the velocity of pricing decisions. Many companies that changed prices once a quarter or less now find they face the daily—or even hourly—challenge of determining what prices to display on their website or to transmit to e-commerce intermediaries. Many companies are only now beginning to struggle with this increased price velocity—and it is likely that the pace of price changes will continue to accelerate. For example, one article estimated that in 2018 Amazon changed 2.5 million prices every day (Mehta, Detroja, and Agashe 2018). All indications are that online pricing is likely to become even more dynamic in the future.
- The internet makes available an immediate wealth of information about customer behavior that was formerly unavailable or only available after a considerable time lag. This includes information on not just who bought what but also who clicked on

what, and who looked at what and for how long. This information can be captured and analyzed by companies both to support cross-selling and up-selling and to understand customer behavior and segmentation.

- The internet provides the ability to experiment rapidly and effectively with pricing alternatives and different pricing models. As McKinsey pricing consultants point out, “Traditional price-sensitivity research can cost up to \$300,000 for each product category and take anywhere from six to ten weeks to complete. . . . On the Internet, however, prices can be tested continually in real time, and customers’ responses can be instantly received” (Baker, Marn, and Zawada 2001, 121). Many e-commerce companies experiment frequently with different prices and different pricing approaches.
- Even in cases where a customer does not buy online, the internet may provide deeper information about costs and competitive prices. This has been particularly true in big-ticket consumer purchases, such as home mortgages and automobiles. A 2019 Cox Automotive survey found that car buyers spend 61% of their shopping time online rather than in a dealership and 80% visited at least one third-party site (Cox Automotive 2019). This means that more and more buyers arrive at a dealership with full information on what is available and what the dealer paid for a car. In this environment, sellers need to be able to use intelligent targeted pricing to remain profitable.

Because the internet has had such an important influence on pricing, we pay special attention to the problems of optimizing prices on the internet throughout the book.

1.1.4 Machine Learning Reaches Maturity

One of the consequences of the rapid evolution of the computational environment described in Section 1.1.2 was a revival of interest in and rapid development of a number of computational approaches that once went under the name of artificial intelligence (AI). Artificial intelligence is a somewhat vague term used for computational agents that can perceive the environment and make decisions in order to attain a goal. Artificial intelligence had experienced a surge of interest and funding from both investors and government research agencies in the mid-1980s into the early 1990s. At the time, the results failed to live up to the absurdly high expectations set by the technology’s advocates, and both interest and funding waned for more than a decade—the so-called AI winter. The fortunes of AI began to revive around the turn of the millennium as increased computational capabilities enabled faster and more efficient implementation of existing algorithms and stimulated the development of new algorithms.

The subfield of artificial intelligence that is most relevant to pricing and revenue optimization is machine learning, which can be defined as the ability of computers to detect patterns in data and make recommendations from data with a minimum of domain-specific modeling.⁸ Machine learning is now commonplace within most large organizations. It has proven to be extremely effective at *supervised learning* tasks that involve classifying objects into different categories. A widely publicized example is Google’s Inception, which can

identify objects, people, and animals in photographs with extremely high levels of reliability. Machine learning approaches have proven equally or even more proficient than traditional methods at more prosaic tasks such as evaluating credit risk or determining whether a transaction is fraudulent. Machine learning has become the weapon of choice for such applications and is increasingly supplanting previous approaches.

Because of this success, machine learning is often offered as a panacea for any complex business decision. For pricing, this sometimes manifests itself in the view that a neural net should simply be trained to determine the best price for any type of product under any circumstances and then turned loose to set and update prices over time. Unfortunately, things are not quite so simple. In particular, training a neural net generally requires observations that span the range of combinations of possible decisions and environmental factors with enough observations to enable statistically reliable recommendations of price for all future combinations that might occur. For most companies, this is still infeasible because of the high dimension—the large number of variables—of the pricing problem. But if we combine machine learning with some straightforward and plausible assumptions about how people respond to prices—for example, higher prices lead to lower demand—we can develop approaches that are both feasible and reasonably efficient.

All four factors that have accelerated the field of pricing and revenue optimization—the success of revenue management, the evolving computational environment, the rise of e-commerce, and the increasing sophistication of machine learning—point in the same direction, toward a future in which pricing will be increasingly dynamic and supported by a wealth of information and sophisticated algorithms. Both the time available to set prices and the allowable margin of error will continue to decrease while the complexity of pricing decisions will increase. Time-consuming offline analyses will become increasingly irrelevant—their results will be obsolete before they can be completed because the world moves too quickly. Automated pricing and revenue optimization systems will increasingly be required to deal with the speed and complexity of pricing decisions.

1.2 THE FINANCIAL IMPACT OF PRICING AND REVENUE OPTIMIZATION

Of course, the most compelling reason for a company to improve its pricing and revenue optimization capabilities is to make more money. “For most companies, better management of pricing is the fastest and most cost-effective way to increase profits” (Marn and Rosiello 1992, 85). So concluded a pioneering study by McKinsey and Company, which concluded that a 1% improvement in profit would, on average, result in an improvement in operating profit of 11.1%. By contrast, 1% improvements in variable cost, volume, and fixed cost would produce operating improvements of 7.8%, 3.3%, and 2.3%, respectively. This analysis has often been replicated: a 2018 book gave an example of a case in which a 5% improvement in price would lead to a profit improvement of 50%, while 5% improvements in unit costs, sales volume, and fixed costs would lead to profit gains of 30%, 20%, and 11.5%, respectively (Simon and Fassnacht 2018).⁹ While other studies have had different results, the overall pattern is clear: improving pricing can have a significant positive impact on profit and may be a more profitable area for investment than cost reduction.

When it comes to the benefits of implementing an actual system, the passenger airlines typically claimed between 8% and 11% revenue increase from the initial implementation of revenue management systems (Smith, Leimkuhler, and Darrow 1992).¹⁰ A *Harvard Business Review* article noted that some retailer early adopters achieved “gains in gross margins in the range of 5–15%” from the use of optimization-based assortment and pricing optimization systems (Friend and Walker 2001, 138). Similar gains from markdown optimization have been reported elsewhere—an A/B test measured an increase of 6% in revenue from distressed inventory at the fast-fashion retailer Zara (Caro and Gallien 2012). More recently, a 2019 Gartner report noted that “Pricing Optimization and Management” software “has been observed to deliver a rapid return on investment when well-implemented and enthusiastically adopted” (Lewis 2019, 8). The same report noted that vendors report increased revenue of 1–2% and increased margins of 2–10%.¹¹

Putting it all together, we can see there is a strong case for many companies to consider investing in pricing and revenue optimization. Not only is improving pricing already the “fastest and most cost-effective way to increase profits,” but it is gaining in importance as the velocity and complexity of pricing decisions inexorably increase. Furthermore, a new generation of information technology provides the information and algorithmic power needed to analyze and exploit market opportunities. Finally, pricing is often the corporate function that can generate the most return with the least investment.

1.3 ORGANIZATION OF THE BOOK

The structure and dependencies of the remaining chapters are illustrated in Figure 1.1. The topic of each chapter is outlined briefly next.

Chapter 2 discusses pricing and revenue optimization as a corporate process and contrasts it with other approaches to pricing. It stresses that PRO is a highly dynamic process, dependent on continual feedback to ensure that pricing decisions are kept in line with changing market realities. It also introduces the concept that the core of pricing and revenue optimization lies in the formulation of pricing and availability decisions as constrained optimization problems.

Chapter 3 discusses the mathematical representation of customer response to price. It introduces the idea of a price-response function and shows how a price-response function is based on the distribution of willingness to pay across a population. The chapter presents several common forms of price-response function and shows how the price-response function can incorporate competitive pricing. Chapter 4 shows how price-response functions can be estimated using historical data on prices and demand. It also discusses data-free approaches to estimating a price-response function.

Chapter 5 shows how optimal (profit-maximizing or revenue-maximizing) prices can be calculated given a price-response function. It defines *incremental cost* and how it influences pricing. The chapter also shows how optimal prices can be calculated using explicit optimization as well as using an adaptive reinforcement learning approach. In addition, Chapter 5 introduces the concept of an efficient frontier as a way to visualize the trade-offs between competing objective functions such as maximizing profit and maximizing revenue or market share.

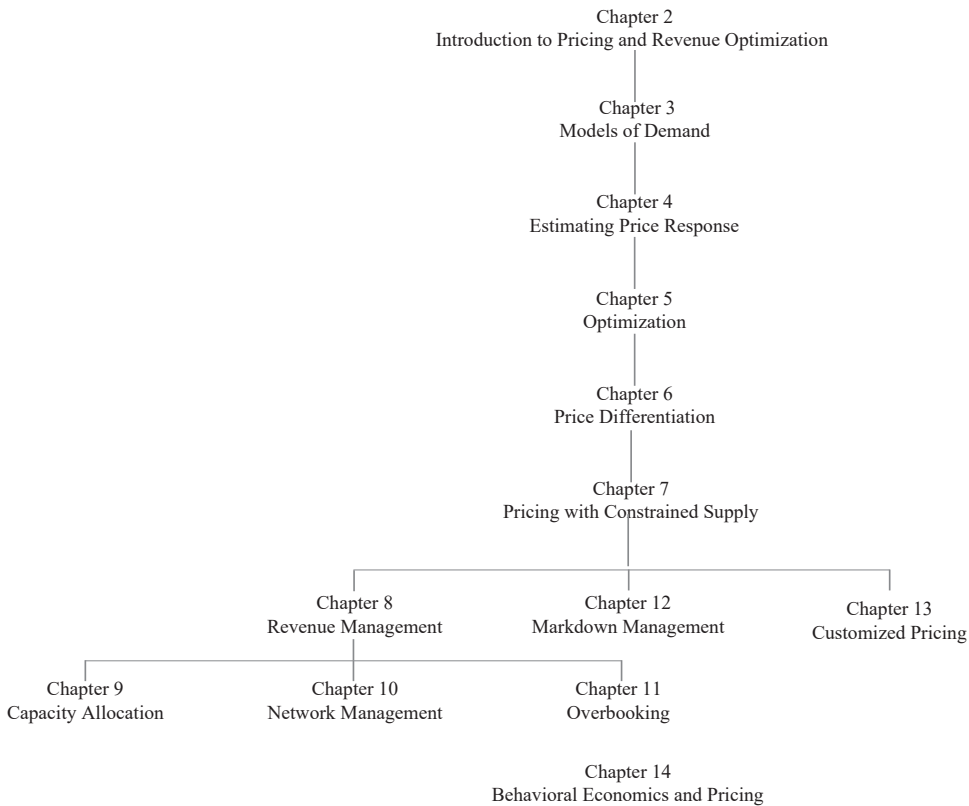


Figure 1.1 Relationships among the remaining chapters.

Price differentiation is one of the key concepts in pricing and revenue optimization. Chapter 6 discusses how markets can be divided into different market segments such that a different price can be charged to each segment. Tactics for price differentiation include virtual products, product lines, group pricing, channel pricing, and regional pricing. The idea of *multipricing* is the source of both the benefits of PRO and the challenges involved in managing a large portfolio of prices. Chapter 6 discusses how to optimize differentiated prices in the face of potential cannibalization.

Chapter 7 introduces another major theme in pricing and revenue optimization—pricing when supply (or inventory) is constrained. Supply constraints are ubiquitous. They may arise from the intrinsically constrained capacity of a service provider such as a hotel, restaurant, barbershop, or trucking company; they may be due to limited inventory on hand; or they may be due to bottlenecks in a supply chain that restrict the rate at which a good can be produced or transported. In any case, constrained supply significantly complicates optimal pricing. Chapter 7 also introduces the key concept of an *opportunity cost*—the incremental contribution lost because of a supply constraint. Chapter 7 also gives examples of pricing with constrained supply in several different industries including airlines, ride-share services, electric power, and theater.

Revenue management has been one of the most important and publicized applications of pricing and revenue optimization over the past three decades. While it has its origin in

the airline industry, it has spread beyond the airlines and is in widespread use at hotels, rental car firms, cruise lines, freight transportation companies, and event ticketing agents. Some of the core techniques are gaining acceptance and use well beyond these industries. Four chapters in this book are devoted to revenue management. The first, Chapter 8, describes the background, history, and business setting of revenue management. Chapter 9 is devoted to capacity allocation, the techniques used to determine which fare classes should be open and which closed at any time for a simple product using only a single product—for example, a single airline flight or a single hotel room night. Chapter 10 extends this analysis to the case of a network, in which an individual product may use many different resources—for instance, multiple connecting flights or multiple-night hotel stays. Chapter 11 discusses overbooking—the question of how many units of a constrained product should be sold when customers may not show or may cancel.

Markdown management, the topic of Chapter 12, is a popular application of pricing and revenue optimization. In a markdown industry, a merchant has a stock of inventory with a value that decreases over time. His problem is to determine the schedule of price reductions to take in order to maximize the return from inventory. Applications are widespread, from fashion goods, to consumer electronics and durables, to theater tickets. Chapter 12 describes basic markdown optimization models and some of the challenges in implementing markdown management in the real world, including the issue of strategic customers who may anticipate the markdown and thus wait for the sale.

Chapter 13 treats customized prices. Customized pricing is common in business-to-business settings where goods and services are sold either through long-term contracts or as part of large individual transactions. In these settings, list prices may not exist or may serve only as guidelines to which discounts are applied. The pricing and revenue optimization challenge is to determine the discount to provide to each customer in order to maximize the *expected profitability* of the deal. This requires estimating the trade-off between the probability of winning the deal and the marginal contribution if the deal is won.

Chapter 14 treats the issue of customer perception and acceptance of pricing tactics. Much of the classical theory of pricing is based on the idea that consumers are rational maximizers of utility and that prices are emotionally neutral signals that guide purchasing decisions. However, both common sense and recent research in behavioral economics have shown that this idea is incomplete at best and misguided at worst. Consumers can care—sometimes deeply—about prices and the way they are presented. In particular, pricing that is perceived as unfair can trigger an emotional rejection. Chapter 14 discusses the implication of consumer irrationality for pricing.

1.4 FURTHER READING

A popular and accessible account of the development of economic thought in the West is Robert Heilbroner's *The Worldly Philosophers* (1999). The classic account of early economic manias is *Extraordinary Popular Delusions and the Madness of Crowds*, by Charles Mackay, originally published in 1841 and reprinted many times since. Mike Dash's book *Tulipomania* (1999) is a very readable account of the incredible boom and bust in tulip prices in the

seventeenth-century Dutch Republic. John Carswell's *The South Sea Bubble* (2001) is the definitive account of the eighteenth-century British stock scandal. *A Conspiracy of Paper* (2000), by David Liss, is an entertaining fictional treatment of the bubble.

For a history of pricing in the West (as opposed to what philosophers, theologians, and economists *thought* about prices) and the emergence of different pricing modalities, see Phillips 2012c.

1. “He who sells anything . . . shall not ask two prices for that which he sells, but he shall ask one” (Plato 2008, bk. 11). “The guardians of the law . . . shall consult with persons of experience, and find out what prices will yield the traders a moderate profit, and fix them” (intro.). Plato not only disapproved of dynamic pricing; he disliked hotel revenue management as well: “But now that a man goes to desert places and builds houses . . . and receives strangers who are in need of the welcome resting-place, and gives them peace and calm when they are tossed by the storm, or cool shade in the heat; and instead of behaving to them as friends, and showing the duties of hospitality to his guests, treats them as enemies and captives who are at his mercy, and will not release them until they have paid the most unjust, abominable, and extortionate ransom—these are the sorts of practises, and foul evils they are, which cast a reproach upon the succor of adversity” (bk. 11).

2. Robert Heilbroner cites a 1639 sermon in which the minister of Boston inveighed against such false principles of trade as “that a man might sell as dear as he can and buy as cheap as he can” and “that he may sell as he bought, though he paid too dear” (1999, 23).

3. Milk prices for the same nine stores were reported for May 2002 in the first edition of this book. Milk was cheaper in 2002, with an average price of \$1.75 versus \$3.23 in 2019, and the organic option was uncommon. However, there was still significant price dispersion in 2002, with prices showing a standard deviation of \$0.20. One way to compare the amount of dispersion for distributions with different means is to use the *coefficient of variation*, which is defined as the standard deviation divided by the mean. The coefficient of variation of (nonorganic) milk prices in 2002 was 11%, while in 2019 it was 16%. This illustrates the general trend for price dispersion to increase as prices increase.

4. The cost factors for computation and storage are from Muehlhauser and Rieber 2014. They have been extrapolated slightly from 2014 to 2019. Other estimates of the decline in costs for storage and computation vary slightly, but all agree that these costs have dropped by a factor of more than 1 million over the period from 1985 to 2019.

5. In October 2019, *Business Travel News* reported that six airlines were piloting a dynamic pricing system that was built on top of the existing fare-class system as a tentative first step toward true dynamic pricing (*Business Travel News* 2019). This would suggest that widespread use of true dynamic pricing seems to be far in the future for the industry.

6. For example, I argue elsewhere that “a key element in the persistence of the upfront market appears to be the institutional and human capital infrastructure that has developed around it” (Phillips 2012c). For more on the up-front price/spot price structure in the television advertising industry, see Phillips and Young 2012.

7. I chose the popular novel *The Institute* for three reasons. First, it was widely available as new from multiple vendors. Second, it was also only available in a single physical edition, ensuring apples-to-apples comparison, and third, as a best seller, it had a large market, which means that changing its price would have a meaningful influence on sales.

8. The definition of machine learning is mine. It emphasizes that machine learning approaches to pricing are distinguished from older approaches by the fact that they require fewer assumptions about the nature of customer response—for example, that some pre-chosen functional form captures customer response to different prices.

9. Similar calculations with the same general conclusion can be found in many books and articles. The calculations need to be taken with a grain of salt because they typically compare the profit implications of a 1% increase in price with a 1% change in other metrics such as a 1% decrease in cost. This does not reflect the fact that pricing optimization involves both raising and lowering prices.

10. It is difficult to obtain more recent figures because every major airline uses some form of automated revenue management, which means that manual baselines for comparison are no longer available.

11. Since these figures were reported by vendors who have a vested interest in advertising substantial improvements from their systems, they are likely to be biased upward.

2

INTRODUCTION TO PRICING AND REVENUE OPTIMIZATION

This chapter introduces the basic concepts behind pricing and revenue optimization. We first look at some of the common pricing challenges faced by organizations—most notably a lack of consistent management, discipline, and analysis across pricing decisions. The chapter then describes three purist approaches to pricing—cost-plus, market based, and value based—and discusses some of their shortcomings. It then introduces pricing and revenue optimization. At the highest level, pricing and revenue optimization is a process for managing and updating pricing decisions in a consistent and effective fashion. At the core of this process is an approach to finding the set of prices that will maximize total expected contribution, subject to a set of constraints. The constraints reflect either business goals set by the organization or physical limitations, such as limited capacity and inventory. While the use of constrained optimization is common to all pricing and revenue optimization applications, the type of problem to be solved depends on the specific characteristics of the market. Markets vary in terms of timing or cadence of pricing decisions, the nature of the goods and services being sold, and the type of customer commitment being priced.

2.1 THE CHALLENGES OF PRICING

For many organizations, *pricing* includes a remarkably complex set of decisions. While most companies have a good idea of the list prices they have established for their products, they are often unclear on the prices that customers are actually paying. A multitude of different discounts, adjustments, and rebates are often applied to each sale. For this reason it is critical to distinguish between the *list price* of a good and its *pocket price*—that is, what a particular customer ends up actually paying. The list price is generic, while the pocket price may be different for each customer. The *price waterfall* was introduced by McKinsey and Company as a graphical way of illustrating the discounts that occur between the list price of a good and its pocket price. A consumer package goods (CPG) example is shown in Figure 2.1. In this case, 12 price reductions or discounts are applied between the list price and the pocket price. These include an 8% competitive discount, a 3% sales special, a 1%

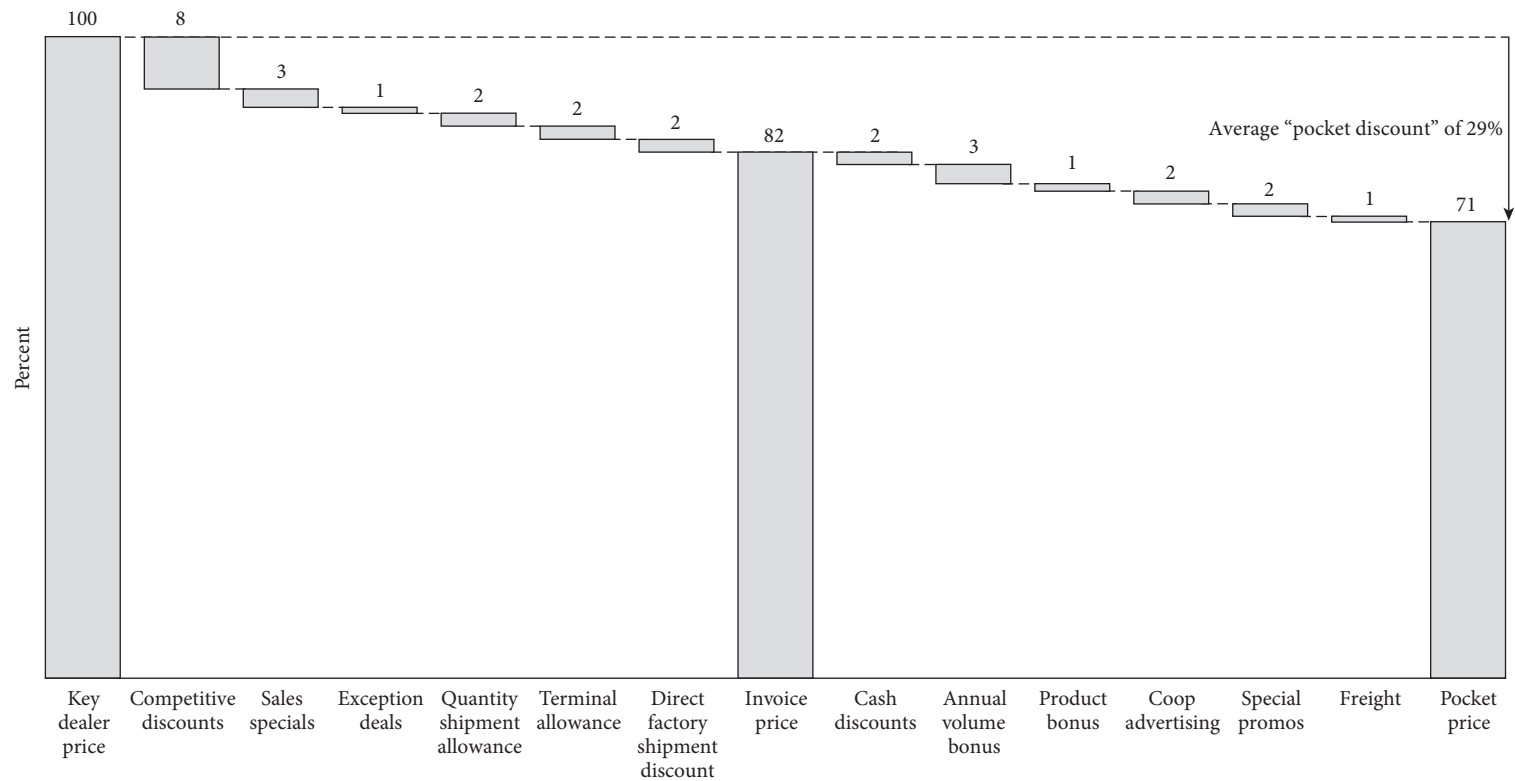


Figure 2.1 Price waterfall for a consumer package goods (CPG) company. *Source:* Courtesy of Mike Reopel of A. T. Kearney.

exception deal, and so on, down to a 1% freight allowance. The net result is that the pocket price for this customer is 29% less than the list price.

The price waterfall illustrates quite neatly that the pocket price paid by an individual customer is often not the result of a single decision, but the cumulative result of many different decisions. In fact, for the majority of companies, many discounts are independently set by different parts of the organization, without consistent measurement or tracking. In Figure 2.1, the competitive discount might have been authorized by the regional sales manager, while the product bonus was determined as part of a general marketing program and the freight allowance was given by the local salesperson in response to a last-minute call by the purchaser. As a result, no one is truly in charge in the sense that no one in the organization is responsible for the fact that the discount offered to this customer was 29%, while that offered to another might be 18%. In fact, not only is no one in charge, but it is often remarkably difficult to determine what the pocket price paid by a particular customer even is. As two McKinsey consultants put it:

The complexity and volume of transactions tend to create a smoke screen that makes it nearly impossible for even the rare senior managers who show an interest to understand what is actually happening at the transaction level. Management information systems most often do not report on transaction price performance, or report only average prices and thus shed no real light on pricing opportunities lost transaction by transaction. (Marn and Rosiello 1992, 86)

Without a consistent process of analysis and evaluation, the probability that a particular pocket price maximizes customer profitability is like the probability that a blindfolded dart player will hit a bull's-eye—not zero but not very high. In fact, the situation can be even worse. Sophisticated buyers often understand a seller's pricing process better than does the seller. A sophisticated procurement department, faced with a price waterfall such as that shown in Figure 2.1, would quickly learn how to divide and conquer to obtain the lowest pocket price. The buyer's procurement agent will seek to extract additional concessions from sales, will argue for relief from strict interpretation of volume purchasing agreements, and will seek extended payment terms from accounts payable. Smart buyers will quickly detect a disorganized or dispersed pricing organization and exploit it to their advantage.

Management often concentrates primarily on invoice prices or list prices. However, the price waterfall illustrates that the majority of important pricing adjustments often take place *after* the list price and even the invoice price. A typical trucking company sells less than 5% of its business at list price—all the rest is sold at a discount. Only a tiny minority of customers purchase a car at the manufacturer's suggested retail price (MSRP). Yet, in many cases, managers spend long hours arguing over and analyzing list prices, despite the fact that list prices have little or no relationship to what most of their customers will be quoted. As Figure 2.1 shows, even companies that focus on the invoice price are still missing much of the important action. In this figure, six different discounts were granted *after* the invoice price.

The distribution of pocket-price discounts given by a CPG company to its various customers over a year is shown in Figure 2.2. In this case, 9% of the customers received a discount of greater than 40%, while 16% received discounts between 35% and 40%, and

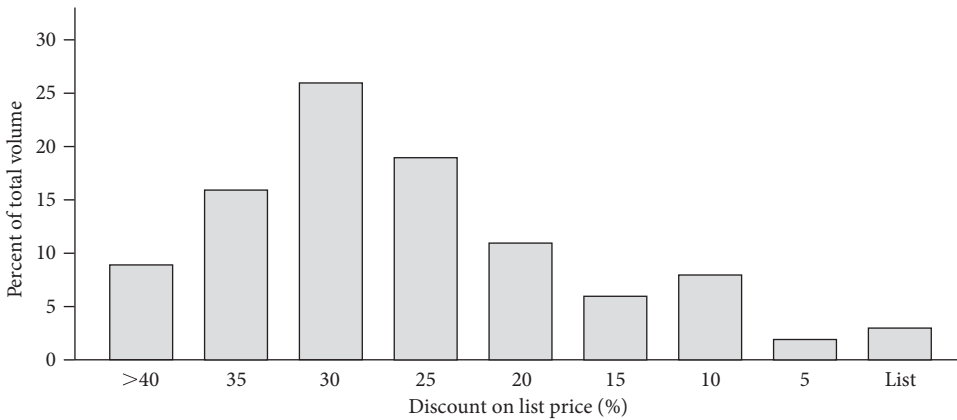


Figure 2.2 Pocket-price distribution for a consumer package goods company.

Source: Courtesy of Mike Reopel of A. T. Kearney.

only 3% paid list price. This distribution represents a fairly typical spread of discounts for the CPG industry. Note that the distribution in itself does not tell us anything about the quality of the pricing decisions being made by the company. However, it immediately demonstrates two facts.

- *The item being sold is not a commodity.* The variation in pocket prices means that customers are willing to pay a range of prices for the item and the company is willing to sell the product at different prices.
- *Only 3% of customers bought at list price.* For this item, setting the list price is not the critical pricing and revenue optimization (PRO) decision. Rather, list price is being set high and discounts are being used to target prices to individual customers. The key decisions are what discounts to offer each customer.

It should be stressed that the existence of a pocket-price distribution such as the one shown in Figure 2.2 does not by itself say anything about the quality of the pricing decisions. A company practicing sophisticated pricing optimization may also show a wide distribution of prices. The question is: *Is the pocket-price distribution the result of a conscious corporate process based on sound analysis, or is it the result of an arbitrary process?*

A key measure of the quality of PRO decision making is the extent to which the pocket price correlates with customer characteristics that are indicators of price sensitivity. In many cases, large customers are more price sensitive than smaller customers. For this reason, many companies offer higher discounts to their larger customers. If customer size is determining discounts, a seller would expect discount levels as a function of customer size to lie within a band something like that shown in panel A of Figure 2.3. The actual mapping of discount level against customer size for the CPG company is shown in panel B of Figure 2.3. The correlation of discount to customer size was only about 0.09 for this company—statistically indistinguishable from random. This means that something other than customer size is determining the level of discounts.

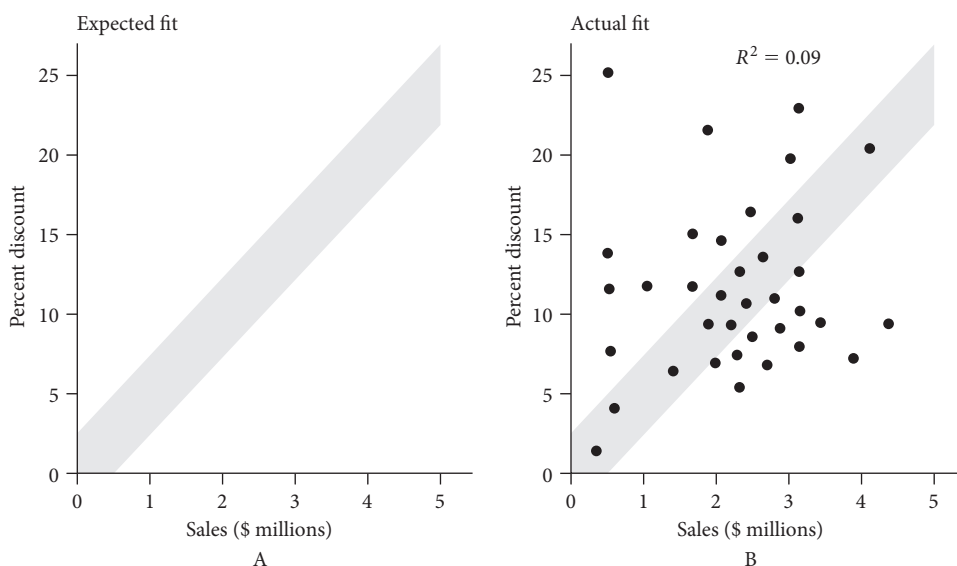


Figure 2.3 Correlation of discount with customer size—consumer package goods example. Panel A shows management expectations; panel B shows the actual distribution. *Source:* Courtesy of Mike Reopel of A. T. Kearney.

Graphics such as the price waterfall in Figure 2.1 and the pocket-price distribution histogram in Figure 2.2 are useful to help companies assess the current state of their pricing. The price waterfall can identify how pricing responsibilities are dispersed across the organization and where sophisticated buyers may be utilizing divide-and-conquer techniques to win higher discounts. The pocket-price histogram can give an idea of the breadth of discounts being given to customers. Graphs such as those shown in Figure 2.3 can show the extent to which discounts correlate to customer characteristics such as size of customer, size of account, and mix of customer business. These graphs often form part of a preliminary diagnostic analysis and can be used to illustrate the need for better pricing decisions.

2.2 TRADITIONAL APPROACHES TO PRICING

Pricing and revenue optimization incorporates costs, customer demand (or willingness to pay), and the competitive environment to determine the prices that maximize expected net contribution. Other approaches to pricing tend to weigh one of these three aspects more than the others, as shown in Table 2.1. *Cost-plus pricing* calculates prices based on cost plus a standard margin. *Market-based pricing* bases prices on competitors' prices. *Value-based pricing* bases prices on estimates of how customers value the good or service being sold. The finance department tends to like cost-based pricing because it guarantees that each sale produces an adequate margin, which seems fiscally prudent. The sales department tends to like market-based pricing because it helps them sell against competition. The marketing department is often the natural supporter of value-based pricing.

TABLE 2.1
Alternative approaches to pricing

Approach	Based on	Ignores	Liked by
Cost-plus	Costs	Competition, customers	Finance
Market based	Competition	Cost, customers	Sales
Value based	Customers	Cost, competition	Marketing

Each of these purist approaches has its strengths and weaknesses, which are discussed in the following sections.

2.2.1 Cost-Based Pricing

Cost-based pricing, also known as *cost-plus pricing*, is perhaps the oldest approach to setting prices and still one of the most popular. It has a compelling simplicity—determine the cost of each product and add a percentage surcharge to determine price. The surcharge is often calculated to reflect an allocation of fixed costs plus a required return on capital. Alternatively, it may simply be based on tradition or a rule of thumb. For example, an old rule of thumb in the restaurant industry is “food is marked up three times direct costs, beer four times, and liquor six times” (Godin and Conley 1987, 58).

The cost-plus pricing approach appears to be objective and defensible. If all competitors in a market have similar cost structures, it would appear to be a reasonable way to ensure consistency with the competition. In addition, cost-based pricing gives the appearance of financial prudence. After all, if all of our products are priced with the right surcharge, the company is guaranteed to make back the cost of production plus fixed costs plus the required return on capital. Everybody, including the shareholders, should be happy. It is not surprising that cost-based pricing, with its dual appeals to objectivity and financial prudence, often appeals to finance departments.

The major drawback of cost-plus pricing is widely recognized: It is an entirely inward-focused exercise that has nothing to do with the market. Calculating prices without any reference to what customers might (or might not) be willing to pay for your product is an obvious folly: Customers do not care about your costs—they care about what they have to pay for the product. Furthermore, cost-based pricing does not support price differentiation—the ability to charge different prices to different customer segments—which is at the heart of pricing and revenue optimization.

Another problem with cost-plus pricing is that the costs used as its basis are often nowhere near as objective as they seem (and as the finance department may believe them to be). The calculation of the variable and fixed costs involved in the production of a complex slate of products involves innumerable subjective judgments. Furthermore, all of the hard cost numbers available to the organization are based on historical performance—production costs in the future may be widely different as the mix of business changes and as production efficiency changes. Basing pricing decisions strictly on costs plus a surcharge can lead to highly distorted prices driving lower-than-expected results. This can yield results that seem puzzling, since the prices were based on seemingly objective costs.¹

Given these drawbacks, it should not be surprising that experts are uniformly harsh toward cost-plus pricing.

- “The problem with cost-driven pricing is fundamental” (Nagle and Müller 2018, 4).
- “Cost-plus pricing is not an acceptable method” (Dolan and Simon 1996, 38).
- “Does the use of a rigid, customary markup over cost make logical sense in the pricing of products? Generally, the answer is no” (Lilien, Kotler, and Moorthy 1992, 207).

Despite nearly uniform condemnation, cost-based pricing is surprisingly resilient. As one author put it, “Cost-plus pricing is a lot like the romance novel genre, in that it’s widely ridiculed yet tremendously popular” (Dholakia 2018). In fact, surveys consistently find that cost-plus pricing is the most widely used approach to pricing across different countries and industries. For example, a 2009 survey across multiple industries in the United States, India, and Singapore found that “the most frequently used strategy was cost-plus pricing (47.2 percent of firms)” (Rao and Kartono 2009, 17). A 2010 survey of British industry found that 44% of respondents priced all or some of their products using cost-plus pricing (Greenslade and Parker 2012).² The only good news about the persistence of cost-based pricing is that it means there is substantial opportunity for improvement.

2.2.2 Market-Based Pricing

Market-based pricing means different things in different contexts. In this book it refers to the practice of pricing based solely on the prices being offered by the competition. It is commonly applied by smaller players in situations in which there is a clear market leader—for example, a small cola brand might set its price based on the price of Coca-Cola. A survey found that 31.7% of firms in the United States, Singapore, and India “match the price set by the overall market or price leader” (Rao and Kartono 2009, 20). Market-based pricing is, of course, also the practice in pure commodity markets, such as bulk chemicals, in which offerings are completely identical and transaction prices are rapidly, perfectly communicated. In these cases, there is no pricing decision per se—all companies take the price as given and adjust their production accordingly. For a commodity, there is no alternative to market-based pricing.

Market-based pricing can also be an effective strategy for a low-cost supplier seeking to enter a new market. For example, Alamo Car Rental started as a low-cost rental car company targeting the price-sensitive leisure market. Alamo’s initial strategy was to ensure it was always priced at least \$1.00 lower than both Hertz and Avis on the reservation system displays used by travel agents. This strategy was effective at meeting the strategic goal of rapid growth and penetration of the leisure market. Similarly, the European airline Ryanair has a strategy of having the lowest fare on the market compared to all competitors serving the same route within two hours of departure (Eisenacher 2019).

While market-based pricing is appropriate in certain cases—in a commodity market, for small players in a market dominated by a large competitor, and as a way to drive market share—it is often used in cases where it is less appropriate. At its most extreme, it means

letting the competition set our prices. Slavishly following competitive prices does not allow us to capitalize on the changing value perceptions of customers in the marketplace. Furthermore, it does not allow us to capitalize on the differential perception that customers hold of us versus the competition. We should charge a higher price to customers who value our product or brand more highly. Monitoring competitive prices and making sure we maintain a realistic pricing relationship with key competitors is always important—but we also need to adjust our position relative to our competitors to reflect current market conditions if we want to maximize profitability.

If all of the competitors in a market use market pricing, it is not clear where the price actually comes from and pathological effects can occur. One of these pathological effects was uncovered on April 18, 2010, when a postdoctoral student working in Michael Eisen's lab at the University of California, Berkeley, searched on Amazon for a book titled *The Making of a Fly: The Genetics of Animal Design*, by Peter Lawrence. He was surprised to find that the book was being offered new by two different independent book sellers on the Amazon marketplace at prices of \$18,651,718.08 by Profnath and \$23,698,655.93 by Bordeebook. Adding insult to injury, both sellers asked an additional \$3.00 for shipping.

Digging in a bit, Eisen found that both sellers were using pure market pricing—that is, each seller was basing its price entirely on the price being offered by the other. Profnath was setting its price at 99.83% of Bordee's price—presumably to be priced just below its rival. However, Bordee was setting its price 27% higher than Profnath. In the absence of orders, the price began to spiral upward. Assuming that Profnath originally priced the book at its list price of \$70, Bordee would respond with a price of \$88.90. Profnath would then slightly undercut the price at \$88.75, and Bordee would respond with a price of \$112.71, which Profnath would undercut at \$112.52. The power of geometric growth meant it would take only about 53 of these cycles to arrive at prices over \$18 million.³

It seems clear that the process that produced the exorbitant book pricing was the result of algorithmic pricing on the part of both sellers using extremely simplistic market-based pricing rules. One lesson is that algorithmic pricing requires sensible guardrails to avoid having prices wander into unacceptable values. However, the more important lesson is that setting prices based entirely on competitor prices can easily lose touch with the marketplace, particularly if all the competitors are also basing their prices on competition. Like the other pure pricing tactics, market-based pricing does not generally lead to an optimal price.

2.2.3 Value-Based Pricing

Like market-based pricing, *value-based pricing* (or *value pricing*) means different things in different contexts. In its broadest sense it refers to the unexceptional proposition that price should relate to customer value. In its narrowest sense, it is sometimes used as a synonym for *personalized* or *one-on-one* pricing, in which each customer is quoted a different price based on her value for the product being sold. In this book it refers to the belief that customer value should be the key driver of price.⁴ Historically, value-based pricing usually referred to the use of methodologies such as customer surveys, focus groups, and conjoint analysis to estimate how customers value a product relative to the alternatives, which is then

used to determine price. This type of value-based pricing is employed most frequently for consumer goods—especially when a new product is being introduced. In a 2003 survey, 52% of companies responding from the United States claimed that they “price this product based on our customers’ perceptions of the product’s value” (Rao and Kartono 2009, 15).

There is nothing wrong with the basic idea behind value-based pricing. If a seller faces no competition *and* the seller can determine the value each customer places on his product *and* charge that value (assuming it is higher than incremental cost) *and* the seller does not need to worry about arbitrage or cannibalization (or about being regulated), then that is what he should do to maximize profit.⁵ This pure approach to pricing has one serious drawback: it is impossible. For one thing, there is no way to discern individual customer value for a product at the point of sale. In addition, the possibility of arbitrage and cannibalization almost always limits the ability to charge different prices for different products. Finally, competitive pressure means that companies almost always have to set prices lower than they would prefer to any group of customers.

The competitive restriction on value-based pricing is worth emphasizing. There is a great difference between the value that a potential buyer might place on our product in isolation and what we can actually get that customer to pay in a competitive market. A customer may value our product or services highly, but she also almost always has alternatives. For example, management consulting firms routinely discuss changing from cost-based pricing (hours worked times rate per hour) to value-based pricing, under the belief that “a good consultant could boost earnings using a value-based model” (McLaughlin 2002, 1). Yet less than 5% of consultants use value-based pricing. Why should this be? Consider a brilliant management consulting organization that can provide services to a client that will lead to improved profitability of \$2 million yet only cost \$500,000 to provide. If the consulting company has a true monopoly, it should be able to close the deal at \$1.95 million, leaving the client \$50,000 ahead. But what if there is another, slightly less brilliant consulting company with the same cost structure that can provide similar services that would lead to improved profitability of only \$1.5 million? That company could counterpropose a project at \$1.4 million, which would be a better deal for the client, who would be \$100,000 ahead. The upshot is that competition can severely restrict the ability of a company to set a price based on value, even when the competition is offering an inferior product. Even the existence of an inferior substitute will mean that a company cannot charge full value.

2.2.4 Summary of the Traditional Approaches

Cost-based pricing, market-based pricing, and value-based pricing are pure pricing approaches—each with serious drawbacks. In reality, most companies are not purists. While a company is likely to have a dominant philosophy, very few companies use a single approach 100% of the time, and companies will modify their approaches to achieve different goals. When Xerox wanted to increase market share, it would place the pricing function in the sales organization—prices would drop, and sales would soar, but unit margins would drop. When Xerox wanted to increase profits, it would move the pricing function into the finance department—prices would increase, as would profits, although sales would drop

(Eric Mitchell, pers. comm., 2003). Other companies are less disciplined—their approach to pricing may change with the flavor of the month: market based when the emphasis is on market share, value based when focusing on the customer comes into vogue, and cost based when profits are required.

Vacillating among pricing approaches may actually be better than strict devotion to one approach. Any company that sticks tenaciously to any one of the three pure approaches would likely soon find itself in deep trouble. What is often seen in reality is a hybrid—companies espouse a particular philosophy but use pieces of all three, supplemented by a considerable amount of improvisation. The upshot is pricing confusion; rarely is a consistent justification or approach applied across all pricing decisions. One of the fundamental ideas behind pricing and revenue optimization is that a company should clearly articulate its goal—that is, what it wants to achieve from pricing. Finally, pricing and revenue optimization as described in this book involves combining costs, competitive pricing, and customer value information in determining the price to offer. In that sense, it represents a blend or balance of all three approaches.

2.3 THE SCOPE OF PRICING AND REVENUE OPTIMIZATION

One of the goals of pricing and revenue optimization is to provide a consistent approach to pricing decisions across the organization. This means that a company needs to have a clear view of all the prices it is setting in the marketplace and the ways in which those prices are set.

2.3.1 The PRO Cube

The goal of pricing and revenue optimization is to provide the right price:

- For every product
- To every customer segment
- Through every channel

In addition, the goal is to update those prices over time in response to changing market conditions. The scope of pricing and revenue management is defined by these three dimensions, which can be represented as a cube, as shown in Figure 2.4. Each element within the cube represents a combination of product, channel, and customer segment. Each element (or *cell*) has an associated price. For example, one element might be

Medium-size turbines sold to large customers in the Northeast via the direct sales channel

Another element might be

Replacement gears sold to small companies via online sales

In theory, a different price might be associated with each cell within the PRO cube. In practice, some cells may not be meaningful. Some products may not be offered through some channels, for example. It is also the case that the prices within the PRO cube may not always be independent of each other. Our ability (or desire) to charge different prices through

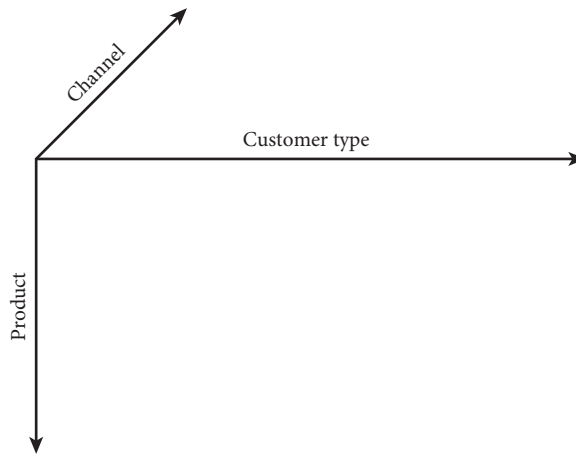


Figure 2.4 Dimensions of the pricing and revenue optimization cube.

different channels may be constrained either by practical considerations or by strategic goals. If we want to encourage small customers to purchase online rather than through our direct sales channel, we might institute a constraint that the online price for small customers for all products must be less than or equal to the direct sales channel price. Considerations such as these need to be incorporated in the business rules applied to the pricing and revenue optimization process as described in Section 2.4.

Companies may offer even more prices than the PRO cube would imply. Certain products might be subject to tiered pricing or volume discounts. Or a company might offer bundles of products at different promotional rates or discounts that are not available for individual products. Each of these bundles or quantity combinations can be treated as an additional, virtual product in the PRO cube.

The PRO cube is a useful starting point for a company seeking to understand the magnitude of the pricing challenge that it faces. Enumerating the combinations of products, market segments, and channels gives a rough estimate of the total number of prices a company needs to manage. Many companies are astonished at the sheer volume of prices they already offer. For example, a bank offered home equity loans with rates that could vary by any combination of credit band, loan size, term, geography, and loan-to-value ratio. When all the possible combinations were enumerated, the bank was surprised to discover that it actually had more than 2 million prices in play for home equity loans at any one time. Such astonishment is often the first step in realizing the need to establish a consistent pricing and revenue optimization process.

2.3.2 Customer Commitments

A core concept in pricing and revenue optimization is the idea of a *customer commitment*. Specifically, a *customer commitment* occurs whenever a seller agrees to provide a customer with products or services, now or in the future, at a price. The elements of a customer commitment include:

- The products and services being offered
- The price
- The time period over which the commitment will be delivered
- The time for which the offered commitment is valid—that is, how long the customer has to make up her mind
- Other elements of the contract or transaction (e.g., payment terms, return policy)
- Firmness of the commitment and risk sharing, including contingencies that might change the price paid by the customer

Some examples of customer commitments include the following:

- A list price is possibly the most familiar form of customer commitment. A list price is a commitment that a buyer can obtain the item by paying the posted price. List price is often (but not always) a nonnegotiable or take-it-or-leave-it price. Common examples are the shelf prices at the grocery store and drugstore, price per gallon displayed on the gasoline pump, and most online retail prices.
- Coupons and other types of promotion are also customer commitments. They allow certain customers to obtain a good at a price lower than list for some period.
- In a business-to-business setting, prices for large purchases are often individually negotiated. The seller may have published list prices, but in many industries items are rarely, if ever, purchased at the list price. Rather, each buyer negotiates an individual discount. The discount that a particular buyer receives can depend on the buyer's purchasing history with the seller, the skills of the purchasing agent and the salesperson, and the desire of the seller to make the sale. Chapter 13 describes an approach to optimizing such customized prices.
- In other business-to-business settings, the buyer and seller negotiate a contract that establishes prices that will be in place for six months, a year, or longer. For example, in less-than-truckload (LTL) freight, shippers agree on a schedule of tariffs that will govern all the shipments they will make over the next year. These schedules are usually expressed in terms of a discount from a standard list tariff. Contracts are individually negotiated between each shipper and each carrier. Contracts may be exclusive (as they often are in package express), or a shipper may establish contracts with two or more carriers, as is common practice in trucking and container shipping.
- Contracts between electronics distributors, such as Ingram Micro and Tech Data, and wholesalers, such as Hewlett-Packard and Sun Microsystems, are often on a tiered-pricing basis, where each tier includes a volume target and an incremental discount for hitting that target. An example is shown in Table 2.2. In this case, the purchaser will receive at least a 5% discount on each unit purchased during the current quarter. If she purchases at least 50,001 units, the discount increases to 6%. And if she purchases 80,001 units, the discount increases to 7%. This type of tiered pricing provides an incentive for the distributor to push the seller's products rather than those of its competitors. Tiered discounts can be based on either units sold or

TABLE 2.2
Tiered pricing

Quarterly units sold	Total discount (%)
0–50,000	5
50,001–80,000	6
80,001–110,000	7
≥110,001	8

revenue. The increased discounts themselves can apply either to incremental sales above the tier or to all sales during the quarter. We examine approaches to setting tiered prices in Section 6.6 on nonlinear pricing.

- Online auction houses such as eBay enable a seller to establish a reserve price for an item she wants to sell. On eBay, the highest bidder above the reserve price will be sold the item. The commitment by the seller is to honor this policy. In addition, the seller may establish a buy-now price that allows immediate purchase. Auctions, either online or offline, are also common for many types of used goods. In an auction setting, the pricing decision on the part of the seller is the reserve and the buy-now price, if applicable.

The forms and types of commitments that sellers make (and buyers expect) vary from industry to industry. As a simple example, airline tickets were historically refundable, meaning that the airline took all the risk of a customer not showing up—a no-show—a risk the airlines sought to mitigate via overbooking. On the other hand, tickets to Broadway shows have historically been nonrefundable, meaning the customer takes on the risk of her no-show decision. As a result, Broadway theaters do not overbook. The details of the types of commitments made in different industries are often the result of complex and contingent historical factors and may change over time—just as airlines are increasingly beginning to adopt nonrefundable ticketing.

In fact, it is not uncommon for individual sellers to utilize different approaches for different products through different channels. For example, an automobile manufacturer is likely to sell the majority of vehicles through dealers, using a combination of list price (in this case, the wholesale price) and promotions. However, General Motors Corporation receives about 11% of its revenue in North America from group sales of cars to corporate, government, and commercial fleets. These sales are generally priced through individual negotiations or bids. Finally, a manufacturer may also be selling some cars directly to the public through a deal with an online channel. In this case, the manufacturer needs to apply pricing and revenue optimization across the entire range of its channels and pricing mechanisms in order to maximize total profitability.

2.4 THE PRICING AND REVENUE OPTIMIZATION PROCESS

Successful pricing and revenue optimization involves two components:

- A consistent business process and organization focused on pricing as a critical set of decisions
- The software and analytical capabilities required to support the process

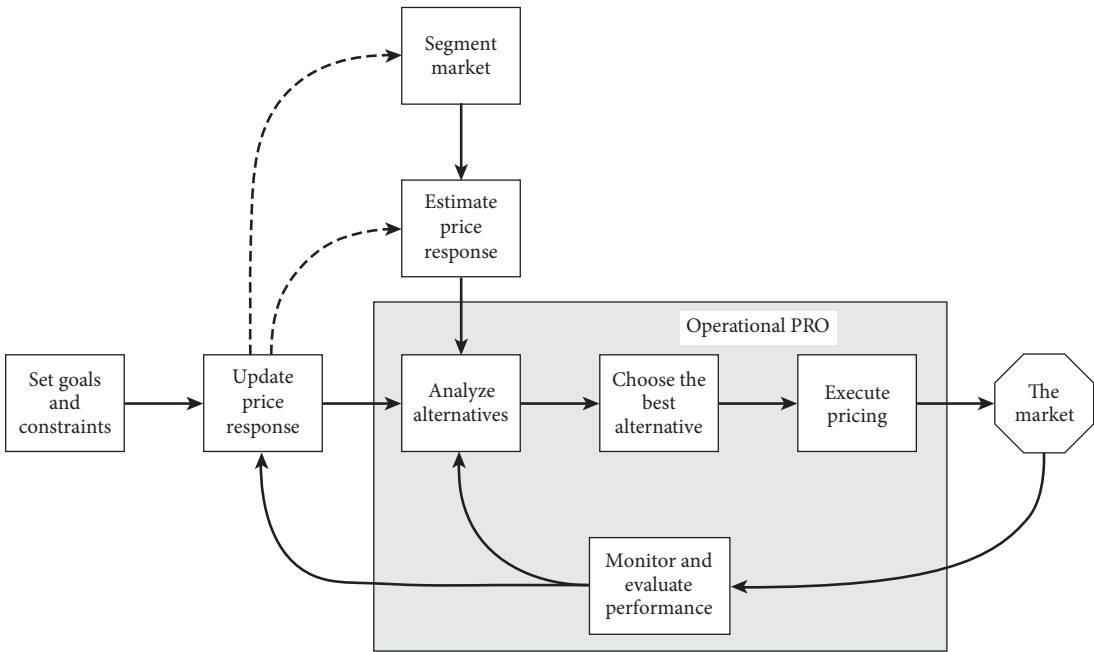


Figure 2.5 The pricing and revenue optimization process.

Much of the interest in pricing and revenue optimization has focused on its use of mathematical analysis. Quantitative analysis is indeed critical in most pricing and revenue optimization settings, but it cannot provide sustainable improvement unless it is embedded in the right process. A diagram of a *closed-loop pricing process* is shown in Figure 2.5. In this figure, the overall pricing and revenue optimization process has been divided into eight activities. Four of these activities are part of operational PRO—that is, they are executed every time the company needs to change the prices it is offering in the marketplace. The remaining four activities—supporting PRO activities—occur at longer intervals. As the figure shows, pricing and revenue optimization is ideally a closed-loop process; that is, feedback from the market must be incorporated into both the operational activities and the more periodic activity of updating market-response functions. Each of the individual activities is discussed below.

2.4.1 Operational PRO Activities

Operational PRO activities work continuously to set and update prices in the marketplace. The timing of the decisions will depend on the application—airline revenue management systems and online retailers can change prices from moment to moment. Other companies may update prices weekly or monthly. Independent of the speed at which prices are changed, the core operational activities will be similar.

Analyze alternatives. This activity is the one most widely identified with pricing and revenue optimization or revenue management. For many companies, it has historically involved the use of spreadsheets to compare pricing alternatives under different scenarios.

Choose the best alternative. The choice of the best alternative requires a clear understanding of the goal for each price in the PRO cube—is it to maximize revenue or maximize price or some combination of the two? When many prices are changing quickly, an algorithm is typically required to solve the underlying optimization problem.

Execute pricing. Finally, the prices that have been calculated need to be communicated to the market. This step is also referred to as *pricing execution*. The mechanics of price transmission and distribution vary greatly from industry to industry, company to company, and even channel to channel within the same company. For an airline or hotel, prices are updated by opening or closing fare classes—a process discussed in more detail in Chapter 8. In other cases, a pricing matrix in a database specifies which prices are available to which customers through which channels. In other cases, new discount guidelines are communicated to the sales force by email. For a company with many products being sold through many channels, communicating the correct prices correctly can be a challenge in itself. A number of pricing execution or pricing management vendors, such as Vendavo, sell software whose primary function is to ensure that complex pricing changes are correctly and efficiently transmitted through all channels. In the face of a complex PRO cube with thousands or even millions of prices, ensuring that each price is correctly calculated and implemented can be a daunting task. One of the important functions of pricing execution systems is to prevent price leakage, in which incorrect prices are published or salespeople price outside established guidelines.

Monitor and evaluate performance. As sellers, when we receive results from the marketplace we need to compare the results with expectations and evaluate overall performance against our goals. Are we achieving the lift we anticipated from a promotion? If not, why not? Was total market demand less than expected? Did the competition react more aggressively than we had anticipated? Were customers less responsive to price than we anticipated? Or was it a combination of all three?

Figure 2.5 illustrates that market feedback occurs at two levels. Analysis of alternatives receives the most immediate feedback. Here, the effects of the most recent actions are monitored so that immediate action can be taken if necessary. The required actions may not always be directly related to pricing and revenue optimization. For example, the markdown management system used by a major retailer indicated that a particular style of pants was selling well below expectations. An investigation found that the pants had been hemmed improperly. Once the problem had been corrected, the pants began to sell at the rate initially expected. In the more common case, a big difference between realized sales and expected sales is likely to be due to an unforeseen change in the marketplace.

The second level of feedback updates the parameters of the market-response functions. If sales of some product are slower than expected, it may indicate that the market is more price sensitive than expected and the future market-response function for that product should be adjusted accordingly.

2.4.2 Supporting PRO Activities

The primary role of the supporting activities is to provide key inputs to the operational PRO activities. The supporting activities occur in a much longer time frame. Typically,

goals and business rules will change quarterly or less frequently. Market segmentations may be updated annually. Price-response functions will be updated much more frequently, typically weekly or monthly in the case of markdown management or customized pricing.

Set goals and business rules. A key initial step in pricing and revenue optimization is to specify the overall goal of the process. Without a clear goal, it is impossible to make consistent decisions and to evaluate decisions in order to improve the process over time. In general, the goal of pricing and revenue optimization is to maximize expected total contribution. However, at certain times, in certain markets, a company may wish to increase market share or attain a volume sales goal. Whatever the goal, it is most important that it be stated clearly and explicitly. As discussed in Section 2.3, the corporate goal for a product, segment, and channel combination determines the form of the objective function that will be used when optimizing that cell of the PRO cube.

Business rules apply whenever there is a restriction that may prevent us from pricing optimally in some market. Is there a minimum per-unit margin requirement we need to meet for all products within a category? Is there a minimum discount we need to offer to certain strategic customers? Do we want our retail list prices to be no lower than our internet prices? Such rules are sometimes referred to as guardrails. Each business rule needs to be implemented as a constraint in the optimization problem at the heart of PRO.

Example 2.1

A heavy-equipment manufacturer wants to make sure that prices in the Northeast are never more than 15% higher than the comparable model being offered by the leading competitor. This decision results in a constraint to be included in the PRO optimization problem: *Our price in the Northeast $\leq$ 115% competitor A's price in the Northeast.*

Note that business rules are *inputs* into PRO—that is, they are decisions made not on the basis of short-term optimization but on the basis of other considerations a company wants to incorporate in its pricing and its willingness to accept the resulting reduction in short-run performance.

Segment the market. An important theme within pricing and revenue optimization is segmenting the market in a way that maximizes the opportunity to extract profit. We discuss techniques for doing this in Chapter 6. The market segmentation needs to be updated periodically to reflect changes in the underlying market. However, this updating should be performed much less frequently than the operational PRO processes. For this reason, the “Segment market” step in Figure 2.5 is shown with a dashed line outside the main process loop.

Estimate price response. For each market segment identified we need to estimate the corresponding price-response functions. How to do this is the topic of Chapter 4.

Update price response. It is not sufficient merely to monitor the performance of PRO actions against expectations. A company needs to update its models to incorporate what it has learned. As long as performance closely matches expectations, the parameters of the models

need not be changed. However, if performance is significantly different from expectations, models need to be updated to reflect the new information.

The most critical message to take away from the pricing and revenue optimization process illustrated in Figure 2.5 is that PRO should be treated like any other critical business process. For a seller to make effective PRO decisions, he needs to clearly identify what he is trying to achieve, the constraints he is facing, and the alternatives available. Based on his understanding of the market and the constraints, he should choose the alternative most likely to achieve his goals. Once this alternative is implemented, he should monitor and measure the results against expectations and update his understanding of the market so that he can make better decisions in the future. The PRO organization, responsibilities, systems, and data-capture methodologies should all be designed to support this process.

We must recognize that however prices are analyzed and calculated, a human being ultimately needs to be responsible for the prices in the market. This does not mean that every price needs to be inspected individually by a human being—a practical impossibility when thousands or millions of prices are changing each day. However, it does mean that prices need to be monitored with appropriate high-level metrics such as the distribution of contribution margin and the distribution of prices relative to competition. Extreme deviations—either high or low—should be audited to ensure they are consistent with the pricing methodology and not the result of calculation errors or bad data.

The PRO process presented in this section is highly general. We return to it in later chapters as we look at how pricing and revenue optimization is applied in specific situations, such as revenue management, customized pricing, and markdown management.

2.4.3 The Time Dimension

Much of the growing interest in pricing and revenue optimization is being driven by the increasing velocity of pricing decisions. Prices in the past changed once a quarter for many industries; now they change once a week, once a day, or even once a minute. Increased frequency and complexity of pricing creates its own momentum. A company that is able to change its prices more rapidly in response to changing market conditions can gain an advantage over its more slow-footed competitors. This creates a strong incentive for other companies to match (or exceed) the frequency of change. E-commerce has been a proven driver of increased price velocity. As discussed in Chapter 1, the adoption of computerized distribution systems—an early form of e-commerce—by the passenger airlines was a direct contributor to the ability of the airlines to develop, manage, and update highly complex fare structures.

Pricing follows different patterns in different industries. The rack (wholesale) price for gasoline fluctuates randomly, largely following the crude oil price, with little obvious trend. Fashion goods are priced high at the beginning of the season and are then subsequently marked down as the season progresses. By contrast, airline prices tend to rise as departure approaches. Different pricing approaches need to be used in each of these markets. Gasoline rack prices are set using dynamic pricing approaches that primarily track changes in the underlying supply and demand imbalance. When supply is low (e.g., Iraqi oil produc-

tion is interrupted) or demand is high (e.g., the Northeast has a cold winter), the price tends to rise. The price of a fashion item falls across the season, since such items become less valuable as the season progresses. On the other hand, the passenger airlines have successfully segmented their customer base into early-booking leisure customers and later-booking business customers. To support this segmentation, ticket prices rise as departure approaches. These different patterns lead to very different optimization approaches in different markets.

Frequent price changes create a challenge for an organization. As price changes become more frequent and the pricing structure becomes more complex, it becomes less and less feasible for a company to apply an in-depth analysis to each pricing decision. The spreadsheet analysis method of making pricing decisions begins to break down. As the complexity and rapidity of pricing decisions continues to accelerate, many organizations become overwhelmed. A common symptom is that despite expending more and more effort on pricing decisions, the company continually seems to be one step behind the market and the competition. At this point, most companies need to adopt a computerized pricing support system.

2.4.4 The Role of Optimization

As the name implies, *optimization* plays a central role in pricing and revenue optimization. As we see in Chapter 5, formulating and solving pricing decisions as constrained optimization problems is at the heart of pricing and revenue optimization. The formulation and solution of these constrained optimization problems draws on techniques from statistics, operations research, management science, and machine learning.

While constrained optimization is at the heart of pricing and revenue optimization, it is also the case that no company in the world actually optimizes its prices. The reason is that determining optimal prices is, in general, impossible—or at least well beyond our current ability. Many pricing and revenue optimization approaches (including those we study) are based on solving stylized representations of the underlying problem. These representations include many of the important features of the real-world problem, but they exclude others. For example, many hotel companies have become reasonably proficient at pricing and revenue optimization. Yet the following is a list of some of the factors often *not* incorporated in hotel pricing and revenue optimization decisions:

- How the price we offer a potential customer now affects her propensity to consider this property (or this chain) in the future
- How group prices should be optimized to trade off the amount received from each group with the number of rooms they will take, considering that we may need to refuse bookings to future independently booking customers
- How the probability that this particular customer will cancel, understay (i.e., check out early), not show, or overstay (i.e., check out late) should influence the price we quote her
- How the price we quote a customer will affect her propensity to overstay or understay

- How the price we quote a customer should be influenced by the prices currently being offered by major competitors in this market
- How the price we display in the system might influence competitors to change their prices
- How a group of properties serving a single location (e.g., all the Marriotts in Manhattan) should be priced to jointly optimize use of all the capacity
- How the expected booking order of future customers by length of stay affects the optimal price to offer this customer

Each of these and many other influences is either often not incorporated or is incorporated in a very basic fashion in most hotel pricing and revenue management systems. The point is not that hotels are poor at pricing and revenue optimization. On the contrary, the industry has become increasingly sophisticated over the past two decades. The point is, rather, that hotels have been able to improve their pricing significantly without becoming perfect.

Every industry that uses pricing and revenue optimization relies on a simplified representation of the underlying problem. The only justification for doing this is that it works. By capturing 75% or so of the real-world complexity, mathematical analysis almost always does better than either unaided human judgment or any of the purist approaches to pricing described in Section 2.2. Of course, companies and vendors continue to invest to improve the sophistication of their pricing and revenue optimization systems. Over time, systems capture more and more aspects of the real world, and the quality of pricing decisions improves. However, it is unlikely that any system or approach will ever generate perfect recommendations. What pricing and revenue optimization really does is use quantitative analysis to make better pricing decisions more quickly. Part of the philosophy of pricing and revenue optimization is that *good prices on time are far better than perfect prices late*. The improvement realized on any individual pricing decision may be small, but the aggregate effect over hundreds, thousands, or millions of pricing decisions can be very large indeed.

An alternative approach to modeling demand and explicitly optimizing price (model-based optimization) is to use a test-and-learn, or learning-and-earning, approach. Under this approach, different prices are tested under different market conditions, and the prices that lead to better outcomes are chosen more often in the future when those (or similar) conditions recur. This approach has the advantage that it does not require explicit modeling of the relationship between price and demand. It has the drawback that it requires a considerable amount of testing to reach acceptable performance. For many companies, price testing may be difficult or impossible. Furthermore, when it is possible, it may require a considerable amount of testing suboptimal prices to find the optimal price, which can be costly. Finally, test-and-learn approaches are no better than explicit modeling in determining the longer-term implications of pricing decisions.

In this book, we discuss both model-based optimization and test-and-learn approaches to pricing and revenue management. Both have their place. Furthermore, it is likely that the most successful next-generation approaches will be hybrids of model-based and test-and-learn approaches. Specifically, such approaches will perform price testing and update

prices guided by parsimonious but realistic assumptions about the underlying structure of customer response. This is, for example, the basis for the data-driven approach to pricing described in Section 5.5.

Finally, I should reemphasize the importance of the feedback loop in Figure 2.5. Pricing and revenue optimization is dependent on rapid and effective updating and monitoring more than almost anything else. The faster and more effectively prices can be adjusted to account for what is currently happening in a market, the better the business results that will be generated.

2.5 SUMMARY

- For many companies, pricing involves a complex set of decisions that is often poorly managed or unmanaged. In many cases, the pocket prices charged to customers are the result of a large number of uncoordinated and arbitrary decisions. The price waterfall and pocket-price histograms can be useful ways of visualizing the current state of pricing management within a company.
- Traditional approaches to pricing include cost-plus, market based, and value based. Of these, cost-plus is the most common. Each pure approach ignores important aspects of the pricing problem. Most companies are not purists but rely on some combination of the approaches, with the emphasis changing over time.
- The scope of pricing and revenue optimization is to set and update the prices for each combination of product, customer segment, and channel—that is, each cell in the PRO cube.
- The approach used for pricing and revenue optimization must be tailored to the specifics of the underlying market structure. From the point of view of the seller, the key concept is what type of *customer commitment* is required in each market. Different types of customer commitments require different analytical approaches.
- Pricing and revenue optimization requires a consistent process for decision making, evaluation, and updating. A typical process is illustrated in Figure 2.5. This process includes operational activities that involve setting and updating prices in response to market and cost changes, as well as supporting activities that provide input to the operational activities.
- Many of the approaches to calculating optimal prices in this book are based on a classic forecast-and-optimize approach in which we derive an explicit model of how we believe that demand will respond to the prices we offer and use the techniques of mathematical optimization to find the prices that best meet our objective function. This approach has the disadvantage that it requires choice of a price-response model and explicit estimation of the parameters—if the wrong price-response model is chosen, then the resulting prices will be less than optimal. Test-and-learn approaches have the appealing quality that they are nonparametric—they do not require the choice of a demand model beforehand. However, they have drawbacks as well. Both types of model are described throughout the book.

2.6 FURTHER READING

The different examples of customer commitments listed in Section 2.3.2 are examples of pricing modalities. A discussion of pricing modalities and why pricing modalities are different in different industries can be found in Phillips 2012c. Examples of pricing modalities in different industries and how prices are set and communicated in different industries can be found in Özer and Phillips 2012, chaps. 3–16.

Some of the common deficiencies in pricing organizations and the challenges of building an effective pricing organization are discussed in Tohamy and Keltz 2008 and in Simonetto et al. 2012. Cudahy et al. 2012 provides more detail on pricing as a closed-loop process.

2.7 EXERCISE

Brain and Company is a consulting group that offers a foolproof pricing and revenue optimization service guaranteed to deliver \$2 million in benefit to a customer. It costs Brain and Company \$500,000 to provide this service. Unfortunately, Brain has a competitor, Dissenture, that provides a similar, but somewhat inferior, service that only delivers \$1.5 million in guaranteed benefit. It also costs Dissenture \$500,000 to provide this service.

- a. When competing with Dissenture, what price does Brain and Company need to charge to guarantee that it wins the business? Assume that neither Dissenture nor Brain will price below cost and that both of them know each other's costs and the customer benefits in each case.
- b. How would Brain's price need to change if it only cost Dissenture \$400,000 to provide its service?

NOTES

1. Several real-world examples of this effect are described in Cooper and Kaplan 1987.
2. In a 2003 survey, 43.7% of responding companies answered that following the rule “Price is made up of direct cost per unit plus a fixed percentage mark-up” is either “Important” or “Very Important” in setting prices (Rao and Kartono 2009).
3. One possible explanation for the policies being used by the two sellers is that Bordee did not have the book in stock but, if it received an order, it planned to purchase a copy from Profnath, while Profnath had the book in stock and just wanted to slightly undercut Bordee. More details on the multimillion-dollar biology book can be found in Eisen 2011.
4. In the software industry, *value-based pricing* often refers to pricing policies that are based on usage or metrics other than the type of computer or the number of users. This practice was pioneered by the software vendor PeopleSoft, which priced its Human Resources module based on the number of employees in the licensed company and its Financial module based on the annual revenue of the licensee (Welch 2002). The goal of such approaches is to segment the market according to value in order to capture higher license fees from high-value customers. While PeopleSoft’s experience was widely regarded as successful, other attempts to implement value-based pricing have been less successful, particularly when competition drives fees down.
5. In the economics literature, determining the value that every customer places on a product and then charging them that value is called *perfect third-degree price discrimination by a monopolist*.

3

MODELS OF DEMAND

This chapter introduces the concept of a price-response function. The price-response function specifies how the demand that a supplier will experience for a good will change as the seller changes his price for that good. I show how the price-response function can be derived from the distribution of willingness to pay among potential customers. We also look at the most common measures of price sensitivity and how they can be estimated from historic data, and at the most common forms used for price-response functions. The chapter also extends these measures to the case when a seller is offering multiple products that may compete with each other. This material provides a foundation for the analytic approaches to pricing that are described in the remainder of the book.

3.1 THE PRICE-RESPONSE FUNCTION

Neoclassical economic theory specifies that, under a set of (seemingly) reasonable assumptions, at any point in time, every consumer (individual or firm) has a maximum willingness to pay for every good on offer by every seller through every channel.¹ If the price of a good is greater than a buyer's willingness to pay, she will not purchase. If the good is for sale at a price lower than her willingness to pay by more than one seller, she will purchase from the one such that the difference between her willingness to pay and the price—her so-called surplus—is maximized. She may not purchase from the seller with the lowest price because her willingness to pay may vary by seller. Thus, one customer might have a higher willingness to pay when purchasing from an online seller because of the convenience of delivery while another may have a higher willingness to pay to purchase in a store where she can examine the item in person before purchasing. A customer may prefer the service offered by a particular airline—or be a member of its loyalty program—so she is willing to pay a bit more to fly on that airline than for a flight at the same time offered by a competing airline. It is also the case that a customer may not know of all sellers who are offering a product or may check the prices of only a subset of those sellers.

TABLE 3.1
Willingness to pay and prices for Example 3.1

	Online (including shipping) (\$)	Chain (\$)	Independent (\$)
Customer A	21.00	16.00	16.00
Customer B	18.00	18.00	18.00
Customer C	17.00	17.00	19.00
Price	19.00	17.50	18.50

Example 3.1

Three customers are considering purchasing the same book from one of three potential sources: an online bookseller, a chain bookstore, and a local independent bookstore. Customer A strongly prefers the convenience of online shopping. Customer B is indifferent between channels and will purchase from the least expensive seller. Customer C likes to support her local independent bookstore. Their willingnesses to pay for purchasing the book from each seller and the price offered by each seller are shown in Table 3.1. In this case, Customer A will purchase online, Customer B will purchase from the chain, and Customer C will purchase from the independent bookseller.

Example 3.1 shows that, when there is a variation in willingness to pay among customers in a marketplace, different sellers can offer the same product at different prices and still all experience positive demand. The milk prices shown in Table 1.1 and the book prices shown in Table 1.2 are real-world examples of such price variation.

There are many different reasons for variation in willingness to pay. As in Example 3.1, different customers can have different preferences among different sellers or channels. One important driver of individual willingness to pay is propensity to shop—a customer who is willing to spend time comparing prices either online or offline will be more price sensitive than a customer who does not shop. Propensity to shop is related to the value that an individual places on personal time. Students and retirees are often found to be highly price sensitive—this can be explained in part by the relatively low opportunity cost of their time. Busy professionals tend to be less price sensitive at least in part because they assign a high opportunity cost to their time. Of course, there are people in all situations who enjoy bargain shopping for its own sake and get a thrill from finding a particularly low price, just as there are those who do not take the time to shop and are willing to accept a higher price in return.

A consequence of the distribution of willingness to pay across customers is that the demand experienced by a seller for a particular product will vary as a function of price; furthermore, the seller's demand as a function of price can be derived from the underlying distribution of willingness to pay. Assume that the three customers of the independent bookseller in Example 3.1 represent his entire market and that his competitors do not respond to changes in his prices. In this case, if the independent bookseller raises his price from \$18.50 to \$19.01, his demand will drop from one to zero. On the other hand, if he lowers his price to \$13.99, all three customers will purchase from him. We can represent the

amount of demand a seller will realize at any price using a *price-response function* $d = f(p)$ where p is price and d is the corresponding demand. The price-response function for the independent bookseller in Example 3.1 is given by

$$d(p) = \begin{cases} 0 & \text{for } p \geq \$19.01 \\ 1 & \text{for } \$17.50 \leq p \leq \$19.00 \\ 2 & \text{for } \$14.00 \leq p \leq \$17.49 \\ 3 & \text{for } p \leq \$13.99 \end{cases}$$

and is shown in Figure 3.1.²

The price-response function is a fundamental input to price and revenue optimization (PRO). There is a different price-response function associated with each element in the PRO cube—that is, there is a price-response function associated with each combination of product, market segment, and channel. Thus, a multichannel seller will generally have a different price-response function associated with online sales than with brick-and-mortar sales.

The price-response function is similar to the market demand function described in standard economic texts. However, there is a critical difference: The price-response function specifies *demand for the product of a single seller as a function of the price offered by that seller*. This contrasts with a demand curve, which specifies how an entire market will respond to changing prices. The distinction is critical because different firms competing in the same market face different price-response functions. In Example 3.1, the price-response function faced by the online seller differs from the function faced by the chain bookstore, which in turn differs from the price-response function faced by the independent bookstore. The differences in these price-response functions result from differences in willingness to pay across customers, which in turn stem from such factors as the varying effectiveness of seller marketing campaigns, perceived differences in quality or customer service, convenience, location, and loyalty, among other factors.

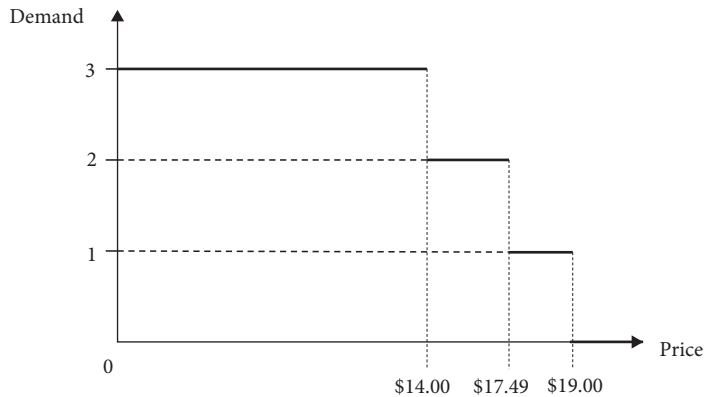


Figure 3.1 Price-response function for the bookseller in Example 3.1.

For our purposes, variation in willingness to pay among customers is the normal situation. However, most readers will be familiar with the concept of *perfect competition* introduced in every basic economics textbook. Perfect competition occurs when sellers offer identical products, all customers are indifferent among sellers, every seller is infinitesimally small, and customers and sellers have perfect information about all prices. In this case, customers will purchase only from sellers offering the lower price and the price will equilibrate to the long-run marginal cost. A commonly cited example is wheat. Each wheat farmer supplies an insignificant fraction of all of the wheat grown in the world. No matter how much an individual farmer produces—1,000 bushels or 10,000—it will be too little to have any effect on the global market price. If he tries to sell whatever he decides to produce above the market price, he will sell nothing, and if he prices below the market price, he will face demand far more than he can produce. In this situation, his motivation is to produce as much as he can at an incremental cost below the market price.

Under perfect competition, the seller has no pricing decision—his price is set by the operation of the larger market. The seller can only improve profit by finding ways to decrease his costs or increase the amount he can produce at an incremental cost below the market price. At any price below the market price, the demand seen by a seller would be equal to the entire demand in the market—the amount D in Figure 3.2. At any price above the market price, he sells nothing.

A seller in a perfectly competitive market has no need for pricing and revenue optimization—indeed, he has no need of any pricing capability whatsoever, because the price is set by the marketplace. This is true of pure commodity markets such as zinc oxide and Brent crude oil and stocks sold on the New York Stock Exchange—in these cases, no one decides on a selling price, because the price is determined by the interplay of supply and demand.

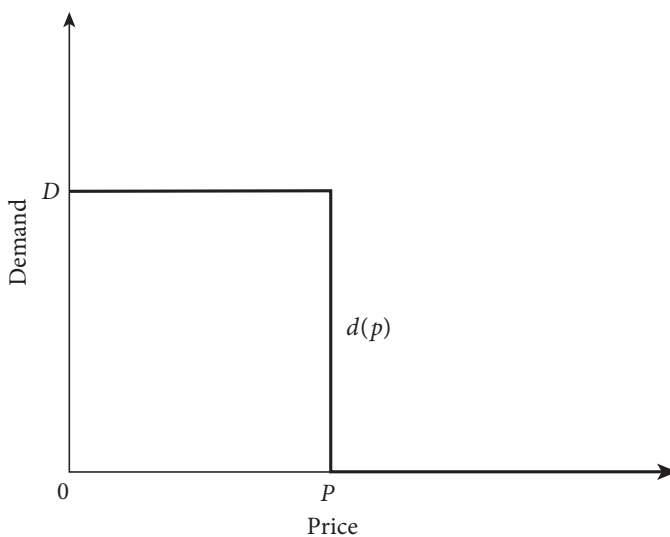


Figure 3.2 Price-response function in a perfectly competitive market.

Sellers decide only how much to produce and sell. In this case, the price-response function is as shown in Figure 3.2.

The vast majority of marketplaces, however (and all of the markets that we consider), are not perfectly competitive. Rather, as a supplier moves his price up or down, he will see his demand change—as in the price-response function shown in Figure 3.1 for the independent bookseller. The primary drivers of the relationship between price and demand are variation in preferences, information, propensity to shop, and differential access to alternatives across customers.

3.1.1 The Price-Response Function and Willingness to Pay

The demand for a product from a seller is the result of thousands, perhaps millions, of individual buying decisions on the part of potential customers. Each potential customer observes the seller's price and decides whether to purchase the product. Those who do not purchase may purchase the same or a similar product from the competition, or they may simply decide to do without. The price-response function specifies the number of customers in a segment who decide to purchase a seller's product through a channel at every price. We usually cannot directly track the thousands or even millions of individual decisions that ultimately manifest themselves in demand for our product—although online sellers can often track most or all of the customers who viewed the prices of their products. Section 4.2.2 shows that information on the number of customers who viewed the price of a product can be an important input to understanding price sensitivity. In any case, the price-response function results from the distribution of willingness to pay across the consumer population.

We assume that, at any time, every customer in the population has a willingness to pay for every product being sold through every channel. Each customer compares her willingness to pay to the prices being offered by one or more competing sellers. If more than one alternative is priced below her willingness to pay, she will purchase the option with the highest surplus, defined as the difference between her willingness to pay and the price. As in Example 3.1, not every customer will choose the cheapest alternative.

Assume that we have a population of customers of size D . The number of customers who will purchase at a price p is equal to the number of customers whose willingness to pay for the product is greater than or equal to p . (For simplicity, we assume that if a customer's willingness to pay is exactly equal to the price, she will purchase.) Define the function $f(x)$ as the density function for willingness to pay across the population, and define D as the *potential demand*—the number of customers who are potentially interested in purchasing the product and are aware of its price. Then, for any price p , $\int_p^\infty f(x) dx = \bar{F}(p)$ is the fraction of the population with willingness to pay $\geq p$, and the demand for the product at price p , which we denote $d(p)$, can be written as

$$d(p) = D \int_p^\infty f(x) dx = D \bar{F}(p), \quad (3.1)$$

where $\bar{F}(p)$ represents the *complementary cumulative distribution function* of the density function $f(p)$. $\bar{F}(p)$ is defined by

$$\bar{F}(p) = 1 - F(p) = \int_p^{\infty} f(x) dx. \quad (3.2)$$

$\bar{F}(p)$ is the probability that a random draw from the distribution with density function $f(x)$ will have a value greater than or equal to p . In our case, it is the probability that a randomly chosen customer will have a willingness to pay greater than or equal to p .

The implication of Equation 3.1 is that a price-response function can be generated from any willingness-to-pay distribution and that every price-response function corresponds to a willingness-to-pay distribution. Conversely, we can derive the density function for the willingness-to-pay distribution from the price-response function according to

$$f(p) = d'(p)/D,$$

which is nonnegative because the price-response function is downward sloping.

Example 3.2

The total potential market for a spiral-bound notebook is $D = 20,000$, and willingness to pay is distributed uniformly between \$0 and \$10.00 as shown in Figure 3.3. This means that

$$f(x) = \begin{cases} 1/10 & \text{if } 0 \leq x \leq \$10 \\ 0 & \text{otherwise} \end{cases}.$$

We can apply Equation 3.1 to derive the corresponding price-response function:

$$\begin{aligned} d(p) &= 20,000 \int_p^{10} (1/10) dx \\ &= 20,000 (1 - p/10) \\ &= 20,000 - 2000p. \end{aligned}$$

The price-response function $d(p) = (20,000 - 2,000p)^+$ is a straight line with $d(0) = 20,000$ and demand equal to zero for prices greater than or equal to \$10.00, as shown in Figure 3.3.

One of the advantages of Equation 3.1 is that it partitions the price-response function into a potential-demand component D and a willingness-to-pay component $f(x)$. This is often a convenient way to model a market. For example, we might anticipate that total demand varies seasonally for some product while the willingness-to-pay distribution remains constant over time. Then, given a forecast of total demand D_t over season t , we can calculate expected demand over the season as

$$d_t(p) = D_t \int_p^{\infty} f(x) dx = D_t \bar{F}(p). \quad (3.3)$$

D_t in Equation 3.3 is a measure of potential demand while $\bar{F}(p)$ is the fraction of potential demand that converts to actual demand at price p . This approach allows us to decompose the problem of forecasting total potential demand from the problem of estimating price response. It also allows us to model influences on willingness to pay and total demand independently and then to combine them. For example, we might anticipate that a targeted

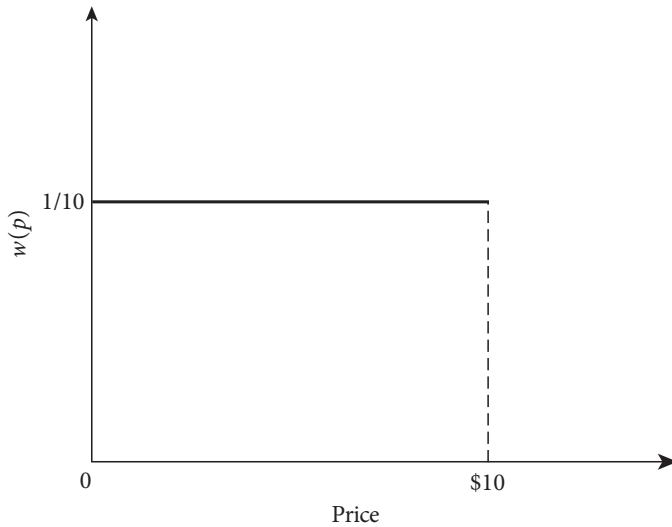


Figure 3.3 Uniform willingness-to-pay distribution.

advertising campaign will not increase the total population of potential customers D_i but that it will shift the willingness-to-pay distribution. On the other hand, if we open a new retail outlet, we might anticipate that the total demand potential for the new outlet will be determined by the size of the population served, while the willingness to pay will have the same distribution as existing stores serving populations with similar demographics. We also find this decomposition useful in the approaches to customized pricing described in Chapter 13.

Customer willingness to pay is not constant but shifts with changes in taste and changes in circumstance. A customer's willingness to pay for a cold soft drink increases as the weather gets warmer—a fact that the Coca-Cola company considered exploiting with vending machines that changed prices with temperature (see Chapter 14 for a discussion of the temperature-sensitive vending machine idea). Willingness to pay to see a movie is higher for most people on Friday night than on Tuesday afternoon. A sudden windfall or a big raise may increase an individual's maximum willingness to pay for a new car. To the extent that such changes are random and uncorrelated among customers, they will not change the overall willingness-to-pay distribution, since increasing willingness to pay on one person's part will tend to be balanced by another's decreasing willingness to pay. On the other hand, systematic changes across a population of customers will change the overall distribution and cause the price-response function to shift. Such systematic changes may be due to seasonal effects, changing fashion or fads, or an overall rise in purchasing power for a segment of the population. These systematic changes need to be analyzed and incorporated into estimating price response.

A disadvantage of the willingness-to-pay formulation is its assumption that customers are considering purchasing only a single unit. This is a reasonable assumption for relatively expensive and durable items. However, for many inexpensive or nondurable items,

a reduction in price might cause some customers to buy multiple units. A significant price reduction on a washing machine will induce additional customers to buy a new washing machine, but it is unlikely to induce many customers to purchase two. However, a deep discount on socks may well induce customers to buy several pairs. This additional induced demand is not easily incorporated in a willingness-to-pay framework—willingness-to-pay models are most applicable to big-ticket consumer items and industrial goods.

3.1.2 Properties of Price-Response Functions

The price-response functions used in PRO analysis have a time dimension associated with them. This is in keeping with the dynamic nature of PRO decisions—we are not fixing a single price that will last in perpetuity but setting a price that will stay in place for some finite period of time. The period might be minutes or hours (as in the case of a fast-moving e-commerce market), days or weeks (as in retail markets), or longer (as in long-term contract pricing). At the end of the period we have the opportunity to change prices. The demand we expect to see at a given price will depend on the length of time the price will be in place. Thus, we can speak of the price-response function for a copy machine model over a week or over a month, but without an associated time interval there is no single price-response function. Given this, every price-response function we consider has the following three properties (unless noted otherwise):

1. It will be *nonnegative*. That is, demand will be greater than or equal to zero at every price.³
2. It will be *downward sloping*. That is, demand decreases (or at least does not increase) as price increases.
3. It will be *continuous*. That is, it will have no gaps or jumps. The price-response function for the independent bookseller in Figure 3.1 has a small number of discrete steps because there are only three potential customers. As the number of customers increases, the price-response function will approach a more continuous function. While price-response functions are not truly continuous because of the discrete nature of customer demand and because prices can only occur at discrete intervals such as \$.01, we often treat them as continuous for mathematical convenience.⁴

Under these three conditions, a price-response function can be written in the form shown in Equation 3.3. Furthermore, every price-response function obeying the three conditions above corresponds to a probability density function on willingness to pay and vice versa. The correspondence between price-response functions and probability density functions is a useful property to which we return when discussing specific price-response functions in Section 3.3.

The statement that price-response functions are downward sloping is a form of the so-called *law of demand*—demand decreases when prices are increased and vice versa. As we have seen, the law of demand is a natural and inevitable consequence of the fact that the price-response function is based on a distribution of customer willingness to pay. However,

it should be noted that downward-sloping price-response functions do not mean that high prices will always be associated with low demand. A hotel revenue management system will experience higher average rates when occupancy is high and lower average rates when occupancy is low. What the downward-sloping property does indicate is that, in any time period, demand would have been lower if prices had been higher, and vice versa. This is consistent with both economic theory (in which consumers maximize their utilities subject to a budget constraint) and to real-life experience.

3.2 MEASURES OF PRICE SENSITIVITY

It is often useful to have a measure of price sensitivity—the rate at which demand will change as the price changes. The three most common measures for price sensitivity are slope, hazard rate, and elasticity. We define each of these measures in terms of the price-response function and show how they can be estimated from data. In order to estimate these properties from data, we assume that we have observations of demand at two different prices $\hat{d}(p_1)$ and $\hat{d}(p_2)$. Here, the caret indicates that the demand is actually being observed from data rather than derived from a model. For the measures described in this section to be accurate, p_1 and p_2 need to be relatively close to each other—say within 5% or so, and a sufficient number of observations are needed to achieve statistical significance.

3.2.1 Slope

The *slope* of the price-response function at a given price is measured by its derivative $d'(p)$. Unless the price-response function is linear, the slope of the price-response function will change as the price changes. The empirical estimate of the slope of the price-response function between two prices is given by the difference in demand divided by the difference in prices:

$$\hat{\delta}(p_1, p_2) = \frac{[\hat{d}(p_2) - \hat{d}(p_1)]}{p_2 - p_1}. \quad (3.4)$$

Because the price-response function is downward sloping, $d'(p)$ will always be less than or equal to zero, which means that a proper estimator $\hat{\delta}(p_1, p_2)$ of the slope should also be less than or equal to zero.

The slope can be used as a *local* estimator of the change in demand that would result from a small change in price. Assume that we have observed the demand at prices p_1 and p_2 . Then, we can use Equation 3.4 to estimate $\hat{\delta}(p_1, p_2)$ and, by the definition of the slope, for some price p close to p_1 , we can write

$$d(p) \approx d(p_1) + \hat{\delta}(p_1, p_2)(p - p_1). \quad (3.5)$$

That is, we can use the estimate of the slope to estimate the demand at a nearby price.

Example 3.3

A semiconductor manufacturer has estimated that the slope of its price-response function at the current price of \$0.13 per chip is $-1,000$ chips/week.

From Equation 3.5, he would estimate that a 2-cent *increase* in price to \$0.15 per chip would result in a reduction in demand of about 2,000 chips per week and a 3-cent *decrease* in price would result in approximately 3,000 chips/week in additional demand.

It is important to recognize that the accuracy of the approximation in Equation 3.5 degrades for larger changes in prices and that the slope cannot be used as an accurate predictor of demand at prices far from the current price. It is also important to realize that the slope of the price-response function depends on the units of measurement being used for both price and demand.

3.2.2 Hazard Rate

The hazard rate of a price-response function at a particular price is equal to minus one times the slope of the price-response function at that price divided by the demand. That is, $h(p) = -d'(p)/d(p)$. Because the slope of demand is negative, the hazard rate is positive. Larger hazard rates mean higher price sensitivity. When the price-response function can be represented in the form $d(p) = D\bar{F}(p)$, the hazard rate can be written $h(p) = d(p)/\bar{F}(p)$; that is, it is equal to the density function of the willingness-to-pay distribution at the price divided by the complementary cumulative distribution function. Given observations of demand at two prices, the corresponding hazard rate estimator is

$$\hat{h}(p_1, p_2) = \frac{-2[\hat{d}(p_2) - \hat{d}(p_1)]}{(p_1 - p_2)[\hat{d}(p_2) + \hat{d}(p_1)]}.$$

3.2.3 Price Elasticity

Perhaps the most commonly used measure of price sensitivity is *price elasticity*, defined as the ratio of the percentage change in demand to the percentage change in price; that is,

$$\varepsilon(p_1, p_2) = -\frac{100[d(p_2) - d(p_1)]/d(p_1)}{100(p_2 - p_1)/p_1}, \quad (3.6)$$

where $\varepsilon(p_1, p_2)$ is the elasticity of a price change from p_1 to p_2 . The numerator in Equation 3.6 is the percentage change in demand, and the denominator is the percentage change in price. Reducing terms gives

$$\varepsilon(p_1, p_2) = -\frac{[d(p_2) - d(p_1)]p_1}{(p_2 - p_1)d(p_1)}. \quad (3.7)$$

The downward-sloping property guarantees that demand always changes in the opposite direction from price. Thus, the minus sign on the right-hand side of Equation 3.7 guarantees that $\varepsilon(p_1, p_2) \geq 0$. An elasticity of 1.2 means that a 10% *increase* in price would result in a 12% *decrease* in demand, and an elasticity of 0.8 means that a 10% decrease in price would result in an 8% increase in demand.

$\varepsilon(p_1, p_2)$, as defined by Equation 3.7, is called the *arc elasticity* between the two prices p_1 and p_2 . It requires both prices to be calculated and, unless the price-response function

displays constant elasticity, it will be different for any choice of p_2 , which means that it is not uniquely defined for a choice of p_1 . We can derive a *point elasticity* at p_1 by taking the limit of Equation 3.7 as p_2 approaches p_1 :

$$\varepsilon(p_1) = -\frac{d'(p_1)p_1}{d(p_1)} = h(p_1)p_1. \quad (3.8)$$

In words, the point elasticity of demand at a particular price is equal to -1 times the slope times the price, divided by demand. It is also equal to the hazard rate times the price. $\varepsilon(p)$ is also called the *own-price elasticity* of demand to distinguish it from cross-price elasticities, which we discuss below. Since $d'(p) \leq 0$, point elasticity will be greater than or equal to zero.⁵ The point elasticity is a useful estimate of the change in demand resulting from a *small* change in price.

Example 3.4

A semiconductor manufacturer is selling 10,000 chips per month at \$0.13 per chip. He believes that the price elasticity for his chips is 1.5. Thus, a 15% increase in price from \$0.13 to \$0.15 per chip would lead to a decrease in demand of about $1.5 \times 15\% = 22.5\%$, or from 10,000 to about 7,750 chips per month.

One of the appealing properties of elasticity is that, unlike slope, its value is independent of the units being used. Thus, the elasticity of electricity is the same whether the quantity of electricity is measured in kilowatts or megawatts and whether the price units are dollars or euros.

Like slope, point elasticity is a *local* property of the price-response function. However, the term *price elasticity* is often used more broadly and somewhat loosely. Thus, statements such as “gasoline has a price elasticity of 1.22” are imprecise unless they specify both the time period and the reference price. In practice, the term *price elasticity* is often used simply as a synonym for *price sensitivity*. Items with “high price elasticity” have demand that is very sensitive to price, while “low price elasticity” items have much lower sensitivity. Often, a good with a price elasticity greater than 1 is described as elastic, while one with an elasticity less than 1 is described as inelastic.

Elasticity depends on the time period under consideration, and, as with other aspects of price response, we must specify the time frame we are talking about. For most products, *short-run elasticity* is lower than *long-run elasticity* because buyers have more flexibility to adjust their buying behavior to higher prices in the long run. For example, the short-run elasticity for gasoline has been estimated to be 0.2, while the long-run elasticity has been estimated to be 0.7. In the short run, the only options consumers have in response to high gas prices are to take fewer trips and to use public transportation. But if gasoline prices stay high, consumers will start buying higher-miles-per-gallon—or even electric—cars, depressing overall demand for gasoline even further. A retailer raising the price of milk by 20 cents may not see much change in milk sales for the first week or so and conclude that the price elasticity of milk is low. But he will likely see a much greater deterioration in demand over time. The reason is that customers who come to shop for milk after the price rise will

TABLE 3.2
Estimated price elasticities for various goods and services

Good	Short-run elasticity	Long-run elasticity
Salt	0.1	—
Airline travel	0.1	2.4
Tires	0.9	1.2
Restaurant meals	2.3	—
Automobiles	1.2	0.2
Chevrolets	4.0	—

still buy milk, since it is too much trouble to go to another store. But some customers will note the higher price and switch stores the next time they shop.

On the other hand, the long-run price elasticity of many durable goods—such as automobiles and washing machines—is lower than the short-run elasticity. The reason is that customers initially respond to a price rise by postponing the purchase of a new item. However, they will still purchase at some time in the future, so the long-run effect of the price change is less than the short-run effect.

It is important to specify the level at which we are calculating elasticity. Market elasticity measures total market response if all suppliers of a product increase their prices—perhaps in response to a common cost change. Market elasticity is generally much lower than the price-response elasticity faced by an individual supplier within the market. The reason is simple: If all suppliers raise their prices, the only alternatives customers have is to purchase a substitute product or to go without. On the other hand, if a single supplier raises its price, its customers have the option of purchasing from the competition.

Table 3.2 shows elasticities that have been estimated for various goods and services. Note that a staple such as salt is very inelastic—customers do not change the amount of salt they purchase very much in response to *market* price changes. On the other hand, the price elasticity of the market-response function faced by any individual seller of salt is quite large, since salt is a fungible commodity in a highly competitive market. This effect can be seen in Table 3.2 in the difference between the short-run elasticity for automobile purchases (1.2) and the much larger elasticity (4.0) faced by Chevrolet models. The table also illustrates that long-run elasticity is greater than short-run elasticity for airline travel (where customers respond to a price rise by changing travel plans in the future and traveling less by plane), but the reverse is true for automobiles (where consumers respond to price rises by postponing purchases).

3.2.4 Cross-Product Measures of Sensitivity

The measures we have considered so far have been own-product measures of price sensitivity—that is, they measure the sensitivity of demand for a product to its own price. When a seller is offering more than a single product, he may also be interested in how changing the price of one product will affect the demand for other products. If lowering the price of one product increases the sales of another, then the two products are said to be *complements*—an example is hot dogs and hot-dog buns. If lowering the price of one product decreases the demand of another, then the two products are said to be *substitutes*—think Coke and Pepsi.

If changing the price of one product has no effect on the demand for another, then the two products are said to be independent. Cross-product measures quantify both the direction and the magnitude of the effects.

Assume that a seller is offering n products with prices $\mathbf{p} = (p_1, p_2, \dots, p_n)$ that have corresponding demands $d(\mathbf{p}) = (d_1(\mathbf{p}), d_2(\mathbf{p}), \dots, d_n(\mathbf{p}))$.⁶ Here, the notation $d_i(\mathbf{p})$ is used to make clear that the demand for product i can depend not only on its own price but on the prices being offered for all the other products. In this case, we let $\partial d_i(\mathbf{p})/\partial p_j$ denote the derivative of the demand for product i with respect to price j . The sign of the derivative depends on the relationship between the two products. If $\partial d_i(\mathbf{p})/\partial p_j > 0$, then products i and j are substitutes. If $\partial d_i(\mathbf{p})/\partial p_j < 0$, then the two products are complements. If $\partial d_i(\mathbf{p})/\partial p_j = 0$, they are independent.

We define the *cross-price elasticity* between product i and product j as

$$\varepsilon_{ij}(\mathbf{p}) = - \left(\frac{\partial d_i(\mathbf{p})}{\partial p_j} \right) \left(\frac{p_j}{d_i(\mathbf{p})} \right). \quad (3.9)$$

The cross-price elasticity between product i and product j is the percentage change in demand for product i resulting from a 1% change in the price of product j . To estimate $\varepsilon_{ij}(\mathbf{p})$, we could run a price test in which we hold all of the prices constant, except for the price of product j . Let $\mathbf{p}_1$ be the vector of original prices and let $\mathbf{p}_2$ be the vector in which all prices are the same, except that price p_j has been replaced by a perturbed price $p_j + \delta$. In this case, we can calculate an estimator of the cross-price elasticity by

$$\hat{\varepsilon}_{ij}(\mathbf{p}_1) = - \frac{[d_i(\mathbf{p}_2) - d_i(\mathbf{p}_1)]p_j}{\delta d_i(\mathbf{p}_1)}.$$

Some typical values that have been estimated for cross-price elasticities are .66 for butter and margarine and .28 for beef and pork (Frank 2015). This would suggest that a 1% increase in the price of butter would lead to a 0.66% increase in demand for margarine.

Note that estimating cross-price elasticity is substantially more difficult than estimating own-price elasticity for two reasons. First, there are a lot more cross-elasticities than own-price elasticities—a seller offering n products has n own-price elasticities and $n^2 - n$ cross-price elasticities. With 50 products, this would mean 50 own-price elasticities and 2,450 cross-price elasticities.⁷ The amount of price-testing required to estimate all 2,450 cross-price elasticities would be well beyond the capability of most sellers. Second, the vast majority of cross-price elasticities are quite small, almost always less than 1 and thus significantly closer to zero than own-price elasticities, which means that more samples are required to generate statistically reliable estimates.

For a seller who is offering a large number of products, calculating cross-price elasticities between all pairs of products is usually impossible. This means that the seller needs to select clusters of products among which he believes that cross-price elasticity may be significant and estimate cross-price elasticities among products in the cluster under the assumption that cross-price elasticity between a product in the cluster and one outside the cluster is negligibly small. For substitutes, the clusters consist of similar products, often defined in

the same category or product family (e.g., big-screen televisions, men's jeans). For complements, the appropriate clusters consist of items that are typically purchased together (e.g., hot dogs and hot-dog buns). Clusters of substitutes are usually derived from a seller's pre-existing product hierarchy, while clusters of complements can be generated by examination of sales records to determine what products are typically purchased together.

A measure related to cross-price elasticity that is sometimes useful is the *diversion ratio*, which measures what fraction of the change in demand for product i that occurs with a change in product j 's price is borne by product j . That is, if we raise the price for product i , what fraction of the reduced demand for product i will translate into increased demand for product j ? If i and j are perfect substitutes—and the only two substitutes for each other—we would expect the number to be close to 1. If i and j are independent, then the diversion ratio between them should be close to 0. The formula for the diversion ratio between i and j is

$$\Delta_{i,j}(\mathbf{p}) = -\frac{\partial d_j(\mathbf{p}) / \partial p_i}{\partial d_i(\mathbf{p}) / \partial p_i} = \frac{\varepsilon_{ij}(\mathbf{p}) d_i(\mathbf{p})}{\varepsilon_{ii}(\mathbf{p}) d_j(\mathbf{p})},$$

where $\varepsilon_{ii}(\mathbf{p})$ is the own-price elasticity of product i . The diversion ratio can be used to determine which product is the buyer's second choice for a particular product. It is also used in evaluating merger proposals with the thought that a high diversion ratio for the products of two companies means that they are viewed as close substitutes by the market.

3.3 COMMON PRICE-RESPONSE FUNCTIONS

In this section we consider five common price-response functions. For each one I provide an overview of the function, explain how to calculate the different measures of price response for that function, and discuss its appropriate application. (More details and additional price-response functions can be found in van Ryzin 2012.) Information about the functions is summarized in Table 3.3.

3.3.1 Linear Price-Response Function

We have seen that a uniform distribution of willingness to pay generates a linear price-response function. The general formula for the linear price-response function is

$$d(p) = D(1 - bp)^+,$$

where $D > 0$ and $b > 0$ and the notation $(x)^+$ indicates the maximum of 0 and x . The linear price-response function is shown in Figure 3.4. The *satiating price*—that is, the price at which demand drops to zero—is $P = 1/b$. The slope of the linear price-response function is $D'(p) = Db$ for $0 < p < P$ and $D'(p) = 0$ for $p \geq P$. The hazard rate is $h(p) = b/(1 - bp)$ for $0 < p < P$ and undefined for $p \geq P$. The elasticity of the linear price-response function is $\varepsilon(p) = bp/(1 - bp)$, which ranges from 0 at $p = 0$ and approaches infinity as p approaches P , dropping again to 0 for $p > P$. For the linear price-response function, the parameter D represents demand when $p = 0$ and $(1 - bp)^+$ is the complementary cumulative distribution function (Equation 3.2) of a uniform distribution with $f(p) = 1/b$ for $0 \leq p \leq P$.

TABLE 3.3
Price-response functional forms and properties

Function	Formula $D(p)$	Parameters	Slope $D'(p)$	Hazard rate $h(p)$	Elasticity $\varepsilon(p)$
Linear*	$D(1 - bp)^+$	$D > 0$, $b > 0$	$-Db$	$b/(1 - bp)^+$	$bp/(1 - bp)^+$
Exponential	De^{-bp}	$D > 0$, $b > 0$	$-bDe^{-bp}$	b	bp
Constant elasticity	$Ap^{-\varepsilon}$	$A > 0$, $\varepsilon > 0$	$-\varepsilon Ap^{-\varepsilon-1}$	ε/p	ε
Logit	$\frac{Ae^{a-bp}}{1 + e^{a-bp}}$	$A > 0$, $b > 0$	$\frac{-Ab e^{a-bp}}{(1 + e^{a-bp})^2}$	$\frac{b}{(1 + e^{a-bp})}$	$\frac{bp}{(1 + e^{a-bp})}$
Probit**	$A \left[1 - \Phi \left(\frac{p - \mu}{\delta} \right) \right]$	$A > 0$, $\sigma > 0$	$-A\phi \left(\frac{p - \mu}{\delta} \right)$	$\frac{\phi((p - \mu) / \delta)}{[1 - \Phi((p - \mu) / \delta)]}$	$\frac{p\phi((p - \mu) / \delta)}{[1 - \Phi((p - \mu) / \delta)]}$

* The slope, hazard rate, and elasticity for the linear price-response function are valid only for $p < D/b$. At prices above D/b , demand is equal to 0.

** $\phi \left(\frac{p - \mu}{\delta} \right)$ denotes the density function, and $\Phi \left(\frac{p - \mu}{\delta} \right)$, the cumulative normal distribution function at the value p , with mean μ and standard deviation δ .

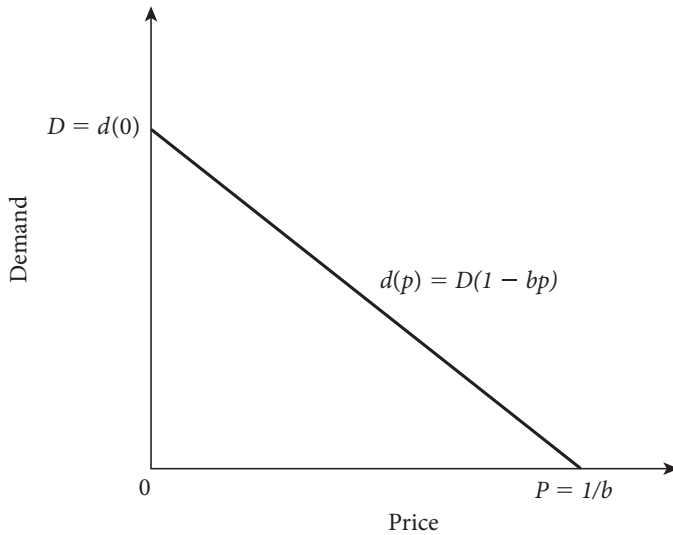


Figure 3.4 Linear price-response function.

We use the linear price-response function in many examples because it is a convenient and easily tractable model of market response. However, it is not a realistic *global* representation of price response. The linear price-response function assumes that the rate of change in demand from a 10-cent increase in price will be the same for all prices. This is unrealistic, especially when a competitor may be offering a close substitute. In this case, we would usually expect the effect of a price change to be greatest when the base price is close to the competitor's price.

3.3.2 Exponential Price-Response Function

The exponential price-response function has the form $d(p) = De^{-bp}$ with $D > 0$ and $b > 0$. The slope of the exponential price-response function is $D'(p) = -bAe^{-bp}$, which can also be written $D'(p) = -bD(p)$. The hazard rate is constant with $h(p) = b$, and the elasticity is linear in price: $\varepsilon(p) = bp$. The exponential price-response function has the property that the logarithm of demand is linear in price—that is, $\ln[d(p)] = \ln[D] - bp$ —thus the parameters of the exponential price-response function can be estimated by applying linear regression using the logarithm of demand as the dependent variable and price as the independent variable.

The exponential price-response function is finite but not satiating—there is no price at which demand goes to 0. The exponential price-response function corresponds to the case in which willingness to pay follows an exponential distribution.

3.3.3 Constant-Elasticity Price-Response Function

As the name implies, the constant-elasticity price-response function has a point elasticity that is the same at all prices. That is,

$$-\frac{d'(p)p}{d(p)} = \varepsilon$$

for all $p > 0$ and some constant elasticity $\varepsilon > 0$. The constant-elasticity price-response function is

$$d(p) = Ap^{-\varepsilon}, \tag{3.10}$$

where $A > 0$ is a parameter chosen such that $d(1) = A$. For example, if we are measuring price in dollars and $d(\$1.00) = 10,000$, then $A = 10,000$. The slope of the constant-elasticity price-response function is $D'(p) = \varepsilon Ap^{-(\varepsilon+1)}$. The hazard rate associated with the constant-elasticity price-response function is $h(p) = \varepsilon p$, and the elasticity is, of course, constant at ε .

An important property of the constant-elasticity price-response function is that the logarithm of demand is linear in the logarithm of price; that is, $\ln[d(p)] = \ln[A] - \varepsilon \ln[p]$. Thus, given historic observations of price and demand, linear regression can be applied to the logarithms of the demands and the logarithm of prices to estimate the parameters of the constant-elasticity price-response function.

Examples of a constant-elasticity price-response function for different values of the elasticity parameter are shown in Figure 3.5. Note that constant-elasticity price-response functions are neither finite nor satiating. Demand does not drop to zero at any price, no matter how high, and demand continues to approach infinity as price approaches zero, which would imply that the potential demand is infinite. Because of this property, the constant-elasticity price-response function does not correspond to any well-defined distribution of willingness to pay.

It is interesting to analyze the impact of price on *revenue* for a seller facing constant-elasticity price response. Let $R(p)$ denote total revenue at price p . Then $R(p) = p \times d(p)$, and

$$R(p) = pAp^{-\varepsilon} = Ap^{(1-\varepsilon)}.$$

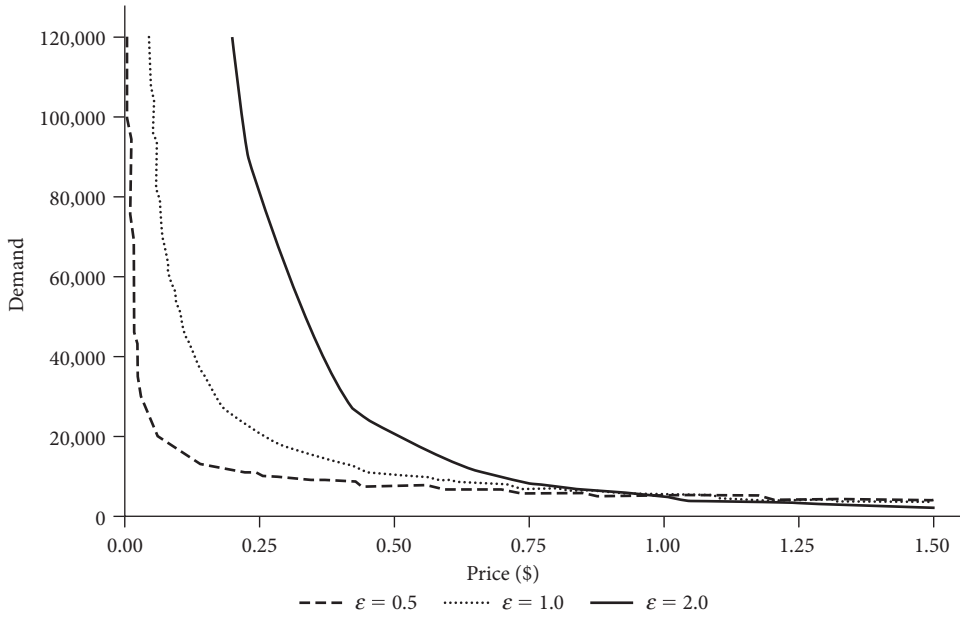


Figure 3.5 Constant-elasticity price-response functions.

Taking the slope of this function gives

$$R'(p) = (1 - \varepsilon)Ap^{-\varepsilon} = (1 - \varepsilon)d(p).$$

Since $d(p) > 0$, the derivative of revenue with respect to price for a constant-elasticity price-response function will be always greater than 0 for $\varepsilon < 1$ (inelastic demand) and will always be less than 0 for $\varepsilon > 1$ (elastic demand). This means that the seller facing constant *inelastic* price response can always increase revenue by increasing price. In fact, profit will also increase, so a seller facing constant inelastic demand should raise his price as high as he can—to infinity, if possible.

On the other hand, if $\varepsilon < 1$, a seller can maximize revenue by setting the price as close to zero—without actually becoming zero—as possible. If $\varepsilon = 1$, then total revenue will not change as the seller changes his price. Because these properties are not found in real-world markets, the constant-elasticity function is problematic as a global price-response function. For this reason, constant elasticity is better suited to represent local price response than to use as a global price-response function. In real-world markets, price elasticity typically increases with increasing price for all goods and services.

3.3.4 Logit Price-Response Functions

Both the linear and the constant-elasticity price-response functions are useful local models of price response. However, they both have rather severe limitations as global models of customer behavior. This can be seen by considering their corresponding willingness-to-pay distributions. The linear price-response function assumes that willingness to pay is

uniformly distributed between 0 and some maximum value. The constant-elasticity price-response function assumes that the distribution of willingness to pay drops steadily as price increases, approaching, but never reaching, zero. Neither of these functions seem to represent customer behavior realistically.

How would we expect customers' willingness to pay to be distributed? Consider the case where we are selling a compact car. Assume that competing models are generally similar to ours and are selling at a market price of about \$22,000. We would expect customers' willingness to pay to be concentrated near the market price. Many customers might be indifferent between our car and competing models. Some might be willing to pay a small premium for our model, and others might be willing to pay a small premium for the competing model. When the market price is \$22,000, not many customers will be willing to pay \$32,000 (or more) for our car. On the other hand, as we drop our price below the general market price, we will initially attract a lot of customers who will be happy to purchase our car at a discount from the market price. There would be very few customers for whom we would need to offer a \$12,000 discount before they would even consider purchasing our model instead of an alternative.

This suggests that we would generally expect customer willingness to pay to follow a bell-shaped distribution with a peak around the general market price. This is, in fact, a characteristic of many price-response functions used in practice. A bell-shaped distribution of willingness to pay generates a price-response function of the general reverse S-shaped (or reverse sigmoid) form shown in Figure 3.6. When a seller prices very low, he sees lots of demand, but demand changes slowly as he increases prices because, even if his price increases by a small amount, his offering is still a great bargain. In the region around market price, demand is very sensitive to his price—small changes in price can lead to substantial changes in demand. At high prices, demand is low and changes slowly as he raises price further. Empirical research has shown that this general form of price-response function fits observed price response in a wide range of markets.

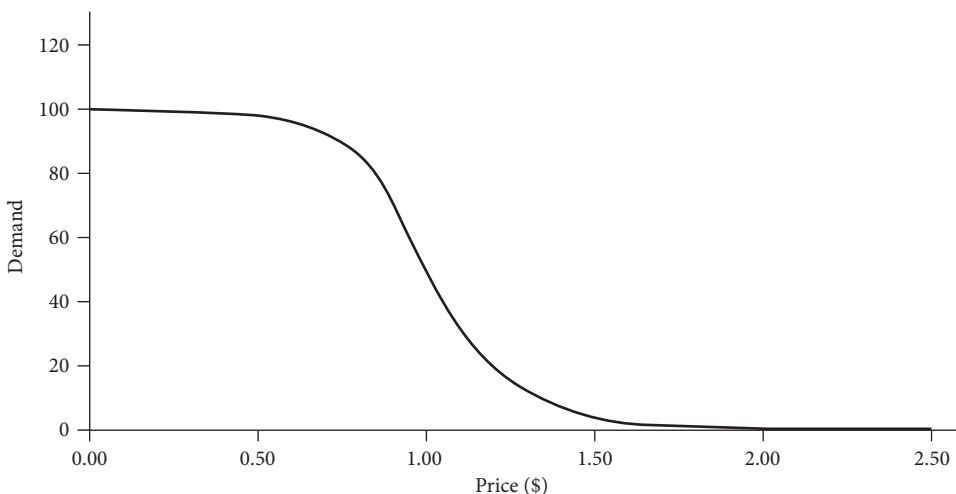


Figure 3.6 Reverse S-shaped, or reverse sigmoid, price-response function.

The most popular reverse S-shaped price-response function is the *logit*:

$$d(p) = \frac{Ae^{a-bp}}{1+e^{a-bp}}. \quad (3.11)$$

Here, a , b , and A are parameters with $A > 0$ and $b > 0$. Parameter a can be either greater than or less than 0, but in most cases we will have $a > 0$. Note that the potential market can be written as $d(0) = Ae^a/(1 + e^a)$. Thus, A scales the total size of the market, while b relates to prices sensitivity, with larger values of b corresponding to greater price sensitivity. The price-response function is steepest at the point $\check{p} = a/b$. In Figure 3.6, $a/b = \$1.00$. This point can be considered to be approximately the market price.

The logit price-response function is shown for different values of b in Figure 3.7. Here we have fixed $\check{p} = \$13,000$, so $a = \$13,000b$ for each function. Demand is very sensitive to price when price is close to $\check{p}$. Higher values of b represent more price-sensitive markets. As b grows larger, the market approaches perfect competition. In other words, the price-response function increasingly approaches the perfectly competitive price-response function in Figure 3.2.

Logit willingness to pay follows a bell-shaped curve known as the *logistic distribution*. The density function for the logistic distribution is similar to the normal distribution, except it has somewhat fatter tails—that is, it approaches zero more slowly at very high and very low values. An example of the logistic willingness-to-pay distribution is shown in Figure 3.8. The highest point (mode) of the logistic willingness-to-pay distribution occurs at $\check{p} = a/b$, which is also the point at which the slope of the price-response function is steepest. This is a far more realistic willingness-to-pay distribution than the distributions associated with either the constant-elasticity or linear price-response functions. For that reason, the logit is usually preferred to linear or constant-elasticity price-response functions when the effects of large price changes are being considered.

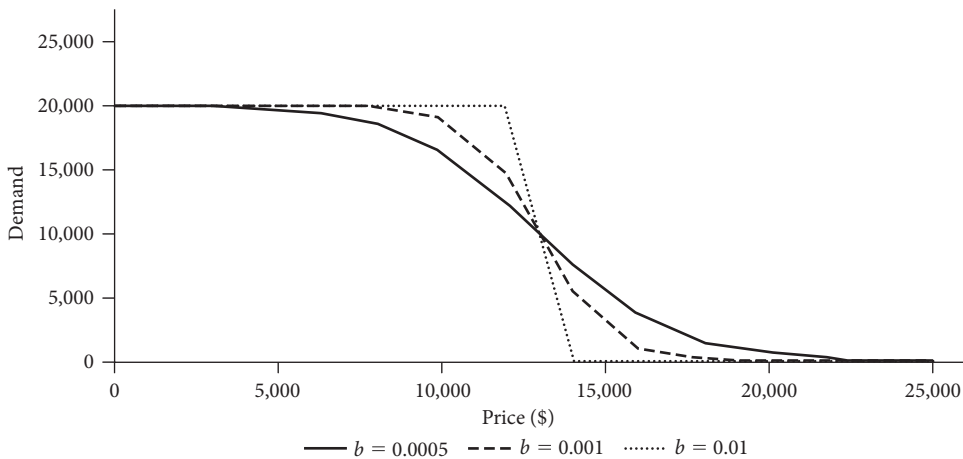


Figure 3.7 Logit price-response functions with $A = 20,000$ and $\check{p} = \$13,000$.

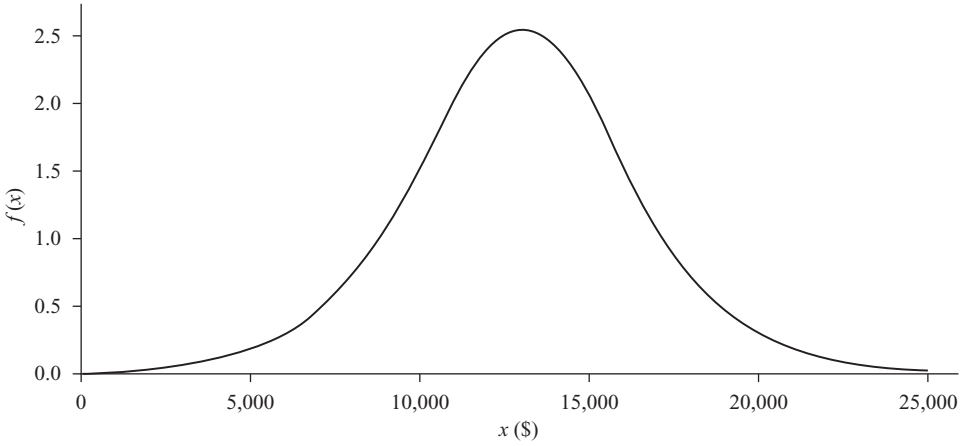


Figure 3.8 Willingness-to-pay distribution corresponding to the logit price-response function in Figure 3.7 with $b = 0.0005$. The y -axis has been scaled by a factor of 20,000.

3.3.5 The Probit Price-Response Function

Given that the most intuitively realistic density function for willingness to pay is bell-shaped, it is reasonable to ask, What about the most common bell-shaped curve of them all—the normal distribution? In fact, normally distributed willingness to pay corresponds to the so-called *probit* price-response function. The probit price-response function can be written as $d(p) = A[1 - \Phi(p; \mu, \sigma)]$, where $A > 0$ is a measure of the overall market size and $\Phi(p; \mu, \sigma)$ is the cumulative normal distribution (which has no closed-form representation) with parameters μ and σ . We must have $\sigma > 0$, but μ can be greater or less than (or equal to) zero.

Those with some familiarity with probability theory will recognize that μ is the mean and σ is the standard deviation of the willingness-to-pay distribution. The slope of the probit is greatest when $p = \mu$, which is intuitive because the mean of the willingness-to-pay distribution is an estimate of the market price, and we would anticipate a high level of price sensitivity at that price. Note that, when price is equal to the mean of the probit, demand will be equal to $d(\mu) = A/2$.

The slope of the probit is given by

$$d'(p) = -A\phi(p; \mu, \sigma).$$

The hazard rate for the probit is given by $h(p) = \phi(p; \mu, \sigma) / [1 - \phi(p; \mu, \sigma)]$, and the elasticity is $\varepsilon(p) = p\phi(p; \mu, \sigma) / [1 - \phi(p; \mu, \sigma)]$. The demand at $p = 0$ is given by $d(0) = A\phi(0; \mu, \sigma)$.

While the probit function has some theoretical appeal because it corresponds to a normal distribution of willingness to pay, the fact that it has no closed-form representation means that the logit function is far more commonly used in practice. Research has shown that the probit and logit functions give very similar results in most cases. However, the probit function is difficult to work with because there is no closed-form version. Also, the two

distributions (and their corresponding price-response functions) behave in a very similar fashion in the region around the market price. In fact, “By judicious adjustment of the coefficients, logit and probit models can be made to virtually coincide over a fairly wide range . . . and it is practically impossible to choose between them on empirical grounds” (Cramer 2003, 26). Since this is so, and the logit is much easier to work with, the logit price-response function is much more widely used than the probit.

3.4 SUMMARY

- A key input into any PRO problem is a price-response function that specifies how demand for a product changes as a function of price. The price-response function is typically nonnegative, continuous, and downward sloping.
- Three common key measures of price sensitivity are the slope, hazard rate, and elasticity of the price-response function, where the slope is defined as the derivative of the price-response function, the hazard rate is the negative of the slope divided by demand, and elasticity is the percentage change in demand that would result from a 1% change in price. All three measures are *local* properties of the price-response function, in that they can be used to estimate the effects of small changes in price but not large changes.
- In most cases, price-response functions can be considered as the measure of the number of people whose maximum willingness to pay (or reservation price) is greater than a certain price. In this case, a price-response function corresponds to a particular distribution of willingness to pay across a population. For example, a linear price-response function corresponds to a uniform distribution on willingness to pay.
- Linear and constant-elasticity price-response functions are both commonly used in analysis. However, both tend to be unrealistic when applied to large changes in price. In such cases, a reverse S-shaped model, such as the logit or the probit, is likely to be more appropriate.

3.5 FURTHER READING

Consumer theory specifies a set of axioms that govern consumer preferences. Assuming that these axioms hold, each consumer will have a utility function that specifies her relative utility among different potential bundles of goods. Given this utility function, the prices of all goods, and her budget, we can derive her willingness to pay for any specific good. Details of the axioms and the derivation of willingness to pay can be found in standard microeconomic textbooks such as Nicholson 2002 and Frank 2015.

A succinct mathematical treatment of different price-response functions and their properties can be found in van Ryzin 2012.

3.6 EXERCISE

The average ticket prices for concerts held by six different artists in 2018 were as follows:⁸

Artist	Ticket price
Bruno Mars	\$176.67
The Eagles	\$159.10
The Rolling Stones	\$155.26
Jay Z/Beyoncé	\$111.42
Guns N' Roses	\$96.46
Kenny Chesney	\$87.21
Dead and Company	\$30.50

Given that the incremental costs of mounting a show are roughly the same for all performers, what do you think accounts for the wide variation in the average ticket price commanded by the artists? Do you believe that all of these artists are seeking to maximize expected revenue from their concerts? If not, why not, and what might they be trying to do? During this same period, the average ticket price to see *Springsteen on Broadway*, in which Bruce Springsteen performed alone in a Broadway theater, was \$508.66. Why do you think this price is so much higher than the other ticket prices listed?

1. The existence of a willingness to pay (which may be less than zero) for every good in the market at every point in time can be derived from a set of simple and apparently intuitive axioms on customer preference. An example is the *axiom of completeness*, which states that, given any two products A and B, a consumer either prefers A to B or prefers B to A or is indifferent between them. Another example is the *axiom of transitivity*, which states that if a consumer prefers A to B and prefers B to C, she must also prefer A to C. With the addition of a few other axioms, we can derive the result that, for any product at any time, a consumer has a maximum willingness to pay for the product. A discussion of the axioms and a description of how they lead to a maximum willingness to pay for every product in the market can be found in any standard microeconomics textbook discussion of *consumer theory*. Similarly, the existence of a business's maximum willingness to pay is a consequence of the standard *theory of the firm* in which firms act to maximize profit. Introductions to consumer theory and the theory of the firm can be found in any standard microeconomics textbook such as Frank 2015 or Mankiw 2017. Deviations from these assumptions are considered *irrational* and their implications are discussed in Chapter 14.

2. Here we have arbitrarily assumed that, if the surplus for two channels is tied, a customer will purchase from the leftmost in the table of the two channels with the same surplus.

3. This means that what we are selling is a *good*—something people are willing to buy—rather than an *illth*—something people are willing to pay to get rid of. This is not a restrictive assumption—we can convert an illth with negative price to a good with positive price by exchanging the buyer and the seller. Thus, instead of assuming that people sell trash (an illth) at a negative price, it is more natural to assume that people buy trash removal (a good) at a positive price.

4. In reality, strict continuity of price-response functions often does not hold. In particular, prices for most items sold in the United States do not vary by less than 1 cent and most items are sold in discrete units. We could easily have the situation where $d(\$5.11) = 5,000$ and $d(\$5.10) = 5,010$. In this case, price response is not technically continuous or invertible—there is no price such that $d(p) = 5,005$. However, as long as these jumps are small relative to overall demand, we can act as if the price-response function is continuous between consecutive prices and round to the nearest penny once we have solved for the right price, knowing that our final price will be off by at most 1 cent.

5. Here we use the convention that own-price elasticities should be positive, hence the minus sign in Equations 3.6, 3.7, and 3.8. This means that the products obeying the law of demand will have positive own-price elasticities. This enables us to associate a term like “more elastic” with higher values of the elasticity, which seems intuitive. However, note that, in the literature, elasticity is sometimes calculated without the minus sign, which means it will be reported as negative.

6. Throughout the book, I use boldface variables to denote a vector. Thus, in the text, $\mathbf{p} = (p_1, p_2, \dots, p_n)$ is a vector with n component prices, and p_i is the i th component price of the vector.

7. The number of potential cross-price elasticities with n products is the number of combinations of different products, or $n^2 - n$. For products whose price is small relative to a consumer's total income, the cross-price elasticity of product i with respect to product j will be the same as the cross-price elasticity of product j with respect to product i . This means that, for example, the percentage change in the demand for margarine resulting from a 1% change in the cost of butter will be the same as the percentage change in the demand for butter resulting from a 1% change in the cost of margarine. In this case, the number of cross-price elasticities that need to be computed is equal to $n(n - 1)/2$. On the other hand, if the price of one or both of the products is large relative to consumer income, then the cross-price elasticities are likely to be quite different. As an example, a 10% drop in housing prices is likely to stimulate far more home purchases with accompanying homeowner's insurance than a 10% drop in homeowner's insurance rates is likely to stimulate new home purchases.

8. Concert ticket prices are as posted by Pollstar (<https://www.pollstar.com>) in August 2018.

4

ESTIMATING PRICE RESPONSE

Chapter 3 introduces the concept of a price-response function that specifies how demand for a product offered to a segment of customers through a particular channel varies as a function of price, and Chapter 5 shows that prices can be optimized, *assuming* that we have appropriate price-response functions in hand. It is clear that a seller would love to have a price-response function for each cell in the pricing and revenue optimization (PRO) cube, but it is not clear where these functions would come from. This chapter describes a number of approaches for estimating price response. We focus primarily on *data-driven approaches* in which future price response is estimated on the basis of an analysis of historical data. A major benefit of data-driven approaches is that they are objective: if future customer behavior is similar to past behavior, then data-driven estimation should provide a good predictor of how demand will respond to price in the future. In addition, data-driven approaches facilitate ongoing evaluation and updating. New observations of demand can be used to evaluate the predictive power of the current model and to update its parameters to reflect changes in market conditions or customer behavior.

The primary obstacle to using data-driven approaches is the availability of data. The ideal data set would contain a large number of observations specifying how past customers responded to a wide range of prices. If a seller does not have access to such a data set—if the number of historical observations is too small or does not include a sufficiently wide range of prices—then an alternative approach such as conjoint analysis or customer surveys needs to be used. Such data-free approaches do not require extensive historical data, but they can be limited in their ability to accurately represent price response across a large number of pricing segments.

In this chapter, we start by considering the type of data required to support a data-driven approach and some different ways that a seller can obtain the necessary data, such as randomized tests and A/B tests. The chapter illustrates regression-based approaches to estimating price-response functions with an extended example. It shows how data on potential demand can be used to improve price-response estimation. We discuss the issues of collinearity and endogeneity in the data and how they can bias the estimation of price

sensitivity. Finally, we discuss the situation in which historical data are not available. In this case, alternative approaches such as surveys or conjoint analyses may be useful.

4.1 DATA SOURCES FOR PRICE-RESPONSE ESTIMATION

Any data-driven approach to estimation requires a data set that includes information about historic sales. At a minimum, the data set needs to include prices and total sales by time period. In some cases, this may be the only data available. However, in many cases there may be more data about such relevant factors as competitive prices or promotions. A particularly powerful piece of data that is often available in online retailing is a measure of *potential demand*—the number of customers who took an action (such as clicking on a link) to observe the price of a product. As discussed later, when potential demand information is available, using it in an estimation of the price-response function can significantly improve accuracy. Other data that may be available include day-of-week, seasonal, and competitive prices, along with other variables. To the extent that these improve the accuracy of estimation, they should be included in the price-response model.

Two conditions are necessary for a data set of historical outcomes to support price-response estimation. First, there must be a sufficient number of observations for the resulting estimates to be statistically significant. In the case of a single price-response function based on price alone, this means at least 40 or more observations. If additional variables are incorporated in the price-response function, then more observations will be required. Second, there must be enough variation among the prices in the data to provide a basis for estimation. If the same price was always offered for the same product, then it is extremely difficult (if not impossible) to estimate a price-response function. If both conditions are not met by the data currently available, then the seller needs to either collect more data or utilize a data-free approach.

There are several ways in which a seller might obtain data to support price-sensitivity estimation. The ideal is to perform randomized price tests. If this is not possible, A/B tests are a good alternative. In cases in which neither randomized tests nor A/B tests are feasible, a seller may need to rely on so-called natural experiments, in which different prices are offered to similar groups for reasons other than price testing. In addition, changes in price at either side of a pricing-segment boundary can be used as part of a regression discontinuity design. In the remainder of this section, we consider each of these approaches to obtaining a data set to support price-response estimation.

The methods described in this section apply to list-pricing situations in which a take-it-or-leave-it price is displayed for a particular product for each customer arriving through a particular channel. This list-pricing modality is the standard practice for most retail markets (both online and offline) as well as consumer products such as travel, hotel, rental cars, and so on. In contrast, in many business-to-business settings and in consumer credit, customers approach a seller that offers the desired product, and an individualized price can be quoted to each customer based on knowledge of the customer's characteristics and desired product. Chapter 13 covers the issue of estimating price response and optimizing prices in such cases.

4.1.1 Randomized Price Testing

The gold standard for statistical inference is *randomized testing*. In a randomized test, members of the population are assigned either to a *control group* or to a *test group* (sometimes called the *treatment group*). There may be one or more test groups. The test groups receive different treatments while the control group does not receive any treatment. In a price test, the control group is quoted business-as-usual prices and the treatment groups are quoted alternative prices. The idea behind randomized testing is, as far as possible, to eliminate any systematic differences among the test groups and the control group that could influence the outcome. If this is done successfully, it is reasonable to infer that differences in outcome observed among the groups can be attributed solely to the differences in treatment.¹ In price testing, the inference is that any systematic differences in demand rate among groups are attributable to the differences in prices.

Ideally, a randomized test should meet four conditions:

- Assignment to the treatment and control groups must be imposed—consumers cannot choose the group that they are assigned to. If individuals are allowed to select their group, then *selection bias* might influence the results: an individual's choice of a particular treatment could be correlated with her response.
- The test and control groups must be balanced with respect to observable characteristics that might influence outcome. For this reason, so-called randomized tests are often not truly randomized: individuals are often systematically assigned to the treatment and control groups in a way that ensures that the groups are balanced with respect to observable characteristics. For an online retailer, this might mean assigning customers to test and control groups so that aspects such as total purchases in the last year are balanced. For a brick-and-mortar retailer, this might mean assigning stores to test and control groups so that similar characteristics are balanced.
- There must be a sufficient number of observations in each group for the differences in outcome to be statistically significant.
- The treatment (e.g., different prices) should be the only difference in the way the control group and the test group are treated—for example, the groups should not be exposed to different marketing campaigns (unless that is part of the treatment being tested).

In a price test with a single test group, customers are assigned either to the test group or to the control group according to the principles listed above. Customers assigned to the control group are quoted the control price p_c , and customers assigned to the treatment group are quoted a test price $p_t \neq p_c$. After a sufficient number of observations, demands from the test group and the control group, which we label $\hat{d}_t$ and $\hat{d}_c$, respectively, are observed. (Recall that the caret above a variable indicates that it is an observation from the data.) Referring to Equation 3.7, the arc elasticity between p_t and p_c can be estimated as $[(\hat{d}_c - \hat{d}_t)(p_c + p_t)] / [(\hat{d}_c + \hat{d}_t)(p_t - p_c)]$. Ideally, price testing would use multiple test groups, each with a different price. These prices and corresponding demands can be considered

points on a price-response function, and the estimation approaches that we consider later in this chapter can be used to estimate the corresponding parameters.

The internet is generally well suited for randomized testing. In an online price test, each successful applicant can randomly be quoted a price chosen from a number of predetermined alternatives. The test ends when a sufficient number of responses have been obtained at each price. Furthermore, online retailers usually (but not always) have access to information on potential demand, which can be used to derive more accurate estimates of price response, as we see in Section 4.2.2.

4.1.2 A/B Testing

In an A/B test, customers with some characteristic are chosen to receive a control price and customers with a different characteristic are chosen to receive a test price. The difference from randomized testing is that selection of customers into the treatment and control groups is not random—although the idea is to choose the test and control groups so that assignment is as close to random as possible. In a well-constructed A/B test, the characteristic used to separate customers into test and control groups is not correlated with the variable of interest—in this case, price response.

Brick-and-mortar retailers perform A/B price tests by offering test prices through a set of test stores. The remaining stores offer pricing as usual. In evaluating the outcome, demand at the test stores should not necessarily be compared to the overall take-up at all other stores. Rather, each test store should be paired with a control store such that members of each pair are as similar as possible to one another in terms of characteristics such as demographics, urban/rural, and competitive environment. This pairing should be done before running the test to avoid selection bias. The goal is to eliminate all other possible factors in order to isolate the influence of price on take-up.

The problem with an A/B test is that, by definition, it is not randomized—changes in competition or the underlying economic environment can influence outcomes at the control store but not at the test stores, or vice versa. These potential influences are the source of many of the objections that can be raised after the fact to the results of A/B tests, such as “The test stores performed better because those regions saw stronger economic growth than the control stores.”

In the absence of true randomization, objections to A/B tests can be hard to answer. One approach to controlling for external influences in A/B tests is *difference-in-differences* (DiD). For a test-store and control-store pair, DiD compares the change in response in test stores before and after the price test to the change in response in control stores. The idea is that the magnitude of the price effect is best measured not by the difference in demand between the test store and the control store during the test period but by the difference between the *changes* in demand before and during the test period.

Example 4.1

A retailer wants to test the price sensitivity of a popular television. The price of the television has been \$550 at all its stores during the month of March. To run

a test, he chooses two stores—Store A and Store B—that are similar in terms of customer demographics. For the month of April, he raises the price of the television set to \$600 in Store A (the test store) and holds the price at \$550 in Store B (the control store). The corresponding sales in each store are shown in Table 4.1.

The difference-in-differences approach relies on the *parallel-trends assumption*: in the absence of the treatment, it is assumed that the test group and the control group would have evolved in parallel. For Example 4.1, this is equivalent to the assumption that sales at Store A would have grown at the same rate as those of Store B if the price had been the same for both stores in April. If the parallel-trends assumption does not hold, then the resulting analysis of test results will be invalid.

One difficulty is that we cannot verify directly whether the parallel-trends assumption has held for an A/B test because we cannot observe what sales would have been for the control set under the treatment. However, it is useful to verify that the parallel-trends assumption held for the test and control sets *before* the test. If so, then it is at least reasonable to assume that it held during the test. If the parallel-trends assumption did not hold before the test, then it is not likely that it held during the test. The idea is illustrated in Figure 4.1. Each of the two graphs in this figure shows sales for a test store and a control store for several weeks prior to and during the test. In the left-hand graph, sales in the test and control move largely in parallel before the start of the test. In this case, it is plausible to claim that the parallel-trends assumption holds and that the difference in sales during the test is due primarily to the treatment. In the right-hand graph, sales in the test and the control groups were moving in different directions prior to the start of the test. It is thus much less likely that the difference in sales during the test is due primarily to the treatment; it would appear that factors other than just price are driving the difference in sales between the two stores.

A criticism of A/B tests is that they may be subject to the so-called *Hawthorn effect*, which occurs when the fact that the staff of a store know that it has been chosen for a test changes their behavior in a way that influences the outcome. If local staff know that their store is being used to test special pricing, they may increase the effort they put into selling the product. This is particularly true if they believe that the results will be scrutinized more closely than usual. This change in behavior may in itself change sales, thereby confounding the results of the test. For this reason, a store's staff should be unaware that it is a test store in an A/B test. If this is not possible, then both the test and the control

TABLE 4.1
Television demand for Store A and Store B in Example 4.1

Month	Test (Store A)		Control (Store B)	
	Price	Demand	Price	Demand
March	\$550	62	\$550	58
April	\$600	60	\$550	61

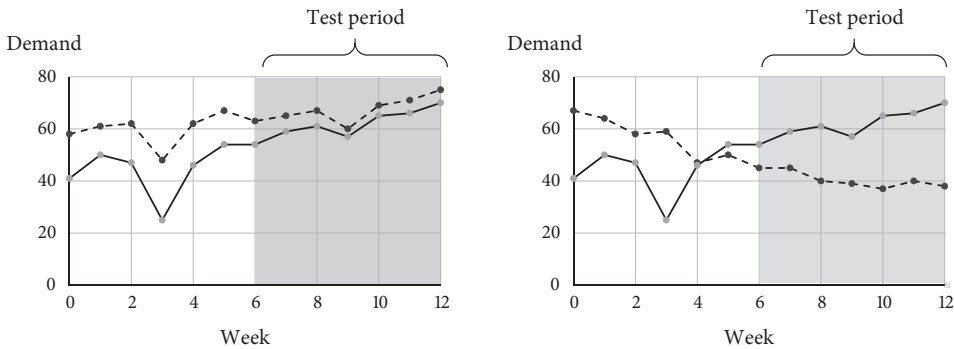


Figure 4.1 Examples in which demand appears to meet the parallel-trends assumption (left graph) and violate the parallel trends assumption (right graph) in the pre-test period.

stores should be monitored to ensure that they are providing consistent treatment to customers.

Proper A/B tests can supply important information about price response, but they also have drawbacks. First, for any product, estimating the price-sensitivity function requires measuring response at more than just two points. Furthermore, sellers want to estimate price sensitivity not just for a single product but for all cells in the PRO cube. This may mean estimating price-sensitivity functions for hundreds or thousands of price segments. In many cases, information technology limitations or corporate policy may limit the ability of a seller to run a sufficient number of controlled price tests to cover all pricing segments.

4.1.3 Switchback Tests

Switchback tests are often used in cases when a seller does not want to offer different prices to customers for the same product at the same time—a practice that many consumers feel is unfair (see Chapter 14). A switchback test is a special form of an A/B test in which the control price is charged to all customers through a certain store or a channel for a period of time, then the price is switched to a test price for a period of time, then it is changed back to the control price (or to another test price) for some period of time, and so on. For example, an online retailer might charge the control price for an hour, then switch to a test price for an hour, and then switch back to the control price for a month. In a switchback test, the customers who arrive during the periods when the control price is charged serve as the control group, and the customers who arrive during the periods when a test price is charged serve as the test group. A retailer who switched back and forth hourly between test and control prices for a week would have 84 hours of test observations and 84 hours of control observations at the end of the week. A brick-and-mortar retailer might switch prices daily or weekly at one of his stores to observe the response.

Switchback tests need to be designed so that time is not a confounding factor—online customers shopping in the morning may have very different price responses than customers shopping in the evening, and customers who do their shopping during the weekend

may be very different from those shopping during a weekday. For this reason, the switch-backs need to be designed so that control and test prices span different days and times.

4.1.4 Natural Experiments

The term *natural experiment* refers to a situation in which different treatments—in our case, different prices—were offered to different groups of customers for reasons other than running a test or optimizing prices. If the differential treatments were imposed randomly, then a natural experiment can supply useful information about price response. An example of a natural experiment would be a software defect or data entry error that caused some customers to be quoted erroneous prices while others were quoted standard prices. If the erroneous prices were quoted to a random set of customers, then the difference in take-up between customers quoted the random prices and those quoted standard prices can be used to estimate price sensitivity. Another example of a natural experiment might be imposition of a local regulation that causes prices to be different in one jurisdiction from a neighboring jurisdiction. In this case, differences in demand on either side of the jurisdictional boundary might be used to estimate price sensitivity.

By definition, natural experiments cannot be planned, and they rarely provide ideal input for analysis. However, it is worthwhile for sellers to be aware of the possibility and to consider any situation in which different prices were offered to similar populations as an opportunity to better understand price response.

4.1.5 Regression Discontinuity Design

A seller who has not varied prices much over time may be able to use a *regression discontinuity design* (RDD) to estimate elasticities at the boundaries of his pricing segments. In most situations, we would expect that price sensitivity varies continuously across a segment boundary, while price jumps from one side of the boundary to the other. We would also expect that customers close to the boundary on either side should share similar price sensitivities, and we can use the price jump across the boundary as a sort of natural experiment to estimate price sensitivity.

Regression discontinuity design was used to estimate the sensitivity of customers to the price of an Uber ride. The researcher used data from the first 24 weeks in 2015. At that time, Uber was applying a *surge multiplier* to prices based on the balance between supply and demand—if demand exceeded the available supply of drivers, a multiplier greater than one would be applied to the standard fare, with the multiplier increasing the more that demand exceeded supply. (More information on how ride-sharing companies calculate dynamic prices can be found in Section 7.6.4.) At the time, Uber calculated the surge price to a high number of decimal points but only displayed the surge price at discrete increments to facilitate the user experience. Thus, Uber might calculate that the ideal surge would be 1.6214, but it would charge the customer 1.6 times the standard fare. In all cases, Uber would round to the nearest surge increment. It is this rounding that provides the price discontinuity that can be exploited—a 1.249 surge would be rounded down to 1.2 while a 1.251 surge would be rounded up to 1.3. It is reasonable to assume that conditions (and the characteristics of

customers) when the calculated surge is 1.249 should be almost the same as those when the calculated surge is 1.251, so the fact that one group got charged 1.2 and the other 1.3 is very close to a randomized experiment. By applying this methodology to a sample of 50 million UberX customers in Uber's four largest cities, the company was able to estimate price elasticities, which fell largely between 0.4 and 0.6. (For more details on this analysis, see Cohen et al. 2016.)

4.2 PRICE-RESPONSE ESTIMATION USING HISTORICAL DATA

Assume we have a data set that includes at least historical demand and prices. It may also contain additional information relevant to demand such as competitive prices and promotional activity. In its most general form, the problem of estimating price response can be specified as finding a functional form and parameters for a model that best fits the data. We consider models of the form

$$d_{ijkt} = f_{ijk}(p_t, \mathbf{x}_t; \boldsymbol{\theta}) + \varepsilon_{ijkt}, \quad (4.1)$$

where

- f_{ijk} is a price-response function for product i sold to customer segment j through channel k ,
- d_{ijkt} is demand for product i sold to customer segment j through channel k during time period t ,
- p_t is the vector of prices for all products sold through all channels to all customer segments,
- $\mathbf{x}_t$ is a vector of *independent variables* (or features) that influence demand,
- $\boldsymbol{\theta}$ is a vector of parameters for the price-response function, and
- ε_{ijkt} is an error term, typically assumed to be drawn from a normal distribution with zero mean.

The specification in Equation 4.1 is general in the sense that it allows demand in any particular cell of the PRO cube to be a function of the prices in all other cells. This allows for the possibility of substitute and complementary products. Equation 4.1 also includes a vector of independent variables $\mathbf{x}_t = (x_{1t}, x_{2t}, \dots, x_{Kt})$. Elements of the vector $\mathbf{x}_t$ can include any information we have available at the time we are setting price that could influence demand and/or price sensitivity. Examples include day-of-week, holidays, seasonality, prices offered by competition, and promotional activity. When we start the estimation process, we have historic values of d_{ijkt} , p_t and $\mathbf{x}_t$ for some set of time periods $t = 1, 2, \dots, T$. The goal of estimation is to find the values of the parameters $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_L)$ such that the function $f_{ijk}(p_t, \mathbf{x}_t; \boldsymbol{\theta})$ best fits or *predicts*² the values of d_{ijkt} for $t = 1, 2, \dots, T$. Regression is the standard approach to this type of problem.

To illustrate the process of price-sensitivity estimation, we draw on an extended example. **We assume a seller has been conducting switchback price tests** in which he has set a

different price each day for nine weeks and observed the resulting demand. In order to compare different approaches on an apples-to-apples basis, the seller has designated the data from the first six weeks (42 days) of the test as the *training set* and the data of the last three weeks (21 days) of the test as the *test set*. The parameters of different models will be estimated using the training set (a process known as *training*) and the performance of the models compared on the test set. The purpose of maintaining separate test and training sets is to prevent *overfitting*, which we discuss in more detail in Section 4.3.2.

The data for the 42 days in the training set are shown in Table 4.2 and the data for the test set in Table 4.3. We assume for simplicity that the seller maintained adequate inventory on

TABLE 4.2
Price and demand data from the example training set

Day	Price (\$)	Sales	Weekend	Potential demand
1	6.45	49	1	52
2	6.55	40	1	41
3	6.65	24	0	31
4	6.75	25	0	34
5	6.85	18	0	27
6	6.95	31	0	49
7	7.05	17	0	37
8	7.15	19	1	52
9	7.25	4	1	22
10	7.15	11	0	37
11	7.05	6	0	14
12	6.95	13	0	29
13	6.85	3	0	6
14	6.75	13	0	14
15	6.65	48	1	61
16	6.55	41	1	44
17	6.45	14	0	15
18	6.55	31	0	33
19	6.65	30	0	35
20	6.75	35	0	42
21	6.85	17	0	27
22	6.95	21	1	36
23	7.05	24	1	54
24	7.15	9	0	32
25	7.25	8	0	42
26	7.15	15	0	43
27	7.05	5	0	9
28	6.95	14	0	23
29	6.85	28	1	41
30	6.75	32	1	45
31	6.65	38	0	46
32	6.55	6	0	7
33	6.45	12	0	13
34	6.55	20	0	23
35	6.65	11	0	16
36	6.75	30	1	41
37	6.85	24	1	37
38	6.95	12	0	19
39	7.05	13	0	34
40	7.15	7	0	20
41	7.25	1	0	26
42	7.35	5	0	28

TABLE 4.3
Price and demand data from the example test set

Day	Price (\$)	Sales	Weekend	Potential demand
1	6.45	52	1	56
2	6.55	49	1	55
3	6.65	34	0	39
4	6.75	15	0	27
5	6.85	17	0	27
6	6.95	17	0	22
7	7.05	6	0	32
8	7.15	24	1	66
9	7.25	15	1	42
10	7.15	13	0	30
11	7.05	20	0	43
12	6.95	18	0	24
13	6.85	18	0	30
14	6.75	12	0	14
15	6.65	39	1	46
16	6.55	31	1	37
17	6.45	25	0	29
18	6.55	20	0	21
19	6.65	34	0	37
20	6.75	22	0	24
21	6.85	23	0	33

hand so that he did not run out of stock on any day during the training period or the test period, which means that the observed sales are equal to demand on each day.

Tables 4.2 and 4.3 include two additional columns. *Weekend* is equal to 1 if the day is Saturday or Sunday and is equal to 0 otherwise.³ *Potential* is a measure of potential demand. For the moment, we ignore these two variables and apply regression to a model where only demand and price are available. In Section 4.2.2, we extend the approach to include additional variables, in particular potential demand.

4.2.1 Regression

Let us start with the simplest interesting model using the data in Table 4.2; that is, let us fit a model of the general form $d(p) = f(p)$, where $f(p)$ is a price-response function. In this case, **the three simplest potential models** that could be used are those described in Section 3.1.4 and listed in the first three rows in Table 3.3—namely,

- Linear: $d(p) = D(1 - bp)^+$
- Exponential: $d(p) = Ae^{-bp}$
- Constant elasticity: $d(p) = Ap^{-\epsilon}$

The goal of the estimation process is to determine the values of the parameters of each of these functions that best fit the historic data—that is, the values of A and b for the linear and exponential functions and the values of A and b for the constant-elasticity function such that the corresponding predictions of demand based on actual price are as close as possible to those shown in Table 4.2. This is a *regression* problem. There are different forms

of regression, depending on the nature of the data and the functional form being fit. Of these, *linear regression* is the simplest and most familiar. Linear regression is a standard function in any statistical software including SAS, R, and the Analysis ToolPak in Excel. (The discussion that follows assumes familiarity with linear regression and, for the homework problems, access to software with linear regression capabilities.)

Linear regression can be applied directly to find the parameters of the linear price-response function. While the exponential and constant-elasticity price-response functions are both nonlinear, they can be transformed into linear relationships by taking the logarithm of each side. The exponential price-response function then becomes

$$\ln[d(p)] = \ln[A] - bp, \quad (4.2)$$

and the constant-elasticity price-response function becomes

$$\ln[d(p)] = \ln[A] - \varepsilon \ln[p], \quad (4.3)$$

where $\ln[x]$ indicates the natural logarithm of the value x . Equations 4.2 and 4.3 are both linear, which means that we can apply linear regression to these transformed versions to estimate the parameters for each function. For the linear price-response function, the x variable is price and the y variable is demand. For the exponential price-response function, the x variable is price and the y variable is the logarithm of demand. For the constant-elasticity price-response function, the x variable is the logarithm of price and the y variable is the logarithm of demand.

Applying linear regression to each of the price-response functions using the data from the training set in Table 4.2 gives us the parameter values shown in Table 4.4, which shows the estimated parameter values and the associated r^2 values estimated for each of the three models based on the training data set.⁴ Note that the product whose price we are testing appears to be highly price-elastic, with an estimated elasticity of 12.8. This is observable from the raw data as demand is generally significantly lower on days when the price is high. Second, based on the r^2 values reported in the table, the three methods appear to have about the same quality of fit to the historic data. However, this can be misleading: The r^2 values computed for the exponential and constant-elasticity price-response functions are based on transformed data. In particular, they are based on how $\ln[d(p)]$ fits to the logarithm of the demand observations rather than to how $d(p)$ fits to actual demand. To perform an

TABLE 4.4
Results from three different price-response models trained on the data in Table 4.2

Model	Estimated parameter values	r^2
Linear	$\hat{D} = 222.59, \hat{b} = 0.13$	.35
Exponential	$\hat{D} = 15.65, \hat{b} = 1.89$	.33
Constant elasticity	$\hat{A} = 27.40, \hat{\varepsilon} = 12.80$	.33

NOTE: A “hat” over a variable in this context indicates that it is a statistically based estimate of the underlying parameter. Thus, $\hat{A}$ is a statistically based estimate of the real parameter A .

TABLE 4.5
Actual demand and predicted demand for linear, exponential, and constant-elasticity price-response models applied to the test data set in Table 4.3

Day	Actual demand	Predicted demand		
		Linear	Exponential	Constant-elasticity
1	52	31.8	32.87	33.16
2	49	28.8	27.22	27.23
3	34	25.9	22.55	22.42
4	15	22.9	18.67	18.51
5	17	20.0	15.46	15.33
6	17	17.0	12.81	12.73
7	6	14.1	10.61	10.60
8	24	11.1	8.79	8.85
9	15	8.2	7.28	7.41
10	13	11.1	8.79	8.85
11	20	14.1	10.61	10.60
12	18	17.0	12.81	12.73
13	18	20.0	15.46	15.33
14	12	22.9	18.67	18.51
15	39	25.9	22.55	22.42
16	31	28.8	27.22	27.23
17	25	31.8	32.87	33.16
18	20	28.8	27.22	27.23
19	34	25.9	22.55	22.42
20	22	22.9	18.67	18.51
21	23	20.0	15.46	15.33

apples-to-apples comparison, we need to convert the predictions of $\ln[d(p)]$ generated by the exponential and constant-elasticity functions to $d(p)$ and perform the comparison on that basis. Following that, we can estimate the quality of model fit based on the demand values of the test set in Table 4.3.

We convert the $\ln(p)$ and $\ln(d)$ values to prices and demands using the identity that $x = e^{\ln(x)}$. Table 4.5 shows the actual demands for each day during the test period along with the demands predicted by each of the three models.

At this point, an obvious question would be which of these three models is better. To answer that question, we need measures of how well the predictions of each model match the observed data. Three metrics are commonly used to measure model fit.

Root-mean-square error (RMSE). RMSE is a measure of the average difference between a forecast value and an observed value. It is defined as

$$\text{RMSE} = \sqrt{\sum_{t=1}^T [d(p_t) - \hat{d}_t]^2 / T}.$$

Here, p_t is the price for period t , and $\hat{d}_t$ is the corresponding demand (the “hat” in $\hat{d}_t$ indicates that it is an observation); $d(p_t)$ is the model prediction of demand based on p_t and T is the total number of observations.

Mean Absolute Percentage Error (MAPE). MAPE measures the average absolute percentage difference between forecast values and observed values. It is defined as

$$\text{MAPE} = 100 \times \left(\frac{1}{T} \right) \sum_{i=1}^T \frac{|d(p_i) - \hat{d}_i|}{\hat{d}_i}. \quad (4.4)$$

MAPE is an average of the percentage errors in the forecasts relative to the observations. A forecast that is 10% high contributes the same amount to MAPE as one that is 10% low.

Weighted MAPE. MAPE has an obvious shortcoming—a high percentage difference between forecast and observation is worse when the magnitude of the item being forecast is large compared to when it is small. If we forecast demand of 1 item for a day and actual demand turns out to be 3, the absolute percentage error is 200%. This is the same percentage error as if we forecast demand of 200 items and demand turns out to be 600. Both of these are treated the same in computing MAPE—however, the second missed forecast seems much worse. If we are dealing with observed values that have a wide range in magnitude, we should be able to do better on a percentage basis in forecasting the larger values and recognize that a 200% miss in forecasting a value of 200 is worse than the same percentage miss in forecasting a value of 1. One way to incorporate this effect is by weighting each entry in MAPE by the magnitude of the observation—in our case, demand. This gives us weighted MAPE, which is calculated by

$$\text{Weighted MAPE} = \frac{\sum_{i=1}^T |d(p_i) - \hat{d}_i|}{\sum_{i=1}^T \hat{d}_i}.$$

Each of the three measures of fit calculated for each model based on the test set data are shown in Table 4.6. The exponential and constant-elasticity models have virtually identical performance on all three measures. The linear model is slightly better than the other two in terms of RMSE and weighted MAPE but slightly worse in terms of MAPE. All three price-response functions are shown graphically in Figure 4.2. In the region of the test—that is, for prices between \$6.45 and \$7.35—the exponential and constant-elasticity functions are essentially indistinguishable, which explains why their performance is so similar.

The situation in Table 4.6 is quite common—test results often give no clear-cut winner among competing price-response functions, and some functions give almost identical results. One possibility in this case is to pick a winner among the three and set the corresponding optimal price. The rightmost column in Table 4.6 shows the optimal price calculated from each of the three price-response models, assuming that the incremental cost per unit is \$6.00. Since the linear model outperformed the other two on RMSE and weighted MAPE, the seller could simply set a price of \$6.76 and move forward. A second option is to do more testing. One outcome of the test is that the optimal price under any of the models

TABLE 4.6
Performance of each price-response model on the test set data in Table 4.3

Model	Root-mean-square error (RMSE)	Mean absolute percentage error (MAPE)	Weighted MAPE	Optimal price (\$)
Linear	10.01	0.81	0.40	6.76
Exponential	10.82	0.73	0.44	6.53
Constant-elasticity	10.87	0.73	0.44	6.52

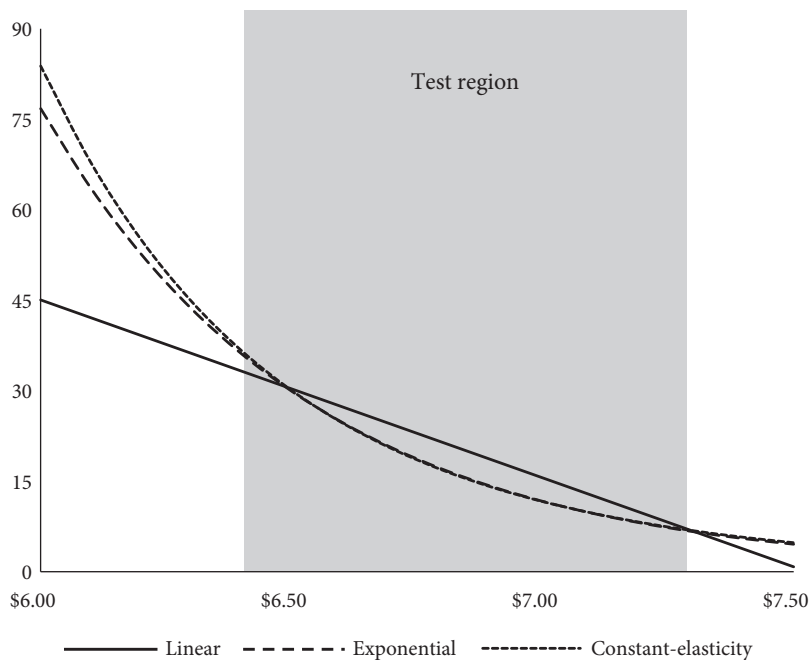


Figure 4.2 Linear, exponential, and constant-elasticity price-response functions with parameters shown in Table 4.4.

falls between \$6.50 and \$6.80. The seller could run another few weeks of tests, considering only prices within that range to refine the parameters of the model.

Another, perhaps more appealing, approach is to incorporate additional variables into the estimation. Table 4.2 includes information on whether a test date was a weekend or weekday and on demand potential that was not included in the model estimation. Both of these variables, along with others that might be available, have the potential to improve model fit (Exercise 1). However, for purposes of price-sensitivity estimation, incorporation of potential demand data is particularly powerful.

4.2.2 Incorporating Potential Demand

In brick-and-mortar retail, sellers can usually only observe realized demand—the number of products that they sold. They do not have a good idea of the number of people who might have actively considered purchasing a product but did not—either because the price was too high or for some other reason. In contrast, online retailers can usually see the number of people who visited a product page and were quoted a price. For example, Amazon provides sellers in its marketplace with statistics on *glance views*, defined as customer visits to a product detail page. Amazon also provides marketplace sellers with their *glance view conversion rates* for each product, defined as the number of orders divided by the number of glance views. Information such as glance views, which we call *potential demand*, can be extremely valuable in enabling more accurate price-response estimates.

It is likely that a brick-and-mortar seller would not have access to information on the potential demand for every product in his store. In this case, the seller can only estimate price response using data on product sales along with additional features such as day of week and seasonality. On the other hand, an online retailer is likely to have access to information on potential demand—for example, that shown in the rightmost column of Table 4.2. When such information is available, we can use it to decompose the price-response function into potential demand (which we assume is independent of price) and a conversion rate, which incorporates customer response to price. In this case, we can write the price-response function in Equation 4.1 as

$$d_{ijkt} = D_{ijkt} \times \rho_{ijk}(p_t, \mathbf{x}_t; \Theta) + \varepsilon_{ijkt}$$

where D_{ijkt} is potential demand during time period t and ρ_{ijk} is a price-response function that estimates the fraction of potential demand that converts into sales as a function of price and the other explanatory variables. For the case of this single product during a single time period, we can drop the subscripts and write the price-response function with potential demand information as $d(p) = D\rho(p) + \varepsilon$, which is very similar to Equation 3.1, recognizing that the fraction of potential demand that will result in a purchase at a price is equal to the fraction of customers who have a willingness to pay greater than the price plus an error term. When we can observe potential demand D , then we can consider the problem as being one of estimating the parameters of the *conversion rate function* $\rho(p)$ based on empirical observations of the conversion rate $\hat{r}_t = \hat{d}_t / \hat{D}_t$ and the prices p_t for values in the training set $t = 1, 2, \dots, T$.

Under the assumption that each customer purchases at most one item, we will have $0 \leq \hat{r}_t \leq 1$ for all t in the training set. (We should remove all observations for which $\hat{D}_t = 0$ from the training set.) Two of the price-response functions introduced in Chapter 3 and listed in Table 3.3 lend themselves particularly well to this setting: the *logit* and the *probit*. As discussed in Chapter 3, the logit corresponds to a logistic distribution of customer willingness to pay and the probit to a normal distribution of customer willingness to pay. While the normal might seem to be the more natural choice, the two functions are quite similar and they predict similar levels of price response when presented with the same data. The logit, however, is much easier to work with and is much more commonly used, and we use it for the remainder of this section.

The logit conversion-rate function has the form

$$\rho(p) = \frac{e^{a-bp}}{1 + e^{a-bp}}. \quad (4.5)$$

We want to determine the values of the parameters a and b so that the resulting function fits as closely as possible to the observed values of $\hat{r}_1, \hat{r}_2, \dots, \hat{r}_T$. After applying some algebra, we can write Equation 4.5 as

$$e^{a-bp} = \frac{\rho(p)}{1 - \rho(p)}.$$

Taking the logarithm of both sides, we obtain

$$a - bp = \ln \left[\frac{\rho(p)}{1 - \rho(p)} \right]. \quad (4.6)$$

The quantity on the right side of Equation 4.6 is called the *log conversion odds*. It is the logarithm of the fraction of quotes that converted at a given price divided by the fraction that did not convert. To estimate the parameters of the conversion-rate function, we regress the observed log conversion odds against the price by applying linear regression to Equation 4.6:

$$\ln\left(\frac{\hat{r}_t}{1-\hat{r}_t}\right) = a - bp_t. \quad (4.7)$$

This gives estimated values of the a and b parameters of $\hat{a} = 36.67$ and $\hat{b} = 5.25$. When we apply the logit function using these parameters to the test data set in Table 4.3, we find that the resulting RMSE = 4.40, MAPE = .19, and weighted MAPE = .08. These values are substantially lower (i.e., *better*) than the fit statistics for the linear, exponential, and constant-elasticity price-response functions shown in Table 4.6, demonstrating that incorporating potential demand information in price-response estimation can significantly improve fit of the price-response function; this result is generally independent of the functional form used (Exercise 2). In fact, sellers that have access to potential demand data have an advantage because the data allow them to decompose demand influences that are not price related and the irrelevant data, or noise, in potential demand from actual customer price response as embodied in the conversion function.⁵

4.3 THE ESTIMATION PROCESS

In Section 4.2, we went through the process of fitting three different model forms (linear, exponential, constant-elasticity) to a training data set and then measuring the quality of the model predictions on a test data set. While this is a toy problem, at a high level the steps are similar to those that would be performed in a real exercise. In particular, a data-driven estimation of price response typically involves the following steps:

1. Obtain historical data on prices and demands and other explanatory variables via one of the approaches described in Section 4.1. **Sufficient historic data must be gathered to support estimation, and the data typically require some preparation to be put in the format required for estimation.**
2. Divide the data into a *training set* and a *test set*.
3. Estimate different models using different variables with the training set, and measure their performance on **the test set**. Choose the best among the candidate models.

Each of these steps is discussed in the following sections.

4.3.1 Data Preparation

Alternatives for obtaining price and demand data to estimate price response are described in Section 4.1. A couple of comments need to be added. First, while the problem of obtaining data is straightforward to describe, experience has shown that obtaining a sufficiently clean data set to perform useful analysis is often the most painful and time-consuming part of the entire process. Often, the raw data are quite messy and unsuited for analysis. As

a simple example, in many cases returns are entered as negative sales on the day they are received. Not only can this result in negative sales for a day, but also it is not clear that this is how returns should be incorporated in price-response estimation in the first place. Most organizations seeking to estimate price response for the first time find that cleansing the data takes far more time and resources than initially planned.

Another common issue is the treatment of **stockouts and sellouts**. A retailer who sells out of a brand of bottles of aspirin by Wednesday does not know what total demand would have been had the aspirin remained in stock. Similarly, an airline that sells all of the seats for a particular flight a week before departure does not know what the total demand for that flight would have been if additional seats were available. This is a problem of *censored data*—for a given price, we do not observe the corresponding demand; rather, we observe sales, which is the minimum of demand and capacity.

There are several approaches to estimating demand in the face of censored data. Perhaps the simplest is a scaling approach. While a seller may not be able to observe potential demand once he has stocked out, it is likely that he can observe historical sales by period as well the period in which the stockout occurred. The scaling approach simply scales the observed sales by one over the fraction of the demand that normally occurred by the stockout time. For example, an airline might sell out a flight one week before departure. Looking at its history of flights that did not sell out, the airline could calculate that bookings up to one week prior to departure typically represented 3/4 of sales for the flight, with the remaining 1/4 booking in the last week. In this case, it could estimate the demand for the flight as 4/3 times sales.

Example 4.2

A retailer stocks 30 bottles of a popular multivitamin every Friday night. His average weekly sales are 24.8 bottles. When he does not stock out, his average sales for each day are as follows: Saturday, 6.2 bottles; Sunday, 4.8 bottles; Monday, 1.6 bottles; Tuesday, 2.2 bottles; Wednesday, 3.4 bottles; Thursday, 3.4 bottles; and Friday, 3.2 bottles. He runs a low-price test and sells his entire stock of 30 bottles by Wednesday evening. Using the scaling model, he estimates that demand for the entire week would have been $(24.8/18.2) \times 30 = 40.9$ bottles.

4.3.2 Choosing a Training Set and a Test Set

The reason for partitioning the historical data into a *training set* and a *test set* is to avoid the problem of *overfitting*, which occurs when some of the variables that have been included in the model actually reduce the predictive power of the model. If we start with a very simple model, we will find that adding more variables will always increase the performance of the model as measured on the training set. That is, if we add *any* external variable to the regressions we performed in the previous section, the performance of every single one of the functional forms would improve—that is, RMSE would decrease. However, at some point, additional variables are being fit only to the noise in the training set data. The situation is illustrated in Figure 4.3: as more and more variables are added to a model, the model error on the training set—whether measured by RMSE, MAPE, or another metric—will always

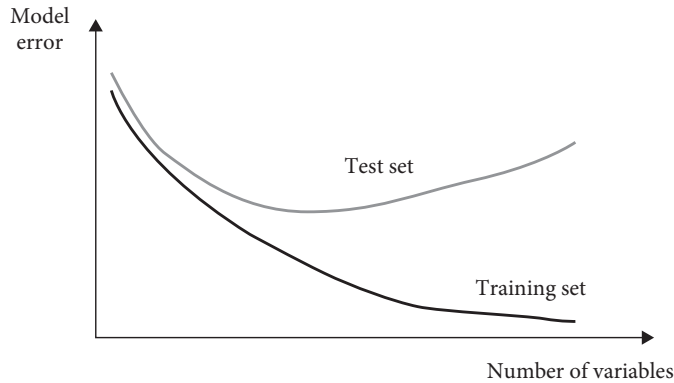


Figure 4.3 Typical behavior of model fit error as measured on a test set and on the training set as the number of variables in the model is increased.

decrease. However, at some point adding more variables begins to reduce the predictive power of the model; at this point the model is said to be *overfit*. If a model is overfit, then it should have a worse fit on the test set than an alternative model, even though it has a better fit on the training set. For this reason, we use the training set to train (i.e., fit) models and the test set to measure their performance.

Opinions differ on how much of the data set should be held out as a test set. Common practice is 33%, but some sources advocate 50%. Perhaps more important than the size is the requirement that the test set and the training set be balanced to the extent possible—that is, the profile of the customers purchasing and the external influences on them should be matched as closely as possible between the two sets.

4.3.3 Model Selection

Once data have been acquired and test and training data sets established, the final phase in estimation is model selection. This includes the selection of the functional form for the model (e.g., linear versus logit), as well as estimation of the parameters for the chosen functional form.

Typically, a functional form is chosen prior to estimation. This means that the primary task is to choose the variables to be included in the model. For example, given the data in Table 4.2, should we include *Weekend* as an explanatory variable in the model? When relatively few exogenous variables are available, we can test including all of them and exclude those that do not improve the model fit. We can create new explanatory variables by transforming exogenous variables (e.g., taking the logarithm) and by crossing variables (e.g., creating a new variable—for example, *Price* times *Weekend*). The goal of model selection is to determine a set of explanatory variables and corresponding parameters that results in the best prediction of observed demand.

To initiate the modeling process, we choose a simple initial model, apply logistic regression to estimate the coefficients of the model on the training set, and then measure the fit of this model on the test data set. This model is called the *champion*. Once we have established

an initial champion, we can determine a final model using a *champion-challenger process*, which proceeds in five steps:

1. Choose a set of variables and variable transformations that has not been tried before.
2. Run a regression using the variables, variable combinations, and variable transformations on the training set. The resulting model is called the *challenger*.
3. Measure the performance of the challenger on the test data. If the challenger performs better than the champion, then the challenger becomes the new champion. If the champion outperforms the challenger, then the champion remains the champion.
4. Go to step 1 and continue until performance is no longer improving as new challengers are evaluated.
5. Once a final champion has been chosen, reestimate the coefficients of the model using all of the available data (training data and test data).

This process is repeated each time a new challenger model is proposed. This means that, at any time, the current champion has demonstrated the best performance of all of the considered models as measured using the test data. It is important to update the test data periodically with recent observations to ensure that model performance is measured against data that reflects the most current market conditions.

The approach of starting from a simple model and adding variables and cross-variables in a way that sequentially improves the model (as measured on the test data set) is known as *forward stepwise regression*. You may wonder, why not take the opposite approach: start by including all the variables and all their transformations and crosses, and then sequentially eliminate those that are not significant—all those with high *p-values* (indicating low statistical significance), for example. This approach is called *backward stepwise regression*. It is a bad idea for three reasons. First, if the number of combinations and variables to be tested is greater than the number of observations in the training set, most regression packages will not converge. This limits the ability to consider variables and combinations. The second reason is that using all possible variables and combinations of variables will lead to a model that is *overfit*—that is, it looks great in fitting the training set but does a terrible job on the test set. In this case, the *p-values* may not give a good signal regarding which variable to remove. Finally, it could be the case that multiple variables are correlated and that no single one of them has a strong influence on demand, so all of them would be excluded. However, if the others had been excluded, one of the remaining variables might have a significant influence and should be included. This is an issue of *collinearity*, which is described in the next section.

4.4 CHALLENGES IN ESTIMATION

The previous section describes a forward stepwise regression process for estimating the coefficients of a price-response model. While the process of choosing explanatory vari-

ables can be complex, the overall idea should be clear: sequentially add new variables and combinations of variables to find the model that best fits the historic data as measured by performance on the test set. Unfortunately, finding a model that simply fits the data well is not sufficient for purposes of pricing optimization. To determine an optimal price, we need to have more than a price-response function that predicts demand; we need a price-response function that accurately represents the *causal link* between price and demand. This is a stricter requirement than predictive accuracy and it requires testing, and potentially modifying our model, for two possible conditions: *collinearity*, in which one or more of the exogenous variables influences price, and *endogeneity*, in which some variable not in the model influences both price and demand. If either of these situations exist, then we will need to modify our model if we want to accurately represent the dependence of demand on price. We consider each in turn.

4.4.1 Collinearity

Collinearity occurs when one or more of the other explanatory variables in the model influences both demand and price. As a simple example, the process of forward stepwise regression might reveal that the previous period's price is significant in predicting this period's demand.

Example 4.3

A seller estimates a linear model based on historical data using both price and unit cost for an item as explanatory variables. He obtains a model that fits his historical data well:

$$d(p, c) = (100 - 1.3p - .24c)^+,$$

where c is unit cost. Furthermore, both coefficients have p -values $< .01$. However, the seller runs a regression of price on cost using the same set of historical data and finds that the equation $p = 1.21c + .02$ has an extremely good fit. This indicates that (not surprisingly) price is highly collinear with cost and that cost should probably be dropped from price-response function.

In Example 4.3, the source of collinearity is clear: cost influences demand through its influence on price; that is, higher costs lead to higher prices, which leads to lower demand. It is unlikely (although not impossible) that cost has a direct influence on demand; it is more likely that the significance of cost in the estimation was the result of cost “reaching through” price to influence demand. In this case, dropping cost as an explanatory variable would almost certainly result in a model that better represented the causal influence of price on demand.

The test for collinearity is straightforward—a linear regression is run using price as the target variable and the exogenous variables as explanatory variables. That is, we test a model of the form $p = a_1x_1 + a_2x_2 + \dots + a_ix_i + \varepsilon$, where p is price, x_i is an exogenous variable, and a_i is the corresponding coefficient. We run a linear regression using the training set observations. If any of the exogenous variables comes up as significant in the regression,

then it is possible that that variable is collinear with price and should be eliminated from the estimation of price response. (More sophisticated approaches to dealing with collinearity are discussed in econometrics texts such as Woolridge 2015.)

Note that collinearity is an issue only when price is correlated with other variables in the model—collinearity among nonprice explanatory variables alone is not a problem. Also, collinearity is only a problem when the source of historical data is not the result of a randomized experiment or A/B test. When prices are assigned randomly, the link between price and other explanatory variables is broken and—assuming that the randomization has been effective—collinearity with price will not be an issue.

4.4.2 Endogeneity

Collinearity is an issue for price-response estimation when price is correlated with other explanatory variables. Endogeneity can be an issue when anticipated demand influences price or when both demand and price are influenced by exogenous variables not included in the model—so-called omitted variables. Airlines provide an excellent example of the phenomenon. Figure 4.4 shows the average fare and the corresponding total demand for 45 past departures of a flight with a 100-seat aircraft assigned. A naïve reading of this data would be that demand for this flight does not obey the law of demand—higher prices seem to result in higher demand. The reality is, of course, that the pattern in Figure 4.4 results from the fact that airline revenue managers are good at anticipating demand and adjusting the price to reflect imbalances between supply and demand (we discuss in detail how they do this in Chapters 8 and 9). As a result, it is higher demand—or more precisely the anticipation of higher demand on the part of revenue managers—that is driving higher prices rather than vice versa.

A (very) naïve approach to the data shown in Figure 4.4 would be to assume that the demand for the flight followed a linear price-response function $d(p_t) = D - bp_t + \varepsilon$, where ε is a mean-zero error term. Applying this model to the data in Figure 4.4 and performing

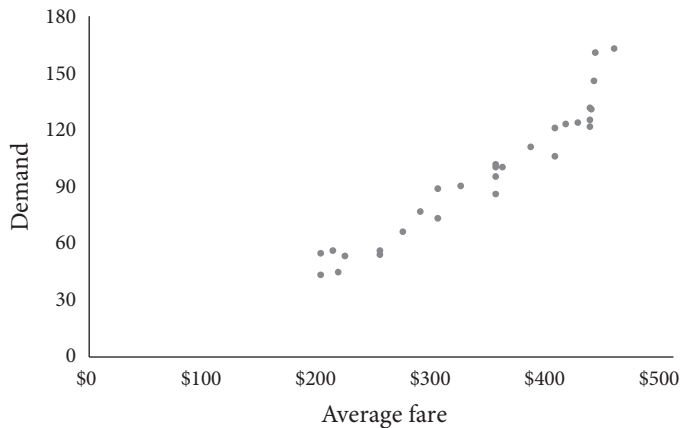


Figure 4.4 Historic average fares and corresponding demands for a flight with 100 seats.

linear regression would give the nonsensical result of an upward-sloping demand curve. A more realistic model might include variables that represent the actions of opening and closing booking classes—call the vector of these variables $\mathbf{x}$. The problem is not simply that $\mathbf{x}$ is omitted from the model; it is that $\mathbf{x}$ is omitted *and* it is correlated both with price (because opening and closing booking classes changes the average price on the flight) and with the demand on the flight (because the revenue managers anticipate demand in making their decisions). When one or more variables have been omitted *and* those variables influence both price and demand, then endogeneity is likely to confound the estimation of price sensitivity.

Detecting endogeneity and deriving methods to develop estimation procedures that account for (or clear) the endogeneity have been among the most active areas of econometric research of the last few years. The lesson for price-response estimation is that all price-response regressions should be tested for endogeneity. The Durbin-Wu-Hausman test is the most popular test and is available in most statistical packages, such as R and Stata. If significant endogeneity is present in the estimation, then you need to consider carefully how to proceed. One approach is to find and include all variables that have been omitted from the model and are correlated with both price and demand. However, in many cases historic values of these variables might not be available. Another approach is to find an *instrumental variable* that is correlated with price but uncorrelated with the error term. Chapter 13 discusses this approach to bid-response estimation in the context of customized pricing.

Finally, the ultimate cure for endogeneity, as for collinearity, is price testing. Data from a properly randomized price will be free from both collinearity and endogeneity.

4.5 UPDATING THE ESTIMATES

The coefficients of a price-response function cannot simply be estimated once and then held constant for all time. Rather, the coefficients need to be updated over time in response to changing market and competitive conditions. Most sellers periodically update the coefficients of their price-response functions at intervals ranging from one month to six months. One consideration is the number of new observations that have been received—there is no gain in updating when there have not been sufficient observations to lead to a statistically significant difference. However, it is also important to reestimate coefficients whenever a substantial change in market or competitive conditions has occurred. A large rise or fall in incremental cost or competitive prices could mean that the price of an item moves to a range in which customer response is very different. It is important to recognize that the estimation techniques described in this chapter are valid only near the range of prices in the historical data. For example, the estimates of the parameters for the models shown in Table 4.4 are valid only for the range of prices in the training set—that is, \$6.45 to \$7.35. If prices significantly outside the test region need to be considered, then the model may not be valid. In this case, the seller should monitor demand carefully and reestimate price-sensitivity coefficients as soon as practical.

In addition to making periodic updates, a prudent seller will continually compare actual demand to the predictions of his model. This approach is consistent with the closed-loop

pricing process shown in Figure 2.5. If the price-response model predicts that take-up in a particular pricing segment at the offered price should be 85% but actual take-up is only 65%, this is a signal that the price-sensitivity coefficients for that segment should be updated. However, it can be the case that price sensitivity within a segment may change in a way that is not reflected in changes in take-up rate. To the extent that it is feasible, the most prudent approach for a seller is to perform periodic price tests to ensure that current estimates of elasticity are still valid and, if they are not, to reestimate the coefficients of the price-response function.

4.6 DATA-FREE APPROACHES TO ESTIMATION

The approaches to price-sensitivity estimation discussed so far rely on the existence of a database containing a significant amount of price and demand history. In some cases, these data may not be available either because the product is too new to have generated a significant number of observations or because there is insufficient price variation in the data to support estimation. In this case, it may be impossible to use data-driven approaches to estimate price sensitivity, and alternative, data-free approaches must be used. The most popular of these approaches include the following:

- *Surveys and focus groups:* Both surveys and focus groups can be used to elicit information about the importance of price relative to other aspects of a product. Both rely on extrapolating response from a relatively small group of customers to the entire population.
- *Conjoint analysis:* Conjoint analysis is a systematic method for eliciting customer trade-offs among price, brand, and other features. Each participant in a conjoint study is given a series of choices between two different possible offerings with different features and prices. For example, a particular choice might be “Would you prefer a 20-year fixed mortgage of \$100,000 at an APR of 5.5% or a 25-year \$100,000 mortgage with an initial fixed APR of 3.5% for a year that converts to a variable-rate mortgage at prime plus 2% after the first year?” For each choice presented, participants specify either which alternative they prefer or that they are indifferent between the two alternatives. To estimate price sensitivity, price must be included along with other attributes such as term, brand, and channel. The idea behind conjoint analysis is that, by structuring the alternatives and the sequence of choices appropriately, the preference structure of each participant can be determined with a relatively small number of questions. If a study includes a sufficient number of participants with backgrounds representative of the population as a whole, the results can be used to construct a price-response function for the entire population. Conjoint analysis has been widely used in designing and pricing new products across a wide range of industries, and there is an extensive literature on how to conduct effective conjoint analyses.

Data-free approaches to price-sensitivity estimation have one important strength: they can be used to elicit information about combinations of features and price that the seller has

not previously offered. This is relevant for new products and may be important if market conditions have changed so dramatically that past customer behavior is not a reliable predictor of future behavior.

However, data-free approaches have their drawbacks. Surveys are subject to *response bias*: recipients who return a survey may not be representative of the population as a whole. For conjoint analyses, it may be difficult to obtain results from groups of participants who are both representative of the population as a whole but large enough to provide statistically reliable results. In addition, participants in surveys and focus groups often give the answer that reflects the choice they think they should make rather than the choice they actually would make in the real world. In many cases, they overestimate their willingness to pay for additional features.

Finally, surveys and conjoint analyses typically present choices in a structured manner (“How important is customer service to you in choosing an online seller?” or “Would you prefer product x to product y?”) that does not necessarily reflect the way purchase decisions arise and are addressed by customers in their daily lives, in which purchase options are not generally evaluated in a pairwise manner. For this reason, focus groups, surveys, and conjoint analyses can give sellers an idea of the general level at which prices should be set, but optimization of the prices across a number of pricing segments typically requires a data-driven approach.

4.7 SUMMARY

- The gold standard for estimating price response is the randomized price test. The goal of randomization is to isolate the effect of price on take-up by choosing test and control populations that are balanced in terms of all other variables that might influence take-up. Ensuring balance may require assigning participants to groups in a fashion that is not truly random.
- Many sellers find price testing to be difficult or impossible. As a result, the data used to estimate price response are usually derived either from (nonrandomized) A/B price tests or from natural dispersion in historical pricing, or both. This dispersion might result from so-called natural experiments (accidents), from pricing discontinuities, or from field pricing discretion.
- To fit a price-response function to a data set including prices and demands, the data is divided into a *training set* and a *test set*. For a particular set of candidate variables, regression is performed using the training set. The fit of the resulting model is measured on the test set using measures such as root-mean-square-error and MAPE. Candidate models are tested by adding, transforming, combining, and removing predictive variables. Each time a new model is estimated, its performance on the test data is compared to the previous model. If the new model is more predictive, it becomes the new champion. This process continues until new models are no longer demonstrating improved performance.
- In estimating the parameters of a price-response function, we are not only interested in the predictive accuracy of the model; we are also interested in the causal link

between price and demand. This means that we need to ensure that the results of a regression exercise do not demonstrate *collinearity*, in which price is influenced by other variables in the model, or *endogeneity*, in which variables that influence both demand and price have not been included in the model.

- If historical data are not available or there is no price dispersion in historic data, then data-free methods such as surveys or conjoint analysis can be used. It is generally advisable that price testing be used to supplement these approaches moving forward so that price response is ultimately based on actual market observations.

4.8 FURTHER READING

Some of the material in this chapter was adapted from my book *Pricing Credit Products* (2018), which discusses the problem of estimating price response in the context of consumer lending.

Two instructive examples of the use of the difference-in-differences (DiD) method in retailing are Gallino and Moreno 2014, which uses DiD to evaluate the effect of a buy-online/pick-up-in-store program for a retail chain, and Blake, Nosko, and Tadelis 2015, which uses DiD to estimate the effectiveness of online advertisement purchases.

Many classic econometric texts on the application of regression go into detail on the process of linear regression as well as such topics as the treatment of collinearity and endogeneity. Examples include Wooldridge 2015 and Greene 2003. A book that takes a modern, machine-learning-based approach to estimation is *Business Data Science*, by Matt Taddy (2019). Chapters 2, 5, and 6 are particularly relevant.

Testing for endogeneity and developing estimation approaches that produce unbiased coefficient estimates in the presence of endogeneity has been an extremely active area of econometric and statistical research in the past decades. In fact, survey of the literature found that the treatment of endogeneity was the primary source of difference among estimates of price sensitivity in similar industries (Bijmolt, Heerde, and Pieters 2005). Davidson and MacKinnon 1993 provides a particularly clear description of endogeneity testing and the use of instrumental variables to account for endogeneity in estimation. A good overview of approaches to estimation in the presence of endogeneity is found in the book *Mostly Harmless Econometrics: An Empiricist's Companion*, by Joshua Angrist and Jörn-Steffen Pischke (2009).

A good, high-level introduction to conjoint analysis is *Getting Started with Conjoint Analysis*, by Bryan Orme (2009). For a survey of applications in marketing, see the survey article by Paul Green and V. Srinivasan (1990).

4.9 EXERCISES

1. For the three price-response functions listed in Table 4.4—linear, exponential, and constant-elasticity—fit a model on the training data in Table 4.2 that includes *Weekend* as an explanatory variable along with *Price*.

- a. What are the coefficients for *Price* and *Weekend* in each model?
 - b. Calculate RMSE, MAPE, and weighted MAPE for the models on the test data in Table 4.3 including *Weekend*. How do the fits of the models including *Weekend* compare to the fits of the previous models? Which model best fits the data?
2. Fit a logistic model to the test data in Table 4.3 using *Price*, *Weekend*, and *Potential Demand* as explanatory variables.
- a. What are the coefficients for each explanatory variable?
 - b. Calculate RMSE, MAPE, and weighted MAPE for the logistic model on the test data in Table 4.3. How does the fit of this model compare to the models estimated in the text and the model estimated in Exercise 1?

NOTES

1. The requirement that the treatment is the only difference between the test and control groups is enforced in medical research by the use of triple-blind trials in which the patient, the administering physician, and those evaluating the results do not know which individuals belong to which group. This prevents patients from changing their behavior according to whether they are receiving the treatment or the placebo. It also prevents physicians from unconsciously providing a different level of care to a patient on the basis of whether that patient has been assigned to the treatment or control group. Finally, it prevents researchers from evaluating outcomes differently, depending on whether a patient was in the control or test group.

2. In this section and in the field of statistical estimation, the terms *predict* and *prediction* are used to refer to the ability of a model to fit historical data—in our case, the ability of our model to generate good values of historic demand d_t given historical prices p_t and historical values of exogenous values x_t . This is different from the everyday use of the terms to refer to the ability to predict future values.

3. This book uses the common convention that the names of variables representing values in the data are italicized. Thus, *Weekend* represents the observed value of whether an observation occurred on a weekend or not.

4. The values in the table and in subsequent analyses of this data were derived using the Regression function in Excel v13.6 running on a Mac. Using a different package or a different computer might give slightly different answers.

5. A difficulty arises if take-up is equal to 1 on a particular day. In this case, $\hat{r}_t = 1$, and the left side of Equation 4.7 is undefined. In this case, it is common practice to replace the left side of the equation with a large positive number.

5

OPTIMIZATION

Chapter 4 introduces the concept of a price-response function, which specifies the demand that a seller will experience for a product at any positive price. This chapter shows how to use a price-response function to generate the optimal price, where optimal can mean either profit-maximizing or revenue-maximizing. We begin by considering the elements of profit, including the appropriate use of costs and ancillary revenues in maximizing profit. In Section 5.2, we derive conditions that must be satisfied for a price to be optimal (again, we consider both revenue-maximizing and profit-maximizing prices) and that can be used to calculate optimal prices. Sections 5.3 and 5.4 present conditions that guarantee that a unique optimal price exists and extend the optimization approach to the case when a seller is setting prices for multiple products simultaneously. Section 5.5 describes a data-driven approach to optimizing a single price—that is, an approach in which we do not assume any particular form for price response. Section 5.6 discusses the issue of competitive response, and finally, Section 5.7 shows how an efficient frontier can help a seller who is interested in two different objective functions—say, maximizing profit and maximizing revenue—visualize the trade-off between them.

5.1 ELEMENTS OF CONTRIBUTION

The goal of pricing optimization is typically to maximize *total contribution*, defined as incremental contribution times sales. *Unit contribution* is the expected incremental revenue from each transaction minus the expected incremental cost associated with each transaction. In the case where we are selling a single unit at a known price and we know the incremental cost associated with the unit, contribution per unit can be written as $p - c$, where p is price and c is incremental cost. Throughout the rest of the book, we typically write unit contribution in this form. However, in many cases, calculating the revenue and costs associated with a customer commitment may not be so straightforward. For example, in many freight industries, the customer commitment is a contract to carry all of the freight tendered for the next year at an agreed-on set of rates. At the time the commitment is made, how much freight will actually be tendered and on which lanes is unknown to both

the shipper and the carrier, and so both the revenue and the cost of service associated with the contract are uncertain. In this case, expected revenue and expected cost of the contract need to be modeled explicitly.

Most industries are not as extreme as freight; however, in many cases, the calculation of the revenue and associated cost may not be entirely straightforward. This section addresses two particular issues in estimating contribution—calculating incremental costs and incorporating ancillary revenue.

5.1.1 Incremental Costs

Cost seems like it should be the simplest component of pricing and revenue optimization. After all, a company may not know the price-response functions it faces, but surely it should know its costs. Unfortunately, things are not quite so simple. With any exposure to the field of accounting, one quickly realizes that the simple question “How much did it cost us to provide this product to this customer through this channel on Thursday?” does not have a unique well-defined answer. Rather, the cost of providing even a simple product depends on what the answer will be used for. As a consequence, accountants have derived a host of product-costing methodologies, including fully allocated costs, partially allocated costs, marginal costs, avoided costs, and financial costs, to name just a few.

While each costing methodology has its place, pricing and revenue optimization decisions are based on the *incremental cost* of a customer commitment, where incremental cost is defined as follows.

The incremental cost of a customer commitment is the difference between the total costs a company would experience if it makes the commitment and the total cost it would experience if it does not.

Calculating incremental cost for a customer commitment depends critically on the context and nature of the commitment. Here are some examples.

- A drugstore sells bottles of shampoo and replenishes its stock weekly. Because selling a unit will result in an additional unit order, the incremental cost for selling a bottle of shampoo is the wholesale unit cost.
- An airline is considering whether to sell a single seat to a passenger on a future flight. The incremental cost is the additional meal and fuel cost that would be incurred from flying an additional passenger plus any commissions or fees the airline would pay for the booking.
- A retailer buys a stock of fashion goods at the beginning of the season. Once he has bought the goods, he cannot return them and cannot reorder. During the season he wants to set and update the prices that will maximize the total revenue he will receive from the fashion goods. Since the cost of the goods is sunk and selling a unit will not drive any additional future sales, his incremental cost per sale is zero.¹
- A distributor is bidding for a year-long contract with a hospital to be the preferred provider for disposable medical supplies. From previous experience, the distributor

knows that this hospital is an expensive customer requiring high levels of customer support and having wide variances in orders and high rates of product returns. The expected incremental cost of the contract includes not only the expected cost of the items the hospital will purchase but also the expected cost of customer service, operating costs and holding costs driven by the wide variance in orders, and the expected costs of returns. In this case, these costs need to be associated with the contract, but some of them, such as the cost of customer service, may not be specifically associated with any particular product.

These examples illustrate some of the key characteristics of incremental cost. These characteristics can be summarized as follows.

- *Incremental cost is forward looking.* It is based on the effect a customer commitment will have on future costs. Costs that have already been taken or that are driven by decisions already made prior to the customer commitment are sunk and do not enter into the calculation of incremental cost.
- *Incremental cost is marginal.* It is the expected cost of making *this* customer commitment. The incremental cost of making this commitment may not be the same as the average cost of similar commitments made in the past.
- *Incremental cost is not fully allocated cost.* Only costs that change as the result of a customer commitment are part of the incremental cost. Overhead, or fixed costs of staying in business, generally should not be allocated to any specific commitment. As a result, the incremental cost of a customer commitment is usually less than the fully allocated cost. Further, the sum of the incremental costs of all commitments will be less than the total operating cost of the company because an unallocated residual fixed cost will remain after all the incremental costs have been totaled.
- *The elements of incremental cost can depend on the type, size, and duration of the commitment.* Specifically, if the customer commitment is a multiunit order, then order-based or setup costs need to be allocated across all the units in the order.
- *The incremental cost of a commitment can be uncertain.* Trucking companies such as Roadway Express and Yellow Freight sell contracts to shippers. Each contract covers the next year and commits the trucking company to carry all the freight tendered by the customer at an agreed-on tariff. The incremental cost associated with one of these contracts is likely to be highly uncertain at the time the commitment is made. First, the amount, timing, and origin and destination of the freight the customer will tender over the next year is uncertain. In addition, the cost of moving the customer's freight will depend on the freight the company will move for other customers, which is also uncertain. This cost will depend in part on what portion of the customer's freight is *backhaul* business, which can utilize excess capacity on existing truck trips, and what portion will be *headhaul* freight, which will require adding new truck trips.

In each case, the calculation of incremental cost requires understanding the nature of the customer commitment and then estimating the additional costs that would be generated by

TABLE 5.1
Incremental cost elements for Acme Hotel

Cost element	Units	Value	Description
Commissions	\$/ \$	15% 0%	Bookings through third-party platforms Bookings through Acme website
Transaction cost	\$/rental	\$31.26	Paperwork, additional cleaning, check-in and check-out
Daily cost	\$/night	\$12.42	Cleaning and supplies

making the commitment—or, equivalently, the costs that would be avoided by not making the commitment. A simple rule of thumb is this: *If changing a price would not change a particular cost, then that cost should not influence the price.*

The methodology for calculating incremental costs is closely related to activity-based costing (ABC). Activity-based costing is a management accounting approach to allocating costs to their underlying causes in order to give a clearer view of the real drivers of cost within an organization. In our case, the underlying cost-drivers are various aspects of the customer commitment. As an example, consider the hypothetical incremental cost elements associated with the Acme Hotel shown in Table 5.1. Three categories of cost are represented: a transaction cost that is generated by every booking, a daily cost that depends on the number of days in the rental, and a commission. The commission is 15% for bookings through online booking platforms such as Expedia and Orbitz and 0 for bookings through Acme Hotel's own website.

Example 5.1

Both Mr. Smith and Ms. Jones want to stay for two nights in the Acme Hotel with costs shown in Table 5.1. The nightly rate available to both Smith and Jones is \$150.00/night. Mr. Smith is booking through an online booking platform, while Ms. Jones is booking through the Acme Hotel website. The incremental cost for Mr. Smith's two-night stay is $\$31.26 + 2(.15 \times \$150 + \$12.42) = \101.10 , while the expected unit cost for the same rental to Ms. Jones is $\$31.26 + 2 \times \$12.42 = \$56.10$.

Note that, in calculating the incremental costs, only cost elements that vary with aspects of the customer stay are included. Costs such as front-desk salary and corporate overhead are not included because they will not change, depending on whether Mr. Smith or Ms. Jones makes a booking.

5.1.2 Ancillary Products and Services

In many industries, particularly service industries, ancillary products and services can be an important source of revenue. An airline passenger may also purchase a beer, a meal, or duty-free goods aboard her flight. Sporting events and concerts sell food, beverages, and “merch” (T-shirts and other souvenirs). Insurance is a highly profitable sideline for automobile sellers, rental car companies, online travel booking companies, and mortgage lenders. Paid warranties are very profitable add-ons to sales of automobiles, appliances, and consumer electronics. These additional sources of contribution from a customer above and

beyond the price she pays are called *ancillary products* or *services*. An ancillary product is one that would not be sold without the sale of the main product—thus, no one without a ticket to the soccer game can buy a beer in the stadium. This differentiates ancillary products from product complements that may be sold together but can also be sold individually, such as hot dogs and hot-dog buns.

Ancillary contribution is the net contribution from sales of ancillary products or services. To maximize expected contribution, a revenue management company needs to include the estimated ancillary contribution for each booking request. This can be challenging because the amount of ancillary contribution that will be received from a prospective booking is typically unknown at the time a pricing or availability decision needs to be made.

To incorporate ancillary contribution into revenue management, we typically use the *expected ancillary contribution* per customer, which we will label π^C . If each attendee at a soccer match purchases, on average, 1.4 beers and the incremental contribution (not price) from each beer is \$1.00, then $\pi^C = \$1.40$. The expected contribution per customer is $p + \pi^C - c$. In general, higher values of π^C will tend to lead to lower optimal prices—thus, an event that expects to make high profits from ancillary revenue should set its ticket prices lower: it can make up for the lost revenue per ticket by the ancillary contribution from the additional customers attracted by the lower price.

Expected ancillary contribution may vary among elements of the pricing and revenue optimization (PRO) cube, and this can provide a rationale for varying price by product, customer segment, and/or channel. For example, many borrowers who take out a consumer loan in the UK also take out insurance against defaulting on the loan because of job loss or injury. Insurance is highly profitable for most lenders, which means that it is a sensible tactic for lenders to offer lower prices to customer segments that the lenders have determined are more likely to take out insurance. As a more extreme example, hotels associated with casinos commonly estimate the gambling revenue that they can expect different customer segments to generate. They then charge lower prices (even 0) or provide expanded availability to customer groups that they believe will generate high levels of gambling revenue.

Example 5.2

Both Mr. Smith and Ms. Jones have previously stayed in hotels in the Acme chain many times. The customer relationship management (CRM) system for Acme shows that Mr. Smith has never ordered room service or eaten in the hotel restaurant in any of his previous stays. Ms. Jones, however, has always ordered dinner from room service, from which the hotel makes an average contribution of \$15 per meal. On this basis, Acme forecasts that Mr. Smith will take no meals at the hotel and that Ms. Jones will order two room-service dinners. In this case, referring back to Example 5.1, the total expected contribution from each of them would be as follows:

	Direct revenue	Expected ancillary contribution	Incremental cost	Expected contribution
Mr. Smith	\$300	0	(\$101.10)	\$198.90
Ms. Jones	\$300	\$30	(\$56.10)	\$273.90

In Example 5.2, incorporating expected ancillary contribution and incremental cost into the calculation reveals that Ms. Jones is expected to be about 38% more profitable than Mr. Smith, even though they are paying the same room rate. The hotel operator could make use of this information by providing a coupon for Ms. Jones for a lower rate. Alternatively, if there is only one remaining room available at the property, a profit-maximizing hotelier would prefer to provide availability to Ms. Jones rather than Mr. Smith. Thus, revenue management decisions such as those discussed in Chapter 8 should be based on maximizing expected contribution.

While ancillary revenue is important in some applications, it is not explicitly included in most of the calculations that follow. However, in cases where expected ancillary contribution can be significant, the contribution term $p - c$ should be understood as including the additional term π^c , representing expected ancillary contribution.

5.2 THE BASIC PRICE OPTIMIZATION PROBLEM

The difference between the price at which a product is sold and its incremental cost is called its *unit margin*. The sum of the margins of all products sold during a time period is called the *total contribution*. We denote the total contribution achieved at a price p by $\Pi(p)$. In most cases, the seller's goal is to maximize total contribution. The unit margin from a product priced at p is $m(p) = p - c$, where $m(p)$ is unit margin and c is incremental cost. The basic single-price optimization problem is

$$\max_p \Pi(p) = (p - c)d(p). \quad (5.1)$$

The fundamental trade-off faced by the seller seeking to maximize total contribution is illustrated in Figure 5.1. If the seller's price is equal to incremental cost, his total contribution will be 0. As the seller increases the price from c , unit margin increases faster than demand decreases so that total contribution increases. At some price, the rate at which demand

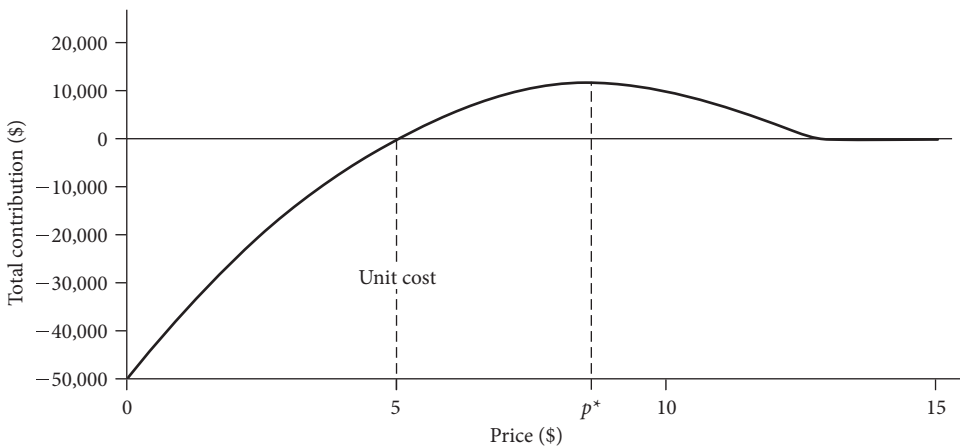


Figure 5.1 Total contribution as a function of price.

decreases balances the rate at which unit margin increases—this is the price that maximizes expected total contribution. In most cases, the total-contribution function $\Pi(p)$ is hill shaped, with a single peak, as shown in Figure 5.1. The top of this peak is the maximum total contribution the supplier can realize in the current time period, and p^* is the price that maximizes the total contribution. Note the fundamental lack of symmetry in the total-contribution curve: The supplier can lose money by pricing too low (below incremental cost), but he cannot lose money by pricing too high—the worst that can happen is demand gets driven to zero. This means that a risk-averse seller would tend to overprice.

5.2.1 Optimality Conditions

The problem in Equation 5.1 is an unconstrained optimization problem, and standard optimization theory tells us that the maximum will occur at the price for which the derivative of $\Pi(p)$ is equal to 0. The derivative of $\Pi(p)$ with respect to price is

$$\Pi'(p) = d'(p)(p - c) + d(p). \quad (5.2)$$

To find the price that maximizes total contribution, we set $\Pi'(p) = 0$, or

$$d'(p)(p - c) + d(p) = 0.$$

Thus, to maximize total contribution, p^* must satisfy

$$d(p^*) = -d'(p^*)(p^* - c). \quad (5.3)$$

We can also establish conditions for a price to maximize total revenue, defined by $R(p) = pd(p)$. Let $\hat{p}$ denote the price that maximizes expected revenue; then the equivalent condition to Equation 5.3 for the revenue-maximizing price is

$$d(\hat{p}) = -d'(\hat{p})\hat{p}. \quad (5.4)$$

Equation 5.4 implies that the price that maximizes revenue is the same as the price that maximizes total contribution if costs were set to 0. This means maximizing revenue is equivalent to maximizing profit in industries in which incremental costs are either zero or very close to zero. This includes some service industries such as movie theaters or sporting events, as well as television and online advertising. Incremental costs are zero (or at least very small) in many cases in which a company has already purchased a fixed amount of nonreplenishable inventory, such as in the case of fashion-goods retailers, who purchase inventory for an entire season at one time. Once the inventory has been purchased, the incremental cost of a sale is zero—and the seller should set a price that maximizes revenue from his fixed stock of inventory.

Example 5.3

A widget maker is looking to set the price of widgets for the current month. His unit cost is \$5.00 per widget and his demand for the current month is governed by the linear price-response function $d(p) = (10,000 - 800p)^+$. (Recall that the notation $(x)^+$ indicates the maximum of 0 and x .) In this case, $d'(p) = -800$.

Substituting into the condition 5.3, we obtain the condition on the optimal price that $(10,000 - 800p^*)^+ = 800(p^* - \$5.00)$, which means that $p^* = \$8.75$. At the optimal price, total widget sales will be 3,000 and total contribution will be \$11,250. If the widget maker were looking to maximize revenue, he would choose the price $\hat{p}$ such that $(10,000 - 800\hat{p})^+ = 800\hat{p}$ or $\hat{p} = \$6.25$ with corresponding sales of 5,000 and revenue of \$31,250.

Conditions 5.3 and 5.4 can be used to define three conditions that can be used to find profit-maximizing and revenue-maximizing prices.

Marginal revenue equals marginal cost. We can rewrite Equation 5.3 as

$$p^* d'(p^*) + d(p^*) = cd'(p^*). \quad (5.5)$$

The term on the left-hand side of Equation 5.5 is *marginal revenue*—the derivative of total revenue with respect to price. This is the rate at which revenue changes as price increases. Typically, marginal revenue is greater than zero at low prices but less than zero at higher prices. The term on the right-hand side of Equation 5.5 is *marginal cost*: the rate at which cost is changing with price. Note that marginal cost is always less than or equal to zero—an increase in price results in lower demand, which in turn leads to lower total costs. Equation 5.5 is a mathematical expression of the well-known principle that *total contribution is maximized when marginal revenue equals marginal cost*.

The equivalent condition for revenue maximization is *total revenue is maximized when marginal revenue equals zero*. This makes sense—as we increase price from zero, total revenue initially increases with price, implying that marginal revenue is positive. However, at some point, demand begins to decrease at a rate such that total revenue will decrease as price is increased. The point at which revenue stops increasing and begins to decrease is the point of maximum revenue and the point at which marginal revenue equals 0.

Figure 5.2 shows the marginal revenue and marginal cost curves for the widget maker in Example 5.3. Marginal revenue decreases linearly. Marginal cost is constant at $-\$4,000$. The contribution-maximizing price lies where these two curves cross—in this case at $p^* = \$8.75$.

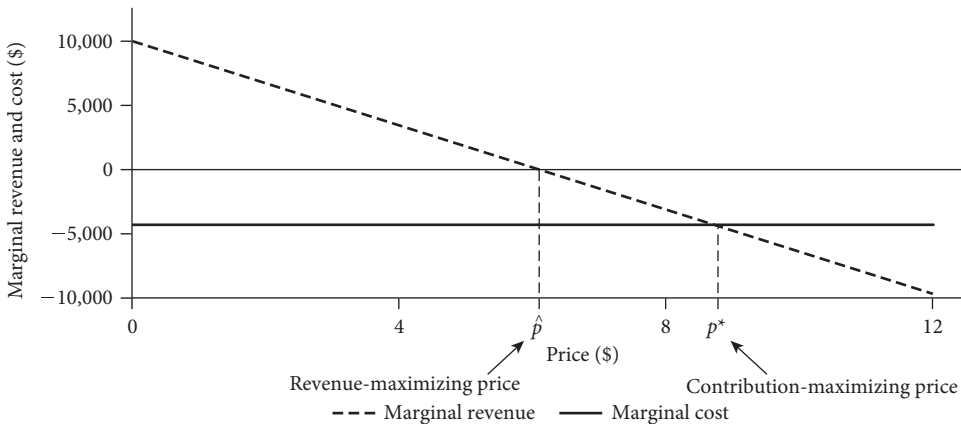


Figure 5.2 Marginal revenue and marginal cost.

The revenue-maximizing price occurs where the marginal revenue line crosses \$0, in this case at $\hat{p} = \$6.25$.

Equation 5.5 provides further useful guidance on price changes that could improve total contribution. If marginal revenue is greater than marginal cost, then the seller can increase his contribution by increasing price. If, on the other hand, marginal revenue is lower than marginal cost, he should decrease his price to increase contribution.

Hazard rate equals inverse margin. Equation 5.5 can be rearranged as $-d'(p^*)/d(p^*) = 1/(p^* - c)$. Recalling that $-d'(p^*)/d(p^*) = h(p)$, where $h(p)$ is the hazard rate associated with the price-response function, we find that the contribution-maximizing price satisfies

$$h(p^*) = \frac{1}{(p^* - c)}. \quad (5.6)$$

Stating that in words, *at the contribution-maximizing price, the hazard rate is equal to the inverse margin*. The equivalent condition for revenue maximization is

$$h(\hat{p}) = \frac{1}{\hat{p}},$$

which means that *at the revenue-maximizing price, the hazard rate is equal to the inverse price*.

Elasticity equals inverse unit margin. Recall from Section 3.2.3 that elasticity is equal to price times hazard rate; that is, $\varepsilon(p) = h(p)p$. We can derive an elasticity-based condition for contribution maximization by multiplying both sides of Equation 5.6 by p^* , which gives us

$$\varepsilon(p^*) = \frac{p^*}{(p^* - c)}. \quad (5.7)$$

In words, *at the contribution-maximizing price, the elasticity is equal to the inverse unit margin rate*, where *unit margin rate* is defined as the unit margin as a fraction of total price.² Equation 5.7 can be rearranged to derive another formula for the optimal price:

$$p^* = \left[\frac{\varepsilon(p^*)}{\varepsilon(p^*) - 1} \right] c. \quad (5.8)$$

Equation 5.8 can be used to calculate the optimal price when both the elasticity and the incremental cost are known.

Example 5.4

A seller faces a constant-elasticity price-response function with an elasticity of 5 and his cost is \$3.00. This means that his profit-maximizing price is given by $p^* = \left(\frac{5}{4} \right) \times \$3.00 = \$3.75$.

The equivalent condition for revenue maximization is

$$\varepsilon(\hat{p}) = 1, \quad (5.9)$$

which means that, *at the revenue-maximizing price, the elasticity is equal to 1*.

5.2.2 Price Elasticity and Profitability

We pause to consider the implications of the relationship between price elasticity and unit margin specified in Equation 5.7. Equation 5.7 can be used to estimate the *implicit elasticity*—that is, the belief that a seller must have about the elasticity of his price response if he believes that his current price is maximizing expected profit.

Example 5.5

A seller believes he is pricing optimally, and his unit margin rate is 20%. This can only be true if the elasticity at his current price is equal to 5.

Imputed price elasticity can serve as a reality check on the credibility of a company's current pricing.

An extremely important implication of Equation 5.7 is that *higher unit margin rates are associated with products that have lower price elasticity*.

Example 5.6

Norton's Salt is an extremely common brand of table salt sold in bulk in grocery stores. The unit cost for Norton's Salt is \$1.50 per one-pound container and it has a price elasticity of 6. La Cloque Rouge Fleur du Sel is a specialty salt harvested by hand from certain salt ponds in the south of France and sold in fancy packaging for use in gourmet cooking. The unit cost for La Cloque Rouge is \$4.20 per one-pound package and it has a price elasticity of 1.5. The contribution-maximizing price for Norton's is $\frac{6}{5} \times \$1.50 = \1.80 , and the contribution-maximizing price for La Cloque Rouge is $\frac{1.5}{5} \times \$4.20 = \12.60 . Thus, the unit margin is \$0.30 per pound for Norton's and \$8.40 per pound for La Cloque Rouge.

Example 5.6 illustrates the importance of price elasticity in determining unit margin. In general, high elasticity is associated with highly commoditized products, which have easy substitution and lots of competition. These products will have low unit margin rates, which means that they typically need to be sold in large volumes to cover fixed production costs and cost of capital. Low elasticity is associated with highly specialized or unique products and services, luxury products, and less competitive markets. These products will typically demonstrate higher unit margin rates, which means that they can profitably be produced in lower volumes. The primary job of marketing and branding can be described as seeking to lower price elasticity by creating product differentiation. In the example, La Cloque Rouge may be chemically indistinguishable from Norton's (or not), but its branding, market positioning, and packaging have distinguished it to the extent that it enjoys a much lower elasticity and, hence, can command a higher unit margin.

5.3 EXISTENCE AND UNIQUENESS OF OPTIMAL PRICES

We have shown that a profit-maximizing price can be determined by solving any one of the optimality conditions in Section 5.2.1 for the value of p^* . However, we have not shown

that such a price actually exists. We have also not shown that, if such a price does exist, it is unique—if hundreds of different prices satisfy one of the optimality conditions, then that condition is not very useful. In addition, as those who remember their calculus will realize, a price that satisfies the first-order conditions is not necessarily a maximizer; it could be a point that *minimizes* profit (or more rarely an inflection point in which profit is neither maximized nor minimized). This section presents conditions under which a price that meets the optimality conditions listed in Section 5.2.1 will be the price that uniquely maximizes profit.

For what follows, we assume three additional properties of price-response functions in addition to the three specified in Section 3.1.2.

1. *Differentiable.* The price-response function has a unique derivative at each point.
2. *Increasing elasticity.* Elasticity increases (or is constant) with increasing price.
3. *Elasticity greater than 1.* Elasticity exceeds 1 for some finite price.

Of these conditions, number 2 may seem the least obvious; however, it is a property of all commonly used price-response functions as well as a real-world phenomenon. You can confirm that all of the price-response functions in Table 3.3 have elasticities that increase with price—except for the constant-elasticity function, which has the same elasticity at every price. The intuitive reason that elasticity should increase with price is that demand decreases as price gets higher so that a small change in price has a much larger percentage impact at a high price than at a low price.³

Finally, we require that the elasticity is greater than 1 for some price. This eliminates the case in which elasticity increases with price but never exceeds 1. In this case, as discussed above, the seller could always increase both revenue and profit by increasing his price. Since this is unrealistic, it must be the case that elasticity exceeds 1 for some price and thus, since elasticity is an increasing function of price, it will exceed 1 for all prices greater than that.

We can now make some statements about existence and uniqueness of optimal prices. *If properties 1–3 above hold for a price-response function that also obeys the four conditions listed in Section 3.1.2 and unit cost is greater than or equal to zero, then*

1. *There exist unique profit-maximizing and revenue-maximizing prices p^* and $\hat{p}$ with $p^* \geq \hat{p}$.*
2. *The profit-maximizing price p^* satisfies $p^* \geq c$ so that the seller does not realize negative contribution at the profit-maximizing price.*

These statements can be proved mathematically, but they can also be seen clearly in Figure 5.3. Elasticity is a continuous, increasing function of price denoted by $\varepsilon(p)$. The revenue-maximizing price $\hat{p}$ occurs where the elasticity is equal to 1. In the figure, this occurs where the elasticity curve intersects 1. The intersection of the elasticity curve with $(p - c)/p$ occurs when $\varepsilon(p) = (p - c)/p$, which, according to condition 5.7, occurs at the profit-maximizing price p^* . It is clear that, for any nonincreasing elasticity curve, there must be one and only one intersection between the elasticity curve and each of the two downward sloping curves, which demonstrates the existence and uniqueness of the two prices $\hat{p}$ and p^* .

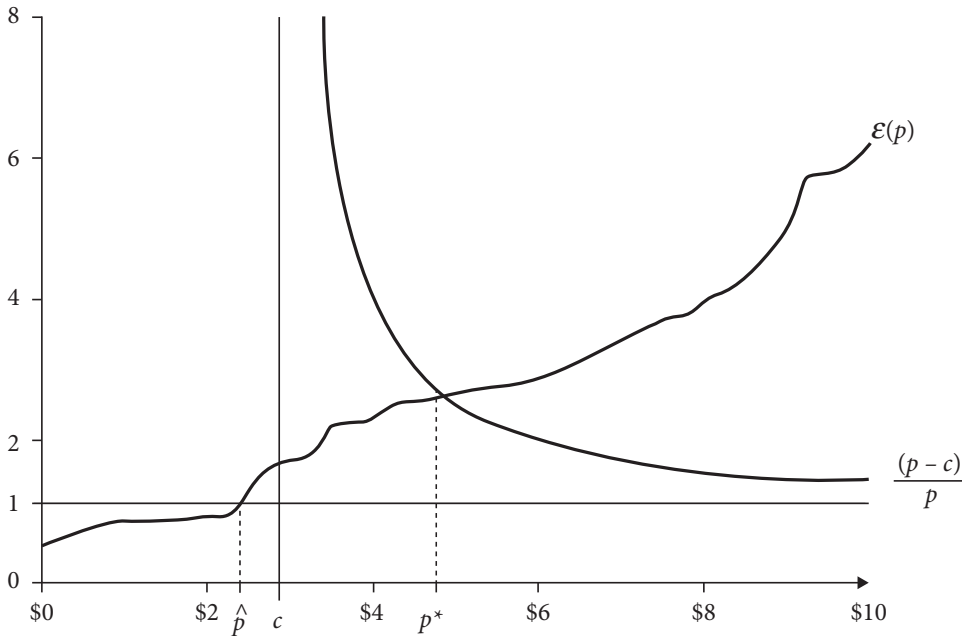


Figure 5.3 Existence and uniqueness of the revenue-maximizing price $\hat{p}$ and contribution-maximizing price p^* .

It is also obvious from Figure 5.3 that the intersection of the elasticity curve with $(p - c)/p$ must occur at a price $p^* > c$. This means that the unit margin at the profit-maximizing price is greater than or equal to 0. Furthermore, $(p - c)/p \geq 1$, which means that the revenue-maximizing price will be less than or equal to the profit-maximizing price, with equality occurring only in the case that $c = 0$. Note that the revenue-maximizing price may be less than the unit cost (as it is in Figure 5.3), in which case a revenue-maximizing seller would lose money on each sale.

One way to check the optimality of p^* is to use the *marginality test*, which states that a particular price is maximizing contribution only if raising the price by a penny or lowering the price by a penny results in reduced contribution. The principle of marginality should be fairly intuitive—if we could increase contribution by changing the price, then the current price would not be optimal—but it is a useful check nonetheless. Table 5.2 shows the results of varying the price in Example 5.3 from \$8.74 to \$8.76. As expected, the price of \$8.75 results in the highest margin of the three alternatives. However, the pattern of change among the prices is instructive: At \$8.74 we are selling eight more units per month, but this is not enough to make up for the lost margin per unit. At \$8.76 we are making a penny more per unit sold, but this is offset by the loss of sales of eight units. The optimal price, \$8.75, exactly balances the gain in units sold from the potential loss of margin from raising prices further.

TABLE 5.2
Impact of price on marginal contribution near the optimal price

Price (\$)	Demand	Unit margin (\$)	Total contribution (\$)
8.74	3,008	3.74	11,249.92
8.75	3,000	3.75	11,250.00
8.76	2,992	3.76	11,249.92

5.4 OPTIMIZATION WITH MULTIPLE PRICES

The optimality conditions presented in Section 5.2.1 pertain to a seller selling a single product. Of course, in many cases, a seller will be setting prices for many products, representing multiple cells in the PRO cube. If demand for products is independent, then the seller can set the price for each product independently using any one of the approaches described so far. However, in many cases, product demands will not be independent—some products may be substitutes for each other, and some may be complements. In this case, the optimal price for a particular product will depend on the prices of other products, and optimal prices for different products cannot be calculated independently.

Assume that a seller is setting n prices given by the vector $\mathbf{p} = (p_1, p_2, \dots, p_n)$ and demand mappings such that $d_i(\mathbf{p})$ is the demand for product i as a function of the entire price vector $\mathbf{p}$. Then the profit optimization problem is to find the vector of prices that solves the optimization problem

$$\max \Pi(\mathbf{p}) = \sum_{i=1}^n d_i(\mathbf{p})(p_i - c_i),$$

subject to $p_i \geq l_i$, where l_i is a lower bound to be imposed on the price of product i and c_i is the incremental cost of product i . Lower bounds are required because an unconstrained optimum might result in some prices that are substantially lower than cost, or even less than 0. Note that the demand for each product depends not only on its own price but also on the price of all other products.

The first-order necessary conditions for the optimal prices in the multiprice case can be written as

$$\frac{\partial \Pi(\mathbf{p})}{\partial p_i} = \frac{\partial d_i(\mathbf{p})}{\partial p_i}(p_i - c_i) + d_i(\mathbf{p}) + \sum_{j=1, j \neq i}^n \frac{\partial d_j(\mathbf{p})}{\partial p_i}(p_j - c_j) = 0.$$

The condition for prices to be optimal can be written as

$$\frac{\partial d_i(\mathbf{p}^*)}{\partial p_i}(p_i^* - c_i) + d_i(\mathbf{p}^*) + V_i(\mathbf{p}^*) = 0,$$

where $V_i(\mathbf{p}^*)$ is defined by

$$V_i(\mathbf{p}^*) = \sum_{j=1, j \neq i}^n \frac{\partial d_j(\mathbf{p})}{\partial p_i}(p_j - c_j).$$

With a little algebra, we can derive the optimality condition that must hold for each product:

$$\varepsilon_{ii}(p_i^*) = \left(\frac{p_i^*}{p_i^* - c_i} \right) \left[1 + \frac{V_i(\mathbf{p}^*)}{d_i(\mathbf{p}^*)} \right]. \quad (5.10)$$

If the price of product i does not influence the demand for any product other than itself, then $V_i(\mathbf{p}^*) = 0$ and Equation 5.10 reduces to Equation 5.3. In this case, the optimal price of product i can be found independently of the prices of any other products using methods described above. If $V_i(\mathbf{p}^*) > 0$, it means that, as a whole, product i has more complements than substitutes and its price should be lower than its independent optimal price because its sales stimulate other sales. If $V_i(\mathbf{p}^*) < 0$, then substitution dominates complementarity and the optimal price of product i is higher than its individual optimal price. The idea is that, as we raise the price, some of the loss from lost sales is recaptured through sales of substitutes.

The problem of setting optimal prices in the face of substitutes and complements requires finding the values p_i^* of that simultaneously satisfy Equation 5.10 for all i . This can be a complex undertaking, but I propose an iterative algorithm that can be used if we know all elasticities and cross-derivatives.

Optimal Multipricing Algorithm

0. Set all prices $p_i^0 = c_i$ for $i = 1, 2, \dots, n$. Choose a convergence criterion $\Delta > 0$ and set the iteration counter $k = 1$.
1. Loop through all products $1, 2, \dots, n$.
 - a. For product i , calculate p_i^k that satisfies

$$\varepsilon_{ii}(p_i^k) = \left(\frac{p_i^k}{p_i^k - c_i} \right) \left[1 + \frac{V_i(\mathbf{p}^{k-1})}{d_i(\mathbf{p}^{k-1})} \right]$$

- b. Calculate the change in price between iterations: $\delta_i(p_i^k) = |p_i^k - p_i^{k-1}|$.
3. If $\max_i \delta_i < \Delta$, stop and set $\mathbf{p}^* = \mathbf{p}^k$. Otherwise set $k \leftarrow k + 1$ and go to step 1.

If complementary and substitution effects are not too strong (as is usually the case), the multipricing algorithm will converge to the optimal prices. However, the algorithm has a number of obvious implementation challenges, notably its assumption that we know not only own-price elasticity but the cross-price derivatives for all combinations of products.

5.5 A DATA-DRIVEN APPROACH TO PRICE OPTIMIZATION

Applying the optimality conditions described in Section 5.2.1 to find the profit-maximizing or contribution-maximizing prices requires that we know the price-response function for a product. This is a strong requirement. Assume that we know nothing about the price-response function for a product, except that it satisfies the conditions listed in Section 5.2.1. Furthermore, we assume that the following two additional conditions hold:

1. We are able to test the demand response at different prices.
2. The product sells sufficiently rapidly that we can quickly observe demand responding to different prices.

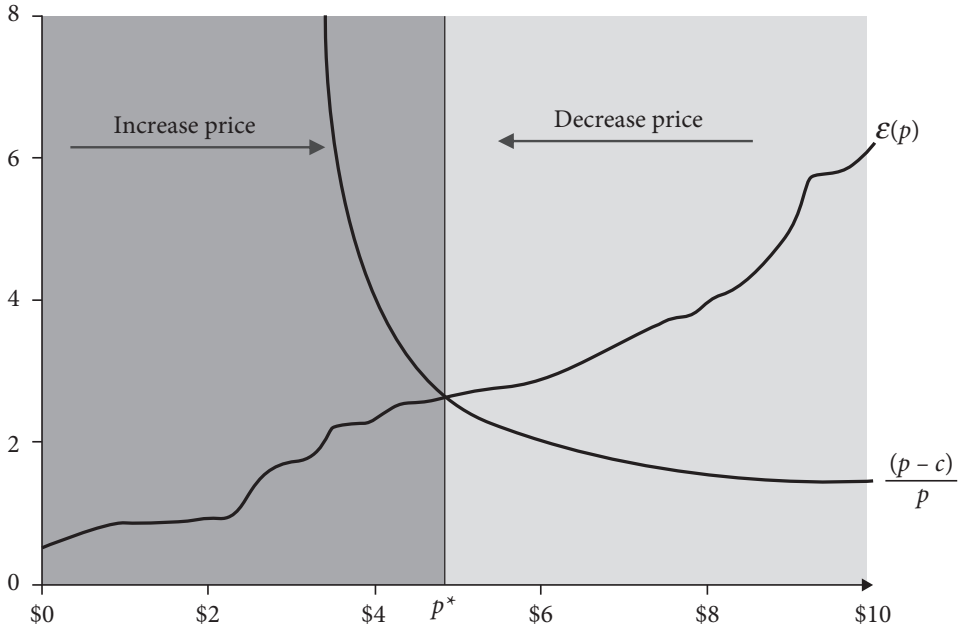


Figure 5.4 Regions for raising, lowering, and holding price in the learning-and-earning algorithm.

In this case, we can use a *multi-armed bandit* approach to determine the optimal price that maximizes a given objective function. This is a test-and-learn (or learning-and-earning) approach in which we observe the payoff from different decisions—in this case, different prices—and gradually favor the prices that generate the highest total contribution. (More details on multi-armed bandit approaches are given in Appendix B.)

To see how this approach can be used to update prices for a seller who seeks to maximize profit, consider Figure 5.4, which graphs elasticity $\varepsilon(p)$ and the inverse unit margin curve $p/(p - c)$ for some product for every $p \geq c$. As in Figure 5.3, the unique profit-maximizing price is where the two curves cross. Note that, if $p/(p - c) > \varepsilon(p)$, then $p < p^*$, and we could move closer to the optimal price (and increase profitability) by increasing the price. If $p/(p - c) < \varepsilon(p)$, then the current price is too high, and we could move closer to the optimal price by lowering the price.

The seller does not know the price-response function and thus does not know the elasticity at any price. However, he has the capability to set different prices and observe demand at those prices. This suggests the following approach to finding the optimal price.

Learning-and-Earning Algorithm for Price Optimization

0. Choose initial parameters $\delta > 0$, $\Delta > 0$, and $\epsilon > 0$.
1. Set $k = 0$, and start with an initial test price $p_0 > 0$.
2. For some small δ , test demand at prices p_k and $p_k + \delta$, which we denote by $\hat{d}(p_k)$ and $\hat{d}(p_k + \delta)$, respectively. Estimate local elasticity by

$$\hat{\epsilon}(p_k) = \frac{[\hat{d}(p_k) - \hat{d}(p_k + \delta)]p_k}{\hat{d}(p_k)\delta}$$

3. Compare $\hat{\epsilon}(p_k)$ to $(p_k - c)/p_k$. Update depending on the relationship between the two:
 - a. If $\hat{\epsilon}(p_k)$ is close to $(p_k - c)/p_k$, say $|(p_k - c)/p_k - \hat{\epsilon}(p_k)| < \Delta$, then p_k is close to p^* . Stop the algorithm and set the price to p_k .
 - b. If $(p_k - c)/p_k - \hat{\epsilon}(p_k) > \Delta$, the current price is too low. Set $p_{k+1} \leftarrow p_k + \epsilon$ and $k \leftarrow k + 1$ and go to step 2.
 - c. If $\hat{\epsilon}(p_k) - (p_k - c)/p_k > \Delta$, the current price is too high. Set $p_{k+1} \leftarrow p_k - \epsilon$ and $k \leftarrow k + 1$ and go to step 2.

The learning-and-earning algorithm described above has the great advantage that it makes no assumption about the functional form of the price-response function. Rather, it estimates the elasticity at the current price and determines whether to raise the price, lower the price, or hold the price the same. While it does not require a form for the price-response function, it does require that the following values be specified:

- δ is the distance between prices that are being tested to estimate elasticity.
- Δ is the stopping condition—the difference between the estimated elasticity and the inverse unit margin.
- ϵ is the amount by which the price is adjusted if the elasticity is not sufficiently close to the inverse unit margin.

These values are called hyperparameters and they determine how quickly (and if at all) the approach will converge to the optimal price. For example, if ϵ is set too small, the price will approach the optimal price very slowly—which means that a lot of time will be spent learning rather than earning. However, if ϵ is set too large, the price may not converge at all: it may bounce back and forth between a price that is below the optimum and one that is above the optimum. For this reason, it is usually ideal to reduce ϵ from iteration to iteration so that the steps get smaller over time. One common approach is to set ϵ as the initial step size and then set the step size on iteration k according to $\epsilon_k = \epsilon/k$. (More details can be found in Sutton and Barto 2018, sec. 2.5.) In general, setting and updating hyperparameters involves quite a bit of tuning—starting with plausible values and adjusting with experience to improve performance.

In theory, the data-driven algorithm would be run once until it converged to an optimal price, which would then remain in place forever. A finite period of learning would be followed by an infinite future of earning. However, things are rarely that simple. In reality, it is likely that the price-response function (and hence the local elasticity) will change over time. Reasons for change include competitive actions, seasonality, life-cycle changes, introduction of substitutes, and changes in consumer tastes, among other factors. With such events, the current price is likely to be no longer optimal. The most common cure for this issue is to initiate testing for a price that is believed to be optimal with some small probability in

each period. If elasticities have changed, then initiate the learning-and-earning algorithm for price optimization again to determine the optimal price under the new conditions.

5.6 COMPETITIVE RESPONSE AND OPTIMIZATION

Chapter 4 shows that if we have access to competitive prices, we can incorporate them in price-response estimation. However, this would seem to take us only partway to nirvana. After all, if we are taking competitive prices into account in setting our prices, we should anticipate that our competitors will take our price into account when they set their prices. If we drop a price, we should anticipate the possibility that competitors will match it, possibly erasing much of the additional demand we might otherwise expect. The results of changing a price would certainly be different, depending on whether our competitors decided to match it.

Attempts to predict competitive response and incorporate it into current pricing decisions generally fall within the realm of game theory, and there is a vast literature on the application of game theory to pricing. While game theory is an important tool in strategic pricing, it is less relevant to the tactical decisions of pricing and revenue optimization. Most pricing and revenue optimization systems do not seek to forecast competitive response and incorporate it explicitly into their recommendations. There are many reasons for this, but I believe that one is particularly important: pricing and revenue optimization is based on playing the best response (i.e., finding the expected contribution-maximizing price) to whatever the competitors are currently doing. Theoretical examinations of markets find that this is almost always a good thing to do and often the best possible action. For example, if price response in a market is well modeled by a multinomial logit function and sellers all set their prices to maximize profit based on the prices set by others, the market will converge to a unique set of equilibrium prices for all sellers. Game theory is not required in this case to determine the profit-maximizing price for any seller (Gallego et al. 2006).

The philosophy of pricing and revenue optimization is to make money by many small adjustments, searching for and exploiting many small and transient opportunities of profit as they appear in the marketplace. It is about making a little more money from each transaction. It is not about finding the great pricing move that will stagger the competition. Many of the price adjustments called for by pricing and revenue optimization are likely to fall below the radar screen of the competition and may not trigger any explicit response whatsoever. Furthermore, if competitors respond to our prices in the future as they have in the past, the price-response functions estimated according to the methods described in Chapter 4 will already incorporate the response of competitors.

This does not mean that competitive response need never be considered in pricing. The potential competitive responses to major bet-the-company bids or substantial changes in pricing strategy need to be carefully considered. For example, in the early 1990s, Hertz Rent-a-Car initiated the use of a sophisticated pricing optimization system. Previously Hertz, like other national car rental companies, had changed prices only rarely. Whenever one of the companies dropped a price, the others were likely to follow immediately—often with even larger drops—precipitating an industry-wide fare war. Hertz took great pains to commu-

nicate to the industry that its new system would be changing prices much more commonly than before—some prices would go up, some would go down, but all would change much more frequently. The communication was successful: Hertz was able to initiate its revenue management program without inciting retaliatory price wars. Once Hertz initiated revenue management, it was able to generate additional revenue through thousands of small adjustments to prices that its competitors were unable to match.

Similarly, online pricing enables much more rapid access to competitive prices via scraping competitor websites. For a third-party seller on Amazon, there is great value to winning the buy box, or becoming a featured offer. However, in most cases a seller can only win the buy box if he offers the lowest price (or at least matches the lowest price) for an item. Because winning the buy box is effective in driving sales, sellers are inspired to match (or undercut) other sellers, which can lead to pricing dynamics that drive the price to a floor. Once the price has reached the floor, it is costly for a seller to raise his price and risk losing the buy box. This situation is similar to a classical Bertrand game, in which two sellers competing to sell an identical item in a marketplace are both forced to lower their price to just above marginal cost.⁴

5.7 OPTIMIZATION WITH MULTIPLE OBJECTIVE FUNCTIONS

In many cases, sellers are not interested in only a single objective. Many times a manager with profit and loss responsibility has been enjoined to maximize both profitability and market share. However, the optimization approaches that we have considered so far in this section can only deal with a single objective function: either maximize profit *or* maximize revenue. As we have seen, in most cases doing both simultaneously is not possible—the price that maximizes profit is not the one that will maximize revenue and vice versa, except in the special case in which incremental cost is zero. In this section, we discuss how a seller can understand the trade-offs between different objective functions and incorporate them into decision making. We use the example of maximizing revenue versus maximizing profit, but the same concepts can be used for any two competing objectives such as profit and market share.

Example 5.7

The CEO of the widget-making company in Example 5.3 decides that the firm's goal for the next month will be to maximize revenue from widget sales as part of the long-term strategy to increase market share. The revenue-maximizing price can be found by solving Equation 5.4. The resulting revenue-maximizing price is equal to \$6.25, with corresponding sales of 5,000 units and revenue of \$31,250. The corresponding per-unit margin is \$1.25, and the total-contribution margin is $5,000 \times \$1.25 = \$6,250$. We can compare this to the maximum margin contribution of \$11,250 and conclude that to maximize total revenue, the company needs to give up $\$11,250 - \$6,250 = \$5,000$ of contribution margin to gain \$5,000 in additional revenue.

The decision that management needs to make in Example 5.7 is whether it is worth giving up a total contribution of \$5,000 per month to buy an additional 2,000 units of

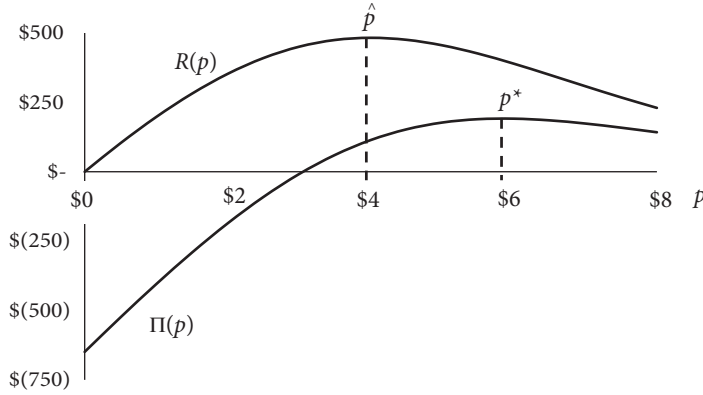


Figure 5.5 Profit and revenue as a function of price.

demand. While this trade-off is clear, it may be that management is interested in some goal other than pure profit maximization or pure revenue maximization. Figure 5.5 shows how profit and revenue change as a function of price for an example seller. In this figure, revenue is maximized at the price $\hat{p} = \$4.00$ and profit is maximized at the price $p^* = \$6.00$, with $\hat{p} \leq p^*$. (Recall that the two prices will be equal only if the incremental cost is zero.) Now, assume that the seller is interested in both revenue and profit. Then, as evident from Figure 5.5, he would only consider prices in the interval between $\hat{p}$ and p^* . For prices below $\hat{p}$, he could increase both revenue *and* profit by increasing price. For prices above p^* , he could increase both revenue and profit by decreasing price. Thus, if profit and revenue are the only two goals he is interested in, he would only consider prices greater than $\hat{p}$ and less than p^* .

This leaves open the question of which price in the range $[\hat{p}, p^*]$ he should choose. If the seller knows his trade-off between profit and revenue, then it is straightforward to accommodate this trade-off. Let us say, for example, that he is willing to give up \$1.00 in profit in order to drive an additional \$5.00 in sales. This trade-off could reflect management's judgment about how the stock market values growth relative to profit in determining his share price. In this case, he would seek to maximize $R(p) + 5\Pi(p)$, where $R(p)$ and $\Pi(p)$ are revenue and profit, respectively, at price p . More generally, let us assume that the seller's trade-off is \$1 of profit in order to gain dollars of revenue. We call α the seller's *revenue-profit trade-off*. The seller is seeking to maximize the objective function $pd(p) + \alpha d(p)(p - c)$, which is equivalent to maximizing $d(p)[(1 + \alpha)p - \alpha c]$. This means that we can write his objective function as

$$\max d(p) \left[p - \left(\frac{\alpha}{1 + \alpha} \right) c \right].$$

This means that the seller can find the price that satisfies his revenue-profit trade-off by finding the profit-maximizing price with the modified unit cost $\left(\frac{\alpha}{1 + \alpha} \right) c$. If $\alpha = 0$, the seller cares only about revenue, and he acts as if his unit cost is zero. As $\alpha \rightarrow \infty$, the seller cares only about maximizing expected profit.

Example 5.8

The widget maker in Example 5.3, with a unit cost of \$5.00 and facing the price-response function of $d(p) = (10,000 - 800p)^+$, decides that he has a revenue-profit trade-off of 10; that is, he would be willing to forgo \$1 of profit to gain \$10 in revenue. In this case, he can determine the corresponding optimal price by setting $d(p) = (10,000 - 800p)^+ = 800[p - (\frac{10}{11})\$5.00]$, which gives $p = \$8.52$. The corresponding revenue is \$27,127.68 and the total contribution is \$11,207.68.

If a seller knows his revenue-profit trade-off, then he can use any of the optimal price criteria listed in Section 5.2.1, substituting the modified cost for the actual cost. For example, the optimal price with a revenue-profit trade-off of α must satisfy the condition

$$\varepsilon(p^*) = \frac{p^*}{p^* - [\alpha / (1 + \alpha)] c}. \quad (5.11)$$

Note that when $\alpha = 0$, Equation 5.11 becomes the revenue-maximizing condition in Equation 5.9, and as α approaches infinity, it approaches the profit-maximizing condition in Equation 5.7. Furthermore, both the direct optimization approach and the data-driven approach described in Section 5.5 can easily be adapted to work with the modified cost in order to find the price that achieves the desired trade-off.

The methods above can be used by a seller to determine the price that meets his revenue-profit trade-off *assuming that he can quantify that trade-off*. In many cases, a seller cannot provide a value for his revenue-profit trade-off. In this case, an *efficient frontier* can help the seller visualize the potential trade-offs between revenue and profit. An efficient frontier, such as the one shown in Figure 5.6, shows all of the possible combinations of revenue and profit (or any two other objectives) that a seller could achieve by varying the price of an item. Let $R(p)$ and $\Pi(p)$ denote the revenue and profit, respectively, realized at price p . The dotted curve in Figure 5.6 shows all of the profit and revenue combinations that the seller could

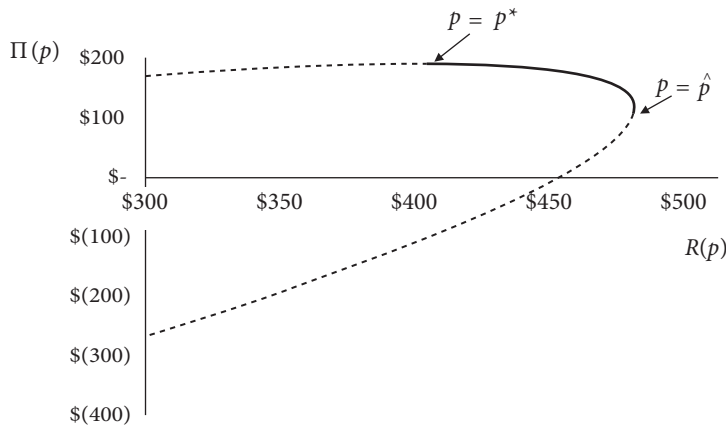


Figure 5.6 A single-product efficient frontier.

realize at different prices. However, the solid part of the curve is the only section of interest. The rightmost point on the solid part of the curve represents the maximum revenue that can be achieved and the corresponding profit: it is the point $(R(\hat{p}), \Pi(\hat{p}))$. The leftmost point on the solid part of the curve represents the maximum contribution that can be achieved and the corresponding revenue: it is the point $(R(p^*), \Pi(p^*))$. The solid curve connecting these two points is the efficient frontier, which consists of the only achievable combinations of profit and revenue that the firm would consider. The dashed points represent combinations of profit and revenue that the seller could achieve at some price, but in every case there is a price on the efficient frontier that generates both more profit *and* more revenue.

The efficient frontier is a convenient and frequently used approach to help visualize the trade-offs involved in pricing. In the case of a single price, the efficient frontier can be generated by starting at the revenue-maximizing price and calculating both profit and revenue for prices at regular intervals up to the profit-maximizing price. These can then be graphed to generate an efficient frontier.

Example 5.9

A seller has an incremental cost of \$4.00 and faces a linear price-response function given by $d(p) = 10,000\left(1 - \frac{p}{12}\right)$. His revenue-maximizing price is \$6.00, and his profit-maximizing price is \$8.00. Revenue and profit for each price at \$0.10 intervals between \$6.00 and \$8.00 are shown in Table 5.3. Plotting these out gives the efficient frontier shown in Figure 5.6.

The situation is more interesting when we consider two or more products. In the case of multiple products, the feasible set of revenue and profit combinations is a *region* rather than a curve as in the example of a single price. An example is shown in Figure 5.7. In this

TABLE 5.3
Efficient frontier for Example 5.9

Price (\$)	Profit (\$)	Revenue (\$)
6.00	10,000.00	30,000.00
6.10	10,325.00	29,991.67
6.20	10,633.33	29,966.67
6.30	10,925.00	29,925.00
6.40	11,200.00	29,866.67
6.50	11,458.33	29,791.67
6.60	11,700.00	29,700.00
6.70	11,925.00	29,591.67
6.80	12,133.33	29,466.67
6.90	12,325.00	29,325.00
7.00	12,500.00	29,166.67
7.10	12,658.33	28,991.67
7.20	12,800.00	28,800.00
7.30	12,925.00	28,591.67
7.40	13,033.33	28,366.67
7.50	13,125.00	28,125.00
7.60	13,200.00	27,866.67
7.70	13,258.33	27,591.67
7.80	13,300.00	27,300.00
7.90	13,325.00	26,991.67
8.00	13,333.33	26,666.67

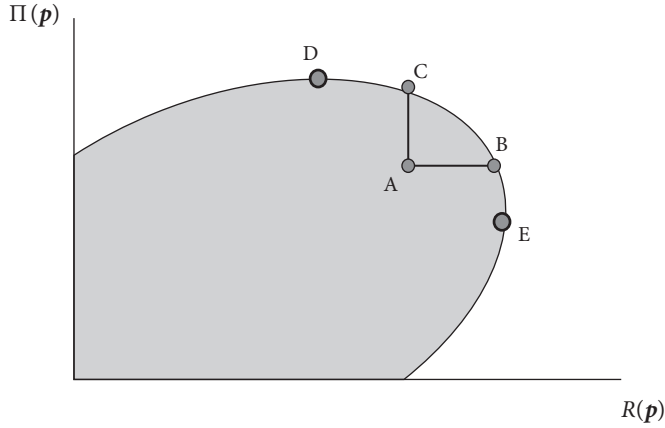


Figure 5.7 An efficient frontier with multiple products. *Source:* Adapted from Phillips 2018.

figure, the shaded area is the set of *feasible* revenue and profit combinations—that is, the combinations of revenue and profit that could be achieved by setting different prices for two different products. In fact, it is only points on the boundary of the feasible region that are of interest to a seller who cares only about revenue and profit. To see this, consider a seller whose current set of prices results in the revenue and profit combination represented by point A in Figure 5.7. This seller could realize the same revenue but higher profit by setting the prices that result in the revenue and profit combination at point B. Alternatively, the seller could realize the same profit but more revenue by pricing at point C. Finally, the seller could realize more revenue and *more* profit by pricing at one of the points on the boundary of the frontier between B and C. In any case, a seller who cared only about revenue and profit would not (knowingly) set prices that resulted in the revenue and profit combination at point A.

In the case of multiple products, the efficient frontier is the set of *undominated* revenue and pricing pairs, which means that for a price on the efficient frontier, additional profit can be achieved only by reducing revenue and vice versa. Points not on the efficient frontier are *dominated*, which means that either one or both of profit and revenue can be increased without decreasing the other. Let $\mathbf{p} = (p_1, p_2, \dots, p_n)$ denote the vector of prices offered by the seller. Then, in Figure 5.7, point D corresponds to the revenue and profit achieved at the profit-maximizing set of prices—that is, $(R(\mathbf{p}^*), \Pi(\mathbf{p}^*))$. Point E is the revenue and profit combination achieved at the revenue-maximizing price, $(R(\hat{\mathbf{p}}), \Pi(\hat{\mathbf{p}}))$. The boundary of the feasible region between these points is the efficient frontier.

As in the case of a single price, the multiprice efficient frontier can help a seller visualize the trade-off between revenue and profit that he faces and can help him choose accordingly. In fact, the slope of the tangent line to the efficient frontier at a point is the ratio between incremental revenue and incremental profit at that point. Thus, in Figure 5.8, line A has a slope of about 0.33. This means that a seller choosing point D values an additional dollar of profit about 3 times as much as an additional dollar of revenue. The slope of line B is about 2.5, which means that a seller choosing point E values each additional dollar of profit only about $1/2.5$ or 40% of the value of an additional dollar of revenue.

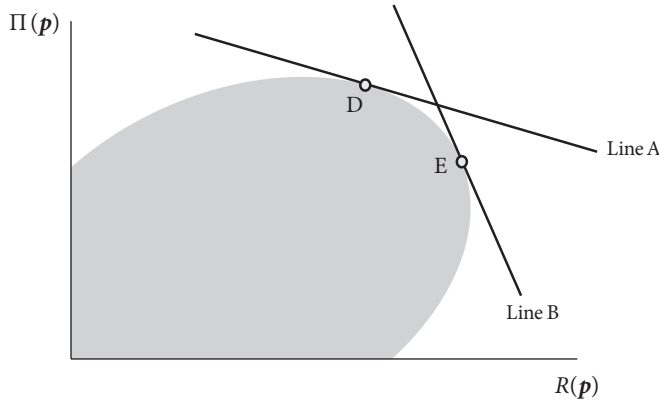


Figure 5.8 Points on the efficient frontier that might be chosen by two different sellers.
Source: Adapted from Phillips 2018.

There are two ways in which the efficient frontier can be calculated with multiple prices. The first is to solve the optimization problem with an objective function of maximizing $\alpha R(\mathbf{p}) + (1 - \alpha)\Pi(\mathbf{p})$, where $0 \leq \alpha \leq 1$ is a parameter that determines the trade-off between incremental profit and incremental revenue. To trace out the efficient frontier, the seller could solve the optimization problem $\alpha R(\mathbf{p}) + (1 - \alpha)\Pi(\mathbf{p})$ for different values of α —say, $\alpha = (0, .1, .2, \dots, .9, 1.0)$ —and then plot the results. The points plotted will be points on the efficient frontier such as the one in Figure 5.8.

An alternative is to maximize profit with different revenue constraints. The idea would be to first maximize revenue and profit to determine both the profit- and revenue-maximizing vectors of prices ($\mathbf{p}^*$ and $\hat{\mathbf{p}}$, respectively) and the profit and revenue corresponding to each vector of prices. Then the maximum revenue that the seller can achieve is $\hat{R} = R(\hat{\mathbf{p}})$, and the revenue he would achieve at the profit-maximizing price is $R^* = R(\mathbf{p}^*) < \hat{R}$. The seller would then choose n equally spaced values of revenue between R^* and $\hat{R}$, where the i th value of revenue is given by $R^i = R^* + \left(\frac{i}{n}\right)(\hat{R} - R^*)$. For each value of $i = (1, 2, \dots, n-1)$, the supplier would solve the optimization problem

$$\begin{aligned} &\text{Maximize } \Pi(\mathbf{p}), \\ &\text{subject to } R(\mathbf{p}) \leq R^i. \end{aligned}$$

The resulting values of $\Pi(\mathbf{p})$ and $R(\mathbf{p})$ are points on the efficient frontier.

5.8 SUMMARY

- The cost used in pricing and revenue optimization is the *incremental cost* of a customer commitment. It is the difference between the total costs a company would incur from satisfying the commitment versus not making it. The incremental cost will vary with the duration and size of the commitment and is not a fully allocated cost. A rule of thumb is that if a change in demand resulting from a change in price

would not change the cost, then the cost should not be included in determining the price.

- For a single product, the following are equivalent conditions for a price to be a profit maximizer:
 - Marginal revenue equals marginal cost.
 - The derivative of total contribution with respect to price is zero.
 - The hazard rate equals the inverse unit margin.
 - Elasticity equals the inverse unit margin.
- For a single product, the revenue-maximizing price is always less than or equal to the profit-maximizing price. They are only equal if the unit cost is equal to zero. The following are equivalent conditions for a price to be revenue maximizing:
 - Marginal revenue equals zero.
 - The derivative of revenue with respect to price is zero.
 - The hazard rate equals the inverse of price.
 - Elasticity equals one.
- Assume that the price-response function for a single product satisfies the following conditions:
 - It is *decreasing in price*.
 - It is *continuous*.
 - It is *differentiable*.
 - It has *elasticity that increases with price*.
 - It has *elasticity greater than 1* for some finite price.

Then there exists a unique profit-maximizing price that is greater than the unit cost. There is also a unique revenue-maximizing price that may or may not be greater than the unit cost.

- The relationship between elasticity and unit margin implies that products facing lower elasticity will have higher unit margin rates than those facing higher elasticity. This means that—all else being equal—sellers should deploy their marketing, advertising, and product improvement budgets in a way that is targeted to reducing price elasticity for their products.
- A *data-driven approach* can be used to set prices that approach the profit-maximizing price p^* by using the following facts: at prices higher than p^* , elasticity is greater than the inverse unit margin, and at prices lower than p^* , elasticity is lower than the inverse unit margin. The data-driven approach tests demand at different prices, estimates the elasticity, and then uses the relationship between elasticity and inverse unit margin to calculate a new price.
- Often, sellers are interested in both profit and revenue. If a seller can quantify his trade-off between profit and revenue, he can determine the price that best meets the trade-off by solving an optimization problem with an objective function that

appropriately weights profit and revenue. If the seller cannot quantify his trade-off, an efficient frontier can help him visualize the possible combinations of revenue and profit that he can achieve.

5.9 FURTHER READING

Activity-based costing (ABC) and the related discipline of activity-based management (ABM) arose in the mid-1980s from the realization that typical cost accounting systems often resulted in a distorted view of the value of different activities primarily because of improper allocation of fixed costs. ABC is a *management accounting* approach designed to help management understand the incremental cost implications of doing more or less of certain activities (e.g., carrying fewer or more passengers), as opposed to *financial account systems* where all costs are allocated according to various regulations (e.g., SEC or FASBI) to provide standardized and consistent reports to regulators and shareholders. The idea was popularized in Kaplan and Cooper 1997. For an updated view of ABC, see Kaplan and Anderson 2007. While some organizations (although by no means all) have adopted ABM, the significance for pricing optimization and revenue management is understanding that the costs that are relevant to pricing decisions are *only* those that would change given the change in demand resulting from the pricing decision.

The Bertrand model was probably the earliest example of game-theoretic reasoning in economics: it was articulated by the French mathematician Joseph Louis François Bertrand in 1883 and developed into a mathematical model by Francis Edgeworth in 1925. Since then, there has been an extensive economics literature on the application of game theory to pricing. For thorough treatments of game theory with a focus on economics, standard references are Fudenberg and Tirole 1991, Myerson 1991, and Gibbons 1992. For a survey of game-theory models in pricing, see Kopalle and Shumsky 2012.

The efficient frontier was introduced by Harry Markowitz (1959) to illustrate the trade-off between risk and return in an investment frontier. For an explanation of how it can be derived and used in this context, see Luenberger 1998, chap. 6. The efficient frontier has been widely used in many situations in which there is a trade-off between two competing alternatives. For more detail on its use in pricing and revenue management, see Phillips 2012b, and for the use of efficient frontiers in consumer credit pricing, see Phillips 2018, chap. 7.

5.10 EXERCISES

1. Consider a seller seeking to maximize contribution. Under what relative values of his current price p , his cost c , and his point elasticity $\varepsilon(p)$ should the seller raise his price to increase contribution? Under what conditions should he lower his price? Keep his price the same?
2. A retailer is currently charging a price of \$147.52 for a Hewlett-Packard OfficeJet printer that costs him \$112.00 per unit. He determines that the point price elasticity of this model of printer is 5.1 at its current price.

- a. If he wants to maximize net contribution, is he better off raising his price, lowering his price, or keeping it the same?
- b. If the elasticity of 5.1 is valid over a range of at least \$20.00 on either side of his current price, what is his optimal (contribution-maximizing) price?
3. An auto manufacturer can manufacture compact cars for an incremental cost of \$5,000 apiece. She faces a logit price-response function for sales in the next month, with parameters $C = 40,000$, $b = 0.0005$, and $\hat{p} = \$12,000$. What price will maximize the total contribution? How many cars will she sell during the month?
4. Frank owns a hot-dog stand. It costs him \$1 to make each hot dog and he faces a linear price-response function $d(p) = (100 - 8p)^+$ for each day.
 - a. Find Frank's contribution-maximizing price and the corresponding contribution. (For this and following questions, allow non-integer units of sales for simplicity.)
 - b. Suppose a big hot-dog franchise offers him a contract. According to the contract, the franchise will provide an unlimited number of prepared hot dogs per day for a fixed cost of K dollars per day. If Frank accepts this contract, what is his optimal price?
 - c. For what values of K should Frank accept the contract if he is maximizing profit?
 - d. Now the franchise offers Frank a new contract that charges a daily fixed cost of \$25. However, they will provide at most 40 hot dogs to Frank every day. Should Frank accept this offer? Explain your answer.

NOTES

1. The problem of finding optimal prices for a good with limited stock whose value is decreasing over time is a markdown pricing problem discussed in detail in Chapter 12.

2. The unit margin rate $L(p) = (p - c)/p$ is also known as the *Lerner index*, which was named after the economist Abba Lerner, who proposed that it be used as a measure of market power. If the price being charged by a firm is denoted by p , then $L(p) = 0$ would indicate that a firm has no market power (price equals incremental cost) and $L(p) = 1$ would indicate very high market power. Note that the condition for a profit-maximizing price can be written as $\varepsilon(p^*) = 1/L(p^*)$.

3. Elasticity increasing with price is equivalent to a property of the underlying willingness-to-pay distribution called *increasing general failure rate (IGFR)*. Any price-response function corresponding to a willingness-to-pay distribution with an increasing general failure rate will have unique profit- and revenue-maximizing prices. Such distributions include the normal, the uniform, the exponential, and the logistic. For more details, including a list of distributions that do and do not satisfy the IGFR property, see Lariviere 2006.

4. The *Bertrand model of competition* is based on two sellers selling an identical product. Both sellers have the same incremental cost, c . Customers observe the prices of both sellers and are indifferent among sellers: if one seller is cheaper, they will all purchase from that seller, and if the sellers offer the same price, demand will split equally between the sellers. Bertrand argued that, if either seller charged a price $p > c$, the other seller's best response would be to charge some price p' , with $p > p' > c$. Since this holds true for both sellers, their prices would inevitably converge toward cost. In fact, prices would equilibrate at $c + \delta$, where δ is the smallest increment of price allowed (e.g., \$0.01).

6

PRICE DIFFERENTIATION

In this chapter we examine one of the most fundamental concepts in pricing and revenue optimization—*price differentiation*. Price differentiation refers to the practice of a seller charging different prices to different customers, either for exactly the same good or for slightly different versions of the same good. It can be a powerful tactic for sellers to improve profitability. It also adds a new level of complexity to pricing, often driving the need to use analytical techniques to improve the calculation and updating of prices over time. Price differentiation can also be dangerous if not managed correctly—it can lead to loss of profit through arbitrage and to negative reactions to pricing that some customers may perceive as unfair.

Tactics for price differentiation include charging different prices to different customers (or groups of customers) for *exactly* the same product, charging different prices for different versions or amounts of the same product, and combinations of the two. It also includes charging different prices for combinations of products than for individual products sold separately. I use the term “price differentiation” rather than the standard economic term *price discrimination* in part to avoid the negative connotations associated with the word “discrimination.” However, I also want to stress that price differentiation includes not only charging different prices to different customers for the same product (group pricing) but also the less controversial strategies of product versioning, regional pricing, channel pricing, and nonlinear pricing.

While “price differentiation” and “dynamic pricing” are often used interchangeably, there is an important distinction between the two concepts. Price differentiation refers to strategies for charging different prices to different customers based on differences in their willingness to pay. Price differentiation usually (but not always) means that different prices are in play for the same (or very similar products) at the same time. Dynamic pricing, on the other hand, refers to changing the price of a product over time. Dynamic pricing is often employed to balance supply and demand—as in the case of the variable-pricing approaches discussed in Chapter 7. Prices may also change over time as a result of changes in the desirability of a product over time—such as the declining appeal of a fashion item as

the end of the season approaches. In this case, all customers may see the same price at the same time, but the price they see changes over time.

The distinction between price differentiation and dynamic pricing is important but not as clear-cut as it may seem. If two customers are being charged different prices for the same product at the same time, that is definitely price differentiation. When a ride-sharing service raises prices for all customers in a location during rush hour, that is dynamic pricing (see Section 7.6.4 for details). However, as we shall see, time can also be used as a segmentation dimension. Airline prices tend to go up as departure approaches in part because airlines know that late-booking customers typically have a higher willingness to pay for a ticket than early-booking customers. In this case, prices may be changing over time both as a form of price differentiation *and* to balance supply and demand (Chapter 8 discusses how airlines change prices over time as a form of price differentiation). Furthermore, a seller can perform price differentiation and dynamic pricing simultaneously—airlines raise and lower prices for the same flight prior to departure (dynamic pricing), but at each point in time they may be offering different prices to customers purchasing the ticket in Europe than in Japan (price differentiation).

There is both art and science to price differentiation. The art lies in finding a way to divide the market into different segments such that higher prices can be charged to the high-willingness-to-pay segments and lower prices to the low-willingness-to-pay segments. There is no one way to segment customers that applies to all possible markets. Instead, a variety of techniques can be applied in different ways, depending on the characteristics of a market, the competitive environment, and the character of the goods or services being sold. The science lies in setting and updating the prices to maximize overall return from all segments.

We start by using a simple example (based on the widget maker from Chapter 5) to illustrate the motivations behind price differentiation.

6.1 THE ECONOMICS OF PRICE DIFFERENTIATION

Let us return to the case of the widget maker in Example 5.3. The widget maker, looking for a way to increase his profitability, might start by taking a close look at his price-response function. He would realize that 10,000 customers are willing to pay some amount greater than zero for his widget, 5,000 are willing to pay \$6.25 or more, 3,000 are willing to pay \$8.75 or more (the contribution-maximizing price), and 1,200 are even willing to pay \$11.00 or more. The marketing department might see an opportunity to improve profitability. After all, 2,000 of the customers who are currently purchasing at \$8.75 would be willing to pay \$10.00 or more for the product. Those customers are getting a great deal at \$8.75, but from the seller's point of view, it is at least \$1.25 in additional revenue left on the table from each sale. Furthermore, 3,000 customers are willing to pay more than the production cost (\$5.00) but less than the sales price of \$8.75. Each of these is a potentially profitable sale lost because the price is too high. What if the seller could determine the maximum amount that each customer would be willing to pay and could charge that amount to everyone willing

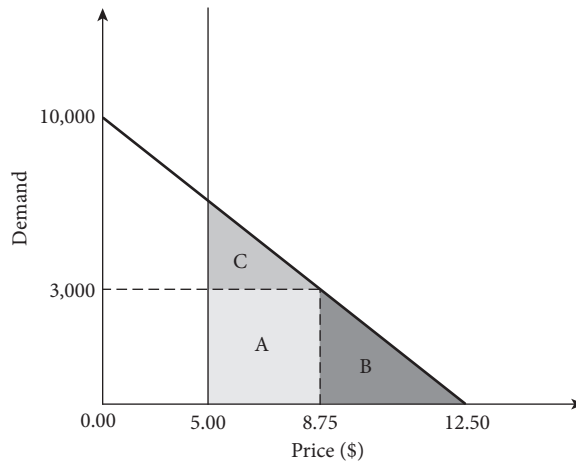


Figure 6.1 Contribution opportunity from price differentiation.

to pay more than the per-unit cost of \$5.00? This would be pricing nirvana—the ultimate in one-to-one pricing.

The total potential opportunity for profit improvement from price differentiation is shown in Figure 6.1. With a single price, the widget maker charges \$8.75 per unit and realizes a total profit of $(\$8.75 - \$5.00) \times 3,000 = \$11,250$. This is the area of region A in Figure 6.1. Better price differentiation would enable the seller to charge each customer who is willing to pay more than \$8.75 per unit her exact willingness to pay. This would provide the additional revenue in region B in the figure. There is an additional opportunity to sell widgets to customers who are willing to pay more than \$5.00, the unit production cost, but less than \$8.75. This potential profit is shown as region C in the figure. The sum of the three regions is the total contribution that the widget maker would realize *if* he were able to charge every potential customer exactly at her willingness to pay. This is an upper bound on what could possibly be realized under any price differentiation program. Charging every customer exactly her willingness to pay is known as *first-degree price discrimination* in the economics literature.¹

While it is unrealistic to assume that this total potential could ever be captured, the sheer magnitude of the potential gain means there is a powerful motivation for sellers to tailor different prices to different customers according to their willingness to pay. Assume that the widget maker finds he can divide his market into two segments. One segment consists of all customers willing to pay more than \$7.00 for widgets. The other segment consists of all the customers willing to pay \$7.00 or less. The corresponding price-response functions are

$$\begin{aligned} d_1(p_1) &= \min[4,400, (10,000 - 800p_1)^+], \\ d_2(p_2) &= (5,600 - 800p_2)^+, \end{aligned}$$

where $d_1(p_1)$ is the price-response function for customers with a high willingness to pay and $d_2(p_2)$ is the price-response function for customers with a lower willingness to pay. These

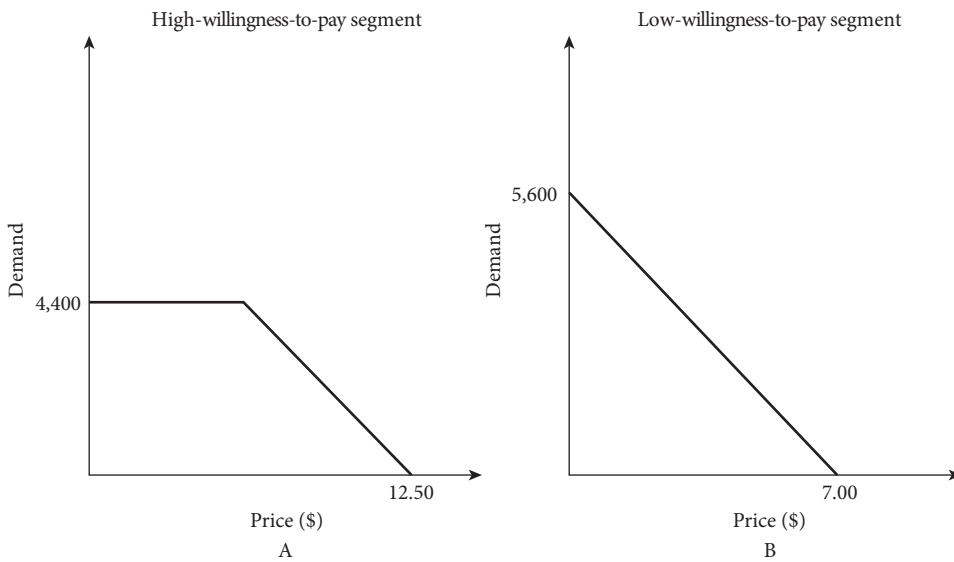


Figure 6.2 Price-response functions for two segments.

TABLE 6.1
Impact of market segmentation

	Unsegmented	Segment 1	Segment 2	Total	Change (%)
Average price	\$8.75	\$8.75	\$6.00	\$8.17	-6.6
Demand	3,000	3,000	800	3,800	26.7
Revenue	\$26,250	\$26,250	\$4,800	\$31,050	18.3
Contribution	\$11,250	\$11,250	\$800	\$12,050	7.0

two price-response functions are shown in Figure 6.2. The sum of these two functions is the original price-response function. The difference is that now the seller can offer a different price to each of the two segments. We assume for this example that the widget maker can perfectly identify customers as belonging to one group or the other and can then offer each customer the appropriate price, without any opportunity for resale or arbitrage between the two groups.

The widget maker can determine the optimal prices for each segment by solving for any of the optimality conditions in Section 5.2.1 twice—once for each segment. The results and a comparison to the unsegmented case are shown in Table 6.1. Dividing his customers between those with willingness to pay greater than \$7.00 and those with willingness to pay less than \$7.00, and charging the optimal prices to the different segments, enables the widget maker to increase his revenues by 18% and his contribution by 7%. It is interesting to note that customers also benefit. The same 3,000 customers who would buy under a single price at \$8.75 still get to buy at \$8.75. In addition, 800 additional customers get to purchase at the new low price of \$6.00. These 800 customers are priced out of the market if the seller can only charge a single price. In this case, price differentiation is a win-win

situation, since the seller is certainly better off and all of the customers are at least as well off as before. However, price differentiation is not always a boon for consumers, as we see in Section 6.5.

6.2 LIMITS TO PRICE DIFFERENTIATION

If price differentiation is such a powerful way for sellers to increase contribution, why is it that sellers do not always do it for all customers? There are three powerful real-world limits to price differentiation.

- *Imperfect segmentation.* The brain-scan technology required to determine the precise willingness to pay of each customer has not yet been developed. The best that can be done is to create market segments such that the *average* willingness to pay is different for each segment.
- *Cannibalization.* Under differential pricing, there is a powerful motivation for customers in high-price segments to find a way to pay the lower price. In the widget example, there is a strong motivation for high-willingness-to-pay customers who are being charged \$8.75 per widget to masquerade as low-willingness-to-pay customers and pay only \$6.00 per widget.
- *Arbitrage.* Price differentials create a strong incentive for *arbitrageurs* to find a way to buy the product at the low price and resell to high-willingness-to-pay customers below the market price, keeping the difference for themselves.

The presence of any one of these factors can drastically reduce or even eliminate the benefits of price differentiation.

Example 6.1

Assume that the widget seller has segmented his market and is charging the two prices shown in Table 6.1. If 300—that is 10%—of the high-willingness-to-pay customers are able to find a way to purchase widgets at the lower cost of \$6.00, it would totally eliminate the benefits of price differentiation.

The deterioration of benefits in Example 6.1 could arise from imperfect segmentation, cannibalization, arbitrage, or a combination of any of them.

6.3 TACTICS FOR PRICE DIFFERENTIATION

The previous section establishes two properties of price differentiation.

1. Successful price differentiation allows sellers to increase profitability by charging different prices to customers with different willingness to pay.
2. Cannibalization, imperfect segmentation, or arbitrage can destroy—or even reverse—the benefits of price differentiation.

The first property means that sellers have a tremendous incentive to find creative ways to position and price their products differently to different market segments. The second means that price differentiation needs to be carefully planned and managed in order to be effective. In this section we look at some of the most common and effective approaches to price differentiation used in different markets.

6.3.1 Group Pricing

Group pricing is the tactic of offering different prices to different groups of customers for exactly the same product at the same time.² The idea is, of course, to offer a lower price to customers with a low willingness to pay and a higher price to those with a high willingness to pay. In practice, pure group pricing requires determining whether a customer belongs to an identified group and using that information to determine which price to offer. Examples of pure group pricing include

- Student discounts
- Senior citizen discounts
- Ladies' Night specials
- Family specials
- Discounts for favored customers
- Favorable terms offered by manufacturers or wholesalers to large retailers such as Walmart
- Lower prices offered to government, educational, and nonprofit organizations by suppliers

Four criteria must hold for group pricing to be successful.

- *There must be an unambiguous indicator of group membership.* Examples of such indicators include a student ID card or a driver's license that lists age. Furthermore, it must be difficult or impossible for members of one group to masquerade as members of another. Otherwise, cannibalization could easily reduce or even erase the benefits of price differentiation.
- *Group membership must strongly correlate with price sensitivity.* Senior citizen discounts are predicated on the knowledge that senior citizens, *on average*, are more price sensitive than the public in general.
- *The product or service cannot be easily traded among purchasers.* This is necessary to avoid *arbitrage*, in which some customers with access to low prices resell to customers who are quoted higher prices.
- *The segmentation must be both culturally and legally accepted.* Group pricing can be extremely controversial. Certain group pricing practices—setting different prices for different races, for example—are illegal in many jurisdictions. Other group pricing practices such as gender-based pricing may or may not be illegal (depending

on the location) but may be extremely unpopular. Note that the social acceptability of many group pricing practices depends not only on the prices themselves but the ways in which the prices are communicated. Thus, a senior citizen discount may be acceptable, while a middle-age surcharge would likely not be acceptable. We examine these issues in depth in Chapter 14.

Taken together, these criteria are so stringent that pure group pricing is relatively rare in direct consumer sales. Group pricing is most common in services. There are a number of reasons for this. Many service suppliers can verify group membership; for example, a movie theater can check a driver's license before applying a senior citizen discount. Most services, from health care to haircuts, are intrinsically nontransferable, so arbitrage is not an issue. Airlines check passenger identification in part as a way to prevent arbitrage. Finally, group pricing is common in consumer and small business lending where the rate for a mortgage or loan might depend on the credit rating of the customer.

Pure group pricing is also common in business-to-business sales. Chapter 13 discusses how businesses can estimate price responsiveness and develop customized prices for different business segments. Customized pricing in business-to-business sales is often combined with some form of product versioning, particularly in industries where individual orders are very complex and/or configurable.

6.3.2 Channel Pricing

Channel pricing is the practice of selling the same product for different prices through different distribution channels. Here are some examples.

- Barnes and Noble sells books for different prices online than in its stores.
- Special web-only fares for airline tickets are available through the internet but not through travel agencies.

As with other price differentiation schemes, there can be more than one reason why a seller might charge different prices through different channels. One is cost: for many companies, selling through the internet is cheaper than selling through traditional channels. In 2002, the average distribution cost for an airline ticket booked through an offline travel agency was \$30.66 compared to \$11.75 for one booked directly on an airline's website (Barnes 2012). This cost difference alone would be motivation for an airline to charge less for tickets purchased on its website than through a travel agency.

However, it is also the case that customers arriving via different channels have different price sensitivities. For personal loans, it has been shown that customers inquiring through the internet are more price sensitive than those contacting a call center, who are in turn more price sensitive than those who apply for a loan at a retail branch. Furthermore, internet customers who access a consolidator website or use a shopping bot tend to be more price sensitive than those who go directly to a bank's website. This is not surprising given the characteristics of the channels—it is generally easier and more convenient to shop and compare prices during a single internet session than by making many phone calls. Thus, differential willingness to pay is also a motivation for channel pricing.

6.3.3 Regional Pricing

Regional pricing is a common price differentiation technique. Here are some examples.

- In Latin America, McDonald's sells hamburgers for higher prices in wealthy neighborhoods than in poorer ones (*The Economist* 2004).
- A round-trip New York–Tokyo ticket purchased in Japan will usually cost more for many airlines than the same ticket purchased in the United States.
- A cold 20-ounce bottle of Coke costs \$1.50 at the local drugstore, \$1.99 at a gas station, \$2.79 at the airport newsstand, and \$3.49 at a hotel gift shop (Simon and Fassnacht 2018, 210).
- In the United States and in Canada, banks often charge different rates for mortgages in different regions of the country.³

In each case, the price difference is based on the supplier's desire to exploit differences in price sensitivity between locations. One limiting factor in regional pricing is typically arbitrage—entrepreneurs can theoretically purchase in the lower-price location, transport to the higher-price location, and resell, keeping the difference. Note that, in each of the cases listed above, arbitrage is difficult or impossible.

In addition to arbitrage, another limiting factor in regional pricing is the potential for region to be perceived as a proxy for other customer characteristics such as income levels or ethnicity. Redlining was the name given to the policies of many U.S. banks not to lend to customers in certain urban neighborhoods—neighborhoods that were typically predominantly African American. Redlining was widely condemned and was ultimately outlawed in the United States.⁴ More recently, a 2015 ProPublica study revealed that Princeton Review charged higher prices for its test preparation services in zip codes that had higher proportions of Asian Americans than other zip codes. Princeton Review admitted differential pricing by region but denied discrimination in “the literal, legal and moral meaning of the word” (Larson, Mattu, and Angwin 2015).

6.3.4 Couponing and Self-Selection

We have seen that group pricing is often both difficult and unpopular—difficult because it requires the seller to categorize customers on the basis of price sensitivity before quoting them a price, and unpopular because it often seems unfair to consumers. It is often much more convenient to differentiate prices in ways that allow customers to self-select. In a self-selection approach, both the list price and a discounted price are available to all customers, but it takes additional time, effort, or flexibility to obtain the discounted price. The idea is that those willing to make the additional effort to get the discount are generally more price sensitive than those who are not. Here are some examples.

- Retailers commonly offer discount coupons through newspapers, direct mail, and email.
- Retailers often offer mail-in rebates for purchasers of a good.

- Movie theaters charge lower prices for a weekday matinee than for a Saturday night show. (This is also an example of variable pricing, as discussed in Section 7.6.5.)
- Brand-name retailers such as Ralph Lauren, the Gap, and Liz Claiborne operate outlet stores in out-of-the-way locations, making merchandise available for a substantial discount. For example, Ralph Lauren operates several branded stores in Manhattan that sell its clothing at full price. It also operates a so-called factory store in Elizabeth, New Jersey, about 20 miles away, in which it sells some of the same items at a deep discount. Local customers are likely to be more price sensitive than Manhattan dwellers. Manhattan dwellers who are willing to make the trek to New Jersey are likely more price sensitive than those who are willing to simply purchase Ralph Lauren from a local store.

The common thread among these examples is that the seller has chosen a mechanism that allows customers to self-select, depending on the value they place on time or flexibility. Any customer can obtain an item at a discount if she is willing to make an additional effort. Research has confirmed that users of coupons are more price sensitive than nonusers of coupons (Narasimhan 1984). Locating outlet stores away from large cities segments the market between those who are willing to spend additional time in transit—for example, driving from Manhattan to Elizabeth, New Jersey, and back—and those who are not. Peak-load, day-of-week, and time-of-day pricing segments the market between those who have the flexibility to change their plans in order to save some money and those who are not willing to do so. Since these mechanisms are based on self-selection, they are far more acceptable to most consumers than mechanisms in which the seller unilaterally selects customers to receive discounts.

6.3.5 Product Versioning

When pure group pricing is not feasible, companies use other strategies to differentiate prices. The most notable of these is designing or developing products (either virtual or real) that may have only minor differences but enable the seller to exploit differences in price sensitivity among customer segments. This can involve developing an inferior variant and/or a superior variant of an existing product. We look at examples of both strategies as well as their logical extension into the creation of a product line.

Inferior goods. Consider the following cases.

- Well-known international brand names such as ExxonMobil and Shell sell excess gasoline in bulk at low prices to so-called off-brand independent dealers who resell it under their own brands.
- A well-known premium wine producer sells some of its production under a different label at about half the price.
- Brand-name vegetable canners sell their products under their own brand but also sell to retailers who sell the product to consumers as a house brand or a generic brand.

While the specifics of each of these cases differ, the motivation on the part of the seller is the same—a desire to sell a product cheaply to customers with lower willingness to pay with-

out cannibalizing sales of the full-price product. This is achieved in each case by creating an inferior version of the standard product.

A particularly extreme example of inferior goods is the category of so-called *damaged goods*. This is a term coined to refer to the situation in which a manufacturer or supplier creates an inferior good by damaging, degrading, or disabling a standard good (Deneckere and McAfee 1996). Since this process usually starts with the standard good, it may actually cost the supplier more to create the inferior good, which it then sells at a lower price. One example is the 486SX processor developed and sold by Intel Corporation.

The 486SX processor of Intel Corporation was initially produced in a curious way. Intel began with a fully functioning 486DX processor, then *disabled the math coprocessor*, to produce a chip that is strictly inferior to the 486DX but more expensive to produce. Nevertheless, in 1991, the 486DX sold for \$588, and the 486SX for \$333, a little over half the price of the chip that is less expensive to produce. (Deneckere and McAfee 1996, 149; emphasis in original)

Another example is the Tesla. According to an article in the *Economist*, “The cheaper models in Tesla’s line-up have parts of their batteries disabled by the car’s software in order to limit their range” (2019, 3). When Hurricane Dorian was heading toward America’s East Coast in August 2019, Tesla, with the “tap of a keyboard in Palo Alto,” was able to temporarily remove the artificial restrictions and “give drivers access to the full power of their batteries” (3). The implication is that the range restriction for the less expensive models was imposed simply to create an inferior product that could be sold at a lower price.

Complex application software packages such as supply chain software or enterprise resource planning (ERP) software are often sold at different prices, depending on the number of features purchased by a customer—the more features purchased, the more expensive the license. In many cases, the software is configured for a particular customer by starting with the complete package and then disabling the features that the customer did not purchase.

The concept of damaging a good to create an inferior good to sell at a lower price may initially seem somewhat bizarre. However, it is really only a special case of the more general category of inferior goods. There can be a tremendous gain from offering an inferior good at a lower price, even if the supposedly inferior product is more expensive to produce. Starting from a standard product and then paying to have it damaged is only a special case of this more general principle.

Superior goods. Spendrups is the largest brewery in Sweden. Traditionally Spendrups brewed medium- or low-priced lagers aimed at the mass market. In the 1980s, it created Spendrups Old Gold, which it advertised as a premium beer and sold in a special, highly distinctive bottle. Although Old Gold did not generally fare better than Spendrups’s other brands in comparative taste tests, Spendrups was able to establish Old Gold as a premium brand and maintain a price 25% to 50% higher than that of its other brands. This is the obvious complement to the inferior-good strategy: creating a superior good to extract a higher price from less price-sensitive customers. When sales of Old Gold began to flag, it was relaunched in a distinctive 10-sided bottle in 2010 to maintain the premium image.

Another example of the superior-good strategy was employed by Proctor-Silex. In 1985, Proctor-Silex priced a top-of-the-line iron at \$54.95, while its next best model was priced

at \$49.95. The only difference between the two was that the top model had a small light indicating when the iron is ready to use. The difference in manufacturing cost between the two models was only \$1.00, yet Proctor-Silex was able to maintain a \$5.00 price difference because, as a Proctor-Silex marketing manager put it, “There is a segment of the market that wants to buy the best, despite the cost” (quoted in Birnbaum 1986, 175). By creating a superior product, Proctor-Silex enabled the less price-sensitive segment of the market to self-select and extracted an additional \$4.00 in contribution margin from each high-end customer. In some ways, a superior-good strategy is safer than an inferior-good strategy because it does not threaten cannibalization of existing sales. Of course, it presumes an ability to create and establish a product that the market perceives as truly superior to the existing product and presumes that there is a customer segment willing to pay a premium for the superior product.

Product lines. Establishing a product line is the natural extension of creating inferior or superior products. A product line is a series of similar products serving the same general market but sold at different prices. For our purposes, we consider *vertical* product lines, where almost all customers would agree that a higher-priced product is superior to a lower-priced one. This applies, for example, to a hotel that charges more for an ocean-view room than a parking-lot-view room—almost all customers would prefer the ocean view. It also applies to personal computers offered by Apple, where each product in the line has higher performance (faster CPU, more memory, etc.) than the product just below it in the line. This can be contrasted to *horizontal* product lines, where different customers prefer different products within the line, even at the same price. Coca-Cola’s offering of Classic Coke, Diet Coke, Cherry Coke, Diet Cherry Coke, Coke Zero, and so on is a horizontal product line because no single product is unambiguously higher quality or more desirable than another.

An example of a vertical product line is shown in Table 6.2, which gives prices for four versions of the QuickBooks financial software offered by Intuit. The versions are the Simple Start version, which costs \$25.00 per month; the Essentials version, which costs \$40.00 per month; the Plus version, which costs \$70.00 per month; and the Advanced version, which costs \$150.00 per month.⁵

In all likelihood, there is only a tiny difference (if any) in the marginal cost of producing and delivering the four different versions of QuickBooks software. Yet the Advanced package is priced six times higher than the Simple Start version. The rationale for the product line and the broad difference is market segmentation—very small businesses that need less functionality and have fewer users are presumably more price sensitive than larger

TABLE 6.2
*Online prices offered by Intuit for the
QuickBooks product family, May 2020*

Product	Price (\$ per month)
Simple Start	25.00
Essentials	40.00
Plus	70.00
Advanced	150.00

TABLE 6.3
*Rates offered by Budget on Expedia for a one-day midweek rental,
 January 2020*

Product	Representative model	Rate (\$)
Economy	Ford Fiesta	68.00
Compact	Ford Focus	70.00
Midsize	Hyundai Elantra	73.00
Premium	Chevrolet Impala	75.00
Full-size	Ford Fusion	75.00
Luxury	Chrysler 300	86.00
Minivan	Chrysler Pacifica	90.00
Elite SUV	Ford Explorer	92.00
Full-size SUV	Chevrolet Tahoe	103.00
Premium SUV	Chevrolet Suburban	123.00

businesses with more users. The establishment of a product line allows Intuit to segment its market via self-selection on the part of its customers.

Another example of product-line pricing is illustrated in Table 6.3, which shows the rates on display through the online travel agency Expedia for a Budget Rental Car one-day rental from the Seattle airport on a midweek day in January 2020. Budget lists 10 different products, with prices ranging from \$68.00 per day for an economy rental to \$123 per day for a premium SUV. By providing a menu of alternatives at different prices, Budget allows customers to self-select: The economy-minded will choose the economy or compact cars, while those who are willing to pay for greater comfort may opt for the full-size or luxury. Furthermore, those traveling with large families may want or need to rent a larger car or an SUV. It is important to stress that the range of rates offered by Budget for these different products to rental car companies is not driven by cost differences. Taking into account the revenue from resale, life-cycle costs do not vary much among different car types. More important, there is little or no difference in the daily incremental cost to Budget for renting out an economy car or a premium SUV. The spread in daily rates is driven almost entirely by Budget's desire to segment its market.

The pricing menus offered by Budget and Intuit are openly communicated and available to all comers.⁶ Consumers get to choose among the alternatives, and the concept of paying more to get more is widely accepted. This makes product-line pricing more acceptable to most customers than group pricing.

For service companies like Budget, creation of a product line has an important side benefit—it creates opportunities for *upgrading*. A rental car company or a cruise line has the ability to oversell lower-quality car types and upgrade customers into higher-quality types. Not only does this provide the company with greater flexibility to manage its inventory, but being upgraded is usually viewed favorably by customers. An important advantage of pricing differentiation by establishing a product line is that consumers generally perceive it as fair.

6.3.6 Time-Based Differentiation

Time-based differentiation is a common form of product versioning. Here are some examples.

- Target.com offers Standard Shipping of 3–5 business days free for items of \$35 or more. For selected items and locations, 2-Day Shipping and Express Shipping of 1-Day are offered at an additional cost.
- Passenger airlines offer discount rates to customers who book a week or more prior to departure. Hotels offer higher rates to customers who “walk up” to obtain a room for that evening.
- Software- and hardware-support contracts charge more for two-hour response than for two-day response.
- Fashion goods cost more during the beginning of the season and are marked down toward the end of the season.

In each of these cases, companies have created differentiated products that allow customers to self-select. In the case of Target, customers who are willing to wait for delivery can pay less. For passenger airlines, customers who have the flexibility to book earlier can pay less. For the passenger airlines, it costs no more to accommodate a passenger who books later than one who books earlier. It may be that the higher price charged by Target for early delivery exactly matches the incremental cost. But it is likely that Target is also using time of delivery as a segmentation variable, relying on the willingness of some of its customers to pay a premium to have the product in their hands sooner, just as the airline is using time of booking to segment customers.

Time-based differentiation plays an important role at passenger airlines, hotels, and rental car companies, in which time of booking and other factors, such as willingness to accept a Saturday night stay-over, are used as indicators of whether a potential customer is traveling for leisure purposes or business purposes. Those traveling for leisure purposes are presumed to be more price sensitive than those traveling for business. This segmentation is the foundation of revenue management in those industries, and Chapter 8 discusses it in detail. The willingness of some customers to wait to purchase fashion goods at a discount is the basis of markdown management, which is treated in Chapter 12.

6.3.7 Product Versioning or Group Pricing?

We have treated product versioning and group pricing as separate strategies for price differentiation. In reality, no clear line separates the two approaches, and many price differentiation strategies contain elements of both. For example, consider the classic airline example (which we explore in detail in Chapter 8) of a round-trip ticket from San Francisco to Chicago costing \$250 if purchased a week in advance and including a Saturday stay-over versus \$750 if purchased at the last minute without restrictions. Is this group pricing or product versioning? Disgruntled customers might argue that it is simply group pricing, since different customers are paying different amounts for exactly the same service—namely, a round-trip coach seat from San Francisco to Chicago. The airline would reply that the two types of tickets are distinct products and that the added cost of the full-fare ticket is fully justified by the flexibility of being able to purchase late and return without staying over a Saturday night.

The reality is, of course, that airline pricing—like many successful examples of price differentiation—includes elements of both group pricing and product versioning. The

airlines consciously created restricted discount fares as an inferior product. They did so, however, as a way to enable them to offer different fares to different customer groups: lower fares to leisure travelers, who are more price sensitive but more flexible; and higher fares to business travelers, who are less price sensitive but also less flexible. Viewed one way, we could say that the airlines created an inferior product as the most efficient and least controversial way to institute group pricing.

The airline example illustrates an important point. Pure group pricing is difficult to pull off in most consumer markets. There are few cases in which consumers can unambiguously be identified as belonging to a particular group and prices can be targeted accordingly with no fear of arbitrage or customer backlash. Airlines have no reliable objective marker to tell them whether a particular customer is flying for business or for pleasure. In the absence of such a marker, they rely on (among other factors) the imperfect criterion of whether a customer can book early. This works well enough, but it is imperfect. For example, plenty of highly price-sensitive leisure customers would be happy to book late. And some business customers are able to book early and take advantage of lower fares. Furthermore, the airlines have lots of empty seats they would like to fill with these customers, even at a very low price. But allowing customers to book at the last minute at a low price would damage the ability of the airlines to segment customers between leisure and business.

6.4 CALCULATING DIFFERENTIATED PRICES

If market segments are completely independent (i.e., there is no cannibalization) and the seller faces no capacity constraints, then calculating differentiated prices is quite simple—the seller simply finds the contribution-optimizing price for each segment. This is the right approach whether the underlying differentiation is based on channel, geography, or pure group pricing.

Example 6.2

An electronics distributor sells portable MP3 players through its exclusive retail outlets and on its website. The unit cost for each MP3 player is \$200. The additional cost per sale through the internet is \$35, including shipping, but \$70 through the retail stores. Through price testing, the distributor has discovered that price elasticity is 2.5 for internet customers versus 2.2 for retail customers. Using Equation 5.8, the distributor calculates the contribution-maximizing price as $(2.5/1.5) \times (\$200 + \$35) = \$392$ for internet sales and $(2.2/1.2) \times (\$200 + \$70) = \$495$ for retail sales.

6.4.1 Optimal Pricing with Arbitrage

Unfortunately for sellers, perfect price differentiation is usually impossible. Cannibalization is likely whenever customers cannot be perfectly segmented according to willingness to pay. Arbitrage is likely whenever a third party can purchase the product at a low price and resell it at a high price. Regional pricing is subject to arbitrage whenever a product can be purchased in a low-price region and transported cheaply to be resold at a higher price

elsewhere. For this reason, global companies often set price bands for various markets to avoid resales from low-price countries that would cannibalize sales in higher-price countries.

Example 6.3

A computer chip manufacturer finds that the contribution-maximizing prices for his chips are \$2.54 in the United States and \$2.43 in Brazil. However, if it costs \$0.08 per unit to ship chips from Brazil to the United States, the potential for arbitrage between the two countries means he will not be able to charge those prices. He needs to set prices for the two countries that do not vary by more than \$0.08.

Regional pricing with arbitrage can be formulated as a constrained optimization problem. Assume that a manufacturer is selling a common product to n different locations. These could be cities or regions, but for purposes of discussion we will consider them to be countries. The delivered cost (including transportation) in country i is c_i . The cost to an arbitrageur to transport the product from country i to country j is a_{ij} .⁷ To avoid arbitrage, the supplier needs to set prices such that $p_j \leq p_i + a_{ij}$ for each i and j . Otherwise, the supplier faces the possibility that an arbitrageur will purchase the product in country i , transport it to country j , and sell it for a price $\hat{p}_j$. If $\hat{p}_j > p_i + a_{ij}$, then the arbitrageur can undercut the supplier in country j while still turning a profit. It is clear that a contribution-maximizing supplier would usually like to avoid this situation.

Let p_i be the price in country i and $d_i(p_i)$ be the price-response function faced by the supplier in country i . Then the problem of optimizing international prices under the possibility of arbitrage can be formulated as

$$\max_p \sum_{i=1}^n (p_i - c_i) d_i(p_i),$$

subject to

$$\begin{aligned} p_j &\leq p_i + a_{ij} && \text{for } i = 1, 2, \dots, n; j = 1, 2, \dots, n; i \neq j, \\ p_i &\geq 0 && \text{for } i = 1, 2, \dots, n. \end{aligned}$$

This problem will not, in general, be a linear program. However, if the price-response function for each country is continuous and downward sloping, it will have a single optimum and will be easy to solve using standard optimization approaches. In many cases, global companies do not solve the full optimization problem but rather determine prices for major markets (e.g., North America, Western Europe, Japan) and then use price bands to determine prices for smaller markets that are close enough to major market prices to eliminate the potential for arbitrage through transshipment.

6.4.2 Optimal Pricing with Cannibalization

Cannibalization presents a somewhat different problem than arbitrage. To analyze differentiated pricing with cannibalization, let us return to the case in which we divided widget cus-

tomers into those with willingness to pay more than \$7.00 and those with willingness to pay less than \$7.00. The corresponding price-response functions, shown in Figure 6.2, assume perfect differentiation—the company can perfectly distinguish between those customers willing to pay more than \$7.00 and those willing to pay less than \$7.00 and can charge the optimal price to each group. What if some of the higher-willingness-to-pay customers find a way to buy at the lower price? Let α be the *cannibalization fraction*—the fraction of higher-willingness-to-pay customers who find a way to buy at the lower price. $\alpha = 0$ corresponds to the case of perfect differentiation, in which none of the high-willingness-to-pay customers buy at the lower price, while $\alpha = 1$ corresponds to no differentiation, in which *all* of the customers buy at the lower price. Values of α between 0 and 1 represent different segmentation efficiencies. Given a value of α , the price-response functions for each of the two segments are given by

$$\begin{aligned}d_1(p_1) &= (1 - \alpha) \min[4,400; (10,000 - 800p_1)^+] \\d_2(p_2) &= (5,600 - 800p_2)^+ + 4,400\alpha\end{aligned}$$

The second term in the equation for $d_2(p_2)$ reflects the fact that 4,400 total customers have a willingness to pay greater than \$7.00 and that a fraction α of them will find a way to buy at the lower price, p_2 .

To find the values of p_1 and p_2 that maximize total contribution, we need to solve the optimization problem

$$\begin{aligned}\max_{p_1, p_2} & (p_1 - c)(1 - \alpha) \min[4,400; (10,000 - 800p_1)^+] \\ & + (p_2 - c)[(5,600 - 800p_2)^+ + 4,400\alpha],\end{aligned}$$

subject to $p_2 \leq \$7.00$. The prices that maximize total contribution in this case are

$$\begin{aligned}p_1^* &= \$8.75, \\ p_2^* &= \min(\$6.00 + \$2.75\alpha, \$7.00).\end{aligned}$$

When $\alpha = 0$ (no cannibalization), $p_2^* = \$8.75$. As α increases, cannibalization increases, and p_2^* increases as well. This corresponds to intuition—the more our low-price product is cannibalizing a high-price product, the higher we need to price the low-price product to maximize total contribution. When $\alpha = 0.36$ (i.e., 36% of the high-willingness-to-pay customers find a way to pay the lower price), the optimal low price has risen to \$7.00. At this point, none of the low-willingness-to-pay customers are buying any more (since their willingness to pay is less than \$7.00) and the only low-price customers are cannibalized high-willingness-to-pay customers.

Figure 6.3 shows optimal total contribution as a function of α . As α increases, total contribution decreases as more and more high-willingness-to-pay customers are cannibalized by the lower price. At a value of α of about 13%, total contribution is \$11,250, the same total contribution that could be achieved with no segmentation. In this simple example, if more than 13% of our high-willingness-to-pay customers will be cannibalized, we are better off offering only a single price than trying to differentiate and offer different prices to the different segments.

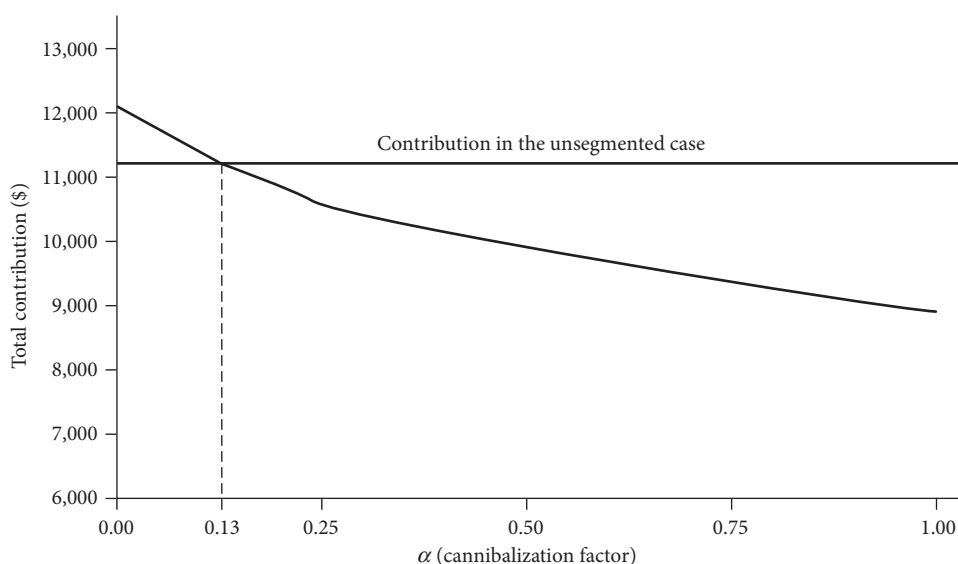


Figure 6.3 Total contribution as a function of the cannibalization fraction α .

While the threshold of 13% is specific to this example, the lesson is general: Even fairly low rates of cannibalization can destroy the benefits of price differentiation. This means that when establishing customer segments and determining the optimal prices, it is critical that we understand the potential for cannibalization and price accordingly within each segment. Otherwise, we can end up in a situation where we are worse off than if we had simply established a single price to the entire market.

6.5 PRICE DIFFERENTIATION AND CONSUMER WELFARE

We have seen that, correctly executed, price differentiation can be good for sellers. But is it good for customers? This is not merely an academic question. Some forms of price differentiation can be unpopular among customers—particularly group pricing, which, as we see in Chapter 14, is often viewed by customers as unfair. This can lead to loss of customers as well as regulatory scrutiny. If a seller can show that a particular differential pricing approach actually makes consumers better off as a whole (or at least not worse off) in comparison with a single price, the seller will at least have an argument to use with regulators.

To look at who gains and who loses from price differentiation, we draw on some concepts from social welfare theory, as illustrated in Figure 6.4. The left-hand diagram in this figure illustrates the case of a seller with unit cost c who is facing a linear price-response function and can set only a single price. We assume that he sets the optimal price p^* with corresponding sales of $d(p^*)$. The seller is not capacity constrained, and so sales equal demand. The shaded area in the figure corresponds to the seller's profit, which is given by $(p^* - c)d(p^*)$. In consumer welfare theory, this amount is called *producers surplus*.

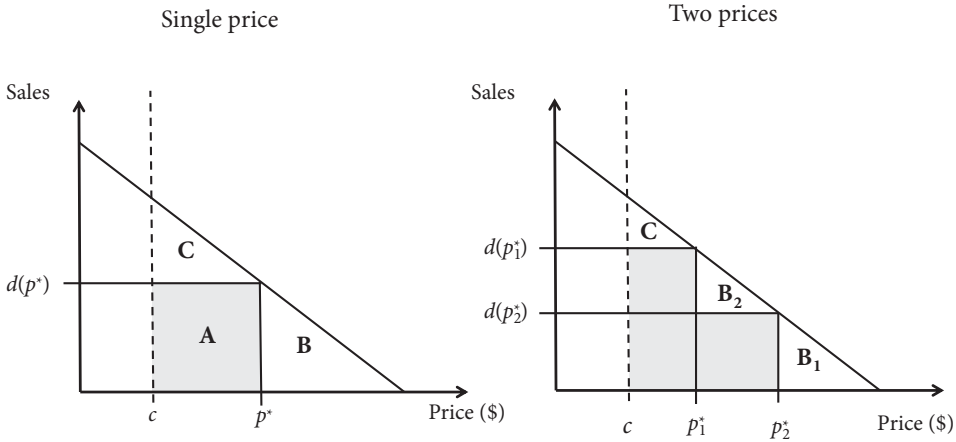


Figure 6.4 Elements of social surplus for a profit-maximizing seller setting a single price (*left*) and two prices with perfect differentiation (*right*).

On the customer side, the analog of producers surplus is *consumers surplus*. For an individual customer who purchases the product, her resulting surplus is equal to the difference between her willingness to pay and the price. The idea is that a consumer who was willing to pay up to \$54 for an item but purchased it at \$45 has enjoyed a surplus of \$9—money she would have been willing to pay for this item but now has available to purchase something else (or to save). The consumers surplus associated with a particular pricing scheme is the sum of the individual surpluses of consumers who purchase the item. The surplus associated with a customer who does not purchase is zero. In the left-hand graph in Figure 6.4, the consumers surplus associated with a single price is equal to the area of the triangle labeled B and the producers surplus (profit) is equal to the area of square A .

The *social surplus* associated with the pricing scheme is the consumers surplus plus the producers surplus—that is, the area of triangle B plus square A . Social welfare theory in its simplest form posits that total economic welfare is maximized when social surplus is maximized.⁸

There is one other area of interest, represented by the triangle labeled C in the left-hand graph in Figure 6.4. A customer whose willingness to pay is between c and p^* will not purchase at the price of p^* and hence realizes a consumers surplus of zero. However, this customer would be willing to purchase at some price greater than c , and a profit-maximizing seller would be willing to sell to them, but because the seller is restricted to a single price, these transactions do not take place. The area of triangle C represents surplus lost because the seller can only charge the single price p^* —it is called *deadweight loss*.

It is instructive to consider what happens to social surplus if the seller decided to sell at a price different from p^* . If he raises his price above p^* , then producers surplus will be reduced, consumers surplus will be reduced, and the deadweight loss will increase, which means that overall social welfare is decreased. On the other hand, if he reduces his price below p^* , producers surplus will be reduced, but consumers surplus will increase and deadweight loss will decrease. Since deadweight loss decreases, social welfare is increased. In the

extreme case, if he sets $p = c$, consumers surplus and social welfare would be maximized, but producers surplus (and deadweight loss) would both be zero. Other than regulation, the most likely motivation for a producer to price at cost would be competition—and, in fact, perfect competition maximizes consumers surplus and social welfare by eliminating producers surplus.

What happens if the seller is able to offer two prices? The situation is shown in the right-hand graph in Figure 6.4. Here, we assume that the seller is able to perfectly segment the market according to willingness to pay and there is no cannibalization or arbitrage. Deadweight loss in this case is the area of triangle C and consumers surplus is the sum of the areas of triangles B_1 and B_2 . Producers surplus is the area of the shaded region (not labeled in the figure). We know that producers surplus (profit) increases with two prices, and it is clear from the figure that deadweight loss has decreased. It can be shown in this case that consumers surplus has decreased and that overall social surplus has increased. Thus, in this case, price differentiation has increased social surplus but at the cost of converting some consumers surplus to producers surplus. The ideal for the profit-maximizing producer would be to charge each customer a price exactly equal to her willingness to pay (*first-degree price discrimination*), in which the producer would capture all of the available surplus— $A + B + C$ in the left-hand figure. Perfect competition and first-degree price discrimination by a monopolist both maximize social surplus, although in the first case the surplus is entirely captured by customers and in the second it is entirely captured by the seller as profit.

Does this mean that price differentiation always reduces consumers surplus? The answer is no: it turns out that price differentiation *can* make consumers better off. Consider the case where the widget maker is able to divide his market into two segments, with the two price-response curves shown in Figure 6.2. As shown in Table 6.1, the seller's optimal policy in this case is to charge \$8.75 to the high-willingness-to-pay segment and \$6.00 to the low-willingness-to-pay segment, with a concurrent increase in contribution of \$800 relative to the best he can do when charging a single price. The total consumers surplus under this new policy is the old surplus, denoted by region C_1 in Figure 6.5, plus the new surplus, C_2 , enjoyed by the customers paying \$7.00. The seller's profit is the sum of regions A_1 and A_2 . Comparison with Figure 6.1 shows that this two-price policy has resulted in both more profit for the seller *and* higher total consumers surplus than the single-price policy. In other words, in this case price differentiation is a win-win for customers and sellers.

The case in Figure 6.5 is notable because some customers are better off and no customer is worse off than with a single price. The customers who paid \$8.75 with the single price still pay \$8.75, so they are unaffected. Customers who pay \$7.00 under the two-price policy are better off since they would not have purchased with a single price. And the seller is better off since he makes more money.

So depending on how it is applied, price differentiation can either increase or decrease consumers surplus. There is one general rule: *If a price differentiation scheme increases profits for a seller but does not result in additional production, it must reduce total consumers surplus.* This rule is most relevant when capacity is constrained—if a theater is already selling out at a single price and it goes to a two-price system that increases revenue, then it has done so by making customers, *on the average*, worse off. Again, this does not mean that no customers are better off under the new scheme: it simply means that the *total* consumers surplus

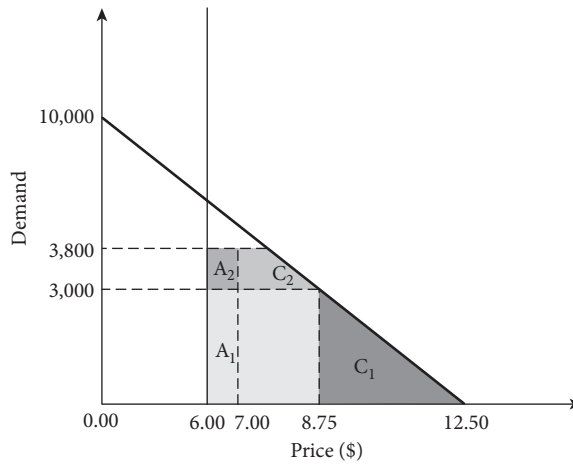


Figure 6.5 Profit and consumer surplus from the two-price case.

is less under the new scheme than under the old. If, on the other hand, a theater that is not full finds that it can increase total revenue by a price differentiation tactic, then the effect of the differentiation on total consumers surplus will depend on the situation. A study of the effects of pricing and consumer behavior for an actual Broadway show concluded that price differentiation increased both attendance and profit by about 5% while having a negligible effect on consumers surplus (Leslie 2004).

Note that social surplus is maximized in two extreme cases. If the producer could use a Vulcan mind meld to determine the exact willingness to pay of every customer *and* he was able to charge every customer exactly her willingness to pay, he would sell his product to every customer with willingness to pay greater than unit cost at that customer's willingness to pay. In this case, the seller would grab every penny of potential surplus. There would be no deadweight loss, but there would be no consumers surplus either. At the other extreme, the seller would sell the product at unit cost. In this case, every customer with willingness to pay greater than cost would purchase. There would be no profit and no deadweight loss, but each consumer would enjoy a high level of consumers surplus. Of course, in the static world represented by a single price-response curve, there is no motivation for the seller to engage in this largesse. In the real world, sellers are pushed toward selling at (or near) cost by the presence or the threat of competition or, possibly, regulation.

6.6 NONLINEAR PRICING

All of the approaches to pricing discussed so far can be characterized as *linear*. This means that the price of a bundle including multiple items is equal to a fixed price per item times the number of items purchased. Thus, if a gallon of milk costs \$3.50 and a loaf of bread costs \$2.50, under linear pricing the cost to purchase two gallons of milk and four loaves of bread would be $2 \times \$3.50 + 4 \times \$2.50 = \$17.00$. Under linear pricing, the cost to the consumer of buying twice as much of an item is simply double the cost of buying a single item, and the cost of buying two different bundles of items is the sum of the prices of each bundle.

While linear pricing is simple and intuitive, a seller can often generate additional revenue or profit through *nonlinear pricing*—that is, charging a price for a bundle of goods that is different (usually lower) than the sum of the prices of the individual items in the bundle. Familiar examples include *volume discounts*, “10% off if you purchase at least five rolls of toilet paper”; *BoGos* (“buy one, get one”) such as “buy two books and get the third one for 50%”; and *bundling discounts*, “buy a grill for full price and get 1/2 off a bag of charcoal.”

Note that nonlinear pricing schemes are a form of price differentiation: under a nonlinear pricing scheme, customers who purchase more of an item or purchase an item in combination with other items will typically pay less *per unit* than other customers who purchase a single unit of the item in question. Yet, unlike group pricing, nonlinear pricing is generally noncontroversial since the same pricing is typically available to all customers. In this sense, it is similar to product differentiation because each customer is typically offered the option to buy a single item at full price or multiple items with a discount.

In this section we consider two of the most common approaches to nonlinear pricing: volume discounts and bundling.

6.6.1 Volume Discounts

Volume discounting—buy more to save more—is a time-honored form of nonlinear pricing. Some examples follow.

- A six-pack of beer purchased from the supermarket costs less than six times the cost of a single bottle, and a case (12 bottles) usually costs less than two six-packs.
- Larger boxes of laundry detergent cost less per ounce than smaller boxes.
- Verizon offers shared data plans for households, including “small” (2 GB) at \$35 per month, “medium” (4 GB) at \$50 per month, and “large” (8 GB) at \$70 per month (Verizon, n.d.).

Volume discounting is as prevalent, if not more prevalent, in wholesale and business-to-business selling. Table 6.4 shows three typical price schedules including volume discounts

TABLE 6.4
Price schedules for Duracell batteries, holiday signs, and Pines wheatgrass products

Batteries		Holiday signs		Pines wheatgrass products	
Pack size	Unit price (\$)	Order size	Unit price (\$)	Order size (\$)	Discount (%)
4	1.49	<50	1.25	<50	0
6	1.25	51–100	1.20	50–74	10
12	1.07	101–250	1.15	75–99	15
24	0.68	251–500	1.10	100–124.99	20
		501–1,000	1.05	125–199.99	30
		1,001–2,000	1.00	200–299.99	35
		2,001–3,000	0.95	>300	40
		3,001–5,000	0.90		
		>5,000	Call for quote		

SOURCE: Battery prices are for Duracell Optimum 1.5V AAA battery, quoted on <https://www.walmart.com> (August 2019); holiday sign pricing from Holiday Signs website, at <https://holidaysignsdirect.com> (August 2019); wheatgrass products pricing from Pines website, at <https://wheatgrass.com> (August 2019).

for three different products offered online: Duracell batteries on Walmart.com, Holiday Signs, and Pines wheatgrass products. Each seller offers significantly lower unit prices to customers who are buying more units. And the price per unit decreases as the number of units in the order increases.

There are at least three reasons why companies offer volume discounts:

- *Transaction or order costs.* If some costs associated with fulfilling an order are independent of size, then the order cost per unit will decrease as the size of the order increases. In this case, it makes sense for the seller to charge a lower price per unit for large orders.

Example 6.4

A software seller has a fixed cost of \$1,000 per installation and a variable cost of \$40 per user. Customer 1 has 10 users and customer 2 has 100 users. The total cost for him to serve customer 1 is \$1,400, or \$140/user, compared with a total cost of \$5,000, or only \$50/user, for customer 2. Assume that both customers have a valuation of \$120/user. Then, if the seller can only charge a single price, he will charge \$120 per copy. At this rate, both customers will buy, but he will lose \$200 on the first customer. If he offers a volume discount such that the price is greater than \$140/user for 10 users or fewer and \$120/user for more than 10 users, the first customer will not purchase and he will make an additional \$200 in profit.

- *Decreasing marginal utility.* In many cases, the marginal utility to a customer decreases as the number of units purchased increases. A hot and thirsty customer walking into a convenience store places more value on the first can of cold soft drink than she does on the second. Decreasing marginal utility is also often seen in business-to-business sales because the incremental value of additional units of office equipment or supplies tends to decline past some point.
- *Price sensitivity increases with volume.* Customers purchasing larger amounts are often more price sensitive than customers purchasing smaller amounts. This is particularly true in business-to-business settings: corporate purchasers who buy large volumes are likely to be following a formal procurement process in which they carefully compare alternatives and will negotiate intensely to gain a lower price. A plumbing contractor who spends millions of dollars on galvanized pipe annually is much more likely to spend time and effort finding a good deal than a homeowner who is buying a single length of pipe at the hardware store. Volume discounting is one way that a seller can differentiate among large-volume, highly price-sensitive customers and smaller-volume, less price-sensitive customers. In this case, volume discounting is a form of product-based customer segmentation.

The discounts shown in Table 6.4 are non-incremental because the discount applies to the entire order. This can lead to negative incremental prices. For example, according to the second column in Table 6.4, a customer ordering 50 signs would pay \$1.25 per unit for

a total cost of \$62.50, whereas if he ordered 51 signs, he would pay \$1.20 per unit for a total cost of \$61.20. The incremental price of the 51st sign is $-\$1.30$. Negative incremental prices are arguably bad things in that they can encourage overconsumption and lost potential revenue. In *incremental discounting*, additional discounts are applied only to additional units—not to the entire order. A common example of incremental pricing in consumer markets is “Buy one, get the second at half price.”

Volume discounts are often applied based on total volume of business over some period, rather than on size of a single order. This is often accomplished in consumer markets by frequent-buyer schemes such as the airlines’ frequent-flyer programs. My local coffee shop keeps track of my purchases and gives me every 10th cup free. The stated purpose of such programs is to reward loyal customers, but they also serve the purpose of lowering the real price seen by frequent purchasers, who are likely to be more price sensitive. This practice is extremely common in business-to-business settings, in which discounts are often applied based on sales volume in a quarter or other period. Electronics distributors resell electronics goods to retailers, government, and educational institutions. A manufacturer (such as Hewlett-Packard) selling through the distributor will offer a progressive end-of-quarter rebate based on how much of its product the distributor has sold during the quarter. For example, Acme Electronics might offer a 0.5% rebate if the distributor sells between \$1 million and \$1.499 million worth of Acme products during the quarter, a 0.75% rebate if the distributor sells between \$1.5 million and \$1.999 million, and a 1% rebate if the distributor sells over \$2 million. In freight transportation, a carrier will often offer a higher discount if a shipper commits to a higher volume of business over some future period.

Combining different terms and discount structures can lead to an extremely wide variety of volume discount schemes. All of these schemes are designed to exploit the decreasing marginal value of purchasers and/or the higher price sensitivity of large purchasers. In the case of decreasing marginal values, a seller could, in theory, capture the maximum contribution by pricing each additional unit at the willingness to pay of the customer for that unit. Of course, this is impossible because the seller is selling to a mixture of customers with different willingness-to-pay values and the exact willingness to pay of each customer is unknown. However, given overall demand curves for purchases of different sizes, a seller can derive a volume discounting scheme that will increase contribution relative to charging a single price.

How should we think about setting volume discounts? Here we consider a simple case of a single customer (or a population of identical customers) with decreasing marginal utility for an item and show how volume discounting can be used to extract additional revenue and profitability. (Warning: While the example is conceptually simple, the algebra can be a bit tedious.)

Consider a single customer (or a population of identical customers) with utility given by $u(x) = A\sqrt{x}$, where x is the number of units purchased and $A > 0$ is a parameter. This utility function is shown in Figure 6.6. We assume for convenience that the customer can purchase fractional units. Note that the customer’s marginal utility is given by $u'(x) = A/(2\sqrt{x})$, so her marginal utility is decreasing for all $x > 0$, which implies that her price sensitivity increases with the number of units she purchases.

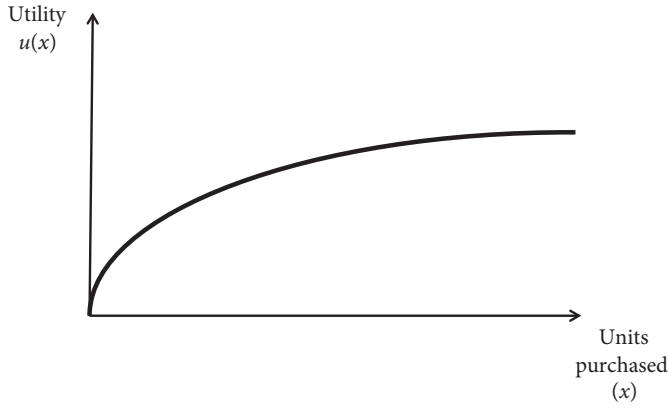


Figure 6.6 Utility of a product to a customer as a function of units purchased.

Now, assume that the product is offered at a single price $p > 0$. Then, for any purchase amount d , the total utility to the customer after the purchase is $u(d) - pd$. We can determine the purchase volume that will maximize her utility by solving the first-order condition $u'(d) = p$. In this case, $p = A / (2\sqrt{x})$, which after rearranging gives

$$d(p) = \frac{A^2}{4p^2}. \quad (6.1)$$

Equation 6.1 is the customer's individual price-response function; that is, for any price p that the seller offers, it shows how many units she will purchase. Note that $d(p)$ is continuous and decreasing in price, consistent with the general requirement for price-response functions.

Assume that the seller has a constant unit cost $c > 0$. With a single price, a profit-maximizing seller will seek to set the price that maximizes $\pi(p) = (p - c)d(p)$. The profit-maximizing price can be found from the standard first-order condition $\pi'(p) = 0$. In this case, $\pi(p) = (p - c)A^2/4p^2$, which means that the first-order condition is

$$\frac{A^2}{4}(2cp^{-3} - p^{-2}) = 0. \quad (6.2)$$

Equation 6.2 has a single finite root at $p^* = 2c$, which means that the profit-maximizing seller will set a price equal to two times his cost. His corresponding sales and profit can be computed by substituting p^* into Equation 6.1: the resulting sales are $d(p^*) = A^2/16c^2$, and the corresponding profit is $\pi(p^*) = A^2/16c$.

We now consider the case in which the seller offers two prices with a single volume discount tier. In this case, the seller sets a triplet x_1, p_1, p_2 , with $p_1 > p_2 > c$. If the customer purchases $x \leq x_1$ units, she will be charged $p_1 x$. If she purchases $x > x_1$ units, she will be charged $p_1 x_1 + p_2(x - x_1)$. The question facing the seller is what values of x_1, p_1 , and p_2 to set to maximize profit.

At first it would seem that the seller has three variables to compute. However, a little reflection should convince you that, for optimality, it must be the case that $x_1^* = x(p_1^*)$ —that

is, the sales threshold for the discount must be equal to the demand at the undiscounted price. If $x_1^* < x(p_1^*)$, then the seller could increase revenue by increasing x_1^* and selling the additional units at p_1 rather than p_2 . If $x_1^* > x(p_1^*)$, the seller is not selling anything at the discount price and he can decrease x_1^* without decreasing sales or revenue. Thus, we can formulate the seller's problem in terms of just p_1 and p_2 . The corresponding optimization problem faced by the seller is

$$\max_{p_1, p_2} \pi(p_1, p_2) = (p_1 - c) \left(\frac{A^2}{4p_1^2} \right) + (p_2 - c) \left(\frac{A^2}{4p_2^2} - \frac{A^2}{4p_1^2} \right),$$

subject to $p_1 > p_2 > c$. The first term of $\pi(p_1, p_2)$ is the profit realized from selling the units at price p_1 and the second term is the profit realized from selling units at price p_2 . We can rearrange terms in the objective function to write

$$\pi(p_1, p_2) = \left(\frac{A^2}{4} \right) (p_1^{-1} + p_2^{-1} - cp_2^{-2} - p_2p_1^{-2}). \quad (6.3)$$

Ignoring the constraints for the moment and differentiating by p_1 gives the first-order necessary condition:

$$-p_1^{-2} + 2p_2p_1^{-3} = 0.$$

So we must have $p_1 = 2p_2$.

We can now differentiate $\pi(p_1, p_2)$ with respect to p_2 , which gives us

$$-p_2^{-2} + 2cp_2^{-3} - p_1^{-2} = 0,$$

or

$$-p_2p_1^2 + 2cp_1^2 - p_2^3 = 0.$$

Substituting $p_1 = 2p_2$ and rearranging terms gives

$$8cp_2^2 = 5p_2^3.$$

Again, the root $p_2 = 0$ is clearly a minimum, so the maximum must be at $p_2 = 8c/5$, which implies that we must have $p_1 = 16c/5$. Substituting back into the objective function in Equation 6.3 and calculating the amounts purchased gives us the results shown in Table 6.5.

The total profit across the two periods is

$$\pi(p_1, p_2) = \frac{100A^2}{1024c} = \frac{25A^2}{256c} = \left(\frac{25}{16} \right) \left(\frac{A^2}{16c} \right) = \left(\frac{25}{16} \right) \pi(p^*).$$

That is, the optimal profit from the volume discount is 56.25% greater than with the single price. This represents a substantial increase, which suggests why volume discounts—often with many discount points—are popular, especially in business-to-business settings.

Results from both the single-price and the volume discount case are shown in Table 6.5. Two things are notable in the volume discount case. First, the first price is higher and the second price is lower than the optimal single price. This is typical. However, it means that,

TABLE 6.5
Results with single price and with volume discount price

	Single-price case	With volume discount			Change (%)
		First price	Second price	Total	
Price	$2c$	$16c/5$	$8c/5$	$2c$	0
Sales	$A^2/16c^2$	$25A^2/1024c^2$	$75A^2/1024c^2$	$25A^2/256c^2$	56.25
Profit	$A^2/16c$	$55A^2/1024c$	$45A^2/1024c$	$25A^2/256c$	56.25

to be effective, volume discounting requires that a large fraction of the customers want to purchase in volume—if only a small number of customers are volume purchasers, the loss from increasing the initial price to enable the volume discount may outweigh the benefits. Second, total sales are significantly higher with two prices than with a single price.

Finally, it is important to remember that, like any market segmentation scheme, volume discounts provide opportunities for arbitrage. Referring to the price schedules in Table 6.4, a 4-pack of Duracell batteries costs $4 \times \$1.49 = \5.96 . An enterprising entrepreneur could purchase a 24-pack of Duracell batteries for $24 \times \$0.68 = \16.32 , repack them as six 4-packs, and sell them for \$5.00 apiece, realizing a profit of $\$30.00 - \$16.32 = \$13.68$, minus any repackaging costs. Many internet resellers make a living by finding and exploiting such opportunities for arbitrage. To avoid such arbitrage (not to mention out-and-out piracy), software companies often require users to register to receive future support and maintenance. In other cases, bulk and volume purchasers are required to sign a contract that prohibits them from reselling for profit. However, in many cases, the potential for arbitrage limits the potential for volume discounting.

6.6.2 Bundling

A bundle is a collection of products, services, or attributes sold as a package in which the price of a bundle is less than the sum of the individual parts. This means that bundling is a form of nonlinear pricing. Bundling may be motivated by the fact that the cost of providing a bundle is less than the cost of delivering the individual components—for example, the shipping cost of 20 items delivered together may be less than 20 times the shipping costs of the individual items. However, as we see in this section, it is also a way to extract additional sales by pricing in a way that increases sales in different market segments.

Bundling can be used in a simple way for *complementary* products—say, hot dogs and hot-dog buns. Consumers may be divided into three groups—those who have hot dogs at home and need buns, those who have buns at home and need hot dogs, and those who have neither and need both to enjoy the full hot-dog experience. If the seller prices buns at \$6 per pack and hot dogs at \$10 per pack, it may be worthwhile for him to offer a bundle of hot dogs and buns at \$15 per pack to sell more hot dogs and buns to the third group without cannibalizing sales to the first two. The reasoning and analysis in this case is similar to those of volume discounts.

A more interesting use of bundling is with *noncomplementary* products. Assume that Microsoft is offering Excel and Word in a business marketplace that consists of four

TABLE 6.6
Willingness to pay for different segments in the bundling example

	Banker (\$)	Trader (\$)	Lawyer (\$)	Consultant (\$)
Excel	400	450	50	200
Word	200	50	450	400
Bundle	600	500	500	600

segments: bankers, traders, lawyers, and consultants. The willingness to pay of each of these segments for each of the two products is shown in Table 6.6. For simplicity, we assume that the size of each market segment is equal to 20,000. What is the optimal price if Microsoft offers each product separately? With some calculation, it is easy to see that the optimal prices are \$400 for Excel and \$400 for Word. At these prices, the bankers and the traders will buy Excel with revenue of $\$400 \times 40,000 = \16MM (million), and the lawyers and consultants will buy Word with revenue of $\$400 \times 40,000 = \16MM for a total of \$32MM.

Now, what if Microsoft bundled Excel and Word together? The willingness to pay of each segment for the bundle is shown in the last row of Table 6.6. One option is for Microsoft to only offer the bundle. In this case, the revenue-maximizing price would be \$500, and all of the segments would purchase for a total revenue of $\$500 \times 80,000 = \40MM . This represents a 25% increase over the unbundled pricing. However, Microsoft could also offer the products individually in addition to the bundle. In this case, the optimal policy would be to price the bundle at \$600 and both Word and Excel at \$450. Bankers and consultants would purchase the bundle, traders would purchase Excel, and lawyers would purchase Word. The total revenue would be $\$600 \times 40,000 + \$450 \times 40,000 = \$42\text{MM}$, a 31.25% increase over the best unbundled pricing.

When does bundling of noncomplementary products work? The most important condition is that customer segments differ in their valuation of the individual items—if each segment valued Excel the same, there would be no opportunity to gain revenue by bundling. In addition, there must be segments who have flipped preferences. That is, there must be one segment whose preference among products (as measured by willingness to pay) is different than at least one other segment. In the example, bankers and traders have a higher willingness to pay for Excel than Word, while the opposite is true for lawyers and consultants. The fact that there are segments whose preferences are *anti-correlated* in this fashion is a precondition for bundling of noncomplementary products to be effective.

Example 6.5

A company offering two products to three customer segments is considering whether to offer a bundled product. Each segment consists of 10,000 customers. Using consumer research, the company establishes that the willingness to pay for each product in each segment is as shown in Table 6.7. Looking at the results, the company can see that the preference order of each segment is the same and can conclude immediately that there will be no gain from offering a bundle. This can be verified by noting that the optimal individual prices for the

TABLE 6.7
Willingness to pay for different segments in Example 6.5

Product	Segment A (\$)	Segment B (\$)	Segment C (\$)
1	50	85	30
2	10	80	20
Bundle	60	165	50

two products are $p_1 = \$50$ and $p_2 = \$80$, generating revenue of \$180,000. If the company offered a bundle, the optimal bundle price would be \$165, generating revenue of \$165,000. This supports the conclusion that there is not an opportunity to increase revenue by bundling in this case.

It should be noted that the problem of optimal bundling—that is, determining the bundle or set of bundles to offer that maximizes revenue (or profit)—becomes extremely difficult when the number of products becomes large. In the case of two products, there are three options—no bundle, bundle only, or bundle plus individual sales. In the case of three products, there are six options: no bundle; three two-product bundles (A,B), (B,C), and (A,C); and a three-product bundle (A,B,C), which could be offered with or without individual sales options. In fact, the number of possible bundles of n items is given by $2^n + 1 - n$, which grows quite quickly—for 20 items, there are 1,048,557 possible bundles that would need to be evaluated.⁹

Optimal bundling of multiple products is a topic of continuing research. However, for practical purposes, the most important insight is that bundling of noncomplementary products is most effective when there are segments whose preferences are anti-correlated—that is, some segments prefer one product while others prefer a different product. In this case, a bundling strategy can often be used to increase sales and profitability. It should also be noted that, like volume discounts, bundles can be subject to arbitrage—if a seller is pricing a bundle below the sum of its constituent prices, an enterprising arbitrageur could purchase a bundle and sell the individual components below their individual prices. For this reason, bundling is often used for items for which arbitrage is difficult, like software or services.

6.7 SUMMARY

- Price differentiation is the tactic of charging different customer segments different prices for the same (or nearly the same) product or service. It can be a powerful way for a seller to increase contribution.
- The effectiveness of price differentiation can be limited by imperfect segmentation, cannibalization, or arbitrage. A poorly conceived or executed price differentiation scheme can result in lower contribution than a single-price scheme.
- There are a wide variety of price discrimination schemes, ranging from group pricing through channel and regional pricing to volume discounting and product versioning. The best approach or combination of approaches depends on the market.

- Product-versioning tactics are better accepted by customers and less susceptible to cannibalization and arbitrage than group pricing. Both inferior and superior products can be used to segment customers.
- The methods for calculating differentiated prices when the supplier does not face capacity or supply restrictions are extensions of the methods introduced in Chapter 5 for a supplier pricing a single product. The possibilities of cannibalization and arbitrage need to be specifically incorporated in order for the prices to be correct.
- The effects of price differentiation on consumer welfare (as measured by consumers surplus) can be either positive or negative, depending on the market and on the differentiation scheme. If a price differentiation scheme does not increase supply, it cannot increase total consumers surplus.
- Nonlinear pricing refers to a wide variety of pricing schemes in which the price for a number of goods is not equal to the sum of the prices of the individual goods. Common examples include volume discounts, “buy one and get one free,” and bundling discounts. Applied correctly, nonlinear pricing can lead to increased sales and profitability, although arbitrage and cannibalization can limit the benefits.

We return to the theme of price differentiation again and again. In particular, price differentiation is the basis of revenue management (Chapter 8), markdown management (Chapter 12), and customized pricing (Chapter 13).

6.8 FURTHER READING

More discussion on strategies for price differentiation can be found in Bodea and Ferguson 2012, Simon and Fassnacht 2018, and Nagle and Müller 2018.

This chapter provides only a high-level treatment of consumers surplus. Fuller treatment can be found in most microeconomics texts, such as Varian 1992 and Nicholson 2002.

There is an extensive literature in both marketing and economics on nonlinear pricing; the classic reference is Robert Wilson’s book *Nonlinear Pricing* (1993). A useful short summary of the topic can be found in Oren 2012, and an overview of tactics for optimal bundling is in Venkatesh and Mahajan 2009.

6.9 EXERCISES

1. *Arbitrage.* A supplier is selling hammers in two cities, Pleasantville and Happy Valley. It costs him \$5.00 per hammer delivered in each city. Let p_1 be the price of hammers in Pleasantville and p_2 be the price of hammers in Happy Valley. The price-response curves in each city are

$$\text{Pleasantville: } d_1(p_1) = 10,000 - 800p_1$$

$$\text{Happy Valley: } d_2(p_2) = 8,000 - 500p_2$$
 - a. Assuming the supplier can charge any prices he likes, what prices should he charge for hammers in Pleasantville and Happy Valley to maximize total contribution? What are the corresponding demands and total contributions?

- b. An enterprising arbitrageur discovers a way to transport hammers from Pleasantville to Happy Valley for \$0.50 each. He begins buying hammers in Pleasantville and shipping them to Happy Valley to sell. Assuming the supplier does not change his prices from those given in part *a*, what will be the optimal price for the arbitrageur to sell hammers in Happy Valley? How many will he sell? What will his total contribution be? (Assume that Happy Valley customers will buy hammers from the cheapest vendor.) What will happen to the total sales and contribution for the supplier? (Remember that he is now selling to the arbitrageur too.)
 - c. The supplier decides to eliminate the arbitrage opportunity by ensuring that his selling price in Happy Valley is no more than \$0.50 above the selling price in Pleasantville (and vice versa). What is his new selling price in each city? What are his corresponding sales and total contribution?
 - d. From among the Pleasantville customers, the Happy Valley customers, and the seller, who wins and who loses from the threat of arbitrage?
2. Assume that it costs Budget \$20 per day for every car it rents out, regardless of model. What is the implied price elasticity of customers for each of the first five car types listed in Table 6.3?
3. For the single-customer demand function specified by Equation 6.1, what is the corresponding hazard rate and price elasticity? What do they imply about the price calculated in Equation 6.2?

NOTES

1. Definitions of first-, second-, and third-degree price discrimination were established by the economist Arthur Pigou in 1920, and the terms are still widely used although they are not exhaustive. *First-degree price-discrimination* refers to the case in which the seller charges each customer exactly her willingness to pay. *Second-degree price discrimination* refers to volume discounting. *Third-degree price discrimination* corresponds to what we call *group pricing*—charging different prices to different customers based on their membership in a group (e.g., senior citizen discounts).

2. I use the term *group pricing* in the sense of Shapiro and Varian 1992. This should be distinguished from use of the term to offer lower rates to groups of customers in industries such as the passenger airlines, hotels, and cruise lines, which is a form of volume discounting.

3. See Phillips 2018 for patterns of regional loan pricing in North America.

4. The classic study of the history of redlining and statistical studies of its effect on minority communities is *The Color of Credit*, by Stephen Ross and John Yinger (2002).

5. See the Intuit QuickBooks home page, at <http://quickbooks.intuit.com>.

6. This does not, of course, mean that Budget and Intuit are not also offering even deeper hidden discounts to favored customers.

7. In many cases, this cost will be symmetrical—that is, $a_{ij} = a_{ji}$. However, this is not always the case. For example, one country may charge a higher import duty on the product.

8. This simple calculation of social surplus is based on the assumption that society values a dollar of producers surplus (e.g., operating profit) the same as a dollar of consumers surplus. There are at least two rationales for this assumption. The first is that corporate profits ultimately flow back to consumers in the form of bonuses or stock dividends. The second rationale is that if people wanted a different distributional outcome—say, more consumers surplus and less producers surplus—the tax rates could be adjusted to tax business more and consumers less. Under this view, producers plus consumers surplus establishes the size of the pie, and the first goal of economic policy should be to make the pie as big as possible. Needless to say, this view has been heavily criticized for, among other things, neglecting the extent to which businesses can influence the political process in their favor.

9. The situation is even more complex if the seller is able to offer multiple bundles; for example, a seller with products A, B, and C could offer (A,B), (A,C), and (A,B,C) as well as only B in isolation. In this case, the number of possible *combinations* of bundles is truly astronomical—on the order of 2^{2^n} , which becomes unmanageably large for even relatively small values of n . In practice, sellers do not combine bundles because of the added complexity of communication to customers.

Chapter 5 shows how a seller can calculate the optimal single-period price when he has the freedom to produce (or order) as much of a good as he wants and he knows with certainty the price-response function he faces. In this case, the seller can maximize expected profit by charging the price at which his marginal cost equals his marginal revenue or, equivalently, the price at which his contribution margin ratio equals one over the elasticity. Things are a bit more complicated in the real world: sellers have the freedom to adjust prices over many periods, they are unsure how customers will respond to different prices, and they face constraints on their ability to satisfy demand. In this chapter, we focus on the influence of constrained supply on optimal prices.

When the supply of a good is constrained, the conditions for optimal prices presented in Chapter 5 do not necessarily give the price that maximizes net contribution or profit. Return for a moment to the example of the widget maker. With only a single period to consider and with perfect knowledge of his market, the optimal actions for him were to set his price at \$8.75 and produce 3,000 units. In this case, he sells all 3,000 units and makes a total contribution of \$11,250. This solution assumes that supply is totally flexible *and* demands are deterministic *and* he is only going to operate during a single period. What should he do if he can only manufacture 2,800 widgets? What price should he set? How does his optimal price change with the supply constraint? These are the sort of questions we address in this chapter.

We start by looking at the nature of supply constraints and the situations in which they occur. We then examine how a seller can determine the optimal price to charge when faced by a supply constraint, and I introduce and describe the important concept of *opportunity cost*. We extend the calculation of optimal prices to the case when a supplier has a segmented market and faces supply constraints. This leads to the tactic of variable pricing, which is used when a supplier has multiple units of constrained capacity and can change prices to balance supply and demand.

7.1 THE NATURE OF SUPPLY CONSTRAINTS

Most sellers face supply constraints at least some of the time. For example, most retailers replenish their stock of inventory at fixed intervals—often weekly. In between replenishment times, they are limited to selling their current inventory—in other words, their supply is fixed. In most cases, retailers stock enough inventory that a stockout is unlikely. A drugstore will typically have enough toothpaste, shaving cream, and aspirin in stock that it will not sell out, except in cases of an extraordinary run on a particular item—bottled water prior to a hurricane, for example. In this case, the seller may technically have a limited amount of inventory on hand, but he does not need to consider a supply constraint when setting price. There are, however, many cases in which a seller needs to consider supply constraints when setting a price.

- *Service providers* almost always face capacity constraints. A hotel is constrained by the number of rooms, a gas pipeline by the capacity of its pipes, and a barbershop by the number of seats (and barbers) it has available.
- *Manufacturers* face physical constraints on the amount they can produce during a particular period. For example, in 2018, the Mack Trucks company supplied all of Australia and New Zealand from a manufacturing facility west of Brisbane, Australia, with a capacity of 250 trucks per month. In any particular month, Mack could only deliver 250 new trucks plus whatever inventory it had at the beginning of the month.
- *Retailers and wholesalers* often sell goods that are not replenishable. These include fashion goods that are typically ordered once or electronic goods that are near the end of their life cycle.
- *Intrinsically scarce or unique items*, such as beachfront property, flawless blue diamonds, van Gogh paintings, and Stradivarius violins, command premium prices because of their scarcity. In these cases, marginal cost is not an important determinant of price because it is meaningless (as in the case of a van Gogh painting) or extremely low relative to the price that the item can command in the market (as in the case of diamonds).

The treatment of a supply constraint depends on its nature. A *hard* constraint is one that cannot be violated at any price. Typically, hotels and gas pipelines face hard constraints. On the other hand, air freight carriers may have the option to lease space on other carriers to carry cargo in excess of their capacity. In other industries, a seller can delay delivery. A freight carrier that can lease additional space or can delay delivery faces a *soft* constraint. Mack Trucks' Australian production limit is also a soft constraint because it can (and does) ask customers to wait for delivery when demand exceeds supply. Whether a supply constraint is hard or soft will, in part, depend on timing. An airline that learns two months in advance that it will be facing very high demand for a particular flight may be able to assign a larger aircraft. However, it will probably not have any option to increase capacity if it only learns about the high demand a week before departure.

7.2 OPTIMAL PRICING WITH A SUPPLY CONSTRAINT

In Example 5.3, the optimal unconstrained price of widgets is \$8.75, with corresponding demand of 3,000 units and a total contribution of \$11,250. This is optimal if the seller is able to manufacture all 3,000 units—or order them from a third party. But what if he cannot supply the 3,000 units demanded? Perhaps he has a supply chain bottleneck that limits his production. Perhaps he is a reseller with a fixed monthly quota he cannot exceed. In either case, the widget maker faces a rationing problem—he needs to determine which customers will be served and which will not. He has three basic options for rationing supply:

1. Do nothing—keep the price at \$8.75 and let customers buy on a first-come, first-served basis until supply is exhausted.
2. Allocate the limited supply to favored customers or allocate to customers using some other mechanism such as a lottery.
3. Raise the price until demand falls to meet supply. He could do this directly by increasing the price or via an auction.

Alternatives two and three are not mutually exclusive—the seller could use a combination of allocations and higher prices to manage a shortage. If he has segmented his market effectively, he could raise his average selling price by allocating most or all of the limited supply to higher-paying customers—this is the basic idea behind revenue management, which we discuss in Chapter 8.

Note that the issue faced by the seller with constrained supply is in part an *allocation problem*—that is, determining which consumers get to possess the desired product or service that he is selling. Options such as first come, first served and lotteries allocate the product to the early or the lucky, respectively. Other options are to choose recipients based on characteristics such as loyalty or volume of business. When the 2011 floods in Thailand drastically cut the global supply of DRAM chips, many sellers chose not to raise prices to the market-clearing level and allocated the limited supply to their most valued customers. Many sold-out concerts use first come, first served as an allocation rule. Both of these approaches appear to be broadly considered fair by customers—although both are also subject to the possibility of arbitrage in which customers who purchase at the lower price may resell at the higher price—a practice known as *scalping* in the case of concert or sporting tickets. The approach discussed in this chapter uses price as the allocation method—goods are sold to those who have the highest willingness to pay.¹ Economists would argue that this is the most efficient approach to allocating scarce goods and, if arbitrage would otherwise occur, it results in exactly the same allocation at exactly the same prices, except that all of the economic rent accrues to the seller rather than scalpers or other middlemen.

Even in the absence of arbitrage, raising the price of a scarce good generally increases expected social welfare. Consider the case of a concert with a uniform general admission price of \$25. There is only one seat remaining for sale, but there are two customers who would like tickets. One has willingness to pay of \$30, and one has willingness to pay of \$75. If the concert promoter holds the ticket price at \$25 and uses first come, first served or a

lottery to allocate the ticket, then there is an equal chance that either customer will get the ticket for \$25, resulting in expected consumers surplus of $0.5 \times (\$30 - \$25) + 0.5 \times (\$70 - \$25) = \$25$, which, added to the producers surplus of \$25, gives a total social surplus of \$50. If the concert promoter raises the price of the last ticket to \$40, then consumers surplus is $\$70 - \$40 = \$30$, which, added to the producers surplus of \$40, gives a total surplus of \$70. Note that, in this case, both expected consumers surplus and producers surplus (profit) increased when the price of the ticket was raised. In many cases, however, using price as the sole mechanism to allocate a scarce good is frowned on by consumers and needs to be used with care. Chapter 14 discusses some of the issues with consumer acceptance of the practice of pricing to allocate scarce supply.

Let us return to the case of the widget maker and assume that he has decided to use price as the mechanism to allocate scarce widgets. Then he can find the price that maximizes his contribution margin by solving the constrained-optimization problem

$$\max d(p)(p - c),$$

subject to

$$d(p) \leq b,$$

where b is the maximum supply available. Let p^* be the optimal unconstrained price. If $d(p^*) \leq b$, the supplier does not need to do any further calculations: p^* is also the optimal constrained price. If, on the other hand, he finds that $d(p^*) > b$, then he needs to charge a *higher* price than p^* in order to maximize contribution. This situation is illustrated in Figure 7.1. Here, the capacity constraint is a horizontal line—the supplier can only produce quantities below that line. At the optimal unconstrained price, p^* , demand $d(p^*)$ exceeds capacity.

When the optimal unconstrained price generates demand that exceeds capacity, the supplier needs to calculate the *runout price*—that is, the price at which demand would exactly equal the supply constraint. In Figure 7.1, the runout price is the price $\bar{p}$ at which the price-response function intersects the capacity constraint. In other words, $\bar{p}$ is the price at which $d(\bar{p}) = b$.

Example 7.1

Assume the widget seller faces the price-response function $d(p) = 10,000 - 800p$ but can only supply a maximum of 2,000 widgets during the upcoming week. Demand at the optimal unconstrained price of \$8.75 is 3,000 widgets. Therefore, the runout price can be determined by finding $\bar{p}$ such that $10,000 - 800\bar{p} = 2,000$ or $\bar{p} = \$10.00$.

The general principle behind calculating the optimal price with a supply constraint is the following.

The profit-maximizing price under a supply constraint is equal to the maximum of the runout price and the unconstrained profit-maximizing price. As a consequence, the profit-maximizing price under a supply constraint is always greater than or equal to the unconstrained profit-maximizing price.

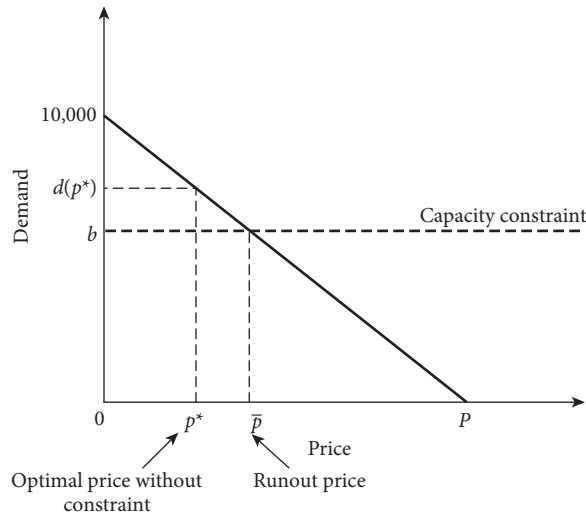


Figure 7.1 Pricing with a capacity constraint.

This principle is the same for the revenue-maximizing price, which is simply the profit-maximizing price with associated cost of 0.

The effects of different levels of capacity constraint on price and total revenue are shown in Table 7.1. For a binding capacity constraint, $b < 3,000$, the runout price can be found by inverting the price-response function:

$$\bar{p} = d^{-1}(b) = \frac{10,000 - b}{800} \text{ for } b < 3,000.$$

Table 7.1 shows the results of reducing capacity incrementally from 3,000 units per month to 1,000 units per month. The optimal price increases steadily as the capacity decreases. As capacity becomes increasingly constrained, there is more and more pressure on the seller to increase the price.

7.3 OPPORTUNITY COST

Everything else being equal, imposing a supply constraint can only reduce contribution. An auto manufacturer experiencing a strike that takes 25% of capacity out of production for a month will likely see lower profits for that month than if all of the capacity were available.

TABLE 7.1
Impact of constrained capacity on optimal price, contribution, and opportunity cost

Available capacity	Optimal price	Contribution at \$8.75	Optimal contribution	Percent change	Marginal opportunity cost
3,000	\$8.75	\$11,250	\$11,250	—	\$0.00
2,500	\$9.38	\$9,375	\$10,950	16.8	\$1.25
2,000	\$10.00	\$7,500	\$10,000	33.3	\$2.50
1,000	\$11.25	\$3,750	\$6,250	66.7	\$5.00

A 200-room hotel that takes 50 rooms out of service for two months to be refurbished is likely to give up some potential revenue. As shown in Table 7.1, the deeper or more binding a supply constraint cuts, the greater the hit the seller is likely to take on his contribution. Taking 75 rooms out of service for two months will lead to less overall contribution than taking 50 rooms out of service for the same period of time.

The reduction in contribution resulting from a supply constraint is called the *opportunity cost* associated with that constraint. It is the difference between the maximum contribution the supplier could realize without the constraint and the maximum contribution with the constraint. Since the constrained contribution can never be greater than the unconstrained contribution, total opportunity cost will always be greater than or equal to zero.

Example 7.2

In Example 7.1, the optimal price for widgets with a supply constraint of 2,000 widgets per week is \$10.00 per widget, with a total contribution of $(\$10.00 - \$5.00) \times 2,000 = \$10,000$. The optimal unconstrained price of widgets is \$8.75, with demand of 3,000 and corresponding total contribution of $(\$8.75 - \$5.00) \times 3,000 = \$11,250$. The total opportunity cost of the supply constraint is $\$11,250 - \$10,000 = \$1,250$.

Total opportunity cost is the maximum amount the seller would be willing to pay to eliminate his supply constraint entirely. In many cases, the more interesting question is how much the supplier would be willing to pay for one additional unit of supply or capacity. This is the *marginal opportunity cost*.

Example 7.3

The marginal opportunity cost for the widget seller facing a constraint of 2,000 widgets per week can be found by solving the constrained-optimization problem with a supply constraint of 2,001 and subtracting the optimal contribution with a supply constraint of 2,000. When the supply constraint is 2,001, the optimal contribution is \$10,002.50. Thus, the marginal opportunity cost of the supply constraint is equal to $\$10,002.50 - \$10,000.00 = \$2.50$.

Optimal contributions, prices, and marginal opportunity costs for the widget example are shown in Table 7.1. Note that as the amount of available capacity decreases (i.e., the supply constraint becomes more binding), the optimal price and the marginal opportunity cost both increase. The marginal opportunity cost is zero when the capacity constraint is not binding—which is the case for any constraint above 3,000 units. This illustrates the following general principle.

Total and marginal opportunity costs are nonzero only when there is a supply or capacity constraint that is binding at the optimal unconstrained price. Otherwise they are zero.

The opportunity cost can be an important input to other corporate decisions. From Table 7.1 we can see that the marginal opportunity cost faced by the supplier is \$2.50 per

widget when supply is limited to 2,000 units. This means that the widget maker would be willing to pay up to \$2.50 in rent for an additional unit of capacity that allowed him to make one more widget at a cost of \$5.00. Alternatively, he would be willing to pay up to \$2.50 + \$5.00 to buy another widget from a wholesaler. Fundamentally, the opportunity cost associated with constrained capacity is an economic measure of how much the company would be willing to pay for additional capacity. By calculating and utilizing this information in its supply decisions, companies can drive higher returns than they could from optimizing their pricing and their supply decisions in isolation.

7.4 MARKET SEGMENTATION AND SUPPLY CONSTRAINTS

Market segmentation can be a powerful tool for increasing profitability when supply is constrained. To see how, let us look at an example.²

Example 7.4

The football game between Stanford and the University of California at Berkeley (a.k.a. the Big Game) is going to be held at Stanford Stadium, which has 60,000 seats. Customers can be segmented into students (those carrying a student ID card) and the general public. We assume that the price-response functions for each of these segments is

$$\text{General public: } d_g(p_g) = (120,000 - 3,000p_g)^+ \quad (7.1)$$

$$\text{Students: } d_s(p_s) = (20,000 - 1,250p_s)^+, \quad (7.2)$$

where p_g is the price charged to the general public and p_s is the student price. What if Stanford can only charge a single price to all? In this case, the aggregate price-response function is $d(p) = (120,000 - 3,000p)^+ + (20,000 - 1,250p)^+$ and the optimal price will be \$20.00. At this price, Stanford would sell exactly 60,000 tickets, grossing \$1,200,000. Note that all ticket sales would be to the general public. The students are priced out of the market, since the highest willingness to pay of any student is $20,000/1,250 = \$16.00$.

Now, what if Stanford could charge different ticket prices to students versus the general public? Then finding the optimal price requires solving the constrained-optimization problem

$$\text{maximize } p_g (120,000 - 3,000p_g)^+ + p_s (20,000 - 1,250p_s)^+, \quad (7.3)$$

subject to

$$(120,000 - 3,000p_g)^+ + (20,000 - 1,250p_s)^+ \leq 60,000.$$

We could, of course, solve this problem directly. However, we gain more insight by using the principle established in Chapter 6: Prices should be set for the two different segments so that the marginal revenues from both segments are equal. When supply is unconstrained,

marginal revenues should all be set to the marginal cost. When supply is constrained, marginal revenues should still be equated, but they need to be set so the supply constraint is satisfied.

Example 7.5

We want to find the revenue-maximizing prices for students and the general public. The marginal opportunity cost is $2p_g - 40$ for the general public and $2p_s - 16$ for students. Equating the two marginal revenues and simplifying gives $p_g = p_s + 12$. In other words, the ticket price for the general public will be \$12.00 higher than the ticket price for students. The other condition that must be satisfied is that the total demand from both students and the general public be equal to the capacity of the stadium; that is, $(120,000 - 3,000p_g)^+ + (20,000 - 1,250p_s)^+ = 60,000$, which, when simplified, gives $3p_g + 1.25p_s = 80$. Solving both conditions simultaneously gives $p_s = \$10.35$ and $p_g = \$22.35$. At these prices, Stanford will sell 52,941 tickets to the general public and 7,059 seats to students, generating revenue of \$1,256,471, a 4.7% increase over the best that could be achieved with a single ticket price.

Note that the profit-maximizing price for the general public is more than double the student price. The stadium is sold out and there are thousands of people who would be willing to pay much more than the student price. Yet the profit-maximizing decision for Stanford is to sell more seats to students at a deep discount. In this case, the effect of market segmentation is that students win while the general public loses. Market segmentation reduces the price of tickets for the more price-sensitive students while increasing the price for the less price-sensitive general public.

The optimal prices assumed that the marginal costs—and ancillary revenues—of serving both students and the general public were the same. Differences in either marginal costs or ancillary revenues between the two market segments would lead to different optimal prices (see Exercise 3). The results also relied on a perfect fence between the student and the general public segments. The optimal prices and revenues would change if student sales cannibalized sales to the general public (see Exercise 1).

7.5 VARIABLE PRICING

Consider the following pricing tactics.

- Disney World charges a different price for its general admission by day and length of ticket. The average cost of a ticket per day for each starting day in January 2020 and length-of-ticket from one to seven days is shown in Table 7.2. Average daily prices range from \$139 for a one-day ticket on January 1 to \$62 for a seven-day ticket starting January 19.
- The San Francisco Opera charges a lower price for weeknight performances than for weekend performances. The most expensive box seats cost \$175 for a Wednesday performance and \$195 for a Saturday performance, while the least expensive balcony

TABLE 7.2
Average daily ticket price for a visit to Disney World in January 2020 by starting date and length of visit

Date	Day	Length of visit (days)						
		1	2	3	4	5	6	7
Jan. 1	Wed.	\$139	\$136	\$128	\$117	\$95	\$80	\$70
Jan. 2	Thurs.	139	132	125	114	93	79	69
Jan. 3	Fri.	139	129	121	111	92	78	68
Jan. 4	Sat.	139	124	117	109	90	77	67
Jan. 5	Sun.	125	118	113	108	89	76	66
Jan. 6	Mon.	125	116	113	108	88	75	65
Jan. 7	Tue.	117	116	113	107	87	74	64
Jan. 8	Wed.	117	120	115	106	87	73	64
Jan. 9	Thurs.	117	120	115	105	86	74	65
Jan. 10	Fri.	125	120	114	104	86	74	66
Jan. 11	Sat.	125	117	111	104	87	75	65
Jan. 12	Sun.	125	114	108	104	87	74	65
Jan. 13	Mon.	117	110	108	105	87	73	64
Jan. 14	Tue.	112	111	110	105	86	73	64
Jan. 15	Wed.	112	136	111	105	86	73	64
Jan. 16	Thurs.	109	119	114	105	86	73	64
Jan. 17	Fri.	121	122	115	105	86	74	63
Jan. 18	Sat.	128	120	113	104	86	74	63
Jan. 19	Sun.	128	116	110	103	86	73	62
Jan. 20	Mon.	121	112	107	103	84	72	63
Jan. 21	Tue.	112	110	107	101	83	71	63
Jan. 22	Wed.	112	112	109	101	83	71	64
Jan. 23	Thurs.	112	114	109	101	83	72	64
Jan. 24	Fri.	112	114	109	101	84	72	63
Jan. 25	Sat.	121	114	109	103	85	73	63
Jan. 26	Sun.	121	112	107	103	85	72	63
Jan. 27	Mon.	112	110	107	103	84	72	63
Jan. 28	Tue.	112	112	109	103	84	72	63
Jan. 29	Wed.	112	114	111	103	84	72	63
Jan. 30	Thurs.	112	116	111	103	84	72	63
Jan. 31	Fri.	121	116	111	103	85	72	64

side tickets cost \$25 for a Wednesday performance and \$28 for a Saturday performance.

- Major League Baseball teams in the United States set different prices for different games, depending on factors such as weekday versus weekend, time of year, opponent, and special events. The pricing schedule for the Colorado Rockies in 2004 is shown in Table 7.3.
- The Gulf Power Company is an electric utility serving more than 428,000 residential customers in northwestern Florida. The majority of these residential customers pay a flat rate of \$0.114 per kilowatt hour for electricity. However, Gulf Power offers residential customers the alternative of purchasing their electricity under its Residential Select Variable Pricing (RSVP) program. Under the RSVP program, each weekday is divided into an off-peak, a shoulder, and an on-peak period. Electricity costs \$0.083 per kilowatt hour (kWh) during the off-peak period, \$0.1046 during the shoulder period, and \$0.189 during the on-peak period. In the summer the weekday on-peak period is from 1:00 p.m. to 6:00 p.m., while in the winter the weekday on-peak period is from 6:00 a.m. to 10:00 a.m. The summer off-peak period is from

TABLE 7.3
Home ticket prices for the Colorado Rockies baseball team, 2004 season

Tier	Description	Average price (\$)
Value	Weekday games against low-drawing opponents in April, May, and September	11.00
Premium	Weekday games against division rivals in April, May, and September	17.00
Classic	Weekend games in May, June, July, and August	19.00
Marquee	Opening day, games against the New York Yankees, fireworks games	21.00

SOURCE: Rascher et al. 2007.

11:00 p.m. to 6:00 a.m. and the winter off-peak period is from 11:00 p.m. to 5:00 a.m. All other hours are shoulder period. In addition to these three periods, with a half hour's notice to RSVP customers Gulf Power can declare a critical period during which electricity costs \$0.777 per kWh. Under the RSVP program, Gulf Power can declare up to 88 hours of critical period per year.³

What do these four pricing tactics have in common? They are all cases in which a seller with constrained capacity adjusts prices in response to anticipated demand in order to maximize return from fixed capacity. We call this tactic *variable pricing*. Industries where variable pricing is employed share four characteristics:

- Demand is variable but follows a predictable pattern.
- The capacity of a seller is fixed in the short run (or is expensive to change).
- Inventory is perishable or expensive to store—otherwise buyers would learn to predict the variation in prices and stockpile when the price is low.
- The seller has the ability to set prices in response to anticipated supply and demand imbalances.

Under these conditions, sellers can use variable prices to shape demand to meet fixed capacity (or supply) by exploiting differences in customer preference. More people want to visit Disney World during the first week in January than during the third week in February. Weekday opera performances are less popular than weekend performances. The Rockies know that a game against the Yankees will generate more demand than a game against the Cincinnati Reds. The highest electricity prices correspond to the periods of highest use and the lowest prices to the periods of lowest use. Variable pricing does not qualify as pure group pricing; rather, it is a form of product differentiation because it allows customers to choose among different products based on price.

Note that variable pricing can either be *static* or *dynamic*. Under static variable pricing, prices are set once in advance and do not change as new information is obtained. The Colorado Rockies' ticket pricing scheme shown in Table 7.3 and the Gulf Power Company RSVP rate plan are both examples of static variable pricing. Under dynamic variable pricing, prices can be changed as new information is received. For example, many sporting teams now change prices as the season advances and tickets become either more or less in demand as a result of team performance.

TABLE 7.4
*Intercepts and slopes for the price-response curves
 in the theme park example*

Day of week	Intercept (D_i)	Slope (m_i)
Sunday	3,100	-62.0
Monday	1,500	-50.0
Tuesday	1,400	-40.0
Wednesday	1,510	-42.0
Thursday	2,000	-52.6
Friday	2,500	-55.6
Saturday	3,300	-60.0

7.5.1 The Basics of Variable Pricing

Consider a hypothetical theme park that can serve up to 1,000 customers per day. The theme park charges a single admission price, and all rides are free after admission. During the summer, demand follows a stable and predictable pattern, with higher demand on weekends than during weekdays. We assume that demands for different days of the week are independent; that the price-response functions are linear, with intercepts and slopes, as shown in Table 7.4; and that the theme park has a marginal cost of zero per customer.

What price should the theme park charge without variable pricing? Since its marginal cost is zero, it maximizes contribution by maximizing revenue. The revenue-maximizing single price can be found by solving the nonlinear optimization problem

$$\max_{p, x} p \sum_{i=1}^7 x_i,$$

subject to

$$\begin{aligned} x_i &\leq D_i + m_i p && \text{for } i = 1, 2, \dots, 7, \\ x_i &\leq C && \text{for } i = 1, 2, \dots, 7, \\ p &\geq 0, \end{aligned}$$

where p is the optimal single price, x_i is attendance on each day, D_i and m_i are, respectively, the intercept and slope of the price-response function for day i , and C is total daily capacity (in this case 1,000). The optimal single price is \$25.00 per ticket. The corresponding attendance and revenue for each day are shown in the columns labeled “Single price” in Table 7.5. With a single price, the theme park serves a total of 4,795 customers during the week, with total revenue of \$119,900. What is striking in Table 7.5 is the wide variation in attendance across the week. The park is full on Friday, Saturday, and Sunday, but it operates at only 25% of capacity on Monday and 40% of capacity on Tuesday. Furthermore, potential customers are being turned away on Friday, Saturday, and Sunday—500 on Saturday alone.

These two conditions—wide variation in utilization and a large number of turndowns—are indications that contribution could be improved through variable pricing. In the case of the theme park, this means charging a different price for each day. When demands are independent—as we have assumed—the optimal daily prices can be calculated by solving

TABLE 7.5

Theme park daily prices, attendances, and revenues under constant pricing and under variable pricing

Day of week	Single price			Variable pricing		
	Price (\$)	Attendance	Revenue (\$)	Price (\$)	Attendance	Revenue (\$)
Sunday	25.00	1,000	25,000	33.87	1,000	33,870
Monday	25.00	250	6,250	15.00	750	11,250
Tuesday	25.00	400	10,000	17.50	700	12,250
Wednesday	25.00	461	11,525	18.01	755	13,590
Thursday	25.00	684	17,100	19.00	1,000	19,000
Friday	25.00	1,000	25,000	27.00	1,000	27,000
Saturday	25.00	1,000	25,000	38.33	1,000	38,333
Total		4,795	119,875		6,205	155,294

independent optimization problems for each day. The resulting daily prices, attendance, and revenues are shown in the columns labeled “Variable pricing” in Table 7.5. With variable pricing, daily prices range from a low of \$15.00 on Monday to a high of \$38.33 on Saturday. The total number of customers served during the week rises 29% from 4,795 to 6,205, and total revenue rises 30% to \$155,294. Note that variable pricing evens out utilization across days of the week and increases overall utilization. Attendance on the low days of Monday and Tuesday increases from 250 and 400 to 750 and 700, respectively. The weekly utilization of the park—its *load factor*—increases from $4,795/7,000 = 68.5\%$ to $6,205/7,000 = 88.6\%$.

The *average* price paid by all customers increases only slightly under variable pricing—from \$25.00 to \$25.03. This is typical—variable pricing can often lead to a major increase in total revenue with only a small increase in the average price paid by customers.

7.5.2 Variable Pricing with Diversion

We have seen that variable pricing can increase operating contribution when demands are independent among periods. However, variable pricing can also shift demand from peak periods to off-peak periods—a phenomenon called *diversion* or *demand shifting*. Diversion occurs when at least some customers are flexible and will shift their consumption among periods if they can save money by doing so. Diversion can be illustrated by an example from Robert Cross (1997). His local barbershop was overcrowded and turning customers away on Saturdays, while Tuesdays were very slow. Some of the Saturday customers were working people who could only come on Saturday; others were retirees and schoolchildren, who could get their hair cut any day of the week. By raising prices 20% on Saturday and reducing them by 20% on Tuesday, the shop induced some of the Saturday customers to shift to Tuesday. As a result of this simple peak-pricing action, turnaways decreased, Saturday service improved, and total revenue increased by almost 20%.

Variable pricing segments the market for a generic product (a haircut) between those with strong preferences for a particular date and those who are relatively indifferent. Since customer preferences are not evenly distributed—more people want to see an opera, attend a theme park, or get their hair cut on a weekend—and capacity is fixed, using price to shift indifferent customers to less popular dates can increase both revenue and utilization and reduce turnaways. In the case of the Colorado Rockies, the four-tiered pricing system

segments the market between customers who badly want to see a popular game (such as one against the Yankees) from those who just want a night out at the ballpark. Furthermore, variable pricing is generally well accepted because it segments via self-selection: customers choose the product and price combination they prefer.

While diversion is an important element of variable pricing, it can be a two-edged sword. Raising the price for peak capacity and lowering it for off-peak capacity will shift customers from the peak to the off-peak period. This is good (at least in moderation) because the overall gain from extracting higher prices from the remaining peak customers outweighs the lower price received for off-peak customers. However, if the price differential becomes too great, we would expect more and more peak customers to defect to the off-peak period. If the barbershop drops Tuesday prices too low and raises Saturday prices too high, it might end up with a Tuesday peak and very slow Saturdays. Ultimately the loss from peak customers who shift to the cheaper off-peak period (cannibalization) can outweigh the benefits of off-peak demand induction and higher peak rates.

While diversion is easy to describe, it is difficult to model. In effect, the different alternatives are substitute products, and each pair could have a different cross-price elasticity. One approach is to presume that every customer has a separate willingness to pay for each alternative. Each customer would compare his willingness to pay for each alternative with the price and purchase the alternative that provided the highest positive surplus—assuming that such an alternative exists.

Example 7.6

Under the separate-willingness-to-pay model, each customer would have a separate willingness to pay for attendance on each day at the theme park. The willingness to pay of each customer can be represented as a vector, so a customer with willingness-to-pay vector of (\$30, \$18, \$22, \$19, \$14, \$18, \$32) would have a \$30 willingness to pay for Sunday attendance, an \$18 willingness to pay for Monday attendance, and so on. We can represent the vector of prices in the same fashion: (\$33.87, \$15.00, \$17.50, \$18.01, \$19.00, \$27.00, \$38.33). The consumers surplus vector for this customer is the difference between her willingness to pay and the price for each day, here (−\$3.87, \$3.00, \$4.50, \$0.99, −\$5.00, −\$9.00, −\$6.33). In this case, the customer would attend on Tuesday because that provides her with the highest surplus (\$4.50). For comparison, the consumers surplus for this customer under a constant price of \$25.00 would be (\$5.00, −\$7.00, −\$3.00, −\$6.00, −\$11.00, −\$7.00, \$7.00), which means she would attend on Saturday. The effect of variable pricing on this customer is to shift her attendance from Saturday to Tuesday.

While the willingness-to-pay point of view helps us understand how an individual consumer might act, it is difficult to extend to a useful model of total market price responsiveness. If we knew the distribution of all customer willingness to pay across the population, we could, in theory, use that information to derive a multidimensional demand function. One problem is that this model would require 7 own-price elasticities (1 for each day of

the week) and 42 cross-price elasticities (1 for each combination of days of the week). It is unlikely that the theme park (or most other companies) would have the data available to estimate a credible model with this many parameters.

We can use a much simpler approach to incorporating diversion in the theme park example—we assume that eight customers will shift from one day to another for every \$1 difference in price. Thus, if the base prices for Monday and Tuesday were \$20 and \$22, respectively, we would first calculate a base demand for each day using its individual linear price-response function and then shift 16 people from Tuesday to Monday. When applied across the whole week, this means that demand will shift from days with higher-than-average prices to those with lower-than-average prices. Under this model, the unconstrained demand the theme park will see on day i is given by

$$d_i = D_i + m_i p_i + \sum_{j=1}^7 8(P_j - p_i),$$

where d_i is the unconstrained demand, p_i is the price, D_i is the demand at zero price, and m_i is the slope of the linear price-response function for day i . Demand for each day now depends not only on the price for that day but on the price for all the other days of the week. We can now formulate the optimal pricing problem as a nonlinear optimization problem:

$$\max_{p, x} p \sum_{i=1}^7 P_i x_i,$$

subject to

$$\begin{aligned} x_i &\leq D_i + m_i p_i + \sum_{j=1}^7 8(P_j - p_i), \\ x_i &\leq C, \\ p_i &\geq 0 \quad \text{for } i = 1, 2, \dots, 7. \end{aligned}$$

The optimal prices, attendance, and revenue under this model are shown in Table 7.6. Base demand is the number of customers calculated using the price-response functions for each day, and net diversion is the net number of customers induced to shift to or from each day by differential pricing. A positive net diversion means that more customers shifted to that day than shifted away; a negative net diversion means the opposite. Variable pricing shifted

TABLE 7.6
*Theme park daily prices, attendances, and revenues using variable pricing
and assuming demand switching*

Day of week	Price (\$)	Base demand	Net diversion	Attendance	Revenue (\$)
Sunday	28.92	1,307	−307	1,000	28,920
Monday	18.12	594	298	892	16,163
Tuesday	19.58	617	216	833	16,310
Wednesday	19.80	680	204	884	17,503
Thursday	21.29	880	120	1,000	21,290
Friday	25.21	1,099	−99	1,000	25,210
Saturday	31.14	1,432	−432	1,000	31,140
Total		6,609	0	6,609	156,536

a total of 838 customers from the peak days of Friday, Saturday, and Sunday to the off-peak days of Monday through Thursday. Furthermore, the theme park served 6,609 total customers during the week, resulting in a 94% utilization and total revenue of \$156,536—a dramatic improvement over charging a single price.

Of course, the theme park operator might not want to set prices exactly as shown in Table 7.6. For simplicity, he may want to reduce the total number of prices and only offer prices in increments of \$5. In this case, it is easy to see that he would still gain significant revenue from a three-tier pricing policy of \$30 on weekends, \$25 on Friday, and \$20 on Monday through Thursday, as opposed to setting a constant price of \$25.

Variable pricing does not need to be implemented within list prices. The theme park could implement two-tier pricing by establishing a \$30 list price but distributing \$10-off coupons redeemable only on weekdays. Even more effectively, the theme park might establish a \$35 price for weekends and Friday, a \$20 price for Monday through Thursday, and distribute \$5-off coupons redeemable only for Monday admissions. In this case, variable pricing has been implemented through a combination of variable list pricing and couponing.

7.6 VARIABLE PRICING IN ACTION

We now consider how variable pricing is used in four different industries—sporting events, airlines, electric power, and ride-sharing. We also look at two industries—movie theaters and restaurants—that meet all the criteria for variable pricing but in which variable pricing is little used.

7.6.1 Sporting Events and Entertainment

Some of the most obvious candidates for variable pricing are sporting events and other forms of entertainment such as concerts and theater. In most cases, sporting arenas, concert venues, and theaters have fixed capacity but demands that vary depending on the offering. A further sign of the potential for variable pricing in these industries is the prevalence of a robust secondary market for tickets. For much of history, this secondary market was driven by scalpers (called ticket touts in the UK) who would purchase tickets at list price with the expectation of selling at a premium. More recently, companies such as StubHub and RazorGator have enabled online resale of tickets. The existence of this secondary marketplace was driven in large part by the fact that concert promoters and sporting teams were unable or unwilling to adjust prices to balance supply and demand.

Around the beginning of this century, sporting teams began to experiment with variable pricing. Variable pricing is particularly attractive in baseball since each team has 81 home games each season that vary widely in terms of fan appeal. The Colorado Rockies were among the earliest adopters, instituting the variable-pricing scheme shown in Table 7.3 in 2004. Over the next 15 years or so, every team in Major League Baseball followed suit, with the New York Yankees being the last to adopt variable pricing in 2017 (Best 2017). While there was some resistance from fans, variable pricing was generally viewed as successful, which has spurred uptake in other leagues including the National Football League, the Na-

tional Basketball Association, the National Hockey League, and the Premier League (soccer) in England, among others.

Many sporting teams employ dynamic variable pricing, establishing an initial set of prices according to a scheme such as that shown in Table 7.3. However, prices can also be updated over time based on information gained as the season progresses—broadly speaking, if the team is doing well, prices will increase; if they are doing poorly, prices will drop. Prices may also be adjusted based on weather, the record of the opposing team, or the emergence of a popular player (for example, it was estimated that, in the 1991–1992 NBA season, Michael Jordan generated over \$50 million in revenue for *opposing teams* through increased attendance and ancillary revenue).⁴

Sporting teams like variable pricing because it increases total revenue without increasing ticket prices. This is attractive in an environment where ticket price increases are likely to be condemned by fans and sportswriters as evidence of greedy owners and players gouging loyal fans. In 2003, the New York Mets adopted a variable-pricing plan under which the price was increased \$10 for the 17 most popular games while lowering or holding prices unchanged for 43 games. “That allowed the Mets, who finished last in their division in 2002, to trumpet the plan as holding ticket price increases to only 4% on average” (Fatsis 2003, 82).

Broadway and other long-run theaters are also an ideal candidate for variable pricing—demand varies in predictable ways across seasons and by the day of the week, and customers purchase advance tickets. Nonetheless, theaters were relatively slow to adopt variable pricing beyond charging more for weekend night shows than matinees and weekday nights. One of the pioneers in dynamic pricing in Broadway theater was Disney’s *The Lion King*. As Disney vice president David Schrader described pricing for *The Lion King*, “First the range of prices across the theater is set according to an estimate of the demand for a particular performance, based on the 15 years of past B.O. [box office] from which to extrapolate annual trends. Second, after tickets go on sale, Disney re-examines each perf [performance] to see how demand is shaping up in reality, and then tweaks pricing accordingly” (Cox 2013, 1). Dynamic pricing was widely touted as one of the reasons that *The Lion King* generated the most revenue of any Broadway show in the least amount of time despite being in one of the smaller Broadway theaters.⁵ With the success of variable pricing at *The Lion King*, many other Broadway theaters, as well as performing arts groups such as the San Francisco Symphony and attractions such as the Indianapolis Zoo, have adopted more dynamic pricing. Ticketmaster, by far the largest distributor of event and sporting tickets in the United States, has adopted dynamic pricing to adjust prices based on expected supply and demand.

7.6.2 Passenger Airlines

Passenger airlines certainly meet the criteria for variable pricing—constrained capacity with demand that fluctuates over time. Southwest Airlines implements variable pricing directly by changing fares for its departures to match demand with the restricted capacity. However, the major nondiscount airlines also practice variable pricing. They do so by opening and closing the availability of various fare classes on different flights. Figure 7.2 shows average fares and load factors on a flight of a major airline from the Midwest to Florida during the winter. For operational reasons, the airline has scheduled the same aircraft

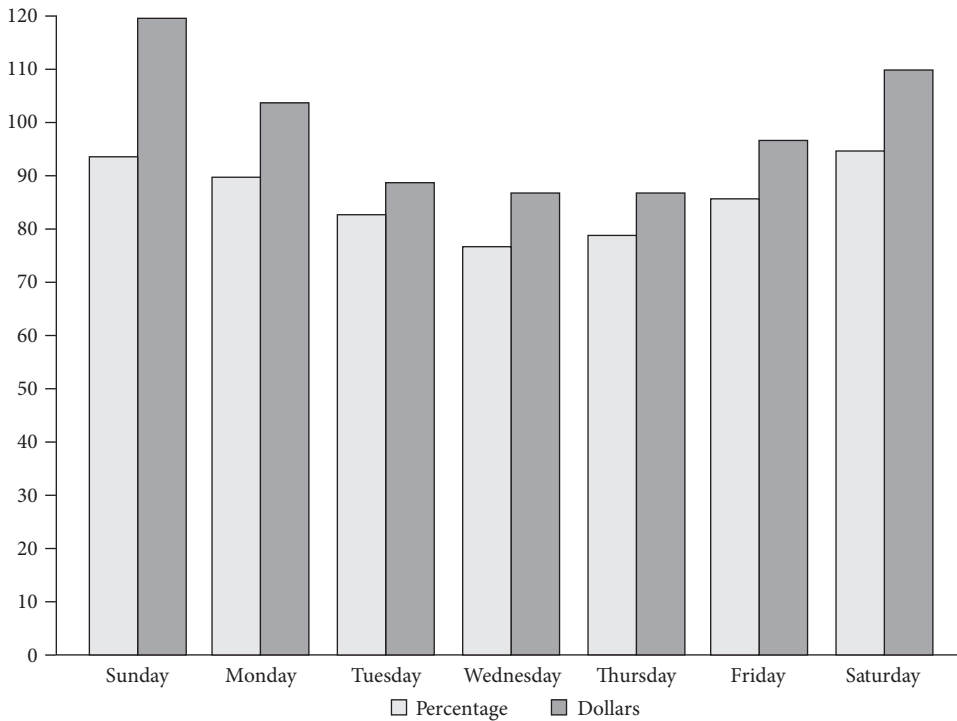


Figure 7.2 Average daily load factors for a Midwest–Florida flight on a domestic US discount carrier. *Source:* Courtesy of Garrett J. van Ryzin.

for the flight each day. The airline is charging a higher average price on high-demand days (Saturday and Sunday) and lower prices on low-demand days (Tuesday, Wednesday, and Thursday). In this case, the airline is adjusting the average fare not by changing the list price but by opening and closing fare classes. When total demand is anticipated to be high, the airline will restrict availability of discount fares, causing the average fare to increase. When total demand is anticipated to be low, the airline will open availability of deep discounts, causing the average fare to drop. It is continually monitoring demand using its revenue management system to determine which classes to open and close on each flight. Chapters 8 and 9 describe in detail how this is done.

Discount airlines use variable pricing even more directly. If demand at Southwest Airlines begins to rise more quickly or more slowly than anticipated, Southwest may change the price. Southwest is not doing this to segment customers between high-elasticity and low-elasticity segments—rather, it is using variable pricing without segmentation to maximize the return from each flight and, to some extent, to smooth demand through demand diversion from high-fare, high-utilization flights to low-fare, low-utilization flights.

7.6.3 Electric Power

Electric power is another obvious candidate for variable pricing. Suppliers have constrained capacity, and electricity is difficult and expensive to store. Demand follows reasonably pre-

dictable patterns, and many customers have the flexibility and willingness to adjust their usage in response to prices. In fact, electric utilities have a further motivation to vary prices: There are tremendous cost differences between generation during peak and off-peak periods. For many utilities, it can be 10 times more expensive to generate power during a peak period than during an off-peak period. Adopting dynamic pricing could reduce generation costs by 8% to 20%, leading to price reductions for all customers, not just those shifting their consumption.⁶

Despite the obvious benefits that electric suppliers and consumers could realize from time-of-day pricing, utilities in the United States have been slow to adopt programs that would adjust the price of electricity in response to changes in demand. Some of the reasons are technical, relating to the unique characteristics of electric power transmission and distribution. However, the most powerful reasons are political and historical. Historically, electric utilities were granted service monopolies over a certain territory and guaranteed a reasonable return on their investments in generation capacity and transmission. In return, they submitted to government regulation of their prices to avoid abuses of monopoly power in pricing. While some jurisdictions are moving to less regulated regimes and seeking to encourage competition, the transition from fully regulated to a more competitive market for electric power has proven to be difficult and in some cases a political disaster. The attempt by California to deregulate its electricity market was a debacle that was one of the drivers behind the California governor recall and election of 2003.

Despite the political barriers, many electric power utilities have instituted some form of variable-pricing program. Most of these programs are aimed at industrial customers, who often have the flexibility to schedule their operations to take advantage of variable prices. In a typical program, the electricity supplier charges the customer for electricity on an hour-by-hour basis, with expected prices for the next day announced 24 hours in advance. This enables industrial customers to reschedule their operations to minimize consumption costs. Programs to charge variable prices to residential customers are much rarer—the Gulf Power RSVP program described earlier is a typical example. It is likely that, despite setbacks, the trend toward more competitive power markets in the United States will continue and that the need for variable-pricing methods will only increase.

7.6.4 Ride-Sharing

Major players in the ride-share industry, such as Lyft, Uber, and Didi Chuxing, have all adopted dynamic pricing. These ride-share operators connect prospective riders with independent drivers who are willing to provide transportation. When a prospective rider opens the app and specifies her destination, the ride-share company quotes a price that can depend on both the origin and the destination. The price for a particular origin and destination combination will change over the course of time in response to current demand and supply conditions, as we shall see.

While business models vary, the ride-share operator typically determines both the *rider price*, or what the rider pays, and the *driver price*, or what the driver will be paid for the trip. The ride-share operator keeps the difference between the rider price and the driver price—the *trip margin*. In most cases, the trip margin is a fixed percentage of the rider price: as of

2019, Uber was advertising that its trip margin percentage was 25%, and Lyft was advertising 20%. This means that, on an Uber trip costing the rider \$20, the driver would get \$15 and Uber \$5.⁷ Ride-share operators typically set a time- and distance-based *base rate* for each trip. These base rates, which may vary by city, establish a minimum rider price and hence a minimum driver payment. However, most ride-sharing operators soon established dynamic pricing in which the rider price and driver price varied by time and by location.

For the ride-sharing operators, the need to balance supply and demand motivated the adoption of peak-load pricing. In particular, in certain times and locations, the demand for rides at the base price exceeds the supply of available drivers—sometimes by a large margin. Demand for rides can vastly outstrip available supply at the end of the workday in business districts in London, Manhattan, and San Francisco; at peak airport arrival times; and when sporting events or concerts let out. When demand exceeds supply, a ride-sharing operator faces the problem of allocating constrained supply. If the operator does not raise the price, many customers would see long arrival times or that no cars are available. As a result, some customers would drop out and supply would be allocated to the customers who were most willing to wait. Alternatively, the ride-share operators could raise the price and allocate the scarce supply using prices. This is the approach that ride-share operators generally adopted.

Of the dynamic pricing schemes in use by ride-sharing companies, Uber's has been the most widely studied. Uber divided each city it serves into hexagons about 0.2 mile on a side, with each hexagon covering an area of about 0.1 square mile. In the initial application of “surge pricing,” the rider fare for every trip originating in a hexagon was subject to a multiplier greater than or equal to 1. If the trip multiplier was greater than 1, then the hexagon was “surging” and the rider fare would be set to the trip multiplier times the base price. So, for example, if the base fare is equal to \$12 for a trip and the surge multiplier were 2.5, the rider price would be $2.5 \times \$12 = \30 plus any fees and tolls. If the trip multiplier were 1, then the hexagon was not surging and the rider would pay the base price plus fees and tolls.

Under this scheme, the problem for Uber is what multiplier to apply at each time at each geographical location and how to update it as conditions change. The multiplier can be changed at relatively short intervals—typically every one to three minutes. We assume for the moment that the same multiplier will be applied to every ride originating in the same region during a time interval. The ride-sharing service typically knows the number of people in the area who have the app open—we denote that *potential demand* by D . Not everyone who opens the app will book a ride, and we can expect that the fraction of open apps that will book will depend on the multiplier applied, with a higher multiplier leading to a smaller fraction of potential demand booking.

The service also knows the number of open cars in the area, S . The ride-sharing service needs to determine the multiplier $p(S, D)$ to set as a function of the current supply and demand situation. If $D \leq S$, then $p(S, D) = 1$. In this case, there is sufficient supply to serve demand even if every open app converted to a booking, so there is no need to raise prices. Because the base price is a minimum, we cannot set $p(S, D) < 1$. If $D > S$, then we are in a situation of excess demand and we want to tamp down demand, so we need to set $p(S, D) > 1$.

The multiplier function $p(S, D)$ needs to satisfy three conditions.

- $D \leq S \rightarrow p(S, D) = 1$. If supply is more than adequate to meet demand, the multiplier is equal to one.
- $\partial p(S, D) / \partial D \geq 0$. The multiplier is increasing in demand.
- $\partial p(S, D) / \partial S \leq 0$. The multiplier is decreasing in supply.

Let's dig a little deeper into the calculation of the multiplier. First, we assume that demand responds instantly to the multiplier. This is realistic; riders tend to respond to fares at the moment they open the app, and if they see a fare that is too high, they may decide to use another mode of transportation or postpone their trip. This will be reflected immediately in the *booking conversion rate*—the fraction of people who book after viewing a fare on their app will decrease as the multiplier increases. We can model this situation as

$$d(p) = D\rho(p), \quad (7.4)$$

where D is the number of people who open the app, $0 \leq \rho(p) \leq 1$ is the fraction of people who open the app who will book a ride given multiplier p , and $d(p)$ is the number of rides that will be booked. We assume that $\rho'(p) < 0$ so that the number of bookings will decrease as the multiplier increases.

This representation is identical to the formulation in Equation 3.1 with the difference that p represents the multiplier instead of the price. This reflects the fact that, from a customer's point of view, the buying experience is much the same: with the ride-hailing app, the customer signals potential interest in purchasing by opening the app. The customer does not decide whether to book until seeing the price, and a higher fraction of customers will book if the price (multiplier) is low than if it is high. It also means that any of the functional forms listed for $\rho(p)$ in Table 3.3, such as linear, logit, and probit, can be used in Equation 7.4 depending on the assumption about the underlying distribution of willingness to pay among riders.

When demand exceeds supply ($D > S$) and supply is fixed, then the ride-sharing operator faces a variable-pricing problem and the optimal (profit-maximizing) multiplier p^* is the one that balances supply and demand. This means that p^* must be such that $D\rho(p^*) = S$, or, equivalently,

$$\rho(p^*) = \frac{S}{D}. \quad (7.5)$$

We can solve for p^* by substituting the appropriate function $\rho(p)$ into Equation 7.5 and solving.

Example 7.7

There are 100 empty cars available in a region at a time when $D = 300$. Assume that the price-response function is linear with $\rho(p) = (1 - .33p)^+$. From Equation 7.5, the multiplier that balances supply and demand is given by $(1 - .33p^*) = \frac{100}{300}$, which gives $p^* = 2$.

For each city, the multipliers for a ride-sharing service like Uber or DiDi are updated for all of the regions within a city every few minutes based on changing supply and demand conditions.

One question about dynamic pricing of this sort is whether it is good for customers. Note that, if a region is in a state of excess demand and prices are not increased, service quality will decrease: riders will see increased time to arrival for their ride or, in the extreme, that no cars are available. While an excess of demand may seem like a good situation for drivers, in actuality it is not. The increased arrival times mean that drivers are being sent from distant regions to satisfy the demand and are thus spending excessive time driving to pickups—time for which they do not get paid. Furthermore, the price has been kept low, so the result is lower driver payments per hour. Riders get good prices, but they need to accept long arrival times, and many will drop out because of excessive arrival times or unavailability of cars. Juan Castillo, Dan Knoepfle, and Glen Weyl (2017) call this a *wild-goose chase equilibrium*, in which drivers spend their time chasing riders. They show that balancing supply and demand using price can lead to an equilibrium that provides more social welfare (as measured by producers plus consumers surplus) than the wild-goose chase equilibrium. Effective variable pricing can make both riders and drivers better off.

This discussion has assumed that supply is fixed—that is, while riders respond to changing prices, drivers do not. This is not entirely true in the short run: drivers do tend to move toward areas of higher surge. It is also likely that, in the longer run, drivers anticipate the time and locations of likely surge (e.g., downtown rush hour) and plan their driving accordingly. However, in the short-run time frame in which drivers make decisions (minutes), supply in a region is largely fixed, meaning that the effect of changing price on immediate supply is relatively small. The impact of incorporating the supply-induction effects of prices into ride-sharing pricing would be to reduce the multipliers somewhat.⁸

7.6.5 Movie Theaters and Restaurants

It is interesting to examine two industries in which the variable-pricing dog has not barked—that is, variable and dynamic pricing has not been widely adopted despite an apparent ideal fit. The first is movie theaters. While movie theaters employ many standard promotional approaches such as senior and student discounts and matinee discounts, they generally do not vary pricing according to the popularity or anticipated popularity of a movie; nor do they change it over time as demand begins to dwindle. The reason they do not is a bit of a mystery. Movie demand has a predictable element to it—demand is higher in the summer and on holidays than other times of the year, and demand is higher on the weekends than on weekday nights. Furthermore, demand tends to decline following the release of a new movie in a fairly consistent and predictable fashion. While there can be unexpected hits and duds, relative demand for different movies is reasonably predictable: the next James Bond movie will certainly outdraw even the most popular documentary released in the same year. Finally, movie theaters meet one of the most fundamental criteria for variable pricing: capacity is fixed and immediately perishable.

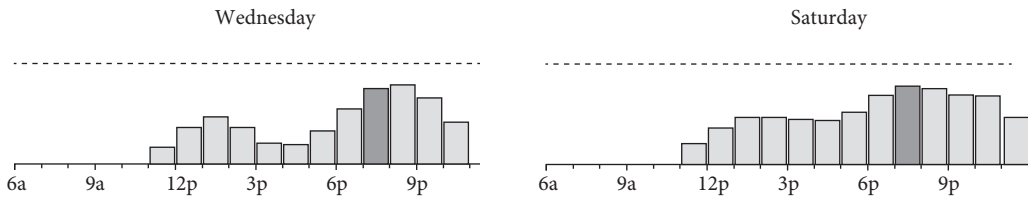


Figure 7.3 Predicted hourly load at Gramercy Tavern restaurant in Manhattan on Wednesday, December 18, 2019 (*left*), and Saturday, December 22, 2019 (*right*).
Source: Google Maps.

There has been some movement toward variable pricing at movie theaters in the US industry—most cinemas across the country slash prices on Tuesday nights. As a result, Tuesday has gone from being the lowest-grossing weeknight to the highest grossing (except for Friday). While this would indicate that there is an opportunity to increase revenue and fill theaters using variable pricing, in fact, “no U.S. chain has tested variable pricing in a serious way,” notes one industry observer (McClintock 2019). There has been some debate about why the industry has been so reluctant when the gains are potentially enormous. The leading theory is that independent theaters and theater chains feel implicitly pressured by distributors who want to keep in-theater prices high for their movies so that the value perception of the movie is not degraded by a low price. It is difficult to confirm or disprove this theory, but, despite the apparent fit, adoption of variable pricing is very limited in US theaters, and there are few indications that the situation is likely to change soon.

Restaurants and bars would seem to be another candidate for variable dynamic pricing. Demand varies rather predictably by time of day, day of week, and season. For example, Figure 7.3 shows Google Maps’ hourly forecast for demand at the New York City restaurant Gramercy Tavern for Wednesday, December 18, and Saturday, December 22, 2019. These forecasts, which were derived entirely from history, show predictable differences in seating demand by both day and hour. The restaurant itself has access to reservations data, which it could use to generate even more accurate forecasts. Finally, of course, restaurants have fixed and perishable capacity.

Despite checking all of the boxes for variable pricing, restaurants and bars do very little of it aside from happy hours, the usual student and senior citizen discounts, and in some cases Groupons. Even extremely popular restaurants typically allocate capacity using scarcity—reservations need to be made well in advance and/or people need to wait in line at popular times—rather than adjusting price over time to balance supply and demand. However, restaurants may present less of a mystery than movie theaters with regard to their reluctance to adjust prices: the use of printed menus makes real-time price adjustment difficult, as does the nature of the dining experience—it would be disconcerting to have prices change between the time you decided on your order and the time you actually ordered.

That said, a development that may lead to higher penetration of variable and dynamic pricing in the restaurant industry is the rise of delivery services such as Uber Eats and Grubhub in the United States and Deliveroo in Europe, which provide apps that allow diners to

order food to be delivered. The popularity of these delivery services has led to the development of so-called virtual restaurants and ghost kitchens that exist solely to serve the delivery market.⁹ It is likely that such establishments selling primarily (or only) through online delivery apps will adopt variable and dynamic pricing—after all, the delivery services already use dynamic pricing for their fees.

7.7 SUMMARY

- Supply constraints are commonplace in many industries. They can arise as a result of limited inventory or because capacity itself is physically constrained. Whenever there is a significant chance that demand might exceed available supply, the supply constraint needs to be explicitly incorporated in determining the optimal price.
- With a single market segment, the optimal constrained price is the maximum of the runout price and the optimal unconstrained price. The optimal price with a supply constraint is always at least as high as the optimal unconstrained price.
- With multiple market segments, the optimal price can be found by solving a constrained-optimization problem. At the optimal price, the marginal revenue will be the same for all segments, but it will not be equal to marginal cost if the constraint is binding.
- A supply constraint has an associated *total opportunity cost*, defined as the additional operating profit the business could achieve if the supply constraint were eliminated. The *marginal opportunity cost* associated with a supply constraint is defined as the additional operating profit the seller would realize if one additional unit of supply were available. Both the total and the marginal opportunity costs associated with a supply constraint are zero if the constraint is not binding.
- Variable prices can be used when capacity is constrained and demand changes over time in a predictable fashion. Variable pricing can increase profitability when demands are independent by adjusting demand to meet supply. It is even more effective when some customers are flexible, because it can shift demand from peak to off-peak periods.
- Optimal variable prices can be found by solving a constrained-optimization problem. The marginal values associated with the supply constraints are the opportunity costs of additional capacity.
- Some industries in which variable pricing is used are sporting events, concerts, passenger airlines, electric power, and ride-sharing. In some of these cases, the variable prices are set once in advance. Other industries, notably ride-sharing platforms and electric power, use dynamic variable pricing in which prices are updated as more recent information about supply and demand is received. This comes close to revenue management, the topic of the next few chapters. Currently, movie theaters and restaurants do not extensively use variable pricing despite apparent suitability.

The pricing and revenue optimization problems addressed in later chapters build on the basic techniques of pricing with constrained supply. Capacity allocation (Chapter 9) is based on opening and closing availabilities to different market segments in order to maximize return from fixed and perishable capacity when willingness to pay increases over time. Network management (Chapter 10) extends this concept to the case when customers are buying multiple resources. Overbooking (Chapter 11) deals with the question of how many total units to offer for sale when capacity is fixed. Markdown management (Chapter 12) addresses the case of setting and updating prices for a fixed stock of perishable goods over time. In each of these problems, much of the complexity of the pricing problem comes from a constraint on supply.

7.8 FURTHER READING

A seminal paper on dynamic pricing with a capacity constraint is Gallego and van Ryzin 1994, which shows that, under very weak conditions, a seller with constrained perishable capacity who is pricing dynamically should set his price to either the profit-maximizing price or the runout price—the price at which he would exactly sell all of his inventory.

Regarding specific industries, variable pricing for baseball is discussed in Rascher et al. 2007, which estimates the potential value for every major league team. The San Francisco Giants were apparently the first to apply dynamic pricing in 2009; see Shapiro and Drayer 2012. For a specific application to the Los Angeles Dodgers, see Parris, Drayer, and Shapiro 2012. An early examination of the potential for dynamic pricing for theater can be found in Leslie 2004. There is a vast literature on dynamic pricing for retail electric power: a brief overview of the issues involved in implementing dynamic pricing in the United States—particularly the barriers that have hindered adoption and the prospects for the future—can be found in Wilson 2012. A more extensive discussion can be found in the papers in Renses, Rodriguez, and Perez-Arriaga 2013. The reasons why dynamic pricing has not been widely adopted by movie theaters are discussed at length in Orbach 2004, Orbach and Einav 2007, and Phillips 2012c.

While restaurants do not widely use variable pricing (with exceptionally rare exceptions), many have adopted various other revenue management–like techniques to maximize revenue per available seat hour (RevPASH). For an overview of these techniques, see Kimes 2004.

7.9 EXERCISES

1. Let us return to the Stanford Stadium pricing problem in Section 7.4, assuming a capacity of 60,000 seats and the price-response functions for students and for the general public as given in Equations 7.1 and 7.2. Assume that 5% of the general public will masquerade as students (perhaps using borrowed ID cards) to save money. Assuming Stanford knows that, what are the optimal prices for student tickets and general public tickets it should set in this case? What is the total revenue,

- and how does it compare to the case without cannibalization? What does this say about the amount that Stanford would be willing to pay for such devices as photo ID cards in order to eliminate cannibalization?
2. An earthquake damages Stanford Stadium so that only 53,000 seats are available for the Big Game. What is the optimal single price and the total revenue? What are the optimal separate prices to charge for students and the general public, and the corresponding total revenue? What is the opportunity cost per seat for the 7,000 unavailable seats in both cases?
 3. *Ancillary Revenue.* Stanford Stadium has been repaired so that it again seats 60,000 people. Now assume that, on average, each member of the general public will consume \$20 worth of concessions, resulting in a \$10 contribution margin, while each student only consumes \$10, resulting in a \$5 contribution margin. Assuming no cannibalization, what are the prices for students and the general public that maximize total contribution margin (including, of course, ticket revenue)?
 4. A barber charges \$12 per haircut and works Saturday through Thursday. He can perform up to 20 haircuts a day. He currently performs an average of 12 haircuts per day during the weekdays (Monday through Thursday). On Saturdays and Sundays, he does 20 haircuts per day and turns 10 potential customers away each day. These customers all go to the competition. The barber is considering raising his prices on weekends. He estimates that for every \$1 he raises his price, he will lose an additional 10% of his customer base (including his turnaways). He estimates that 20% of his remaining weekend customers would move to a weekday to save \$1, 40% would move to a weekday to save \$2, and 60% would move to a weekday to save \$3. Assuming he needs to price in increments of \$1, should he charge a differential weekend price? If so, what should the weekend price be? (Assume he continues to charge \$12 on weekdays.) How much revenue (if any) would he gain from his policy?
 5. The optimal variable prices for theme park admission in Table 7.5 are based on the assumption that admission fees are the only source of revenue for the park. However, the owner determines that visitors to the theme park spend an average of \$12 per person on concessions, generating an average concession margin of \$5 per person. Under the same assumptions about capacity and demand, what is the single-admission price the theme park should charge to maximize total weekly margin (admission price plus concession margin)? What are the individual daily prices he should charge under a variable-pricing policy, assuming independent daily demands? What is the impact on total weekly admissions? What is the impact on total weekly margin from explicitly including concessions in the optimization relative to optimizing prices on the basis of admission revenue alone?
 6. Assume that the theme park owner invested in expanding his park so that he could accommodate up to 1,500 customers each day.
 - a. What single price would maximize his total revenue, assuming he faces independent demands for each day and that the price-response functions per day

- are as specified in Table 7.4? What is his corresponding attendance and revenue per day?
- b. Under the same assumptions, what variable prices should he charge for each day to maximize expected revenue, and what are the corresponding attendance and revenue per day?
7. The theme park owner performs some market research and determines that his customers can best be represented by the following model.
- Base customer demand for each day of the week is linear and is specified by the parameters in Table 7.4.
 - Weekend (Saturday and Sunday) customers will switch to the other weekend day at the rate of 10 customers for every \$1 difference in price. They will not switch to weekdays at any price.
 - Weekday customers will switch to any other day (including a weekend day) at the rate of 8 customers for every \$1 difference in price.

What are the optimal daily prices the theme park should charge?

8. The popular a cappella choral group Here Comes Treble is coming to Columbia University for a concert. The concert will be in Lerner Auditorium, which has C seats. Customers can be segmented into students holding a student ID card and the general public. The price-response function for the general public is $d_g(p_g) = (2,400 - 60p_g)^+$, and the price-response function for students is $d_s(p_s) = (400 - 25p_s)^+$. (Note: If you cannot do part a , you can assume that $C = 1,500$ for the following parts.)
- Columbia plans to sell general admission and student tickets. To find the revenue-maximizing ticket prices, Columbia decides to solve a revenue-maximization problem with the capacity constraint $12p_g + 5p_s \geq 320$, together with the demand nonnegativity constraints $p_s \leq 16$ and $p_g \leq 40$. Given this information, what is the capacity of the auditorium?
 - Given the capacity of the auditorium that you calculated in part a , what is the revenue-maximizing single price?
 - Columbia decides to go through with the plan of charging different prices for general admission and student tickets. Given the capacity of the auditorium you calculated in part a , what are the revenue-maximizing ticket prices for students and the general public?
 - Assume that Columbia reschedules the concert for a larger auditorium with 3,000 seats. What are the revenue-maximizing prices for students and the general public?
9. The popular ride-sharing app Lyber has determined that the general form of rider response to its price multiplier p is given by the logit formula

$$d(p) = D \left(\frac{e^{(a-bp)}}{1 + e^{(a-bp)}} \right).$$

Here, D is the total number of people in an area who have opened the Lyber app, p is the multiplier and base price, $d(p)$ is the number of people in the area who will order a Lyber if the multiplier is p , and a and b are parameters.

- a. Let S be the number of available drivers in the area, and assume that $S < D$. What is the formula for the multiplier that would exactly balance supply and demand in terms of a , b , S , and D ?
- b. Let $D = 1000$, $S = 500$, $a = 10$, and $b = 5$. What value of p would balance supply and demand?

NOTES

1. Note that a seller does not need to be a purist in allocating scarce supply. He can, for example, raise price to less than the market-clearing price and combine that with one of the other allocation approaches. This would allow him to realize some additional profit while still favoring preferred customers.

2. This example is modified from one originally developed by Jeremy Bulow of the Stanford Business School.

3. These rates were in effect on December 12, 2019, according to the Gulf Power rate sheet found at that time through <https://www.gulfpower.com>.

4. The impact of Michael Jordan on revenues is from *Fortune* magazine, based on work originally by Jerry Hausman and Gregory Leonard, who concluded that “superstars are more important than league balance for revenues” (1997, 586). Overall, *Fortune* estimated that Jordan’s impact on home gate receipts was \$165.5 million and on road games, \$30.5 million. This was an era before dynamic pricing—it is likely that the impact would have been substantially greater with optimal dynamic pricing in which games featuring Jordan would have been priced higher than other games (Johnson 1998).

5. Through the 2017–2018 Broadway season, *The Lion King* had grossed \$1.54 billion over its run, compared to \$1.25 billion for second-place *Wicked*. Of course, *The Lion King*’s record-setting financial performance is due to more than just variable pricing; it also benefits from an excellent product that appeals to all ages, association with the Disney brand, strong marketing support, and a great location close to Times Square.

6. For the potential benefits of dynamic pricing for electric power, see Imelda et al. 2018.

7. Ride-share pricing is somewhat more complicated because of various fees and incentives. For example, as of 2018 in the United States, both Lyft and Uber charge a \$1 booking fee for each trip, which is not shared with the driver. Various services—including both Lyft and Uber—enable riders to tip their drivers. In addition, many operators provide various driver subsidies that can increase driver earnings—for example, an operator may pay a driver an extra \$50 for completing at least 20 trips in a week. For this reason, the total compensation for a driver over a period will not necessarily be equal to a fixed fraction of the prices paid by the riders she carried. For base rider pricing, see O’Connell 2018.

8. Ride-sharing is a relatively new industry, and the ride-sharing companies continue to explore new pricing models. For example, in early 2019 Uber began to experiment with “additive surge,” in which prices are modified not by using multiplicative factors but by using “adders.” Under a multiplicative surge of 2.0, a ride with a base fare of \$5.00 would have a fare of \$10.00 and a ride with a base fare of \$15.00 would have a dynamic fare of \$30.00. With a \$6.00 additive surge, the ride with a base fare of \$5.00 would have a fare of \$11.00 and the \$15.00 base fare ride would have a dynamic fare of \$21.00. Garg and

Nazerzadeh 2019 provides an argument that additive surge is superior to multiplicative surge, and Uber 2019 gives a description of how it operates in practice.

9. A *virtual restaurant* is one whose brand appears only online, although it may be tied to a differently named physical restaurant. A *ghost kitchen* is an establishment that exists only to serve the delivery market—there is no associated dining establishment. Confusingly, ghost kitchens are sometimes referred to as virtual restaurants.

8

REVENUE MANAGEMENT

Revenue management (RM) (once called *yield management*) refers to the strategy and tactics used by a number of industries—notably the passenger airlines but also including hotels, rental cars, cruise lines, and others—to manage the allocation of their capacity to different fare classes over time in order to maximize revenue. Revenue management is applicable when all of the following conditions hold.

- The seller is selling a fixed stock of perishable capacity.
- Customers book capacity prior to departure.
- The seller manages a set of *fare classes* (also called *booking classes*), each of which has a fixed price (at least in the short run).
- The seller can change the availability of fare classes over time.

Revenue management can be considered a special case of variable and dynamic pricing with constrained supply. However, the final two conditions give revenue management its special flavor. Revenue management is based not on setting and updating prices but directly on setting and updating the availability of fare classes, where each fare class has an associated fare (price) that remains constant through the booking period. This distinctive feature is a legacy from revenue management's origin. The passenger airlines that pioneered revenue management in the 1980s needed to utilize the capabilities they had at hand. This meant using the booking controls embedded in their reservation systems as the primary mechanism for controlling the fares displayed to customers at any time. Following success at the airlines, revenue management has been adopted by a number of other industries, including hotels, rental cars, freight transportation, and cruise lines—many of whom use the same (or similar) reservation systems as the passenger airlines.

We use *revenue management industries* as a generic term for industries meeting the conditions listed above and *revenue management companies* to refer to companies that make use of booking controls and fare classes. While these companies apply revenue management in its purest fashion, the techniques of revenue management can be applied to any

situation in which a seller needs to determine how to allocate a fixed supply among different channels or customer segments paying different prices.

This chapter serves as an introduction to revenue management. It is a prelude to the next three chapters, which address specific aspects of revenue management in more detail. We start by surveying some of the history of revenue management and then look at revenue management strategy as an application of some of the price differentiation techniques introduced in Chapter 6. In Section 8.2, we discuss how revenue management operates at three different levels: the strategic, the tactical, and that of moment-to-moment booking control. Section 8.3 describes the customer segmentation strategy underlying revenue management. Section 8.4 describes the interactions between the reservation and distribution systems in the travel industry, and Section 8.5 describes how bookings and cancellations are managed within those systems. Understanding this material is a prerequisite for understanding the ideas behind tactical revenue management. Section 8.6 introduces the three elements of tactical revenue management: capacity control, network management, and overbooking, which are described in detail in Chapters 9, 10, and 11, respectively. Section 8.7 describes various metrics that are used to measure the effectiveness of revenue management. Section 8.8 discusses the role of incremental cost and ancillary profit in revenue in various industries, and Section 8.9 reviews the current status of revenue management adoption in various industries.

Since this chapter is introductory, it may feel a bit lacking in algorithms and computations. This is made up for in the following chapters, each of which addresses a specific aspect of tactical revenue management in detail.

8.1 HISTORY

Prior to 1978, the airline industry in the United States was heavily regulated. Both schedules and fares were tightly controlled by the Civil Aeronautics Board (CAB). Fares were held sufficiently high to guarantee airlines a reasonable return on their extensive capital investments in airplanes. In 1978, Congress passed the Airline Deregulation Act. The act specified that the industry would be deregulated over four years, with complete elimination of restrictions on domestic routes and new service by December 31, 1981, and the removal of all fare regulation by January 1, 1983. The result was a shock from which it took many years for the airlines to recover.¹

One of the reasons Congress passed the Airline Deregulation Act was to inspire and encourage creative new entrants into the passenger airline business. One of the first new airlines to arise after deregulation was PeopleExpress. PeopleExpress was not unionized, and it offered a bare-bones service, with passengers paying extra for baggage handling and onboard meals. As a result, its cost structure was significantly lower than those of American and the other major airlines, such as United and Delta. By offering fares up to 70% below the majors, PeopleExpress filled its planes with passengers from a previously untapped market segment: price-sensitive students and middle-class leisure travelers who were induced to travel by the existence of fares far lower than anything the industry had seen

before. PeopleExpress built its business initially by entering underserved markets, where its competition was primarily bus or car travel. PeopleExpress experienced four years of phenomenal growth and, in 1984, began service on Newark–Chicago and New Orleans–Los Angeles routes—key markets for American Airlines.

The choice American Airlines faced was stark. If it matched People's low fares, it might retain its customer base but would not be able to cover its costs. If it did not match People's fares, most of its customer base would be siphoned off by the low-price competitors. And in the newly deregulated environment, nothing could keep a successful PeopleExpress from moving in on all of American's core markets. American Airlines seemed doomed—its only choice appeared to be between a slow death and a rapid one. However, Robert Crandall, CEO of American Airlines, formulated a counterattack: American would compete with PeopleExpress on low fares and simultaneously sell some of its seats at a higher fare as well.

In January 1985, American Airlines announced its Ultimate Super Saver Fares program. Ultimate Super Saver Fares matched PeopleExpress prices, with two key differences:

- For a passenger to qualify for an Ultimate Super Saver discount fare on American, she would need to book at least two weeks before departure and stay at her destination over a Saturday night. Passengers not meeting this restriction would be charged a higher fare. In contrast, PeopleExpress put no restrictions on its discount fares—every passenger paid a low fare.
- American restricted the number of discount seats sold on each flight to save seats for full-fare passengers who would be booking within the last two weeks prior to departure. PeopleExpress allowed every seat to be sold at a low fare.

American's two-pronged approach was carefully thought out and turned out to be the perfect counter to People's challenge. The booking restrictions ensured that the vast majority of the discount passengers buying American were leisure passengers who were able to book early and who were more price sensitive. The later-booking passengers who paid full fare were primarily business travelers who were less price sensitive but needed seat availability at the last minute. Both groups of passengers preferred American's superior service to People's bare-bones approach. In effect, American Airlines had segmented the market between leisure and business travelers and used differentiated pricing to attack a competitor.

The impact of American's actions was dramatic. American announced the new fares and new structure in January 1985. By March of that year, PeopleExpress was struggling, and by August it was on the verge of bankruptcy. In September, Texas Air bought PeopleExpress for less than 10% of the market value it had enjoyed a year earlier. PeopleExpress's flamboyant CEO, Donald Burr, put the blame for his airline's demise squarely on American's superior yield management capability: "We had great people, tremendous value, terrific growth. We did a lot of things right. But we didn't get our hands around the yield management and automation issues" (Cross 1997, 129).

This was the origin of revenue management (or *yield management* as it was called at the time). American's success prodded other major airlines in the United States to develop revenue management capabilities of their own. Throughout the remainder of the 1980s

and well into the 1990s, carriers such as United, Delta, and Continental invested millions of dollars in implementing computerized revenue management systems and establishing revenue management organizations. As world aviation markets were increasingly deregulated, carriers in Europe and Asia also began to adopt revenue management. Hotels and rental car companies followed the airlines in adopting revenue management. Marriott was a pioneer in hotel revenue management, and Hertz and National were pioneers in rental car revenue management. Vendors such as PROS and Talus Solutions developed and sold commercial revenue management software packages. The next wave of adopters included cruise lines, passenger trains, and various modes of freight transportation. Development and investment in revenue management continues in many of these and other industries today.

8.2 LEVELS OF REVENUE MANAGEMENT

Successful revenue management requires consistent execution at three levels, as shown in Table 8.1. *Revenue management strategy* is the identification of customer segments and the establishment of products and prices targeted at those segments. Once products and prices have been established, *revenue management tactics* require setting and updating limits on how much of a particular product can be sold at a particular fare to each segment for some period of time—say, a day or a week. *Booking control* is the moment-to-moment determination of which booking requests should be accepted, which should be rejected, and how capacity for different products should be updated in the face of bookings and cancellations. For airlines, hotels, rental car companies, and cruise lines, booking control is performed by the reservation system. Tactical revenue management is the brains of the process. It is where future demand is forecast, optimization algorithms are run, and booking limits are set and updated.

We first consider the strategy of revenue management and how it applies in different industries. We then turn to a somewhat detailed discussion of booking control—the real-time face of revenue management. With this background in place, the next three chapters deal with elements of tactical revenue management—capacity control, network management, and overbooking.

8.3 REVENUE MANAGEMENT STRATEGY

Revenue management strategy consists of the identification of customer segments and the establishment of products targeted at those segments. A fundamental element of revenue management strategy at many hotels, rental car companies, and airlines is the distinction between leisure customers and business customers first recognized by American Airlines.

TABLE 8.1
The three levels of revenue management decisions

Level	Description	Frequency
Strategic	Segment market and differentiate prices	Quarterly or annually
Tactical	Calculate and update booking limits	Daily or weekly
Booking control	Determine which bookings to accept and which to reject	Real time

TABLE 8.2
Characteristics of leisure and business airline-passenger segments

Leisure	Business
Highly price sensitive	Less price sensitive
Book earlier	Book later
More flexible on departure and arrival times	Less flexible
More accepting of restrictions such as Saturday night stayovers	Less accepting of restrictions

TABLE 8.3
Typical airline market product segmentation

	Business-oriented product			Leisure-oriented product	
	Unrestricted business	Corporate discount	Discount business	Regular leisure	Web-only discount
Refundable?	Fully	Partially	Partially	Partially	Nonrefundable
Changeable?	Fully	Fully	With fee	With fee	Not changeable
Restrictions	None	Booked direct through corporate	Booked at least one week before departure	Booked at least two weeks before departure	Booked at least two weeks before departure, web only
Inventory	Last seat	Somewhat restricted	Restricted	Restricted	Very limited
Price	Highest	High	Moderate	Low	Very low

The different characteristics of leisure and business customers are shown in Table 8.2. The airlines used these characteristics to segment their market and create virtual products oriented toward the different segments. A typical approach is shown in Table 8.3. Here, an airline has identified five customer segments, three business and two leisure, with at least one product targeted toward each segment. The leisure products have various restrictions (early purchase, Saturday night stay) that make them unattractive or unavailable to many business travelers. This is a classic application of the segmentation technique described in Section 6.3.5 by which a seller artificially creates an inferior product to sell at a lower price in order to segment the market.

Product versioning is not the only price differentiation tactic used by the airlines. They also use most of the other tactics described in Section 6.3. Airlines sell products targeted to many segments such as government, senior citizens, groups, tour operators, and cruise lines. International airlines are great believers in regional pricing. They will sell the same tickets for different prices in different countries (even after adjusting for exchange rates) to exploit differences in price sensitivity. Airlines are also channel pricers; ticket prices may vary among online platforms and on their website.

Other revenue management companies followed the airlines in segmenting their market, creating virtual products and establishing a wide range of prices. Hotels and rental car companies establish and manage products and different prices oriented toward various groups: corporate, leisure, and business segments. As a simple example, many hotels in Hawaii maintain *kama'aina* rates for local residents.² These rates are often far lower than the standard rates, and hotel revenue managers make them available during slack periods to fill their properties. Only customers who can prove Hawaiian residency qualify for a

kama'aina rate. This is a classic example of group pricing. By confining availability to locals, hotels ensure that the low kama'aina rates do not cannibalize their high-paying business from the mainland United States and Asia.

Cruise lines do not have late-booking business customers, but they do sell to the *incentive* segment—bulk purchases of cruise capacity by companies to be used as incentives for their employees. For example, a medical supply company might purchase 20 berths on a particular sailing to reward the 20 top North American sales managers. This incentive business is booked earlier and is sold at lower fares than individual cruise tickets. Cruise lines also charge different rates for customers from different cities. Managing availability by city is one way they can maximize contribution from each sailing.³

8.4 THE SYSTEM CONTEXT

American Airlines was able to outmaneuver PeopleExpress in part because it had developed a *computerized reservation system* called SABRE that allowed it to save seats for later-booking business passengers—a capability PeopleExpress lacked. When it was turned on in 1964, SABRE was a technological marvel. Not only did it replace the colored index cards in revolving trays that American and other airlines had used since the 1930s to manage their reservations; it gave American Airlines the ability to distribute its products and fares globally and to receive bookings from anywhere in the world.

Airline distribution has evolved considerably over the past few decades (and continues to evolve), but the distribution systems and reservation systems continue to play an important role. A high-level schematic diagram of airline distribution is shown in Figure 8.1.

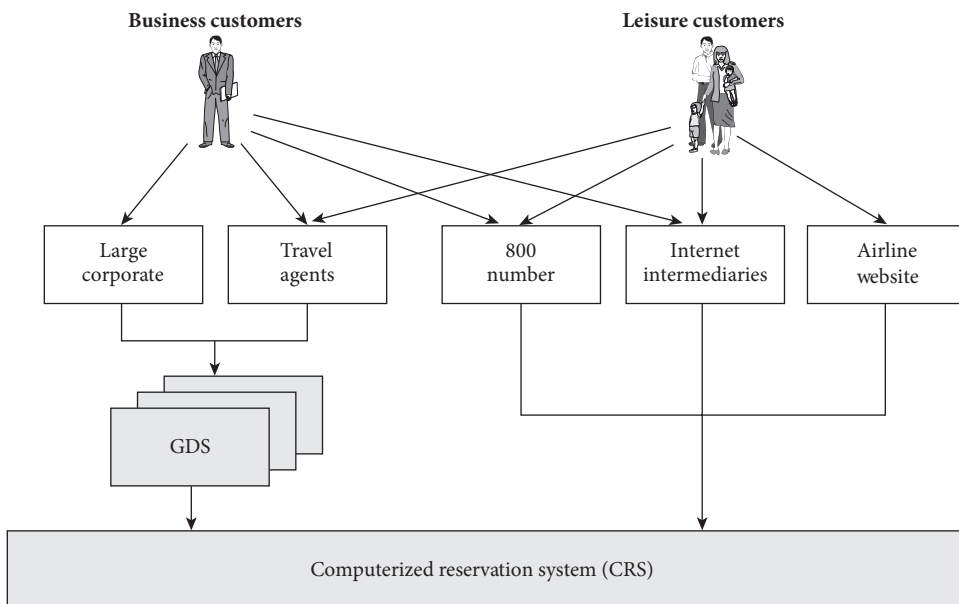


Figure 8.1 Airline distribution channels.

Consumers have a variety of channels for purchasing tickets ranging from online platforms such as Expedia and Orbitz, an airline's own website, travel agencies, and in-house corporate travel management. The global distribution systems transmit price and availability information from the airline to each of these channels and send booking and cancellation information from the channels back to the airlines. Every airline has a reservation system that enables it to keep track of all of the current bookings that it has in the system and their status. As of 2019, four major global distribution systems are in use—Amadeus, Worldspan, Travelport, and China's TravelSky. Most airlines distribute through all four systems because individual travel agencies (including corporate travel) are typically associated with only one of the systems. The same global distribution systems provide similar services for other travel-related industries such as hotels, cruise lines, and rental cars.

This system background is important because airlines did not choose to pursue a revenue management strategy and then design reservation systems to support that strategy. Rather, reservation systems and global distribution systems were developed first and revenue management capabilities were added later. SABRE went operational in 1964—14 years before the airlines were deregulated and 21 years before American first did battle with PeopleExpress. The computerized reservation systems were among the most sophisticated and complex transaction-processing systems of their day. But they were not very flexible. Since they were built on large mainframes using 1960s-vintage software, major modifications were (and are) expensive and time consuming. As a result, the way airlines and other travel companies manage their inventory is still dictated by the capabilities of these systems. And at the heart of these systems are the booking control mechanisms that determine moment by moment which fare classes are available for sale and which are not.

8.5 BOOKING CONTROL

Booking control is the real-time face of revenue management. The primary function of booking control is to determine whether each booking request received should be accepted or rejected.⁴ Secondary functions are to update availabilities for different products when bookings and cancellations occur. As shown in Figure 8.1, booking requests stream into the reservation system from many different channels. Within a very short time (typically less than 200 milliseconds), the reservation system needs to send a message saying whether or not that request can be accepted. Given the short time available, reservation systems rely on very simple, mechanical procedures for determining whether to accept a request.

When a booking request is received—say, for a seat for a future flight or a hotel room for some future dates—the request is assigned to a fare class. This assignment may be based on the time and characteristics of the request (e.g., whether it meets the qualifications for a deep discount), the channel through which the request was received, the market segment of the customer (e.g., group versus individual), or combinations of all of these. The fare classes are typically assigned letters, so we can speak of a B-Class or M-Class booking request.⁵

The reservation system includes a *booking limit* for each fare class on each product. When a booking request is received, the reservation system checks the booking limit for the

associated fare class. If there is sufficient availability, the request will be accepted; if not, the request will be rejected.

Example 8.1

An airline receives a B-Class request for three seats on Flight 137 from Houston to Miami, departing in two weeks. The current B-Class booking limit for this flight is two. Because there is insufficient availability, the request is rejected.

At each time, the booking limits currently in place determine which booking requests will be accepted and which will be rejected. When a new booking is accepted or a previously accepted booking cancels, the reservation system automatically updates the booking limits for that product. This is all done in milliseconds, consistent with the need of the reservation systems to communicate availabilities to customers in real time.

8.5.1 Allotments

An obvious way to manage bookings would be to divide the available capacity into discrete chunks and to allocate each chunk to a fare class. This is known as the *allotments* approach, and the size of the chunk allocated to each fare class is called its allotment. Under this approach, bookings are accepted in a class until the allotment for that class is exhausted.

Example 8.2

A 100-seat aircraft is being managed using allotments: 30 seats have been allotted to deep-discount bookings (B-Class) with a \$125 fare, 45 seats to full-fare coach (M-Class) with a \$200 fare, and 25 seats to business class (Y-Class) with a \$560 fare. Two weeks before departure, 25 B-Class bookings, 45 M-Class bookings, and 10 Y-Class bookings have been accepted. The remaining allotments are 5 seats for B-Class, no seats for M-Class, and 15 seats for Y-Class.

While the allotments approach is easy to understand, it has a major drawback: it does not work very well. In particular, it can result in high-fare customers being rejected while lower-fare customers are still being accepted. In Example 8.2, if the M-Class allotment closes before the B-Class allotment, the airline will be accepting customers paying \$125 while rejecting \$200 bookings. This is a prime revenue management sin—an airline cannot maximize revenue by rejecting high-fare customers to save seats for low-fare customers. For this reason, revenue management companies nest their inventory so that high-fare customers have access to all of the inventory available to lower-fare customers.

8.5.2 Nesting

Nesting was developed to avoid the situation in which high-fare bookings were rejected in favor of low-fare bookings. To describe nesting, we number fare classes so that 1 is the highest (most expensive) fare class and n is the lowest. We define b_i as the booking limit for class i . With nested booking controls, booking limits are always nondecreasing; that is,

$$b_1 \geq b_2 \geq \dots \geq b_n. \quad (8.1)$$

Furthermore, at each time, every fare class has access to all of the inventory available to lower fare classes. This avoids the possibility of accepting low-fare bookings while rejecting high-fare bookings.⁶ The booking limit for the highest class, b_1 , is an upper bound on the total number of bookings that will be accepted. If the airline is not expecting any no-shows or cancellations, then b_1 would equal the capacity of the flight. When there is a possibility that bookings will cancel or not show, then it is usually optimal to *overbook*—that is, to accept more bookings than available capacity, in which b_1 , and possibly the booking limits for other classes, might be greater than the capacity of the flight. Chapter 11 addresses how to calculate overbooking limits.

We can also describe nesting in terms of *protection levels*. The protection level for class i is the total number of seats available to class i and all higher classes. Let y_j be the protection level for class $j = 1, 2, 3, \dots, n-1$. The relationship among booking limits and protection levels with nested booking controls is illustrated in Figure 8.2. From this figure it should be evident that

$$y_j = b_1 - b_{j+1} \quad \text{for } j = 1, 2, 3, \dots, n-1 \quad (8.2)$$

and $y_n = b_1$; that is, the protection level for the lowest fare class is equal to the total booking limit for the flight. Combining Equation 8.2 with condition 8.1, we can see that protection levels decrease for higher fare classes; that is,

$$0 \leq y_1 \leq y_2 \leq \dots \leq y_{n-1},$$

as can be seen in Figure 8.2.

8.5.3 Dynamic Nested Booking Control

What happens when we accept a booking when limits are nested? Define the *availability* of a fare class as the booking limit for that class minus the booking limit of the next lower class. The availability of the lowest fare class is its booking limit. Thus, if a flight has two fare classes with booking limits $(b_1, b_2) = (100, 30)$, the availability for class 2 is 30 and for class 1 is 70. The standard practice in the airline industry is to decrement the availability of

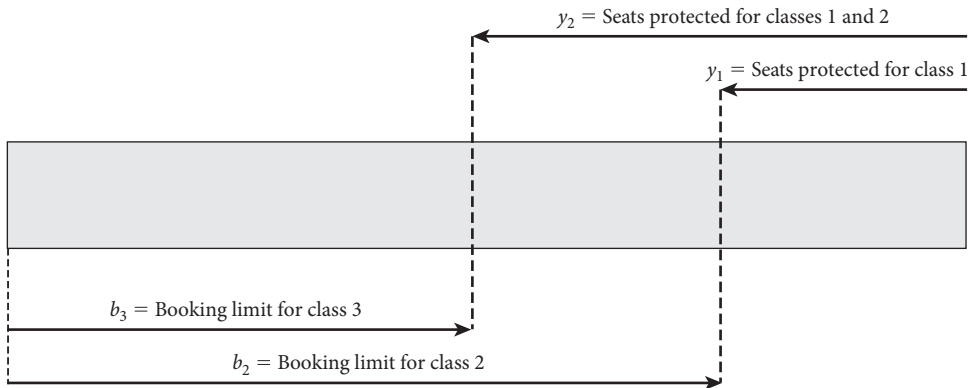


Figure 8.2 Nested booking limits and protection levels.

the booking class from which the booking occurred, from which the booking limits can be reconstructed. Thus, if the airline with booking limits (100, 30) accepted a five-seat booking in fare class 2, its new booking limits would be (95, 25). If it accepted a five-seat booking in fare class 1, its new booking limits would be (95, 30).

This approach is called *standard nesting* and it is the most widely used among the airlines. However, an alternative is to decrement the availability of lower fare classes as well when a higher-fare booking limit is accepted. In this case, accepting a five-seat booking in fare class 1 would result in new booking limits of (95, 25). This approach is called *theft nesting* and it is sometimes found in practice.

Example 8.3

A flight has a total booking limit of 100 and five fare classes with booking limits of $(b_1, b_2, b_3, b_4, b_5) = (100, 73, 12, 4, 0)$. Note that this flight is currently not accepting any bookings from class 5. Any booking class with a limit of 0 is said to be *closed*. We can derive the corresponding protection levels using Equation 8.2: $(y_1, y_2, y_3, y_4, y_5) = (27, 88, 96, 100, 100)$. The availabilities for the fare classes are $(a_1, a_2, a_3, a_4, a_5) = (27, 61, 8, 4, 0)$. Assume that a booking request arrives for 3 seats in class 3. Because the booking limit in class 3 is 12, this request will be accepted. Under standard nesting, only the availability in class 3 will be changed, so the new availabilities will be $(a_1, a_2, a_3, a_4, a_5) = (27, 61, 5, 4, 0)$, which corresponds to booking limits of $(b_1, b_2, b_3, b_4, b_5) = (97, 70, 9, 4, 0)$. Under theft nesting, the total seat capacity drops from 100 to 97, and the booking limits are adjusted to $(b_1, b_2, b_3, b_4, b_5) = (97, 70, 9, 1, 0)$, which corresponds to protection levels of $(y_1, y_2, y_3, y_4, y_5) = (27, 88, 96, 97, 97)$.

Note that the protection levels for the higher classes are preserved when a booking is accepted for a lower class. Even after the request for three seats was accepted, 27 seats are still protected for class 1 demand under both theft nesting and standard nesting. Although both theft nesting and standard nesting are in use, most systems use standard nesting. If bookings strictly occur in fare class order—that is, booking class n books before booking class $n - 1$ —then theft nesting and standard nesting will give the same results. Some evidence indicates that when bookings do not occur in strict fare order, standard nesting tends to perform better than theft nesting.⁷ Table 8.4 shows the dynamics of bookings accepted and rejected under standard nesting and theft nesting. Note that theft nesting rejects more lower-fare bookings than standard nesting in this case, meaning that more capacity remains for later, hopefully higher-fare bookings.

8.5.4 Managing Cancellations*

Since almost 30% of airline bookings are canceled before departure, the booking management process at a passenger airline needs to update booking limits when cancellations occur. The obvious way to do this would be to treat a cancellation as an increase in capacity; that is, when a single-seat booking cancels, increase all the positive booking limits by 1. This is, in fact, the way that many reservation systems treat cancellations.

TABLE 8.4
Dynamics of booking limits and protection levels under standard nesting and theft nesting

	Booking limits					Protection levels					Request	Action
	1	2	3	4	5	1	2	3	4	5		
Under standard nesting												
1	100	73	12	4	0	27	88	96	100	100	2 seats in class 5	Reject
2	100	73	12	4	0	27	88	96	100	100	5 seats in class 2	Accept
3	95	68	12	4	0	27	83	91	95	95	1 seat in class 2	Accept
4	94	67	12	4	0	27	82	90	94	94	1 seat in class 4	Accept
5	93	66	11	3	0	27	82	90	93	93	3 seats in class 3	Accept
6	90	63	8	3	0	27	82	87	90	90	4 seats in class 3	Accept
7	86	59	4	3	0	27	82	83	86	86	2 seats in class 1	Accept
8	84	57	2	3	0	27	82	81	84	84	4 seats in class 3	Reject
9	84	57	2	3	0	27	82	81	84	84	1 seat in class 3	Accept
10	83	56	1	3	0	27	82	80	83	83	2 seats in class 2	Accept
11	81	54	1	3	0	27	80	78	81	81	2 seats in class 2	Accept
Under theft nesting												
1	100	73	12	4	0	27	88	96	100	100	2 seats in class 5	Reject
2	100	73	12	4	0	27	88	96	100	100	5 seats in class 2	Accept
3	95	68	7	0	0	27	88	95	95	95	1 seat in class 2	Accept
4	94	67	6	0	0	27	88	94	94	94	1 seat in class 4	Accept
5	94	67	6	0	0	27	88	94	94	94	3 seats in class 3	Accept
6	91	64	3	0	0	27	88	91	91	91	4 seats in class 3	Reject
7	91	64	3	0	0	27	88	91	91	91	2 seats in class 1	Accept
8	89	62	1	0	0	27	88	89	89	89	4 seats in class 3	Reject
9	89	62	1	0	0	27	88	89	89	89	1 seat in class 3	Accept
10	88	61	0	0	0	27	88	88	88	88	2 seats in class 2	Accept
11	86	59	0	0	0	27	86	86	86	86	2 seats in class 2	Accept

There is a problem with this approach, however. Say we have a booking limit of three seats in the lowest booking class and we accept a booking for four seats in some higher class. Then, the lowest class would close. Now, assume that 10 seconds later the booking for four seats is canceled (perhaps the original request was an error). Under the approach described earlier, the four seats would be added back to the open booking classes and the lowest booking class would remain closed. This seems wrong. After all, we have the same available capacity and the same forecasts for future demand that we had 10 seconds before—why should our booking limits change?

Another way of viewing the issue is that, under the process described so far, the closure of a booking class is irreversible, at least until the revenue management calculates new booking limits.⁸ For this reason, this procedure is called an *irreversible* process. Recognizing the shortcomings of an irreversible process, some reservation systems enable a *reversible* process. The trick in creating a reversible process is to allow the booking limits to go below zero. In a reversible process, new bookings are subtracted from all booking limits and cancellations are added back to all booking limits. As before, a booking request is accepted only if the corresponding booking limit is greater than the number of seats requested. This means that a class with a negative booking limit is closed. With a reversible process, a cancellation can result in availability going from zero (or less than zero) to positive, opening a previously closed class.

TABLE 8.5
Dynamics of booking limits and protection levels

Event	Irreversible approach					Reversible approach				
	1	2	3	4	Action	1	2	3	4	Action
1 Request 2 seats in class 3	73	12	4	0	Accept	73	12	4	0	Accept
2 Request 2 seats in class 3	71	10	2	0	Accept	71	10	2	-2	Accept
3 2 bookings cancel	69	8	0	0		69	8	0	-4	
4 Request 5 seats in class 3	71	10	0	0	Reject	71	10	2	-2	Reject
5 Request 3 seats in class 2	71	10	0	0	Accept	71	10	2	-2	Accept
6 Request 2 seats in class 3	68	7	0	0	Reject	68	7	-1	-5	Reject
7 3 bookings cancel	68	7	0	0		68	7	-1	-5	
8 Request 2 seats in class 3	71	10	0	0	Reject	71	10	2	-2	Accept
9 Request 2 seats in class 3	71	10	0	0	Reject	69	8	0	-4	Reject
10 2 bookings cancel	71	10	0	0		69	8	0	-4	
11 Request 1 seat in class 2	73	12	0	0	Accept	71	10	2	-2	Accept
12 Request 1 seat in class 3	72	11	0	0	Reject	70	9	1	-3	Accept
Final state	72	11	0	0		69	8	0	-4	

These two approaches to treating cancellations are illustrated in Table 8.5, which shows a sequence of booking requests and cancellations and the corresponding evolution of the booking limits for four booking classes under both approaches. The initial set of booking limits is (73, 12, 4, 0). The second entry in each row of Table 8.5 is either a booking request or a cancellation. The other entries in each row are the booking limits just prior to the event under both approaches and the action (if any) that the airline would take. Since an airline cannot reject a cancellation, no actions are associated with cancellations.

From Table 8.5 we can see that the reversible approach sometimes opens previously closed booking classes. This means that it often accepts more low-fare bookings than the irreversible approach. While this may seem appealing, the other side of the coin is that the irreversible approach saves more seats for later high-fare bookings. Generally, it is felt that the reversible approach provides higher revenue, since a booking request that immediately cancels will not change the booking limits. (For an example of this, see event numbers 2 and 3 in Table 8.5 and the resulting changes in booking limits under the two approaches.) However, which approach is used is often determined by the capabilities of its reservation system rather than by considerations of relative benefit.⁹

8.6 TACTICAL REVENUE MANAGEMENT

The job of tactical revenue management is to calculate and periodically update booking limits. These booking limits are transmitted to the reservation system, which then uses them to determine which booking requests to accept and which to reject based on the logic described in Section 8.5.

To introduce the tactical revenue management problem in its most general form, it is useful to define *resources*, *products*, and *fare classes*.

- *Resources* are units of capacity managed by a supplier. Examples of resources are a flight departure, a hotel room night, and a rental car day. Each resource is typically

TABLE 8.6
Revenue management in four different industries

	Passenger airlines	Hotels	Rental car companies	Cruise lines
Resource	Flight leg	Property	Location	Sailing
Resource unit	Seat	Room night	Rental day	Berth
Resource types	Service classes	Room types	Car types	Berth classes
Number of types	1–3	1 to many	3–10+	5–10
Products	Itineraries	Arrival date and length of stay	Arrival date and length of rental	Cruise
Products per resource	Many for hub-and-spoke airline	Many	Many	Few

constrained—the flight departure has a limited number of seats, the hotel only has a limited number of rooms it can sell, and a rental car company has only so many cars it can rent out.

- *Products* are what customers seek to purchase. A product may require use of one or more resources. A seat on Flight 130 from St. Louis to Cleveland on Monday, June 30, is a product that uses only a single resource. A two-night stay at the Sheraton Cleveland Airport for a customer arriving on March 19 and departing on March 21 is a product that uses two resources: a room night on March 19 and a room night on March 20.
- One or more *fare classes* are associated with each product. Each fare class is a combination of a price and a set of restrictions on who can purchase the product and when. Different fare classes can be used to establish different virtual products, for group pricing, for regional pricing, or for combinations of all of these.

Table 8.6 shows how the resource and product approach applies in four different revenue management industries. A *resource unit* is the smallest unit of capacity that can be sold: a single seat for an airline, a room night for a hotel. A company may sell several different *types* of resource: airlines sell coach, business, and first-class seats; rental car companies offer subcompact, compact, midsize, and luxury cars; and a hotel may sell five or six different room types. While resources are the assets a company needs to manage, products are what customers actually want to *buy*. An airline's products are direct and connecting flight combinations. A hotel's products are combinations of arrival date and length of stay. A customer reserving a two-night stay starting next Wednesday is buying a different product than one reserving a two-night stay starting next Thursday.

The resource, product, and fare class terminology enables us to formulate the tactical revenue management problem in a very general manner. A seller faces a tactical revenue management problem when he:

- Manages a set of resources with fixed and perishable capacity
- Controls a portfolio of products consisting of combinations of one or more of the resources

- Can set and update the price of each product. In classical revenue management, prices are set and updated by opening and closing discrete fare classes. In other applications, prices can be set and updated directly.

The tactical revenue management problem discussed in this and the next three chapters is how a seller should choose which fare classes should be open for sale and which should be closed for each product at each moment to maximize expected total net contribution.

The fact that revenue management is based on opening and closing fare classes does not make a big difference from the customer's point of view—customers just see the fare changing over time. The focus on opening and closing fare classes is partly due to the design of the reservation systems. It is also based on the competitive dynamics of the airline market. Specifically, most airlines believe they need to match the advertised, or headline, fares offered by their competition in key markets to drive demand for bookings. However, as bookings arrive, revenue management enables them to shape demand to the limited capacity of each flight. Revenue management supplements rather than replaces pricing.

8.6.1 Components of Tactical Revenue Management

As we have seen, the point of tactical revenue management is to determine which fare classes should be open and which should be closed for all products to maximize return from a fixed set of resources. Tactical revenue management can be decomposed into three constituent decisions.

- *Capacity allocation.* How many customers from different fare classes should be allowed to book?
- *Network management.* How should bookings be managed across a network of resources, such as an airline hub-and-spoke system or multiple-night hotel stays?
- *Overbooking.* How many *total* bookings should be accepted for a product in the face of uncertain future no-shows and cancellations?

A general view of the relative importance of each of these three decisions in different industries is shown in Table 8.7.

Capacity allocation is important whenever a company sells the same unit of constrained capacity or inventory at two or more different prices. In this case, the company needs to determine how many units to sell at the lower price(s) and how many to reserve for sale at a higher price. The major passenger airlines need to determine how many seats to sell to early-booking groups and discount customers and how many to reserve for later-booking business customers. Rental cars and hotels face similar challenges. Sporting teams need to determine how many tickets to sell as season tickets or on discounts or promotions and how many to hold to sell at full price. Air freight carriers have agreements with different customers that specify different rates—the carriers want to make sure that high-paying customers do not find themselves shut out because capacity was entirely reserved by lower-paying customers. On the other hand, resort hotels, discount carriers such as

TABLE 8.7
Relative importance of revenue management problems among industries

Industry	Capacity allocation	Network management	Overbooking
Passenger airlines			
Hub-and-spoke	****	***	****
Point-to-point	**	N/A	****
Hotels			
Business	***	****	***
Resort	**	**	*
Rental cars	***	****	***
Cruise lines	****	*	*
Passenger rail	**	****	*
Air freight	***	**	***
Event ticketing	****	N/A	N/A

* = of little importance, ** = can be important in some cases, *** = usually important, **** = of great importance, N/A = not applicable

Southwest, and passenger trains may maintain only a small number of prices for the same inventory. For these companies, capacity allocation tends to be less important than other aspects of revenue management.

Network management is important for companies that sell products consisting of combinations of resources. It is the single most important component of revenue management for passenger railways and for rental cars and business hotels, where managing length of stay usually has more impact on total revenue than managing rate classes. It is important for airlines that operate hub-and-spoke systems but not as important as capacity allocation or overbooking. It is of little or no importance in industries in which each product sold uses only a single resource, such as cruise lines, point-to-point airlines, sporting events, and theater.

Overbooking is important whenever bookings are allowed to cancel or not show with little or no penalty. Since airline tickets were historically sold in this way, overbooking has been a major component of airline systems. Overbooking has become somewhat less important as the fraction of nonrefundable tickets has increased, but it is still an important component of revenue management at most airlines, as it is at business hotels and rental car companies. On the other hand, most resort hotels and cruise lines do not overbook because of the high perceived cost of denying service to a customer who may have planned her entire vacation (or even honeymoon) around staying at a particular resort or departing on a particular cruise. Sporting events and theaters generally have nonrefundable tickets and usually do not overbook.

Over the course of the next three chapters, we examine each of these three constituent problems independently, but in fact they are closely entwined and, for maximum profitability, they need to be solved together.

8.6.2 Revenue Management Systems

The primary job of a revenue management system is to calculate and update the booking limits within the reservation system. Typically, a revenue management system is separate from the reservation system but linked to it, as shown in Figure 8.3. The revenue management system receives a feed of bookings and cancellations from the reservation system and,

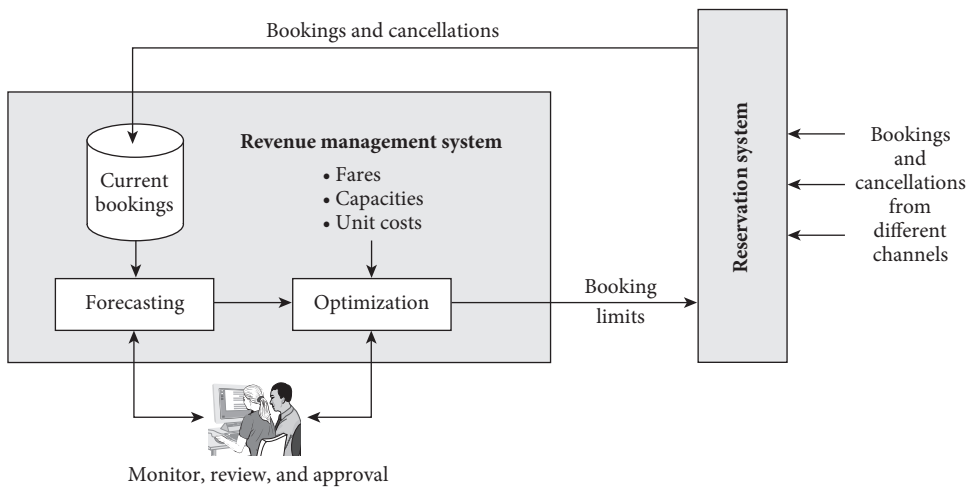


Figure 8.3 Schematic overview of a typical revenue management system.

in turn, calculates booking limits that are transmitted periodically to the reservation system. The revenue management system will then send messages to update the booking limits in the global distribution systems.

The revenue management system includes a database with information on current bookings on all flights. It also includes a database incorporating fares for all product–fare class combinations, capacities on all flight legs, and passenger variable costs by product. This information is typically extracted from other computer systems, specifically the pricing system, the scheduling system, and various accounting systems.

The *forecasting* module generates and updates forecasts for all product–fare class combinations for all future dates. Generally, an initial forecast is generated about one year prior to departure—the time when most airlines first allow bookings for a future flight. This forecast is updated periodically as bookings and cancellations are received over time. Typically, a forecast is updated monthly when a flight is six months or more from departure and more frequently as departure approaches. Forecasts are typically updated daily for flights that are within two weeks or so of departure.

The forecasts generated by the revenue management system are *probabilistic*; that is, they predict both a mean and a standard deviation for future demand. Uncertainty plays a major role in the calculation of optimal booking limits.

Example 8.4

A typical forecast generated by a revenue management system would be that the number of expected future B-Class bookings on Flight 47 from Chicago to Grand Rapids, Michigan, departing at 9:30 two weeks from today is 13 passengers with a standard deviation of 9, while the number of expected M-Class bookings is 22 passengers with a standard deviation of 10.

These forecasts are based on current and recent bookings for this flight, historic booking performance for this flight, and general demand trends for all flights. Most systems

also forecast cancellations and no-shows to support overbooking calculations—a topic discussed in Chapter 11. Standard forecasting techniques are used to generate the initial forecast and to update the forecasts as bookings and cancellations are received over time.

The probabilistic forecasts are key inputs used by the optimization module to generate the booking limits. As described in Chapter 9, booking limits are calculated by estimating the economic trade-off between accepting more discount bookings now versus having more capacity available to serve future bookings. Typically, booking limits are automatically recalculated whenever the demand forecasts are updated.

Revenue management systems recommend booking limits for each product–fare class combination. Effective revenue management requires that these recommendations be continually monitored and reviewed. For example, Delta Airlines employs a group of more than 30 revenue management analysts who continually monitor the forecasts and booking limits generated by Delta’s revenue management system. Monitoring and adjusting forecasts are particularly important parts of the process. For example, the revenue management system will not know that the Super Bowl will be held in New Orleans this year. It is the job of the revenue management analyst to adjust the forecasts for January flights in the New Orleans market to reflect the additional demand generated by the Super Bowl.

8.6.3 Updating Booking Limits

Most RM companies set initial booking limits based on resource capacities, the demand forecasts for each booking class, and the economic trade-offs among the classes. Most reservation systems begin to accept bookings about a year prior to a flight departure or room rental date. At this time an airline sets initial booking limits for fare classes on those departures. The initial booking limits are loaded into the reservation system and decremented as bookings are accepted. Periodically, demand is reforecast and new booking limits are calculated based on the new forecast of demand and remaining unbooked capacity. This is known as a *reoptimization* or an *update*. Updates can be triggered in three different ways.

- *Periodic updates* occur at scheduled intervals. These intervals are typically long (say, monthly) when the flight is far from departure and booking requests are rare. As the flight approaches departure and the booking pace increases, periodic updates are scheduled more frequently—daily or even more often during the last two weeks prior to departure.
- *Event-driven updates* are triggered by events such as a booking class closing, a change in aircraft, and an unanticipated spike in demand.
- *Requested updates* may be launched at any time by a flight controller or revenue manager based on competitive actions, changes in fares, anticipated changes in future demand, or any other reason.

Each time booking limits are updated for a flight, the new booking limits are loaded into the reservation system and the booking management process resumes immediately starting from the new limits.

8.7 REVENUE MANAGEMENT METRICS

Before deregulation in 1983, airlines could only change their fares after a lengthy regulatory review process. Without direct control over fares, the point of most airline marketing was to fill the planes. The Civil Aeronautics Board always set fares sufficiently high that more passengers meant more money and therefore more profit. Furthermore, little or no price differentiation was allowed, so there was no scope to perform revenue management. In this environment, the airlines used *load factor* as their primary performance measure. Load factor is defined as the number of seats sold on a flight divided by the number of available seats. A 150-seat flight that departs with 105 passengers on board has experienced a $105/150 = 70\%$ load factor. Load factor was a sensible performance measure in the regulated world. Prices were regulated at a level such that if an airline filled about 75% of its seats on each flight—the *break-even load factor*—it would meet its total costs. Each passenger above the break-even load factor represented additional profit. A marketing program was considered successful if it led to increased load factors.

Following deregulation, finding appropriate performance metrics for a flight became more complicated. In a multifare world, load factor could no longer pass muster as the most important performance metric for a flight. The reason is easy to see—in response to the rise of discount carriers, the major airlines had introduced deep discounts that induced vast amounts of leisure demand on many flights. It would be easy to maximize load factor for these flights—simply allow deep-discount customers to book all the seats. But this would be counter to the entire purpose of tactical revenue management, which is to maximize revenue by reserving seats for late-booking business passengers. Since load factor ignores revenue, it is the wrong metric for a revenue management world. In fact, it is worse than that: policies designed to maximize load factor would lead to planes filled with deep-discount passengers and massive denial of availability to business passengers—the airlines' highest-paying customers.

To get a better view of the revenue picture, airlines began to look at *yield* in addition to load factor. Yield is simply revenue per passenger mile. Like load factor, it can be measured at any level from flight to market to entire airline. It was sometimes stated that the goal of managing bookings should be to increase yield—hence the term *yield management*, which was the original name for what is now more commonly called *revenue management*. Unfortunately, yield alone is a very imperfect metric since it ignores flight capacity. The strategy for increasing the yield from a flight is the exact opposite from that for increasing load factor—reject all early-booking demand and accept only late-booking, high-yield passengers. This policy would maximize yield but would be disastrous for the airline, since it would result in flights that were empty, except for a few high-paying business passengers.

What the airlines really needed was a metric that combined the capacity focus of load factor with the revenue focus of yield and recognized that the goal of revenue management is to maximize the return from resources. For an airline, these resources are seats, and the airline needs to maximize *revenue from available seats*. Since prices (and costs) are at least somewhat proportional to distance, it helps to normalize by distance and to measure

revenue per available seat mile, or RASM. An airline that flies a 100-seat aircraft on an 1,800-mile flight (about the distance from Chicago to San Francisco) with a total revenue of \$50,000 achieved an RASM of $\$50,000 / (100 \times 1,800) = \0.28 for that flight. Note that RASM is indifferent to how the revenue is distributed among passengers. Fifty passengers paying \$1,000 each (a 50% load factor) would result in the same RASM as 100 passengers paying \$500 each (a 100% load factor). This is the right focus from the revenue management point of view—a policy that has resulted in \$51,000 from the flight is more successful than one that resulted in \$50,000, even if the first policy resulted in a lower load factor.¹⁰ However, load factor can still be useful: if a flight is achieving a high RASM with a low load factor, it would be good to look for ways to find more customers for the flight even at low fares, as long as existing sales are not cannibalized. And if the load factor is extremely high for a flight, the airline might experiment with raising some of the fares to see if it can continue to fill the flight at higher average fares. If successful, these tactics would lead to increased RASM.

Revenue per available seat mile can also be expressed in terms of net yield and load factor:

$$\text{RASM} = \text{net yield} \times \text{load factor}$$

Revenue per available seat mile is the leading metric currently used by airlines to measure effectiveness of pricing and revenue management. Similar approaches are also employed at other RM companies: *revenue per available room night* (often abbreviated *REVPAR*) at hotels, *revenue per available rental day* at rental car companies, *revenue per berth* for cruise lines, and *revenue per available seat hour* (RevPASH) for restaurants. In each case, the metric emphasizes that the goal should be to maximize the revenue—or, more accurately, contribution—throughput per unit of capacity.

The RASM metric enables comparison of revenue performance across markets and within a single market over time. Speaking broadly, flights that are achieving high RASM are generating a higher return on the company's assets than those with low RASM. And RASM also provides a useful performance benchmark among different airlines. Those airlines achieving high RASMs are generating more revenue from the utilization of their assets than those airlines with lower RASMs.

While RASM is an important high-level metric for measuring the overall effectiveness of revenue management, it does not address the need to measure the individual effectiveness of the different components of revenue management, such as capacity allocation and overbooking. To do this, airlines typically use a portfolio of metrics in addition to RASM, which is likely to include the following.

- *Voluntary and involuntary denied-boarding rates* measure the fraction of passengers who show up for a flight but are denied boarding either by volunteering (voluntary) or by being chosen (involuntary). The calculation of these rates is described in Section 11.8.
- *Spoilage rates* measure the number of empty seats at a flight departure that could have been filled but, because some booking requests were denied, went empty. This is a measure of the effectiveness of overbooking and is described in Section 11.9.

- *Dilution rates* measure the extent to which early low-fare bookings take seats that could have been sold to future high-fare bookings—an example of cannibalization. Dilution rate is a measure of capacity control effectiveness, and it is described in Section 9.6.
- *Revenue opportunity metrics* measure the effectiveness of capacity control against a hypothetical “perfect seat allocation” alternative, as described in Section 9.6.

In each case, tracking trends is usually more important than the absolute numbers. It is often difficult to say if the revenue opportunity metric or RASM achieved by a particular flight departure is good or bad given the multitude of market and competitive factors that can influence it. But a consistent downward trend in revenue opportunity metrics or RASM usually signals a situation that requires attention.

Revenue management metrics are still evolving. By using a portfolio of metrics, a revenue management company can get a good view of its overall performance in generating revenue from its markets relative to competition and how that performance is changing over time. It can evaluate the performance of its revenue management program and determine areas for improvement. It can evaluate individual revenue managers by measuring the number of times their interventions improved overall performance versus the times when intervention degraded performance. However, a considerable amount of work remains to be done on developing improved revenue management metrics. Areas of current research include developing metrics to evaluate the performance of network management and developing metrics that separate the effects of pricing decisions from those of revenue management decisions.

8.8 INCREMENTAL COSTS AND ANCILLARY REVENUE IN REVENUE MANAGEMENT

Chapter 5 notes that the goal of pricing optimization is usually to maximize expected contribution, incorporating both the incremental cost and expected ancillary revenue associated with a customer commitment. The same is true of revenue management—decisions on which bookings to accept and reject should be made on the basis of expected contribution. Revenue management was developed in the passenger airlines industry at a time when the incremental cost of carrying a passenger was extremely low relative to the fares. The earliest generation of airline revenue management systems ignored incremental costs (or assumed they were zero) and focused on maximizing expected revenue—hence the name revenue management. As a result, much of the revenue management literature is couched in terms of maximizing revenue instead of maximizing contribution.

For airlines, this can be a reasonable assumption—while it depends on the length of the flight and other factors, the incremental cost of an additional passenger is on the order of \$10 to \$20 in the United States, which is typically (but not always) much lower than the fare. Nonetheless, incremental costs can be an important consideration when fares are low, and they can be quite important in other revenue management industries. Table 8.8 shows some of the most important elements of incremental cost and a subjective estimate of the

TABLE 8.8
Incremental cost elements in some revenue management industries

Industry	Incremental cost elements	Relative importance
Passenger airlines	Commissions, incremental fuel, meal costs, passenger fees	Low
Hotels	Commissions, room cleaning, wear and tear, amenities, heating and electricity, check-in and check-out	Low
Rental cars	Commissions, check-in and check-out, wear and tear	Moderate
Cruise lines	Commissions, food, cleaning costs	High
Container shipping	Repositioning, handling, incremental fuel	High
Sporting events and theater	Commissions	Very low

TABLE 8.9
Ancillary products and services in some revenue management industries

Industry	Example ancillary products and services	Importance
Passenger airlines	Duty-free purchases, meals, luggage, leg room	Medium but growing; high for some low-cost carriers
Hotels	Food and beverage, minibar, spa fees	Medium
Rental cars	Insurance, gasoline	Medium
Cruise lines	Gambling, onboard sales, excursion fees	Medium to high
Freight transportation	Sorting, call before delivery, special handling	Medium
Sporting events, theater	Food and beverage, merchandise sales	High
Casino hotels	Gambling	Very high

importance of incremental cost in several revenue management industries. Incremental costs are relatively high—and thus important to consider—in cruise lines and container shipping. Food is a major expense for cruise lines, and the amount of food they purchase for a sailing depends on the number of passengers booked. At the other end of the spectrum, incremental costs are relatively low (and therefore generally unimportant) in theater and sporting events. It generally costs little more to stage a play in front of a full theater versus an empty one—not counting, of course, the psychic cost to the actors of playing before an empty house. In between these two extremes fall hotels, rental car companies, and passenger airlines. In all cases, it is most important to estimate and incorporate incremental costs into revenue management decisions when the incremental cost is high relative to the fare.

Ancillary revenue can also be important in revenue management. Over the past decade or so, the airlines have unbundled and begun to charge separately for services that were previously provided for free, such as luggage and onboard meals. As a result, global airline ancillary revenue grew from \$32 billion to \$59 billion from 2010 to 2016, representing about 9% of airline revenue. Ancillary revenue has become an important part of the business model of some airlines—it generated 45% of the revenue for low-cost Spirit Airlines in 2018.¹¹ In fact, ancillary revenue plays an important role in most revenue management industries: Table 8.9 shows the major sources of ancillary revenue associated with different revenue management industries and their relative importance in profitability. Casino hotels are an interesting extreme case: in many cases, the potential gambling revenue can substantially outweigh the room rate for a customer, which means that it can be profitable

for the casino to offer rooms for very low rates (or even free), with the knowledge that they will make up any lost room revenue in gambling revenue (Kuyumcu 2002).

In Chapters 9, 10, and 11, revenue management problems are often formulated in terms of maximizing revenue rather than maximizing contribution. This is a remnant of the origin of revenue management in the airlines where incremental costs were historically much lower than fares and could be ignored in pricing and booking acceptance decisions. However, for a company to maximize profit, it should adopt revenue management policies that maximize expected contribution defined as fare (or price) plus expected ancillary profit minus incremental cost.

8.9 REVENUE MANAGEMENT IN ACTION

By most measures, revenue management has been a great success. From its origins in the 1980s, revenue management has advanced to the point that it is now *de rigueur* for all major passenger airlines as well as most major hotel chains, rental car companies, and cruise lines. For most major airlines, it makes no sense to ask, “What benefit are you realizing from your revenue management system?” because the alternative of returning to setting and updating booking limits manually is unthinkable. In the airlines, hotels, rental car companies, and a number of related industries, revenue management has gone from a strategic investment that provided a step change in capabilities and profitability to a fundamental capability necessary to compete—the table stakes.

It is a sign of the effectiveness of revenue management that low-cost carriers such as Southwest and EasyJet have invested in developing (or purchasing) automated revenue management systems and have built substantial revenue management departments. While they continue to compete primarily on the basis of lower costs, they have learned that revenue needs to be maximized as well to survive in the fiercely competitive passenger airline industry.

Furthermore, revenue management has expanded well beyond the airlines to other service industries that share the characteristics of constrained and immediately perishable inventory. These industries include spas and health clubs, parking garages, toll roads, self-storage units, theme parks, trucking, air cargo, passenger and freight railways, parking lots, and multifamily rental units (apartments). In each of these industries, there has been academic research into the application of revenue management, one or more companies providing revenue management software, or both.

In any case, the most important development in most revenue management industries has been the rapid growth and ultimate dominance of the internet as a channel for sales. The internet has influenced the development of revenue management and pricing optimization in four important ways:

- The internet has provided unprecedented fare visibility to consumers. Instead of relying on travel agents or 800 numbers, price-conscious shoppers surf the net for bargains on their own time, 24 hours a day, 7 days a week. This increases the pressure for airlines and others to continually manage their prices and availabilities.

- While the internet has been a disaster for the traditional travel agency distribution channels, it has led to the rise of new online intermediaries, such as Expedia, Travelocity, and Priceline. These intermediaries and their various competitors are all striving to be the dominant retailer of travel products on the internet. Naturally, the airlines are leery of the prospect of any intermediary gaining a dominant position, so a consortium of US airlines created Orbitz to be their own online retailer. For the foreseeable future, airlines will need to manage pricing and availability through a variety of online retailers as well as through their own websites and the traditional channels.
- The internet has also created new opportunities for the airlines to create and sell more “inferior” products with fewer features included. For example, Priceline allows customers to bid for travel without knowing the exact departure time or airline they are purchasing. Priceline has explicitly marketed this as an inferior product that the airlines can safely sell at a high discount without cannibalizing their mainstream products. Other airlines and hotels are experimenting with selling distressed inventory (i.e., empty capacity close to departure) at deep discounts. The need to manage these additional discount products adds further complexity to revenue management.
- Revenue management was initially shaped by the design of the global distribution systems the airlines developed in the 1960s. These systems maintain a relatively small number of fare classes for each product, and the focus of revenue management has been on which of these classes to open and close at different times. Travel agents were trained in the parlance of booking classes and product restrictions that the airlines had established. But the internet is much more suited to real-time pricing than to availability management. Online shoppers do not care how many Y-Class or M-Class seats are available; they want to know how to get from point A to point B as cheaply and conveniently as possible. While many industries—notably online retail—are able to adjust prices in a fully dynamic fashion, many revenue management industries (hotels, airlines, rental cars) are still stuck in a world where the need to manage booking classes limits their flexibility to perform fully dynamic pricing.

In any case, revenue management is here to stay. Until the flexible airplane and hotel are invented, airlines and hotels will need to maximize net contribution from fixed and perishable capacity. The capability to determine who gets to purchase at what fares will be an important driver of market success for the foreseeable future.

8.10 SUMMARY

- *Revenue management* is the name given to the ways by which a number of industries maximize expected contribution from their constrained resources. It is applicable to a seller who meets the following conditions:
 - The seller is selling a fixed stock of perishable capacity.

- The seller offers a set of fare classes, each of which has a fixed price (at least in the short run).
- The seller can change the availability of fare classes over time.
- Customers book capacity prior to usage.
- High-fare customers book later than low-fare customers.

Revenue management is used in its purest form by passenger airlines, hotels, rental car companies, cruise lines, passenger trains, apartment rental managers, and various forms of freight companies. Many of the core concepts also apply in other industries.

- Revenue management needs to be implemented at three levels. At the strategic level, it requires identifying customer segments and creating virtual products and other ways of differentiating prices. At the booking-control level, it requires determining in real time whether or not booking requests should be accepted or rejected. At an intermediate level, tactical revenue management periodically recalculates and updates the booking limits used for booking controls.
 - The primary revenue management segmentation at the passenger airlines is between early-booking, price-sensitive leisure travelers and later-booking, less price-sensitive business customers. The airlines, hotels, and rental car companies use time-based product differentiation to create products oriented toward each of these segments. In addition, revenue management companies typically have products tailored to other segments, such as groups and large corporations. Finally, they may also use channel pricing and regional pricing to further segment the market. This can lead to pricing structures of great complexity.
 - In general, a revenue management company controls a set of constrained resources that can be combined to create products. Each product has a range of discrete fare classes with different associated fares. These fare classes reflect the price differentiation tactics being used by the seller and the set of products and fares the seller has established. Tactical revenue management requires updating booking limits on these fare classes over time.
 - Tactical revenue management consists of three interrelated problems:
 - *Capacity allocation*: How should fare class booking limits be set for a single-resource product?
 - *Network management*: How should bookings for multi-resource products be controlled?
 - *Overbooking*: How many total bookings should be allowed for a resource?
- These three problems are not independent. However, they have different levels of importance within different revenue management industries.
- The main job of a revenue management system is to set initial booking limits and to perform the periodic reoptimization of the booking limits. A revenue management system includes a forecasting module that calculates probabilistic forecasts of future

demand and an optimization module that uses those forecasts along with other information to determine the optimal booking limits.

- Early programs for revenue management aimed to maximize expected revenue. In an environment where incremental costs are low relative to prices, this is a reasonable assumption. However, as prices have fallen and revenue management has been adopted by new industries, it has become increasingly important to estimate incremental costs and include them in the objective function. Further, ancillary products play an important role in many revenue management industries, and their expected contribution needs to be incorporated into revenue management decisions.
- The best measure of overall revenue management performance is contribution per available unit of resource. The metric most typically used at the airlines is RASM—revenue per available seat mile. Equivalent metrics exist in other revenue management industries. For example, REVPAR—revenue per available room—is the preferred metric for revenue management performance in the hotel industry.
- Revenue management has been a highly effective tactic at the major airlines since the mid-1980s. However, the major airlines are increasingly finding themselves challenged by low-cost carriers. While the major airlines have been able to maintain higher revenues per available seat mile than the low-cost carriers, this revenue advantage has not been able to overcome the cost advantage of the low-cost carriers. It is likely that, at least in the airlines, revenue management will increasingly evolve toward dynamic pricing, with less emphasis on capacity allocation.

The next three chapters address each of the three tactical revenue management problems in more detail. Chapter 9 deals with capacity allocation, Chapter 10 with network management, and Chapter 11 with overbooking.

8.11 FURTHER READING

Robert Cross's book *Revenue Management: Hard-Core Tactics for Market Domination* (1997) includes the story of PeopleExpress versus American Airlines, as well as other examples of revenue management adoption in other industries. Boyd and Bilegan 2003 discusses the role of distribution systems in the development of airline revenue management. Barnes 2012 gives an account of the history of pricing and revenue management in the airline industry, and Boyd 2007 provides a first-hand account of the early days of revenue management in different industries.

Forecasting is a key element of most revenue management systems, but I do not address the topic in this book. Chapter 9 of Talluri and van Ryzin 2004b describes some of the leading forecasting techniques used to support revenue management.

The travel distribution ecosystem is complex and continues to evolve. Barnes 2012 discusses historical developments.

The following are some references on applications of revenue management in industries other than airlines and hotels:

- Rental cars: Carroll and Grimes 1996; Geraghty and Johnson 1997; Anderson, Davison, and Rasmussen 2004
- Casino hotels: Kuyumcu 2002
- Cruise lines: Lieberman 2012
- Golf courses: Kimes 2000; Kimes and Schruben 2002
- Restaurants: Kimes 2004; Kimes, Phillips, and Summa 2012
- Spas: Kimes and Singh 2009
- Theme parks: Heo and Lee 2009
- Toll roads: Göçmen, Phillips, and van Ryzin 2015
- Trains (passenger and freight): Kraft, Srikar, and Phillips 2000; Armstrong and Meissner 2010

8.12 EXERCISE

A restaurant has 25 tables, each of which can seat up to four people. Typically, it can do three seatings per table during the dinner period from 6:00 to 10:00 p.m. The restaurant has a fairly predictable demand pattern, with Fridays and Saturdays the busiest nights and Sundays and Mondays the least busy. The restaurant is always booked to capacity on Fridays and Saturdays and sometimes on Wednesdays and Thursdays but virtually never on Sunday or Monday. It has implemented a revenue management program under which it will sometimes turn down booking requests for one or two people on Fridays and Saturdays if it believes it can fill the restaurant with bookings for larger parties of three and four people. What performance measure should it use to evaluate the effectiveness of its revenue management program?

NOTES

1. Some would argue that the global airline industry has still not fully recovered from the shock of deregulation; see, for example, Berry and Jia 2010.
2. *Kama'aina* is the Hawaiian word for “native.” As another example of this type of segmentation, during the off-season, Disney World offers discounts to Florida residents who can produce a driver’s license as proof of residence.
3. Lieberman 2012 discusses the “air-sea cruise line management process” in detail.
4. Airlines, hotels, and rental cars need to deal with both booking requests and availability requests. The *booking request* is a request for an actual booking. An *availability request* is a check on what fares are available for a particular product. Many availability requests are from customers who are just shopping. For simplicity, the book speaks of booking requests.
5. As a consequence of the fact that fare classes are typically labeled with a letter of the alphabet, some reservation systems limit the number of booking classes per product to a maximum of 26.
6. This is true at least when fare classes have independent demands. If a lower fare class can cannibalize demand from a higher fare class, then it may be optimal to close the higher fare class earlier than the lower one. Section 9.3 discusses this possibility.
7. Discussion of theft nesting and standard nesting can be found in Talluri and van Ryzin 2004b and Gallego and Topaloglu 2019.
8. There is one exception: most reservation systems will reopen the highest class when cancellations occur on a totally booked flight.
9. Once again, different reservation systems use slightly different approaches to update booking limits and protection levels in the face of cancellations, and there is no consensus on the best approach.
10. From an overall pricing and revenue optimization (PRO) point of view, *net contribution per available seat mile* (NCASM) would be an even better metric. For the example flight, assume there is a per-passenger operating cost of \$50. Then, 100 passengers paying \$500 each would result in an operating margin of $\$50,000 - 100 \times \$50 = \$45,000$, for an NCASM of \$0.25, while 50 passengers paying \$1,000 apiece would result in an operating margin of $\$50,000 - 50 \times \50 , for an operating margin of \$47,500 and an NCASM of \$0.26. The higher operating margin per available seat mile (OMASM) accurately reflects the superior operating margin of the second case, which is not reflected in RASM. However, for RM companies where the fares are much higher than the operating costs, RASM is close enough to NCASM to provide an effective metric.
11. Ancillary revenue plays a particularly important role for low-cost airlines and, in many cases, is a core element of their business model.

Capacity allocation is the problem of determining how many seats (or hotel rooms or rental cars) to allow low-fare customers to book when there is the possibility of future high-fare demand. Capacity allocation is particularly important to passenger airlines because many airlines have based their revenue management strategy on segmenting the market between early-booking, low-fare leisure customers and later-booking, higher-fare business customers, as discussed in Chapter 8. Capacity allocation is also an issue for any revenue management company that has the opportunity to restrict early low-price bookings in order to reserve capacity for later higher-price bookings. The techniques described in this chapter are used *mutatis mutandis* in many industries, including rental car companies, hotels, cruise lines, freight transportation, and made-to-order manufacturing, among many others.

We begin by analyzing the two-class model in some detail. We then move on to the multiclass problem, with an emphasis on the widely used expected marginal seat revenue (EMSR) heuristics. Next we consider the dynamic multiclass problem when fare classes are nested. We then relax the independence assumption and examine how booking limits could be set when demand in different fare classes can depend on which classes are currently open. We look at a data-driven approach that does not depend on estimating parameters of a demand distribution. Finally, we see how the performance of a capacity allocation program can be measured and evaluated.

9.1 THE TWO-CLASS PROBLEM

In the two-class capacity allocation problem, a flight (or other product, such as a hotel room night) with fixed capacity serves two classes of customers: discount customers who book early and full-fare customers who book later. Discount customers each pay a fare $p_d > 0$, and full-fare customers each pay a higher fare $p_f > p_d$. In the basic two-class model, we assume that all discount booking requests occur before any full-fare passengers seek to book. We have a limited number of seats. The static two-class capacity allocation problem is to determine how many discount customers (if any) we should allow to book. Or, equivalently, how many seats (if any) we should protect for full-fare customers.

The standard approaches to the capacity allocation problem described in this section are formulated as if the goal were to maximize expected revenue—that is, incremental costs and ancillary contribution are zero. This streamlines the discussion and is consistent with the revenue management literature. But, in reality, companies will be (or should be) maximizing expected total contribution, including incremental cost and ancillary contribution, as discussed in Section 8.8. The approaches in this section can be made consistent with maximizing total contribution by substituting *net contribution* wherever *price* or *fare* appears in an algorithm or equation.

The goal of two-class capacity allocation is to determine a *discount booking limit*—the maximum number of discount bookings that will be allowed. When there are only two classes, the protection level for full-fare bookings is equal to the capacity minus the booking limit: $y = C - b$, where C is capacity as described in Section 8.5. It is straightforward to calculate the booking limit given the protection level and capacity, and vice versa.

9.1.1 The Decision Tree Approach

At the heart of the capacity allocation problem is the trade-off between setting the booking limit too high and setting it too low. If we set the discount booking limit too low, we will turn away discount customers but not see enough full-fare demand to fill the plane. The plane will depart with empty seats even though we turned away bookings. This is called *spoilage*, since the empty seats become spoiled inventory at the moment the flight departs. On the other hand, if we allow too many discount customers to book, we run the risk of turning away more profitable full-fare customers. This is called *dilution*, since we diluted the revenue we could have received from saving an additional seat for a high-fare booking. The challenge of capacity allocation is to balance the risks of spoilage and dilution to maximize expected revenue.

The trade-offs in the two-class problem are shown as a decision tree in Figure 9.1 for a flight with total capacity C . $F_f(x)$ is the probability that full-fare demand is less than or equal to x , and $F_d(x)$ is the probability that discount demand is less than or equal to x . Assume we have currently set a booking limit of b with $0 \leq b < C$. The decision we face is whether to increase the booking limit by one seat (or, equivalently, to reduce the protection level by one seat). What would be the change in expected revenue if we increase the booking limit from b to $b + 1$? If discount demand d_d is less than or equal to b , there will be no effect on expected revenue. (Recall that booking classes are nested, so all of the seats that are not sold to discount customers are available to full-fare customers.) In this case, the net change in expected revenue of increasing the booking limit by one seat is 0, as shown on the top branch of the tree in Figure 9.1.

Only when discount demand is greater than b —which occurs with probability $\bar{F}_d(b) = 1 - F_d(b)$ —do things get interesting.¹ In that case, increasing the booking limit from b to $b + 1$ results in an additional discount booking. (For simplicity, throughout the chapter, we assume no cancellations or no-shows so that the additional booking will result in a paying passenger.) The net effect on revenue depends on full-fare demand. If full-fare demand is greater than $C - b$, which happens with probability $\bar{F}_f(C - b) = 1 - F_f(C - b)$, the additional discount booking will displace a full-fare passenger. This is *dilution*, and the resulting change in total revenue is $p_d - p_f < 0$. The other possibility is that discount demand is greater

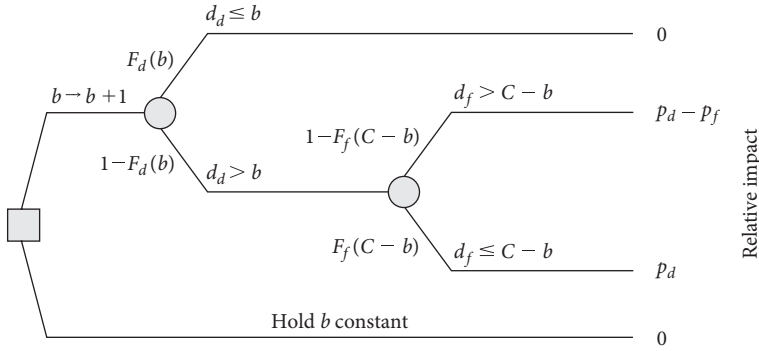


Figure 9.1 Two-class capacity allocation decision tree.

than b and full-fare demand is less than the protection level, $C - b$. In this case, we accept an additional discount passenger but do not displace a full-fare passenger. The gain is p_d . In this case, increasing the discount booking limit has reduced spoilage.

The expected change in revenue from changing the booking limit from b to $b + 1$ is the probability-weighted sum of the possible outcomes, or

$$\begin{aligned} E[h(b)] &= F_d(b) \times 0 + \bar{F}_d(b) [\bar{F}_f(C - b) (p_d - p_f) + F_f(C - b) p_d] \\ &= \bar{F}_d(b) [p_d - \bar{F}_f(C - b) p_f] \end{aligned} \quad (9.1)$$

If the term on the right-hand side of Equation 9.1 is greater than zero, then we can increase expected profitability by increasing the discount booking limit from b to $b + 1$. On the other hand, if the term is less than zero, increasing the discount booking limit will lead to decreased expected revenue.

The key term in Equation 9.1 is $p_d - \bar{F}_f(C - b) p_f$. As long as this term is greater than zero, we should keep increasing the discount booking allocation until it becomes less than or equal to zero or $b = C$, whichever comes first. With a booking limit of b , there will always be *at least* $C - b$ seats available for full-fare passengers. If discount demand turns out to be less than b , even more seats will be available for full-fare passengers. $\bar{F}_f(C - b)$ is the probability that full-fare demand will exceed the protection level. It is shown as a function of b in Figure 9.2. The term $\bar{F}_f(C - b)$ is small when the discount booking limit is small (that is, the protection level is large) and gets larger as we increase the booking limit (that is, as we decrease the protection level). If $p_d < \bar{F}_f(C) p_f$, then we should not allocate *any* seats for discount passengers—we are better off saving all seats for full-fare passengers. If $p_d > \bar{F}_f(C) p_f$, then we are better off allocating at least one seat for discount passengers.

This marginal analysis suggests the following simple hill-climbing algorithm for calculating the optimal booking limit b^* .

Algorithm for Computing Optimal Discount Booking Limit

1. Set $b = 0$.
2. If $b = C$, set $b^* = C$ and stop. Otherwise, go to step 3.

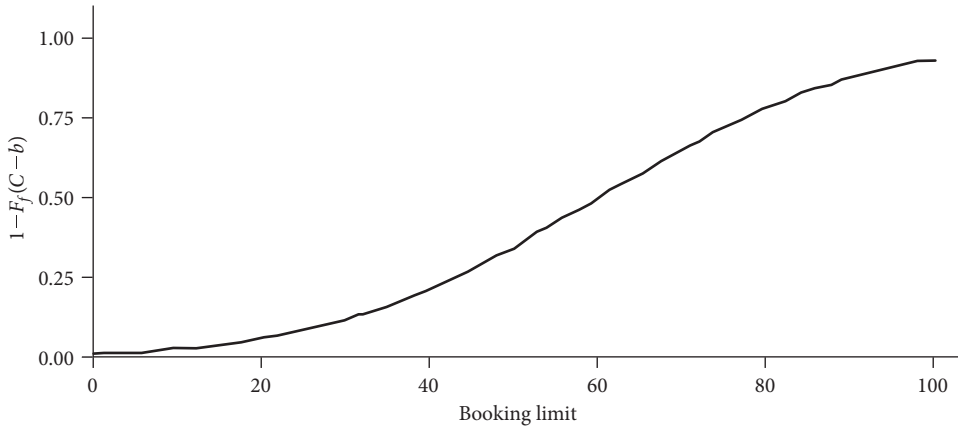


Figure 9.2 Probability that full-fare demand will exceed the protection level as a function of booking limit for a 100-seat aircraft.

3. Calculate $E[h(b)] = p_d - \bar{F}_f(C)p_f$.
4. If $E[h(b)] \leq 0$ or $F_d(b+1) = 1$, set $b^* = b$ and stop.
5. If $E[h(b)] > 0$ and $F_d(b+1) < 1$, set $b \leftarrow b+1$ and go to step 2.

This algorithm involves adding one seat at a time to the discount booking limit (or, equivalently, removing one seat at a time from the full-fare protection level) and continuing as long as adding one more seat increases expected revenue. However, there is a simpler way to get to the same point.

9.1.2 Littlewood's Rule

The sequential algorithm finds the largest integer value of b^* such that $\bar{F}_f(C - b) > p_d/p_f$. The value of b^* that meets this condition can be approximated by solving the equation

$$\bar{F}_f(y^*) = p_d/p_f, \quad (9.2)$$

or, equivalently, $F_f(y^*) = 1 - p_d/p_f$, where $y^* = C - b^*$ is the optimal protection level for full-fare bookings. Equation 9.2 is an approximation because it is unlikely that the value for which y^* holds exactly will be an integer; however, if $\bar{F}_f(y^*)$ is a continuous distribution (say, the normal or log-normal), then we can solve 9.2 exactly and round to find a reasonable integer solution. If we are using a discrete distribution, then we set y^* to the smallest value of y such that $\bar{F}_f(y^*) \leq p_d/p_f$.

Condition 9.2 for the optimal two-class discount booking limit was first described in 1972 by Kenneth Littlewood, an analyst at British Overseas Airways Company (BOAC, a predecessor to British Airways), and is known as *Littlewood's rule*. The quantity on the left-hand side of Equation 9.2, $\bar{F}_f(y^*)$, is the probability that full-fare demand will exceed the protection level. Littlewood's rule states that, to maximize expected revenue, the probability that full-fare demand will exceed the protection level should equal the fare ratio p_d/p_f . If the discount fare is 0, Littlewood's rule states that we should set the protec-

tion so high that there is no chance that full-fare demand would exceed it. In other words, the optimal discount booking limit would be 0. This is intuitive—if discount passengers are not paying us, we should not give them any seats! As p_d increases, the optimal number of seats to protect decreases until, when $p_d = p_f$, it is optimal to set the discount booking limit to C . This is also intuitive—if discount passengers are paying as much as full-fare passengers, there is nothing to be gained by turning away a discount booking when there are still empty seats.

Almost always, the function $\bar{F}_f(y)$ will be a *strictly decreasing* function of the protection level y . In this case, $\bar{F}_f$ will be invertible and we can write Littlewood's rule as

$$y^* = \min[\bar{F}_f^{-1}(p_d/p_f), C], \quad (9.3)$$

where $\bar{F}_f^{-1}$ refers to the inverse complementary cumulative distribution function of full-fare demand. At the point where $y^* = \bar{F}_f^{-1}(p_d/p_f)$, the risks of protecting too much capacity for full-fare demand and protecting too little are exactly balanced.

Note that, according to Equation 9.3, the optimal protection level does not depend on the aircraft capacity *except* that it cannot be larger than the capacity. This means that we do not need to recalculate the protection level if we change capacity as long as the fares do not change. The idea is illustrated in Example 9.1.

Example 9.1

An airline has set a discount booking limit of 80 seats on a 150-seat aircraft for a certain flight departure. The corresponding protection level is 70 seats. If the airline substituted a 100-seat aircraft for that flight and neither the fares nor the full-fare demand forecast changed, then the optimal protection level would still be 70 seats. The optimal discount booking limit with the smaller aircraft would be $100 - 70 = 30$ seats. If the airline decides to go even smaller and assign a 50-seat aircraft to the flight, the optimal decision would be to protect all 50 seats for full-fare passengers.

One fact you may find surprising about Littlewood's rule is that the optimal protection level—and, hence, the optimal discount booking limit—does not depend in any way on the forecast of discount demand. The distribution of discount demand F_d does not appear in Equation 9.2. Doubling (or halving) the discount demand forecast will not change the optimal protection level. The economic trade-offs embodied in Littlewood's rule only depend on the two fares and the distribution of expected full-fare demand.

Example 9.2

The mean full-fare demand for a flight with a 100-seat aircraft is 50 with a standard deviation of 100. For any fare ratio, the corresponding optimal booking limit is the largest value of b for which $\bar{F}_f(C - b) \leq p_d/p_f$. Table 9.1 illustrates the key calculation in the optimal two-class discount booking limit algorithm for a 100-seat flight. For a fare ratio of 0.4, the optimal booking limit is 25; for a ratio of 0.5, it is 50; and for a ratio of 0.6, it is 76. Since discount demand does

TABLE 9.1
The value of $1 - F_f(C - b)$ as a function of b when total capacity is 100 seats and the mean full-fare demand is 50 with a standard deviation of 100

b	$1 - F_f(C - b)$	b	$1 - F_f(C - b)$	b	$1 - F_f(C - b)$	b	$1 - F_f(C - b)$
0	0.309	24	0.397	49	0.496	75	0.599
1	0.312	25	0.401	50	0.500	76	0.603
5	0.326	26	0.405	51	0.504	80	0.618
10	0.345	30	0.421	55	0.520	85	0.637
15	0.363	35	0.440	60	0.540	90	0.655
20	0.382	40	0.460	65	0.560	95	0.674
23	0.394	45	0.480	70	0.579	100	0.691

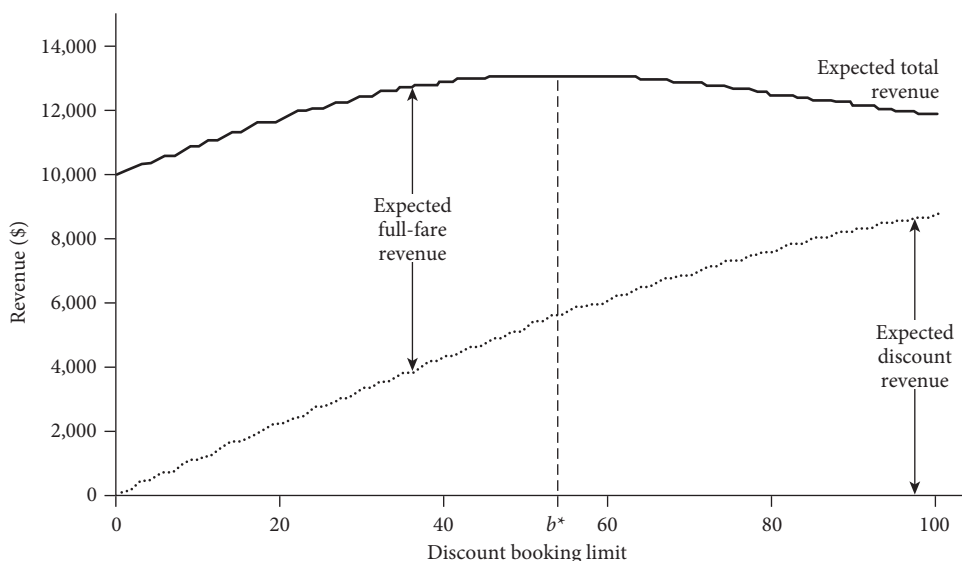


Figure 9.3 Expected discount, full-fare, and total revenues as a function of booking limit.

not enter into Littlewood's rule, these limits are independent of the discount demand forecast.

In Example 9.2 the booking limit increases with the fare ratio, as expected. Notice that if the fare ratio is less than 0.309, we would not allow any discount passengers to book—we would be better off protecting the entire plane for full-fare passengers. On the other hand, if the fare ratio is greater than 0.691, we would open the entire flight to discount passenger bookings. In either case, the airline might want to consider revisiting its fare structure—in the first case the discount fare is probably too low, and in the second case it is probably too high relative to the full fare.

The trade-offs involved in setting the discount booking limit are shown in Figure 9.3 for a 100-seat flight with mean full-fare demand of 50 and standard deviation of 25—the same values as in Table 9.1. The discount fare is \$100, and the full fare is \$200, so the fare ratio is 0.5. The top curve is total expected revenue—the sum of the expected revenue from full-

fare passengers and discount passengers. The dotted curve is expected discount revenue, and the difference between expected total revenue and expected discount revenue is expected full-fare revenue. As we increase the discount booking limit from zero, the expected revenue from discount passengers increases, with very little decrease in expected full-fare passenger revenue. As we continue to increase the discount booking limit, we begin to displace more and more full-fare passengers. The point at which the decrease in expected full-fare passengers exactly balances the gain from expected additional discount passengers is the optimal booking limit calculated by Littlewood's rule.

To summarize, Littlewood's rule determines the optimal discount booking limit in the case when discount demand books before full-fare demand. The optimal booking limit depends only on the forecast of full-fare demand, the aircraft capacity, and the fare ratio. It is not affected by the forecast of discount demand. However, the forecast of discount demand has a big impact on the sensitivity of expected revenue to the booking limit. When the forecast of discount demand is low, expected revenue is usually not very sensitive to the booking limit, within reasonable ranges. But when the forecast of discount demand is high, expected revenue becomes very sensitive to the booking limit. The lesson is that it is most critical to calculate precise booking limits when we forecast high levels of discount demand.

9.1.3 Littlewood's Rule with Independent Normal Demands

The two-class discount booking limit algorithm will find the optimal booking limit given any distribution of full-fare demand. But there is a particularly simple way to calculate the optimal discount booking limit when full-fare demand is normally distributed.² In this case,

$$F_f(x) = \Phi\left(\frac{x - \mu_f}{\delta_f}\right),$$

where $\Phi(x)$ is the standard cumulative normal distribution and μ_f and δ_f are the mean and standard deviation, respectively, of full-fare demand. Littlewood's rule implies that we want to find the value of b such that

$$\Phi\left(\frac{C - b - \mu_f}{\delta_f}\right) = 1 - \frac{p_d}{p_f},$$

which means that the optimal booking limits and protection levels must satisfy

$$b^* = [C - \delta_f \Phi^{-1}(1 - p_d/p_f) - \mu_f]^+ \quad (9.4)$$

and

$$y^* = \min[\mu_f + \delta_f \Phi^{-1}(1 - p_d/p_f), C], \quad (9.5)$$

where $\Phi^{-1}(x)$ is the *inverse cumulative normal distribution*.³ (Note that the solutions to 9.4 and 9.5 are unlikely to be integers, so rounding will usually be required.)

Figure 9.4 shows $\Phi^{-1}(1 - p_d/p_f)$ as a function of the fare ratio p_d/p_f . From this figure and Equation 9.5, we can derive a number of important characteristics of the optimal

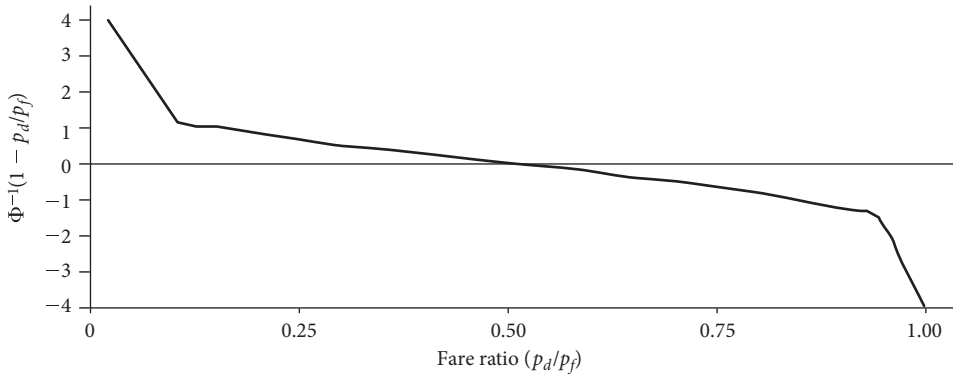


Figure 9.4 Inverse cumulative normal function of $1 - p_d/p_f$

discount booking limit. First, $\Phi^{-1}(1 - p_d/p_f)$ is a decreasing function of the fare ratio. This means that b^* is an *increasing* function of the fare ratio. Note that when $p_d = p_f/2$, then $\Phi^{-1}(1/2) = 0$, which implies that $y^* = \min(\mu_f, C)$; that is, it is optimal to set the protection level to expected full-fare demand when the fare ratio is $1/2$. For $p_d/p_f < 1/2$, it is optimal to protect *more* seats for full-fare demand than we expect to receive, and for $p_d/p_f > 1/2$, it is optimal to protect *fewer* seats than expected full-fare demand. In other words, the fare ratio determines whether we should save more or fewer seats than expected demand for full-fare passengers.

Example 9.3

A flight has a capacity of 100 seats and two fare classes. The full fare is \$300, and full-fare demand is normally distributed with a mean of 70. If the discount fare is \$150, the airline should protect exactly 70 seats for full-fare customers and set a booking limit of 30 for discount-fare demand. If the discount fare is \$180, the airline should protect fewer than 70 seats for full-fare demand; if the discount fare is \$120, the airline should protect more than 70 seats for full-fare demand. The optimal protection levels for each case can be determined from Equation 9.5.

Equation 9.5 specifies that the optimal protection level is equal to the expected full-fare demand μ_f plus the standard deviation of full-fare demand δ_f times a term. Recall that $\delta_f > 0$ is a measure of the uncertainty in our forecast of full-fare demand. A higher value of δ_f means a flatter distribution and more uncertainty. What impact does it have on the optimal booking limit? First, notice that $\delta_f = 0$ implies that $b^* = [C - \mu_f]^+$. This means that if we knew full-fare demand with certainty, we would protect seats for full-fare demand unless $\mu_f > C$, in which case we would reserve the entire capacity of the plane for full-fare passengers and set $b^* = 0$. Because we know $d_f = \mu_f$ that when $\delta = 0$, this policy will maximize expected revenue.

Of course, things would be too easy (and there would be no need for sophisticated revenue management) if companies knew future full-fare demand with certainty, so we need to consider the case when the standard deviation is greater than zero.⁴ From Equation 9.5,

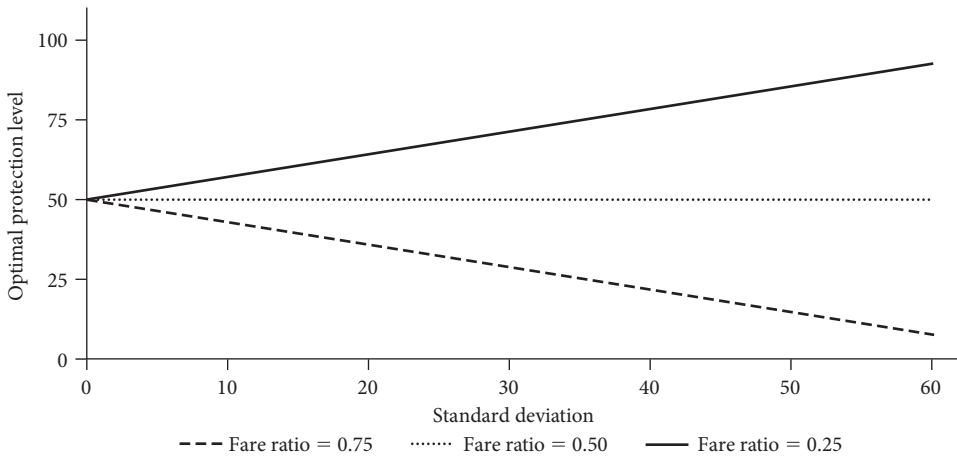


Figure 9.5 Optimal discount booking limit for a 100-seat flight as a function of the standard deviation of full-fare demand.

TABLE 9.2
Dependence of protection level y^* on the standard deviation of full-fare demand
when full-fare demand is normally distributed

Fare ratio	Dependence of protection level
$p_d/p_f > 0.5$	y^* is less than mean full-fare demand and decreases with δ_f
$p_d/p_f = 0.5$	y^* equals mean full-fare demand
$p_d/p_f < 0.5$	y^* is greater than mean full-fare demand and increases with δ_f

we can see that y^* is a linear function of the standard deviation when full-fare demand is normally distributed and the fares are constant. The effect of the standard deviation on the optimal protection level when mean full-fare demand is 50 is shown in Figure 9.5 for different values of the fare ratio, p_d/p_f .

The dependence of the optimal protection level on full-fare demand standard deviation and the fare ratio is summarized in Table 9.2. When the fare ratio is exactly 1/2 or the standard deviation of full-fare demand is zero, then the optimal protection level for full-fare demand is equal to mean full-fare demand. If the fare ratio is less than 1/2, the optimal protection level will be greater than μ_f , and increasing the standard deviation will increase the protection level. If the fare ratio is greater than 1/2, the optimal protection level will be less than μ_f and will fall as the standard deviation increases.

9.1.4 Relation to Newsvendor Problem

Littlewood's rule has a close relationship to the *critical fractile* (or *critical ratio*) solution to the optimal replenishment problem for perishable goods that is studied as part of operations management. Both capacity allocation and replenishment of perishable goods are themselves special cases of a classic problem known as the *newsvendor problem*. The newsvendor problem well predates revenue management; it was first identified and studied as early as 1888.

The newsvendor problem considers the situation in which a purchaser needs to order a quantity of some good in the face of uncertain demand. The problem got its name from the example of a newspaper salesperson who needs to determine how many newspapers to purchase at the beginning of the day to satisfy the day's uncertain demand. The key to this decision is that the newsvendor faces different costs if he purchases too much or too little. If he buys too many newspapers, then at the end of the day he will have unsold, worthless copies that he discards. If he buys too few, he will sell out of papers and turn away potentially profitable customers. The purchaser might also be a retailer deciding how many units of a fashion good to order, or it might be a manufacturer ordering raw materials.

The solution to the newsvendor problem can be expressed in terms of the *overage cost* and the *underage cost* of the decision. The overage cost is the cost per unit of purchasing too many items; the underage cost is the unit cost of purchasing too few. Assume that the newsvendor buys newspapers for 20 cents apiece and sells them at 25 cents. His overage cost is 20 cents per unsold paper; his underage cost is the 5-cent profit that he forgoes for each missed sale. (Note that the overage cost is an actual monetary cost, while the underage cost is an opportunity cost.) If U is the underage cost and O is the overage cost, the optimal order quantity for the newsvendor is the value of Y such that

$$F(Y) = \frac{U}{U + O}, \quad (9.6)$$

where $F(y)$ is the cumulative distribution on demand.

Example 9.4

When newspapers cost 20 cents and are sold for 25 cents, Equation 9.6 implies that the number of newspapers the vendor should purchase would be the smallest value of Y such that $F(Y) \geq 5/(20 + 5) = 0.2$.

Finding the optimal discount booking limit in the two-class case is analogous to the newsvendor problem. The airline needs to determine how many seats to protect (i.e., “order”) to satisfy unknown full-fare demand. The airline's unit overage cost from protecting too many seats is a displaced discount fare: $O = p_d$. The underage cost is the revenue from a lost full-fare booking minus the discount fare: $U = p_f - p_d$. Substituting into Equation 9.6 means that the airline should set the protection level such that

$$F(y) = \frac{U}{U + O} = \frac{p_f - p_d}{p_f} = 1 - \frac{p_d}{p_f},$$

which is Littlewood's rule.

9.2 CAPACITY ALLOCATION WITH MULTIPLE FARE CLASSES

Littlewood's rule finds the optimal discount allocation in the case of two classes when the lower fare class books first and demands are independent. However, as discussed in Section 8.3, airlines are prolific price differentiators, often offering tens if not hundreds of different

fares on a flight, such as special discount fares, corporate fares, and group fares in addition to the standard discount fares and full fare. An airline needs to manage the availability of all of these fares to maximize total contribution. The same holds true for any revenue management company selling multiple fare classes. More specifically, they need to set the nested booking limits within a booking control structure, as described in Section 8.5.2. This is the *multiclass capacity allocation* problem, and, predictably, it is more difficult to solve than the two-class model.

To analyze the multiclass model, we make the following assumptions.

- A product has n fare classes, each with an associated fare. Fare classes are numbered in descending fare order—that is, $p_1 > p_2 > \dots > p_n$. A fare class is called higher than another if it has a higher fare—class 1 is the highest class, and class n is the lowest class.
- Demand in each class is an independent random variable. We denote the demand in fare class i by d_i . Demand in class i can be described by a probability density function $f_i(x)$ defined on integer $x \geq 0$, where $f_i(x)$ is the probability that demand for fare class i will be x . $F_i(x)$ denotes the probability that $d_i \leq x$, and $\bar{F}_i(x)$ denotes the probability that $d_i > x$.
- Demand books in increasing fare order. That is, the lowest-fare passengers (those paying p_n) book first, followed by the second-lowest-fare passengers and so on, so that the highest-fare passengers (those paying p_1) book last.
- There are no no-shows or cancellations.
- The objective function is to maximize expected revenue. (As noted before, the approaches and algorithms in this section can be modified to maximize expected total contribution by replacing price with net contribution.)

The time convention for the multiclass model is shown in Figure 9.6. During the first period, the airline sees only booking requests from the lowest-fare passengers, those paying p_n . At the end of this first booking period, the airline has accepted some number of bookings from these passengers, say, $x_n \geq 0$. During the second period, passengers paying the next highest fare start to book, and so on, until the last period, in which the highest-fare passengers—those paying p_1 —seek to book. Note that the indices of the fare classes decrease

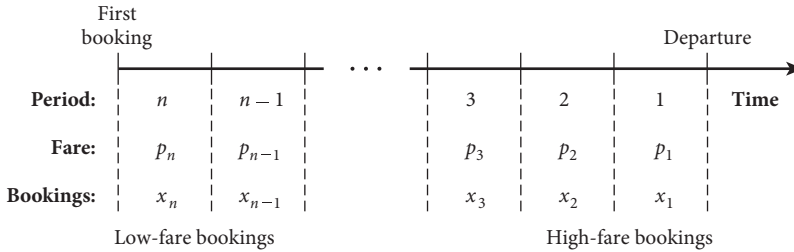


Figure 9.6 Booking process assumed in simple multiclass model.

as the time of departure approaches. (It may help to think of a countdown to departure as a way to remember this convention.)

The airline's decision at the start of each booking period j is how many bookings of type j it should accept. It knows how many bookings it has already accepted—namely, $\sum_{i=j+1}^n x_i$. Since there are no no-shows or cancellations, $C_j = \left[C - \sum_{i=j+1}^n x_i \right]$ is the unbooked capacity remaining at the beginning of period j . The airline also knows the demand distributions for future booking periods. Given this information, the airline needs to find the maximum number of bookings from class j to accept in order to maximize expected revenue. We call this quantity the *class j booking limit* and denote it by b_j . In this model, the airline only needs to calculate one booking limit at the start of each period, since it will receive bookings from only one fare class during the coming period. The airline can wait until it sees how many bookings it accepts in this period before setting the booking limits for the next higher class.

9.2.1 The Decision Tree Approach

To solve the multiclass booking limit problem, we work backward. At the beginning of the final period, the airline has C_1 seats remaining. Since there are no no-shows or cancellations, it is clearly optimal for the airline to allow all remaining seats to be booked. So $b_1 = C_1$. Next we consider period 2—the penultimate period before departure. At the beginning of period 2, the airline has C_2 seats remaining. The problem the airline faces at the beginning of the second period is exactly the two-fare-class problem we solve in Section 9.1. Therefore, the airline can maximize revenue by applying Littlewood's rule (Equation 9.2): The optimal booking limit is the smallest value of $0 \leq b_2 \leq C_2$ such that $F_1(C_2 - b_2) < 1 - p_2/p_1$.

Things get more complicated when we move backwards one period to calculate b_3 . Conceptually, the trade-off is similar to the two-class case: If we increase the booking limit in period 3, one of two things can happen. Either we receive an additional type 3 passenger booking or we do not. If we do not, the net impact on total revenue of increasing the booking limit will be zero. If we do book an additional customer in period 3, we will receive a fare of p_3 , but we run the risk of displacing a later-booking customer who would pay a higher fare—either p_1 or p_2 . There is no easy way to calculate either the probability that we will displace a booking or the conditional probability of which type of booking will be displaced. However, the fundamental trade-off remains the same as in the two-class case—we want to set the booking limit that balances the risks of spoilage and dilution.

The period 3 problem is illustrated in Figure 9.7, where q_2 is the probability that increasing the class 3 booking limit will displace a class 2 passenger and q_1 is the probability that it will displace a class 1 passenger. From this tree we can see that the optimal class 3 booking limit depends on the probability that an additional class 3 booking would displace a higher-fare booking and the fare of the booking that would be displaced. If we knew the displacement probabilities q_1 and q_2 , it would be straightforward to roll back the decision tree to solve for the optimal b_3 (Exercise 1). Unfortunately, there is no easy way to calculate the displacement probabilities for three or more fare classes, and there is no closed-form solution for the optimal booking limits. As a result, the solution techniques for finding the optimal booking limits for three or more fare classes require the use of dynamic programming.

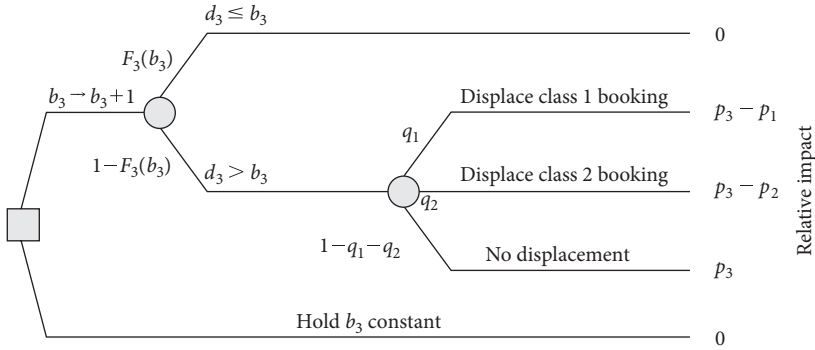


Figure 9.7 Capacity allocation decision tree for period 3 prior to departure.

Dynamic programming approaches to optimal multiclass capacity control are outside the scope of this book; the reader who is interested in learning about such approaches will find useful references in Section 9.8.

While the techniques for finding the optimal booking limits and protection levels in the multiclass problem are not mathematically difficult, they can be computationally intensive, particularly when they need to be applied to thousands of flights in a limited period of time. For this reason, airlines, hotels, rental car companies, and their revenue management brethren typically do not calculate the *optimal* multiclass booking limits. Rather, they utilize a heuristic to determine good but not necessarily optimal booking limits. The most popular of these heuristics are the expected marginal seat revenue (EMSR) approaches, especially EMSR-a and EMSR-b. These are very fast and almost always find booking limits that generate expected revenue very close to the theoretical optimum.

9.2.2 Expected Marginal Seat Revenue (EMSR) Heuristics

Expected marginal seat revenue (EMSR) comes in two flavors: EMSR-a and EMSR-b. Both EMSR-a and EMSR-b were introduced by Peter Belobaba in his PhD dissertation and a series of subsequent papers (Belobaba 1987, 1989). Both are based on approximating the multiclass problem by a series of two-class problems and applying Littlewood's rule to obtain a solution.

EMSR-a. Expected marginal seat revenue—version a (EMSR-a) is the best-known heuristic for the capacity allocation problem. It is based on the idea of calculating protection levels for the current class relative to each of the higher classes using Littlewood's rule. These protection levels are then summed to create the protection level for the current class. For example, if we are setting a protection level for class 3, EMSR-a would calculate a protection level for class 3 against class 1 (which we denote by y_{31}) using the form of Littlewood's rule in Equation 9.3:

$$y_{31} = \bar{F}_1^{-1}(p_3/p_1)$$

and a protection level for class 3 against class 2, which we denote by y_{32} according to

$$y_{32} = \bar{F}_2^{-1}(p_3/p_2)$$

The EMSR-a protection level for class 3 is the sum of these two protection levels; that is, $y_3^a = \min(y_{31} + y_{32}, C)$ or $y_3^a = \min(\bar{F}_1^{-1}(p_3/p_1) + \bar{F}_2^{-1}(p_3/p_2), C)$, where the superscript a in y_3^a denotes that it is a protection level calculated by EMSR-a. The booking limit for class 3 can then be written $b_3^a = [C_3 - y_3^a]^+ = [C_3 - \bar{F}_1^{-1}(p_3/p_1) - \bar{F}_2^{-1}(p_3/p_2)]^+$.

EMSR-a can be generalized to any number of fare classes. For $j \geq 2$, the EMSR-a protection level is given by

$$y_j^a = \min \left[\sum_{i=1}^{j-1} \bar{F}_i^{-1}(p_j/p_i), C \right].$$

If demands are normally distributed for each fare class, the EMSR-a protection level for class j is

$$y_j^a = \min \left[\sum_{i=1}^{j-1} \mu_i + \delta_i \Phi^{-1} \left(\frac{p_j}{p_i} \right), C \right],$$

which is quite easily computed using standard cumulative normal distribution tables. It is easy to confirm that EMSR-a reduces to Littlewood's rule in the case of two fare classes.

EMSR-b. *Expected marginal seat revenue—version b (EMSR-b)* assumes that a passenger displaced by an additional booking would be paying a fare equal to a weighted average of future fares. EMSR-b creates an artificial class with mean demand equal to the sum of the mean demands for all the future periods and a fare equal to the average expected fare from future bookings. It then uses Littlewood's rule to calculate the booking limit of the current class j with respect to the artificial class. EMSR-b assumes that the expected fare of the displaced booking is equal to the expected fare of future demand. This is an approximation because limits will also be set on higher-fare booking, so the mean of the demand *accepted* in a higher booking class will generally not be equal to the mean of the unconstrained demand.

To formalize EMSR-b, we assume as before that demands in different classes are independent, with the demand in class i having a mean of μ_i and a standard deviation of δ_i . Assume we are at the beginning of period $j \geq 2$ and want to calculate the booking limit b_j . EMSR-b then proceeds as if we were solving a two-class problem. If we assume that demands for the different classes are normally distributed, then EMSR-b assumes that demand for the artificial class is normally distributed with mean demand μ , average fare p , and standard deviation δ given by

$$\mu = \sum_{i=1}^{j-1} \mu_i, \quad p = \sum_{i=1}^{j-1} p_i \mu_i / \mu, \quad \delta = \sqrt{\sum_{i=1}^{j-1} \delta_i^2}. \quad (9.7)$$

(The formulas for the mean and the standard deviation are standard for the sum of any independent random variables. See Appendix B for more details.) We can use the version of Littlewood's rule for normal distributions in Equation 9.5 to write

$$y_j^b = \min \left[\mu + \sigma \Phi^{-1} \left(\frac{p_j}{p} \right), C \right]$$

as the formula for the EMSR-b protection level for classes higher than class j when demands are normally distributed. Again, it is easy to confirm that EMSR-b reduces to Littlewood's rule in the case of two fare classes.

Comparison of EMSR heuristics. It is difficult to make comprehensive statements about the relative merits of the two EMSR heuristics. There are cases when EMSR-a gives higher booking limits than EMSR-b and cases when the opposite is true. More pertinently, there are cases when EMSR-a results in higher revenue than EMSR-b, and vice versa. Both appear to give solutions close to the optimal booking limits in most realistic cases, and both generally capture a high percentage of the optimal revenue. One set of studies has shown that EMSR-b was consistently within 0.5% of the optimal revenue, while EMSR-a led to revenues that were more than 1.5% lower than optimal in some cases (Belobaba 1992). Other studies (and much practical experience) has shown that both EMSR heuristics tend to perform well in almost all realistic cases and that neither strictly dominates the other, although EMSR-b seems to perform slightly better than EMSR-a in practice.

Guillermo Gallego and Huseyin Topaloglu (2019) report on a series of simulations to compare the performance of the two EMSR heuristics on an aircraft with four booking classes. They assumed that demands for the four classes followed independent normal distributions. The fares, mean, and standard deviations of demand along with the corresponding optimal protection levels (y_j^*), EMSR-a protection levels (y_j^A), and EMSR-b protection levels (y_j^B) for the four booking classes $j = 1, 2, 3, 4$ are shown in Table 9.3. The performance of the two EMSR heuristics against the optimal protection levels was tested for capacities ranging from 80 seats to 160 seats using 500,000 simulation runs for each capacity. Table 9.4

TABLE 9.3
Fares, demand data, and resulting protection levels for EMSR-a and EMSR-b heuristics

Class (j)	Fare (p_j)	Demand statistics		Protection levels		
		Mean (μ_j)	Standard deviation (δ_j)	Optimal	EMSR-a	EMSR-b
1	\$1,050	17.3	5.8	9.7	9.8	9.8
2	\$950	45.1	15.0	54.0	50.4	53.3
3	\$699	39.6	13.2	98.2	91.6	96.8
4	\$520	34.0	11.3			

SOURCE: Gallego and Topaloglu 2019. Used with permission.

TABLE 9.4
Revenue from the optimal policy and the two EMSR heuristics as a function of total capacity

Capacity	Optimal revenue (\$)	EMSR-a: % optimal	EMSR-b: % optimal
80	67,505	99.90	100.00
90	74,003	99.94	100.00
100	79,615	99.60	99.98
110	84,817	99.65	99.98
120	84,963	99.73	99.99
130	94,860	99.85	99.99
140	99,164	99.94	99.99
150	102,418	99.99	100.00
160	104,390	100.00	100.00

SOURCE: Adapted from Gallego and Topaloglu 2019.

shows the results. The expected revenue using the optimal booking limit is shown along with the percentage of the expected optimal booking limit achieved by the two heuristics. The heuristics performed extremely well, capturing more than 99% of the optimal revenue in each case. This is consistent with previous studies of the same sort.

It is widely felt that the additional revenues from solving the optimal booking control problem do not generally justify the extensive additional computations that would be required relative to using an EMSR heuristic. As a result, EMSR-b has become the standard for calculating booking limits for a single resource.

9.2.3 Bid Pricing

We have seen that optimal nested booking limits have the property that if one fare class is open, then all higher fare classes will be open. This means that one of three conditions must hold when the allocations for a resource are optimized:

- All fare classes are open.
- All fare classes are closed.
- There is an open fare class such that all higher classes are open and all lower classes are closed.

This suggests the following rule:

Accept a single-seat booking request if its associated fare is greater than or equal to the fare in the lowest open class. Otherwise reject it.

Example 9.5

A flight departure has six fare classes. Two weeks before departure, fare classes 1, 2, and 3 are open, while classes 4, 5, and 6 are closed. The class 3 fare is \$250 and the class 4 fare is \$210. If the next booking request has a fare greater than or equal to \$250, it will be accepted; otherwise it will be rejected.

This simple rule to control bookings is called *bid pricing*, and the minimum acceptable fare is called the *bid price*.⁵ In bid pricing, the fare of a booking request is compared to a *hurdle rate* (the bid price) for a product and accepted if and only if the fare exceeds the bid price. When the bid price is zero, any booking request with positive net contribution will be accepted—this is equivalent to having all fare classes open. When the bid price is set to any amount higher than the highest fare, no booking requests will be accepted—this is equivalent to having all fare classes closed. The seller can use the bid price as an intensity control, raising the bid price when he wishes to reduce the rate of sales and increase average price, and lowering the bid price when he wishes to increase the sales rate at a lower average price.

Bid pricing has an appealing simplicity. It requires only calculating and updating a single number—the bid price. It avoids the complexity involved in recalculating multiple booking limits. *If* bid prices are continually updated *and* each request is for a single unit of resource (e.g., a single airline seat or a single hotel room night) *and* demand for different

fare classes were independent, a system under which booking requests are evaluated only against a bid price could capture at least as much revenue as the standard booking limit approaches.

The bid price criteria should remind you of the rule we established in Section 7.3 that a request for a constrained resource should be accepted only if the associated price exceeds the opportunity cost of the resource. There is a reason for this—the bid price for a resource *is* the opportunity cost for that resource. The decision tree approach described in Section 9.1.1 makes this explicit: we should accept a booking only if its fare exceeds the expected revenue we could realize from that unit of capacity in the future—that is, its opportunity cost. Bid pricing is based on the following equivalence.

$$\begin{aligned}\text{Bid price for a resource} &= \text{fare in the lowest open fare class} \\ &= \text{opportunity cost for the resource} \\ &= \text{lowest fare we should accept for a booking request}\end{aligned}$$

Bid pricing is simply an example of the rule that a unit of a constrained perishable resource should be sold only if the price received is greater than the opportunity cost.

Despite the appealing simplicity of bid pricing, most revenue management companies do not use pure bid pricing systems. Rather, they rely on the booking limits and protection levels described in Section 8.5.2. If bid pricing is so good, why is it not used more often? There are three main reasons.

- Bid prices give only *marginal* guidance—that is, a bid price can only tell us whether we should accept a booking request for a single seat. The bid price does not give a definitive answer to the question of whether or not we should accept a booking request for more than one seat. Booking limits enable more precise control of multiseat booking requests: for example, an airline would be able to accept a booking request for three seats in a particular class while rejecting requests for four or more seats.
- The bid price for a resource jumps upward any time a booking is accepted and decreases any time an existing booking is canceled. For bid prices to provide optimal control, they would need to be calculated at least every time a booking or cancellation takes place. This would be computationally infeasible given the volume of bookings and cancellations that an airline, rental car company, or large hotel chain needs to process. Booking limits can remain in place for a longer period of time, with the limits updated as bookings and cancellations occur according to the mechanisms described in Section 8.5.
- As described in Chapter 8, booking limits were built into the airline reservation systems from their inception. The nested booking controls described in Section 8.5 were the only way available to airlines and other revenue management companies to implement capacity allocation.

Despite these practical concerns, bid pricing is an important concept. Since the bid price on a leg approximates the opportunity cost for a resource, it is a good estimate of the additional revenue that could be achieved by acquiring an additional unit of resource. Bid

prices also play an extremely important part in network revenue management, as we see in Chapter 10.

9.3 CAPACITY ALLOCATION WITH DEPENDENT DEMANDS

The models we have considered so far assume that demand in each fare class is independent of demand in the other fare classes. Perhaps even more surprisingly, they assume that demand in each fare class is independent of whether other classes are opened or closed. This is equivalent to assuming that the fare classes perfectly segment the market (see Chapter 6). Standard capacity allocation models, such as Littlewood's rule and the EMSR heuristics, assume no cannibalization—opening a discount class has no effect on full-fare demand. Full-fare passengers are either unwilling or unable to purchase at the discount fare, and none of the discount passengers would purchase the product at the full fare.

The assumption of fare class independence does not reflect the real world. In reality, there are no such things as full-fare customers and discount customers; there are only potential buyers searching for the alternative that best meets their needs. In fact, the assumption of independence fails even at the most basic level of airline segmentation. Some business travelers book early to take advantage of discount fares, just as some leisure travelers would be willing to pay full fare for travel but are more than happy to purchase at a lower discount fare. This means that some fraction of the customers who are refused when we close the discount fare will go on to purchase at the full fare and that the two demands are not really independent. Rather, the full-fare demand will depend on whether the discount fare class is opened or closed. When we close discount fares, we would expect more full-fare demand because some of the discount-fare customers will go ahead to purchase at full fare when the discount fare is not available.

As revenue management systems became more widely used, revenue managers—the airline personnel who monitor fare class allocations for a set of flights—noticed that closing a discount fare class would often lead to increased demand in higher classes. This phenomenon was termed *buy-up* or *sell-up*. By the same token, opening a discount fare class would often lead to decreased demand in higher classes—cannibalization. Revenue managers would close discount classes earlier than recommended by the revenue management system because they knew that they would gain benefits from buy-up.

Buy-up and cannibalization are not separate phenomena. As Bill Brunger, vice president of revenue management for Continental Airlines, put it in conversation with me, “Passengers do not come with Y-Class or M-Class stamped on their foreheads.” Rather, customers search for the best combination of fare and travel option that meets their needs. The internet has increased visibility of all airline fares and their accessibility to all customers. As a result, the difference between full fares and discount fares has grown tremendously—for some flights, the average full fare can be five times or more the average discount fare. This provides a tremendous incentive for customers to shop for lower fares.

The assumption of fare class independence was built into revenue management from the beginning. The reasons for this were twofold. First, the most pressing need facing the airlines was to quickly develop systems that enabled them to respond to the challenge of low-cost competitors and to manage the exploding number of fares in the marketplace.

Rapidly implementing relatively simple approaches was more important than waiting to develop a perfect solution. The second reason was that, in many cases, the assumption of perfect segmentation was not too bad. The booking fences between discount and full-fare segments were fairly effective. Furthermore, for international airlines, regional pricing was a major part of revenue management strategy. In the 1980s, the fare paid in Japan for a ticket would often be 50% more than the same ticket purchased in Europe or the United States, and in the 1990s, one European airline estimated that 100% of its profit was due to its ability to sell tickets in certain overseas markets at fares higher than in its own market. Before the rise of the internet, this geographical discrimination was quite powerful and not subject to high levels of cannibalization. However, the internet has eroded many of the barriers among booking classes, and the need to incorporate demand dependence into revenue management has become more pressing.

9.3.1 Demand Dependence with Two Fare Classes

We can modify the marginal analysis approach of Figure 9.1 to find the optimal booking limit under imperfect segmentation with two fare classes. We assume as before that the discount fare is p_d and the full fare is p_f , with $p_f > p_d$. For this analysis, we use new definitions of demand:

$$\begin{aligned} d_d &= \text{Total discount demand,} \\ d_f &= \text{Total full-fare demand assuming all discount bookings are accepted.} \end{aligned}$$

In addition, we define α as the fraction of rejected discount demand that will seek to book a full fare, with $0 \leq \alpha \leq 1$. The actual full-fare demand the airline sees will be a function of the discount booking limit and can be written as

$$\hat{d}_f = d_f + \alpha[d_d - b]^+$$

This means that increasing the booking limit b by one seat will decrease expected full-fare demand by α if discount demand exceeds the booking limit and will have no effect on full-fare demand otherwise.

We can relate this model to the willingness-to-pay approach introduced in Chapter 3. Specifically, this model is equivalent to assuming that there are d_d customers with a willingness to pay for the discount product that is greater than p_d and are able to book early. Of these, αd_d also have a willingness to pay for the full-fare product that is greater than p_f ; if the discount product is not available, they will buy the full-fare product. Finally, there are d_f customers who have a willingness to pay for the full-fare product that is greater than p_f and who are unwilling or unable to book early, so their willingness to pay for the discount product is lower than p_d . Of course, we would expect d_d , d_f , and α to change if we changed the fares p_d and p_f .

The decision tree for the two-class model with demand dependence is shown in Figure 9.8. The only difference from the basic two-class allocation model in Figure 9.1 occurs when discount demand is greater than the booking limit *and* full-fare demand is less than the protection level, $C - b$. In this case, increasing the booking limit by 1 leads to an additional discount booking, but there is a probability of α that this increased discount booking cannibalizes a future full-fare booking. The net impact on expected profitability is $p_d - \alpha p_f$.

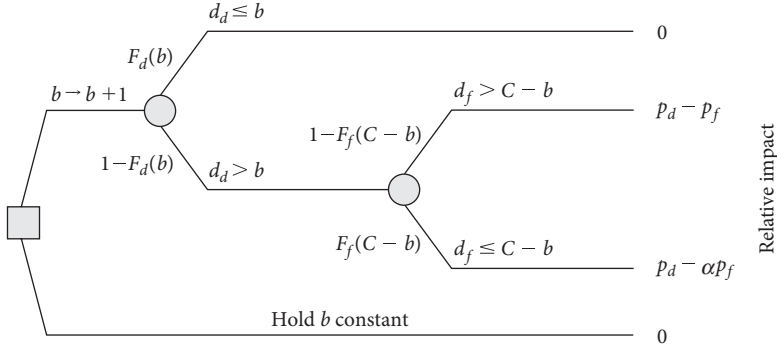


Figure 9.8 Two-class capacity allocation with demand dependence.

It is apparent that the expected marginal impact of increasing the booking limit by 1 in this case is given by

$$\begin{aligned} E[h(b)] &= F_d(b) \times 0 + \bar{F}_d(b) [\bar{F}_f(C - b)(p_d - p_f) + F_f(C - b)(p_d - \alpha p_f)] \\ &= \bar{F}_d(b) [p_d - \alpha p_f - (1 - \alpha) \bar{F}_f(C - b) p_f]. \end{aligned} \quad (9.8)$$

The change in expected profit will be determined by the sign of the term $p_d - \alpha p_f - (1 - \alpha) \bar{F}_f(C - b) p_f$. If this term is greater than zero, increasing the booking limit by one will lead to increased expected revenue. If it is less than zero, increasing the booking limit will lead to decreased expected revenue. Note that if $\alpha \geq p_d/p_f$, the term on the right will be less than or equal to zero for all values of b . This corresponds to the case where cannibalization overwhelms demand induction. In this case, we know immediately that the optimal booking limit is $b^* = 0$.

Assume now that $p_d/p_f \geq \alpha$. Then, from Equation 9.8 we can see that we are looking for the protection level y^* that solves

$$\bar{F}_f(y^*) = \frac{p_d - \alpha p_f}{(1 - \alpha) p_f}. \quad (9.9)$$

Note that Equation 9.9 reduces to Littlewood's rule when $\alpha = 0$.

We can also write Equation 9.9 as

$$\bar{F}_f(y^*) = \frac{1}{1 - \alpha} \left(\frac{p_d}{p_f} - \alpha \right). \quad (9.10)$$

The expression $\left[\frac{1}{1 - \alpha} \left(\frac{p_d}{p_f} - \alpha \right) \right]^+$ is called the *modified fare ratio*, and Equation 9.10 is the same as Littlewood's rule, using $\frac{1}{1 - \alpha} \left(\frac{p_d}{p_f} - \alpha \right)$ in place of the fare ratio p_d/p_f . Since the modified fare ratio is less than the actual fare ratio, the impact of demand dependence is to reduce the discount booking limit (equivalently, increase the full-fare protection level) from the case with no demand dependency.

When full-fare demand is normally distributed, the optimal protection level is

$$y^* = \min \left\{ \mu + \delta \Phi^{-1} \left[\frac{1}{1-\alpha} \left(1 - \frac{p_d}{p_f} \right) \right], C \right\}.$$

9.3.2 Modified EMSR Heuristics

Both EMSR-a and EMSR-b can be modified to incorporate imperfect segmentations. The modifications are based on the assumption that when a class closes, rejected demand for that class may buy up to the next highest class—but not to any higher classes. For example, if we close class 3, some class 3 demand may seek to buy in class 2 but not in class 1. We can easily derive modified versions of the EMSR heuristics under this simple assumption.

Define α_j for $j = 2, 3, \dots, n$ as the fraction of class j customers who would purchase in the next higher class, $j-1$, if class j is closed. Thus, α_3 is the fraction of class 3 customers who would “buy up” to class 2 if they find class 3 closed when they request a booking. α_j is referred to as the *buy-up factor* for class j . Note that there is no buy-up factor for the highest fare class. Then, for the case of normal demand distributions, EMSR-a and EMSR-b can be extended to incorporate buy-up factors as follows:

$$\begin{aligned} \text{EMSR-a with buy-up: } y_j &= \mu_{j-1} + \delta_{j-1} \Phi^{-1} \left[\frac{1}{1-\alpha} \left(1 - \frac{p_j}{p_{j-1}} \right) \right] \\ &\quad - \sum_{i=1}^{j-2} \left[\delta \Phi^{-1} \left(1 - \frac{p_j}{p_i} \right) - \mu_i \right], \\ \text{EMSR-b with buy-up: } y_j &= \mu + \delta \Phi^{-1} \left[\left(\frac{1}{1-\alpha} \right) \left(1 - \frac{p_j}{p} \right) \right], \end{aligned}$$

where μ , p , and δ for the modified EMSR-b algorithm are as defined in Equation 9.7.

Modified EMSR-a calculates the protection levels for class j against every higher class under the assumption that buy-up will only occur to the next highest class, $j-1$, if we close j . It then subtracts the sum of all these protection levels from the remaining capacity to determine the booking limit for class j . EMSR-b assumes that buy-up will occur to the aggregate class consisting of all the higher classes.

These two modifications to the EMSR heuristics have been studied by Peter Belobaba and Lawrence Weatherford (1996), who showed that the modified EMSR heuristics provide additional revenue when buy-up is present. The modified EMSR heuristics are easy to implement, and versions of them have been used in practice. However, they have some disadvantages. They require buy-up factors to be estimated and stored for each flight. They also require redefining the demand for each fare class as the demand that would be received if the next higher class is open. In their simplest form, they do not incorporate the possibility of buy-up from class j to classes higher than $j-1$.

9.3.3 Capacity Control with a Consumer Choice Model

The EMSR heuristics modified to incorporate buy-up in the previous section are somewhat unsatisfying. For one thing, they are a heuristic grafted onto another heuristic. They

are not derived from a fundamental model of consumer choice. Rather, they assume that passengers *do* have M-Class and Y-Class stamped on their heads and that only some of the M-Class passengers will buy up to Y-Class if M-Class is closed. “Buy up” is not really a good description of the underlying phenomenon—at heart, what is happening is that consumers are faced with the decision of purchasing a ticket from among the options that they have available to them.

This section presents a more general approach to incorporating consumer choice into capacity control models. There are challenges in implementing this approach fully into revenue management systems, which we consider. However, the approach provides insights into the underlying problem and, in particular, illustrates the important concept of *inefficient sets*.

For this approach, we drop the assumption that bookings occur in fare order. Instead, we assume that, at any time, more than one fare class may be open. Fare classes are distinguished by various restrictions, with low-price fare classes being more restricted than higher-price fare classes. The lower-fare classes are inferior products being sold at lower prices. The decision facing the airline at each time period is which *set* of fare classes to open. At each time period, customers arrive and observe which fare classes the airline has made available. Each customer either purchases from one of the open fare classes or does not purchase. The question facing the airline is which set of fare classes to open at which time.

As before, we assume that time periods are indexed so that time runs from the current period t counting down to departure at time 0. There are n fare classes indexed so that the prices are ordered according to $p_1 > p_2 > \dots > p_n > 0$. There is a fixed capacity of C , and bookings are firm—there are no cancellations or no-shows. For this model, we assume that there is *at most* one booking request in each period. For simplicity, we assume that the probability of a booking request is the same in each period and is equal to λ with $1 \geq \lambda > 0$.

At each time period, the airline needs to determine what *set*, $S(t)$, of booking classes to offer. There are 2^n possible sets that can be offered corresponding to all possible combinations of individual booking classes being opened or closed. If all booking classes are closed, the flight is closed for booking. Let $S(t)$ indicate the set that is open at time t , and define $p_j(S(t))$ as the probability that a customer arriving at time t chooses class j given that $S(t)$ is the set of open classes, with $p_0(S(t))$ representing the probability of no purchase. Obviously $p_j(S(t)) > 0$ only if class j is in set $S(t)$; otherwise $p_j(S(t)) = 0$.

These concepts are best illustrated by an example. Assume that a flight has three booking classes, Y, M, and K, with fares \$800, \$500, and \$450, respectively. Y fare is unrestricted, while M and K have various restrictions—say, a 14-night stay and a Saturday-night stay, respectively. There are eight possible sets that the airline could offer on this flight. Assume that customers for this flight fall into five segments—two business and three leisure. One of the business segments can accept no restrictions; the other can purchase 14 days in advance but not stay over Saturday. All of the leisure segments can purchase 14 days in advance, but Leisure Segment 1 cannot accept a Saturday-night stay. Leisure Segments 2 and 3 can both accept both restrictions, but Leisure Segment 3 is not willing to pay more than \$470 for the flight. The situation is summarized in Table 9.5 along with the probability that a booking

TABLE 9.5
Segments and their characteristics for the example

Segment	Probability	Willing and able to purchase		
		Y-Class	M-Class	K-Class
Business 1	0.1	Yes	No	No
Business 2	0.2	Yes	Yes	No
Leisure 1	0.2	No	Yes	No
Leisure 2	0.2	No	Yes	Yes
Leisure 3	0.3	No	No	Yes

request belongs to each of the five segments. Note that a customer segment will purchase a product only if they can accept the restrictions *and* they are willing to pay the associated fare.

From the information in Table 9.5, we can define the following values:

$$Q(S) = \sum_{i=1}^n P_i(S) = 1 - P_0(S)$$

and

$$R(S) = \sum_{i=1}^n P_i(S) p_i.$$

Here, $Q(S)$ is the probability that a customer will purchase one of the products in S given that the set S is on offer, and $R(S)$ is the average revenue from an arriving request given that set S is on offer. $P_0(S)$ is the probability of no purchase in the period. The values of $Q(S)$ and $R(S)$ for the example are shown in Table 9.6. We assume that each booking request purchases in the least-expensive fare class available that meets the customer's requirements.

We now establish the following definition. A set T is said to be *inefficient* if there exist probabilities $\alpha(S)$ such that

$$Q(T) \geq \sum \alpha(S) Q(S) \quad \text{and} \quad R(T) < \sum \alpha(S) R(S),$$

where the summations are over all sets. Here, each $\alpha(S)$ is a probability, which means that each $\alpha(S) \geq 0$ and that $\sum \alpha(S) = 1$. There is an important special case in which a set T is inefficient if there exists a set U such that $Q(T) \geq Q(U)$ and $R(T) < R(U)$.

This has been a long setup, but the first payoff is the observation that *inefficient sets can never be optimal*. This means that they can be dropped from consideration. Table 9.7 illustrates the calculation of $Q(S)$ and $R(S)$ for each of the sets in the example from Tables 9.5 and 9.6. To clarify how these are calculated, consider the set $\{M, K\}$. If the airline offers this set, then every customer except those in the Business Segment 1 will purchase. Business 2 and Leisure 1 will purchase from M-Class for a total probability of $P_M(S) = .4$ of an M-Class purchase. Leisure 2 and Leisure 3 will both purchase from M-Class for a total probability of $P_K(S) = .5$. The total probability of purchase is the sum of these, so $Q(S) = .9$. The expected revenue is thus .4 times the M-Class fare plus .5 times the K-Class fare: $R(S) = .4 \times \$500 + .5 \times \$450 = \$425$.

In Table 9.7, only Set 1, which includes only Y-Class; Set 5, which includes Y-Class and K-Class; and Set 7, which includes all three classes, are efficient. The remaining four sets are

TABLE 9.6
Choice probabilities $P_j(S)$, probability of purchase $Q(S)$, and expected revenue $R(S)$
for all sets corresponding to segments and probabilities in Table 9.5

Set (S)	$P_Y(S)$	$P_M(S)$	$P_K(S)$	$Q(S)$	$R(S)$	Efficient?
{Y}	0.3	0	0	0.3	\$240	Yes
{M}	0	0.4	0	0.4	\$200	No
{K}	0	0	0.5	0.5	\$225	No
{Y, M}	0.1	0.6	0	0.7	\$380	No
{Y, K}	0.3	0	0.5	0.8	\$465	Yes
{M, K}	0	0.4	0.5	0.9	\$425	No
{Y, M, K}	0.1	0.4	0.5	1.0	\$505	Yes

TABLE 9.7
Booking class purchased by each segment for the example in Table 9.5

Set	Class	Segment				
		Business 1	Business 2	Leisure 1	Leisure 2	Leisure 3
{Y}	Y	X	X			
{M}	M		X	X		
{K}	K				X	X
{Y, M}	Y	X				
	M		X	X	X	
{Y, K}	Y	X	X			
	K				X	X
{M, K}	M		X	X		
	K				X	X
{Y, M, K}	Y	X				
	M		X	X		
	K				X	X

inefficient and should never be offered—thus, the airline should never offer only M and K classes together without Y class.

The intuition about efficiency is that, if a set is inefficient relative to another set—that is, $Q(T) \geq Q(S)$ and $R(T) < R(S)$ for some sets T and S —then S provides more expected revenue than T and has a lower expected opportunity cost because it has a lower probability of consuming a seat. If T is inefficient against a combination of sets, then the same holds true against the policy of choosing one of the sets in the combination with probability $\alpha(S)$.

The second payoff is that, for two *efficient* sets, S_1 and S_2 , $Q(S_1) \geq Q(S_2)$, implies that $R(S_1) \geq R(S_2)$ and *vice versa*. This means that the efficient sets can be ordered in terms of expected sale probability and expected revenue. This ordering is shown in Figure 9.9 for the extended example. The points corresponding to the three efficient sets are labeled. The lines connecting them represent an efficient frontier—combinations of expected sales and expected revenue that could be achieved by randomizing among the three efficient sets. The points for the inefficient sets fall below the efficient frontier—they are dominated by weighted combinations of the efficient sets.

Figure 9.9 also provides a key as to how choice-based capacity control could be embedded in a dynamic booking control system. When a flight is not expected to book up, it is optimal to choose a set that provides a high probability of selling a seat—in Figure 9.9, that is the set $\{Y, M, K\}$. As capacity is expected to become scarcer, we would move down the

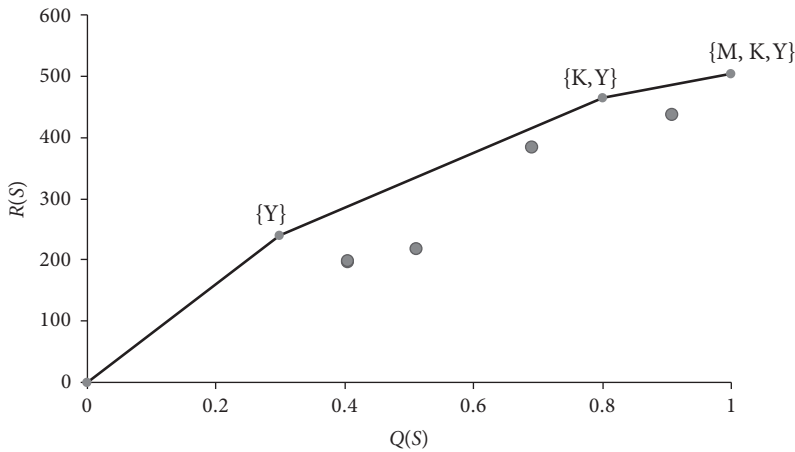


Figure 9.9 Efficient and inefficient sets for choice-based capacity control example.
Source: Talluri and van Ryzin 2004b. Used with permission.

efficient frontier. Thus, the next action would be to close M-Class and offer only {Y, K}. If capacity becomes highly scarce, then we would close all of the booking classes except Y-Class. As in traditional capacity control, we are managing availability by opening and closing booking classes; the difference is that we are not closing in order of fare class.

Note that under this approach, the probability that a booking inquiry would purchase from booking class j if set S is offered— $P_j(S)$ —will change over time as departure approaches. For an obvious example, the probability that 10 days before departure a customer will book in a fare class that has a 14-day booking restriction is 0. However, as we have seen, the mix of business and leisure passengers also changes as departure approaches. This means that the airline needs to forecast and estimate values of $Q(S)$ and $R(S)$ that depend on time to departure.

9.4 A DATA-DRIVEN APPROACH TO CAPACITY CONTROL

The methods for setting booking classes described in Section 9.3 assume that we have an estimate of the distribution of demand for each booking class. Assuming that demands for booking classes follow a normal distribution, these methods require an estimate of the mean and the standard deviation of demand for each booking class. They also assume that demands are independent—early demand for discount bookings does not give us any information about later full-fare demand. In all cases, we need to choose distributions of demand with parameters derived by applying statistical estimation to historic observations. This section discusses a data-driven approach that does not make assumptions about the distributions or independence of demand and allows us to go directly from the data to booking limits without estimating the parameters of an underlying distribution.

Let us assume that we have historic observations of demand by booking class for each of T preceding departures of a flight (same flight number, same time, same day of week).

As usual, we assume n booking classes with fares $p_1 > p_2 > \dots > p_n$, and we assume no cancellations or no-shows. We seek to calculate the booking limits for future flight $T + 1$, which we denote as $b_1, b_2, \dots, b_n$. Since there are no no-shows or cancellations, we know that $b_1 = C$.

The classical approach to calculating (and the approach used by most airlines, hotels, etc.) is to make a probabilistic forecast of demand based on historical demand and apply an algorithm similar to those described in Sections 9.1 through 9.3. However, the data-driven approach tackles the problem by asking: *What set of booking limits would have maximized revenue if we had applied them to flights 1, 2, ..., T?* Those would then be the booking limits that we would apply to flight $T + 1$. The logic is that if booking-class demands for flight $T + 1$ are being drawn from the same distributions as those for flights 1 through T , then the set of booking limits that maximizes total revenue across those flights would be good ones to apply to flight $T + 1$.

To formalize our statement, let d_i^t be the total demand for booking class i on flight t for $1 \leq t \leq T$ with booking limits $b_1, b_2, \dots, b_n$. Then, the bookings that would be realized in class i on flight t would be

$$\begin{aligned} x_n^t &= \min[d_n^t, b_n], \\ x_{n-1}^t &= \min[d_{n-1}^t, b_{n-1} + b_n - x_n^t], \\ x_i^t &= \min[d_i^t, b_i + b_{i+1} + \dots + b_n - x_{i+1}^t - x_{i+2}^t - \dots - x_n^t] \quad \text{for } i = 1, 2, \dots, n-2 \end{aligned}$$

In words, the number of bookings accepted in each booking class on each flight is the minimum of the demand for that booking class and the difference between the sum of the booking limits for all lower classes and the bookings accepted in those classes. Let us denote $x_i^t = \phi(d_i^t, \mathbf{b})$, where $\mathbf{b} = (b_1, b_2, \dots, b_n)$ is the vector of booking limits. The optimal data-driven booking limits can be found by solving the optimization problem

$$\mathbf{b}^* = \arg \max \sum_{t=1}^T \sum_{i=1}^n b_i \phi(d_i^t, \mathbf{b}), \quad (9.11)$$

subject to $b_1 = C, b \geq 0$.

In the case of two booking classes, the optimization problem in Equation 9.11 is linear and can be solved using standard linear programming approaches. In the case of multiple booking classes, the problem is not linear, but it is generally not difficult to solve given the simple constraint structure.

Example 9.6

A flight has a capacity of 100 seats and three booking classes: B-Class with a fare of \$160, M-Class with a fare of \$240, and Y-Class with a fare of \$700. As usual, B-Class books before M-Class, which books before Y-Class. The flight has been in operation for the last 10 weeks, and the demands for each of the three fare classes for the last 10 Monday flights are shown in Table 9.8. To find the optimal booking limits for this flight, the airline solves the optimization problem in Equation 9.11 using the demands shown in Table 9.8. The results are booking limits of 39 for B, 69 for M, and 100 for Y. With these limits, the *average* loads per flight and revenue per flight are 39 and \$6,240 for B-Class, 28.4 and \$6,816 for M-Class,

TABLE 9.8
*Booking demands by fare class for the
 previous 10 departures of the flight in
 Example 9.6*

	Demand by booking class		
	B	M	Y
1	60	25	22
2	72	36	30
3	74	42	31
4	56	33	21
5	50	30	22
6	44	27	17
7	63	31	31
8	80	45	49
9	70	40	30
10	41	22	21

and 25.6 and \$17,920 for Y-Class, with an average revenue of \$30,976 per flight. This could be considered a forecast of the performance of the next flight.

In some ways, the data-driven approach to capacity allocation is quite appealing. It does not require arbitrary assumptions about the distribution of demand for a particular booking class–flight combination or about the independence (or lack of it) in demands for flight–booking class combinations. It collapses the two steps of “forecast” and “optimize” required by classical approaches into a single “optimize” step. Although it requires observations of total demand by booking class and flight that are often not available, this is not a disadvantage relative to the forecast-and-optimize approach, because forecasting demand also typically requires estimates of historical demand (not just loads).

Despite these apparent advantages, a pure data-driven approach has not (to my knowledge) been implemented at any revenue management company in 2019. There are several likely reasons for this. The first is that most revenue management companies have implemented some flavor of the classical “forecast, then optimize” systems, which are generally working well enough that there is no financial incentive to rip them out to implement an entirely new approach. Second, forecast-then-optimize approaches have the advantage that they allow revenue managers to update forecasts to account for anticipated changes in demand. For example, a hotel owner who knows that the Super Bowl will be held in his city next year would have his revenue manager increase the demand forecasts for those dates well before the date of the game. The revenue management system would then reoptimize capacity controls given the new forecasts. It is difficult (but not impossible) to incorporate such user knowledge as overrides in a data-driven system. It is also not clear how to incorporate consumer choice into data-driven models. Addressing these issues is a topic of current research.

9.5 CAPACITY ALLOCATION IN ACTION

We have discussed several techniques for calculating booking limits. These techniques need to be used within the context of a booking control structure, as described in Section 8.5. In

practice, booking limits are periodically calculated offline for a flight and transmitted to the reservation system, where they are used to control bookings. The combination of periodic recalculation of booking limits and the nested booking limits within the reservation system defines a *dynamic booking control process*.

Dynamic Booking Control Process

1. Calculate the initial booking limits for a flight.
2. When a booking request for class j is received, compare the number of seats requested with the current booking limit for the class.
3. If the number of seats requested exceeds the booking limit for class j , reject the request and go to step 6.
4. If the number of seats requested is within the booking limit for class j , accept the request. Depending on the capabilities of the reservation system, update booking limits using either standard nesting or theft nesting.
5. When a booking cancels, either (a) increment the booking limits for classes with positive availability by the number of seats canceled (irreversible) or (b) increment to booking limits for *all* classes by the number of seats canceled (reversible).
6. Periodically recalculate all booking limits based on total bookings and demand expected in each class until departure. (This recalculation is often called a *reoptimization*.)

This dynamic booking control process makes no assumption about the nature of the reoptimization approach being used or the frequency with which reoptimizations occur. In fact, there is considerable variation among companies in the optimization approaches used and the frequency and situations under which they are applied. There is a general belief that frequent reoptimization is important—particularly when demand is changing rapidly or departure is approaching and bookings and cancellations are being received at a high rate. Often, the trade-off is between using a more complex algorithm and reoptimizing less often versus using a simpler algorithm and reoptimizing more often. These issues are the topics of ongoing research and are by no means settled.

9.6 MEASURING CAPACITY ALLOCATION EFFECTIVENESS

As noted in Chapter 8, revenue per available seat mile (RASM) is the metric most widely used to measure the overall effectiveness of revenue management at an airline. RASM is a good high-level measure, but it does not tell the entire story. It is often not immediately clear why RASM is high or low in a particular market. Are we achieving a high RASM in a particular market relative to the competition because we have superior service? Superior marketing? Better revenue management? All of these? Or have we simply assigned too few seats to the market? RASM by itself does not address the important need to isolate and measure the effects of revenue management separately from other influences in a market.

The need to understand the effects of revenue management is important because revenue management software systems are expensive, usually requiring millions of dollars of investment in software and hardware. Revenue management programs are also expensive to operate—a major airline may have a department of 50 or more people dedicated to running the revenue management system and reviewing and updating its forecasts and recommendations. How can management be convinced that this expensive and somewhat esoteric department is yielding results that justify this investment? A metric specifically targeted at measuring revenue management effectiveness is required.

One popular approach to measuring revenue management effectiveness is the *revenue opportunity model* originally developed by American Airlines (Smith, Leimkuhler, and Darrow 1992). The revenue opportunity model assumes we have a record of all the booking requests we received for a flight—those we accepted and those we rejected. We then measure the revenue that could be achieved from a specific flight under two extreme cases. In the base case, bookings are accepted in the order in which they are received until the capacity constraint is reached or demand is exhausted, whichever comes first. The revenue achieved in the base case is the flight's performance without any revenue management—we can call it the “No RM Revenue.” In the perfect revenue management case, bookings are accepted in decreasing order of revenue until demand is exhausted or the capacity constraint is reached. In this case, no booking is accepted that would displace a higher-fare future booking. This represents a case in which we achieved as much revenue as possible given the demands for different fare classes. We call the revenue calculated this way the “Perfect RM Revenue.” Of course, this level of performance is not achievable because it would require perfect foresight on future demand.

It should be clear that the revenue achieved under perfect revenue management will always be greater than or equal to the revenue achieved under no revenue management. If the flight is full, the revenue achieved under any actual revenue management program will generally lie between these extremes—better than no revenue management but never better than perfect revenue management.⁶ The *total revenue opportunity* on a flight is the difference between the revenue achieved from perfect revenue management and the revenue achieved from no revenue management. Then the *revenue opportunity metric* (ROM) is the revenue actually achieved from that flight minus the revenue that would have been achieved under no revenue management expressed as a percentage of the total revenue opportunity.

Example 9.7

Senilria assigned a 100-seat aircraft to a particular route with four booking classes. The fares for the four booking classes were $F = \$1,000$, $Y = \$800$, $M = \$600$, and $B = \$200$. Using its sophisticated revenue management system, Senilria set booking limits of (32, 45, 70, 100) for the flight. Actual demand for this flight by booking class came in at $F = 25$, $Y = 30$, $M = 20$, and $B = 50$. To determine the ROM for the flight, calculate as follows:

$$\begin{aligned} \text{Perfect RM revenue} &= 25 \times \$1,000 + 30 \times \$800 + 20 \times \$600 + 25 \times \$200 \\ &= \$66,000 \end{aligned}$$

$$\text{No RM revenue} = 30 \times \$800 + 20 \times \$600 + 50 \times \$200 = \$46,000$$

$$\begin{aligned} \text{Realized revenue} &= 25 \times \$1,000 + 25 \times \$800 + 13 \times \$600 + 32 \times \$200 \\ &= \$59,200 \end{aligned}$$

Thus, Senilria realized a ROM of $(\$59,200 - \$46,000) / (\$66,000 - \$46,000) = 66\%$ on this flight departure.

Example 9.7 illustrates the calculation of ROM using a single flight departure. In practice, ROM is only meaningful when applied to a large number of flights. And even then, *changes* in ROM are probably more meaningful than the absolute measure itself. For example, for the case of the 90-seat flight with results shown in Table 9.4, the expected revenue that was achieved using an optimal capacity allocation algorithm was \$74,003. For the 500,000 replications of this flight, the first-come, first-served revenue was \$61,215 and the maximum possible revenue was \$79,458. This means that the ROM achieved from using an optimal algorithm was $(\$74,003 - \$61,215) / (\$79,458 - \$61,215) = 70.1\%$. Since this result was achieved using an optimal algorithm with the same means and standard deviations that were used to generate the actual loads, it represents an upper bound to the average revenue that could be achieved in practice.

ROM has been used by airlines, hotels, and other RM companies to measure the effectiveness of their revenue management programs. American Airlines claimed that its systemwide ROM increased from 30% in 1988 to 49% in 1990 as the result of better revenue management systems and processes (Smith, Leimkuhler, and Darrow 1992). ROM has also proven useful in evaluating the performance of different forecasting and optimization algorithms as well as in measuring the performance of revenue managers. However, it can be sensitive to changes in market conditions. For example, the ROM for a flight with very low demand will always be close to 100%, since every booking will be accepted. On the other hand, a flight with high uncertainty in high-fare demand (a high standard deviation) will tend to have a lower ROM than one with the same expected high-fare demand but less uncertainty (a lower standard deviation). This means that changes in ROM for a flight or group of flights may be due to changes in the underlying structure of demand rather than changes in revenue management effectiveness.

9.7 SUMMARY

- The capacity allocation problem is how to regulate booking requests for a product consisting of a single resource over time in order to maximize revenue or contribution. For most revenue management companies, this is solved by periodically setting and updating booking limits for a nested booking control structure. The problem is typically posed assuming that bookings occur in order of increasing fare.
- At the heart of the two-class problem is the trade-off between accepting too many discount bookings, which cannibalizes future full-fare demand, and accepting too few discount bookings, which means that the plane departs with empty seats. Littlewood's rule finds the booking limit that balances these two risks to maximize total expected contribution.

- The underlying trade-off when there are more than two fare classes is the same as in the two-class problem: accepting too many early discount bookings may cannibalize future full-fare bookings, whereas accepting too few may lead to empty seats. However, when there are more than two fare classes, there is no easy closed-form solution for finding the optimal booking limits. Rather, airlines and other revenue management companies typically rely on various heuristic approaches, of which EMSR-a and EMSR-b are the best known.
- The basic EMSR approaches to multiple-fare class booking control do not incorporate the fact that opening or closing a fare class will change the anticipated demand in other fare classes for the same product. EMSR heuristics are often modified to incorporate the phenomenon of dependent demand, also known as *buy-up* and *sell-down*.
- Incorporating a fully flexible model of consumer choice is complex because the combinations of booking classes that could be offered is 2^n , where n is the number of classes. However, the number of efficient sets, which are the only ones that need to be considered, may be much smaller than the number of possible sets. Furthermore, the efficient sets can be nested in order of increasing revenue and potential demand in a way that facilitates nesting.
- An alternative to explicit optimization or heuristics is a data-driven approach in which the booking limits that would have maximized expected revenues for past flights is calculated and applied to future flights.
- Typically, the decision of whether to accept a particular booking request is based on whether there is sufficient availability for that class in the booking system. The function of the revenue management system is to periodically update the booking limits in the system based on current bookings and changes in the market.
- A common way for airlines and other revenue management companies to measure the effectiveness of their capacity allocation is by using the revenue opportunity metric, or ROM. ROM measures the revenue actually achieved as a percentage of the maximum revenue that could have been achieved if the supplier had perfect forecasts of future demand. ROM is not an absolute measure, because underlying uncertainty in demand ensures that 100% ROM can never be achieved. But it is a good relative measure of how well different flights are being managed.

Capacity allocation is the first of the three constituents of revenue management that we study. Network management, which is treated in Chapter 10, extends the ideas from this chapter to the case in which a company is selling products that consist of more than one resource. Overbooking, treated in Chapter 11, extends the idea of capacity allocation still further, to the situation in which bookings can cancel or not show.

9.8 FURTHER READING

As noted in Section 9.2, there is no closed-form solution to the multiclass booking problem. Curry 1990, Wollmer 1992, and Brumelle and McGill 1993 describe three different solution approaches utilizing dynamic programming. A comparison among these three

approaches is given in Li and Oum 2002. The equivalence between Littlewood's rule and the critical fractile approach is described in Netessine and Shumsky 2002.

The seminal paper on incorporating choice in optimal capacity allocation is Talluri and van Ryzin 2004a. The example in Section 9.3.3 was adapted from Talluri and van Ryzin 2014. A good survey of research on incorporating choice into capacity allocation is Strauss, Klein, and Steinhardt 2017, which provides many additional references to research.

9.9 EXERCISES

1. Roll back the decision tree in Figure 9.7 to determine a formula for the period 3 booking limit that would maximize expected net revenue given the displacement probabilities q_1 and q_2 .
2. Granite State Airlines serves the route between New York and Portsmouth, New Hampshire, with a single-flight-daily 100-seat aircraft. The one-way fare for discount tickets is \$100, and the one-way fare for full-fare tickets is \$150. Discount tickets can be booked up until one week in advance, and all discount passengers book before all full-fare passengers. Over a long history of observation, the airline estimates that full-fare demand is normally distributed, with a mean of 56 passengers and a standard deviation of 23, while discount-fare demand is normally distributed, with a mean of 88 passengers and a standard deviation of 44.
 - a. A consultant tells the airline it can maximize expected revenue by optimizing the booking limit. What is the optimal booking limit?
 - b. The airline has been setting a booking limit of 44 on discount demand, to preserve 56 seats for full-fare demand. What is its expected revenue per flight under this policy? (Hint: Use a spreadsheet.)
 - c. What is the expected gain from the optimal booking limit over the original booking limit?
 - d. A low-fare competitor enters the market and Granite State Airlines sees its discount demand drop to 44 passengers per flight, with a standard deviation of 30. Full-fare demand is unchanged. What is the new optimal booking limit?
3. Granite State Airlines also serves the route between Washington, D.C. (Reagan), and Portsmouth, New Hampshire, with a single flight daily. The airline sells both discount-fare and full-fare tickets. The airline has assigned a 100-seat aircraft to the flight. Using Littlewood's rule, the airline has determined that the optimal discount-fare booking limit for a flight departing in two weeks is 40.
 - a. What is the protection level for the flight?
 - b. The revenue manager has learned that the annual Yorkshire Terrier Fancier's Convention will be in Portsmouth in two weeks. As a result, he increased the forecast of mean discount-fare demand by 50% over its previous value. The full-fare forecast remained the same. What are the new booking limits and protection levels for this flight?

TABLE 9.9
*Booking demands by fare class for the
 previous 12 departures for the Senilria flight
 described in Exercise 4*

	Demand by booking class		
	B	M	Y
1	184	84	78
2	88	36	36
3	132	64	58
4	102	50	36
5	60	20	20
6	176	86	76
7	126	54	44
8	154	74	70
9	182	82	82
10	160	74	72
11	130	60	54
12	88	38	28

- c. Right after learning about the Yorkshire Terrier Fancier's Convention, the revenue manager learned that, because of maintenance problems, there will also be a switch of aircraft on the flight. Instead of a 100-seat aircraft, the flight will be served by a 120-seat aircraft. What are the new booking limits and protection levels for this flight?
 - d. The full fare for the flight is \$200, and the discount fare is \$100. What is the expected full-fare demand for this flight, assuming that both discount-fare and full-fare demand follow normal distributions?
4. Senilria Airlines has started a new flight from Rome to Palermo, Sicily, once a week on Monday mornings and has operated the flight for the last 12 weeks. Senilria offers three booking classes on the flight: B-Class with a fare of €70, M-Class with a fare of €120, and Y-Class with a fare of €200. The demand for each fare class for each of the previous 12 flight departures is shown in Table 9.9.
- a. Assume that Senilria has assigned a 200-seat aircraft to this flight for the initial 12 flights and for the next flight. Using the data-driven approach, what would be the recommended booking limits for each of the three classes for the 13th departure? What is the expected revenue?
 - b. Assume that Senilria is considering assigning a 225-seat aircraft to the Rome-to-Palermo flight for the next departure. Given a capacity of 225 seats, what are the recommended booking limits for each of the three classes for the 13th departure? What is the expected revenue? What is the additional revenue from the 25 seats of extra capacity?

NOTES

1. Here we use the notation $\bar{F}(x)$ to denote the *complementary cumulative distribution (CCDF)* associated with the cumulative distribution function $F(x)$. It is defined by $\bar{F}(x) = 1 - F(x)$. More details can be found in Appendix B.
2. We use the *discrete* version of the normal distribution here, so some of the expressions that follow need to be treated with care. See Appendix B for details.
3. That is, if $\Phi(y) = x$, then $\Phi^{-1}(x) = y$. Some properties of the inverse normal distribution are discussed in Appendix B.
4. Airlines typically find that the standard deviation of demand ranges from 25% to 75% or more of the expected demand. This is equivalent to saying that the *coefficient of variation*, defined as σ/μ , ranges from 0.25 to 0.75.
5. The term “bid price” is unfortunate because it is easily confused with the idea of bidding in an auction. This is not the situation in revenue management usage—the bid price is the minimum price that a seller would accept for another unit of a resource (such as a seat on a flight). The closest concept in auction theory is the *reserve price*, which is the minimum amount a seller would accept for an item. However, the terminology is standard, so we use it.
6. It is possible for achieved revenue to be lower than the no-revenue-management case if too many early-booking requests were rejected in anticipation of future bookings that did not materialize.

10 NETWORK MANAGEMENT

When every product offered by a supplier uses only a single resource, the supplier can maximize total profitability by maximizing the profitability of each resource independently. For example, a Broadway show can maximize total revenue over a season by setting prices and availabilities that maximize revenue independently for each performance.¹ However, a seller that sells products that use multiple resources—such as an airline selling tickets on connecting flights or a hotel selling multiple-night stays—faces a much more complex problem. The seller can no longer maximize total contribution by maximizing contribution from each resource independently. Rather, he needs to consider the interactions among the various products he sells and their effect on his ability to sell other products. This is the challenge of *network management*—the topic of this chapter.

Chapter 9 describes the techniques used to maximize expected profitability for a single resource. This is all that is needed by airlines such as EasyJet that only allow single-leg bookings.² These airlines—like the Broadway show—can maximize total profitability by maximizing the profitability of each resource independently. However, this approach will not maximize system profitability for airlines that offer passengers the opportunity to buy tickets for two or more connecting flight legs. Hotels and rental car companies also sell multiple-resource products, since hotel guests can stay for more than one night and rental car customers can rent for more than a day. *Network management* is applicable to any company controlling a set of constrained and perishable resources and selling products that consist of combinations of those resources.

Section 10.1 introduces the network management problem and the types of industries in which it applies. The section also demonstrates why the problem is complex and why a simple and intuitive greedy heuristic does not work. The subsequent sections present various solution approaches to the network management problem. One intuitive approach is linear programming. However, linear programming is a computationally intensive procedure that can only be run on a periodic basis. In the interim, most airlines use some version of *virtual nesting* to manage booking requests. We examine virtual nesting and the concept of *net leg fare* (sometimes called *net local fare*), which is a consistent way to rank (and hence nest) network booking requests. Section 10.4 shows how *bid pricing* can be used in network

management, and Section 10.5 discusses some of the issues involved in implementing network management.

10.1 WHEN IS NETWORK MANAGEMENT APPLICABLE?

Network management is important in any revenue management industry that sells products consisting of more than one resource. As shown in Table 8.7, there are some revenue management industries, such as cruise lines and event ticketing, in which network management does not play a major role.³ By contrast, in the industries shown in Table 10.1, managing products that consist of combinations of resources is a critically important aspect of revenue management.

Network management is very important to business hotels, rental car companies, and multiunit housing—and indeed to any company that rents out capacity for periods of varying duration. Network management is also important to airlines that offer connecting service. This can be seen from Figure 10.1, which shows the distribution of passengers by

TABLE 10.1
*Examples of multi-resource products from various
revenue management industries*

Industry	Resource unit	Multi-resource product
Passenger airline	Seat on a flight leg	Multileg itinerary
Hotel	Room night	Multinight stay
Rental car	Rental day	Multiday rental
Passenger train	Leg	Multileg trip
Container shipping	Leg	Multileg routing
Multifamily housing	Unit/month	Multimonth rental

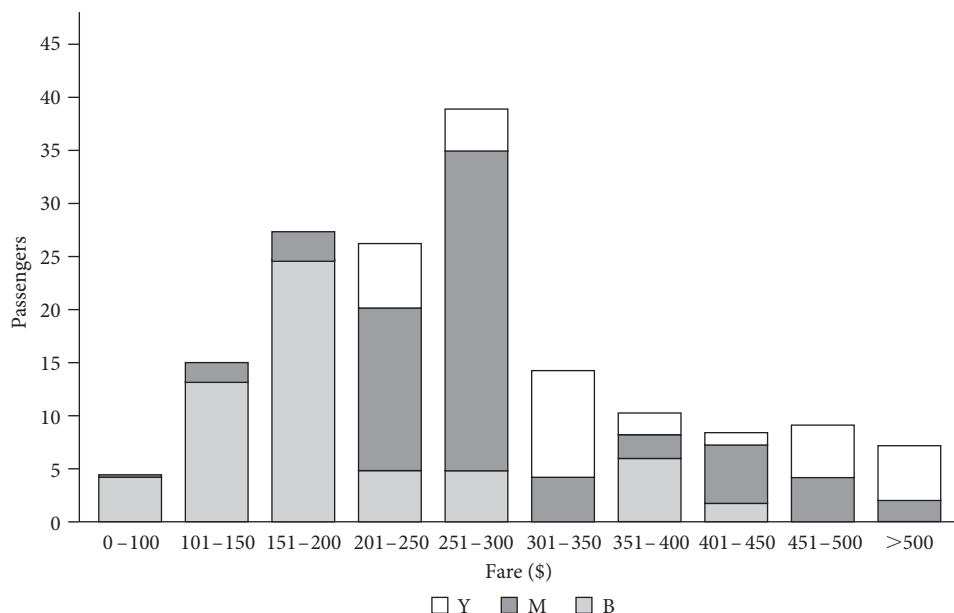


Figure 10.1 Distribution of total fares by fare class on an example flight.

fare class and total fare paid on a flight from San Francisco to Chicago. There are three fare classes on this flight: full fare (Y), discount (M), and deep discount (B). In general, Y-Class passengers tend to pay more than M-Class passengers, who tend to pay more than B-Class passengers. However, fare class is not a perfect proxy for revenue. Some B-Class passengers pay more than \$400 and some Y-Class passengers pay less than \$250. Managing this flight based on fare class alone will not maximize revenue. If this airline closes B-Class, it will still be accepting Y-Class (and M-Class) passengers who are paying less than \$250 while rejecting B-Class passengers willing to pay more than \$400 for a seat. Furthermore, managing by fare class does not allow the airline to take advantage of the spread of fares within each class. If capacity on this flight were constrained, the airline might want to accept only passengers paying more than, say, \$200.

Consider an airline that has San Francisco and Chicago as hubs. This means that the San Francisco-to-Chicago flight carries passengers traveling from many origins to many destinations. Only a small fraction of the passengers are actually flying from San Francisco to Chicago; the majority are connecting in either San Francisco or Chicago (or both). For example, the highest fares on the flight (those with total fare over \$3,000) might be flying from Tokyo to Chicago via San Francisco or from San Francisco to Paris via Chicago. Other passengers might be flying from San Francisco to Chicago to Grand Rapids or from Fresno to San Francisco to Chicago, and so on. Each origin-destination combination can be purchased in different fare classes, adding even more complexity. The fare for an M-Class Tokyo-to-Chicago passenger is likely to be higher than a Y-Class San Francisco-to-Chicago passenger. Yet leg-based capacity allocation would not allow the airline to accept the M-Class Tokyo-to-Chicago passenger while rejecting the Y-Class San Francisco-to-Chicago passenger. The goal of network management is to improve revenue by managing the mix of products as well as the mix of fare classes being sold for each product.

Airlines have estimated that network management adds 0.5% to 1% of additional revenue on top of the revenue gains from optimal capacity allocation and overbooking. This may not sound like much, but for a carrier such as British Airways with revenue of about 13 billion British pounds (\$16 billion) in 2018, it represents \$80 million to \$160 million in additional operating profit. Network management is even more critical for hotels and rental car companies. In these industries, managing the mix of bookings by length of stay (or length of rental) provides more leverage than managing by fare class alone. For a hotel, it is usually more important not to accept short-stay bookings that displace future longer-stay booking requests than it is to worry about the rate mix of any individual booking length. For this reason, unlike the airlines, the first successful hotel (and rental car) revenue management systems included network management capabilities.

10.1.1 Types of Network

In the 1960s, major airlines in the United States began to establish *hub-and-spoke* networks as a way to better leverage their operations. Figure 10.2 shows a hub-and-spoke network with 2 *hubs* and 13 *spoke cities*. Each spoke city is connected to one or more of the hubs. Arrivals and departures at the hubs are timed so that passengers arriving from cities to the west of the hub can connect to cities to the east, and vice versa. The hub-and-spoke

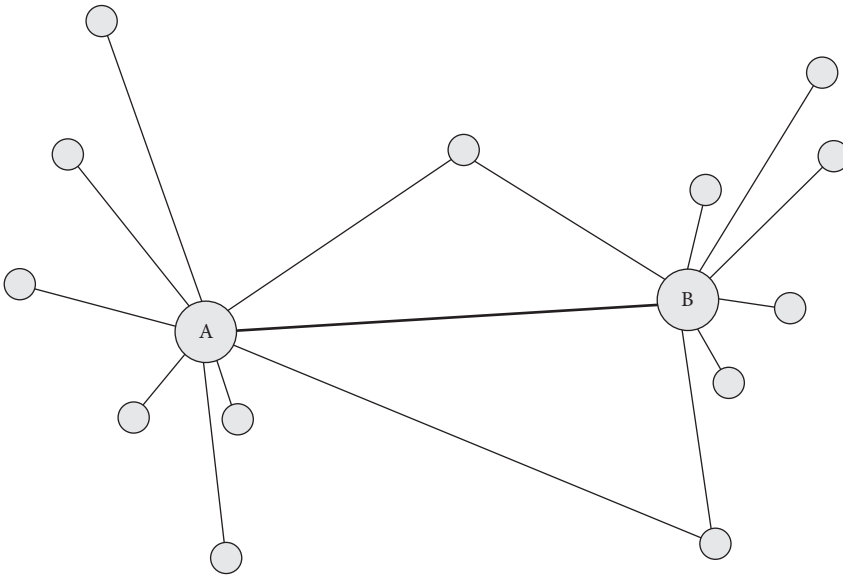


Figure 10.2 Example of a hub-and-spoke network. Nodes A and B are hubs; the other nodes are spoke cities.

network allows an airline to offer a large number of products with a relatively small number of flights. With 20 eastbound flights into a hub connecting with 20 westbound flights, an airline can offer 440 products (all possible connections plus the direct flights in and out of the hub) with only 40 total flights. While hub-and-spoke airlines offer some direct flights between spoke cities, the vast majority of their traffic touches at least one hub—as either an origin, a destination, or a connecting point. Airlines operating hub-and-spoke networks include the majority of global carriers including American, Delta, United, Air France, British Airways, and Singapore Airlines, among many others. During the 1980s, most large airlines organized their operations around the hub-and-spoke model and saw their connecting traffic soar. In the United States, about 35% of passengers travel on connecting flights (Berry and Jia 2010). While rushing to make a connection may not be popular with passengers, the efficiency of hub-and-spoke networks means that they are here to stay.

The resources for a hotel are room nights, and its products are combinations of arrival date and length of stay. A hotel essentially runs a *linear network* in which each room night connects to the preceding night and to the following night. Such a network is shown in Figure 10.3. Passenger railway networks are also linear; for example, Amtrak's California Zephyr starts at Emeryville, California, and terminates in Chicago, making 33 intermediate stops along the way. A passenger boarding the Zephyr at Emeryville can get off at any one of 34 stops (the 33 intermediate stops plus Chicago); a passenger boarding at the second stop (Martinez, California) can get off at any one of 33 stops; and so on. Each of the 34 legs is a resource, and each combination of origin and destination is a product. The typical passenger railroad has a linear network similar to the hotels and rental car companies, but with the length of the network equal to the number of legs being offered.

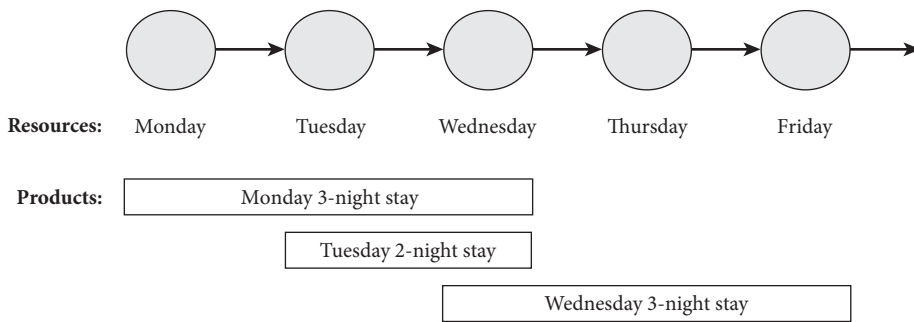


Figure 10.3 A linear network, typical of hotels, passenger rail, and rental car companies.

Each product in a network has one or more associated fare classes. The network management problem is to determine which booking requests to accept for every possible combination of product and fare class at every time. We use the term *origin-destination fare class* (or ODF) for a product–fare class combination.⁴ While ODF is an airline term, we use it to refer to a combination of product and fare class in any industry. Thus, for a hotel, a four-night *rack-rate*⁵ booking arriving June 13 is a different ODF than a three-night discount booking arriving on June 14. The goal of network management is to manage and update the availability of all ODFs over time to maximize expected contribution.

The number of products being offered by a hotel or rental car company can be very large—much larger, in fact, than the number of rooms in the hotel. A hotel accepting reservations for customers arriving for the next 365 days with stays from 1 up to 15 days in length is offering $15 \times 365 = 5,475$ different products. In fact, the number of products can be even larger, since each room type corresponds to a different product dimension. A particular hotel might have four different types of room (for example, standard, deluxe, deluxe view, and suite), for which it can charge different rates. Since a customer can book any one of these room types for any arrival date and length of stay, the number of products being offered by the hotel is actually $4 \times 15 \times 365 = 21,900$. In other words, a hotel may well be offering 100 times more products than it has rooms! On the other hand, cruise lines typically offer only one or two options for embarkation and disembarkation per sailing. The different products on a particular sailing are determined by the different cabin types, for example, upper-deck outside versus lower-deck inside.

10.1.2 A Greedy Heuristic for Network Management—and Why It Fails

Consider a 100-room hotel facing the *unconstrained demand* pattern for some future week shown in Figure 10.4. (Unconstrained demand is the total number of room nights the hotel could sell if it accepted every booking request.) The pattern shown in Figure 10.4 is typical of a midtown business hotel—demand is low on the weekend and peaks during the middle of the week. In this example, unconstrained demand exceeds capacity on Wednesday. This means that the hotel will need to reject some bookings or it will be oversold on Wednesday (ignoring no-shows and cancellations). The network management problem faced by the

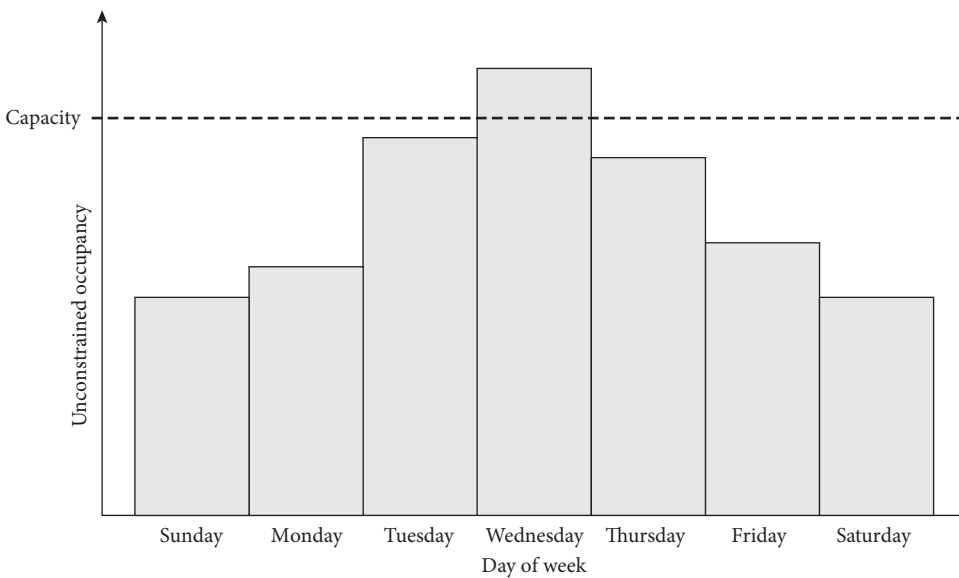


Figure 10.4 Typical unconstrained occupancy pattern for a midtown business hotel.

hotel is to decide which bookings to accept and which to reject *considering both room rate and length of stay* in order to maximize expected contribution margin.

Let us say that the hotel offers two rates: a rack rate of \$200 per night and a discount rate of \$150 per night. The hotel manager might look at the demand pattern in Figure 10.4 and conclude (correctly) that the hotel's constrained resource is Wednesday night capacity. The manager might reason that the best way to maximize operating profit is to limit the number of discount-rate customers booking any product that includes Wednesday night to protect availability for rack-rate customers. In other words, the hotel might manage bookings based strictly on fare class. This approach might improve revenue relative to a first-come, first-served policy, but it has an obvious drawback. Rack-rate customers arriving Wednesday for a one-night stay pay \$200. But discount-rate customers arriving Tuesday for a three-night stay pay \$450. If the hotel limits discount bookings to save room for full-fare bookings, it runs the risk of turning down a \$450 customer to save room for a \$200 customer—a prime revenue management sin. Clearly a more sophisticated approach is required if the hotel is to maximize expected revenue.

When there is only a single constrained resource, the manager could arrange all of the ODFs that use the constrained resource by total rate paid and solve the multiclass problem using EMSR or a similar heuristic.⁶ This would mean that the lowest class on Wednesday would be one-night discount customers paying \$150, the next highest class would be one-night rack customers paying \$200, followed by two-night discount stays for either Tuesday–Wednesday or Wednesday–Thursday paying \$300, followed by two-night rack customers paying \$400, and so on. By establishing classes in this fashion and setting the appropriate booking limits, the hotel would maximize expected operating contribution as long as unconstrained demand exceeded capacity *only on Wednesday night*.

Sorting all the ODFs on each leg in fare order and setting availabilities based on total fare is called the *greedy heuristic* for network revenue management. The greedy heuristic is optimal if there is only a single constrained resource (that is, only a single bottleneck). It works perfectly in the hotel example if unconstrained demand for hotel rooms exceeds capacity *only* on Wednesday night. However, the greedy heuristic breaks down quickly when more than one resource is constrained. This would be the case for the hotel if unconstrained demand exceeded 100 rooms on both Wednesday and Thursday nights. In this case, it is no longer optimal for the hotel to order fare classes strictly by total revenue. The hotel should accept a two-night Wednesday–Thursday discount booking paying \$300 if it believes that Thursday is not going to sell out. But if it believes that Thursday is going to sell out, it may be better off rejecting the \$300 two-night booking in favor of waiting for two \$200 one-night rack-rate bookings for Wednesday night and Thursday night.

The implication is that when multiple resources are constrained, the right ordering of ODFs on a leg can depend on the fares and demands for all the other ODFs as well as the capacities of all the other resources in the network. In fact, as we accept bookings over time and update our forecasts of future demand, the ordering of ODFs on each leg may change as well: there is no stable optimal ordering of ODFs when multiple resources in a network can be constrained.

The failure of the greedy heuristic can be illustrated with the simple network shown in Figure 10.5, where an airline offers two flights—Flight 1 from San Francisco (SFO) to Denver (DEN) and Flight 2 from Denver (DEN) to St. Louis (STL)—that have been scheduled so that passengers can connect from Flight 1 to Flight 2 at Denver. For simplicity, let us assume that the airline offers a single fare for each product and that the San Francisco–to–Denver fare is \$200, the Denver–to–St. Louis fare is \$160, and the San Francisco–to–St. Louis fare is \$300. The airline has exactly one seat left on each of the two flights and has a customer who is willing to pay \$300 for a ticket from San Francisco to St. Louis. Should the airline let her book? Or should the airline reject the booking in the hopes of booking local passengers on both legs? Let p_1 be the probability that the airline will receive at least one future booking request for Flight 1 and p_2 be the probability that the airline will receive at least one future request for Flight 2. Then the expected value of refusing the connecting customer and letting only the local customers book is $200p_1 + 160p_2$. If this amount is greater than \$300, then the airline is better off refusing the connecting customer and relying on local traffic to fill the last seats. If, on the other hand, $200p_1 + 160p_2 < 300$, then the airline should go ahead and allow the connecting passenger to book.

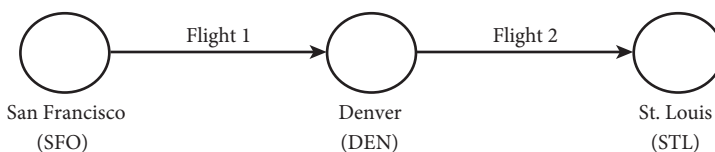


Figure 10.5 A simple hub-and-spoke network.

The probabilities p_1 and p_2 are the airline's forecasts of future local demand on the two flights. Whether or not the airline should accept the through passenger depends on these forecasts. This can be generalized to a full network: There is no unambiguous answer to the question of whether an airline should prefer connecting traffic to local traffic on a given leg. The right answer depends on the forecasts of future demand for all ODFs. Ultimately, when many resources are constrained, we need to optimize over the entire network to determine which bookings we prefer on any leg in the network.

10.2 A LINEAR PROGRAMMING APPROACH

Linear programming (LP) is a natural approach to allocating scarce resources to competing product demands. In fact, if we take the gigantic leap of assuming that future demands are known with certainty, linear programming provides an exact solution to the network management problem. Since uncertainty is a key element of revenue management decision making, a solution derived while ignoring uncertainty is not wholly satisfactory. However, the linear programming formulation of the network management problem is worth studying for three reasons. First, it solves the capacity allocation component of network management and provides insight into the nature of the optimal solution. Second, the solution to the network management linear program provides a good starting point for a fully optimal solution. In many hotels and airlines, linear programming is used to determine an initial solution, which is then adjusted to account for the uncertainty in future demand. Finally, the linear program generates marginal values of capacity as a by-product. These marginal values approximate the opportunity costs of the capacity and can serve as an estimate of network bid prices, which is discussed in Section 10.4.

To formulate the network management problem as a linear program, assume we know the future demand for every ODF with certainty. Furthermore, assume that this demand will not be influenced by which ODFs we open and close (in other words, we have independent demands on each flight and among flight legs). We have m resources and n ODFs utilizing combinations of those resources with $n \geq m$. We use the subscript i to index resources and the subscript j to index ODFs. Each resource i has a constrained capacity $C_i > 0$. Each ODF has a known demand $d_j > 0$ and a net contribution margin $p_j > 0$. Let $x_j \geq 0$ be the amount of ODF j we will allow to book (our allocation). Then the *deterministic network management problem* is to find the values of x_j for $j = 1, 2, \dots, n$ that maximize total net contribution subject to the constrained capacities of the resources.

To complete the formulation, we need to represent which resources are used to produce each ODF. To do this, we define the incidence variable a_{ij} as follows:

$$a_{ij} = \begin{cases} 1 & \text{if resource } i \text{ is used in ODF } j \\ 0 & \text{otherwise.} \end{cases} \quad (10.1)$$

To illustrate the use of the incidence variables, consider the simple flight network shown in Figure 10.5. There are two resources in this network: (1) Flight 1 from San Francisco to Denver and (2) Flight 2 from Denver to St. Louis. With three products and two fare classes, the airline is offering six ODFs:

1. San Francisco to Denver full fare
2. San Francisco to Denver discount
3. Denver to St. Louis full fare
4. Denver to St. Louis discount
5. San Francisco to St. Louis full fare
6. San Francisco to St. Louis discount

Using this numbering, $a_{11} = 1$ because the San Francisco-to-Denver full-fare ODF uses flight 1, while $a_{13} = 0$ because the Denver-to-St. Louis full-fare ODF does not utilize Flight 1. The values of all the incidence variables for this simple example are shown in Table 10.2.

It is instructive to compare the hub-and-spoke network in Table 10.2 with the pattern of incidence variables for a linear network. Consider a hotel that offers one-night, two-night, and three-night products only. The resources for the hotel are the room nights, and the products are combinations of arrival night and length of stay. Table 10.3 shows the values of the incidence variables (a_{ij}) for three different lengths of stay and a week of resources. In theory, the network management problem for a hotel stretches out indefinitely into the future. In practice, a hotel will only accept bookings for some limited period into the future (often a year), limiting the number of resources and products that need to be managed. Furthermore, in theory a hotel offers an infinite number of products, since customers can book any length of stay. In practice, lengths of stay longer than 14 nights are extremely rare at most hotels, and hotels usually manage them as a single product.

TABLE 10.2
*Values of the incidence variables (a_{ij})
for the two-flight example*

Resource (i)	ODF (j)					
	1	2	3	4	5	6
1	1	1	0	0	1	1
2	0	0	1	1	1	1

TABLE 10.3
Values of the incidence variables (a_{ij}) for a hotel

Resource	Arrival day														
	Sunday			Monday			Tuesday			Wednesday			Thursday		
	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
Sunday	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
Monday	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
Tuesday	0	0	1	0	1	1	1	1	1	0	0	0	0	0	0
Wednesday	0	0	0	0	0	1	0	1	1	1	1	1	0	0	0
Thursday	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1
Friday	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1
Saturday	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

NOTE: Numbers under the arrival day indicate length of stay in days.

10.2.1 The Deterministic Network Linear Program

We can now formulate the deterministic network management problem as a linear program:

$$\max_{x_j} \sum_{j=1}^n p_j x_j, \quad (10.2)$$

subject to

$$\sum_{j=1}^n a_{ij} x_j \leq C_i \quad \text{for all } i, \quad (10.3)$$

$$x_j \leq d_j \quad \text{for all } j, \quad (10.4)$$

$$x_j \geq 0 \quad \text{for all } j. \quad (10.5)$$

The variables $x_1, x_2, \dots, x_n$ are the total seats allocated to each ODF. The objective function 10.2 stipulates that the goal is to accept the demand that maximizes total net contribution. There is one constraint of the form of (10.3) for each resource. These constraints ensure that the airline can physically accommodate all the demand that it accepts. The demand constraints in (10.4) stipulate that sales of each ODF are limited by demand for that ODF. Finally, (10.5) ensures that all allocations are greater than or equal to zero.

Example 10.1

An airline is operating the two-flight airline network in Figure 10.5 and offering a full fare and a discount fare for each of the three itineraries. There are six ODFs using two resources, and the ODF-resource mapping is as shown in Table 10.2. The airline has assigned a 100-seat aircraft on Flight 1 from San Francisco to Denver and a 120-seat aircraft on Flight 2 from Denver to St. Louis; demands and fares are as shown in Table 10.4. Notice that the unconstrained demand is 160 on the San Francisco–to–Denver flight and 170 on the Denver–to–St. Louis flight. Since these demands exceed the flight capacities for the two flights, we will need to turn some passengers away. The linear program for this example is

$$\begin{array}{llllllll} \text{maximize} & 150x_1 & +100x_2 & +120x_3 & +80x_4 & +250x_5 & +170x_6 & \\ \text{subject to} & & & & & & & \\ & x_1 & +x_2 & & & +x_5 & +x_6 & \leq 100 \\ & & & x_3 & +x_4 & +x_5 & +x_6 & \leq 120 \\ & x_1 & & & & & & \leq 30 \\ & & x_2 & & & & & \leq 60 \\ & & & x_3 & & & & \leq 20 \\ & & & & x_4 & & & \leq 80 \\ & & & & & x_5 & & \leq 30 \\ & & & & & & x_6 & \leq 40 \\ & x_1, & x_2, & x_3, & x_4, & x_5, & x_6 & \geq 0 \end{array}$$

which can be solved easily using Excel's SOLVER or any other LP program.

TABLE 10.4
Two-flight network management example fares and demands

Number	ODF	Fare (\$)	Demand
1	San Francisco to Denver full fare	150	30
2	San Francisco to Denver discount	100	60
3	Denver to St. Louis full fare	120	20
4	Denver to St. Louis discount	80	80
5	San Francisco to St. Louis full fare	250	30
6	San Francisco to St. Louis discount	170	40

TABLE 10.5
Two-flight network management example solution

Number	ODF	Fare (\$)	Allocation	Revenue (\$)
1	San Francisco to Denver full fare	150	30	4,500
2	San Francisco to Denver discount	100	40	4,000
3	Denver to St. Louis full fare	120	20	2,400
4	Denver to St. Louis discount	80	70	5,600
5	San Francisco to St. Louis full fare	250	30	7,500
6	San Francisco to St. Louis discount	170	0	0
Total				24,000

The optimal allocations for Example 10.1 and the total net contribution are shown in Table 10.5. Note that in this case, we took all of the full-fare passengers for each product. We also took *some* (but not all) of the local discount traffic in both markets and none of the San Francisco–to–St. Louis discount through traffic. At first glance, it might seem surprising that we did not take any of the San Francisco–to–St. Louis discount traffic, since those passengers were paying the second-highest fare in the system—more than any of the local full-fare passengers and much more than any of the local discount passengers. But the reason we rejected them is obvious when we look at the total network: The *sum* of the fares paid by the local passengers was greater than the San Francisco–to–St. Louis discount fare. Given our demand forecasts, we gain more contribution for the system by rejecting the discount through passengers in favor of local demand.

10.2.2 Structure of the Optimal Solution

Notice that in the optimal solution shown in Table 10.5, ODFs fall into three categories:

1. ODFs for which we did not accept *any* bookings (San Francisco to St. Louis discount)
2. ODFs for which we accepted *some* but not all of the demand (Denver to St. Louis discount and San Francisco to Denver discount)
3. ODFs for which we accepted *all* of the demand (all of the full-fare ODFs)

In the optimal solution to the deterministic network management problem there will be, at most, one fare class for each product for which we accept some but not all demand.⁷ All of the other fare classes for that product will be all or nothing: we will either accept all of their demand or none of it. Acceptance of fare classes for a product proceeds in fare order:

For each product there is a set of fare classes (possibly empty) for which we accept all of the demand and a set of fare classes (possibly empty) for which we accept none of the demand. The fares for the fully accepted classes are all higher than the fares for those classes for which we accept none of the demand. Finally, there may be one class for which we accept only some of the demand. This class will have a fare *higher* than the fares of the rejected classes but *lower* than the fully accepted classes.

Example 10.2

An airline operates on the flight network shown in Figure 10.5 with the same total demand by product as in Example 10.1, but with five classes per product, with the fares and ODF demands shown in the fourth and fifth columns of Table 10.6. This airline is offering 15 ODFs. The optimal allocations are shown in the corresponding column of Table 10.6. For the San Francisco–to–St. Louis product, it is optimal to accept all passengers who pay more than \$200 and reject those who pay less. The airline should accept exactly 7 of those paying exactly \$200, out of a total demand of 10. Similarly, for the Denver–to–St. Louis product, the airline should accept everybody paying more than \$75 and reject everybody paying less than \$75 while accepting 13 out of 30 passengers willing to pay exactly \$75. For the San Francisco–to–Denver product, the airline should accept everyone paying more than \$130 and reject all requests whose associated fares are lower than \$100.

Example 10.2 shows that we never want to be accepting bookings for a lower fare class when we are rejecting them for a higher fare class *for the same product*. For the example in Table 10.6, there is no combination of demands, capacities, or fares for which it would be optimal to accept San Francisco–to–St. Louis \$170 bookings while rejecting San Francisco–to–Denver \$180 bookings. On the other hand, it is optimal in the example to reject \$190 San Francisco–to–St. Louis bookings that use the San Francisco–to–Denver leg while

TABLE 10.6
Expanded network solution

Number	Product	Class	Fare (\$)	Demand	Allocation	Revenue (\$)
1	San Francisco–Denver	A	180	10	10	1,800
2		B	160	20	20	3,200
3		C	140	15	15	2,100
4		D	130	18	18	2,340
5		E	100	27	0	0
6	Denver–St. Louis	A	130	15	15	1,950
7		B	110	20	20	2,200
8		C	90	15	15	1,350
9		D	80	20	20	1,600
10		E	75	30	13	975
11	San Francisco–St. Louis	A	260	20	20	5,200
12		B	240	10	10	2,400
13		C	200	10	7	1,400
14		D	190	15	0	0
15		E	170	15	0	0
Total						26,515

accepting \$130 San Francisco–to-Denver bookings. In network management, fare classes should be nested by product but not by resource.⁸

10.2.3 Strengths and Weaknesses of the Linear Programming Approach

Efficient linear programming solution packages, such as CPLEX, have made the linear programming formulation solvable, even for many large airlines. To account for uncertainty, expected demand is often used in the demand constraints (10.4), and various adjustments are made to account for overbooking and for the dynamic nature of bookings and cancellations. On the other hand, working with a large linear program is often cumbersome and produces a large number of outputs (the ODF allocations) that can be difficult to interpret and implement. A large hub-and-spoke airline may have many ODFs with very low levels of demand. For example, a major hub-and-spoke airline offers connecting service between many small cities that have very small demand—say, Portland, Maine, to Chicago and San Francisco to Santa Barbara, California. This may be a perfectly valid set of connections and therefore a product that the airline wants to offer, but the expected demand for this product would likely be fewer than one passenger per day and the expected demand per fare class for this product would be even smaller. Many hub-and-spoke airlines offer thousands of such ODFs with very low levels of demand—less than one expected booking per month in many cases. But these ODFs still need to be managed, and linear programming would require calculating and maintaining a booking limit for each of them. This is a tremendous amount of computational and implementation overhead to manage a large number of ODFs that collectively may only represent a small fraction of total demand and revenue.

There are other issues with linear programming as a network revenue management approach. One is that it is not immediately clear how to incorporate demand uncertainty. One obvious approach is to replace the deterministic demands for each ODF, d_j , with the *mean* demand for that ODF, μ_j . But analogy with the single-leg case will show that this will not provide an optimal solution: maximizing revenue based on expected demand does not maximize expected revenue.

Finally, the astute reader will notice that the network linear program has been formulated in terms of calculating *allotments* for different ODFs. As Section 9.4 shows, the allotment approach breaks down when bookings do not arrive in strict fare order. This is an even more pressing issue with network management than with leg-based management—it is extremely unrealistic to expect that passengers booking on different ODFs would oblige us by booking in strict order of value to the airline. In fact, passengers flying high-fare, multileg itineraries often book earlier than those flying lower-fare, single-leg itineraries. In theory at least, we could deal with the dynamics of network management by rerunning the linear program each time a booking is accepted or a cancellation takes place. However, the size and complexity of the corresponding linear program render this approach impractical. What we really need is a way to extend the concept of nesting to booking requests that utilize multiple legs. The most common way to do this in many travel-related industries is by *virtual nesting*, which is discussed in the next section.

10.3 VIRTUAL NESTING*

Virtual nesting is rather technical and specific to airlines and a number of other travel-related industries, and this section can be skipped by readers without a particular interest in travel-related industries.

Virtual nesting was the earliest approach to network management. It was first applied by American Airlines (Smith, Leimkuhler, and Darrow 1992), and variations on the basic idea are in use at many airlines today. Virtual nesting was developed to allow an airline to apply the preexisting, leg-based control structures in its reservation system to the network management problem with minimal changes. As the name implies, it allows ODFs to be nested. Because it is extremely flexible and can be implemented relatively easily, some variation of virtual nesting is used by practically every airline that has implemented network management.

The first step in airline virtual nesting is to define a set of *buckets* on each flight leg. Each bucket represents a range of fare values. The second step is to map each ODF into a bucket on each of its legs based on an estimate of the ODF's value to the airline. The process of mapping ODFs into buckets on legs is called *indexing*. At the end of the indexing process, each bucket contains ODFs of similar value. (For the moment, we will defer the question of how we might calculate the value of an ODF.) We will use the convention that the lowest-numbered bucket on each leg contains the highest-value ODFs. The buckets are nested so that bucket 1 has access to the entire capacity of the leg, bucket 2 has access to all the inventory except that protected for bucket 1, and the lowest-value bucket has access only to its own allocation. Each bucket has a booking limit and a protection level that is updated every time a booking is accepted or a cancellation occurs. This means we can apply all of the mechanics for leg-based revenue management to buckets.

Example 10.3

An airline operating the simple network shown in Figure 10.5 offers a Y-Class fare and an M-Class fare for each product. This is a total of six ODFs. The airline has established three buckets on each leg. Then one possible indexing is illustrated in Figure 10.6. In this case, the airline maps SFO-STL Y-Class passengers into bucket 1 on Flight 1 but into bucket 2 on Flight 2. This means that the SFO-STL Y passengers will have access to the entire capacity of Flight 1, but they only have access to the bucket 2 allocation of Flight 2, which is less than the capacity of the plane.

A schematic diagram of a virtual nesting system is shown in Figure 10.7. The reservation system now has an additional function that maps booking requests to buckets as they arrive. Once a booking has been mapped to buckets on each leg, the reservation system checks availability of the appropriate bucket on each leg to determine whether the booking should be accepted. This function is closely analogous to the booking control function when booking requests are managed one leg at a time, as described in Section 8.5. In fact, most network management systems utilize the fare class booking structures within the reservation systems as their buckets. The function of the revenue management system in a

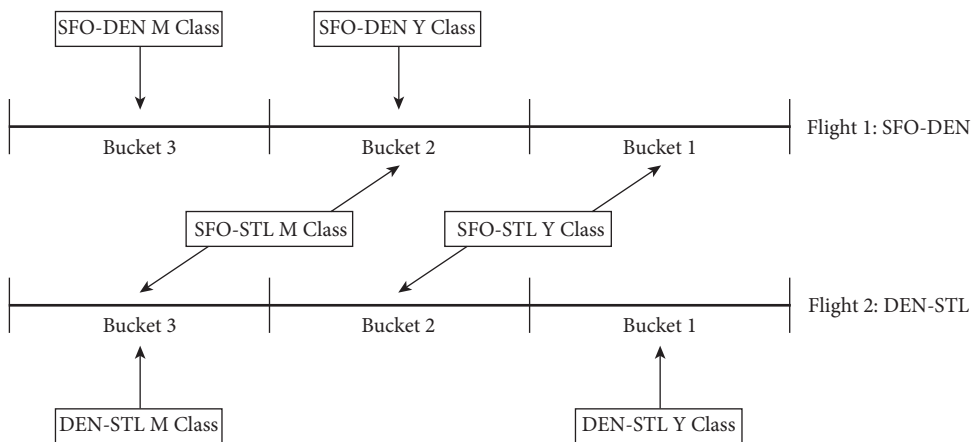


Figure 10.6 Example mappings of ODFs for a two-flight network.

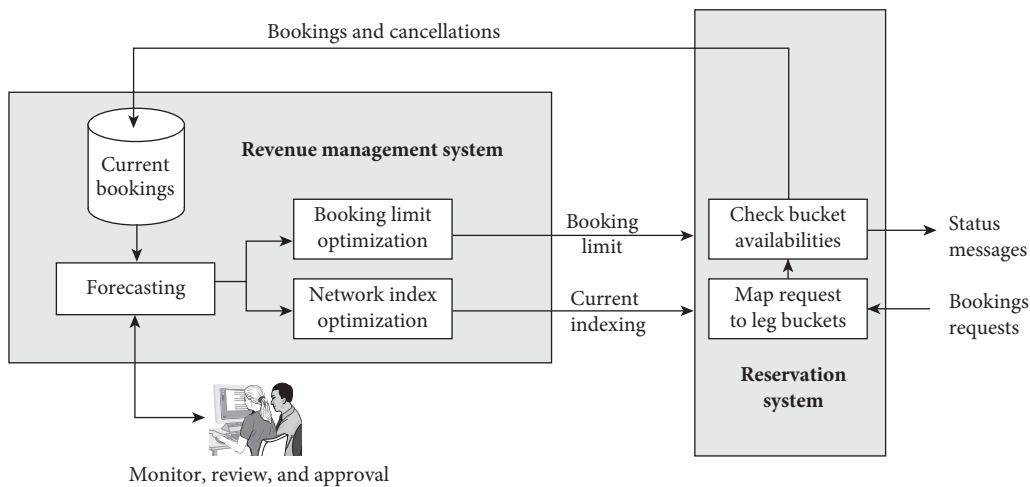


Figure 10.7 Schematic view of a virtual nesting architecture for network management.

virtual nesting approach is both to maintain and communicate the latest indexing and to calculate and update booking limits for each bucket on each leg.

10.3.1 Virtual Nesting Booking Control

The management of bookings under virtual nesting proceeds in a fashion closely analogous to single-leg booking control.

Virtual Nesting Booking Control

1. Initialization

- a. Establish an indexing that maps every ODF into a bucket on each leg in the ODF.

- b. For each leg bucket, estimate the mean and standard deviation of total forecasted demand.
 - c. Using the mean and standard deviation of demand by bucket, average contribution by bucket, and leg capacity, use EMSR or a similar approach to determine booking limits and protection levels for each bucket on each leg.
2. *Operation*
- a. When a booking request for an ODF is received, check each leg in the ODF. If there is sufficient capacity in the corresponding bucket on each leg, accept the request. Otherwise reject it.
 - b. If the booking is accepted, reduce bucket availabilities on all legs in the booked ODF.
 - c. When cancellations occur, increase bucket availabilities for each leg in the canceled ODF.
3. *Reoptimization*
- a. Periodically reforecast demand for each ODF and update expected bucket demand by leg.
 - b. Periodically rerun EMSR or other algorithm to recalculate nested booking limits for each bucket based on the new forecasts and capacity remaining on each leg.

Note that virtual nesting could be described as *leg-based revenue management plus indexing*. The connection to leg-based revenue management is a strength—it allows airlines to use the well-established (and hard-to-change) nested fare class structures within their reservation systems.

Another strength of virtual nesting is its flexibility. As an example, simply mapping all ODFs from the same fare class into the same bucket on each leg (i.e., mapping all M-Class bookings into an M-Class bucket on each leg and all Y-Class bookings into a higher Y-Class bucket) will give the same results as managing each leg independently. More usefully, an airline can use virtual nesting to provide preferential availability to an ODF simply by mapping it to higher buckets on its constituent legs. This can be useful, for example, if the airline is running a marketing campaign and wants to ensure that sufficient availability will be maintained for certain discount fares—even if it may not be optimal to do so from a revenue maximization point of view.

10.3.2 Indexing

Virtual nesting can be described as booking control plus indexing. We have seen how booking control works within virtual nesting. This leaves the question of how ODFs should be mapped to their constituent legs—in other words, how they should be indexed. Both the indexing scheme chosen and the frequency with which it is updated strongly influence not only the additional revenue that will be achieved but also the complexity of the revenue management system. The remainder of this chapter is devoted primarily to describing different indexing approaches and discussing the implications of each.

One immediate thought might be to map an ODF into buckets on each leg based on its total fares; that is, we would map the ODFs with the highest total fare into the highest bucket, those with slightly lower total fare into the second bucket, and so on for every leg. This is equivalent to the greedy heuristic for network management, and, as discussed in Section 10.1.2, the greedy heuristic fails when multiple resources in a network are constrained because it ignores opportunity cost in determining how an ODF should be bucketed on one of its legs. Specifically, an ODF with a high total fare but a high opportunity cost on its other legs should potentially be bucketed below an ODF with lower total fare but no opportunity cost on its other legs. This type of bucketing can never be achieved by the greedy heuristic.

The failure of the greedy heuristic suggests that we need a way of indexing ODFs that includes their opportunity costs. As it turns out, the best way to value an ODF for bucketing on a resource is on the basis of its total fare *minus the opportunity cost (if any) on other resources it uses*. This quantity, known as the *net leg fare*, is defined as follows:

The net leg fare for ODF i on leg k is equal to the total fare for ODF i minus the sum of opportunity costs on all resources other than k (if any) in ODF i .

The term *net leg fare* (also called *net local fare*) is a bit unfortunate because of its airline connotation. The same calculation holds for hotels, where the resources are room nights, and for rental car companies, where the resources are rental days.

Example 10.4

A hotel has estimated that the opportunity cost for a room on Tuesday night, January 23, is \$80 and the opportunity cost for Wednesday night, January 24, is \$105. The hotel receives a booking request for two nights, arriving January 23 and departing January 25, with an associated total rate of \$220. The net leg fare associated with that booking request is $\$220 - \$105 = \$115$ for the night of January 23 and $\$220 - \$80 = \$140$ for the night of January 24.

Note that the net leg fare is calculated by subtracting the opportunity cost on *other resources* from the total fare. It is a local measure of the incremental contribution of one more booking in the ODF to system profitability, including the opportunity that would be foregone from other legs (if any) in the ODF. Only if the fare associated with the ODF exceeds the opportunity cost for all of the resources it uses should it be accepted. For this reason, it is the right value to use to map products into buckets on each leg.

Example 10.5

Consider the three-city network shown in Figure 10.5. Assume that the opportunity cost on Flight 1 (SFO-DEN) is \$250 and on the Flight 2 (DEN-STL) leg is \$800. Each leg has three buckets, defined as follows:⁹

- Bucket 1: Net leg fares \$1,000
- Bucket 2: $\$600 \leq \text{Net leg fares} < \$1,000$
- Bucket 3: Net leg fares $< \$600$

TABLE 10.7
Net leg fares (NLF) and buckets for six ODFs in the network from Figure 10.5

Product	Class	Total fare (\$)	Flight 1		Flight 2	
			NLF (\$)	Bucket	NLF (\$)	Bucket
SFO-DEN	M	400	400	3	—	—
SFO-DEN	Y	650	650	2	—	—
DEN-STL	M	550	—	—	550	3
DEN-STL	Y	750	—	—	750	2
SFO-STL	M	1,300	500	3	1,050	1
SFO-STL	Y	1,800	1,000	1	1,550	1

NOTE: The opportunity cost on Flight 1 is \$250, and the opportunity cost on Flight 2 is \$800.

Table 10.7 shows how six different ODFs would be mapped into buckets in this case. Note that SFO-STL M-Class passengers are mapped into the highest bucket on Flight 2 but the lowest bucket on Flight 1. This means that on Flight 1 we will be protecting seats for local M-Class passengers from the SFO-STL M-Class passengers; but on Flight 2 the situation is reversed.

10.3.3 Static Virtual Nesting

We are most of the way to a practical approach for network management. We can calculate the net leg fare for each ODF on each of its legs based on its total fare minus the opportunity cost on other legs. This provides an indexing scheme that allows us to map the ODF into a bucket on each leg. If each of the buckets has sufficient availability, we accept the booking; otherwise we reject it. The remaining piece of the puzzle is how to calculate and update the opportunity cost for each leg. It is this calculation that differentiates virtual nesting approaches.

One obvious way to estimate the future opportunity cost for a product is to base it on past experience. This is, in essence, equivalent to forecasting opportunity cost.

Example 10.6

A Tuesday 2:00 flight from San Francisco to Dallas had opportunity costs of \$180, \$130, \$190, \$205, and \$95 at departure. One way to estimate the opportunity cost for next Tuesday's flight would be to average the past five opportunity costs; that is, the estimate of next Tuesday's opportunity cost would be $(\$180 + \$130 + \$190 + \$205 + \$95)/5 = \160 .

The method of using historic opportunity costs to map ODFs into buckets is called *static virtual nesting*. With static virtual nesting, an airline (or other revenue management company) estimates an opportunity cost for each resource based on the sales history of that resource.¹⁰ If the resource rarely sells out (e.g., a Tuesday night in a midtown hotel in Detroit), the opportunity cost will be small. If, on the other hand, the resource sells out with regularity, its opportunity cost will be much higher.

Example 10.7

Flight leg 135 has historically sold out 30% of the time, and its average opportunity cost when it sold out is \$750. An estimate of the expected future opportunity cost for this leg would be $0.3 \times \$750 = \225 .

There are other approaches to static virtual nesting, but they all have one thing in common: indexing is based entirely on past experience. This means that the indexing for a particular leg will not change during the booking period for a flight. In static virtual nesting, an airline might recalculate its indexing on a monthly or quarterly basis. In the interim, ODFs would always be mapped into the same buckets on each leg.¹¹ For our two-leg airline, the mapping in Table 10.7 would be used for all departures of Flights 1 and 2 within this period. However, this mapping is based on the expectation that Flight 2 is likely to sell out and therefore is assigned a high displacement cost. While this may be true on average, it is unlikely to be true for every departure. For example, we may be almost certain that a particular departure of Flight 2 will not sell out. This means that the expected opportunity cost for this departure of Flight 2 would be close to 0. But by using the mapping in Table 10.7, the airline is mapping connecting passengers into buckets that are too low for this departure.

Another shortcoming of static virtual nesting is implicit in the name: since it is static, it does not respond to changes in the course of the booking process. An airline's initial forecast of the opportunity cost for a leg might change considerably between first booking and departure. (As we have seen, there is a correspondence between opportunity cost and the boundary between open and closed classes. The assumption that a leg's opportunity cost does not change between first booking and departure is equivalent to assuming that no buckets will open or close during the time that the flight is open for booking.) For the two-flight example, we could certainly have the situation where an unanticipated surge in bookings (possibly a large group) resulted in Flight 1's actually closing to future bookings while Flight 2 remained open in bucket 1 but closed in buckets 2 and 3. Using the mapping in Table 10.7, the only ODFs that fall in bucket 1 on Flight 2 are SFO-STL through passengers. But under static virtual nesting, these booking requests will be rejected on Flight 1—which is closed in all buckets. So the system is effectively closed to all bookings, even though there are seats available on Flight 2 that could be profitably filled with local passengers.

It seems apparent that virtual nesting could be improved if we could update opportunity costs more often based on bookings and cancellations as they occur as well as on changing expectations of total demand for a flight. It turns out that, if we update opportunity costs often enough, they can almost serve as the only controls we need for network management. This is the idea behind bid pricing. Section 9.2.3 shows how bid pricing applies to a single leg. We see next how it can be applied to a network.

10.4 NETWORK BID PRICING

Section 9.2.3 describes how we should accept a booking request for a single resource only if the fare associated with the request exceeds the bid price for that resource. Perhaps not

surprisingly, there is an equivalent condition for products consisting of multiple resources: we should only accept a multiple-resource booking if the fare exceeds the sum of the bid prices for the constituent resources. This condition can be written as follows:

Accept a booking for a product (ODF) only if the product's price is greater than the sum of the bid prices on the constituent resources.

The idea behind network bid pricing is to use this condition as the basis of the revenue management system. In an airline bid-pricing system, the revenue management system calculates a bid price for each leg (we see later how this is done) at each time before departure. When a booking request is received, its fare is compared to the sum of the bid prices for its constituent legs. If the fare for the request is greater than or equal to the sum of the bid prices, the request is accepted; otherwise, it is rejected. That is all there is to bid pricing, at least from a booking control point of view.

In the extreme, this approach would allow us to dispense with the need for any sophisticated booking control structures and to manage bookings on bid price alone. However, we also need to allow for the fact that not all booking requests will be for a single seat. The easiest way to manage multiseat bookings is to keep track of the remaining unbooked seats on each leg and to use that number as a total booking limit.¹² Without no-shows and cancellations, the total booking limit would be equal to the unbooked capacity. If we anticipate no-shows and cancellations, the total booking limit could be calculated using one of the overbooking techniques described in Chapter 11. Managing bookings in this fashion is called *network bid price control*.

Network Bid Price Control

1. Calculate an initial bid price and a total booking limit for each leg.
2. When a booking request is received, calculate the *product bid price* as the sum of the bid prices for all the legs in the product and the *product availability* as the *minimum* of the total booking limits on all the legs used by the product.
3. If the fare for the booking exceeds the product bid price *and* the number of seats requested is less than the product availability, accept the request and go to step 4. Otherwise reject the request and go to step 2.
4. Decrease the total booking limit on each resource within the product by the number of seats in the booking and go to step 2.
5. When a booking cancels, increment the total booking limits for all flights in the booking.
6. Periodically reoptimize all flight booking limits and all bid prices.

Example 10.8

Consider the solution to Example 10.2 shown in Table 10.6. If we define \$75 as the bid price for the Denver-to-St. Louis leg and \$125 as the bid price for the San Francisco-to-Denver leg, we will see that we have established a set of bid

prices consistent with the optimal solution to the network management problem. The bid prices provide the optimal thresholds between the fare classes we want to accept for each product and those we want to reject. For the example, these thresholds are \$75 and \$125 for the two single-leg products, plus $\$75 + \$125 = \$200$ for the San Francisco–to–St. Louis through product. By using the bid prices for the two legs plus a total booking limit for each leg in the dynamic bid price control structure, we could achieve a result very close to the optimal network solution in Table 10.6. And we have done so much more economically, by using just four total controls—two bid prices plus two booking limits—instead of the 15 allocations in Table 10.6.

Network bid price control is a remarkably simple approach to revenue management. It requires only two controls per leg—a bid price and a total booking limit. However, real-world considerations complicate the picture substantially. First, we have not yet considered how bid prices can be calculated. Furthermore, bid prices need to be updated continuously over time. If calculating optimal network bid prices is time consuming or complex (as it is), the airline will require an interim control scheme to handle bookings between reoptimizations. In addition, network bid price controls require additional controls if multiple-seat booking requests are to be handled correctly. Finally, any network management approach for airlines needs to be implemented in conjunction with the existing reservation system and distribution system infrastructure, and most existing reservation and distribution systems are not designed to easily enable pure bid price management.

The system shown in Figure 10.8 is known as an *availability processor* architecture. It is an implementation of a pure bid-pricing approach to network management. Booking requests arrive at an online processor known as the availability processor. The availability

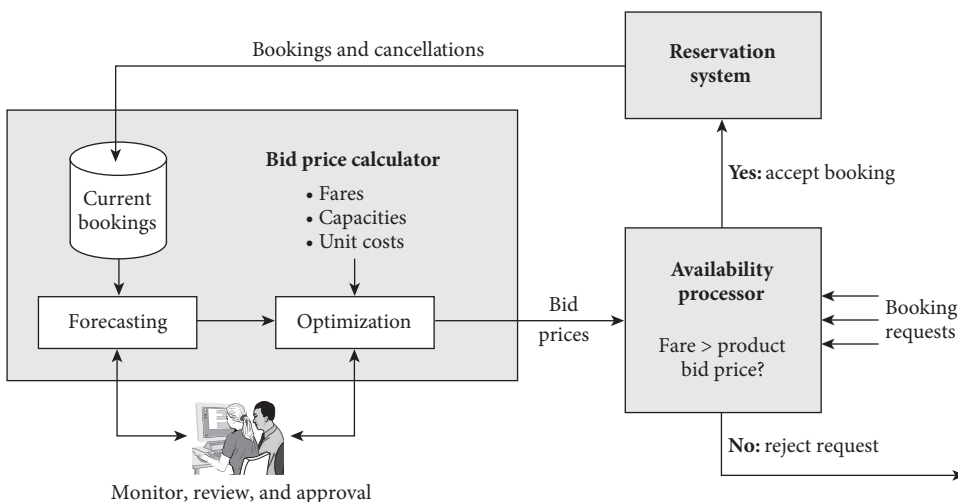


Figure 10.8 Schematic view of an availability processor architecture for network management.

processor calculates the fare associated with each request and compares it to the current bid price for the requested product. If the fare is less than the bid price, the request is rejected. If the fare is greater than the bid price for the requested product, the booking is accepted and communicated to the reservation system.

10.4.1 Bid Prices and Opportunity Costs

Section 9.2.3 notes that the bid price for a resource was equal to the opportunity cost for that resource and could be calculated as the fare of the lowest open fare class. Not surprisingly, network bid prices are also related to the opportunity costs of the resources in the network. We can extend this observation to a network by noting that the following six quantities are (approximately) equivalent.

- *Bid price*: the minimum price we should accept for a customer on a leg
- *Opportunity cost*: the revenue we would gain from an additional seat on a leg
- *Displacement cost*: the revenue we would lose if we had one seat less on a leg
- Marginal value of the capacity constraint in the network linear program
- Boundary between the highest closed and lowest open fare classes in a single-leg problem
- *Closed bucket boundary*: boundary between the highest closed and lowest open buckets on a leg in a virtual nest

To be sure, these six quantities will not always be exactly equal. For example, the revenue gained from an additional seat on a leg will not necessarily be exactly the same as the revenue lost from removing a seat. The marginal value of the capacity constraint in the deterministic network linear program will not generally equal the values obtained by approaches that explicitly incorporate uncertainty. However, the near equivalence among these quantities supplies ideas for how bid prices can be calculated as well as providing insights into why bid pricing works.

10.4.2 Calculating Bid Prices*

So far, we have not considered how bid prices might be calculated for a large network. You may not be surprised to find out that calculating bid prices can be quite complex—so much so that developing faster and more accurate bid-pricing algorithms is a very active area of research. While understanding the details of bid price calculation is not critical, we will briefly look at two commonly used approaches. This section is somewhat technical and can be skimmed or skipped.

Calculating bid prices by linear programming. Section 10.2 shows how the deterministic network management problem can be formulated as a linear program, where the outputs of the linear program are allotments for all the ODFs in the network. The value of the objective function at the optimal solution is the maximum revenue the company could achieve from its fixed stock of capacity. One way to calculate the bid price on a leg is to solve the network linear program twice: the first time with the actual capacities and the second time

with the capacity of the leg reduced by one seat. The difference in the total contribution between the two runs would be the displacement cost on the leg, which, as we have seen, is an estimate of the bid price on the leg.

Of course, this would be a cumbersome way to estimate bid prices, since it requires solving a large linear program twice for each leg in the network. Fortunately, this is not necessary. Deterministic displacement costs for all the legs in the network can be more practically estimated using linear programming in two different ways by using the marginal values associated with the capacity constraints as bid prices.

A key to calculating bid prices is to understand their relationship to the *dual variables* of the underlying optimization problem. Consider the linear program specified in Equations 10.2–10.5. Associated with each constraint is a dual variable that corresponds to the improvement in the objective function associated with an incremental relaxation of the corresponding constraint. Thus, for the linear program in Equations 10.2–10.5, there are two sets of dual variables:

- λ_i is the dual variable associated with the constraints in Equation 10.3: $\sum_{j=1}^n a_{ij}x_j \leq c_i$. This constraint guarantees that the number of accepted bookings for products that use a resource do not exceed the capacity of that resource.
- μ_j is the dual variable associated with the constraints in Equation 10.4: $x_j \leq d_j$. These constraints guarantee that the number of accepted bookings for a product do not exceed the demand for that product.

These values can be computed directly by solving the *dual* of the deterministic linear program, which is

$$\text{Minimize } \sum_{i=1}^m C_i \lambda_i + \sum_{j=1}^n d_j \mu_j,$$

subject to

$$\sum_{i=1}^m a_{ij} \lambda_i + \mu_j \geq p_j \quad \text{for } j = 1, 2, \dots, n,$$

$$\lambda_i, \mu_j \geq 0 \quad \text{for all } i, j.$$

This problem can be solved directly using a standard linear programming approach, in which case λ_i is the bid price associated with resource i .

Calculating bid prices by sequential estimation. The method of sequential estimation is based on the observation that the bid price on a leg should be equal to its closed bucket boundary in a virtual nest, where the closed bucket boundary is defined as the boundary between the lowest open bucket and the highest closed bucket on the leg. It should be clear that the closed bucket boundary should approximate the bid price on a leg. After all, we will accept a local booking for a single seat only if its fare exceeds the closed bucket boundary, which is the definition of bid price. Furthermore, we will accept a multileg booking request for a single seat only if its net leg fare is greater than the closed bucket boundary on each leg in the ODF. You can confirm that this is the same as ensuring that its total net fare exceeds the sum of the closed bucket boundaries.

This means we can calculate bid prices for a network by finding a set of closed bucket boundaries that is consistent across the network. The challenge arises because the closed bucket boundary (bid price) on any leg depends on the closed bucket boundaries on all other legs via the net leg fares of multileg ODFs. If we change the bid price on a leg, we will change the net leg fares for multileg ODFs on all connecting legs. This will in turn change the buckets into which those ODFs are mapped on the other legs. When we reoptimize availabilities on those legs, the new mapping may well change the bid prices on those legs, which will change the ODF mappings on the original leg. A consistent set of bid prices is one in which the closed bucket boundaries on all legs are locally optimal given the closed bucket boundaries on all other legs.

Sequential estimation calculates bid prices by starting with an initial estimate of all bid prices and then updating the bid price on each leg until the bucket boundaries are consistent across the entire network. To see how this works, assume we have a network consisting of two legs. We initially map all the ODFs into buckets by assuming that each leg has a bid price of 0 (recall that this is equivalent to the greedy heuristic). Given this indexing, we can estimate the total demand by bucket on each leg. Then, for leg 1, we can use EMSR to calculate the bucket allocations on leg 1 given the current indexing. Once we have calculated the allocations on leg 1, we can reestimate the bid price on leg 1 as its closed bucket boundary. Given this new estimate of the leg 1 bid price, we can recalculate the net leg fares on leg 2 and rebucket all of the ODFs on leg 2 accordingly. We can then apply EMSR on leg 2, determine its closed bucket boundary, and use that as the new bid price for leg 2. Given this new leg 2 bid price, we can rebucket the ODFs on leg 1 and continue. Under the right conditions, this procedure will converge to a situation where the mappings and allocations on each leg are consistent with each other.

We can extend this approach to a full network as follows.

Sequential Estimation Algorithm to Calculate Bid Prices

1. Establish a set of buckets on all legs, and calculate probabilistic demand forecasts for all buckets. Set all leg bid prices to 0.
2. Loop over all legs, $k = 1, 2, \dots, N$.
3. For the current leg k and every ODF i that includes leg k , calculate the net leg fare as the fare for ODF i minus the sum of the current bid prices for *all other* legs in the ODF.
4. Map all the ODFs on the current leg into buckets based on their net leg fares for the ODFs. Once this is complete, use EMSR to determine allocations by bucket.
5. If all buckets are open, set the new bid price for leg k to 0. If all buckets are closed, set the new bid price for leg k to the highest bucket boundary. Otherwise, set the new bid price to the boundary between the lowest open bucket and the highest closed bucket.
6. Continue to the next leg until all legs have had new bid prices calculated.

7. When all leg bid prices have been recalculated, check the change between the new bid price and the old bid price on each leg. If the two bid prices are sufficiently close for all legs, stop—the current set of bid prices is optimal.
8. Otherwise go to step 2.

This sequential estimation algorithm generally converges to a set of consistent bucket boundaries across a network. When bookings are accepted or forecasts change, the sequential estimation algorithm can be restarted using the previous set of bid prices.

10.4.3 Dynamic Virtual Nesting*

Dynamic virtual nesting is the marriage of bid pricing and virtual nesting. It combines the conceptual elegance and intuitiveness of bid pricing with the practical orientation of virtual nesting. Under dynamic virtual nesting, new bid prices are frequently recalculated for all legs in the network based on current bookings and forecasts of future demands. The recalculation of bid prices could be purely periodic (e.g., nightly or weekly), it could be event driven (e.g., recalculated whenever a flight closes), or it could be ordered by a revenue analyst. In most cases, all of these can trigger a recalculation. Every time new resource bid prices are calculated, new net leg fares are calculated for each leg in the network. These net leg fares are used to define a new indexing. This indexing is put in place until the next recalculation of bid prices.

The strength of dynamic virtual nesting is that it updates bid prices to reflect the current situation on each leg—both current bookings and expected future demand—while utilizing virtual nesting to take advantage of leg-based control structures. Of course, an important determinant of the effectiveness of the approach is how often bid prices are recalculated. If bid prices are not recalculated very often, then dynamic virtual nesting may not be much of an improvement over static virtual nesting. If bid prices are continually updated (say, every second), then dynamic virtual nesting will approximate real-time network bid price control for single-seat bookings.

The first dynamic virtual nesting revenue management system was implemented in 1989 by Hertz rental car company and Decision Focus Incorporated (Carroll and Grimes 1996). The first airline dynamic virtual nesting system was developed jointly by Decision Focus Incorporated and the Scandinavian Airlines System in 1992. Since then, the approach has been widely applied across many revenue management industries.

10.4.4 Strengths and Weaknesses of Bid Pricing

Bid prices are more than tools to manage bookings. They are also useful pieces of information in themselves. As opportunity costs, they specify the value of an additional unit for each resource in the network. If a flight leg has a bid price of \$250, this indicates that the airline could realize (approximately) \$250 in additional revenue from adding another seat to that flight. Since the bid prices are marginal signals, this logic cannot be extended to determine the effect of adding more than one seat: A bid price of \$250 on a flight leg does

not mean that an airline could expect to gain \$2,500 from adding 10 seats.¹³ However, even with this limitation, the bid prices can provide useful input into capacity decisions. A flight leg that consistently has a high opportunity cost at departure would be an excellent candidate to be considered for assignment to a larger aircraft, while a flight leg that consistently departs with a bid price of 0 is an excellent candidate to be considered for downsizing. A rental car company with the option to move its cars around should consider moving cars from locations with consistently low bid prices to those with consistently high bid prices.

Example 10.9

The Rent-a-Lemon rental car company rents cars at both the San Francisco airport and the San Jose airport. The two airports are 40 miles apart, and it costs \$10 per car to “hike” cars from one airport to the other. For a date two weeks in the future, the bid price at the San Francisco airport is \$73 and the bid price at the San Jose airport is \$22. The expected net gain from moving a car from San Jose to San Francisco would be $\$73 - \$22 - \$10 = \41 .

Bid pricing is particularly well suited to hotel applications where there is one bid price for each room type for each future night. The product bid price is the sum of the bid prices for the nights stayed. Hotel and rental car bid prices can be conveniently displayed using a calendar, as shown in Figure 10.9. The calendar shows the bid prices (labeled *b*) and the forecast unconstrained occupancies (labeled *d*) for a hotel as they might be displayed by a revenue management system for a future month. A customer who wants to rent a room for the nights of March 23, 24, and 25 would need to pay at least the sum of the bid prices for these nights ($\$199.93 + \$178.25 + \$122.20 = \500.38) to be allowed to book. On the other hand, a customer who wants to rent for March 12, 13, and 14 would be allowed to book if he were paying more than $\$42.34 + \$32.34 + \$54.37 = \129.05 . The lower product bid price reflects lower anticipated demand during the second period than the first. The dark-shaded dates are those with a bid price greater than \$150, those with medium shading have a bid price between \$100 and \$150, and those with no shading have a bid price less than \$100. The pattern shown in Figure 10.9 is typical of a metropolitan business hotel with a midweek peak.

Bid pricing is an appealing and intuitive approach to network revenue management. However, a word of caution is in order. The product bid price is the *minimum* price we should accept for a product given remaining capacity and anticipated future demand by fare class. It does not tell us how we should actually be *pricing* the product. A hotel manager looking at the bid price calendar in Figure 10.9 should not make the mistake of thinking that he should set the rack rate on Wednesday, March 16, to \$122.47. In this sense, the term *bid price* is unfortunate since, as we have seen, the bid price is better regarded as an opportunity *cost*. In the extreme case, a bid price of zero for a product tells us nothing about how we should price the product. Bid price control is a mechanism for ensuring that we do not accept any business whose margin does not exceed its opportunity cost—it does not tell us whether we are priced correctly in the market.

March						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
28 $d = 85$ $b = \$84.34$	1 $d = 93$ $b = \$92.07$	2 $d = 112$ $b = \$153.12$	3 $d = 108$ $b = \$112.34$	4 $d = 99$ $b = \$92.57$	5 $d = 65$ $b = \$54.30$	6 $d = 80$ $b = \$62.33$
7 $d = 91$ $b = \$88.47$	8 $d = 102$ $b = \$122.00$	9 $d = 135$ $b = \$172.15$	10 $d = 120$ $b = \$142.34$	11 $d = 92$ $b = \$95.67$	12 $d = 53$ $b = \$42.34$	13 $d = 44$ $b = \$32.34$
14 $d = 67$ $b = \$54.37$	15 $d = 85$ $b = \$72.48$	16 $d = 110$ $b = \$122.47$	17 $d = 97$ $b = \$99.97$	18 $d = 93$ $b = \$92.34$	19 $d = 72$ $b = \$55.18$	20 $d = 66$ $b = \$54.54$
21 $d = 86$ $b = \$89.11$	22 $d = 104$ $b = \$130.02$	23 $d = 157$ $b = \$199.93$	24 $d = 140$ $b = \$178.25$	25 $d = 122$ $b = \$122.20$	26 $d = 95$ $b = \$100.69$	27 $d = 85$ $b = \$85.18$
28 $d = 84$ $b = \$84.33$	29 $d = 92$ $b = \$93.44$	30 $d = 114$ $b = \$155.67$	31 $d = 100$ $b = \$101.01$	1 $d = 82$ $b = \$78.77$	2 $d = 60$ $b = \$53.92$	3 $d = 75$ $b = \$62.74$

Figure 10.9 Bid price calendar.

10.5 NETWORK MANAGEMENT IN ACTION

The first step taken by most airlines in revenue management was to use the approaches described in Chapter 9 to maximize expected revenue on each flight leg independently. In most cases, this approach provided tremendous benefits from not managing bookings at all—that is, allowing customers to book on a first-come, first-served basis. However, hub-and-spoke airlines quickly realized that this approach was not capturing the full potential revenue from their systems.

Because of the need to work within existing reservation systems, the airlines generally adopted network management by instituting increasingly sophisticated forms of virtual nesting, often following the progression implied in Table 10.8. The first step was usually some form of ad hoc virtual nesting, under which a few high-value ODFs were mapped into higher buckets than their fare classes would imply. At some point, most hub-and-spoke airlines instituted some form of static virtual nesting, under which all ODFs would be mapped into buckets on each leg. These mappings would be updated quarterly or monthly based on forecasts of future demand and opportunity cost. Finally, many airlines worked to make these mappings dynamic—that is, updated frequently for each flight based on bookings and cancellations.

TABLE 10.8
Approaches to virtual nesting

Approach	Description
Greedy heuristic	Set all leg opportunity costs to 0 (not effective)
Ad hoc virtual nesting	Use judgment, or ad hoc methods, for index
Static virtual nesting	Use history to estimate leg opportunity costs
Dynamic virtual nesting	Update leg opportunity costs periodically
Pure bid pricing	Update bid prices after each booking or cancellation

The transition from leg-based revenue management to network management at the airlines involved more than just new software; the focus of revenue managers also needed to change. Under leg-based capacity control, *flight controllers* were responsible for monitoring forecasts and fare class allocations for some set of flights. For example, a flight controller for Delta Airlines might be responsible for all flights between the Atlanta hub and cities in Florida. As the name implies, flight controllers by and large took a flight-centric view of the world, looking to maximize return from their portfolio of flights. When there was a significant amount of connecting traffic, this flight-based approach could not maximize total system return.

Network management required a realignment of responsibilities. Now, there is no individual forecast for, say, M-Class bookings on a particular Atlanta-to-Orlando flight. Rather, the Atlanta-to-Orlando flight has 20 or so buckets. Each of those buckets can contain demand from hundreds of different ODFs. Thus, one bucket might include Detroit-Atlanta-Orlando Y-Class demand, Atlanta-Orlando B-Class demand, Denver-Atlanta-Orlando Y-Class demand, and so on. Furthermore, depending on demand dynamics, the mapping of ODFs to buckets might change weekly or even daily on the same flight. In essence, it is virtually impossible for a controller to manage forecasts and bucket availabilities on a flight basis. Instead, under network management the focus of the revenue management organization needs to shift from flights (or resources) to products. Instead of a flight controller focused on Atlanta-to-Florida flights, there might be a *market manager* focused on destination-Florida markets. The focus of this market manager would be on understanding the market from all domestic origins to Florida. She would be expected to ensure that all inbound forecasts to Florida incorporate the latest market intelligence. For example, each year she would be expected to know the academic schedule for colleges in the Northeast to ensure that the Fort Lauderdale destination forecasts included spring break demand.

Unlike the airlines, hotels and rental car companies incorporated network management into their revenue management systems right from the start. This was not due to superior technical sophistication on their part but simply to the fact that capacity allocation on a single-resource basis did not work. In fact, early experiments adapting existing leg-based airline revenue management systems to work for hotels were disastrous. The first successful hotel revenue management systems, such as those for Marriott and Hyatt, and the first successful rental car systems, such as those built for Hertz and National, all incorporate length-of-stay or length-of-rental control.

Network management is on the cutting edge of pricing and revenue optimization practice. Existing systems and approaches are a compromise between what is theoretically

correct and what is practical. The need to work with preexisting reservation systems and to respond quickly to market changes means that the use of heuristics, ad hoc solutions, and good-enough algorithms are commonplace. Research into better algorithms and approaches to network revenue management continues. A topic of particular research interest is incorporating consumer choice into network management.

Finally, it should be noted that the success of discount airlines means that the traditional hub-and-spoke airlines are being forced to move from a capacity control environment to a dynamic pricing environment. In this environment, the key decision is not which fare classes should be open or closed at any time, but what prices should be on offer for each product. This requires somewhat new approaches, but the basic idea that capacity should never be sold for a fare lower than its opportunity cost is still valid.

10.6 SUMMARY

- Network management is an issue for any revenue management company that sells products consisting of a combination of two or more of the resources it controls. Examples include airlines that offer connecting services, hotels renting rooms for multiple nights, and passenger trains offering tickets that include multiple legs. Network revenue management is important for hub-and-spoke airlines, hotels, rental car companies, passenger trains, and freight transportation providers. It is generally not important for cruise lines, event ticketing, and point-to-point airlines.
- A key challenge in network management is the size and complexity of the problem. A large hub-and-spoke network may have millions of product–fare class combinations it needs to manage for future departures. The size and complexity of the problem is a major consideration in implementing solutions.
- Ordering all ODFs by total fare and setting capacity limits on that basis is called the *greedy heuristic* for network management. It does not work well when more than one resource is constrained and is not used in practice.
- Linear programming can be used to determine the optimal allocation of products to buckets when demand is deterministic. In real-world applications, linear programming formulations need to be modified to account for demand uncertainty.
- Most real-world network management systems use some form of virtual nesting. Under virtual nesting, a set of buckets (analogous to fare classes) is established for each resource. When a request for an ODF is received, it is mapped into buckets on each constituent resource using a predetermined indexing scheme. If each of these buckets has sufficient availability, the request is accepted; otherwise it is rejected. Different ways of defining and updating the indexing will lead to different virtual nesting approaches.
- Pure bid price control is based on establishing and updating a bid price for each resource. The bid price for a resource is the minimum fare that would be accepted for selling a unit of that resource. It is equivalent to the opportunity cost for a resource.

A booking request for an ODF is accepted only if its associated fare is greater than the sum of the bid prices on all constituent resources *and* there is sufficient remaining capacity in each resource to accommodate the request.

- While pure bid price control is conceptually appealing, it is not widely used in practice since it would require very frequent reoptimization to be robust. Furthermore, it requires extensive (and expensive) modification of the reservation systems on which most revenue management companies rely.
- Dynamic virtual nesting is a combination of bid pricing with virtual nesting. Bid prices are recalculated frequently during the booking period for a product. The new bid prices are then used to define a new indexing, at which point bucket availabilities also need to be updated.
- Network management was incorporated into the revenue management programs for hotels and rental car companies from the beginning. Most airlines, on the other hand, initially managed capacity on a leg-by-leg basis and then introduced network management, often by way of increasingly sophisticated and dynamic virtual nesting approaches. The transition from leg-based management to network management required organizational and business process changes as well as software system changes.

10.7 FURTHER READING

For those who want to go deeper into different network revenue management algorithms, two good sources are Talluri and van Ryzin 2014 and Gallego and Topaloglu 2019, chaps. 2, 7.

The relationship between the opportunity cost of capacity on a flight and the bid price is noted in Simpson 1989 and Phillips 1994. The relationship between the dual variables of a linear program and the opportunity costs of the associated constraints and the derivation of the dual linear program are discussed in most textbooks on linear programming, such as Luenberger and Ye 2008.

Regarding individual companies that implemented network revenue management systems, Marriott's experience is discussed in Robert Cross's book *Revenue Management* (1997). Hertz's system is described in Carroll and Grimes 1996 and National's in Geraghty and Johnson 1997.

10.8 EXERCISES

1. How many products can Amtrak offer on the California Zephyr, assuming a single fare class? (Remember that every combination of possible boarding and deboarding stations defines a different product.)
2. Consider an airline with a single hub in the Midwest. Ten flights arrive from cities in the West at the hub every day and connect with nine flights departing for cities in the East. How many products can the airline offer, assuming a single fare class? (Hint: Do not forget the nonstop products.)

3. The airline offering the flights and fares shown in Table 10.4 decides to raise the San Francisco–to–St. Louis discount fare from \$170 to \$225. It estimates that discount demand at this new fare will be 20. Given that all other fares and demands remain the same, what is the new optimal set of allocations and total revenue for the network?
4. In Example 10.1, determine which bookings would be accepted using the greedy heuristic and the corresponding system revenue. What was the percentage improvement in revenue from optimizing the network?
5. CU Airlines operates two flights, one from New York to Phoenix and another one from Phoenix to San Francisco. It sells a full fare and a discount fare for each leg, and a single fare for the combined leg from New York to San Francisco. The airline has assigned a 120-seat aircraft for the first flight and a 100-seat aircraft for the second flight. The demand for each fare is assumed to be random and normally distributed. The fares and demands for these flights are shown in the following table.

Number	ODF	Fare	Demand mean	Demand std. dev.
1	NYC-PHX full fare	\$300	30	6
2	NYC-PHX discount fare	\$200	70	20
3	PHX-SFO full fare	\$250	20	5
4	PHX-SFO discount fare	\$150	50	15
5	NYC-SFO fare	\$300	40	15

- a. Formulate the problem as a deterministic network linear program using the means. What is the optimal allocation and revenue?
- b. To incorporate the uncertainty of demands, the chief science officer wants you to use virtual nests for each leg. For each flight, you will have three buckets—one for each ODF. The average contribution to each flight from the NYC-SFO fare is distributed proportionally according to the discount fares for each flight. How would you set the buckets for each flight? Use EMSR-b to calculate the protection levels for each bucket on each flight. How do the protection levels compare to the allocation you obtained in part *a*?

NOTES

1. This is not strictly true. The theater might still be able to gain additional revenue by considering the possibility of diversion from, say, weekend performances to weeknights in its pricing, as described in Section 7.6.4.

2. Customers can book individual flight legs on EasyJet that they can treat as connecting flights (e.g., they can book a flight from London to Rome and a flight three hours after the Rome arrival to Athens if they wish), but this is different from booking a connecting flight from London to Athens via Rome, since EasyJet cannot manage availability for such customers based on true origin and destination. As discussed in this chapter, the economics of offering connecting flights are so compelling that a number of airlines such as Southwest and Ryanair that initially allowed only single-flight bookings have since begun to offer connecting services.

3. This does not mean that network revenue management never plays a role in these industries. For example, cruises may have multiple stops that allow passengers to board and depart at different ports. And baseball teams may offer package deals for groups of games. In both cases, the seller is offering a product using different resources, and network management may come into play.

4. This is standard terminology. However, it can be misleading, since the real unit being managed is itinerary fare class, not origin-destination fare class. An M-Class passenger traveling from San Francisco to Denver on Flight 18 at 8 a.m. is in a different ODF than an M-Class passenger traveling from San Francisco to Denver on Flight 21 at 9 a.m., even though they share the same origin, destination, and fare class.

5. The highest nightly rate offered by a hotel is called the *rack rate*.

6. Recall that *heuristic* refers to an approach to an optimization problem that generates a solution that is not guaranteed to be optimal.

7. This requires that all of the fare classes for a product have different fares.

8. As with single-leg capacity allocation, this property holds true when demands are uncertain and independent but may not hold with dependent demands.

9. In Example 10.5, we assume for simplicity that the buckets on each leg are the same, but this does not need to be true in general.

10. Static virtual nesting is sometimes called *displacement-adjusted virtual nesting*, or DAVN.

11. The exception is when an ODF fare changes: in this case, the airline would recalculate the net leg fares for that ODF based on the new fare and update the indexing scheme accordingly.

12. Note that multiunit bookings are much more of an issue for airlines than for many other revenue management companies. The vast majority of car rental bookings and hotel bookings are for a single car and a single room, respectively, and the issue of managing multiunit bookings is accordingly much less important in these industries.

13. In general, additional revenue is a concave function of additional capacity—that is, the additional revenue from adding n seats will be less than n times the bid price for $n > 1$.

11 OVERBOOKING

Overbooking occurs whenever a seller with constrained capacity sells more units than he has available (or believes he will have available). The reason that sellers engage in such a seemingly nefarious practice is to protect themselves against no-shows and cancellations. A *cancellation* is defined as a booking that a customer terminates by notifying the seller at some point prior to the service date. A *no-show* occurs when a buyer with a booking does not cancel but simply fails to show up.

American Airlines estimated that about 50% of its reservations resulted in either a cancellation or a no-show (Smith, Leimkuhler, and Darrow 1992). Without overbooking, many of these cancellations and all of the no-shows would result in an empty seat—even when there are other customers willing to fly. In this situation, overbooking becomes critical. Consider a flight with 100 seats that consistently faces demand much higher than 100. Assume that customers have a 13% no-show rate. If the airline never overbooks, it will, on average, leave with 13 empty seats on every flight while denying reservations to customers who want to fly. This would be an enormous waste. The airline would be in the same situation as a manufacturer who had a plant that could only run at 87% of capacity. Without the ability to overbook, not only would the airline lose potential revenue; it would be paying to purchase, maintain, and support huge amounts of useless capacity.

No-shows and cancellations are not only an airline issue; they are a feature of any service industry in which bookings can be canceled (or not honored) without penalty or where the penalty is small relative to the opportunity cost of the resource booked. Primary care medical appointments, for example, typically have a no-show rate of 15–20% in the United States, and accordingly, most clinics overbook. And, of course, many of the usual revenue management industries such as hotels, rental cars, and freight transportation face similar issues. In all of these industries, overbooking is used as a way to improve capacity utilization in the face of cancellations and no-shows.

The literal-minded might argue that overbooking is not really part of pricing and revenue optimization because it does not directly affect price. However, overbooking is inextricably intertwined with capacity allocation: to determine how many seats to offer to different booking classes on a flight, we certainly need to know how many seats we will be

offering in total. For that reason, it is generally considered to be an integral part of revenue management, and we treat it as such.

This chapter starts by giving a short history of overbooking. It then characterizes industries in which overbooking is used and introduces four different approaches to overbooking: a deterministic heuristic, risk-based policies, service-level policies, and hybrid policies. We see how booking limits can be determined under each policy in the case of a single price and an uncertain level of no-shows. We then examine a data-driven approach to overbooking similar to the data-driven approach to capacity allocation in Section 9.4. We consider dynamic overbooking in the face of cancellations and multiple fare classes. Finally, we look at some extensions to the basic problem and some alternatives to overbooking for managing cancellations and no-shows.

11.1 BACKGROUND

11.1.1 History of Overbooking

Overbooking is closely identified with the passenger airline industry (particularly by anyone who has had the experience of being denied a seat on an overbooked flight). From the earliest days of commercial aviation in the United States, airlines adopted the policy that a customer with a reservation could cancel at any time before departure without paying a penalty and that a customer who had purchased a ticket for a flight would not be penalized if she did not show up for that flight. As a result, an airline ticket was like money in that it could be used at full face value for a future flight or redeemed for cash. (This policy should be contrasted with Broadway shows, opera, and rock concerts, where unused tickets are generally not refundable.) This raised a problem for the airlines—how many passengers should they allow to book on each flight? If they limited bookings to the capacity of a flight, many fully booked flights would leave with empty seats. As mentioned, American Airlines estimated in 1990 that about 15% of the seats on sold-out flights would be empty because of no-shows and cancellations if they only booked up to capacity (Smith, Leimkuhler, and Darrow 1992).

By the 1960s, no-shows were becoming a major problem. According to Kalyan Talluri and Garrett van Ryzin, “In 1961, the Civil Aeronautics Board (CAB) reported a no-show rate of 1 out of every 10 passengers booked among the 12 leading carriers. . . . The CAB acknowledged that this situation created real economic problems for the airlines” (2004b, 131). As a result, the airlines were allowed to overbook. This was effective in increasing loads but, predictably, meant some passengers were refused boarding on a flight for which they held a ticket—so-called denied boardings (DBs). When a flight was oversold—that is, the number of passengers showing up exceeded the seats on the flight—the airline would pick customers to bump, that is, rebook on a later flight. If the flight was much later, the bumped passengers were provided with a meal; if it was the next day, they were provided with overnight accommodation. In addition, the airline paid a penalty to each bumped passenger. At one point this penalty was equal to 100% of ticket value. When a passenger is bumped against her will, it is known as an *involuntary denied boarding*. In 1966, the Civil

Aeronautics Board estimated that the involuntary denied-boarding rate was about 7.7 per 10,000 boarded passengers.¹

Because airlines needed to overbook to remain financially viable, the practice became universal. The risk of being bumped from an oversold flight was seen by consumers as yet one more irritating aspect of flying, along with delayed flights and lost luggage. However, the airlines were reluctant to own up to overbooking. In the 1960s, flight controllers at American Airlines recognized that overbooking was critical to financial performance. They continued to overbook despite periodically reassuring senior management that they were not doing so. When denied boardings occurred, the flight controllers blamed them on “system error.”² Airline overbooking continued largely without constraints until 1972, when Ralph Nader was denied a boarding on an Allegheny Airlines flight. Nader sued and won a judgment of \$25,000 based on the fact that Allegheny did not inform customers that they might be bumped against their will. Following this judgment, the Civil Aeronautics Board ruled that airlines must inform their customers that they engage in overbooking.

In the late 1970s, following a suggestion by economist Julian Simon (1968), the airlines began to experiment with a *voluntary denied-boarding* policy. In Simon’s vision, the airlines would conduct a sealed-bid reverse auction to find enough passengers willing to be bumped at different levels of compensation. While the airlines initially objected that such an auction would be impractical, ultimately, they adopted a variant of Simon’s scheme, in which an overbooked airline asks for volunteers to be bumped in return for compensation (usually a voucher for future travel). If enough volunteers are not found, the compensation level may be increased once or twice. If enough volunteers are still not found, the airline will choose which additional passengers to bump. The volunteers are termed *voluntary denied boardings* and those chosen by the airline *involuntary denied boardings*. This approach to overbooking has been successful, both from the point of view of the customers and from the point of view of airlines. Airline research shows that voluntary denied boardings are often happy to get bumped in return for compensation. Not only has it made customers happier; the volunteer policy has resulted in a drastic drop in the involuntary denied-boarding rate for major US airlines—from 7.7 in 10,000 passengers in 1985 to 0.13 per 10,000 in the third quarter of 2018. In fact, involuntary denied boardings have become almost extinct at some airlines—Delta managed to board over 37 million passengers in the third quarter of 2018 without a single involuntary denied boarding. On the other hand, voluntary denied boardings have increased substantially from a rate of about 20 per 10,000 in 1995 to 43.5 per 10,000 in the third quarter of 2018.

Overbooking is also practiced by hotels and rental car companies, although statistics on its prevalence are hard to come by, since, unlike the airlines, other industries are not required to report statistics on denied service. The typical hotel practice is to find accommodations for a bumped guest at a nearby property—preferably one in the same chain. Compensation such as a discount coupon for a future stay is sometimes (but not always) offered. In many cases, there is no consistent policy across a chain or group of hotels, and reimbursement policies are determined by each property manager. A rental car overbooking is usually experienced by the customer as a wait—a customer who arrives to find no car available needs to wait until a car is returned, cleaned, and refueled. In rare situations,

a location may be so overbooked that there is no prospect of a car for every customer. In this case, the manager will either send booked customers to competitors or move cars from another location if possible to satisfy the additional demand. Again, compensation in the form of a discount coupon may or may not be offered, depending on the situation and company policy.

11.1.2 When Is Overbooking Applicable?

Overbooking is applicable in industries with all the following three characteristics.

- Capacity (or supply) is constrained and perishable, and bookings are accepted for use of future capacity.
- Customers are allowed to cancel or not show.
- The cost of denying service to a customer with a booking is *relatively* low.

It should be understood that the cost of denied service includes intangible elements, such as customer ill will and future lost business, as well as any direct compensation, as discussed in Section 11.1.3. If the total denied-service cost is sufficiently high, it is not in the seller's interest to overbook, since the cost of denying service will overwhelm any potential revenue gain. For example, airlines would be unlikely to overbook at all if they had to pay a million dollars to every denied boarding.

The three characteristics just listed are the classic requirements for overbooking. Under these circumstances, companies overbook to hedge against the possibility of excessive cancellations or no-shows. However, there is another situation in which overbooking may come into play—when the amount of capacity that will be available is uncertain. Both hotels and rental car companies face this issue because of the risk of overstays and understays: Customers may stay longer or depart earlier than their reservations specify. If 10 customers depart early, then a hotel will have 10 additional empty rooms the next night. If it booked only to expected capacity, these rooms would go empty, even in the absence of no-shows or cancellations.

Television broadcasters also face a problem of uncertain capacity. Broadcasters in the United States sell the vast majority of their capacity for the season starting in September during the upfront market during the previous May. Each buyer is sold a schedule of slots and guaranteed a certain number of impressions (“eyeballs”) by the broadcaster. If the schedule sold delivers the guaranteed number of impressions (or more), then that is the end of the matter. But if the schedule does not deliver the guaranteed number of impressions, the broadcaster needs to supply the advertiser with additional slots until the guarantee is met—a practice known as *gapping*. But the broadcaster does not know in advance how many people will actually watch a particular show. A new show may be an unexpected dud or an unexpected hit. This is similar to the airline overbooking problem, with the difference that the broadcaster knows the demand he must meet but is unsure what the supply (impressions) will be.³

Cruise lines and resort hotels generally avoid overbooking. In these industries, the cost of denied service is simply too high. A couple who arrives at the dock with all their luggage

TABLE 11.1
Overbooking in different industries

Industry	Importance of overbooking	Treatment of overbooked customers
Passenger airlines	Very high	Compensation and accommodation on other flights
Business hotels	High	Reaccommodation at other hotels, usually without additional compensation
Rental cars	High	Wait or reaccommodation at other companies
Air freight	High	Reaccommodation on a later flight
Made-to-order manufacturing	Medium	Back-order or delay delivery
Cruise lines	Low	Typically do not overbook
Resort hotels	Low	Typically do not overbook
Sporting events, concerts, and shows	Low	Nonrefundable tickets with very little overbooking

for a two-week cruise is unlikely to be easily mollified when they find out they will not be able to board because the cruise is overbooked. Instead of overbooking, cruise lines and resort hotels manage the risks of cancellations and no-shows by a combination of nonrefundable deposits and higher prices.⁴ Most industries that sell nonrefundable bookings (or tickets) do not overbook. Theater tickets and tickets to sporting events are examples. In this case, the risk to the customer of purchasing a ticket is mitigated by the fact that bookings in these industries are transferable to others—unlike airline tickets.

Table 11.1 shows the relative importance of overbooking in various revenue management industries. It can be argued that overbooking has had the greatest financial impact of any aspect of passenger airline revenue management. For example, American Airlines estimated that it achieved benefits of \$225 million in 1990 from overbooking (Smith, Leimkuhler, and Darrow 1992)—more than from any other element of its revenue management program, including capacity allocation and network management. Overbooking is important for hotels and rental car companies, both of which face uncertain supply (due to overstay and understay) as well as no-shows and cancellations. It is also quite important for air freight, where no-shows and cancellations are commonplace.

11.1.3 The Cost of Denied Service

A key consideration in setting an overbooking limit—and indeed in whether to overbook at all—is the cost of denied service. This cost varies from industry to industry and depends on how customers who are denied service are treated. This varies from industry to industry and from company to company within an industry. A broadcaster that cannot meet the number of impressions he has guaranteed to an advertiser needs to provide additional advertising spots until he has fulfilled the guarantee. A rental car outlet that does not have a car available when a booked customer arrives must either ask the customer to wait until a car is available or provide a vehicle from a competing company.

In the passenger airline industry, the cost of denied service is called *denied-boarding cost*. In the United States, the treatment of overbooked passengers is regulated by the Department of Transportation. As of July 2004, the DOT requires that an airline with an overbooked

flight first seek customers willing to take a later flight in return for compensation. As noted previously, airlines will typically increase the compensation level one or two times if they cannot find enough volunteers at the initial level. For these volunteers, the denied-boarding cost is simply the amount of compensation that the airline provides. If the airline cannot find enough volunteers, it will need to bump one or more passengers (i.e., take some *involuntary denied boardings*). In the United States, the Department of Transportation has specified *minimum* compensation that must be provided to each passenger who is involuntarily denied a boarding. As of July 2020, each person involuntarily denied boarding must be provided an alternative ticket to the destination plus additional compensation as follows:⁵

- If the arrival delay is less than one hour: no compensation
- If the arrival delay is between one and two hours: 200% of one-way fare, with maximum of \$675
- If the arrival delay is more than two hours: 400% of one-way fare, with maximum of \$1,350

These rules do not apply to charter flights, scheduled flights with planes that hold 30 or fewer passengers, when an airline has substituted a smaller plane for the one it originally planned to use, or when an aircraft needs to shed weight to meet weight and balance constraints. The rules are also minimum guidelines—the airlines are free to pay additional compensation if they so choose. Usually, the only additional compensation the airlines provide is meal vouchers if the delay is more than two hours, and a hotel voucher if the bumped passenger is accommodated on a flight that leaves the next day.

The airlines are unusual in that several governments have mandated a procedure and compensation level for bumping passengers. Companies in other industries are generally free to deal with denied-service situations as they see fit. However, denied-service cost in all cases includes one or more of four components:

- The *direct cost* of the compensation to the bumped passenger—this could be a certificate for future travel or a future hotel room night.
- The *provision cost* of meals and/or lodging provided to a bumped passenger.
- The *reaccommodation* cost of a customer who is denied service. For an airline, this is the cost of putting the customer on another flight to the same destination; for a hotel, it is the cost of alternative accommodation for the night.
- The *ill-will cost* from denying service. This may be hard to calculate but is usually an estimate of the expected lost future business from the bumped customer.⁶

Denied-service cost will vary depending on the situation. For a passenger airline, bumping a passenger from the last flight of the day will usually incur the additional cost of lodging her overnight. Reaccommodation cost for an airline depends on whether the bumped passenger is rebooked on one of its own flights or on a competing airline's flight. Of course, the ill-will cost for a voluntary denied boarding is much less than for an involuntary denied boarding—in fact, in many cases, a volunteer has been happy to have the opportunity to

take a later flight in return for a flight voucher—perhaps resulting in a goodwill benefit (although airlines do not consider such benefits in their overbooking calculations.) The relatively low cost of voluntary denied boardings has led to their proliferation—as noted above, the voluntary denied-boarding rate among US major carriers was 43.5 per 10,000 customers in the third quarter of 2018 as opposed to a total denied-boarding (all involuntary) rate of 7.7 per 10,000 customers in 1985.

11.1.4 A Model of Customer Booking

To describe and compare different approaches to overbooking, we initially use the following simple model of booking dynamics.

- A supplier plans to accept bookings for a fixed capacity, C .
- The supplier sets a booking limit b before any bookings arrive.
- The supplier continues to accept bookings as long as total bookings are less than the limit b . Once the limit is reached (if it ever is), the supplier stops accepting bookings.
- At the time of service (e.g., the departure time for a flight), customers arrive. Booked customers who arrive are called *shows*; those who fail to show are called *no-shows*.
- Each show pays a price of p .
- The supplier can accommodate up to C of the shows. If the number of shows is less than or equal to C , they are all accommodated. If the number of shows exceeds C , exactly C of the shows will be served, and the rest will be denied service. Shows that are denied service are each paid denied-service compensation of $D > p$.⁷

The supplier's problem is to determine the total number of bookings to accept. We call this number the *booking limit* and denote it by b .

This model is rich enough to illustrate the fundamental trade-offs and algorithms used to determine the total booking limit. However, it makes three heroic assumptions that are relaxed in later sections.

- The model ignores cancellations by calculating the booking limit based entirely on no-shows. This makes the booking limit easier to calculate. Furthermore, it means that the booking limit b can be static—that is, it does not need to change over the booking period. If bookings can cancel prior to departure, then the optimal booking limit is likely to change over time as departure approaches. Section 11.7.1 addresses the calculation of dynamic booking limits.
- The model assumes that each customer will pay the same price p . However, as described in Chapter 9, airlines, hotels, and rental car companies all offer different prices for the same unit of capacity. This can significantly complicate the calculation of the optimal total booking limit. Section 11.7.2 addresses overbooking with multiple fare classes.
- The model assumes that only those customers who arrive (i.e., the shows) pay. Bookings themselves are costless. This is the historic airline, hotel, air freight, and rental

car situation, in which bookings could be canceled without penalty and tickets were fully refundable. However, many of these industries are moving all or partway toward nonrefundable or partially refundable prices. Section 11.7.3 addresses the effect of nonrefundable or partially refundable prices.

11.2 APPROACHES TO OVERBOOKING

Once a company decides it is going to overbook, it needs to decide what it wants to achieve with its overbooking policy and how it is going to achieve it. Most companies follow one of four approaches:

- A simple *deterministic heuristic* that calculates a booking limit based only on capacity and expected no-show rate
- A *risk-based* policy that involves estimating the costs of denied service and weighing those costs against the potential revenue to determine the booking levels that maximize expected total revenue minus expected overbooking costs. A risk-based booking limit can either be calculated by explicit optimization or using a data-driven approach. We look at both approaches.
- A *service-level* policy that involves managing to a specific target—for example, targeting no more than one instance of denied service for every 5,000 shows
- A *hybrid* policy in which risk-based limits are calculated but constrained by service-level limits

The different policies are closely analogous to the different replenishment policies followed by different retailers. Some stores set their replenishment levels so as to ensure that stockouts do not exceed a certain frequency: a service-level policy. Others specifically trade off the cost of a potential stockout with the cost of holding more inventory to determine the replenishment level that maximizes expected profit: a risk-based policy. Just as retailers differ in their stocking policy, so do companies differ in their overbooking policies. For example, at one point, the Hertz rental car company used a risk-based policy to set total booking limits (Carroll and Grimes 1996), while National Car Rental used a service-level policy (Geraghty and Johnson 1997). Sections 11.3–11.6 describe each of these four ways to calculate a total booking limit using the model of customer booking described in Section 11.1.4.

11.3 A DETERMINISTIC HEURISTIC

A hotel has observed that its historic show rate has averaged 85%. Furthermore, this rate has been consistent over time. A reasonable policy might be for the hotel to set its total booking limit b so that if it sells b rooms and experiences the average show rate, it will fill exactly C rooms. That is, it would set b such that $C = 0.85b$, or $b = C/0.85$.

This approach to calculating a total booking limit can be written as $b = C/\rho$, where C is capacity and ρ is the show rate. Despite its simplicity, the deterministic heuristic turns out

to be a reasonable approximation to the optimal booking limit in many cases. It is still used to calculate overbooking limits by many companies in cases in which the cost of denied (or delayed) service is unclear (for example, medical clinics).

Example 11.1

For a hotel with 250 rooms and an expected show rate of 85%, the deterministic heuristic gives a booking limit of $250/0.85 = 294$ rooms.

Note that applying this approach does not mean that the hotel will be overbooked 15% of the time or that 15% of bookings will experience an overbooking.

11.4 RISK-BASED POLICIES

The deterministic heuristic described in the previous section is certainly simple to calculate, but it obviously does not consider the relative costs of an additional customer versus denied service. Under a risk-based policy, the booking limit is set by balancing the expected cost of denied service with the potential additional contribution from more sales. The first step in calculating a risk-based booking limit is to specify the objective function.

11.4.1 The Risk-Based Objective Function

Denote the number of passengers who show up at departure by s . The number of shows is a random variable that depends on both the booking limit and the total demand for bookings. Since each show pays price p , total revenue is $p \times s$. If shows exceed capacity, then the supplier must deny service to $s - C$ customers. Each customer denied service results in a denied-boarding cost of D . For the moment, we assume that this denied-boarding cost is the same for each passenger, which means that denied boardings are either all voluntary or all involuntary. The total denied-boarding cost is 0 if $s \leq C$ and is $D(s - C)$ if $s > C$. The net revenue is then

$$R = ps - D(s - C)^+. \quad (11.1)$$

(Recall that $(s - C)^+$ denotes the maximum of $s - C$ and 0.) A company pursuing a risk-based policy wants to set its booking limit so as to maximize the expected value of net revenue as defined in Equation 11.1, where the number of shows is a random variable that depends on the booking limit, which the company can choose.

Total revenue as a function of the number of shows is plotted in Figure 11.1. Revenue increases linearly until the capacity limit is reached, at which point it begins to decrease. Two things are apparent immediately from Figure 11.1. First, the number of shows will always be less than or equal to the booking limit, so it is never optimal to set the booking limit less than the capacity. Thus, we know that the optimal booking limit b^* must satisfy $b^* \geq C$. Second, if the price is higher than the denied-boarding cost (that is, $p > D$), net revenue would continue to increase even when shows exceed the booking limit. In this case, the seller should accept every booking and set $b^* = \infty$. Since this case is not very realistic, we assume that $D > p$.

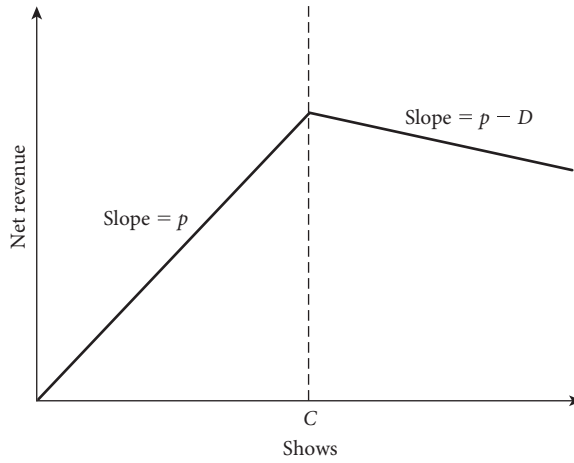


Figure 11.1 Net revenue as a function of shows.

Calculating the optimal risk-based booking limit is complicated by the fact that the number of shows is a function of three different factors:

- The booking limit b
- The total demand for bookings, d
- The number of bookings that ultimately show, s

We are interested in the number of *bookings at departure*, which we denote by b_d . Bookings at departure is the minimum of the booking limit and the demand for bookings—that is, $b_d = \min(d, b)$. The total number of shows is the number of bookings at departure minus no-shows—that is, $s = b_d - x$, where x denotes the number of no-shows. We denote the *show rate*, ρ , as the fraction of bookings at departure we anticipate will show—that is, $\rho = E[s/b_d]$ —while the *no-show rate* is the fraction of bookings at departure we anticipate will not show—that is, $1 - \rho$. Airline no-show rates typically range between 10% and 20% for major airlines; however, there is significant variation. In general, international and transcontinental flights have much lower no-show rates than do shorter domestic flights. Bad weather can cause no-show rates to soar to 30% or more. Resort hotels tend to experience lower no-show rates than business hotels, and cruise lines have very low no-show rates.

11.4.2 Modeling Shows

Risk-based models of overbooking all require a probabilistic representation of no-show behavior. The simplest assumption is that each booking has an identical probability, $0 < \rho \leq 1$, of showing and that show decisions are independent. Then the number of shows given n bookings follows a *binomial distribution*.⁸ Let $g(s|b)$ be the probability of s shows given a booking limit of b —which also means b bookings at departure. Then

$$g(s|b) = \binom{b}{s} \rho^s (1 - \rho)^{b-s} \quad \text{for } s = 0, 1, \dots, n. \quad (11.2)$$

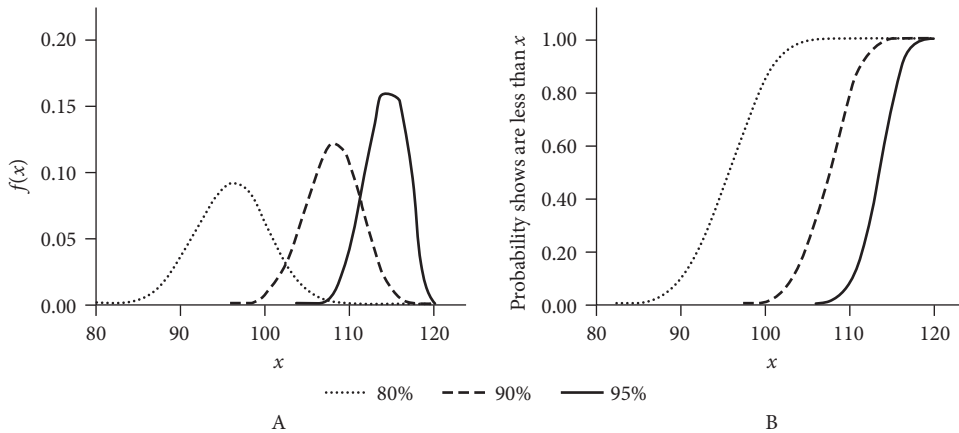


Figure 11.2 Distribution of shows given 120 bookings. (A) Probability density function for different show rates; (B) corresponding cumulative distribution functions.

with mean $E[(sb)] = \rho b$ and variance $\text{var}[(sb)] = \rho(1 - \rho)b$. Under these assumptions, the number of no-shows x will also follow a binomial distribution with parameters $1 - \rho$ and b . More information on the binomial distribution can be found in Appendix B.

The density function of shows is given in panel A of Figure 11.2 for $n = 120$ bookings and show rates of 80%, 90%, and 95%. Panel B of Figure 11.2 gives the corresponding cumulative distributions. Each cumulative distribution shows the probability that shows will be less than x , given 120 total bookings. This can be interpreted as the probability that a flight with x seats will be overbooked if it has 120 bookings at departure, given different no-show rates. Panel A shows that both the mean and the standard deviation of the number of shows depend on the show rate. The mean number of shows is the show rate times the number of bookings: 96, 108, and 114 for show rates of 80%, 90%, and 95%, respectively.

Notice that the bulk of the probability on shows for any show rate between 80% and 95% lies between 80 and 110 when there are 120 bookings at departure. Thus, for a show rate of 80%, it is almost certain (a 99.97% chance, to be exact) that the number of shows will be between 80 and 110. We can use panel B of Figure 11.2 to estimate the probability that we will not have an oversold flight if we have 120 bookings at departure for aircraft with different seating capacities. As shown, if we have 120 bookings for a flight with 100 seats, we will almost certainly have at least 1 denied boarding if our show rate is 90% or 95%. On the other hand, we will only have about an 8% chance of being oversold if the show rate is 80%. These types of relationships are critical in determining the optimal booking limit.

11.4.3 A Simple Risk-Based Booking Limit

The decision tree approach can be used to derive the conditions for an optimal total booking limit in a fashion similar to the way it is used to derive the condition for an optimal protection level in Section 9.1.1. Recall that capacity is C , the price is p , and the denied-service

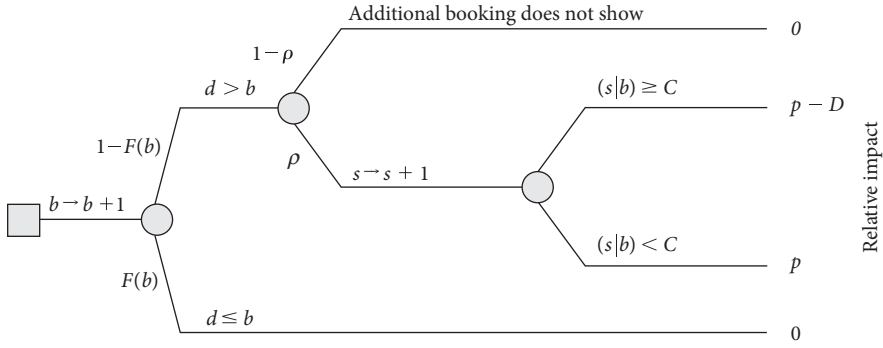


Figure 11.3 Overbooking decision tree.

cost is D , with $D > p$. We define $F(b)$ as the probability that bookings will be less than or equal to b , and $(s|b)$ as the number of shows given that bookings are b .

Figure 11.3 illustrates how the overbooking decision in this case can be illustrated as a decision tree in a format similar to the capacity allocation decision tree in Figure 9.1. Here, the decision is whether to increase the booking limit from b to $b + 1$. If the booking limit is increased but there is no additional booking, then there is no change in total revenue—this is the lowest branch on the tree. If there is an additional booking that no-shows, then there is also no change in total revenue—this is the top branch on the tree. If the booking shows, there are two possibilities: if shows given b are less than C , then the additional booking will result in an additional fare p , and if shows given b are greater than or equal to C , then the additional booking will result in a fare minus the denied-boarding cost, or $p - D$. Rolling back the decision tree and setting the marginal cost to zero gives the optimality condition

$$\Pr\{(s|b^*) \geq C\} = \frac{p}{D}, \quad (11.3)$$

where b^* is the optimal risk-based booking limit.

The condition in Equation 11.3 is formulated as an equality, but because the booking limit is discrete, it is unlikely that there is a value of b for which Equation 11.3 holds exactly. A way to account for the discrete nature of the booking limit is to start with the case when the booking limit is set to capacity—that is, $b = C$ —and then sequentially increase the booking limit by increments of one seat. Assume that we increase the booking limit by one seat so that $b = C + 1$. Let us assume that demand is very high relative to capacity, so that we will always receive another booking when we increase the booking limit by one. If this booking does not show—which will happen with probability $1 - \rho$ —there is no impact on revenue at all. If the booking does show—which happens with probability ρ —there are two possibilities. The first possibility is that all C of the other bookings show, in which case we have a denied boarding with associated cost $p - D$. The second possibility is that at least one of the C bookings does not show, in which case we can accommodate the new booking and gain a payment of p . Note that the probability that all C of the initial bookings will

show is equal to ρ^C , and thus the expected change in revenue from increasing the booking limit from C to $C + 1$ is

$$\Delta R(C, C + 1) = \rho[(1 - \rho^C)p - \rho^C(p - D)] = \rho(p - \rho^C D)$$

Here, $\Delta R(C, C + 1)$ denotes the expected total revenue change from changing the booking limit from C to $C + 1$. Overbooking by at least one seat will increase expected revenue if the term on the right side is greater than 0 or, equivalently, if

$$\frac{p}{D} > \rho^C. \quad (11.4)$$

Example 11.2

An airplane has 100 seats and a historical show rate of 90% and faces very high demand for the flight. The fare is \$250 and the denied-boarding cost is \$500. In this case, $\frac{p}{D} = .5$ and $\rho^C = .9^{100} = 0.000027$. Given that condition 11.4 is satisfied, it is profitable for the airline to overbook by *at least* one seat. The increase in expected revenue from increasing the booking limit from C to $C + 1$ is $\Delta R(C, C + 1) = .9 \times [\$250 - .000027 \times \$500] = \224.99 .

Example 11.2 illustrates the strong economic incentive for airlines and other sellers with constrained capacity to overbook. With a 90% show rate, the probability that a departure with 100 seats that is booked to capacity will leave with at least 1 empty seat is 99.997%. In this case, the denied-boarding penalty that would be required to make it unprofitable for the airline in Example 11.2 to overbook is much larger than the likely actual denied-boarding cost (see Exercise 7).

We have seen how to determine if it is profitable to overbook by at least one seat, but the question remains of how to find the optimal booking limit—the one that will maximize expected revenue. By extending the logic from before, we can write

$$\Delta R(b, b + 1) = \rho[p - \Pr\{(s|b) \geq C\}D], \quad (11.5)$$

where $\Delta R(b, b + 1)$ is the change in expected revenue from increasing the booking limit from b to $b + 1$, $\Pr\{(s|b) < C\}$ is the probability that the shows given the booking limit b are less than the capacity C , and $\Pr\{(s|b) \geq C\} = 1 - \Pr\{(s|b) < C\}$ is the probability that the shows given the booking limit b are greater than or equal to C .

We can rearrange Equation 11.5 to derive the relationship

$$\Delta R(b, b + 1) > 0 \Leftrightarrow \Pr\{(s|b) \geq C\} < \frac{p}{D}. \quad (11.6)$$

That is, expected revenue will always increase as long as the probability that shows are less than capacity is less than the ratio of the price to the denied-boarding cost, which means that we can start by setting the booking limit equal to capacity and then sequentially increase the booking limit by one and test against condition 11.6. We stop when increasing the booking limit by another seat would lead to a booking limit that violated 11.6. This suggests the following algorithm for computing the optimal booking limit:

Simple Risk-Based Overbooking Algorithm

1. Set $b = C$.
2. Compute $\Pr\{(s|b) \geq C\}$.
3. If $\Pr\{(s|b) \geq C\} \geq \frac{p}{D}$, set $b^* = b$ and STOP.
4. Otherwise set $b \leftarrow b + 1$ and go to step 2.

This algorithm will find the booking limit b^* that maximizes expected revenue. It starts by setting $b = C$. For the current value of b , it determines the change in revenue that would occur by increasing b by 1. The optimal booking limit is the minimum value of b for which increasing it by 1 would lead to a decrease in revenue.

The results so far can be summarized as follows: *The optimal booking limit when demand is much higher than capacity is the smallest value of b for which $\Pr\{(s|b) \geq C\} \geq \frac{p}{D}$.*

Example 11.3

Consider a flight with 100 seats, a fare of \$120, and a denied-boarding cost of \$300. Then, $p/D = 0.4$. Assume that the probability that each booking will show is .92. Then, Table 11.2 shows the values of $\Pr\{s|b\}$, passenger revenue, expected denied boardings, expected denied-boarding costs, and expected net revenue for values of the booking limit b from 100 to 109. The optimal booking limit is $b = 108$, which satisfies the condition for an optimal booking limit.

The only part of the algorithm not yet discussed is the calculation of the probability $\Pr\{(s|b) \geq C\}$. For a given value of b , this is the complementary cumulative binomial function. Given a booking limit of b , Equation 11.2 gives the probability of s shows for any value of $s < b$. The probability that s is greater than or equal to C for some value of $b \geq C$ can be found by summing the individual probabilities—that is,

$$\Pr\{(s|b) \geq C\} = \sum_{s=C}^b \binom{b}{s} \rho^s (1-\rho)^{b-s} \quad (11.7)$$

This can be a bit of a pain to calculate; however, standard cumulative binomial distribution tables enable the required values to be quickly accessed. Perhaps more conveniently,

TABLE 11.2
Calculating booking limits for Example 11.3

b	$\Pr\{s b > C\}$	Passenger revenue (\$)	Expected DBs	Expected DB costs (\$)	Expected net revenue (\$)
100	0.000	12,000.00	0.00	0.00	12,000.00
101	0.000	12,120.00	0.00	0.00	12,120.00
102	0.002	12,240.00	0.00	0.59	12,239.40
103	0.009	12,360.00	0.01	3.38	12,356.62
104	0.029	12,480.00	0.04	12.13	12,467.87
105	0.071	12,600.00	0.11	33.32	12,566.68
106	0.140	12,720.00	0.25	75.40	12,644.60
107	0.239	12,840.00	0.49	147.00	12,693.00
108	0.359	12,960.00	0.85	254.71	12,705.29
109	0.489	13,080.00	1.34	401.41	12,678.59

NOTE: DB = denied boarding.

for sufficiently large values of b , the cumulative binomial distribution can be approximated by the cumulative normal distribution. Specifically, the binomial distribution in Equation 11.7 has mean ρb and standard deviation $\sqrt{\rho(1-\rho)b}$. Under appropriate conditions, we can use the normal approximation:

$$\Pr\{(s|b) > C\} \approx 1 - \Phi\left(\frac{C - \rho b}{\sqrt{\rho(1-\rho)b}}\right). \quad (11.8)$$

This is quite convenient since it allows us to use standard normal distribution tables to estimate the probability. However, the approximation is only valid within certain bounds. A common rule of thumb is that the approximation in (11.8) is only valid if $(1-\rho)C > 9$; that is, the no-show rate times the capacity is greater than 9. For a departure with 150 seats and a typical no-show rate of .1 or so, we would have $(1-\rho)C = 15$, which meets the criterion. For much smaller flights (or hotels) or much higher no-show rates, the normal approximation may not be valid and tables of the binomial distribution may be needed to calculate the overbooking limit.

Like the two-class booking limit problem discussed in Section 9.1.4, the problem of setting a risk-based overbooking limit has a close relationship to the newsvendor problem. Recall that the solution to the newsvendor problem requires ordering an amount of inventory Y so that

$$\Pr\{\text{demand is greater than or equal to } Y\} = \frac{U}{U+O},$$

where U is the underage cost, O is the overage cost, and $U/(U+O)$ is the *critical fractile* or *critical ratio*. In setting a risk-based booking limit b , we are determining how many bookings to order—that is, the total number of bookings to allow. The underage cost is the opportunity cost of an empty seat—that is, p —and the overage cost is the denied-boarding fee for a show—that is, $D-p$. Therefore, the critical ratio is p/D . The event that triggers an overage cost is “shows exceeding capacity,” which in our simplified model is $\Pr\{(s|b) \geq C\}$. Thus, Equation 11.3 is equivalent to the risk-based model.

11.4.4 A Risk-Based Model with Demand Uncertainty*

To make the approach to overbooking more realistic, we need to consider the fact that the demand for a flight is not infinite. In this case, increasing the booking limit from b to $b+1$ will only result in an additional booking if demand is greater than b and the additional booking shows. Let $f(d)$ be the probability that booking demand is exactly equal to d . The cumulative distribution $F(d) = f(1) + f(2) + \dots + f(d)$ is the probability that demand is less than or equal to d , and the complementary cumulative distribution function, $\bar{F}(d) = 1 - F(d)$ is the probability that the demand for bookings is greater than d . If we increase the booking limit from b to $b+1$, the probability that we will experience an additional booking is $\bar{F}(b)$.

For a given booking limit, the expected number of denied boardings will be given by $E[(s|b) - C]^+$, where $(s|b)$ is the number of shows given booking limit b . Thus, the change in total expected revenue from increasing the booking limit from b to $b+1$ is

$$\Delta R(b, b+1) = p\bar{F}(b) - D\{E[(s|b+1) - C]^+ - E[(s|b) - C]^+\}. \quad (11.9)$$

We can see that $\Delta R(C, C+1) = p\bar{F}(C) - DE[(s|C+1) - C]^+$. But $E[(s|C+1) - C]^+ = \bar{F}(C)p^{C+1}$ so that $\Delta R(C, C+1) = \bar{F}(C)(p - Dp^{C+1})$. This will be greater than zero if $\bar{F}(C) > 0$ and if condition 11.6 holds. This means that, even when demand is uncertain, there is still a strong motivation to set a booking limit higher than capacity—that is, to overbook.

The optimal booking limit with demand uncertainty is the smallest value of b for which $\Delta R(b, b+1)$ defined in Equation 11.9 is less than zero. This can be computed iteratively starting from $b = C+1$ as in the previous model, and it can also be approximated numerically.

11.4.5 A Data-Driven Approach

The approaches explained in Sections 11.4.3 and 11.4.4 can be described as forecast and optimize because they both start with probabilistic forecasts of demand and shows given demand and then optimize to find the booking limit that maximizes total expected revenue from a flight, including denied-boarding costs. In the spirit of the approach to capacity allocation described in Section 9.4, we can consider a data-driven approach to overbooking. Assume that we are trying to determine the booking limit to apply to a particular flight. We assume (not unreasonably) that the same flight has been operated at the same time to the same destination with the same equipment for some time in the past. We also assume, somewhat less reasonably, the total booking demand for every historical departure of the flight—that is, how many bookings would occur if no booking limit had been applied. Finally, we assume, very unrealistically, that we know how many of the bookings would show for every possible booking limit. This is not unreasonable for booking limits less than those applied historically because the airline can observe which passengers showed and which did not. But it is unrealistic to assume that an airline would know who would show and who would not for bookings it did not accept. In short, for any booking limit b that could have been applied to flight t , we know the number of bookings that would show at departure t —denote this by $s_t(b)$. Note that $s_t(b)$ is a nondecreasing function of b . Assume that the airline has observations for historical departures operated on day $t = 1, 2, \dots, T$ and it is looking to determine the booking limit for the departure scheduled for day $T+1$. As before, we assume that the price for a ticket on this departure is p , the capacity is C , and the denied-boarding cost is D .

We can compute the net revenue that we would have achieved from departure t if it had capacity C and we had applied booking limit b with $b \geq C$ as

$$R_t(b) = \min[s_t(b), C]p + \max[s_t(b) - C, 0]D.$$

If we applied the same booking limit to all of the historic departures, the average net revenue would have been

$$\bar{R}(b) = \left(\frac{1}{T}\right) \sum_{t=1}^T R_t(b).$$

Under the data-driven approach, we would choose the booking limit for departure $T+1$ that maximizes $\bar{R}(b)$ —that is, $b_{T+1}^* = \arg \max \bar{R}(b)$. The idea is illustrated in Example 11.4.

Example 11.4

Senilria Airlines runs a 20-seat flight from San Francisco to San Bernardino, California, every Monday morning at 10:00 a.m. It has operated this flight for the last 10 weeks and did not apply any booking limits, which allowed it to observe the show/no-show behavior of every booking. Table 11.3 shows the order of bookings for each departure of the flight and whether each booking showed; a value of 1 indicates that the booking showed, a value of 0 indicates that the booking did not show, and a dash indicates no booking. Thus, departure 1 had 30 bookings of which 26 showed and departure 3 had 16 bookings of which 14

TABLE 11.3
Bookings and no-shows for Senilria Airlines flight

	Flight									
	1	2	3	4	5	6	7	8	9	10
1	1	1	1	1	0	1	1	0	1	1
2	1	1	1	1	1	1	1	1	1	1
3	1	1	1	0	1	1	1	1	1	1
4	0	1	1	1	1	1	1	1	0	0
5	1	1	1	1	1	1	0	1	1	1
6	1	1	1	0	1	1	0	1	1	1
7	1	0	0	1	1	1	1	1	1	1
8	1	1	1	0	0	1	1	1	1	1
9	1	1	0	1	1	1	1	1	1	1
10	1	1	1	1	1	1	1	1	0	0
11	1	1	1	1	1	1	1	0	0	1
12	1	1	1	1	0	1	0	1	1	1
13	0	1	1	0	1	1	1	1	0	0
14	1	1	1	1	1	1	1	1	1	1
15	1	1	1	1	1	1	1	0	1	1
16	1	1	1	1	1	0	1	1	1	1
17	0	1	—	1	1	1	1	1	1	1
18	1	1	—	1	1	1	1	1	1	1
19	1	1	—	1	1	1	1	1	1	1
20	1	1	—	1	0	1	—	1	1	1
21	1	1	—	1	1	1	—	1	1	1
22	1	1	—	1	1	1	—	1	0	1
23	1	1	—	1	1	1	—	1	0	1
24	1	0	—	1	1	1	—	1	1	0
25	1	1	—	0	1	1	—	1	1	1
26	1	1	—	0	1	0	—	1	1	1
27	0	0	—	1	1	1	—	1	1	1
28	1	0	—	1	1	1	—	0	1	—
29	1	1	—	1	0	0	—	1	1	—
30	1	1	—	1	1	1	—	0	1	—
31	—	1	—	1	1	1	—	1	1	—
32	—	1	—	1	1	—	—	1	1	—
33	—	—	—	1	1	—	—	1	0	—
34	—	—	—	1	1	—	—	0	1	—
35	—	—	—	1	1	—	—	1	1	—
36	—	—	—	1	1	—	—	1	1	—
37	—	—	—	0	1	—	—	—	1	—
38	—	—	—	0	—	—	—	—	1	—
39	—	—	—	1	—	—	—	—	—	—
40	—	—	—	1	—	—	—	—	—	—

TABLE 11.4
Shows per historic flight for different booking limits

Booking limit (b)	Flight									
	1	2	3	4	5	6	7	8	9	10
20	17	19	14	16	16	19	16	17	16	17
21	18	20	14	17	17	20	16	18	17	18
22	19	21	14	18	18	21	16	19	17	19
23	20	22	14	19	19	22	16	20	17	20
24	21	22	14	20	20	23	16	21	18	20
25	22	23	14	20	21	24	16	22	19	21
26	23	24	14	20	22	24	16	23	20	22

TABLE 11.5
Net revenue per flight and average revenue for Senilria Airlines departures in Example 11.4

Booking limit (b)	Departure										Average
	1	2	3	4	5	6	7	8	9	10	
20	\$3,400	\$3,800	\$2,800	\$3,200	\$3,200	\$3,800	\$3,200	\$3,400	\$3,200	\$3,400	\$3,340
21	\$3,600	\$4,000	\$2,800	\$3,400	\$3,400	\$4,000	\$3,200	\$3,600	\$3,400	\$3,600	\$3,500
22	\$3,800	\$3,700	\$2,800	\$3,600	\$3,600	\$3,700	\$3,200	\$3,800	\$3,400	\$3,800	\$3,540
23	\$4,000	\$3,400	\$2,800	\$3,800	\$3,800	\$3,400	\$3,200	\$4,000	\$3,400	\$4,000	\$3,580
24	\$3,700	\$3,400	\$2,800	\$4,000	\$4,000	\$3,100	\$3,200	\$3,700	\$3,600	\$4,000	\$3,550
25	\$3,400	\$3,100	\$2,800	\$4,000	\$3,700	\$2,800	\$3,200	\$3,400	\$3,800	\$3,700	\$3,390
26	\$3,100	\$2,800	\$2,800	\$4,000	\$3,400	\$2,800	\$3,200	\$3,100	\$4,000	\$3,400	\$3,260

showed. On the basis of this history, Senilria wants to determine the booking limit that would maximize expected revenue from the next departure given a fare of \$200 and a denied-boarding cost of \$300.

Table 11.4 shows the number of shows that each departure would have experienced given levels of the booking limit b from 20 to 26. Booking limits above 26 do not need to be considered because at a limit of 26, every departure with at least 20 shows is full. Finally, Table 11.5 shows the net revenue that would be achieved for each flight. Based on this history, the value of b that would maximize expected revenue is $b^* = 23$ with a corresponding expected revenue of \$3,580. This can be compared with the expected revenue from not overbooking of \$3,340, a gain of \$240 or a 7.2% increase.

While more than 10 historical observations would typically be required, Example 11.4 illustrates the strength of the data-driven approach: it enables us to go directly from the underlying data to the decision without the need to make assumptions about underlying distributions or estimating coefficients. The approach does not require independence of shows and could easily be adapted to the case of multiple-person bookings. Furthermore, it will adapt over time to changes in the marketplace.

The shortcomings of the approach are also evident. In particular, it requires information on all bookings and shows that is generally not available. An airline can fully observe which of its bookings show, but it cannot observe whether bookings it did not accept would have shown. In fact, once a departure has reached its booking limit, it is not usually possible to

determine how many additional customers would have booked, much less which of those would have shown.

One way to overcome the information shortage is through experimentation. Senilria could set the booking limit to 26 for several departures to observe the corresponding pattern of shows. This has a cost—based on the information in Table 11.5, overbooking at 26 costs an average of \$80 per flight relative to not overbooking at all, and it costs \$320 relative to the optimal overbooking limit of 23. This is a classic trade-off between exploration and exploitation; the challenge in using a data-driven approach in this setting is to experiment periodically by setting the booking limit artificially high for some departures to observe the pattern of shows and then use that information to set the optimal booking limits for future departures. Periodic experimentation is required to ensure that the underlying pattern of demand has not changed in a way that would imply a different limit.

11.5 SERVICE-LEVEL POLICIES

In the absence of any other considerations, properly calculated risk-based booking limits maximize expected short-run profitability. From this, it would seem to follow that risk-based booking limits would be universally used. However, this is not the case. Many airlines and other overbookers do not explicitly trade off denied-boarding costs and customer revenue to set booking limits. Rather, they try to determine the highest overbooking limit that keeps denied-service incidents within management-specified levels. Reasons that a company might prefer a service-level policy over a risk-based policy include the following.

- Some components of denied-service cost, such as ill will, might be viewed as difficult or impossible to quantify. In view of this, management might consider a service-level policy to be safer than a risk-based policy.
- Risk-based booking limits can lead to wide variation in booking limits—and potential numbers of denied boardings—from flight to flight. Under a risk-based policy, an airline might set overbooking levels ranging from 10% to more than 50% of capacity for different flights departing from the same airport during the day. Instead of accepting this wide variation, the airline might be comfortable with simply setting a constant overbooking limit over all flights to smooth staffing needs and to ensure that no single flight ever experiences a massive overbooking situation.
- Corporate management may feel that a risk-based booking limit calculation is a mathematical black box that they cannot understand and do not have confidence in. On the other hand, service-level limits are easy to understand, the results are easy to measure, and they give comfort that denied-service levels will not be out of line with those experienced by competitors.

Whatever the reason (or combination of reasons), service-level policies are commonplace. Examples of companies that use (or have used) service-level policies include airlines such as American (Rothstein and Stone 1967), rental car companies such as National (Geraghty and Johnson 1997), and many hotels and resorts.

A typical service-level policy is to limit the fraction of booked customers who are denied service. For example, an airline might set a policy that the fraction of bookings that result in a denied boarding should be approximately 1 in 10,000. Recalling that (slb) refers to shows given booking limit b and that C refers to capacity, this policy would be equivalent to setting b such that

$$\frac{E[(slb) - C]^+}{E[(slb)]} = \frac{1}{10,000}, \quad (11.10)$$

or, equivalently, $E[(slb) - C]^+ = 0.0001 \times E[(slb)]$.

An alternative service-level policy would be to specify that the number of denied-service incidents should be some specified fraction of *customers served* rather than of total bookings. This is the way in which airlines report denied boardings to the Department of Transportation. Under this policy, the company would set b so that

$$E[(slb) - C]^+ = q E[\min((slb), C)] \quad (11.11)$$

where q is the target denied-service fraction and $E[\min((slb), C)]$ is expected boardings (accommodated customers) given a booking limit of b .

It should be noted that applying the policies implied by Equations 11.10 and 11.11 on a flight-by-flight basis (or a rental-day-by-rental-day basis) will result in conservative booking limits. That is, the fraction of denied-service incidents that will actually occur will almost certainly be less than q . This is because the booking limits calculated by Equations 11.10 and 11.11 will result in an average denied-service rate of q on flights with demand sufficiently high that the booking limit is relevant. Some flights will have such low demand that they will never sell out. Including bookings on these flights will improve the denied-service statistics at a system level.

11.6 HYBRID POLICIES

Many airlines, hotels, and rental car companies are not purists when it comes to overbooking. In many cases, they use a hybrid policy under which they calculate booking limits using both risk-based and service-level policies and use the minimum of the two limits. This allows them to gain some of the economic advantage from trading off the costs and benefits of overbooking while still ensuring that metrics such as “involuntary denied boardings per 10,000 passengers” remain within acceptable bounds—however these bounds are set.

11.7 EXTENSIONS

11.7.1 Dynamic Booking Limits

Suppliers overbook to compensate for both cancellations and no-shows. However, so far we have only considered no-shows in our calculation of booking limits. Without cancellations, a supplier can simply calculate an optimal booking limit at the beginning of the booking period and hold it constant until departure. The supplier does not need to update

the booking limit or change it over time. When cancellations are factored in, the situation becomes more complex. Specifically, the optimal booking limit for a supplier who allows bookings to cancel prior to departure will change over time. In this case, the supplier needs to calculate a *dynamic booking limit*.

In the most general case, a supplier will face both cancellations and no-shows. This is certainly the case for airlines, hotels, and rental car companies. American Airlines has estimated that about 35% of all bookings will cancel before departure, compared to 15% of bookings at departure that will not show. While cancellation rates are typically higher than no-show rates, cancellations are less costly than no-shows since they allow the opportunity to accept a booking to fill the space freed up by the cancellation.

A common model of cancellations is to estimate a dynamic cancellation fraction $r(t)$, where t is the number of days until departure. Assume an airline has accepted $b(t)$ bookings at time t . Then it would expect that, on average, $r(t)b(t)$ of those bookings will cancel, while $[1 - r(t)]b(t)$ of them will convert to bookings at departure. With a show rate of ρ , the airline would expect that an average of $\rho[1 - r(t)]b(t)$ of the current bookings will show. One tempting (and common) approach is to use $\rho[1 - r(t)]$ as a dynamic show rate and to apply standard risk-based or service-based models to determine the current booking limit.

Example 11.5

An airline is using a dynamic version of the deterministic booking heuristic in Equation 11.1, in which the booking limit is set to the capacity divided by the show rate—that is, $b = C/\rho$, where C is the capacity and ρ is the expected show rate. The airline has an average no-show rate of 13% for a flight assigned an aircraft with 120 seats. Twenty days before departure, the expected cancellation rate for this flight is 60%. The airline therefore sets a booking limit of $b = 120/[(1 - 0.6) \times 0.87] = 345$. Five days before departure, the expected cancellation rate is 20%, and the corresponding booking limit is $b = 120/[(1 - 0.2) \times 0.87] = 172$.

Bookings tend to firm toward departure—the fraction of current bookings that will cancel decreases as t approaches 0. This means that $r(t)$, the expected cancellation rate, is typically an increasing function of t . It may not, however, be a continuous function. For example, many airlines specify that certain discount tour and group bookings cannot be canceled later than 14 days prior to departure without penalty. This often leads to a booking cliff, with the number of bookings crashing dramatically on the same day as all the tours and groups simultaneously cancel their unsold allocations. The booking limit needs to adjust accordingly to reflect the fact that reservations on the books within 14 days of departure are far more likely to convert to shows than those on the books prior to that period.

A typical dynamic booking limit is shown in Figure 11.4. Here, the heavy curve shows the total booking limit at each time before departure, while the lighter line shows actual bookings. The booking limit starts out high to allow for the possibility of a high proportion of future cancellations. As the departure date of the flight approaches, the booking limit decreases as bookings become more firm. Bookings initially start out low and increase toward

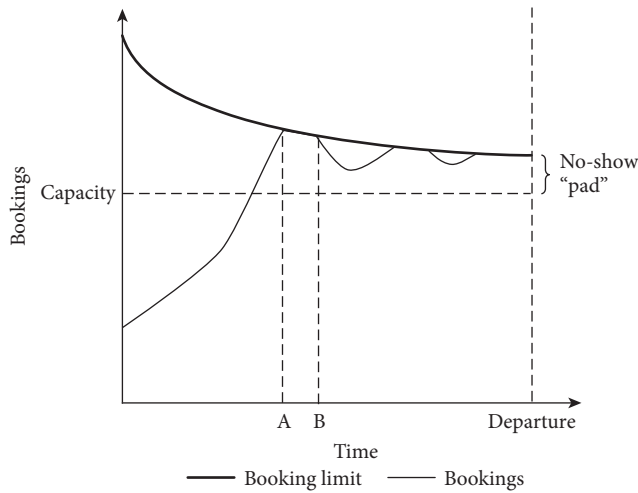


Figure 11.4 Example of a dynamic booking limit.

departure. In Figure 11.4, total bookings hit the booking limit at time A. At that point, the airline stops accepting bookings. Note that for some period, accepted bookings can actually exceed the booking limit. In Figure 11.4, a sufficient number of cancellations occur so that bookings fall below the limit at time B. At time B, the airline starts accepting bookings again, until the booking limit is reached again, at which time the flight would again be closed. In the figure, the flight closes three times and reopens twice.

The difference between the booking limit at departure and the capacity of the aircraft is the amount the airline has overbooked specifically to accommodate no-shows. It is where the airline would like bookings to be, considering future cancellations and no-shows (assuming the airline is following a risk-based overbooking policy). If the airline has done a good job managing bookings over time *and* gotten a bit lucky, it will reach this point. Of course, demand may be so low that bookings never reach the limit. Or they may reach the limit at some point, but a higher or lower cancellation rate than expected may result in bookings at departure being higher or lower than the ideal. Cancellations add yet another complication to the already dicey game of managing bookings to maximize profitability.

Figure 11.4 illustrates an important aspect of dynamic booking limits: the booking limit tends to be high early on, just when actual bookings are likely to be low. Except for flights that have extraordinary levels of early-booking demand, the total booking level is not likely to be binding when there is still a long time until departure. This means that using computational resources to calculate exact total booking limits that have little chance of being binding and to update them frequently months before departure is not necessary. However, as departure approaches, the optimal booking limit decreases and the number of bookings generally increases. This means it becomes more and more worthwhile to compute an exact booking limit as departure approaches. Typically, an airline may only update the total booking limit for a flight monthly when it is six months or more until departure. As departure approaches, the airline will increase the frequency of recalculation. Most airlines

will be recalculating the total booking limit every day during the last week before departure. Similar approaches are used by hotels, rental car companies, freight carriers, and other suppliers that practice overbooking.

11.7.2 Overbooking and Capacity Allocation

The models we have studied so far have been based on the assumption of a single fare class. However, as Chapter 9 shows, the same unit of capacity can often be sold to customers from many different fare classes at different prices. If customers are booking in reverse fare order *and* bookings can cancel or not show, then the airline faces a combined overbooking and capacity control problem. Say we have n fare classes, with $p_1 > p_2 > \dots > p_n$ with class n booking first, followed by class $n - 1$, and class 1 booking last. The problem faced by the airline is the same in spirit as the capacity allocation problem treated in Chapter 9—that is, the airline needs to set booking limits (or, equivalently, protection levels) for each fare class. The difference now is that these booking limits need to incorporate the fact that some of the bookings from each class are likely to cancel or not show.

The problem of finding optimal booking limits for multiple fare classes—the *combined overbooking and capacity allocation problem*—is extremely difficult to solve in general. It is complicated by the fact that not only are different booking classes likely to have different fares, but they are also likely to have different cancellation and no-show rates. In fact, the general problem of combining overbooking with capacity allocation is so difficult that many companies use some variant of the following heuristic.

Combined Overbooking and Capacity Allocation Heuristic

1. Compute a total booking limit for the entire plane using either the deterministic heuristic, a risk-based approach, or a service-level approach. Call this limit B . No matter what approach is used, B will be greater than or equal to capacity.
2. Use a capacity allocation approach such as EMSR-a or EMSR-b to determine protection levels. (Recall that optimal protection levels are not based on capacity.)
3. From the protection levels calculated in step 2, determine booking limits for each fare class as if the capacity were equal to B .
4. Update B and the protection levels as bookings and cancellations occur.

Example 11.6

An airline is selling three fare classes—full fare, standard coach, and discount—on a flight with 100 seats. Three weeks before departure, the airline sets a total booking limit of 115. Using EMSR-b, the airline calculates protection levels of 35 for full fare and 60 for full fare and standard coach. The airline then sets booking limits of $115 - 60 = 55$ for discount bookings, $115 - 35 = 80$ for standard coach, and 115 for full fare.

A question not yet addressed is how the total booking limit B should be calculated. The risk-based approaches discussed in Section 11.4 assume that the same fare will be received

for each additional seat that is filled. This is the right assumption when there is a single fare. When there are multiple fares, it is not so clear what the increased revenue would be from filling an additional seat. A common heuristic is to use an estimated fare that is a weighted average of the fares, where the weights are proportional to the mean demands in each fare class. In other words, the average fare is

$$p = \frac{\sum_{i=1}^n p_i \mu_i}{\sum_{i=1}^n \mu_i},$$

where μ_i is the mean demand in fare class i . Substituting p into the risk-based algorithms will enable the calculation of a booking limit.

This approach—sometimes called the *pseudocapacity* approach—is widely used. It has the distinct advantage of allowing the supplier to mix and match overbooking and capacity management approaches; any approach to calculating the total booking limit (e.g., risk based or service level) can be combined with any approach to calculating the booking limits for different fare classes. Research has shown that it generally provides a good solution as long as no-show rates do not vary widely among classes—however, it is by no means optimal.

An alternative to the pseudocapacity approach would be to use a data-driven approach such as that described in Section 11.4.5. In theory, this would allow us to determine a booking limit that maximizes expected revenue without making assumptions about the mix of customers that would be accepted or rejected. Note, however, that the number of decisions that need to be considered is much greater because we need to update the dynamic booking limit over time, which implies that much more experimentation would be required to understand the implications of different limits at different times. Developing hybrid approaches that enable taking advantage of experimentation but use realistic structural assumptions is a promising area of research.

11.7.3 Other Extensions

We have assumed that the denied-boarding cost is a constant. In reality, the denied-boarding cost per passenger is likely to be an increasing function of the number of oversales. Consider a flight that is oversold by 15 passengers. The airline might be able to persuade five people to volunteer to take another flight for \$200 apiece. It might convince another seven volunteers for \$600 apiece. It then may have to choose three more involuntary denied boardings, with an associated cost of \$850 apiece (including ill-will cost). In this case, total denied-boarding cost is the piecewise linear function of the number of oversales shown in Figure 11.5.

An airline cannot know in advance exactly how many volunteers it will induce at each level of compensation for a particular flight departure. This means not only that the denied-boarding cost is not constant but also that it is a random variable. Increasing the booking limit b leads to an increase in the expected number of denied boardings $E[((sb) - C)^+]$, which in turn leads to an increased expected denied-boarding cost per passenger. Instead of a constant value of denied-boarding cost D , the airline needs to calculate an expected

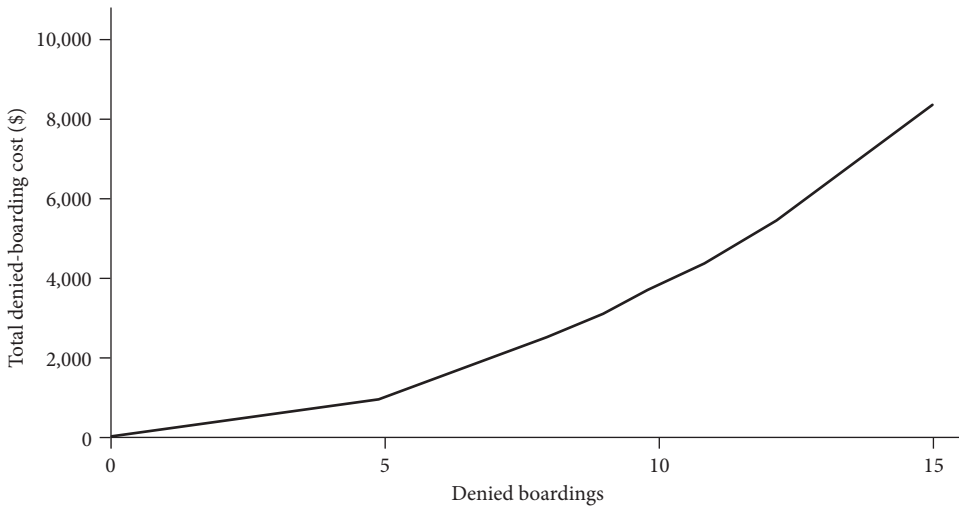


Figure 11.5 Nonlinear denied-boarding cost.

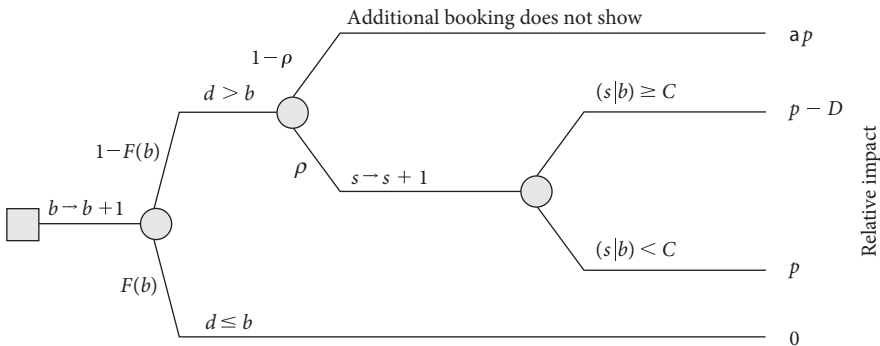


Figure 11.6 Calculating the optimal total booking limit with a no-show penalty.

denied-boarding cost as a function of the booking limit. Increasing denied-boarding costs tend to *reduce* the optimal booking limit relative to constant denied-boarding costs.

We have also assumed that tickets are totally refundable and that no-show customers do not pay any penalty. This was standard practice in the airline industry for many years. However, airlines, hotels, and rental car companies are increasingly selling partially refundable bookings and charging penalties for no-shows and cancellations. A partially refundable ticket or no-show penalty changes the underlying economic trade-offs and therefore also changes the optimal booking limit. Assume the airline charges a penalty of α times the price for each no-show, where $\alpha < 1$. Thus, if $\alpha = 0.25$, the airline would collect \$25 for a passenger who purchased a \$100 ticket but did not show. The marginal impact of changing the booking limit in the case when the airline collects αp from each no-show is shown in Figure 11.6. This tree is identical to the tree in Figure 11.3, with the exception that the top branch, which represents an additional no-show from increasing the booking limit by 1,

now has a payoff of αp . We can use this tree to calculate the booking limit that maximizes expected net revenue using the same approach as before (see Exercise 1).

Finally, in many industries the risk of no-shows and cancellations is counterbalanced to some extent by the possibility of *walk-ups*: customers without a reservation who show up just prior to departure wanting to buy a ticket.⁹ Companies sell capacity to walk-ups only if it is available after all shows are accommodated. Since walk-up customers have not made bookings, they are not entitled to payment if they are not served. Walk-ups are highly desirable customers: not only can they be used to fill seats that would otherwise go empty, but they can usually be charged high prices. Thus, hotels will often charge walk-ups the rack rate—even if the hotel is not close to being full—under the belief that a walk-up customer has a high willingness to pay since the cost of finding an alternative is relatively high.

It is easy to see that the possibility of walk-ups reduces the optimal total booking limit. If a hotel knew that it would have 10 high-paying walk-up customers arriving every day, it would set its booking limits as if it had 10 fewer rooms. Of course, like all elements of future customer demand, the number of walk-ups is uncertain at the point when the booking limit needs to be set. However, in cases where walk-ups are important, companies will forecast expected walk-up demand and incorporate its effect explicitly in calculating booking limits (see Exercise 3).

Each of these complications adds to the complexity of calculating the total booking limit. However, in any circumstances, the basic idea is the same. We want to find the booking limit such that the additional revenue we would expect from increasing the booking limit balances the additional expected denied-boarding cost we would incur.

11.8 MEASURING AND MANAGING OVERBOOKING

Prior to the deregulation of the airlines, airline management focused on *load factor* as the key performance indicator, both for individual flights and on a system level. The load factor of a flight is the ratio of the number of passengers on a flight (its load) to its capacity. A flight departing with 100 seats and 86 passengers has an 86% load factor. Load factor can be measured at any level, from a single flight to a market served by many flights to an entire airline. Up into the 1970s and even beyond, load factor was the main metric that airlines tracked. Marketing programs were evaluated based on changes in load factor—a program that increased load factors was successful, one that did not increase them was ineffective. Persistently low load factors on a flight were considered a signal that a smaller aircraft should be assigned to the route or that the flight should be rescheduled to a different time or even be dropped.

Unfortunately, load factor fails as a measure of overbooking policy. The booking policy that maximizes load factor is simple—accept every booking request. This may result in hundreds or thousands of denied boardings, but the planes will be full and load factors will be sky-high.

To avoid the hordes of denied boardings that would result from an unconstrained booking policy, airline management often instructed booking controllers to minimize denied boardings. This created a basic conflict: Any booking policy that increases load factor is

likely to result in an increase in denied boardings, and policies that decrease denied boardings are likely to reduce load factors. As we have seen, an optimal risk-based booking policy neither maximizes load factors nor minimizes denied boardings. Rather, it finds the booking limit that best balances the risks of spoilage with the risk of denied boardings, to maximize expected net revenue. Therefore, airline overbooking policies are often evaluated on two metrics:

- *Spoilage rate*: the number of empty seats at departure for which a booking was denied, expressed as a fraction of total seats on the departure¹⁰
- *Denied-boarding rate*: the number of denied boardings, expressed as a fraction of total shows for a departure (involuntary and voluntary denied boardings are usually tracked separately)

Both the spoilage rate and the denied-boarding rate need to be measured against targets. These targets can be calculated with the same forecasts used to calculate the booking limit in the first place.

Example 11.7

An airline sets a total booking limit for a flight with a 100-seat aircraft, expected demand of 110 passengers with a standard deviation of 55, a no-show rate of 10%, and a discount fare of \$200 and a full fare of \$350. If we are using the optimal risk-based booking limit calculated in Section 11.4.4, we would expect a spoilage rate of about 0.06 and a denied-boarding rate of about 0.02. Of course, the spoilage and denied-boarding rates for a specific flight departure will most likely be different, but the average overall flight departures with the same characteristics should be close to 0.06 and 0.02. If the rates are significantly different, we should seek the reasons why. Are our demand or no-show forecasts consistently in error? Are booking controllers intervening too often to raise or lower the booking limits?

It should be stressed that performance needs to be evaluated against targets in both directions. For example, it might seem that a denied-boarding rate of 1% would indicate better performance than 2%. But if the lower denied-boarding rate was achieved at the cost of a higher spoilage rate, it may mean that our overbooking policy was not sufficiently aggressive and we lost profitable opportunities to fill additional seats. If the lower denied-boarding rate did not coincide with a higher spoilage rate, we still need to understand the reason why it differed from our target. It may mean that our demand and no-show forecasts need to be adjusted for future flights.

11.9 ALTERNATIVES TO OVERBOOKING

Needless to say, involuntary denied service is not popular with customers. Customers hate arriving weary at their hotel at midnight only to be told that the hotel is overbooked and they will be bused to another hotel 10 miles away. Overbooking is also unpopular with sup-

pliers. Not only does it create unhappy customers; it is a continual source of stress for gate agents, desk clerks, or whoever needs to deliver the bad news to the customer that she is going to be denied service. Airline gate agents often feel as if the sole purpose of overbooking is to make their lives more difficult, and airline revenue managers spend a considerable amount of time explaining to them the need for overbooking and the financial benefits it brings. Furthermore, overbooking as a policy is inefficient—an airline can only find out how many of its customers are sufficiently flexible to take an alternative flight just prior to departure. If an airline knew earlier that a high fraction of its bookings were flexible, they would be more willing to overbook than if all of their bookings were inflexible.

In this environment, it is not surprising that airlines and other overbookers have actively searched for alternative ways to manage the uncertainty intrinsic in allowing customers to cancel and not show other than waiting until the last few minutes before departure. The following are some of the approaches that have been utilized or proposed:

- *Standbys.* A standby booking is one that is sold at a discount and gives the customer access to capacity only on a space-available basis. Customers with standby tickets arrive at the airport and are told at the gate whether they will be accommodated on their flight. If they cannot be accommodated on their flight, the airline books them at no charge on some future flight (usually also on a standby basis).
- *Bumping strategy.* If the fares for late-booking passengers are sufficiently high, an airline could pursue a bumping strategy—that is, if unexpected high-fare demand materializes, the airline would overbook with the idea that it can deny boardings to low-fare bookings in order to accommodate the high-fare passengers. For a bumping strategy to make sense, the revenue gain from boarding the full-fare passenger must outweigh the loss from bumping the low-fare booking, including all penalties and ill-will cost that might be incurred. Historically, airlines were reluctant to overbook with the conscious intent of bumping low-fare passengers to accommodate high-fare passengers. However, with the average full fare now equal to seven or more times the lowest discount fare on many routes, the bumping strategy may make more economic sense.
- *Flexible products.* A flexible product guarantees that the buyer will receive one out of a set of closely substitutable services. The seller decides the exact service close to (or at) the time of fulfillment on the basis of demand information acquired during the booking period (Gallego and Phillips 2004). An example would be purchasing a non-stop coach ticket from London to New York City on a date three weeks from today from an airline that has nonstop departures at 9:00 a.m., noon, and 5:00 p.m. The night prior to departure, the airline would contact the customer to inform her which flight she has been assigned to. Similarly, a traveler could book a room in Chicago from a large hotel chain (such as Marriott) that has multiple properties in Chicago. The night before arrival, the traveler would receive a text informing her which property she will be staying at. Typically, the flexible product would be sold at a discount to the fixed product in which the exact flight or exact property is specified.

One advantage of the flexible product is that it allows the seller to identify flexible customers prior to the date of service delivery. This enables them to better match

demand to limited supply. For an airline, under standard overbooking, a customer who arrives for the noon flight cannot be offered the option of the 9:00 a.m. flight, but flexible customers on the overbooked noon flight can be shifted to the 9:00 a.m. flight. This is an advantage for both the seller and the customer over standard overbooking.

Flexible products have not been widely adopted in the airline or hotel industries. However, they are used in air freight and ocean shipping where many shippers only care about departure time and delivery time and not the itinerary taken by the cargo. They are also used by internet advertisers. The German tour operator Germanwings starts with a large set of alternatives and allows customers to pay to remove specific alternatives from the set (Gallego and Stefanescu 2012).

- *Opaque products* are similar to flexible products in that sellers offer customers a bundle of substitutable products (often from different sellers). Which product the customer is assigned is revealed only when she has made her purchase. Opaque selling is used in the travel industry by Hotwire and Priceline; for example, a customer can reserve a rental car from a major provider and will only learn if it is Hertz, Avis, or National after making the reservation. One rationale for companies to participate in opaque products is the desire to offer discounted products while maintaining their brand image.
- *Options*. A logical approach to managing the adverse effects of uncertain future events is through the use of options. It is not surprising that options have been proposed as an alternative to overbooking, although they are not widely used. *Callable options* for airline seats were proposed in Gallego, Phillips, and Sahin 2008. A callable option includes the option for the seller to recall the capacity at a prespecified price before the service is delivered. The callable services could either be sold at a discount or with a recall price premium to compensate the customer for the potential inconvenience of having the service recalled. For example, a customer could purchase a discount ticket for \$200 with a recall option of \$50. If the airline found that it had excess full-fare demand for \$500 for that flight, it could call the option, in which case the customer holding the option would pay nothing but would receive a payment of \$50. The benefit for the airline is that it receives a payment of \$500 minus the \$50 call payment, or \$450 for a seat that would otherwise be sold for \$200. By offering both callable options and regular seats, a seller can segment the market between flexible and inflexible customers and thereby provide a source of additional riskless revenue.
- *Last-minute discounts*. Historically, the price of airline bookings has tended to increase as departure approaches, as airlines seek to exploit the fact that later-booking customers tend to be less price sensitive than early-booking customers. However, increasingly airlines have been using last-minute deep discounts to sell capacity that would otherwise go unused. For example, the company Last-Minute Travel (<https://www.lastminutetravel.com>) specializes in selling deeply discounted capacity for flights that are nearing departure and hotel check-in dates that are only a few days in the future. To limit cannibalization, many airlines, hotels, and rental car companies

only offer last-minute discounts through opaque channels. In a similar vein, classical concerts and operas often sell standing-room-only tickets at a deep discount once all seats have been sold.

- *Cancellation and no-show penalties.* Many airlines, hotels, and rental car companies have instituted penalties for customers who cancel or do not show. These penalties can range from 10% or less of the price all the way to 100% for a nonrefundable ticket. These penalties can significantly change the economics of overbooking. (See Exercise 1.)

Each of these approaches has drawbacks. Standby tickets usually need to be priced at extreme discounts in order to sell. Furthermore, while standbys may help reduce spoilage, they do nothing to help the situation in which the airline has accepted too many bookings and needs to deny some boardings. In fact, they make the situation worse, since both the passengers denied boarding and the standbys will be demanding seats on some future flight to the same destination.

Cancellation and no-show penalties might seem to be an obvious answer to the overbooking problem, since such penalties should decrease the number of cancellations and no-shows as well as reduce the revenue risk to airlines. However, two points need to be made. First, an airline with a cancellation penalty will still have a motivation to overbook as long as it faces a significant number of cancellations and no-shows. An airline with non-refundable tickets may still face a combined late cancellation and no-show rate of 10% or more. In this case, the airline can still increase revenue by overbooking. Second, it is not always easy for an airline to institute cancellation or no-show penalties. Customers whose travel plans are somewhat unsure will value the ability to cancel or change their flights without penalty. Instituting a cancellation penalty may induce these customers to purchase from a competitor who does not penalize cancellations.

Flexible products, opaque products, and options appear to be attractive alternatives for managing demand to better accord with supply constraints; however, their use is currently somewhat limited save for the distribution of opaque products through suppliers such as Hotwire and Priceline.

11.10 SUMMARY

- Overbooking is the policy of selling more units of a resource than are expected to be available. It is employed as a tactic by industries in which bookings can be canceled and/or some customers book but do not show up (are no-shows), and in which the cost of denying service is not too high relative to the fare. Industries in which overbooking is common including airlines, nonresort hotels, rental cars, freight, medical clinics, and universities. It is less common or nonexistent in resort hotels and cruise lines.
- The cost of denied service consists of four elements:
 - The *direct cost* of compensation to the customer
 - The *provision cost* of meals or other extras

- The *reaccommodation cost* of providing an alternative service
- The *ill-will cost* from denying service
- Four general approaches to overbooking are common:
 - A *simple heuristic* in which the capacity is adjusted by the show rate
 - *Risk-based approaches* in which a supplier trades off the cost of additional denied-service costs with the potential of additional booking revenue to find the booking limit that maximizes expected total revenue (including expected denied-service costs). The optimal risk-based booking limit can be calculated in two ways:
 - *Forecast and optimize*: Probabilistic forecasts of demand and shows are used within an optimization framework to determine the optimal booking limit.
 - *Data-driven*: The optimal booking limit is based on the net revenue minus denied-service costs that would have been achieved had the booking limit been applied to similar flights in the past.
 - *Service-level policies* in which the booking limit is set to meet a goal on the rate of denied service
 - *Hybrid policies* in which an optimal risk-based booking limit is calculated subject to service-based limits
- The overbooking problem is further complicated by the fact that booking limits will change during the time between when a flight opens to first booking and when it departs. In general, the booking limit decreases as the flight departure date approaches because there will be fewer cancellations.
- To achieve maximum contribution, overbooking needs to be solved simultaneously with capacity allocation. One heuristic is to determine a total booking limit, including overbooking, for a flight and then use that pseudocapacity as an input to the capacity allocation approaches discussed in Chapter 9.
- Two metrics that are used to measure the effectiveness of overbooking are the *spoilage rate* and the *denied-service* (or *denied-boarding*) *rate*. Spoilage measures the seats that could have been filled but were left empty as a result of a booking limit that was too low, and denied service measures the number of passengers who showed but were denied boarding on a flight because shows exceeded available capacity.
- Because overbooking (or at least involuntary denied service) is unpopular with customers, airlines and other travel providers have experimented with alternative approaches including standbys, bumping, flexible products, opaque products, options, last-minute discounts, and penalties. Each of these has its benefits and costs relative to traditional overbooking.

11.11 FURTHER READING

The history of overbooking at American Airlines is told in Rothstein 1985. A review of the literature is given in McGill and van Ryzin 1999. Key early research papers in airline overbooking include Robinson 1995 and Chatwin 1998. Papers that discuss integrating capacity

control and overbooking include Subramanian, Stidham, and Lautenbacher 1999; Feng and Xiao 2001; and Karaesmen and van Ryzin 2004. Models for overbooking in medical clinics can be found in Muthuraman and Lawley 2008 and Zacharias and Pinedo 2014.

A good overview and discussion of alternatives to overbooking, such as flexible products, opaque products, standby, and so on, can be found in Gallego and Stefanescu 2012 and the references therein.

11.12 EXERCISES

1. Solve the decision tree in Figure 11.6 to determine a formula for the total booking limit that maximizes expected net revenue when there is a no-show penalty of αp . Does the no-show penalty generally result in higher or lower booking limits than the case when tickets are entirely refundable?
2. A flight has 100 seats and a passenger fare of \$130. The number of no-shows is independent of total bookings and is given by a normal distribution with a mean of 22 and a standard deviation of 15. The denied-boarding cost is \$260 per denied boarding. Use a table of cumulative normal distribution values or the NORMINV function of Excel to determine the optimal booking limit in this case. Recall that the booking limit needs to be an integer.
3. Capacity, fare, denied-boarding cost, and the no-show distribution are the same as in Exercise 2. However, there is now a 0.6 probability that there will be a walk-up customer for the flight. Assuming there is a seat available on the flight after shows are accommodated, the airline will charge the walk-up customer \$200. What is the optimal booking limit for this flight?
4. A flight has 100 seats and a passenger fare of \$130. The denied-boarding cost is \$390 per denied boarding, and the no-show rate is 0.16. Demand for this flight is extremely high; in fact, for any booking limit $b < 200$, bookings will always hit the booking limit. What is the optimal total booking limit in this case? What is the corresponding expected net revenue? How much does the airline gain from overbooking in this case?
5. A low-cost airline sells only nonrefundable tickets. Customers pay in full at the time of booking. If they cancel or miss their flight for any reason, no portion of the price is refunded, and they cannot board another flight without buying a new ticket. Despite this policy, the airline still experiences a no-show rate of 5%. That is, if it sells 100 tickets on a 100-seat aircraft, it will, on average, depart with 5 empty seats. Should this airline overbook? If so, what kind of policy should it adopt? If it should not overbook, why?
6. A rental car company will have 100 cars available for rent on a particular day. It expects that demand for that day will be very high, so demand is certain to be higher than 300 bookings. If it expects a 15% no-show rate, what booking level should it set if it wants to ensure that the probability that a booking will be denied service is

0.0002? (You can use the normal approximation to the binomial described in Appendix B to estimate your answer.)

7. In Example 11.2, what is the minimum value of the overbooking penalty D that would be required for the expected revenue change from overbooking by one seat to be less than 0?

NOTES

1. The figures for voluntary and involuntary denied boardings here and elsewhere in this chapter are from various editions of the *Air Travel Consumer Report*, published by the US Department of Transportation Office of Aviation Enforcement and Proceedings, available at <https://www.transportation.gov/individuals/aviation-consumer-protection/air-travel-consumer-reports>.

2. The story of “system error” is from Rothstein 1985, which provides a history of overbooking at American Airlines.

3. For more background on the television advertising market, see Phillips and Young 2012.

4. In addition, many cruise lines sell trip insurance, which will refund the price of the cruise to the buyer in case she needs to cancel.

5. For the latest DOT rules on treatment of involuntary denied boardings, see US Department of Transportation 2020.

6. In certain circumstances, the ill-will cost can be much higher. In April 2017, when United Airlines could not find enough volunteers on an oversold flight from Chicago to Louisville, Kentucky, security personnel forcibly removed a struggling passenger, Dr. David Dao, who hit his head on an armrest while he was being removed and was bleeding profusely as he was dragged out of the plane. The event was filmed by other passengers on their cell phones and immediately went viral, creating an enormous backlash toward United. United Airlines later settled with Dao for a reported \$140 million (Economy 2019).

7. This booking model assumes that all shows buy a ticket. Those who are denied service pay the price p but receive denied-service compensation of D . The *net denied-service cost* to the supplier for each denied service is then $p - D$. The case in which those who are denied service do not buy a ticket but pay a fee of f can be represented in this model by setting $D = p + f$.

8. The assumption that no-shows are independent is often unrealistic—many bookings are for multiple individuals (e.g., a family traveling together), who tend to show or no-show together. Also, events like bad weather or extremely heavy traffic may drive large numbers of no-shows. However, a number of studies (starting with Thompson 1961) have shown that a binomial distribution is reasonable, assuming that very large groups are treated separately. Background on the binomial distribution is given in Appendix B.

9. Walk-ups are also known as *go-shows* in the passenger airlines and as *walk-ins* in the rental car and hotel industries.

10. It is important to note that only the seats on a flight for which a booking was denied were spoiled. Seats for which there was no booking request are not considered spoiled. A flight with five denied booking requests that left with seven empty seats had five spoiled seats. On the other hand, a flight with three denied booking requests that left with one empty seat had one spoiled seat.

12

MARKDOWN MANAGEMENT

Nobody but a fungoid creature from another galaxy with no familiarity with earthly ways would ever pay list price for anything.

—Dave Barry, *Dave Barry's Money Secrets*

“Waiting for the sale” is a time-honored strategy for savvy shoppers. The fashion-conscious may splurge on clothes shortly after the spring fashions arrive at Macy’s, Nordstrom, Bloomingdale’s, or Saks, but the budget-minded know they can save by waiting. And the longer they wait, the more they save; many items will be marked down by 70% or more by the end of the season. Waiting for the sale is not without risk. There is always the chance that the store will sell out of a desired item before it is marked down. Nonetheless, an increasing number of customers are willing to take the risk in order to realize the savings. Dave Barry may be overstating the case, but in many categories retail list price is becoming a ceiling, with most sales taking place at a discount. More and more customers refuse to buy at full price if they feel that a sale will be coming.

Retailers use a wide variety of mechanisms to discount their merchandise. Broadly speaking, these can be classified into promotions and markdowns. A *promotion* is a temporary reduction in price. Examples of promotions include a Memorial Day sale, Amazon’s Prime Day, a temporary two-for-one sale, an introductory low price, and a limited-time coupon for a 10% discount. The common link among these is that they are all temporary. In contrast, a *markdown* is a permanent reduction, usually to clear inventory before it becomes obsolete or needs to be removed to make way for new stock. When an item is marked down, both the seller and the buyer know that the price for that particular item will never go back up.

The difference between markdowns and sales promotions is illustrated in Figure 12.1, which shows the price over four months of a woman’s sweater (panel A) and a television set (panel B), both expressed as a percentage of list price. The television set was promoted three times, with its price returning to list following each promotion. On the other hand, once the price of the sweater was reduced, it never returned to earlier levels. After about three weeks at list price, it was marked down 20%, followed by three further markdowns. At the end of the four-month period, the sweater was selling for about 20% of its list price, while the television was priced at list.¹ Note that markdowns and promotions are not mutually exclusive; a temporary promotion can be applied to an item that is also marked down.

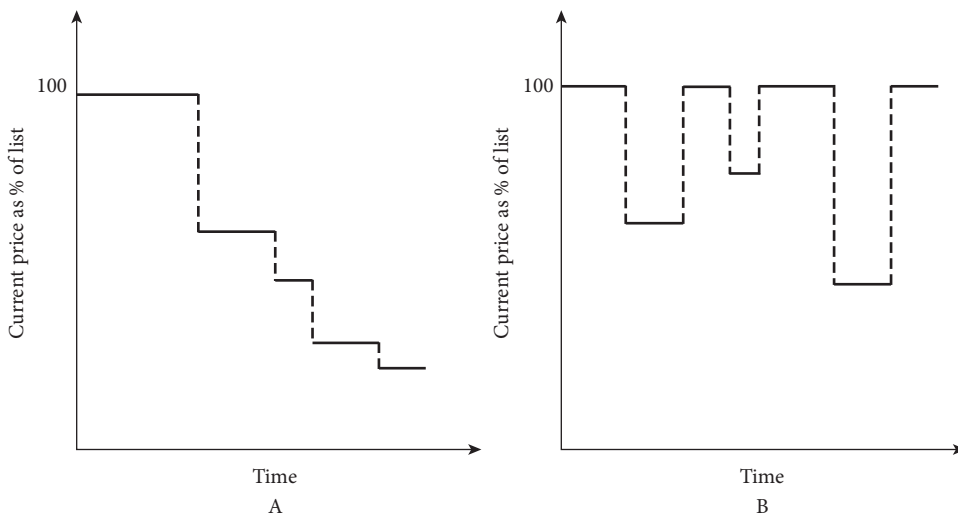


Figure 12.1 Example retail price tracks over four months for (A) a sweater and (B) a television set.

This chapter discusses the tactics behind *markdown management*. The point of markdown management is to determine the timing and magnitude of markdowns that move the inventory while maximizing revenue. For many years, retailers relied on their judgment or simple rules of thumb to determine when and how much to mark down their goods. However, over the past 20 years, an increasing number of retailers have begun to use more sophisticated analytic tools to help guide this process, and companies such as SAS and Oracle offer commercial markdown optimization software.²

We begin with at some background and how, for certain goods and services, markdowns segment the market and provide a simple method by which retailers can profit from intertemporal price differentiation. The markdown management process that retailers follow and some of the business issues that can constrain them are outlined. We formulate the markdown problem as a constrained optimization problem and consider the most common approaches to solving the problem. We examine the implications of strategic customers—those who anticipate future prices—on markdown policies. Finally, we look at the use of markdown management systems and the experiences of companies using markdown optimization.

12.1 BACKGROUND

The practice of marking down distressed inventory is probably as old as commerce itself. Nonetheless, in the United States, markdowns and promotions were relatively rare prior to 1950. Sales were associated with holidays such as Christmas and Thanksgiving, or they occurred as periodic price wars between rivals such as Macy's and Gimbels in Manhattan. In fact, most retailers tried hard to avoid markdowns. In the 1950s, managers at the Emporium, a leading San Francisco department store, were told:

High markdowns benefit no one. Not the store. Not the manufacturer. Not the customer. The store loses in value of assets or inventory. The manufacturer loses in future sales and by loss of prestige of his product. The customer is getting merchandise that is not up to standard, at a low price it is true, but, remember—she would rather have fresh, new merchandise that she can be proud of and with real value at regular price than pay less for questionable merchandise. (Emporium 1952, 3)

Many retailers considered markdowns to be *prima facie* evidence of mistakes in purchasing, pricing, or marketing. The official line at the Emporium was “Like accidents, most markdowns are a direct result of carelessness” (Emporium 1952, 4). On the other hand, a few retailers have always recognized that markdowns are more than just a way to clean up after mistakes. As early as 1915, one of the leading textbooks on retailing noted:

This method of marking up goods and making reductions as necessary seems to be the most effective way to sell some goods, especially those that are subject to changes in style. Besides bringing handsome profits to the concern directly, the spectacular cuts in prices . . . furnish excitement for readers of the store’s advertising, and help to draw crowds who may buy other goods as well as those advertised. (Nystrom 1915, 242)

Since the 1960s, there has been a steady increase in both the size and the frequency of retail discounts. Figure 12.2 shows the growth in average discount from list price paid for items purchased at both department stores and specialty stores from 1967 through 1997. In 1967, the average price paid for all items purchased at department stores was at a 6% discount from list; by 1997, the average discount was 20%. The comparable figures for specialty stores were 10% and 28%. Both the fraction of goods sold at a discount and the

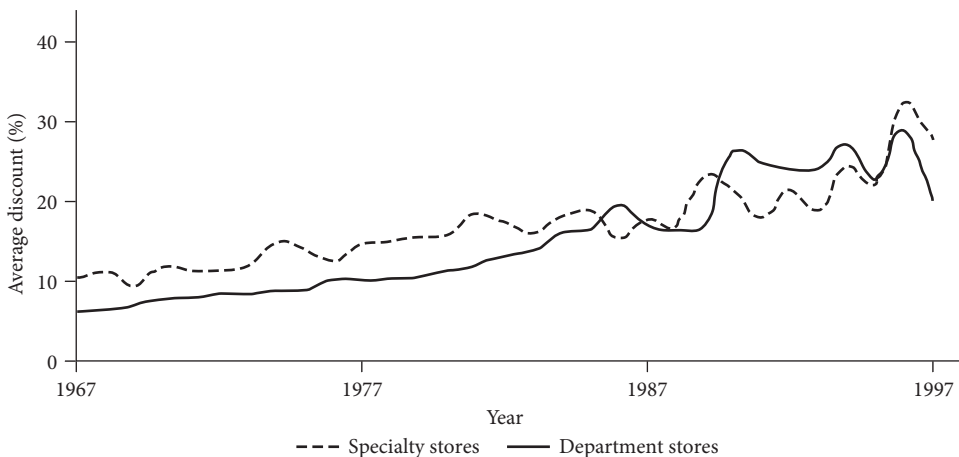


Figure 12.2 Average discount from list price for goods sold at department and specialty stores from 1967 through 1997. Discounts include both markdowns and promotions. Data are based on stores with annual sales greater than \$1 million to 1982 and greater than \$5 million thereafter. *Source:* Data from National Retail Merchants Association 1968, 1977, 1987; National Retail Federation 1998.

average discount inexorably increased from 1967 through 1997. Sometime in the 1970s, purchasing goods at a discount went from being an exception to being the rule. Markdowns as a percentage of retail sales were 8% in 1971, rising to 35% in 1996. By 2006, 78% of all apparel sold by major chains in the United States was marked down (Levy et al. 2007).

The reasons for the steady rise in the average discount illustrated in Figure 12.2 have been much discussed. Here are some of the reasons that have been proposed.

- Increased customer mobility, leading to more intense competition among retailers.
- The rise of discount chains and outlets, many with “everyday low pricing” strategies.
- A “vicious,” or self-reinforcing, cycle under which customers expect discounts and will not purchase unless retailers provide them, thereby creating an expectation of future discounts.
- An increasing interest by consumers in variety rather than standard items. The market share of colored and patterned sheets relative to white sheets increased from 12% in 1970 to 60% in 1988. Sales of patterned men’s shirts also increased relative to standard white and blue (Pashigian 1988). Since nonstandard items have shorter shelf lives than standard items, merchants have to discount more often and more deeply to move old inventory and make room for new inventory.
- The increased use of *markdown money*, by which manufacturers reimburse retailers for some portion of the discounts provided to move slow-selling items. This has made it easier and more economical for retailers to resort to more aggressive discounting. Markdown money shifts some of the risk of unsold inventory from retailers to manufacturers.

It is worth noting that a few sectors have resisted the trend toward pervasive discounting. Discounting is still relatively rare in prescription drugs, movie tickets, and various staple food and home items. As a result, for these items, even relatively small promotions can often drive dramatic sales increases. Furthermore, over the last 25 years there have been a rising number of everyday-low-price retailers, such as Walmart and Dollar General, who discount rarely, if at all. Nonetheless, the consensus is that no matter the cause, the increase in retail discounting is likely to continue, and, for the majority of retailers, planning and executing discount programs will consume an increasing portion of their time and energy. Since retail is a notoriously thin-margin business, properly planned markdowns can play a make-or-break role in determining profitability.

12.1.1 Reasons for Markdowns

Let us return to the price histories for the sweater and the television set shown in Figure 12.1. Why was the sweater marked down but not the television set? If you asked the retailer, he would probably mention two reasons. First, the sweater is a fashion good—its perceived value decreases over time, and it needs to be marked down to reflect this declining value. On the other hand, the value of the television does not decline over time. The second reason is that the sweater needs to be sold to make room for next season’s line of clothes. If it is

not sold by a certain date (called the *out date*), it will be removed from the store and sent to an outlet store or sold for 20% or less of its original price to a jobber—someone who will try to sell it for more. Markdowns are necessary to move the goods and clear the space for next season's clothes. On the other hand, there is no pressing need to clear the televisions—that is, unless the current model is being discontinued or made obsolete by the imminent arrival of a new model.

As these examples show, *from the seller's point of view*, markdown products typically share two characteristics:

1. Inventory (or capacity) is fixed.
2. The inventory must be sold by a certain out date or its value drops precipitously.

These two reasons provide the *seller* with motivation to mark down. However, there are additional reasons why the value of an item to *buyers* might decrease over time:

1. *Time of use.* An overcoat purchased in September can be worn during the upcoming winter months, unlike the same overcoat purchased in April. Many consumers assign value to wearing the coat during the upcoming winter, with that value declining as winter progresses.
2. *Fashionability.* Some consumers place a high value on getting a fashionable item early in the season, while others may care much more about buying at the lowest price. This applies not only to clothing but also to short-life-cycle goods such as consumer electronics and video games, where some buyers value being among the first to own the latest gadget or game.
3. *Deterioration.* The quality of a good may deteriorate over time. Day-old bread does not taste as good as fresh bread. Clothes that have been on display for months can get a pawed-over look that makes them less appealing.
4. *Obsolescence.* Consumers may believe that a better-quality version of a product will be available at some future date. As time passes, it increasingly makes sense to wait for the new product rather than buy the current version. To induce customers to buy now rather than wait may require deeper and deeper discounts. This is an especially important effect in consumer electronic goods, computer equipment, and automobiles.

Note that the first three reasons involve customer segmentation and price discrimination. *Time of use* segments the market between those customers who need an overcoat now (and are willing to pay more for it) and those who can wait. *Fashionability* segments the market into fashionistas, who place a high value on being au courant, and value-conscious customers, who are willing to wait. And deterioration allows the seller to offer an inferior product at a lower price. As we have seen, customer segmentation and price differentiation are powerful ways for a seller to increase revenue, especially when his inventory is constrained. Thus, it should be no surprise that they play a major role in the use of markdowns as a selling strategy. It seems reasonable that prices would drop for items whose quality deteriorates over time—everyone's willingness to pay for day-old bread is less than that for fresh bread.

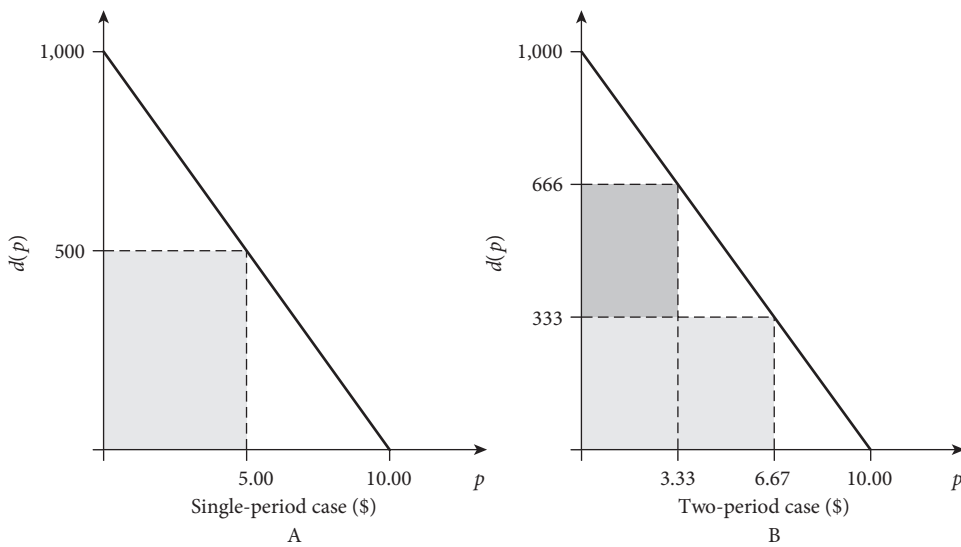


Figure 12.3 Optimal prices, sales, and total revenue in (A) the single-period case and (B) the two-period case. In the two-period case, the optimal first-period price is \$6.67, and the optimal second-period price is \$3.33.

While deteriorating quality is often found in markdown products, we can use a simple model to show how markdowns can segment a market even when quality does not deteriorate.³ Consider a seller facing the linear price-response function $d(p) = 1,000 - 100p$, shown in Figure 12.3. Recall that this linear price-response function is equivalent to a population of 1,000 potential buyers whose willingness to pay is uniformly distributed between \$0 and \$10. Assume that the good being sold has a marginal cost of zero. (As we shall see, this is often a reasonable assumption in markdown optimization.) Section 5.3 shows that, given a single period, the price that maximizes the seller's revenue is $p^* = \$5$, with corresponding sales of 500 units and total revenue of \$2,500, which is equal to the shaded area in panel A of Figure 12.3. At this price, everyone in the population with a willingness to pay greater than or equal to \$5 purchases the product, and those with a willingness to pay less than \$5 either do not purchase or purchase from a competitor.

So far, so simple. But what happens if, after all the buyers with a willingness to pay higher than \$5 had purchased the product, the seller had the opportunity to change the price? In this case, it should be apparent that the seller can gain additional revenue by charging a lower price in the second period. If the seller charges a second-period price higher than \$5, he will not sell anything in the second period, since all customers with a willingness to pay higher than \$5 have already purchased. But if he charges a lower price—say, \$2.50—he can realize some additional revenue. In addition to the 500 units he sold at the first-period price of \$5, he would sell an additional 250 units in the second period at a price of \$2.50. His second-period buyers are those whose willingness to pay is between \$2.50 and \$5. By charging a lower price in the second period, the seller can increase his total revenue by 24%, from \$2,500 to \$3,125.

In fact, the seller can do even better. If he sets his first-period price at \$6.67 and his second-period price at \$3.33, he can achieve total revenue of \$3,333.33. This solution is shown in panel B of Figure 12.3. He sells 333 units in the first period and 333 units in the second period, for a total contribution of $333 \times \$6.67 + 333 \times \$3.33 = \$3,330$ —an increase of 33% over the single-price case! If he had more periods with more opportunities to change the price, he could make even more revenue by selling at progressively greater discounts in each period (see Exercise 1).

If the price-response function is linear, the optimal price in the second period of the two-period model will always be one-half of the optimal price in the first period. This result depends on a number of assumptions that are unlikely to be true in practice—particularly the global linearity of the price-response function and that customers do not anticipate the markdown. However, it is interesting that there are at least two cases of products that undergo a single markdown and the discount is 50% of the full price: baked goods and Broadway tickets. Day-old bread is often sold at half off the full price. And Broadway theaters have the option to sell tickets that otherwise would not be sold through a TKTS outlet in Times Square.⁴ Perhaps coincidentally, the price of a ticket purchased at TKTS is often one-half the full price.

How does a second-period markdown result in additional revenue? There was no change in market conditions between the two periods, and the pool of customers did not change. The additional revenue came entirely from *customer segmentation*. The seller has used time of purchase and decreasing prices as a way of segmenting customers between those with a high willingness to pay and those with a lower willingness to pay. In this case, the markdown is an example of time-based segmentation introduced in Chapter 6.

However, the simple model is based on several questionable assumptions. As previously noted, it assumes a single pool of customers who are all indifferent between buying in the first period and buying in the second period. In most retail situations, we would expect that at least some buyers would have a lower willingness to pay for the item in the second period, because they benefit from having the item sooner (as with a coat at the beginning of winter rather than at the end of winter), or because the item deteriorates (as with day-old bread), or because it becomes obsolete (as with a laptop computer). Of course, the presence of customers with decreasing willingness to pay will only increase the incentives for the seller to decrease price in the second period (see Exercise 2).

The second assumption underlying the two-period model is that customers do not anticipate the second-period markdown. If customers knew (or even suspected) that the second-period price would be lower, we would expect at least some of them to wait for the sale. This is cannibalization, and cannibalization reduces and can even eliminate the benefits of segmentation. One way to incorporate cannibalization in the two-period model is to specify a fraction of customers who will wait to buy in the second period even though their willingness to pay exceeds the first-period price. In other words, these customers would be willing to pay the first-period price, but they wait for the second period because they anticipate the markdown. A *cannibalization fraction* of 10% means that 10% of the customers who would be willing to pay at the first-period price wait to purchase in the second period at a lower price instead.

TABLE 12.1
Effect of cannibalization in the two-period model

Cannibalization fraction (%)	First-period price (\$)	Second-period price (\$)	Total revenue (\$)
0	6.67	3.33	3,290
20	6.87	3.75	3,125
40	7.06	4.12	2,914
60	7.22	4.44	2,778
80	7.37	4.74	2,632

Table 12.1 shows how optimal first- and second-period prices and total revenue change as the cannibalization fraction increases from 0 to 100%. As you might expect, increasing cannibalization results in decreased revenue. Furthermore, as cannibalization increases, both the first-period price and the second-period price increase. However, as long as the cannibalization fraction is less than 100%, the retailer still gains revenue relative to the single-period revenue of \$2,500 by marking down in the second period. In this model, the retailer can improve revenue from following a markdown policy even if a majority of customers are willing to wait for the sale.

12.1.2 Markdown Management and Demand Uncertainty

The previous section shows how markdown management could improve profitability when demand is deterministic and the seller knows both the size of the total market and the distribution of customer willingness to pay. Of course, this assumption of perfect knowledge is highly unrealistic. It is easy to see that the seller's uncertainty about demand provides an additional motivation for pursuing a markdown policy. Consider a merchant with an inventory of a particular style of sweater who is unsure if the sweater will be a top seller during the upcoming season. If the sweater is a top seller, he will be able to sell his entire inventory at \$79; if it is not a top seller, then the market price will only be \$59. If he has two periods in which to sell the sweaters and offers them at \$59 during the first period, he will sell them all during the first period. But if he offers them at \$79 during the first period, he may sell them all at the high price; if not, he can lower the price to \$59 in the second period and sell the remaining stock. In this case, a markdown policy enables the seller to maximize revenue in the face of demand uncertainty.

The use of markdowns to resolve demand uncertainty is exemplified in dramatic fashion by the *Dutch auction*, a term that derives from the use of this mechanism in Dutch flower markets. The most famous of these markets is in the town of Aalsmeer, where more than 4 billion flowers are sold every year. In a Dutch auction, the seller sets a maximum and a minimum price for the product he is seeking to sell.⁵ During the period of the auction (5 minutes in the case of the Aalsmeer flower auction), the price drops steadily from the maximum price to the minimum price. Each potential buyer has a button that she can press at any time during the auction. When the first buyer pushes her button, she has bought the lot at the current price and the auction is over. If the price drops to the seller's specified minimum without any takers, the lot goes unsold.

In some ways, the Dutch flower auction is like a fashion season in fast-forward. The two markets share important similarities: Both flowers and fashion goods are perishable. In

both cases, the seller is often uncertain about the value that buyers will place on his good. For fashion goods, the uncertainty arises from the fickle tastes of customers and the inability of the seller to know whether any specific product will be considered hot during the current season. In the flower market, the uncertainty is due to the daily changing supply and demand for each type of flower. In both markets, the pattern of lowering prices over time can be seen as a way for sellers to extract the highest price for their constrained and perishable supply in the face of demand uncertainty.

12.1.3 Markdown Management Businesses

Three criteria are all required for a markdown opportunity to be present.

- The item for sale must be perishable.
- The supply must be limited.
- The desirability of the item must hold constant or decrease as it approaches its perishability date.

As we have seen, fashion goods, baked goods, Broadway theater tickets, and wholesale flower sales meet these criteria and commonly employ markdowns. Here are some other businesses in which markdowns are common.

- *Holiday items.* These include Halloween candy; Christmas trees, decorations, and cards; and fireworks for the Fourth of July. Such items usually begin to be marked down just before the holiday, with much deeper discounts once the holiday is over.
- *Tours.* A tour package typically includes both airline transportation and lodging. Although tours constitute less than 10% of leisure travel in North America, they represent a major portion of vacation travel in Europe and Asia. A typical tour package might be six days on the Costa del Sol (in Spain), in which the tour price covers both round-trip air transportation and hotel accommodations. Bus travel to and from the airport in Spain and meals might also be included. Tour operators start selling tours a year or so ahead. As time passes, the tour operator continually monitors the bookings for different tours. If a tour is performing worse than expected (i.e., bookings are low), the tour operator may lower the price. This is a markdown management problem because the price is always lowered (never raised) and the stock of inventory is perishable.
- *Automobiles.* Like fashion goods, most automobile manufacturers operate on an annual selling season. New models arrive every September and must be sold within a year to make room for next year's model. Although some models may be considered hot and not require discounting, other models will not sell quickly enough at the list price to clear the lot. The manufacturer and the dealer will use a mixture of promotions and markdowns to move the inventory. In general, the average selling price of an automobile tends to decline fairly steadily over the season. This gives automobile pricing the flavor of a markdown management problem, although the markdowns are typically realized through various promotional vehicles rather than as a reduction in the list price.
- *Clearances and discontinuations.* Almost any seller can be in a situation where he has inventory he needs to clear by a certain date. When Maytag announces that it will be

shipping a new and improved model of washing machine in three months, retailers have three months to sell their existing inventory before it becomes obsolete. Sequential markdowns are used to maximize revenue from this existing stock. The problem occurs at least occasionally for almost all retailers (and many wholesalers) but most frequently for those selling short-life-cycle goods such as computers and home electronics. In many cases, sellers of short-life-cycle goods mark down some portion of their total inventory every day.

Note that revenue management industries meet the first two criteria, but not the third. That is, while seats on a flight are limited and perishable, their desirability typically *increases* for most customers as departure approaches.

12.1.4 Constraints on Markdown Policies

While markdowns are frequently very profitable, business are often constrained in the markdowns that they can take, either by business rules or technical constraints. Some of the most common constraints are listed below.

- *Price ladders.* Discounted prices may need to be at discrete points (“rungs”) on a ladder either as discount percentages (for example, a five-rung discount ladder with 10%, 25%, 50%, 60%, and 80% as possible discounts from the list price) or as discrete price points (for example, a four-rung price ladder with potential prices of \$29.99, \$24.99, \$19.99, and \$10.99). In either case, markdowns need to be at one of the rungs on the ladder. Ladders keep pricing simple for both the retailer and the customer. In addition, they enable the retailer to take advantage of various penny-ending effects as discussed in Chapter 14.
- *List price period.* The list price may need to be in effect for some minimum period—typically several weeks—at the start of the season. The primary motivation for this rule is to maintain the value of the item in the minds of customers and to provide some integrity and rationale for the list price.
- *Maximum and minimum price reductions.* Since changing the price of an item in a brick-and-mortar store has some cost, minimum price reductions are used to avoid many small markdowns. Online retailers also use minimum price reductions to avoid incremental discounts of a few cents, which could confuse or annoy customers. On the other hand, most retailers want to avoid extremely large markdowns. An enormous initial markdown may suggest to customers that an item is inferior and may actually harm sales. Maximum reduction restrictions are often imposed on the *first* markdown taken.
- *Collections price together.* All items in a given line (e.g., Liz Claiborne sweaters) might need to have the same markdown cadence in a given store; that is, they must all be marked down the same percentage at the same time.
- *Maximum number of markdowns.* Often the total number of markdowns during a season is limited in a retail store to minimize the workload created by the need for re-tagging merchandise.

- *Minimum final price.* Typically, the final “fully marked down” price must exceed a certain percentage of the list price (say, 25%). This is to prevent extreme markdowns that devalue the merchandise. Instead of taking an extreme markdown, many retailers will send items to an outlet store or sell them to jobbers.
- *Regional markdown coordination.* In many cases, retail chain executives require that all stores in a specified region (e.g., Manhattan, London, Southern California) must have the same markdown cadence. The reason is to prevent customers from being motivated to shop at different locations in the same region to realize the best discount.
- *Residual inventory.* Sometimes, sufficient inventory must remain unsold to provide minimal stocking levels for outlet stores. This may seem a bit odd, since the rationale behind establishing outlet stores is to dispose of unsold merchandise from retail stores. But for some retailers, outlet stores have become significant profit centers in their own right, and these retailers want to ensure that there is sufficient unsold inventory from their retail stores to fully stock the outlet stores.

These restrictions can significantly reduce the benefits that a retailer might receive from markdowns on his inventory. Furthermore, they can complicate the computation of the optimal markdown policy by requiring the addition of constraints.

12.2 MARKDOWN OPTIMIZATION

We have seen *why* sellers might use a markdown policy. We now look at some approaches to determine *how* they should mark their inventory down over time. These models are based on the following assumptions.

- A seller has a fixed inventory (or capacity) without the opportunity to reorder.
- The inventory has a fixed expiration date (the *out date*), at which point unsold inventory either perishes or is sold at a small salvage value.
- Initially, the price of the good is set at list price. The seller can reduce the list price one or more times before the expiration date.
- Only price reductions are allowed. Once the price has been reduced, it cannot be increased again.
- The seller wants to maximize total revenue—including salvage value—from his fixed inventory.

Markdown optimization typically assumes that marginal costs are zero and that the goal of the seller is to maximize total revenue. This may seem surprising since the seller paid for the goods. However, at the point a seller is considering markdowns, his costs are sunk—his inventory has already been bought and paid for. Furthermore, since the seller cannot reorder, increased sales do not result in additional future orders. Therefore, the incremental cost of an additional sale is zero, and we can state the objective function in terms of maximizing revenue from a fixed inventory. This is an example of the principle articulated in

Section 5.1.1: because the markdown prices do not change the costs paid by the seller, the costs should not influence the markdown prices.

Let p_1 denote the initial (list) price, and assume that we have T markdown opportunities before the out date. For example, a typical season for a fashion item might be 15 weeks. Assuming that the item might be marked down at most once per week, we would have p_1 as the price during the first week, p_2 as the price during the second week, and so on through the final price of p_{15} .⁶ We assume a salvage value of r that is received for all unsold items following the last markdown. This would be the price a retailer would receive from a jobber for unsold clothes or the price he would expect to sell them for at an outlet store. If inventory entirely perishes at the end of period T —as in the case of a tour operator or a Broadway show—then $r = 0$. It should be obvious that we would never set the price during any markdown week to be less than r .

Let x_t be the unsold inventory at the beginning of period t . Then x_1 is the initial inventory, which is given. We assume that demand in any period is a function of the price in that period, and we let $d_t(p_t)$ denote the price-response function in period t . The price-response functions $d_t(p_t)$ satisfy all the standard properties listed in Chapter 3—that is, they are downward sloping, continuous, and so on. Sales in period t are denoted by q_t . Since sales in any period cannot be more than inventory, we have $q_t = \min[d_t(p_t), x_t]$. We initially consider deterministic models, in which x_t will uniquely determine $d_t(p_t)$ and hence q_t . We then turn our attention to (more realistic) probabilistic models, in which both $d_t(p_t)$ and q_t are random variables.

The starting inventory in each period is equal to the starting inventory from the prior period minus prior-period sales; that is, $x_{t+1} = x_t - q_t$ for $t = 1, 2, \dots, T-1$. This process is illustrated in Figure 12.4. The retailer starts with an inventory x_1 and chooses a price p_1 . He realizes sales of $q_1 = \min[d_1(p_1), x_1]$, and his inventory at the start of the second period is

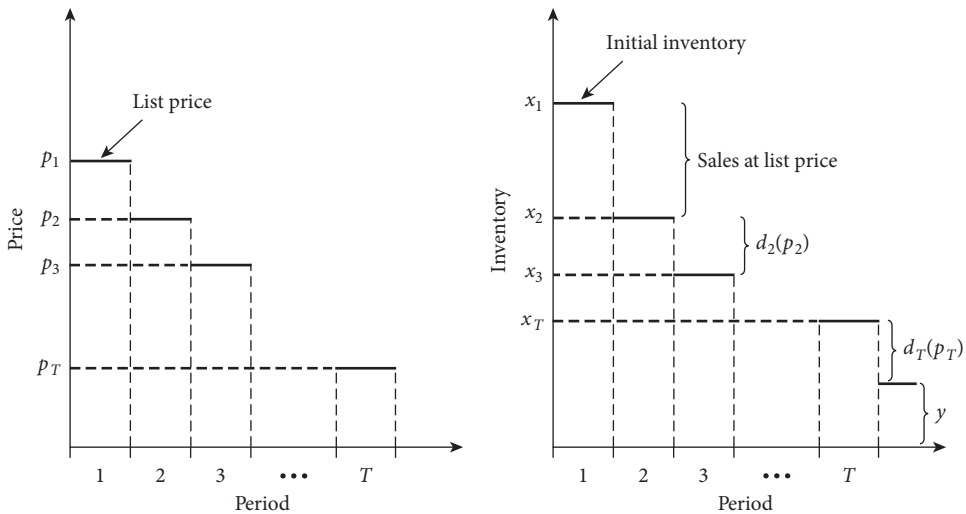


Figure 12.4 Dynamics of markdown optimization.

$x_2 = x_1 - q_1$. He chooses his second-period price p_2 and the process continues. At the end of the season, he has unsold inventory $y = x_T - q_T$, for which he receives a salvage value of r per unit. Thus, his total revenue from his fixed inventory is $R = p_1 q_1 + p_2 q_2 + \dots + p_T q_T + ry$. In a deterministic model, the seller finds the set of prices that maximize R . In a probabilistic model, he finds the prices that maximize the expected value of R .

12.2.1 A Deterministic Model

Since we know that sales in each period need to be less than starting inventory in that period, one might think that we need a set of T constraints specifying that $q_i \leq x_i$ for $i = 1, 2, \dots, T$. However, with a little reflection, it should be clear that we require only one constraint—namely, that the sum of sales for all periods needs to be less than or equal to the starting inventory. With this in mind, we can write the deterministic markdown optimization problem as

$$\max_{p_1, p_2, \dots, p_T} \sum_{t=1}^T p_t d_t(p_t) + ry,$$

subject to

$$\sum_{t=1}^T d_t(p_t) \leq x_1,$$

$$y = x_1 - \sum_{t=1}^T d_t(p_t),$$

$$p_t \leq p_{t-1} \quad \text{for } t = 2, 3, \dots, T,$$

$$p_T \geq r.$$

We can simplify the formulation even further by substituting for y in the objective function, which gives a new objective function:

$$\max_{p_1, p_2, \dots, p_T} \sum_{t=1}^T (p_t - r) d_t(p_t) + rx_1.$$

Since rx_1 is simply a constant, we can eliminate it from the objective function and write the markdown management problem (MDOWN) as

$$\max_{p_1, p_2, \dots, p_T} \sum_{i=1}^T (p_i - r) d_i(p_i), \tag{12.1}$$

subject to

$$\sum_{t=1}^T d_t(p_t) \leq x_1, \tag{12.2}$$

$$p_t \leq p_{t-1} \quad \text{for } t = 2, 3, \dots, T, \tag{12.3}$$

$$p_T \geq r. \tag{12.4}$$

The interpretation of the markdown management problem is straightforward. The objective function 12.1 states that the goal of the retailer is to maximize total revenue. This is expressed as maximizing the total increment over the salvage value received for each unit sold. Constraint 12.2 states that the retailer cannot sell more than his first-period inventory. The constraints in 12.3 guarantee that the markdown prices are decreasing over time, and constraint 12.4 guarantees that the final price is greater than the salvage value.⁷

Example 12.1

A retailer has a stock of 160 men's sweaters, and he has four months to sell the sweaters before he needs to clear the shelf space. The merchant wishes to establish a list price at the beginning of the first month and then mark the sweaters down at the beginning of each of the next three months. Sweaters unsold at the end of the fourth month will be sold to an outlet store for \$5 apiece. Furthermore, by dint of his long experience, the seller knows that demand in each of the four months can be represented by independent linear price-response functions:

$$\begin{aligned}d_1(p_1) &= (120 - 1.5p_1)^+, \\d_2(p_2) &= (90 - 1.5p_2)^+, \\d_3(p_3) &= (80 - 1.5p_3)^+, \\d_4(p_4) &= (50 - 2p_4)^+, \end{aligned}$$

where p_i is the price in month i for $i = 1, 2, 3, 4$. Formulating this problem as in MDOWN (Equations 12.1 through 12.4) gives the results shown in Table 12.2. If the retailer had to maintain a single price during the entire four months, his price-response function would be the sum of the four individual price-response functions—that is, $d(p) = d_1(p) + d_2(p) + d_3(p) + d_4(p)$ —and his price optimization problem would be to find the price p that maximizes $pd(p) + \$5[160 - d(p)]^+$. The optimal solution to this problem is $p = \$34.72$ with $d(p) = 133$, so the total contribution is $133 \times \$34.72 + 27 \times \$5 = \$4,752.76$. The optimal markdown policy resulted in a revenue increase of about 7% over the use of the optimal single price.

If a seller really knew the price-response functions for each week in the selling season, he could plug them into MDOWN and solve for the optimal markdown schedule at the beginning of the season, and his work would be done. Since the functions are deterministic, sales

TABLE 12.2
Optimal solution for example deterministic markdown problem

Month	Price (\$)	Discount (%)	Demand	Revenue (\$)
1	42.50	0	56	2,380.00
2	32.50	23.5	41	1,332.50
3	29.17	31.4	36	1,050.12
4	15.00	64.7	20	300.00
Unsold	5.00	—	7	35.00
Total	—	—	160	5,097.62

will be exactly as predicted, and there would be no need to deviate from his initial planned markdown schedule. In reality, of course, things are not so simple. Without access to the ever-elusive crystal ball, sales are likely to deviate from expectations. The most effective approach is to incorporate this uncertainty explicitly in calculating prices—an approach we take in the next section. However, the deterministic problem formulation can be used as the basis of a simple dynamic markdown algorithm.

Deterministic Markdown Pricing Algorithm

1. Solve MDOWN to determine the initial price of p_1 .
2. Observe sales q_1 during the first period. Starting inventory for the second period is $x_2 = x_1 - q_1$.
3. Solve MDOWN for $p_2, p_3, \dots, p_T$ using a starting inventory of x_2 . Put p_2 in place as the price for the second period.
4. At the beginning of each subsequent period, the current inventory is the previous period's starting inventory minus last-period sales—that is, $x_t = x_{t-1} - q_t$. Solve MDOWN for $p_t, p_{t+1}, \dots, p_T$, using a starting inventory of x_t . Use the value of p_t as the price during the upcoming period t .

Example 12.2

Consider the previous example and assume that during the first month, the retailer sells 70 sweaters instead of the 56 he anticipated. This means he has a starting inventory of only 90 sweaters at the beginning of month 2. Using $x_2 = 90$, he can use the price-response functions $d_2(p_2)$, $d_3(p_3)$, and $d_4(p_4)$ to recalculate prices for months 2, 3, and 4. The new optimal prices are $p_2 = \$34.00$, $p_3 = \$30.67$, and $p_4 = \$16.50$. He therefore sets his price for month 2 at \$34.00. At the end of month 2, he can observe remaining inventory and solve once more to determine the price he will charge in month 3.

This algorithm does not explicitly incorporate future uncertainty into its calculation of the current price. However, it is adaptive in the sense that it recalculates the price in each period, recognizing that sales to date are likely to be different from those originally anticipated. While this is less than ideal, this general approach can lead to improvements over human judgment. One study computed the results of a deterministic algorithm similar to MDOWN applied to 60 fashion goods at a women's specialty apparel retailer in the United States and compared the predicted results to what actually occurred when the store used its usual markdown policy. To conform with the business practices of the retailer, the algorithm only changed the discount level if the new discounted price was at least 20% lower than the previous price. Furthermore, each style had to be offered at list price for at least four weeks prior to the initial markdown. The deterministic algorithm increased total revenue by about 4.8% relative to standard store practice. Much of this increase came from taking smaller markdowns sooner than usual store policy (Heching, Gallego, and van Ryzin 2002).

12.2.2 A Simple Approach to Including Uncertainty

A seller is facing his last markdown opportunity. He has a fixed inventory x and must set the price p for which it will be sold during the upcoming period. At the end of that period, any remaining unsold inventory will be sold for the salvage price r (which may be 0). If he knows the price-response function for this period, his problem is a simple single-period special case of MDOWN. But now we add uncertainty to the mix. Specifically, we assume that demand given price, $D(p)$, is a random variable whose distribution depends on p . What price should the seller set to maximize expected revenue? The problem faced by the seller is

$$\begin{aligned} \max_p E[R(p)] &= pE[\min\{D(p), x\}] + r(x - E[\min\{D(p), x\}]) \\ &= (p - r)E[\min\{D(p), x\}] + rx, \end{aligned} \quad (12.5)$$

where the first term on the right-hand side of Equation 12.5 is expected revenue from first-period sales and the second term is salvage revenue.

Equation 12.5 is not too difficult to solve if we can model how the expectation $E[\min\{D(p), x\}]$ depends on p . One simple possibility is to assume that $D(p)$ follows a uniform distribution between 0 and $a - bp$ for some a and b . Then, if $a = 200$ and $b = 10$, demand would be uniformly distributed between 0 and 180 when $p = \$2.00$ and uniformly distributed between 0 and 150 for $p = \$5.00$. Expected first-period revenue, salvage revenue, and total revenue for different prices are illustrated in Figure 12.5 for the case of $a = 200$, $b = 10$, a salvage value $r = \$5.00$, and an inventory of $x = 60$. In this case, the optimal price is \$13.29, with corresponding expected sales revenue of \$440.89, expected salvage revenue of \$134.23, and expected total revenue of \$575.02.

We can extend this formulation of the two-period markdown problem to create a simple algorithm that incorporates uncertainty. The idea is similar to the deterministic algorithm

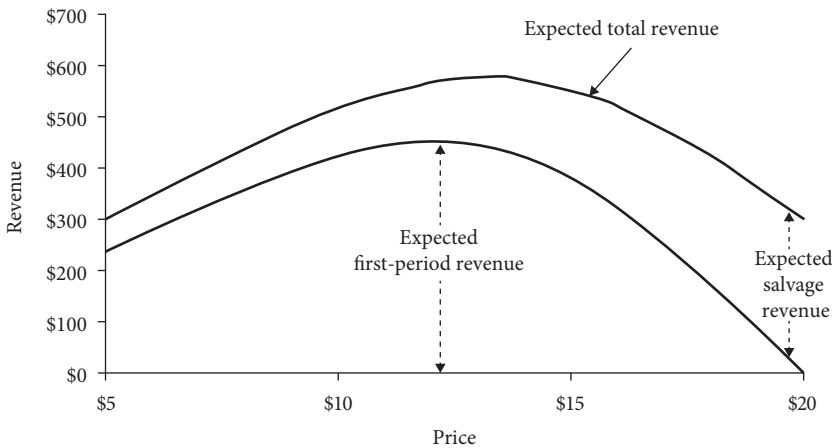


Figure 12.5 Expected revenue as a function of price in the single-period markdown model with salvage.

in the previous section, in which we solved the single-period deterministic problem to establish a markdown price for the next period, observed sales during that period, and then solved the same problem again at the beginning of the next period, using the new inventory value to get a new price. This time we follow a similar philosophy using the probabilistic formulation in Equation 12.5, but with a twist. At the beginning of each period, we estimate a probability distribution on future demands as if we are *going to hold the price constant for all remaining periods*. That is, for any price in period t , we define the random variable $\hat{D}(p_t) = \sum_{i=t}^T D_i(p_t)$.

Example 12.3

A seller faces two pricing periods, with the out date occurring at the end of period 2. $D_1(\$5.00)$, the demand in period 1 given a price of \$5.00, follows a normal distribution with mean of 50 and variance of 25, and $D_2(\$5.00)$ follows a normal distribution with mean of 30 and variance of 10. Then $\hat{D}(\$5.00) = D_1(\$5.00) + D_2(\$5.00)$ follows a normal distribution with mean of 80 and variance of 35.

In the simple probabilistic algorithm, we calculate the price for the upcoming period as if it were the final markdown period. Then we find the current price p_t by solving Equation 12.5 using $\hat{D}(p_t)$ as our probabilistic forecast of demand. This is equivalent to assuming that the price we choose for the current period will be held constant for the remainder of the selling season. This leads to the simple probabilistic algorithm.

Simple Probabilistic Markdown Pricing Algorithm

1. Set an initial price of p_1 .
2. Observe sales x_1 during the first period. Starting inventory for the second period is $x_2 = x_1 - q_1$.
3. At the start of each period t , calculate the distribution of $\hat{D}_t(p_t) = D_t(p_t) + D_{t+1}(p_t) + \dots + D_T(p_t)$.
4. Find p_t that maximizes $(p_t - r)E[\min\{\hat{D}_t(p_t), x_t\}] + rx_t$. Set this value as the price for the current period.

This approach was tested for six items in eight stores of the Chilean fashion retail chain Falabella and compared against actual sales for the 1995 autumn-winter season. The recommendations from the probabilistic algorithm generated revenue 12% higher than that obtained by the product managers (Bitran, Caldentey, and Mondschein 1998). It is tempting to compare the 12% improvement in this study with the 4.8% improvement reported from the deterministic approach described in Section 12.2.1 and attribute the improvement to the incorporation of uncertainty in the algorithm. However, this would ignore the fact that the improvements came from applications in two very different settings and that the improvements were reported based on differences from previous store practices, which may have been very different in the two situations.

12.2.3 Exhaustive Search

The previous two sections described approaches to setting markdown prices using explicit optimization. In both cases, the optimization problems were specified without incorporating any of the business considerations described in Section 12.1.4. One way to incorporate such business considerations into the markdown problem would be by adding constraints to the optimization problems in 12.1 through 12.4 and 12.5. As long as the number of the constraints is relatively small and remains stable over time, this is not much of a problem. However, if the number of constraints is large or is changing over time, then it can become a problem. Specifically, explicit optimization approaches tend to work well when the number of constraints is small and the constraints have a simple structure. As the number and complexity of constraints increases, then explicit optimization approaches tend to take more time—often much more time—to find a solution. In the case where business considerations require many constraints to be imposed and these considerations change over time, *exhaustive search* may be a superior alternative to explicit optimization.

To understand how exhaustive search works, assume that our current price is p_1 , our current inventory is I , and we have three periods left and four possible prices at which we can sell in each period with $p_1 > p_2 > p_3 > p_4$. We also assume that there is a salvage price $r < p_4$ at which all unsold inventory can be sold, with the possibility that $r = 0$. We want to determine what price to charge in the next period ($t = 2$), knowing that we will be able to change the price at most twice more. Possible markdown price paths are shown in Figure 12.6. Each arrow represents a potential price transition. To execute exhaustive search, we calculate the demand at each node in the graph in Figure 12.6 at the specified price so that the demand at price-level i in week t is given by $d_t(p_i)$. These demands can be estimated using a demand model such as the one described in Section 12.3.

Once these demands have been calculated, we move forward in time and estimate the expected revenue for each possible path. For each pathway, we start with the initial inventory. When we arrive at a node, we calculate sales as the minimum of the demand at that node

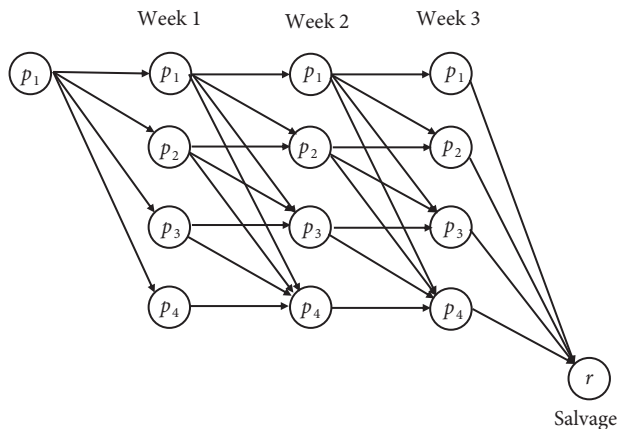


Figure 12.6 Possible markdown paths with three periods and four prices on a price ladder.

and the remaining inventory. Revenue at a node is the product of the sales and the price associated with that node. Each path terminates at the salvage node and any remaining inventory is sold at the salvage price—which will be zero if the inventory is totally perishable. Note that, for exhaustive search as for explicit optimization, we calculate a markdown price for all future periods, but we only need to commit to the price for the upcoming period. If sales are higher or lower than expected, we can recalculate the future markdown path based on updated inventory.

Example 12.4

Consider a retailer with 95 units left in stock of a television that is being replaced by a newer model in three weeks and needs to be marked down. The television set currently sells at \$500, and the seller has a pricing ladder with four prices: \$500, \$450, \$400, and \$350. Unsold televisions will be sent to an outlet where any remaining inventory will be sold for \$300—the salvage price. The demand for the television set is independent of week with $d(\$500) = 10$, $d(\$450) = 20$, $d(\$400) = 40$, and $d(\$350) = 60$, where each number is demand per week that will be realized at that price. We can calculate his optimal policy using enumeration. Starting in the first period, if he sets the initial price at \$500, he will sell 10 televisions for first-period revenue of \$5,000 and have a remaining inventory of 85; if he sets the initial price at \$450, he will sell 20 televisions for revenue of \$9,000 and have a remaining inventory of 75, and so on. When we have completed this process, we can add up the revenue per period and the salvage revenue for each policy to calculate the total revenue per policy as shown in Table 12.3. As can be seen from the table, the optimal policy is to drop the price immediately to \$450 and then drop it again to \$400 for the second and third weeks. This policy sells all of the television sets and realizes expected revenue of \$39,000. Note that the only immediate decision facing the seller is the next price to set. In this case, he would set the price to \$450, observe demand during the week, and perform another optimization at the end of the week to determine the next price to charge.

The complexity of the exhaustive search method comes from the dimensionality of the problem. The markdown schedule shown in Figure 12.6 has 20 possible markdown policies; a full 13-week schedule with 10 rungs each week would have 497,420 possible markdown schedules. While this does not necessarily present a particular computational challenge in itself, it can begin to present an obstacle when markdowns need to be computed for millions of items across hundreds of stores. To deal with this challenge, systems based on exhaustive search use various methods to prune the tree—that is, eliminate branches before evaluating them further. For example, if one combination of prices for the first two weeks results in less revenue and less remaining inventory than another combination, then the first combination can be eliminated from further consideration.⁸

The strength of the exhaustive search method arises from the fact that additional constraints tend to make exhaustive search easier because they eliminate the need to evaluate pathways that do not meet those constraints. For example, assume that in Example 12.4, the

TABLE 12.3
Revenues by week, salvage revenue, and total revenue for markdown policies applied to Example 12.4

Policy (\$)			Revenue (\$)				
Week 1	Week 2	Week 3	Week 1	Week 2	Week 3	Salvage	Total
500	500	500	5,000	5,000	5,000	19,500	34,500
500	500	450	5,000	5,000	9,000	16,500	35,500
500	500	400	5,000	5,000	16,000	10,500	36,500
500	500	350	5,000	5,000	21,000	4,500	35,500
500	450	450	5,000	9,000	9,000	13,500	36,500
500	450	400	5,000	9,000	16,000	7,500	37,500
500	450	350	5,000	9,000	21,000	1,500	36,500
500	400	400	5,000	16,000	16,000	1,500	38,500
500	400	350	5,000	16,000	15,750	0	36,750
500	350	350	5,000	21,000	8,750	0	34,750
450	450	450	9,000	9,000	9,000	10,500	37,500
450	450	400	9,000	9,000	16,000	4,500	38,500
450	450	350	9,000	9,000	19,250	0	37,250
450	400	400	9,000	16,000	14,000	0	39,000
450	400	350	9,000	16,000	12,250	0	37,250
450	350	350	9,000	21,000	5,250	0	35,250
400	400	400	16,000	16,000	6,000	0	38,000
400	400	350	16,000	16,000	5,250	0	37,250
400	350	350	16,000	19,250	0	0	35,250
350	350	350	21,000	12,250	0	0	33,250

NOTE: Optimal values are shown in *italics*.

seller wanted to impose a constraint that there should be at most one markdown before the end of the season. By eliminating markdown plans with multiple markdowns, this would reduce the number of alternatives that need to be evaluated from 20 to 16. In general, adding constraints makes explicit optimization more difficult but exhaustive search easier.

12.3 ESTIMATING MARKDOWN SENSITIVITY

Key inputs to all of the optimization approaches described in Section 12.2 are estimates of demand (or expected demand) as a function of the price in each period—the functions denoted $d_i(p_i)$ and $E[d(p)]$ in Section 12.2. Any of the techniques for measuring price sensitivity described in Chapter 4 can be used to estimate the parameters of these functions. However, markdown management systems often use forecasting approaches that focus on the *change* in demand from an expected path that was induced by a markdown.

Sales and prices from an actual markdown schedule are illustrated in Figure 12.7. This figure shows price (as a percentage of list) and weekly sales for a woman's blouse across a 30-week selling season that started in week 23 and lasted through week 52. In this case, there were three markdowns: a 30% markdown in week 32, a further 20% taken off in week 40, and another 20% reduction in week 48. The final price was 30% of the list price. Sales of the blouse grew for the first few weeks and then began to decline until the first markdown was taken, at which point sales surged. Each of the three markdowns was accompanied by a jump in sales followed by a decline. This pattern is typical—markdowns can induce sudden and substantial sales gains. Markdown optimization systems typically seek to represent this phenomenon by separately modeling the natural time track of demand—the *product life cycle*—and the incremental effect of markdowns and then combining the two.

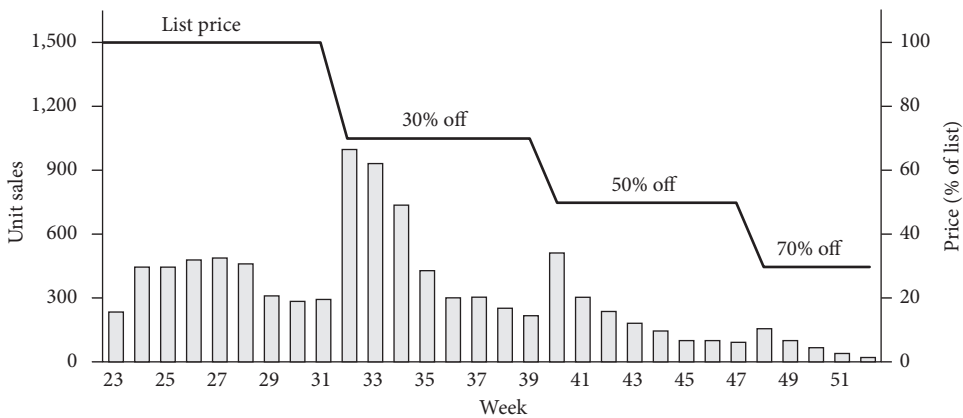


Figure 12.7 Example of weekly sales and price under a markdown policy. *Source:* Adapted from Levy and Woo 1999.

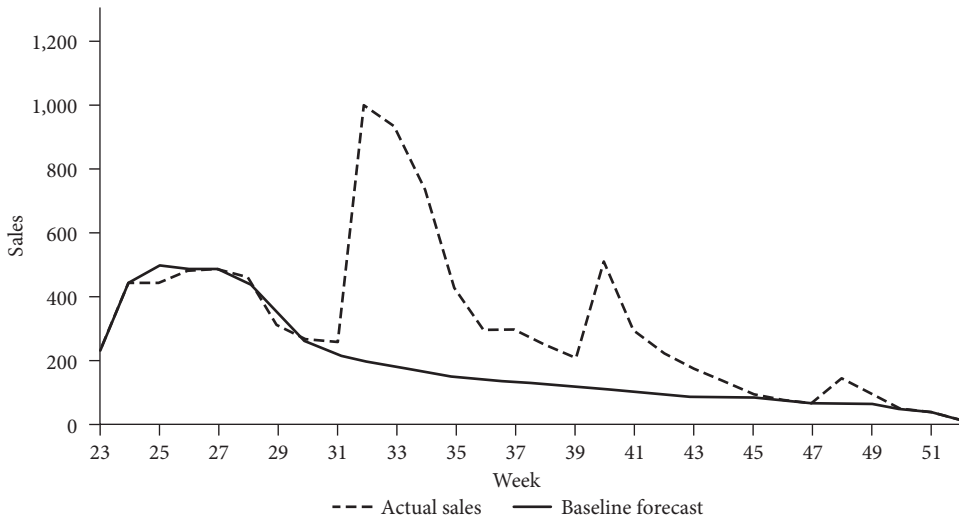


Figure 12.8 Example of weekly sales and baseline product life cycle forecast corresponding to Figure 12.7.

The first step in estimating markdown sensitivity is to calculate a *baseline product life cycle* forecast. This is a forecast of how sales would have proceeded through the season without any markdowns—that is, how the item would have sold if the price remained constant at list price. The baseline product life cycle can be estimated by aggregating sales of fashion goods for periods in which no markdowns have been taken. Figure 12.8 shows the baseline product life cycle along with actual sales for the markdown schedule in Figure 12.7. One way to estimate price sensitivity is to attribute all sales in excess of the baseline product life cycle to the effect of markdowns and to fit an appropriate price-response function. For the example in Figure 12.8, the sales in excess of the baseline for weeks 32 through 39 would be attributed to the 30% markdown taken in week 32, the sales in excess of the baseline for weeks 40 through 47 would be attributed to the 50% total discount in place during that

TABLE 12.4
*Baseline forecast and actual sales for weeks 32 through 39
 in Example 12.5*

Week	Baseline forecast	Actual sales
32	200	1,100
33	182	941
34	165	739
35	152	430
36	142	302
37	134	281
38	126	249
39	118	211
Total	1,219	4,253

period, and the sales in excess of the baseline for the remainder of the season would be attributed to the 70% total discount initiated in week 48.

Example 12.5

For the example in Figure 12.8, the baseline product life cycle forecast of sales and the actual sales for weeks 32 through 39 are as shown in Table 12.4. Forecast sales of 1,219 at the list price and actual sales of 4,253 at the 30% discount provide two points that can be used to establish a price-response function.

Note from Table 12.4 that the seller experienced a $(4,253 - 1,219)/1,219 = 249\%$ increase in sales from a 30% decrease in price, corresponding to a markdown elasticity of $249/30 = 8.3$. Such unusually high levels of elasticity are often found in markdown and promotions situations. They do not imply that the elasticity of sales with respect to list price is anywhere near as high under usual circumstances. Typically, markdown elasticities are higher than list price elasticities both because some customers wait to purchase in anticipation of the markdown and because advertised markdowns increase excitement and sales.

In addition to product life cycle and markdown sensitivity, demand forecasts for markdown management typically include seasonality. Seasonality represents the natural variation in sales for different product categories over the course of year. This can be driven by the seasonality of the items themselves—bathing suits sell more in the summer than in the winter—as well as events such as holidays and Black Friday. Seasonality can be modeled using a factor that depends on the week of the year.

We can put these elements together to derive a simple demand model. Let $t = 1, 2, \dots, 52$ represent the week of the year and τ represent the number of weeks since a product was introduced.

$$d(t) = LC(\tau) \times SEAS(t) \times PD(t). \quad (12.6)$$

In this equation,

$d(t)$ = Demand for an item (what we are trying to forecast) in week t

$LC(\tau)$ = Natural life-cycle demand for an item as a function of weeks since it was introduced

$SEAS(t)$ = Seasonality as a function of week of the year

$PD(t)$ = Price discount effect in week t

The price discount effect represents the increase in demand that will be realized from a markdown in a period. It can be modeled in several ways, but one approach is to set

$$PD(t) = e^{\beta[(p-p_t)/p]},$$

where p is the full price, p_t is the actual (marked-down) price, and $\beta > 0$ is the markdown elasticity. Note that $p_t = p$ corresponds to no markdown, in which case $PD(t) = 1$, which means that demand is entirely determined by the product life cycle and seasonality. As the markdown discount increases, p_t decreases and $PD(t)$ accordingly increases, reflecting the ability of the markdown to drive additional demand.

Equation 12.6 incorporates the three most important elements that determine demand in a markdown situation. However, other factors may need to be incorporated to make the model fully realistic. For one thing, additional promotions—such as taking an additional 10% off of everything—may occur during the markdown period. Such promotions may require an additional factor to be added. Also, when inventory of an item gets low, sales often drop as a result of the *broken assortment* effect: When inventory of an item is low in a physical store, the remaining items may become harder to find. Furthermore, some items—like a sweater—may be priced at the item level, but, as inventory becomes low, certain sizes will sell out before the inventory is totally depleted. Finally, some consumers may want to purchase a certain minimum quantity of certain items but are unable to do so if inventory is too low. The result is that sales will drop independently of price if inventory drops below a certain level. Both promotions and broken assortment effects can be modeled by adding factors to Equation 12.6.⁹

12.4 STRATEGIC CUSTOMERS AND MARKDOWN MANAGEMENT

In his 1885 novel, *Au Bonheur des Dames* (*The Ladies' Paradise*), Emile Zola chronicled the changes that a new form of commerce was bringing to Paris. The action in the novel revolves around the new department store *Au Bonheur des Dames*. The fictional store is based closely on *Le Bon Marché*, a pioneering store that brought mass merchandising and promotions to Paris for the first time. In the novel, two friends, Madame Marty and Madame Bourdelais, are shopping together when Madame Marty, on a whim, buys a red parasol for 14 francs, 50 centimes. Madame Bourdelais chides her, “You shouldn’t be in such a hurry, in a month’s time you could have got it for ten francs. They won’t catch me like that!” (Zola [1885] 1995, 245).

Madame Bourdelais is the first recorded example of a *strategic customer*—a customer who anticipates future pricing action on the part of a seller and acts accordingly. Specifically, modern customers have learned from past experience that most fashion goods and other markdown items will be marked down. This creates a motivation to wait for the sale—a phenomenon that has become so pervasive that, as noted above, only a small fraction of fashion goods are still sold at list price. Waiting for the sale is not just a phenomenon for brick-and-mortar stores; it is also commonplace in e-commerce where retailers

not only apply markdowns but also employ regularly scheduled promotions such as Cyber Monday and Amazon's Prime Day. One survey showed that 77% of customers in India who purchase online "sometimes" or "always" wait for a sale (Gautam, Shubham, and Mishra 2018). Customers have learned that goods being sold on the internet are just as likely to be marked down as they are in physical stores.

The fact that customers can anticipate a markdown and that some of them are willing to wait for it complicates markdown pricing. So far, the discussion has assumed that customers do not anticipate a markdown—they purchase when the price first goes below their willingness to pay. If customers truly behaved this way, the logic behind a markdown strategy would be unassailable—the model of sequential downward pricing introduced in Section 12.1 would maximize revenue. But a customer who believes that she could purchase the item in a future period at a lower price and is willing to wait might do so even if the current price is less than her willingness to pay. Countering her tendency to wait is the possibility that the item may not be available in the next period if it sells out in the current period.

We can formulate the decision faced by a customer in the following way. Assume that the current price of the item is p_H and that the customer believes, based on the past actions of the sellers, that, in the next period, the item will be marked down to some lower price $p_L < p_H$. Her willingness to pay for the item is w and, for simplicity, we assume that she is indifferent between purchasing the item today or purchasing in the next period. We further assume that $w > p_H$ so that she would be willing to purchase in the first period if she knew that the item would not be available in the second period. The final element in her decision is whether the item is likely to be available for purchase if she waits. Denote her judgment of the probability that the item will be available in the second period as ρ . Then, a risk-neutral customer will purchase in the first period if and only if her surplus from a first-period purchase is greater than her expected surplus from waiting—that is, if $w - p_H \geq \rho(w - p_L)$. Rearranging terms, we see that she will purchase in the first period if $\rho \leq (w - p_H)/(w - p_L)$.

Example 12.6

Sharon is considering whether to buy a ticket for a Broadway show for full price at \$400 or go to the TKTS outlet on the day of the show in hope that a ticket will be available at half price for \$200. Her willingness to pay for a ticket is \$500. She will purchase now if her probability that the ticket will be available at TKTS is less than or equal to $(\$500 - \$400)/(\$500 - \$200) = 1/3$. If Sharon believes that the probability that a ticket for the show will be available at TKTS is greater than $1/3$, then she will take the chance and seek to buy at half price.

Example 12.6 illustrates the important role that consumer expectations play in determining the profitability of a markdown strategy. Specifically, the higher the likelihood that a good will be sold at a markdown in the future, the more customers will be willing to wait. In Example 12.6, the ticket seller wants customers to believe that tickets will be scarce in the future so they will buy now. However, to segment the market and sell at differential prices, the seller would like to sell second-period seats at a discount. If the seller sells too many seats

at a discount, it encourages high-willingness-to-pay customers to wait rather than purchase at full price. How do we determine the right policy in this case?

We can explore this issue with a simple game-theoretic model. Assume that a seller sells to two classes of customers: those with a high willingness to pay (w_H) for his product and those with a low willingness to pay (w_L), with $w_H > w_L > 0$. Assume that the number of high-willingness-to-pay customers is D_H and the number of low-willingness-to-pay customers is D_L . The seller can sell in either period at one of two prices: a high price p_H or a low price p_L , with $p_H > p_L > 0$. For the moment, we assume that p_H and p_L are given—that is, they are not determined by the seller. Finally, we assume that $w_H > p_H > w_L > p_L$, which means that the high-willingness-to-pay customers would be willing to purchase at either price, while the low-willingness-to-pay customers would only be willing to pay at the lower price. The seller has two decisions: how much of the good to purchase (x) and the price to charge in each of the two periods. We assume that customers are aware of the seller's decisions and act to maximize their expected utilities.

Assume that the seller has x units on hand. What prices should he charge in each period? There are three possibilities:

1. Charge the high price in both periods—denote this policy by (p_H, p_H) .
2. Charge the low price in both periods: (p_L, p_L) .
3. Adopt a markdown policy: (p_H, p_L) .

We do not need to consider the policy of raising the price (p_L, p_H) since everyone would purchase in the first period and it achieves the same revenue as (p_L, p_L) .

Under the first policy, only the high-willingness-to-pay customers will purchase and the seller will realize revenue of $R(p_H, p_H) = p_H \min(x, D_H)$. Under the second policy, both high- and low-willingness-to-pay customers will purchase at the lower price and the corresponding revenue will be $R(p_L, p_L) = p_L \min(x, D_H + D_L)$. Let us assume that $x \leq D_H + D_L$ so that supply is less than or equal to total demand. In this case, $R(p_H, p_H) \geq R(p_L, p_L)$ if $x \leq (p_H/p_L)D_H$. In other words, if supply is sufficiently scarce, it is better to price high than price low. If supply is sufficiently abundant, then it is better to price low than price high.

But is a markdown policy ever optimal? Whether a markdown is optimal depends on the expectations of the high-willingness-to-pay customers. Assume that the seller adopts the markdown policy (p_H, p_L) . Then, as noted above, high-willingness-to-pay customers will purchase in the first period if they feel that the probability that the item will be available at the lower price in the second period is lower than $(w_H - p_H)/(w_H - p_L)$. If all of the high-willingness-to-pay customers wait until the second period to purchase, they will be competing with the low-willingness-to-pay customers for the scarce supply. Assuming that every customer has an equal probability of purchasing the scarce supply, the probability that any customer will be able to purchase is equal to the supply divided by the total demand, or $x/(D_L + D_H)$. This means that a markdown policy will only be effective if

$$x \leq \frac{(D_L + D_H)(w_H - p_H)}{(w_H - p_L)}. \quad (12.7)$$

If condition 12.7 holds, under a markdown policy (p_H, p_L) , the high-willingness-to-pay customers will all purchase in the first period at the high price, while the low-willingness-to-pay customers will wait to purchase in the second period at the lower price. If condition 12.7 does not hold, all of the high-willingness-to-pay customers will wait and seek to purchase at the lower price. If $x \leq D_H$, there is no motivation for the seller to adopt a markdown policy since he can sell all of his inventory in the first period at the high price. Define Z as the quantity on the right side of the inequality (Equation 12.7)—that is, $Z = (D_L + D_H)(w_H - p_H)/(w_H - p_L)$. Now assume that $D_H < x < Z$. In this case, by adopting a markdown policy, the seller can achieve revenue of $p_H D_H + p_L(x - D_H)$. This is clearly higher than the revenue of $p_H D_H$ that he would achieve from the high-price policy (p_H, p_H) . It is also higher than the revenue of $p_L x$ that he would achieve from the low-price policy (p_L, p_L) . So, in this case, a markdown policy is optimal. Finally, when $x \geq Z$, the revenue from a high-price policy is $p_H D_H$ and from a low-price policy is $p_L x$; which of these two policies generates higher revenue will depend on the specific values of p_H, p_L, x , and D_H . The conditions under which these different policies are optimal are summarized in Table 12.5.

Example 12.7

A theater owner has 100 unsold seats left and has established the price of tickets for his show at \$100 per seat. He has the option to sell unsold seats for \$50 on the day of the show. Alternatively, he could drop the price to \$50 immediately. He knows that there are two types of customers for the show: *theater lovers* who are willing to pay up to \$150 for a seat and *bargain hunters* who are willing to pay \$75 for a seat. He believes there are 50 theater lovers and 200 bargain hunters in the market. This means that $Z = \frac{(200+50)(150-100)}{(150-50)} = 125$. Since his remaining inventory of 100 seats is greater than the theater-lover demand of 50 but less than Z , his optimal policy is a markdown policy. Under this policy, he will achieve revenue of $50 \times \$100 + 50 \times \$50 = \$7,500$. Note that, if he had 130 seats left, his optimal policy would be to discount immediately, in which case his revenue would be $130 \times \$50 = \$6,500$.

In Example 12.7, increasing the available seating capacity from 100 to 130 actually reduces the revenue that the theater owner can generate. This occurs because, with the greater capacity, high-willingness-to-pay customers realize that the seller will ultimately reduce the

TABLE 12.5
Optimal policies and associated revenue with strategic customers as a function of inventory

Inventory (x)	Optimal policy	Revenue
$x \leq D_H$	High price: (p_H, p_H)	$p_H x$
$D_H < x \leq Z$	Markdown: (p_H, p_L)	$p_H D_H + p_L(x - D_H)$
$Z < x \leq D_L + D_H$	High price: (p_H, p_H) or Low price: (p_L, p_L)	$p_H D_H$ or $p_L x$

NOTE: $Z = (D_L + D_H)(w_H - p_H)/(w_H - p_L)$.

price and that their chance of scoring a ticket at a discount is sufficiently high to make it worthwhile for them to wait. This illustrates that it is in the seller's interest to communicate when supply is constrained—"Only two seats left at this price" or "Available only while supply lasts"—to encourage buyers to purchase now under the belief that the future price will not be discounted.

This model of strategic customers is quite simple and is based on possibly unrealistic assumptions—namely, that both the seller and all customers have full information about willingness to pay, supply, and demand. However, despite these simplifications, the model provides a number of key insights. First of all, the optimality of a markdown policy depends on the relationship of supply (or capacity) to high-willingness-to-pay demand. If high-willingness-to-pay demand is less than (or not much greater than) supply, it may make sense to keep the price high—as with a hot toy at Christmas. If supply is much greater than high-willingness-to-pay demand—and customers know it—it may be better simply to lower price now rather than delay sales. It is the intermediate case in which it makes sense for the supplier to adopt a markdown policy.

The simple strategic customer model assumes that all customers know what policy the seller will follow—in particular whether he is going to mark down the price in the second period. This assumption is not as unrealistic as it seems—for seasonal fashion goods, sellers often have rather predictable markdown cadences and customers can use the seller's past markdown behavior to determine the likelihood that the seller will mark down current inventory. This suggests that it is in the interest of the seller to create doubt in the customer's mind over whether he will be marking down current inventory—specifically to reduce the probability in the customer's mind that inventory will be put on sale. Research has shown that there can be cases in which a seller should pursue a mixed strategy—that is, sometimes not mark down inventory even when it would maximize short-run revenue for him to do so, or even withhold inventory from a markdown to reduce the probability that a customer who waits for a sale will be able to purchase at a discount. The point of both of these strategies is to artificially reduce the probability that customers who wait will be able to find inventory on sale, therefore encouraging them to purchase now at full price.

12.5 MARKDOWN MANAGEMENT IN ACTION

Automated markdown management and optimization systems were first developed and deployed at scale in the 1990s. One motivator was a dramatic expansion in retail capacity in the 1990s, which was followed by the need to focus on improving operations. Another reason was the arrival of the 800-pound gorilla of brick-and-mortar retailing: Walmart. With its wildly successful everyday-low-pricing policy, Walmart forced its competitors to improve every facet of their operations, from purchasing and logistics through pricing and promotions. While retailing has always been a thin-margin business, the need to wring every penny of profit from the market has never been greater.

The first widely publicized success of markdown optimization was at ShopKo. In 2000, ShopKo, which operated 141 discount stores under the ShopKo name and another 229 smaller, rural discount stores under the Pamida brand, installed a markdown optimiza-

tion system developed by Spotlight Solutions. A pilot program showed a 14% increase in revenue from using the markdown optimization software. The success of the ShopKo pilot was widely reported, including in a *Wall Street Journal* article, helping to spur interest in the potential for such solutions. In addition to ShopKo, retailers that have announced the adoption of markdown optimization systems include the Gap, JCPenney, Home Depot, Bloomingdale's, Sears, and Circuit City.

Markdown managements systems were initially developed for traditional retailers, who typically order their inventory well in advance—often six months or so—and then have it delivered by ocean shipping. For these retailers, there is usually no opportunity to replenish stock during the season (at least at a regional level), which means that ordering inventory for the season is a one-shot deal. As we have seen, this creates a strong incentive to order substantial amounts of inventory at the beginning of a season and then rely on a markdown strategy to maximize revenue from the inventory. One of the most successful innovations in retailing in the past 30 years is *fast fashion*, pioneered by the Spanish retailer Zara. In contrast to traditional retailers, Zara and other fast-fashion retailers, such as H&M and Forever 21, use a much more rapid supply chain to allow multiple replenishments during a season. This allows them to purchase much less initial inventory, thereby reducing the need for markdowns—the fraction of items that go on markdown at Zara is less than 20%, compared to 40% at traditional retailers. Furthermore, Zara realized 85% of list price on marked-down inventory compared to 25% on the part of traditional retailers. Nonetheless, Zara sponsored the development of a markdown optimization system tailored to its specific needs that is credited for increasing revenue from marked-down inventory by 6% (Caro and Gallien 2012).

It should be noted that markdown optimization is equally applicable to online and offline fashion retailers. Online retailers have the same need to clear excess inventory as their offline counterparts, and many of them use markdown optimization systems to maximize the return from their inventory.

In theory, markdown optimization increases revenue by optimizing the timing and depth of each markdown decision. In practice, a substantial portion of the benefits from markdown optimization comes from two sources. First, human decision makers typically wait too long to begin marking down inventory and then take very large markdowns. The reason seems to be that the people making markdown decisions tend to be the same people who bought the merchandise in the first place and are reluctant to appear to admit a mistake by marking down. As the pioneering American retailer E. A. Filene put it:

One of the few certainties in retailing is that some of the merchandise bought enthusiastically in the wholesale market, where it looks eminently saleable, will, when it reaches the store, prove stubbornly unsaleable. But despite this inevitability, *buyers, like the rest of humanity, are reluctant to admit and address their errors.* They find endless excuses for the slow sale of merchandise: the weather's still too hot; the weather's still too cold. Easter's late this year; Easter was early this year. It hasn't been advertised; the ad was lousy; the ad ran on the wrong day; the ad ran in the wrong newspaper.

The danger in these rationalizations is that they usually increase markdowns. A coat that does not sell early in the fall may still tempt customers if it is marked down early by as little as 25 percent. But as the season ends, it must be marked down far more drastically to tempt a customer who, having done without a new coat for so long, may otherwise reasonably decide to wait until the next season. (Quoted in Harris 1982, 132)

Markdown optimization systems have generally resulted in retailers' taking smaller discounts earlier than they used to do, usually substantially increasing overall revenue (Namin, Ratchford, and Soysal 2017); E. A. Filene's observation from 1910 has been vindicated, over 100 years later.

A second important benefit from markdown management systems has been their ability to tailor markdown schedules to the particular characteristics of individual regions or stores. In many cases, retailers had simply used the same markdown cadence for all stores in a chain: a particular style of sweater would be marked down by the same amount at the same time in Boston as in Los Angeles. Markdown management systems have been an important catalyst for enabling different markdown schedules for different stores. Canadian apparel retailer Northern Group Retail Ltd. implemented a markdown management solution from ProfitLogic (now part of Oracle) that it credited for helping move the company away from chainwide discounting to an approach "more attuned to regional needs, weather patterns, and other trends" (McPartlin 2004, 50). In another example, a men's retail clothing chain traditionally put swimsuits on sale in September in all its stores nationwide. This made sense in northern locations such as Boston, Chicago, and New York, where the summer season was ending, but it made little sense in Orlando and Miami, where the tourist season was just beginning. The adoption of a markdown management system was the catalyst that enabled the chain to establish different markdown schedules in New York and Miami.

12.6 SUMMARY

- Under a markdown policy, the price of an item is sequentially decreased until either it sells or a selling period expires. Markdowns are widely used in a variety of industries—notably fashion goods and short-life-cycle products. The use of markdowns has been growing steadily in the United States, at least since World War II.
- A markdown policy is effective when the item for sale is perishable, supply is limited, and the desirability of the item decreases as it approaches its expiration date.
- A markdown policy is also a way for a seller to maximize expected revenue when supply is constrained and he is uncertain about the distribution of customer willingness to pay.
- Markdowns enable a seller to segment his customers between those who are willing to pay more to buy early and those who are willing to wait in order to save money.
- When price-response functions are known, the markdown optimization problem can be formulated and solved as a mathematical program.

- When price-response functions are uncertain, the markdown optimization problem can be formulated and solved as a dynamic program. This requires starting in the last period and working backward in time to determine the optimal policy for the present.
- Business constraints such as a limit on the number of markdowns that can be taken or minimum time intervals between sequential markdowns can significantly add to the complexity of the markdown problem. When the markdown problem is highly constrained, it can be more efficient to solve it using exhaustive search than to apply explicit optimization.
- Another consideration in markdown management is the presence of *strategic consumers* who anticipate the possibility of a markdown and may wait to take advantage of a lower price. When strategic consumers are present, it may be profitable for a seller to destroy unsold inventory rather than mark it down or to adopt a mixed strategy and randomize the decision to mark down.
- Many of the benefits of real-world markdown management implementations have come from taking smaller discounts earlier than customary practice and from the ability to derive markdown cadences tailored to the specific situations of individual stores within a chain.

12.7 FURTHER READING

More discussion of markdown optimization can be found in Smith 2009, which describes results at some retailers. Valkov 2006 describes other successful implementations. Namin, Ratchford, and Soysal 2017 compares the results of different markdown approaches to human-managed markdowns. Ramakrishnan 2012 has an extensive discussion of practical considerations in modeling markdowns, including the incorporation of business constraints and the use of exhaustive search. Hamermesh, Roberts, and Pirmohamed 2002 presents a case study of the challenges faced by ProfitLogic, one of the earliest commercial providers of markdown optimization software.

Markdown pricing in the presence of strategic customers has been a topic of considerable research over the last decade or so. For a survey, see Gönsch et al. 2013. On using a mixed strategy, see Gallego, Phillips, and Sahin 2008. On withholding inventory, see Liu and van Ryzin 2008. Özer and Zheng 2016 considers the effect of potential customer regret on the decision of whether to mark down inventory.

12.8 EXERCISES

1. Extend the two-period markdown model in Section 12.1.1 to three periods. That is, assume that the price-response function is $d(p) = 1,000 - 100p$, marginal cost is 0, and customers purchase as soon as price falls below their willingness to pay. What three prices, p_1 , p_2 , and p_3 , will maximize total revenue? What if there are four periods and four prices? What is the general formula for n prices?

2. Now extend the two-period markdown model in Section 12.1.1 to the case where customers have a lower willingness to pay for the good in the second period. Specifically, assume that each customer's willingness to pay in the second period is 75% of her willingness to pay in the first period. All other assumptions remain the same.
 - a. What is the optimal price and corresponding total revenue for the seller, assuming he can charge only a single price in both periods?
 - b. What are the optimal prices and corresponding total revenue, assuming he can charge different prices in the two periods?
3. A department store has 700 pairs of purple capri stretch pants that it must sell in the next four weeks. The store manager knows that demand by week for the next four weeks will be linear, with the following price-response functions:

$$\text{Week 1: } d_1(p_1) = 1,000 - 100p_1$$

$$\text{Week 2: } d_2(p_2) = 800 - 100p_2$$

$$\text{Week 3: } d_3(p_3) = 700 - 100p_3$$

$$\text{Week 4: } d_4(p_4) = 600 - 100p_4$$

Assume that the demands in the different weeks are independent—that is, that customers who do not buy in a given week do not come back in subsequent weeks.

- a. What is the optimum price the retailer should charge per pair if he can only set one price for all four weeks? What is the corresponding revenue?
 - b. Assume he can charge a different price each week. What are the optimum prices by week he should charge? What is the corresponding revenue?
4. Consider a seller who wants to sell a product in two periods when all customers are strategic: they all fully anticipate the actions of the seller and choose to purchase in the first period, wait to purchase in the second period, or do not purchase at all. There are two classes of customers: low-willingness-to-pay customers have a maximum willingness to pay of \$4, while high-willingness-to-pay customers have a maximum willingness to pay of \$10. All customers maximize expected value and, if demand exceeds supply in either period, the limited supply will be allocated randomly to all customers without regard to their willingness to pay. In either period, the seller can set a price of either \$2 or \$7. The seller must choose what price to charge in each period as well as the total amount of the product to purchase. The unit cost of the product is \$1 and the salvage value is \$0.
 - a. Assume that there are 10,000 low-willingness-to-pay customers and 5,000 high-willingness-to-pay customers. How much should the seller order and put on sale to maximize profitability? What price should he charge in period 1 and in period 2? What is the resulting revenue?
 - b. Now, assume that the seller can charge any price he wants in both periods—that is, he is not constrained to \$2 or \$7. What price would he charge and how much would he order?

5. The questions in this problem are variations on Example 12.4. The best way to answer them is to build a spreadsheet model that calculates the revenue for the different policies under different parameter values. Each subproblem starts with the values as in the example—that is, the subproblems are independent, not cumulative.
- a. Consider the markdown problem in Example 12.4. What is the optimal policy if everything else is the same as in the example, but the salvage value is \$0 per television?
 - b. What is the optimal policy if the starting inventory is 120 instead of 95?
 - c. What is the optimal policy if the seller wants to have at least 10 television sets remaining at the end of the third week to provide sufficient inventory for his outlet store?
 - d. What is the optimal policy and corresponding revenue if each markdown generates additional demand of 5 television sets on top of the standard demand? Assume that the additional demand occurs once, in the period when the markdown occurs. That is, if the seller adopts the policy (\$500, \$450, \$450), he will generate an additional 5 units of demand in week 2; if he adopts the policy (\$500, \$450, \$350), he will generate an additional 5 units of demand in week 2 and in week 3.

NOTES

1. These examples and others can be found in Warner and Barski 1995.
2. In this section, I use the terms *markdown management* and *markdown optimization* interchangeably.
3. This model was first proposed by in Lazear 1986.
4. TKTS is a nonprofit consortium of theaters operating a discount-ticket booth. Obtaining tickets at the discount prices requires waiting in line, and there is no guarantee that the show you want will be available when you get to the counter. Thus, TKTS creates an inferior product that can be sold at a lower price with minimal cannibalization of full-price tickets.
5. The term *Dutch auction* has also been used for a somewhat different method for selling multiple products in financial markets and on eBay, leading to some confusion.
6. There is no requirement that the periods need to be of equal length. If, for example, the retailer had a policy that all items had to be sold at list price for at least three weeks before being marked down for the first time, the first period would be three weeks long, and there would be only 13 total pricing periods for the season.
7. Strictly speaking, constraint 12.4 is not required, since the optimal solution to MDOWN will never result in a markdown price that is less than the salvage value.
8. For a more in-depth description of exhaustive search applied to markdown management, including additional examples of path reduction rules, see Ramakrishnan 2012.
9. The broken assortment effect was first described by in Smith and Achabal 1998. For details on how additional effects such as promotions and the assortment effect can be incorporated in modeling markdown demand, see Ramakrishnan 2012.

13

CUSTOMIZED PRICING

13.1 BACKGROUND AND BUSINESS SETTING

The pricing tactics we have studied so far all assume that buyers and sellers interact in the same basic fashion. The seller offers a stock of goods to different customer segments, possibly through different channels. Customers arrive, observe a price, and decide whether to purchase. The seller decides what price to offer for each product to each customer type through each channel. The seller monitors sales of his goods and periodically updates the prices and/or availabilities he is offering. This is the *list-pricing* modality and it is characterized by the fact that the seller publishes to customers a price or set of prices that are generally final and nonnegotiable. All the methodologies covered so far, including peak load pricing, revenue management, and markdown management, assume list pricing.

List pricing may seem quite general—in fact, it is what first comes to mind when people think about pricing. However, there is another important pricing modality—*customized pricing*—that is of equal or greater importance in many industries. In customized pricing, potential customers approach the seller, one by one, and the seller quotes each one a price. Often (but not always) each potential buyer wants something different—either a different bundle of products or services, a different quantity, or a different variation on a basic product. In customized pricing, the seller can quote a different price for each request. A price list may exist, but it is usually an upper bound on the price quoted. The price quoted to a buyer can be based on knowledge of the individual buyer, the product(s) requested by the buyer, and other factors, such as general market conditions and competitive offerings. Following are some examples of customized pricing.

- *Auto lending.* e-Car is an online auto lender. A customer arriving at the e-Car website fills out a loan application with information about the customer's identity. The loan application also specifies the term and amount of the loan, along with whether the loan would be used to purchase a new car, purchase a used car, or refinance an existing car loan. Using this information, e-Car will pull a credit score for the customer,

estimate the riskiness of the loan, and determine whether to extend a loan to the customer.

If an applicant is judged too risky, e-Car will reject her application. If an applicant is judged creditworthy, e-Car will offer her a loan and quote an annual percentage rate (APR) for the loan along with the monthly payment. The customer has 45 days to accept the loan. If the customer does not accept the loan within 45 days, the offer expires and the customer is considered lost.

e-Car can use all of the information in the application including the customer's credit score, the amount and term of the loan, and whether it is for a used car, a new car, or a refinance in determining the rate to quote to the customer. Since e-Car can quote different rates to different customers, this is a customized-pricing problem.¹

- *Trucking services.* Businesses that purchase trucking services usually solicit bids from a number of competing carriers, such as Old Dominion and YRC. Each carrier will submit a bid detailing a price schedule at which it will carry freight for the customer for some period of time, usually a year. Each carrier faces the problem of determining what price schedule to bid.
- *Telecommunications services.* A telecommunication company sells a menu of telecommunications services to corporate, educational, and government customers. It can determine a price to bid for each customer depending on the size and type of customer, the services it requires, and the amount of business it might generate. It has categorized its customers according to the division in Table 13.1. For each customer in each category, the telecommunication company bids annually to provide services for that customer in the following year.
- *Automotive fleet sales.* A metropolitan police department wants to purchase 50 police cars. It solicits bids from Ford, General Motors, and Chrysler. The bid solicitation describes the performance and options required. Each of the three manufacturers needs to determine what price to bid.
- *Medical equipment.* A hospital has decided to purchase a magnetic resonance imaging (MRI) machine to improve its diagnostic capabilities. It sends copies of a request for proposal (RFP) to the five leading MRI manufacturing companies. The RFP specifies the required capability as well as the criteria the hospital will choose to determine the winning bidder. Each of the manufacturing companies needs to determine what price to bid.

TABLE 13.1
Global telecommunications company customer categories

Segment	Annual revenue	Customer size	Transactions	Sales organization
Global	\$8 billion	Large global	20–30/month	14 corporate sales staff
Growth	\$8 billion	Medium	7,000/month	4,000 regional sales staff
Metro	\$3 billion	Small	50,000/day	Inbound and outbound telesales

The customized-pricing modality has three characteristics that differentiate it from list pricing.

1. *Customers approach the seller prior to seeing a final price.* In the cases described above, the customer takes the initiative and describes what she wants to purchase prior to seeing the price.
2. *The seller can quote a different price to each customer.* The customized price can (and should) reflect the best information—including customer-specific information—that the seller has at that time. Of course, the seller will usually not have perfect information about the preferences of any individual customer or about competitive prices. But, as we will see, he can use statistical reasoning to increase his expected profitability for each bid.
3. *In customized pricing, the seller can track lost business.* In a customized-pricing situation, the seller will either win—that is, get the business—or lose. In some (but not all) situations, a losing seller can find out which of his competitors won the business. However, even when this information is not available, the buyer knows that a potential customer has inquired about his product and decided not to purchase. This is in contrast to list pricing, in which the seller can only observe how many units he sold—not how many customers considered buying his product but decided against it. This lost bid information is important because it can form the basis for statistical estimation of customer bid-response functions.

The existence of a *bid* or a *price quote* is a common element in customized pricing. In business-to-business settings, the quote is often in response to an RFP or a request for quote (RFQ) issued by a buyer. In business-to-consumer situations, such as unsecured consumer loans and mortgages, the final price quote usually occurs after a customer has made a phone call or filled out an application. In any case, the price quote occurs only after the seller knows something about the buyer and what she wants. This contrasts with list pricing, in which potential customers find prices by accessing a price list (online or in a catalog) or reading a label in a store and the prices need to be set before the seller has information about any individual customer.

List pricing and customized pricing often coexist, in the sense that a seller will use a list price to set a ceiling, while most customers are quoted a price at a discount from the list. For example, the *manufacturer's suggested retail price* (MSRP) is usually an upper limit on the price for, say, a new car, with the selling price ultimately determined by some combination of the customer's ability to negotiate and the dealer's willingness to sell at a lower price.² In a similar vein, shipping companies maintain voluminous tariff schedules that specify list prices for shipping different types of freight between every zip code pair in the United States. However, only a small fraction of sales (usually less than 5%) takes place at list price. The rest moves at discounted rates negotiated individually with customers.

In the remainder of this chapter, we formulate the customized-pricing decision as an optimization problem and see how that problem can be solved to maximize the expected

contribution margin from each bid. A key element in the customized-pricing problem is the *bid-response function*, which specifies the seller's expectation of how each customer will respond to his bid price. We see how bid-response functions can be estimated for different customers who are seeking to purchase different products. We then look at how the customized-pricing model can be enhanced to deal with other pricing settings as well as objectives other than maximizing expected contribution. Finally, we examine how these computations fit into a general process for customized pricing.

13.2 CALCULATING OPTIMAL CUSTOMIZED PRICES

To motivate the formulation of the customized-pricing problem, we put ourselves in the place of a seller who has been asked to bid on a potential deal. Here, *deal* means a particular piece of business, which may be a single order or a contract to provide future products and services. *Bid* means the price we offer.³ This is a simplified setting because it assumes the buyer has dictated all elements of the deal and that the only decision on the part of the seller is what price to bid. In the real world, a seller may also have to decide on other elements in his bid, such as delivery date, contract terms and conditions, and even product functionality. Furthermore, we assume initially that the seller is submitting a single take-it-or-leave-it bid. Of course, in many situations, not only price but every element of the deal may well be on the table and there may be more than one round of negotiation.

The first thing we need to determine is our goal for this bid. That is, what is our objective function? For now, we assume that we want to maximize expected contribution from the bid. For a given price, p , expected contribution can be expressed as

$$\begin{aligned} \text{Expected contribution at price } p = & (\text{contribution at } p) \\ & \times (\text{probability of winning bid at } p) \end{aligned} \quad (13.1)$$

The first term on the right-hand side of Equation 13.1 is the *deal contribution*—the margin we will realize if we win the bid. If the bid is to sell d units of an item with unit cost of c , then our deal contribution would be $(p - c)d$. The calculation of deal contribution would be more complicated if we were bidding to supply a complex bundle of products and services, each of which has a different unit cost, or if we were negotiating a contract for future services in which the volume and mix of services to be supplied were uncertain.

The second term on the right-hand side of Equation 13.1 is the probability that we will win the deal at a given price. Our uncertainty about winning at a given price derives from two sources.

- *Competitive uncertainty.* We do not know what our competitors will bid. In fact, in many cases we may not know the identity or even the number of competitors we face.
- *Preference uncertainty.* We usually do not know exactly what criteria (both conscious and unconscious) the buyer will use to evaluate competitive bids. Nor do we know for sure how the buyer values our brand and our product/service offerings relative to the competition.

In most situations, both preference uncertainty and competitive uncertainty will be present: we will not know exactly what our competitors are going to bid; nor will we know exactly how the buyer will choose among competing bids. Of course, if we knew what our competitors were bidding and how the buyer would choose among bids, our problem would be relatively simple—we would submit the highest-price bid that would win (assuming that we would be profitable at that price). However, in the real world, sellers rarely, if ever, have access to that level of information—at least not legitimately.

There is, however, one important situation in which preference uncertainty is absent—when we know that the buyer is going to choose the lowest bid. Many government agencies are required by law to purchase from the lowest bidder. In these situations, the agency often issues extremely detailed specifications for the goods to be purchased so that price is the only difference among bids. In a seller auction (also called a reverse auction), the buyer initially selects a set of acceptable suppliers and commits to purchase from the lowest bidder. In most seller auctions, each supplier has the opportunity to observe the bids of other suppliers and then to submit a lower bid if he desires to do so. The ability to rebid based on the bids of other suppliers is not addressed here.

13.2.1 Single-Competitor Model

Imagine you are an auto manufacturer bidding against a single competitor to sell 50 pickup trucks to a county park district. The park district will buy all 50 trucks from a single supplier and is committed by law to pick the lowest bidder. Each supplier has been asked to submit a single sealed bid. The bids are final, and the lower of the two bids will win. Your production cost per truck is \$10,000. Based on past experience, your belief about what your competitor will bid can be described by a uniform distribution between \$9,000 and \$14,000, as shown in Figure 13.1. What should you bid to maximize expected profitability?

Call your bid p and the competing bid q . You will win if your bid is less than the competing bid—that is, if $p < q$. (For simplicity, we ignore the possibility of a tie.) Let $\rho(p)$ be

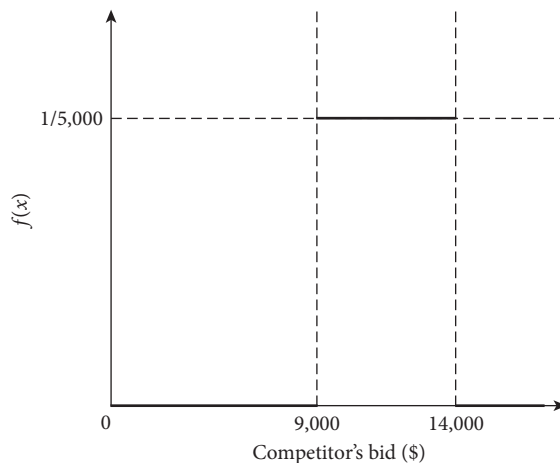


Figure 13.1 Uniform probability distribution on a competitor's bid.

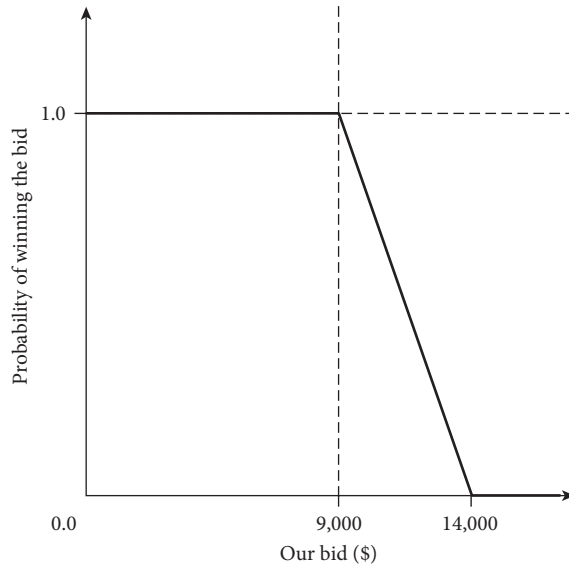


Figure 13.2 Probability of winning the bid as a function of price when our belief about the competitor's bid follows the uniform distribution in Figure 13.1.

the probability that you would win if you bid price p . Then $\rho(p) = 1 - F(p)$, where F is the cumulative distribution function of the competing bid—that is, $F(x)$ is the probability that the competing bid will be less than x . Your probability of winning the bid as a function of price is shown in Figure 13.2. If you bid below \$9,000, then you will win for sure, since you know that the competitor will bid more than \$9,000. Your probability of winning the bid decreases linearly between \$9,000 and \$14,000 and is zero if you bid above \$14,000.

The expression $\rho(p)$ denotes the *bid-response function* for this deal: Figure 13.2 shows the bid-response function in the case described above. For each deal, the bid-response function specifies the probability of winning the deal as a function of our bid. It is the customized-pricing analog of the price-response function introduced in Chapter 3. Where the price-response function specifies total expected demand as a function of list price, the bid-response function specifies the probability of winning an individual bid as a function of bid. Like the price-response function, the bid-response function is downward sloping; that is, the probability of winning the bid decreases as the bid price increases. The bid-response probability is less than or equal to 1 for all prices and decreases to 0 at high bids. For the example, the bid-response function is given by

$$\rho(p) = \begin{cases} 1 & \text{for } p < \$9,000, \\ 2.8 - p/5,000 & \text{for } \$9,000 \leq p \leq \$14,000, \\ 0 & \text{for } p > \$14,000, \end{cases} \quad (13.2)$$

and the price that maximizes expected profitability can be found by solving

$$p^* = \max_p \left[50 \left(2.8 - \frac{p}{5,000} \right) (p - 10,000) \right]. \quad (13.3)$$

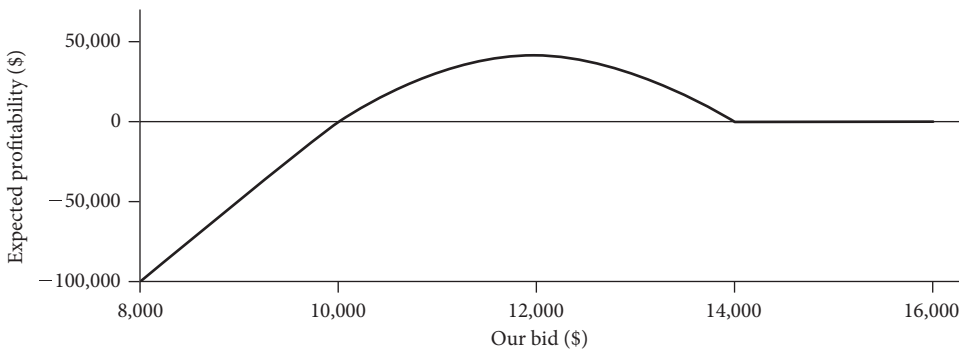


Figure 13.3 Expected profitability as a function of price bid in the example.

The objective function in Equation 13.3 is maximized at $p^* = \$12,000$. At this price, the probability of winning the bid is $2.8 - 12,000/5,000 = .4$. The margin per unit if the deal is won is $\$12,000 - \$10,000 = \$2,000$ and the expected margin is $.4 \times 50 \times \$2,000 = \$40,000$.

Figure 13.3 shows how expected profit varies as a function of bid. The expected profit function has the familiar hill shape. At a bid price below \$9,000, we are certain to win the deal, but at a loss. Above \$9,000, the probability of winning the bid decreases. Expected profit is positive for any bid above the cost of \$10,000. Above \$10,000, unit margin increases with increasing price, but this increase is counterbalanced by the decreasing chance of winning the deal. At \$12,000, these two effects balance, and expected contribution is maximized. Above \$12,000, decreasing chances of winning the deal overwhelm the increased margin. At any bid above \$14,000, you are certain to lose the deal, so expected contribution is \$0.

The optimization problem in Equation 13.3 is identical to the problem confronting a seller facing a deterministic price-response function of $50(2.8 - p/5,000)^+$ and a unit cost of \$10,000. In both cases, the optimal price is \$12,000. However, there is an important difference between the two situations. A seller facing a deterministic price-response function of $d(p) = 50(2.8 - p/5,000)$ and charging \$12,000 per unit will sell exactly 20 units and will make a profit of \$40,000. But in the bid-response case, by bidding \$12,000 per unit, the seller will either win the bid and sell 50 units, for a total contribution of \$100,000, or the seller will lose the bid and realize nothing. Changing the bid does not change the number of units sold; it changes the probability of selling all the units. At the optimal bid of \$12,000, the seller has a 40% chance of winning the deal and making \$100,000 and a 60% chance of losing the deal and making nothing.

13.2.2 Multiple Competitors

We would naturally expect the addition of more competitors to reduce the probability of winning a deal. Otherwise, we would be willing to pay additional companies to compete against us—behavior that is never seen in real life. We can see the effect of multiple competitors using the simple example of the county park district. What if we faced two competitors instead of one bidding for the order of 50 pickup trucks? To simplify matters, let us

assume we have identical beliefs about how both competitors will bid. That is, we believe that both competitor 1 and competitor 2 will each price according to the uniform distribution shown in Figure 13.1.

We also assume that the two competing bids will be independent—that is, that information about one competitor's bid would not change our assessment of the bid distribution of the other competitor. While the independence assumption may generally be reasonable, there are situations where it is not appropriate. One case would be when we believe that our two competitors are colluding to coordinate their bids. The second case would be when the two competitors are not necessarily colluding but are basing their bids on joint information that we do not share. If both of the competitors share a common cost (say, from a common supplier) and that cost is unknown to us, learning what one of them planned to bid might enable us to estimate the common cost and therefore to come up with a better estimate of what the other competitor would bid. In that case, the bids would not be independent.

We will win the deal if and only if our bid is below the lower of the bids submitted by competitor 1 *and* competitor 2. For a given p , the probability that our bid will be lower than the bid from competitor 1 is $1 - F(p) = 2.8 - p/5,000$, which, by assumption, is also the probability that our bid will be lower than the bid from competitor 2. Since competitors 1 and 2 are submitting independent bids, the probability that p will be lower than both competing bids is $\rho(p) = (2.8 - p/5,000)^2$. This bid-response function is shown in Figure 13.4. You can see that the addition of the second competitor decreases our chance of winning the bid for all realistic prices.

With two competitors, our expected contribution is given by $50(2.8 - p/5,000)^2 (p - 10,000)$. The price that maximizes expected revenue is $p^* = \$11,333.33$ per unit, with a corresponding probability of winning the bid of 28.4% and an expected total profit of \$18,963. Note that the additional competition creates pain for us in two ways: our optimal bid is lower, so total contribution is lower if we win the deal, *and* our probability of winning the

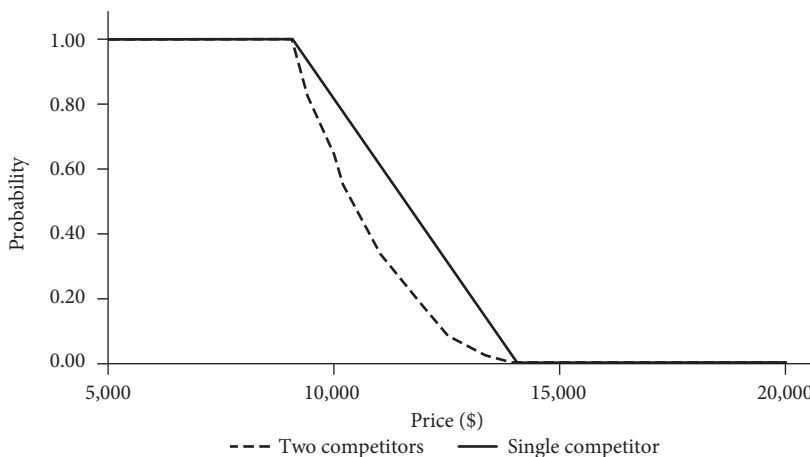


Figure 13.4 Our probability of winning the bid as a function of price with one competitor and with two identical competitors.

deal is lower, even at our lower bid. The overall effect is dramatic: Our expected contribution from the deal is reduced by more than 50% from \$40,000 to less than \$19,000. This is a graphic illustration of the power of competition to reduce both the price and the profitability of a deal. Obviously, there is a strong incentive for the buyer to increase the number of bidders—or at least make the bidders believe there are more competitors!

As a matter of reference, the probability that a bid p is the lowest when there are n competitors submitting independent bids is given by

$$g(p) = [1 - F_1(p)] \times [1 - F_2(p)] \times \dots \times [1 - F_n(p)], \quad (13.4)$$

where $F_i(x)$ is the cumulative distribution function on the price bid by competitor i —that is, the probability that competitor i will bid less than x .

Equation 13.4 gives the probability that our bid will be the lowest in any situation where competitors are submitting independent bids. If the buyer is choosing a winner based strictly on lowest bid, then this also gives our probability of winning at any price; that is, $\rho(p) = g(p)$. However, if the buyer is taking factors other than price into account when choosing a winner, then our probability of winning the bid will not be the same as our probability that we have submitted the lowest bid.

13.2.3 Nonprice Factors in Bid Selection

So far, we have assumed that the buyer makes her purchasing decision based entirely on price. This is the exception rather than the rule. Price is almost always a factor in bid selection, but it is rarely the *only* factor. It is much more common for a buyer to make her choice based on a combination of factors. Sometimes the buyer will lay out her decision criteria in an RFP, perhaps like this example:

- Previous experience with supplier—10%
- Quality of proposal—5%
- Technical fit of solution—25%
- Price—60%

In this case, the buyer is declaring that price will be the most important factor in her decision but that other factors will also be considered.⁴ In most cases, the buyer does not provide even that much information. From the sellers' point of view, the buyer is going to make her decision based on some combination of objective factors, such as price and solution fitness, and subjective factors, such as the brand image of the seller, the buyer's experience (or lack of it) with the different sellers, and even the personal relationship the buyer might have with a seller's sales representative. These elements need to be incorporated into the process of optimal customized pricing.

To make this more specific, consider the case of a large insurance company deciding to purchase 150 laptop computers for a new branch office. The information technology (IT) department invites Lenovo and Hewlett-Packard (HP) to bid on this business. Lenovo and HP both know they are the only suppliers invited to bid. However, both of them also know that the company has only purchased Lenovo computers in the past, that its overall expe-

rience with Lenovo has been good, *and* that the purchasing agent has a good relationship with the Lenovo sales representative. In this case, both HP and Lenovo are likely to believe that Lenovo has an edge on this deal. In the extreme case, HP might believe that Lenovo has a lock on the deal and the buyer is only going through a competitive bidding process to satisfy its own internal procedures and to keep Lenovo honest. If HP truly believes this, then it is facing a bid-response function such that $\rho(p) = 0$ for any value of $p \geq c$; that is, it stands no chance of winning the deal. HP might still bid—if only to keep pressure on Lenovo and to signal HP's willingness to sell to the buyer in the future—but it would need to use these strategic considerations to determine which price to set.

In the more likely scenario, HP might feel that Lenovo has an edge but that HP has a shot at winning the business if it significantly beats Lenovo's price. How can this situation be addressed? One possibility is for HP to estimate Lenovo's *bid premium* on this deal. The HP sales representative might believe that HP needs to bid at least \$200 per laptop less than Lenovo to win. This bid premium reflects the edge that Lenovo has won through its strong history and good relationship with the customer.

It is easy to incorporate this type of premium into the calculation of the optimal customized price. Let us assume for a moment that HP's cumulative distribution on Lenovo's bid is $F(x)$; that is, for any x , HP believes that the probability that Lenovo will submit a total bid less than x is $F(x)$. If Lenovo and HP were at parity—that is, neither one had a premium relative to the other—then the buyer would be choosing a supplier based strictly on price and, as described in Section 13.2.1, the bid-response function $\rho(p)$ would be equal to $1 - F(p)$. The effect of a \$200-per-unit premium for Lenovo is to shift this bid-response function to the left by $150 \times \$200 = \$30,000$, as shown in Figure 13.5. In other words, HP needs

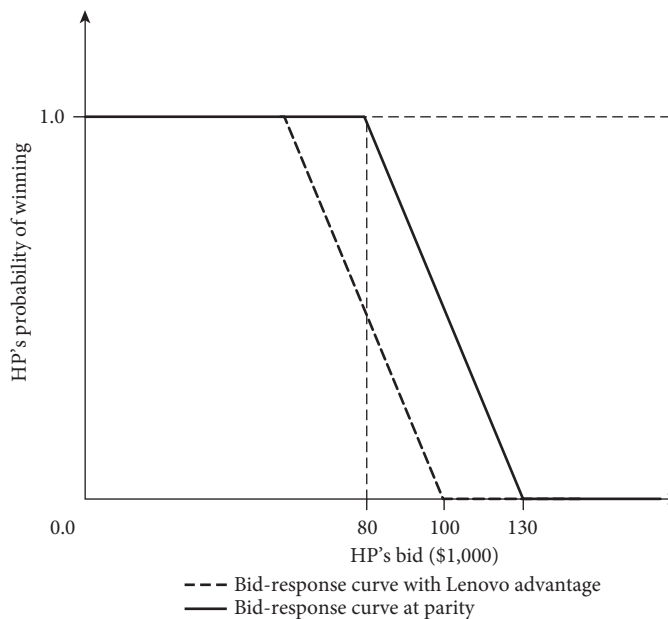


Figure 13.5 Shifted bid-response function for HP when Lenovo enjoys a premium.

to bid \$30,000 lower on the deal to achieve the same probability of winning as it would without the premium. For any potential bid p , HP's probability of winning is the probability that Lenovo's total bid is greater than $p + \$30,000$ and HP's bid-response function is $\rho(p) = 1 - F(p + 30,000)$. As shown in Figure 13.5, the net effect is that, for any bid it might make, HP faces a lower probability of winning the deal than if it were at parity with Lenovo. In this situation, HP can calculate its optimal bid by solving Equation 13.4 using the shifted bid-response function (see Exercise 1).

Let us pause here to consider several points. First, it is unlikely that HP would know Lenovo's exact bid premium. In fact, HP might be uncertain about which seller actually enjoys a premium; that is, HP may not know whether the buyer favors HP or Lenovo or is truly neutral between the two vendors. In this case, the bid premium itself is a random variable. Second, the buyer has a strong motivation to mislead the sellers about her preferences—or at least to be selective with her communication. It is in the buyer's best interest for each bidder to *believe* that the other bidder is preferred and therefore enjoys a premium. Thus, the buyer may overstate her satisfaction and loyalty to Lenovo when talking to the HP sales representative and to deny any loyalty to Lenovo when talking to the Lenovo sales representative. Of course, both Lenovo and HP should have savvy sales representatives who know enough to take such statements with the requisite grains of salt.

Third, regardless of the posturing and misrepresentation that may take place, at the end of the day each seller must take all the information it has available to it and craft the best bid that it can (or decide not to bid). Information is likely to be asymmetrical among sellers: As the incumbent, Lenovo should know much more about the buyer and her needs than HP does. Since different sellers have different levels of information, the bid-response functions are subjective and specific to each seller. In our example, Lenovo may submit a bid that it believes has a 70% chance of winning, while HP submits a bid that it believes has a 45% chance of winning. Obviously, these estimates are inconsistent with each other, but that is no more unusual than two people with different beliefs or information assessing different probabilities for a given team to win a football game.

13.2.4 The General Problem

The *customized-pricing problem* for a particular deal is

$$\max_p \rho(p)m(p), \quad (13.5)$$

where $\rho(p)$ is the bid-response function, and $m(p)$ is deal contribution at price p . For our purposes, $\rho(p)$ will be a continuous, downward-sloping function of p , and $m(p)$ will be a continuous, upward-sloping function of p . This means that the expected contribution from each deal will be a hill-shaped function of price as shown in Figure 13.3. Assume that satisfying the order will have an incremental cost of c . Then, we set the derivative of (13.5) with respect to price to 0 and apply some algebra to obtain

$$\frac{p^*}{p^* - c} = \frac{\rho'(p^*)p^*}{\rho(p^*)}. \quad (13.6)$$

The expression on the right-hand side of Equation 13.6 is the *bid-response elasticity*—that is, the percentage change in probability of winning the bid that will result from a 1% change in price.⁵ It is the customized-pricing version of own-price elasticity. A bid-price elasticity of 2 means that a 1% *increase* in price will lead to a 2% *decrease* in the probability of winning the deal. The expression on the left-hand side of Equation 13.6 is the inverse of the contribution margin ratio. In other words, at the optimal customized price, the bid-response elasticity is equal to the inverse of the contribution margin ratio:

$$\text{Bid-response elasticity at optimal price} = 1/(\text{contribution margin ratio})$$

Equation 13.6 is the customized-price version of the condition for optimal list prices in Equation 5.7, which demonstrates the similarity of the customized-pricing optimization problem to the list price optimization problem. It also means that all of the conditions for existence and uniqueness of an optimal list price described in Section 5.2.1 can be applied to customized pricing.

Note that using probabilities to represent our chances of winning a deal does not imply that either the buyer or our competition is making decisions by flipping a coin or using any other type of random process. Indeed, we would expect the competition to be setting the prices they will bid based on rational analysis of the opportunity—including their best information on what they believe we will bid. The uncertainty in the process arises entirely from our lack of information on the behavior of the other players in the bidding game: the buyer and the competition.

13.3 BID RESPONSE

The most challenging part of optimizing customized prices is determining an appropriate bid-response function. If we had a bid-response function already in hand, finding the price that maximizes expected contribution is relatively simple—just solve the optimization problem in (13.5). So how do we estimate the bid-response function to use for a particular deal? There are at least three possibilities.

- We could use *bottom-up modeling* to derive a bid-response function from our probability distributions based on how we believe our competitors will bid and the selection process we believe the buyer will use. This is what is done in Sections 13.2.1 and 13.2.2.
- We could convene the people within the company who have knowledge of this particular deal, experience with the customer, understanding of the competition, and experience with similar bidding situations and derive a bid-response function based on their *expert judgment*. This approach is often used when a deal is particularly large, important, or unique. For example, the telecommunication company with the market segmentation shown in Table 13.1 might use this approach for preparing bids for the relatively small number of customers in the Global market segment.
- We could use *statistical estimation* based on the historical patterns of wins and losses we have experienced with the same (or similar buyers) in similar past bidding

situations. For the Growth and Metro segments in Table 13.1, it would be impossible for the telecommunication company to convene a panel of experts to independently strategize and prepare a unique bid for thousands of bids per month. We describe an approach that uses statistical analysis of prior bids to estimate an optimal price for each bid moving forward.

Customized pricing requires two steps: estimating the bid-response function and then optimizing to find the price that maximizes the expected contribution for each bid. In this section, we consider estimation of the bid-response function.

13.3.1 Estimating the Bid-Response Function

We can use historical information about bids won and lost at different prices to estimate a bid-response function if three conditions hold.

1. The historical record includes bids for similar products to similar customers.
2. The current market conditions and products being offered are similar to those in the historical record.
3. There is a sufficient number of bids in the historical record to estimate a statistically significant bid-response function.

Under these conditions, many different functions can be used to represent bid response, and these functions can be fit to historical data in a number of different ways. This chapter does not try to cover all the different bid-response functions; nor does it cover all the possible estimation approaches, which would involve a long detour into statistical issues that are well outside the scope of this book. Rather, the chapter gives an overview of the most popular approach (maximum likelihood estimation) to estimating the most commonly used bid-response function (the logit function). Along the way, it mentions some of the alternatives and provides references to more extensive discussions of the pros and cons of different approaches.

The starting point for the bid-response estimation process is a bid-history database. A simple example of a bid-history database is shown in Table 13.2, which shows the bid prices and the results of 40 bids for a hypothetical item. Note that the bids range from \$9.10 to \$9.80. For the first bid, the bid price was \$9.20 and the outcome was a win (W). On the other hand, for the third bid, the bid was \$9.70 and the outcome was a loss (L).

We can visualize the relationship between price and wins and losses by assigning each win a value of 1 and each loss a value of 0. The outcomes of the 40 bids in Table 13.2 as a function of price are shown as a scatter plot in Figure 13.6. Graph A in Figure 13.6 shows the actual data; graph B in Figure 13.6 shows the same data with the points jittered—that is, a small random number has been added to each value. The sole purpose of jittering is to separate points representing multiple values; in analyses, only theunjittered values are used. It is apparent by looking at the chart that lower bids are more likely to win, but there is not a strict relationship between price and outcome: some bids at \$9.50 were won and some bids at \$9.30 were lost. We want to find the bid-response function that best captures our

TABLE 13.2
Sample bid prices and results

Bid	Price	Result	Bid	Price	Result
1	\$9.20	W	21	\$9.70	L
2	\$9.35	W	22	\$9.40	W
3	\$9.70	L	23	\$9.30	W
4	\$9.50	W	24	\$9.40	L
5	\$9.30	L	25	\$9.35	L
6	\$9.40	W	26	\$9.60	L
7	\$9.45	L	27	\$9.20	W
8	\$9.60	W	28	\$9.40	W
9	\$9.25	W	29	\$9.25	W
10	\$9.65	L	30	\$9.50	W
11	\$9.18	W	31	\$9.32	W
12	\$9.60	L	32	\$9.10	W
13	\$9.20	L	33	\$9.80	L
14	\$9.40	L	34	\$9.65	W
15	\$9.20	W	35	\$9.70	L
16	\$9.75	L	36	\$9.35	W
17	\$9.25	W	37	\$9.30	W
18	\$9.50	L	38	\$9.60	L
19	\$9.50	L	39	\$9.40	W
20	\$9.25	W	40	\$9.50	W

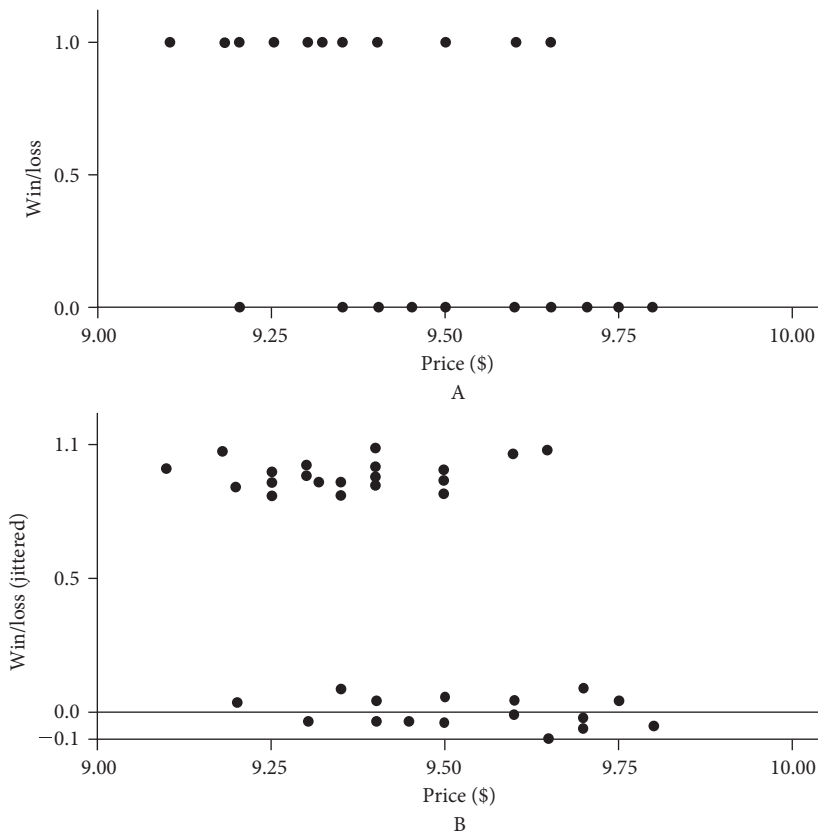


Figure 13.6 Wins and losses as a function of price. (A) The raw data; (B) the same data jittered by adding a small random number to each value.

probability of winning a bid at each price based on the results of these 40 bids—recognizing before we start that a bid-response function estimated on such a small amount of data is unlikely to have much statistical significance.

Fitting a bid-response function to data such as those in Table 13.2 is a problem of statistical estimation similar to the problem of estimating a price-response model discussed in Chapter 4. In Chapter 4, the independent variables in the estimation of the price-response function are price and a set of exogenous variables, and the dependent variable is either total demand or the fraction of potential demand that would convert to actual demand. For customized pricing, the independent variable is price and all of the characteristics of a particular deal, and the dependent variable is the won/lost indicator, which can be either 0 or 1. This means that we want to fit a function that incorporates information about each deal to predict a binary dependent value. This is a problem of *binary* or *Bernoulli regression*.

The first question we need to answer is: What kind of function do we want to fit to the data in Table 13.2? One possibility is a linear model, which would correspond to a uniform probability density function on winning the bid, such as the one in Figure 13.1. The linear bid-response function that best fits the data in Table 13.2 can be determined using linear regression: it is $\rho(p) = 14.8 - 1.51p$, shown in graph A of Figure 13.7 along with the jittered win/loss data.

The linear price-response function in graph A of Figure 13.7 is not too unreasonable. It is continuous and downward sloping. Since it has a constant slope of -1.51 , this price-response function implies that each increase of \$0.01 in price would result in a decreased probability of winning the bid of about 0.015. However, there are problems with a linear bid-response function. First, for prices above \$9.81 it predicts that the probability of winning is less than 0, and for prices less than about \$9.14 it predicts win probabilities greater than 1. Since probabilities must be between 0 and 1, a realistic bid-response function would need to be capped at 0 and 1 as shown in graph B of Figure 13.7.

Even when capped, however, a linear price-response function is unrealistic. Inspection of Figure 13.7 indicates that the linear model (capped or uncapped) is not a very good fit to the data. For one thing, the linear model specifies that the change in probability from a \$0.01 price is the same for any price between \$9.14 and \$9.81. However, we would usually expect that the effect of changing our price would be greater when our probability of winning the bid is around 50% than when it is very high or very low. This would imply that we should consider an alternative form for the bid-response function.

Consider the *bid-reservation price*—the maximum price at which the buyer would accept our bid. (The bid-reservation price is the customized-pricing equivalent of customer willingness to pay.) If the bid-reservation price is \$9.50, then we will win the deal if we bid at any price of \$9.50 or less, and we will lose the deal if we bid above \$9.50. Of course, as a seller we do not know the buyer's bid-reservation price—if we did, we would simply bid at that price (assuming, of course, that it is above our variable cost). This means we need to treat the bid-reservation price as a random variable, say, x , with a corresponding distribution $f(x)$ and cumulative distribution function $F(x)$. At any price p , our probability of winning the bid is simply the probability that $p \leq x$; that is, $\rho(p) = 1 - F(p) = \bar{F}(p)$. Note that

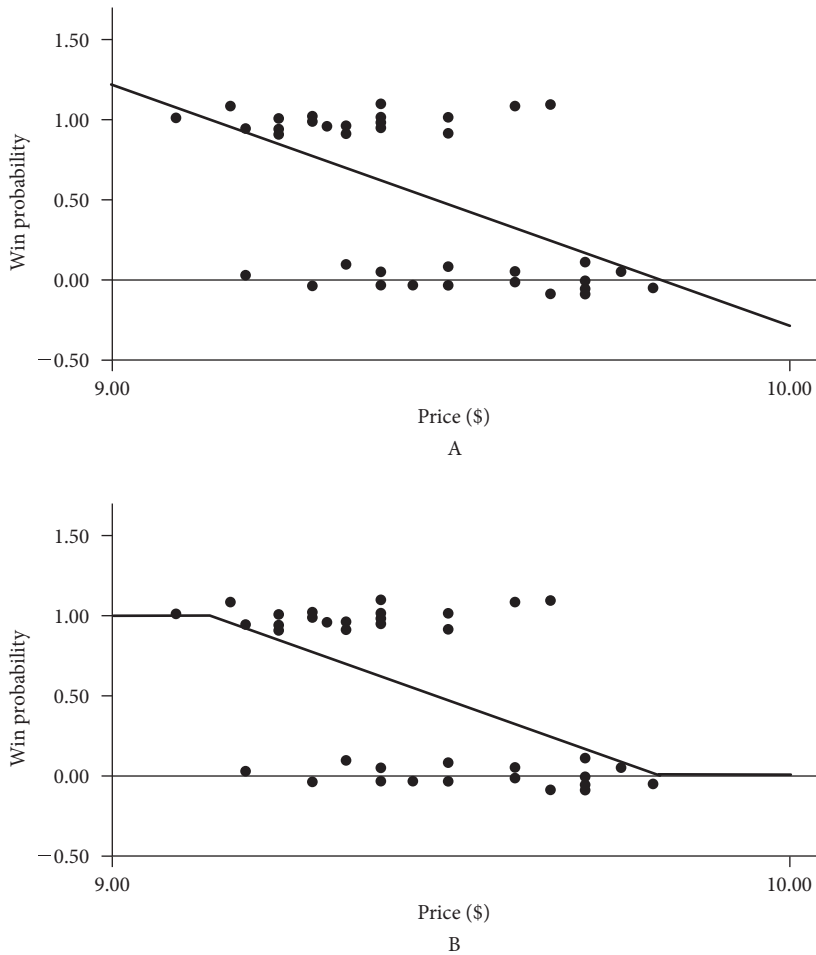


Figure 13.7 Linear bid-response functions fitted to the win/loss data from Table 13.2. (A) A pure linear bid-response function; (B) the same function capped at 0 and 1. The jittered win/loss values are also shown.

the cumulative distribution function incorporates both our preference uncertainty and our competitive uncertainty about the deal.

The logit bid-response function. What form would we expect the bid-response function, $\rho(p)$, to have? If $f(x)$ is uniform, then $\rho(p)$ will be linear. But in most cases, we would expect $f(x)$ not to be uniform but to be more or less bell shaped. When $f(x)$ is bell shaped, the corresponding bid-response function $\rho(p)$ has a reverse S shape—similar to the price-response function shown in Figure 3.6. When our bid price is very high or very low (relatively speaking), the effect of a small price change on our chance of winning the bid will be relatively small. However, when our price is in the more competitive intermediate range, we would expect that small changes in our price would have a larger effect on our chances of winning. Among the alternatives, by far the most common and convenient functional form for a bid-response function is the logit.

The logit bid-response function is given by

$$\rho(p) = \frac{e^{a-bp}}{1 + e^{a-bp}}. \quad (13.7)$$

For a realistic bid-response function in which the probability of the bid is a decreasing function of price, we must have the parameter $b > 0$. Various properties of the logit are described in Section 3.3.4, and its slope, hazard rate, and elasticity are shown in Table 3.3. The logit provides a smooth, realistic bid-response function, with the probability of winning the bid approaching 0 as price increases indefinitely. Note that Equation 13.7 is identical to Equation 3.11, with $A = 1$. This highlights that customized pricing is similar to list pricing, but with a sequential series of market-of-one pricing decisions rather than a single price that a population of customers will all see.

Estimating a logit bid-response model. Section 4.2.2 describes how to use linear regression to estimate a logit distribution when the target variable is the fraction of potential demand that converts at a price. In that case, the target variable is the conversion rate for each period, which is between 0 and 1, and we can use the transformation $\ln[r_t/(1 - r_t)]$ to convert the underlying model to a linear model. However, in the case of bid-response estimation, the target variable is the win indicator for each bid i , with $W_i = 1$ indicating that we won the bid and $W_i = 0$ indicating that we lost the bid. In either case, the transformation $\ln[W_i/(1 - W_i)]$ is undefined, which means that we cannot convert the data to a linear model and use linear regression to estimate the parameters as we did with the logit model in Chapter 4. Fortunately, there is a very straightforward approach to the problem of *logistic regression* called *maximum likelihood estimation* (MLE).

We specify the problem as follows. We have a sequence of historical bid observations $i = 1, 2, \dots, n$. For each bid, we have the price that we bid, p_i , and the corresponding outcome, W_i . We want to find the values of the parameters a and b such that Equation 13.7 best predicts the observed values of W_i given the price bid p_i . For any values of the parameters a and b , the probability of winning bid i is $\rho(p_i) = e^{a-bp_i}/(1 + e^{a-bp_i})$, and the probability of losing bid i is $1 - \rho(p_i) = 1/(1 + e^{a-bp_i})$. We want to find the values of a and b that result in predictions of $\rho(p_i)$ that are each as close to the corresponding values W_i as possible.

We can write the probability of realizing the outcome that we actually realized for bid i , given the parameters a and b and the price p_i as

$$\begin{aligned} L_i(a, b) &= \rho(p_i)W_i + [1 - \rho(p_i)](1 - W_i) \\ &= \frac{W_i e^{a-bp_i}}{1 + e^{a-bp_i}} + \frac{1 - W_i}{1 + e^{a-bp_i}}. \end{aligned}$$

$L_i(a, b)$ is called the *likelihood* of observing outcome i if we assume parameters a and b . Note that $L_i(a, b) = \rho(p_i)$ if we won bid i , and $L_i(a, b) = 1 - \rho(p_i)$ if we lost bid i . Now, assume that we have a set of n bids and their outcomes. Assuming that all of the bids are independent, the *joint likelihood* of observing the particular pattern of wins and losses that we actually achieved for bids $i = 1, 2, \dots, n$ is

$$L(a, b) = \prod_{i=1}^n L_i(a, b) = \prod_{i=1}^n \left[\frac{W_i e^{a-bp_i}}{1 + e^{a-bp_i}} + \frac{1 - W_i}{1 + e^{a-bp_i}} \right], \quad (13.8)$$

where the symbol $\prod_{i=1}^n$ means that we multiply the n individual elements together. MLE finds the values of a and b that maximize the likelihood of the actual observation. That is, MLE solves the nonlinear optimization problem $\max_{a,b} L(a, b)$, where $L(a, b)$ is given by Equation 13.8.

With a large number of observations, the likelihood associated with even a very good predictor will be very low. For example, assume we had a model that could predict, with 90% accuracy, whether we would win or lose a bid—far higher than we will ever be able to achieve in practice. For a sample of 1,000 bids, the expected likelihood for such an estimator would be $0.9^{1,000} = 1.75 \times 10^{-46}$. Working with such tiny numbers presents all sorts of numerical problems. Therefore, the usual approach is to maximize the natural logarithm of the likelihood. Since the logarithm is an increasing function of the probability, the values of a and b that maximize likelihood will also maximize the logarithm of likelihood (and vice versa). So, in MLE, we find the values of a and b that solve the optimization problem

$$\max_{a,b} \ln[L(a,b)] = \sum_{i=1}^n \ln \left[\frac{W_i e^{a-bp_i}}{1 + e^{a-bp_i}} + \frac{1 - W_i}{1 + e^{-a-bp_i}} \right]. \quad (13.9)$$

This is a much easier problem to work with than maximizing the likelihood function itself. Not only does it enable us to work with much more reasonable numeric values; it is additive, which makes it easier to work with in general. Note that, since the likelihood is always less than 1, the log likelihood will always be less than 0. Maximizing the log likelihood means finding the values that bring the value of Equation 13.9 as close to 0—the least negative value—as possible.

Solving Equation 13.9 for the data in Table 13.2 gives us the values of $a = -80.576$ and $b = 8.506$ as the maximum likelihood estimators for the logit parameters. The corresponding logit bid-response function is shown in Figure 13.8. Note that these parameter values imply that we would expect to win 50% of our bids if we set a price of $80.567/8.506 = \$9.47$. The corresponding total log likelihood is -19.752 . We can compare this to the log

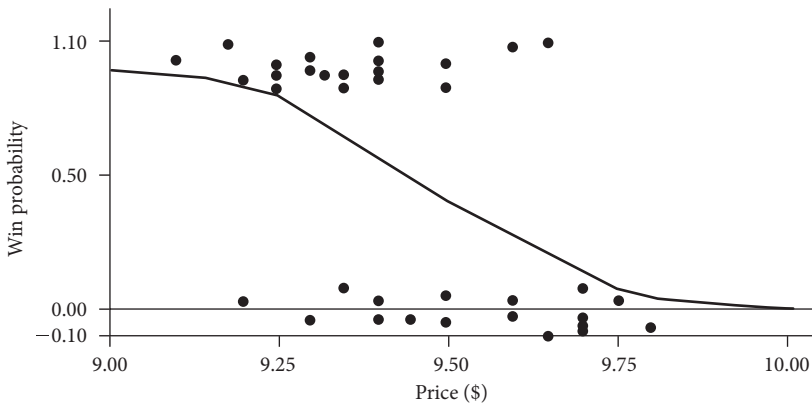


Figure 13.8 Logit bid-response function fit to the values in Table 13.2 by maximizing log likelihood. The parameters of the function are $a = -80.576$ and $b = 8.506$.

likelihood that would be obtained using an estimator independent of price—that is, $\rho(p_i) = .575$ for all values of i . The log likelihood for the constant estimator is -27.274 . Using the logit response model has resulted in an improvement in log likelihood of about 30% over a constant estimator. In case this does not seem sufficiently impressive, we can look at it this way: it improves the likelihood by a factor of 1,808!

13.3.2 Extension to Multiple Dimensions

We have seen how the parameters of the logit bid-response function can be estimated in the case of a single product being sold to a single customer segment through a single channel. In this case, we only need to estimate values for two parameters: a and b . However, in most cases we would expect there to be differences in customers, channels, and products. Some of the differences that can occur include the following.

Channel Variation

- Internet versus call center
- Retail versus wholesale
- Direct versus indirect

Product Variation

- Size of order
- Products ordered
- Configuration of products
- Ancillary services (e.g., extended warranty)

Customer Variation

- Size of customer
- Location
- Size of account
- Business
- New or repeat

To maximize profitability, a company would like to use as many of these variables as possible in setting its prices. A new, small customer with a large order entering through the internet should get a different price from a repeat large customer with a small order entering through a direct channel, and so on. In the extreme, we want to estimate different bid-response functions—and hence different optimal prices—for every element of the pricing and revenue optimization (PRO) cube in Figure 2.5.

Customized pricing is commonly used for highly configured products, in which case the product dimension of the PRO cube can be quite rich. One example is the heavy truck manufacturing industry:

The 2008 Model 388 Day Cab heavy truck sold by Peterbilt allows the customer to choose his desired options for thirty-four different components of the truck ranging from the rear-wheel mudflap hangers which can be straight, coiled, or tubular to fifteen different choices for rear axles to seven different transmissions to the choice between an aluminum or steel battery box. Multiplying the number of options in the Peterbilt list would imply that they could be combined in 75×10^{16} possible ways—each corresponding to a different truck. (Phillips 2012a, 467)

The fact that each customer is likely to be ordering a different truck gives wide scope for the price to be customized. The question is how to determine the influence of different combinations of product characteristic, channel, and customer characteristic to determine the best price to offer for each transaction. To do this, we need to have information from historical bids about the customer segment, channel, and product(s) associated with that bid as well as the price and whether we won the bid. This richer set of information is illustrated in Figure 13.9.

We incorporate additional dimensions in the bid-response function by using a multivariate logit function of the form

$$\rho(p_{ijk}) = \frac{e^{a_{ijk} - b_{ijk}p_{ijk}}}{1 + e^{a_{ijk} - b_{ijk}p_{ijk}}}. \quad (13.10)$$

Here, p_{ijk} is the price we offer to a customer of type i , purchasing a product of type j , through channel k , and a_{ijk} and b_{ijk} are the parameters of the corresponding logit function. Given a sufficient number of bids, we could use MLE to estimate these parameters, establishing a separate bid-response function for each cell in the PRO cube.

The bid-response function specification in Equation 13.10 has a drawback: estimating all the values of a_{ijk} and b_{ijk} requires running MLE independently for each cell in the PRO cube. However, reliable estimation of the parameters of the logit bid-response function requires at least 200 historical observations. In general, we cannot expect to have this many observations for every cell. If we had 50 different products being sold to 10 different customer types through five channels, then our PRO cube would have $50 \times 10 \times 5 = 2,500$ cells. For each cell to have 200 historical observations, we would need a historical-bid database with at least $2,500 \times 200 = 500,000$ entries. This approach would obviously be infeasible for the

W/L	Price	Customer characteristics	Product characteristics	Market characteristics
0				
1				
1				
1				
0				
1				
0				
1				
1				
0				
0				
1				

Figure 13.9 Typical bid-history table template.

Peterbilt example above where there are 75×10^{16} potential products—not even including the customer or channel dimensions. Most companies do not have anywhere near this much historical-bid data. This means that we have to reduce the dimensionality of the problem.

One way to reduce dimensionality is to define a single value of a and a single value of b for each product, customer, and channel value, rather than for each combination. In this case, changing our notation slightly, our model becomes

$$\rho_{ijk}(p) = \frac{e^{a_i + a_j + a_k - (b_i + b_j + b_k)p}}{1 + e^{a_i + a_j + a_k - (b_i + b_j + b_k)p}},$$

where a_i and b_i are the parameter values estimated for customer type i , a_j and b_j are those estimated for a product of type j , and a_k and b_k are those estimated for channel type k . This reduces the number of parameters required for the 50-product, 10-customer-type, 5-channel case from 5,000 to $2 \times (50 + 10 + 5) = 130$. The historical data required to derive a statistically reliable estimate of this smaller parameter set is much reduced.

To see how this works, consider the case of a company that sells three grades of printer cartridges—silver, gold, and platinum—to three types of customers: government agencies, educational institutions, and retailers. The seller has kept records on the outcome of each bid that it has made historically and the amount of business that it has done in the previous year in a format similar to that shown in Table 13.3.

The seller wishes to use this historical data to estimate a bid-response function. Define the following variables⁶ to represent the information in Table 13.3:

- $GRADES = 1$ if grade is silver, 0 otherwise.
- $GRADEG = 1$ if grade is gold, 0 otherwise.
- $GRADEP = 1$ if grade is platinum, 0 otherwise.
- $PRICE = 1$ for the price bid.
- $SIZE = 1$ for the order size.
- $NEW = 1$ if customer is new, $NEW = 0$ if it is a previous customer.

TABLE 13.3
Sample historic-bid data available to a printer ink cartridge seller

Bid	Won or lost	Order characteristics			Customer characteristics		
		Price bid per unit	Grade	Size	New or existing	Type	Business level
1	W	\$7.95	S	1,259	N	Ret	\$0
2	L	\$9.50	P	655	E	Gov	\$153,467
3	L	\$9.75	P	790	E	Gov	\$1,402,888
4	W	\$8.50	G	840	N	Ret	\$0
5	L	\$8.47	G	833	E	Ret	\$452,988
6	L	\$9.22	P	540	E	Edu	\$55,422
7	W	\$8.61	G	850	N	Ret	\$0
...	...	...	...	...	...	...	...

NOTE: The grade of the product ordered is S for silver, G for gold, or P for platinum. There are three types of customer: Edu is an educational institution, Gov is a government entity, and Ret is a retailer. Business level is the total amount of revenue that the cartridge seller accrued from the customer in the previous year.

- $RET = 1$ if customer is a retailer, $RET = 0$ otherwise.
- $GOV = 1$ if customer is a government agency, $GOV = 0$ otherwise.
- $EDU = 1$ if customer is an educational institute, $EDU = 0$ otherwise.
- $BUSLEVEL = 1$ for the total amount of business that has been done with the customer in the last 12 months.

The variables specifying the product grade, the type of customer, and whether the customer is new or old are called *categorical variables*. When there are two or more mutually exclusive categories such as $GRADES$, $GRADEG$, and $GRADEP$, exactly one of the variables will be equal to 1 with the others equaling 0 (assuming that there are no other alternatives). This means that $GRADES + GRADEG + GRADEP = 1$ and the variables are *collinear*, which means that if we know the values of two of the variables, we could calculate the value of the other; for example, if we know $GRADES$ and $GRADEG$ we can calculate $GRADEP = 1 - GRADES - GRADEG$. As a consequence of collinearity, when there are n mutually exclusive alternatives for a feature, we only need to include $n - 1$ categorical variables. This is obvious in the case when there are only two categories such as new or old: we do not need a variable named OLD ; we can simply let $NEW = 0$ represent the case of an existing customer. When there are more than two alternatives, we can either choose which of the options to leave out or, with most regression software packages, allow the software to choose automatically. For this reason, we exclude $GRADEP$ and OLD from the formulation here.

Define the following linear function:

$$\begin{aligned}
 g(p, \mathbf{x}) = & \beta_0 \\
 & - \beta_1 \times PRICE \\
 & + \beta_2 \times GRADES + \beta_3 \times GRADEG \\
 & + \beta_4 \times SIZE \\
 & + \beta_5 \times NEW \\
 & + \beta_6 \times RET + \beta_7 \times GOV \\
 & + \beta_8 \times BUSLEVEL.
 \end{aligned} \tag{13.11}$$

The corresponding logit bid-response function is given by

$$\rho(p, \mathbf{x}) = \frac{e^{g(p, \mathbf{x})}}{1 + e^{g(p, \mathbf{x})}}.$$

The next step is to estimate the values of the coefficients $\beta_0, \beta_1, \dots, \beta_8$ that best fit the historical data. The same procedure of maximum likelihood estimation discussed in Section 13.3.1 can be applied to Equation 13.11, with the only difference being that, in this case, we are estimating nine coefficients, $\beta_0, \beta_1, \dots, \beta_8$, instead of two.

Equation 13.11 is not the only possible formulation of a bid-response model given the available data. For example, Equation 13.11 includes the term $\beta_4 \times SIZE$, which specifies

that the effect of order size on winning a bid is linear, all else being equal. It is not unusual for this effect to be significant—customers placing large orders tend to be more price sensitive than those placing smaller orders. However, it may be that this effect is not linear but tends to get weaker the larger the order. To represent this possibility, we might try replacing the term $\beta_4 \times SIZE$ with a term $\beta_4 \times \ln(SIZE)$. (Recall that $\ln(SIZE)$ denotes the natural logarithm of the variable $SIZE$.) In addition, we might want to try a model in which the product of order size and price influences the probability of winning a bid, in which case we could add another term of the form $\beta_9 \times PRICE \times SIZE$.

The process of determining the best model to fit the data is discussed in Section 4.3.3—the same process holds for fitting bid-response data as price-response data. Generally, a modeler will use a forward stepwise regression process, which means starting with a simple model such as that in Equation 13.11 and then experimenting with adding transformed variables such as $\ln(SIZE)$ and crossed variables such as $PRICE \times SIZE$. For each model, measures of model performance on historical data are calculated, and once multiple models have been tested, the model demonstrating the best performance on these measures is selected.

13.3.3 Binary Model Performance Measures

Section 4.2 describes three measures of model fit: root-mean-square error (RMSE), mean absolute percentage error (MAPE), and weighted MAPE. RMSE can be applied to binary models without any modification; however, neither MAPE nor weighted MAPE can be directly applied to binary models because they both require dividing by the outcome, which is often 0. Fortunately, there are two additional model-fit measures that are applicable to binary regression problems.

- *Concordance* (also called area under the curve, or AUC) measures the frequency with which a model estimated a higher conversion probability for accepted bids than for rejected bids. A concordance of 1 means the model is a perfect predictor—that is, there is some value of $\hat{p}$ such that every bid for which the bid-response model estimated $p(p_i) > \hat{p}$ was accepted and every bid for which $p(p_i) < \hat{p}$ was rejected. A concordance of .5 means that the model was no better than random chance and a concordance less than .5 means that the model was worse than chance. The higher the value of the concordance, the better job the model does of predicting success. In practice concordance is typically between .5 and 1.0, with a higher value indicating a better-fitting model.
- *Log-likelihood* is calculated according to Equation 13.9. It measures the probability of observing the pattern of wins and losses in the test data set if the underlying model were correct. Log-likelihood is always less than 0, with larger values (i.e., less negative) indicating a better-fitting model.

Example 13.1

Assume that our test data set includes only five bids (a woefully inadequate number for any meaningful analysis) and that the predicted win probability

TABLE 13.4
Model predictions and actual bid outcomes

Number i	Predicted win probability $\rho(p_i)$	Win? W_i	Squared error $[\rho(p_i) - W_i]^2$	Likelihood $\rho(p_i)W_i + [1 - \rho(p_i)](1 - W_i)$
1	.9	1	.01	.9
2	.8	1	.04	.8
3	.6	0	.36	.4
4	.3	1	.49	.3
5	.2	0	.04	.8

$\rho(p_i)$ from our challenger model and the actual outcome W_i are as shown in Table 13.4. The squared error for each bid is $e^2 = [\rho(p_i) - W_i]^2$, and the likelihood for each bid is $L_i = \rho(p_i)W_i + [1 - \rho(p_i)](1 - W_i)$. The mean squared error of the model on the test data set is the average of the squared errors, which is .188, and the RMSE is the square root of this number, which is .44. The likelihood of the observed outcome is the product of the individual likelihoods: $L = L_1 \times L_2 \times \dots \times L_5 = .069$; the log-likelihood is the natural logarithm of the likelihood, or $\ln(.069) = -2.67$. To calculate the concordance, we look at all pairs of bids in which one was accepted and the other was not (*disparate pairs*). There are six disparate pairs in Table 13.4. A disparate pair is *concordant* if the predicted win probability for the win was greater than or equal to the win probability for the loss. For example, the pair (1, 3) is concordant because the predicted win probability for bid 1 (which was won) is greater than the predicted win probability for bid 3 (which was lost). On the other hand, the pair (3, 4) is *discordant* because the predicted win probability for bid 3 (which was lost) is higher than the predicted win probability for bid 4 (which was won). Of the six disparate pairs, five were concordant and one was discordant, so the concordance is $5/6 = .833$.

Each of these measures can be useful in comparing the performance of different models on the same test data set. In addition, concordance and RMSE can be useful in comparing models across data sets. Log-likelihood is not useful for comparing model performance on different test data sets because it depends strongly on the number of entries in the data set.

13.3.4 Endogeneity

Endogeneity can be an issue in estimation, as seen in Section 4.4.2. Endogeneity arises when unrecorded characteristics influence both the final price and the customer willingness to pay. This is a common problem in customized pricing when the final prices are set by negotiation with a salesperson. In this case, the local salesperson may observe customer or market attributes that are not recorded in the data and use those attributes in negotiating the final price.⁷ For example, a salesperson, noticing that a particular customer seems very eager to purchase, holds the line to achieve a higher price, while another customer seems

less eager to purchase and the salesperson drops the price to close the deal. To the extent that the salesperson is correct in his estimation of customers' willingness to pay, the price offered depends on customer willingness to pay through variables not recorded in the data. In this case, endogeneity will bias the coefficient of price in the estimation process.

The bias introduced by endogeneity can go either way. If sales staff systematically adjust prices toward customers' willingness to pay (as they should to increase profitability), then a naïve estimation procedure that ignores endogeneity will underestimate customer price sensitivity. If, however, sales staff are using their pricing discretion on the average to adjust prices away from customer willingness to pay, naïve estimation will lead to price sensitivity being *overestimated*. If sales staff are varying prices in a way that is uncorrelated with customer willingness to pay, then they are just adding noise to the process and the estimate of price sensitivity will be unbiased.

When the variation of prices in historical data is due partially or entirely to sales force discretion, it is prudent to test for the presence of endogeneity and, if necessary, correct for it. One common approach to deal with endogeneity in customized pricing is to use a control-function approach, which proceeds in four steps:

1. Estimate an *expected price* for each bid. This is the price that would have been bid in the absence of local knowledge of unrecorded characteristics. One approach to estimate this expected price is to average the prices offered for similar bids during the same time period. For example, the expected price might be calculated as the average price of all bids or of similar products to similar customers. An alternative is to build a model that predicts expected price for each bid as a function of customer, product, and channel characteristics.
2. Calculate the differences between the expected prices estimated in step 1 and the actual prices bid.
3. Run a regression with *Outcome* as the target variable that includes the differences calculated in step 2. If these differences enter the regression as significant, this indicates that endogeneity is present.
4. If endogeneity is present, then the difference between the actual price and the expected price should be included as an explanatory variable in the regression.

Although this may sound complicated, the basic idea is simple. In step 1, we estimate the prices that would have been offered for each bid if the sales staff did not possess unobserved bid-specific information. This expected price is an *instrumental variable* because it is plausibly correlated with the bid price, but it is not correlated with the customer's idiosyncratic willingness to pay. If the difference between expected price and the price actually offered is not significant, then endogeneity is not present and the results of a regression using *Price* as an independent variable will not be biased. If, however, the difference between predicted price and actual offered price is a significant predictor of *Outcome*, then this difference should be included as an explanatory variable in the regression. The coefficient of this variable will be a better estimate of price sensitivity than the coefficient of *Price* estimated through simple regression.

13.4 EXTENSIONS AND VARIATIONS

We have seen how a seller can find the customized price that will maximize expected contribution from a prospective deal in a simple bidding situation, assuming that the seller faces no constraints on the price he can bid. There are two obvious variations on this basic model:

- The seller might want to do something other than maximize expected contribution.
- The seller might be constrained in the price he can offer for this deal.

In the first case, the seller is pursuing a *strategic goal* that involves something other than maximizing expected deal contribution. If it is a goal that can be imposed on a deal-by-deal basis, then he can do this by simply maximizing a different objective function than Equation 13.5 for each deal. For example, if he wants to maximize expected revenue, he could substitute the objective function $R(p) = p(p)p$ as an alternative to maximizing expected profit. The properties of revenue- and profit-maximizing prices are similar to those discussed in Chapter 5—the revenue-maximizing price is lower than the profit-maximizing price for each bid. Furthermore, the profit from a bid may be negative at the revenue-maximizing price. It is also possible that the seller is interested in multiple objectives, such as maximizing profit and maximizing sales. In this case, the efficient frontier described in Section 5.7 can be used to understand the trade-off between competing objectives.

The other case occurs when the seller is subject to one or more *business rules* that limit his choice of prices. These business rules could arise externally (it may be illegal to charge different prices to certain groups of customers) or internally (the company may wish to keep prices through one channel lower than prices through another channel, for example). In both cases, the effect is to make the customized-pricing problem more complex than solving the unconstrained optimization problem in Equation 13.5. In particular, business rules usually require adding constraints to the basic problem.

In addition, there are many cases in which the bid setting itself is more complicated than a simple optimization—bids may involve many different bundled products and services and may involve several rounds of negotiation. These variations require extensions to the simple model, which Section 13.4.2 discusses.

13.4.1 Strategic Goals and Business Rules

Consider the following situations.

- A package express company responds to 100,000 bid requests per year. Each request is for a contract to supply package express services for the next 12 months. The company currently seeks to maximize expected contribution for each bid, and it wins approximately 40% of the time. In 2021, senior management for the company declares that the company has an important new initiative with the goal “to become the dominant provider of shipping services to online retailers.” As part of this initiative, the CEO tells the executive vice president of sales that he now wants the sales force to win at least 75% of its bids to online retailers while maintaining a minimum level of profitability.

- Gammatek is a distributor that purchases various types of electronic equipment from manufacturers and sells the equipment to corporate, government, educational, and retail customers throughout North America. Gammatek has five divisions: CPUs, printers, storage devices, peripherals, and consumer electronics. It has about 23,000 customers and bids on about 400,000 pieces of business a year, with about 25% of the bid requests coming in to the call center and the remainder via the internet. Since Gammatek can set a different price for each bid, it has adopted a customized-pricing process. At the beginning of the 2021 fiscal year, Gammatek's president announces that each division will have a minimum average margin target that it will be expected to meet for the coming year. These targets were adopted because of the perception that stock analysts will be watching margins particularly closely in the coming year because of a soft economy and are likely to reduce the rating of any company with eroding margins.
- The Bank of Albion in the UK receives about 1 million requests per year for consumer loans. For each loan, the bank quotes an APR based on the size and term of the loan, customer credit score, and channel through which the request was received. Its newest marketing campaign advertises that a "typical" £5,000 loan has an APR of 13.9%. This means that, by British law, the average APR for £5,000 loans issued by the bank must be less than or equal to 13.9%.

In each of these three cases, the seller needs to do something beyond simply determining the prices that maximize expected contribution for each bid. The package express company needs to calculate prices that enable it to hit its market-share target for the online retail market. It can meet this *strategic goal* by solving the customized-price optimization problem with a different objective function. Gammatek and the Bank of Albion want to maximize expected contribution, but they now face business rules that constrain the prices they can charge. In the case of Gammatek, the constraints arise from the internally imposed margin targets; in the Bank of Albion's case, the constraint is an externally imposed legal requirement. In both cases, the business rules can be addressed by constraining the basic customized-pricing problem.

Let $\rho_i(p)$ and $f_i(p)$ be the bid-response function and the contribution, respectively, for a particular deal, i . One way for the package express company to meet its market-share target would be to solve the constrained optimization problem

$$\max_p \rho_i(p) f_i(p),$$

subject to

$$\rho_i(p) \geq 0.75, \tag{13.12}$$

whenever it is bidding to an online retailer. (Bids to all other customers would still be unconstrained.) Constraint 13.12 guarantees that the company will have *at least* a 75% probability of winning each bid to an online retailer. If the unconstrained optimal price would result in a probability of winning the bid that was greater than 75%, the constraint will not make any difference—the optimal price will be the same. On the other hand, if the

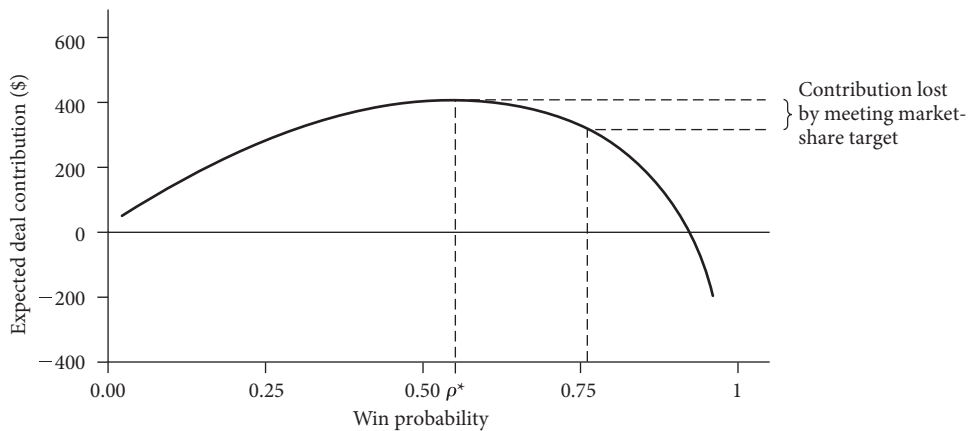


Figure 13.10 Applying a minimum market-share constraint.

unconstrained optimal price would have resulted in a probability of winning of less than 75%, constraint 13.12 will ensure that the calculated price yields a win probability of 75%. In this case, the constrained price will be *lower* than the unconstrained price.

Figure 13.10 shows expected contribution as a function of the probability of winning the bid in a typical bid-response situation. The unconstrained optimum contribution is, of course, at the top of the hill. The win probability corresponding to the unconstrained optimum price is denoted by p^* , which is equal to about 0.55 in the figure. If the market-share target is less than p^* , then the constrained optimum price is the same as the unconstrained optimum price. If the market-share target is greater than p^* , then the constrained optimum price will be lower than the unconstrained optimum. As shown in Figure 13.10, the expected contribution from this bid will also be lower—we are buying market share by pricing lower than the contribution-maximizing price. Figure 13.10 shows the amount of lost contribution if the seller enforces a target win probability of 0.75 for this bid. This difference is the opportunity cost associated with the constraint that enforces the business rule.

To guarantee that it meets its minimum margin targets, Gammatek will need to apply a constraint to its prices. Gammatek's unit margin contribution for each sale is $m = (p - c)/p$. Let m^* be the minimum margin required for a particular line of business. (There would be a different value of m^* for each line of business.) Then one way to meet the minimum margin requirement is to ensure that $(p - c)/p \geq m^*$ for every bid, or, equivalently, that $p \geq c/(1 - m^*)$ for every bid. This constraint can be applied directly to the customized-pricing problem.

The minimum margin target has an effect opposite to that of the minimum market-share target: For each bid, it will result in prices that are greater than or equal to the unconstrained optimal prices. However, like any constraint, if it is binding, it will result in a lower expected contribution than the unconstrained case.

Portfolio constraints. You may have noticed that we fudged in applying the minimum margin constraint at Gammatek. The corporate goal was for each division to meet a margin target. This target would logically be applied to the total margin achieved by the business—

the total contribution for the division divided by the total revenue for the division. However, we applied the constraint that $p \geq c/(1 - m^*)$ to every deal. This is too restrictive! The division could accept a lower-than-target margin on some deals as long as there are sufficient higher-margin deals so that the division average exceeds the target. Specifically, the division could possibly make additional money by pricing below the margin target to some very price-sensitive customers while pricing higher to some other customers who are less price sensitive.

It is clear that the package express company is in a similar situation—it does not need to have a win probability of 75% or higher on every bid in order to win 75% of its bids. It could well be optimal for the package express company to accept a 70% probability of winning a particularly competitive bid if it anticipates that it will be bidding on a less competitive bid in the future with an 80% chance of winning. Finally, it is not clear at all how to incorporate the mean APR constraint into the customized-pricing problem at the Bank of Albion, other than forcing every bid to take place at the mean, which would hardly seem to qualify as PRO.

The upshot is that certain business rules cannot be applied independently to each bid as it arrives. Rather, they need to be treated as *portfolio constraints* to maximize expected profit over all bids. That is, these rules cannot be applied one bid at a time; they must be applied across the portfolio of all bids. This requires that the seller maintain a forecast of expected business for each combination of customer, product, and channel—in effect, for each cell in the pricing and revenue optimization cube. To see this, let d_i be the number of bids we anticipate from cell i in the PRO cube and p_i be the corresponding optimal price. (In other words, each index i corresponds to a different combination of product, customer type, and channel.) Then the problem faced by the Bank of Albion can be formulated as

$$\max \sum_i \rho_i(p_i) f_i(p_i) d_i,$$

subject to

$$\frac{\sum_i \rho_i(p_i) d_i p_i}{\sum_i \rho_i(p_i) d_i} \leq p^*,$$

where p^* is the advertised typical rate. The constraint guarantees that the average price over all bids will be less than or equal to p^* , but it does not constrain every single price to be less than or equal to p^* .

From a practical point of view, the existence of one or more portfolio constraints significantly complicates the customized-pricing process. Instead of being able to calculate bid prices for deals one by one as they arrive, prices need to be precalculated for the entire PRO cube. Precise treatment of portfolio constraints also requires the forecasting of anticipated demand for each combination of product, customer type, and channel (the d_i in the earlier example). For these reasons, it is worth carefully considering whether it is important to incorporate portfolio constraints into the estimation of customized prices, or whether bid-by-bid approximations will suffice.

Infeasibility. Business rules are usually applied for good reasons. However, there can be a dangerous tendency for business rules to accrue over time, particularly if obsolete

rules are never removed. If unchecked, the accumulation of business rules can lead to the customized-pricing problem becoming overconstrained or even infeasible. For example, a computer equipment wholesaler may want to impose two business rules:

- “We need to win at least 40% of our bids in the small-business segment.”
- “We need to maintain a margin of at least 12% on sales in our printer division.”

Taken individually, each goal might be quite reasonable. However, if printers represent a large fraction of the company’s sales to the small-business segment, the two goals might be mutually incompatible—namely, it might be impossible for the company to win 40% of its bids in the segment while maintaining a margin of 12% or more on its printer sales. This is the problem of infeasibility. The likelihood of infeasibility rises as the number and variety of business rules in place at one time increases.

Infeasibility is often dealt with in customized-pricing software by forcing the user to rank the business rules in the order of importance. In case of infeasibility, the software will start by relaxing the least important rule first and continue to relax rules until it can find a solution. This is a common *technical solution*. However, the larger issue is that any company using customized pricing needs to actively manage business rules to keep them from being overapplied and accruing over time. As we have seen, adding business rules can never improve expected contribution. Therefore, they should be added only when necessary. In particular, it is important to have a process for periodically reviewing the rules that are in place and replacing those that are obsolete or no longer applicable.

13.4.2 Variations on the Basic Bidding Model

So far, we have looked at customized pricing in its simplest setting—when sellers submit a single bid for a known product to a buyer. Although this setting is realistic in some markets, in many others there are significant variations that have implications for how the sellers should set their prices. This section touches on a few of the most important examples.

Contract versus single purchase. In many cases, a buyer will request bids for a contract for future goods or services, typically for a fixed period, such as the next six months or a year. This is a very common way of purchasing freight, telecommunications, and other services. A five-year information technology (IT) outsourcing or cloud computing deal would also fall into this category. This adds a twist to customized pricing: a seller must make its bid without knowing the precise level or mix of business it will actually realize if it wins the bid.

Consider a video game manufacturer soliciting bids from a trucking company to distribute its video games nationwide for the next 12 months. The trucking company needs to decide what discount from its basic tariff it should offer. If the manufacturer’s newest game is a hit, then the trucker may ship a huge volume of games over the next few months—so much so that it might require additional trucks to be added to the fleet to accommodate the increased business. If, on the other hand, the game is a bust, the additional business from this customer would be simply incremental and not require any additional investment. A trucking company needs to incorporate this uncertainty into its bid.

In this case—when the level or mix of demand from a potential customer is uncertain *and* the profitability of an account depends on the level or mix of demand—the seller needs to forecast the demand he anticipates from that customer and use that information in calculating the price to bid. Furthermore, if there is a wide range of possible outcomes, he may need to explicitly incorporate the uncertainty in the business level in his bid.

Multiple line items. It is extremely common that a bid includes more than one item. As a very simple example, a park district may submit an RFQ that asks for a bid on 10 medium-size automobiles and 10 pickup trucks as part of the same bid. At the more complex end of the spectrum, a bid to construct a power plant may include thousands of individual line items. As an intermediate example, the heavy truck offered by Peterbilt described in Section 13.3.2 has 75 different options. For each option (e.g., engine type), there are two or more alternatives, each with a different list price. A customer configures the truck by choosing the options she wants, and the manufacturer must determine the overall discount to offer based on the fully configured truck.

When multiple line items are involved in a bid, the tactic a bidder should use to determine his bid price depends on how the buyer is going to select a supplier. Here are the three broad possibilities.

- *Single bundled price.* The buyer may commit ahead of time to purchase from a single bidder and require only a single, all-inclusive price from each bidder. This is the case with the configurable heavy truck purchase, for example. In this case, the price for the entire bundle can be determined using the standard optimal customized-pricing approach.
- *Itemized prices with single supplier.* In this case, the purchaser commits to purchase from a single bidder but requires itemized prices. This may seem identical to the previous case, but there are a number of potential pitfalls. First, even though the purchaser will only be paying a single price, his choice of supplier may be influenced by some of the itemized prices. For example, automobile purchase decisions are often more influenced by the base price of the car than by the cost of the options. Some purchasers are more likely to purchase the same car if it is presented as a \$19,000 car with \$5,000 worth of options than if it is presented as a \$20,000 car with \$3,800 worth of options. This is an issue of price presentation, which is treated in Chapter 14.

A more important issue with itemized prices is that they give the purchaser more scope and leverage for negotiation, particularly in a competitive bid. When there is only a single price, there is only one dimension for negotiation. When there are many line-item prices, a savvy buyer will negotiate each one along with the total discount. For this reason, sellers prefer to offer a single bundled price, while buyers almost always prefer line-item prices, even when they will only be purchasing from a single supplier.

- *Itemized prices and multiple suppliers.* In this case, the purchaser requests itemized prices and reserves the right to purchase different items from different bidders. For

example, the park district might reserve the right to purchase the cars from one company and the trucks from the other. Obviously, the motivation for the purchaser is to select the cheapest supplier for each item (cherry picking). In this case, the strategy for the seller is to treat each item as an individual bid and to calculate the optimal price for each line item independently of the others. Of course, a supplier might want to provide an extra discount if the purchaser will purchase everything from him.

The tactics for dealing with these different settings are fairly clear: What is most important for a bidder is to understand what game he is playing before he submits his bid. That is, if itemized prices are requested, it is critical to know whether the purchaser is committed to a single supplier or is reserving the option to purchase from more than one bidder.

Negotiation. In many customized-pricing situations there is an element of negotiation—the first deal proposed is not necessarily the final one, and there may be several rounds of negotiation before the final price is determined. Negotiation may take place on price (“Cut the total price by 1%, and I’ll sign right now”), on product bundle (“Throw in the extended warranty and we have a deal”), or both. Clearly, this is a different setting from the single-bid process that we have considered, and recommended prices need to be modified accordingly.

There is a substantial literature on the art and science of negotiation, not to mention a small industry devoted to helping people become better negotiators. However, from the point of view of pricing and revenue optimization, a typical approach is to give the salesperson a range within which to bid. How such a range can be incorporated into sales force guidance is illustrated in Figure 13.11 for the example of the park service vehicle bid. Expected contribution is within 4% of the optimal for any amount bid between \$11,600 and \$12,400. The salesperson negotiating the deal might be provided with this range of prices and expected to use his negotiating skill to obtain a price as high as possible within the range. The savvy salesperson will choose an initial bid at the high end of the range to give himself room to bargain. This allows the company to provide guidance to a salesperson in the negotiation process while ensuring that the final price achieved is sufficiently close to optimal.

A price range such as the one shown in Figure 13.11 is often used to support a combined centralized/decentralized approach to negotiated pricing. In this approach, headquarters can determine the target price and the upper and lower bounds for each deal. The local salesperson has the freedom to negotiate the best price he can within these bounds. The idea is to gain the best of both worlds: to use all of the data available centrally to calculate bid-response functions and bounds while enabling the salesperson to adjust the price to the characteristics of a specific deal and the characteristics of an individual customer. Ideally, the local salesperson will have immediate knowledge of the local competitive situation or the needs of each customer that can be used to determine a better price than could be generated centrally.

Does setting price ranges centrally improve results over the two alternatives of completely centralized and completely decentralized pricing? In theory, negotiation within

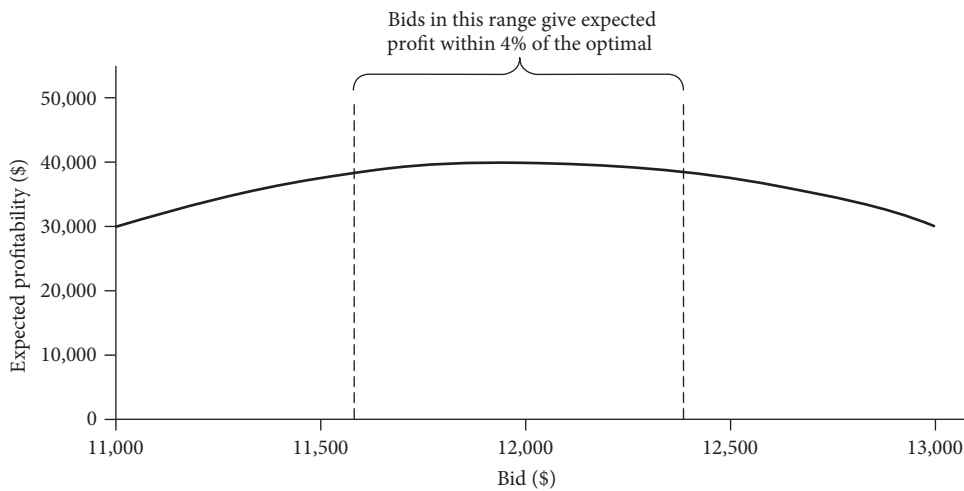


Figure 13.11 Range of bids within which expected contribution is within 4% of the optimum for the example with the profitability curve shown in Figure 13.3.

bounds should benefit the seller, since a savvy salesperson should be able to evaluate the eagerness of a customer to purchase and adjust the asking price accordingly. This is what a study of auto-loan pricing found—loan officers who had authority to negotiate rates within a range were able to increase the profitability of loans over the price list that had been centrally established (Phillips, Simsek, and van Ryzin 2015). On the other hand, an internal study by Nomis Solutions found that loan officers for a Canadian bank were actually reducing the overall profitability of the portfolio by negotiating prices that were consistently lower than optimal. The difference between the two situations can be best explained by incentives—the auto loan negotiators were incentivized (indirectly) on profitability, while the Canadian loan officers were incentivized in large part by portfolio growth and by a recommender score. This motivated them to offer lower prices than those that would have maximized profit.⁸

Fulfillment. In many contract bidding situations, a buyer may not want to commit all of her business to a single supplier. For example, a shipper may not want to commit all of her business for the next year to a single trucking company. Instead, the shipper may want to have contracts with two different trucking companies to provide herself with flexibility in case of a dispute with one of the companies or in case one is shut down temporarily by a strike. Furthermore, having contracts with two different trucking companies may save the shipper money if she picks the cheaper trucker for each shipment. In other business-to-business relationships, a purchaser will choose two or more competitors as *preferred suppliers*. As specific needs arise during the year, the purchaser will choose among the preferred suppliers for each order.

In this case, the determination of the price to bid needs to consider not only the probability of winning—that is, becoming a preferred supplier—but also the amount (and mix) of business that will come his way during the term of the contract. Generally, both the

bidder's probability of becoming a preferred supplier and the amount of business he will receive from the contract will increase as he decreases his price. Both considerations need to be incorporated into the estimated contribution.

13.5 CUSTOMIZED PRICING IN ACTION

Effective customized pricing requires a disciplined business process both to calculate and administer prices and to ensure that the parameters of the bid-response functions are updated to capture changes in the market. A high-level sketch of such a process is shown in Figure 13.12. There are three core steps in the customized-pricing process.

1. *Calculate bid-response function.* Using available information about the customer, the product (or products) she wishes to purchase, and the channel through which the request was received, the seller estimates how the probability of winning this customer's business varies as a function of price. This requires retrieving (or calculating) the parameters of the bid-response function that apply to this particular combination of segment, channel, and product.
2. *Calculate deal contribution.* The seller estimates the deal contribution function relating contribution from this deal to its price.
3. *Determine optimal price.* Combining the bid-response function and the deal contribution function, the seller calculates the price that maximizes expected contribution (or other goal) subject to applicable business rules.

These three core steps may be executed each time a bid request is received and a price needs to be calculated. Alternatively, the PRO cube might be populated ahead of time with

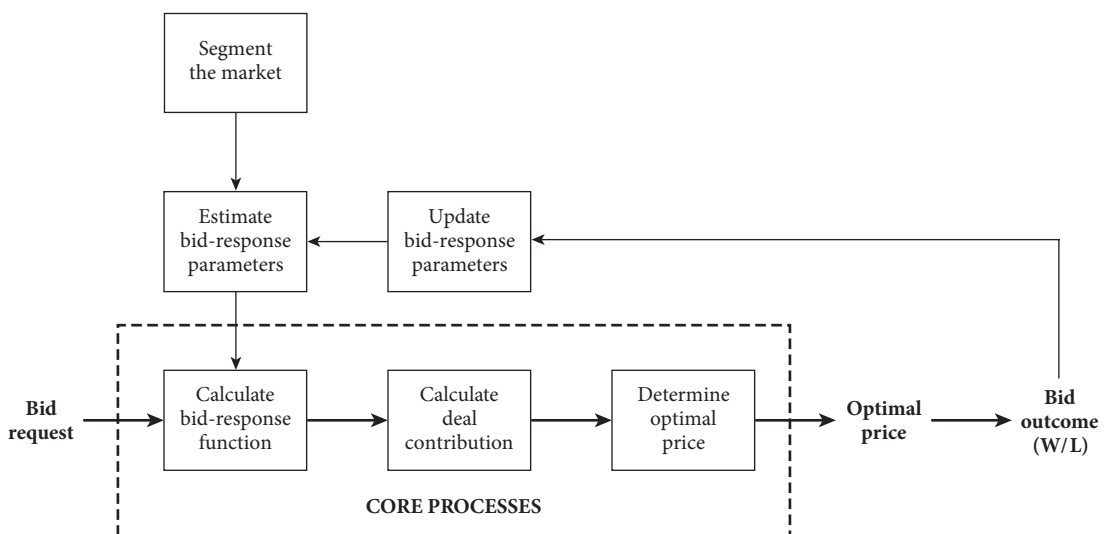


Figure 13.12 The customized-pricing process.

optimal prices that are accessed whenever a bid request is received. This requires executing the three core steps in batch mode prior to processing any bids. Which approach is better depends on the volume of bids to be processed and the time available to respond to each one. A lending company that needs to respond rapidly to thousands of customer inquiries about consumer loans every day is more likely to prepopulate a database with prices. On the other hand, a heavy equipment manufacturer that responds to 40 or 50 RFPs per week, with several days available to respond to each, is more likely to optimize the price for each bid independently.

Whichever approach is used, the core steps need to be supported by three additional steps that are executed less frequently.

1. *Segment the market.* The seller determines the market segments he will use to differentiate his prices. This establishes the structure of the PRO cube and needs to be done prior to estimating any bid-response parameters. Typically, the market segmentation does not need to be updated often—perhaps annually or semiannually. The exception is when the seller seeks to serve a new market or type of customer. Note that the extent to which a seller will be able to segment his market will be limited by the amount of historical bid-response data that he has.
2. *Estimate bid-response parameters.* The seller estimates the parameters of the bid-response function when the market is initially segmented. The bid-response parameters need to be reestimated whenever the segmentation itself is changed.
3. *Update bid-response parameters.* The seller updates the parameters of the bid-response function on a routine basis—usually weekly or monthly—to ensure that changes in the underlying market are captured. Updating the bid-response parameters is not much different from estimating them in the first place. In the simplest approach, the results of the most recent bids (i.e., those since the last update) are appended to the bid-history file in Figure 13.9, and the same approach is used to estimate the updated parameters using the new, larger database of bids. However, this approach would give increasingly old bid results the same weight as much newer bids in estimating the parameters. For this reason, most companies drop old bids—for example, those over a year old—from their bid-history file. Alternatively, other companies use Bayesian updating approaches that incorporate all bid information but weight newer bids more than older bids.

13.6 SUMMARY

- Customized pricing occurs when customers approach a seller individually and describe their desired product (or products) prior to purchasing. Customized pricing is commonplace in business-to-business marketplaces as well as some consumer marketplaces, notably lending and insurance.
- In a customized-pricing setting, the seller knows the desired product for each customer, the channel through which the customer approached, and some of the

characteristics of the customer. Furthermore, the seller can use this information to set a customized price for each bid.

- An important aspect of customized pricing is that the seller obtains information about not only bids that he won but also bids that he lost. He can use this information to estimate a *bid-response function* that determines his probability of winning a bid based on the characteristics of that bid and the price that he charges.
- When a seller quotes a price for a bid, he faces two types of uncertainty: *competitive uncertainty* regarding competing bids and *preference uncertainty* regarding the criteria the purchaser will use to choose a winner. In some cases, the buyer is committed to purchasing from the lowest bidder—in this case, the seller can calculate a *bid-response function* based entirely on his uncertainty regarding competitive bids.
- In the more common case, a seller needs to estimate a bid-response function based on the historical characteristics and prices of bids that he has won and lost. This is a problem of binary regression, and the most popular approach to estimating a bid-response function is to use *logistic regression*.
- A seller who has estimated a bid-response function can use it to determine the price that maximizes his expected profitability from each bid, which is calculated as the probability of winning the bid at a price times the profitability of the bid at that price if it is won.
- Application of optimal customized prices is complicated by the fact that many sellers are subject to business requirements such as minimum margin requirements and minimum market-share requirements. These are typically imposed as constraints in the customized-pricing optimization problem. Applying inconsistent constraints can lead to *infeasibility*, in which no set of prices can simultaneously satisfy all the constraints.
- Optimal customized pricing is typically implemented in an ongoing business process in which the optimal price for each bid is determined based on the characteristics of the product, the channel, and the customer at the time of the bid. The parameters of the bid-response function are periodically updated based on the performance of recent bids.

13.7 FURTHER READING

Customized pricing is discussed further in Bodea and Ferguson 2012 and Phillips 2012a. Applications of customized pricing to medical devices are described in Agrawal and Ferguson 2007 and Dubé and Misra 2017. Phillips 2018 describes at length the application of optimized customized pricing to various forms of consumer credit.

A classic text on logistic regression is Hosmer, Lemeshow, and Sturdivant 2013, which includes much of the mathematical details. A shorter introduction is Pampel 2000, and a useful practical guide is Kleinbaum and Klein 2010.

The issue of centralization versus decentralization in customized pricing, including the auto lending study and the Canadian lending study, are discussed in Phillips 2014 and Phillips 2018, chap. 8. A broader discussion of the trade-offs between a centralized and a decentralized pricing organization can be found in Simonetto et al. 2012.

The closely related topics of negotiation and auctions both have a vast literature. Raiffa, Richardson, and Metcalfe 2013 is a great starting point and source for negotiations. Steinberg 2012 surveys the auction literature from the seller's point of view, and Milgrom 2004 bridges theory and practical applications. Elmaghraby and Keskinocak 2003 reviews the literature on online supplier auctions.

The printer cartridge example was adapted from Phillips 2012a, which also describes the calculation of efficient frontiers in customized pricing. The discussion of endogeneity in customized pricing was adapted in part from Phillips 2018.

13.8 EXERCISES

1. HP is bidding against Lenovo for an order of 150 laptops. HP's unit cost is \$1,000 per laptop, and, based on previous experience, HP's belief is that Lenovo's bid will be uniformly distributed between \$1,200 and \$1,500 per unit.
 - a. Assuming that HP and Lenovo are at parity (e.g., neither supplier enjoys a premium on this deal), what is HP's optimal price per unit to bid? What is HP's corresponding probability of winning the bid and expected contribution?
 - b. Assume that Lenovo is an incumbent supplier to this customer and, as a result, HP believes that Lenovo enjoys a \$200 per unit premium for this deal. Assuming that HP has the same distribution on Lenovo's bid as previously stated, what is HP's optimal price per unit to bid? What is HP's corresponding probability of winning the bid and its expected contribution?
2. The printer cartridge manufacturer with the data shown in Table 13.3 has used historical data to estimate the coefficients in the model in Equation 13.11 and has come up with the model $p(p, x) = a + bPRICE + cGRADES + dGRADEG + eSIZE$

TABLE 13.5
Model predictions and actual bid outcomes for telecommunication problem

Number	Predicted win rate		Win?
	Model 1	Model 2	
1	.6	.2	0
2	.8	.5	1
3	.5	.3	0
4	.7	.9	1
5	.2	.4	0
6	.3	.3	0
7	.7	.8	1
8	.8	.6	1
9	.7	.6	0
10	.4	.8	1

$+ f_{NEW} + g_{RET} + h_{GOV} + i_{BUSLEVEL}$. What win probability would this model predict for each of the bids in Table 13.3

3. Table 13.5 shows the take-up probability predicted by two different bid-response models and the actual outcome of 10 historical bids for a telecommunications company.
 - a. What are the RMSE, concordance, and log-likelihood of each model based on the data shown?
 - b. Which model performed better on these data?

NOTES

1. The e-Car example is adapted from Phillips 2018.
2. The exception is the rare case when a hot new model is introduced and demand at the MSRP far exceeds supply. In this case, dealers may decide to charge more than the MSRP. However, this practice tends to be unpopular with customers. For example, a California dealership quoted a price \$10,000 above the MSRP for the extremely popular Corvette Stingray to widespread complaints of price gouging (*Automotive News* 2013).
3. In this chapter, I occasionally talk about the *bid price*, by which I simply mean the price associated with the bid. This should not be confused with the *bid price* used in revenue management, as defined in Section 9.2.3.
4. How these factors might actually translate into a decision between competing suppliers and the extent to which the buyer is being truthful about her selection criteria are separate issues.
5. As in the definition of price elasticity in Equation 3.8, the minus sign guarantees that the elasticity will be positive, since $p'(p)$, the derivative of the price-response function, is less than 0.
6. Following common practice, I capitalize and italicize explanatory variables such as *GRADEP* and *PRICE*.
7. In this case, the word “negotiation” does not necessarily indicate the back and forth usually associated with the term. It may be that the customer does not know that the price is actually negotiable and therefore treats the quote as fixed.
8. The studies on auto lending and Canadian lending are discussed in Phillips 2014.

14 BEHAVIORAL ECONOMICS AND PRICING

The foundation for the standard model of price response is the assumption that consumers are *rational* utility-maximizing agents. Before making a purchase, a rational consumer gathers information and then systematically and dispassionately evaluates her alternatives. A rational consumer has a willingness to pay (which may be less than zero) for every item on offer to her by every seller. She chooses the set of purchases that maximizes her expected utility (measured in monetary units) from the purchases minus the total amount she must pay. Variation of willingness to pay among customers generates a price-response function as described in Chapter 3. In a similar vein, neoclassical economics assumes that firms choose among alternatives to maximize expected profit; uncertainty on the part of sellers about how different customers value their product offerings relative to the competition gives rise to the bid-response function described in Section 13.3.

The idea of rationality is based on a handful of plausible axioms about how consumers and firms behave, and it provides an intuitive and compelling basis for price optimization. However, it is not difficult to find examples of behaviors that are not consistent with the assumption of consumer rationality. Consider the following two examples.

Coca-Cola makes customers hot. In a 1999 interview with the Brazilian magazine *Veja*, Douglas Ivester, chairman and chief executive officer of the Coca-Cola Company, announced that his company was testing a new vending machine that would change the price of soft drinks in response to outside temperature. When it was hot outside, a cold can of Coke would cost more than when it was cold outside. As he explained:

Coca-Cola is a product whose utility varies from moment to moment. In a final summer championship, when people meet in a stadium to enjoy themselves, the utility of a chilled Coca-Cola is very high.¹ So it is fair it should be more expensive. The machine will simply make this process automatic. (Martinson 1999)

While Douglas Ivester might have thought this was fair, others emphatically did not agree. The *San Francisco Chronicle* (1999) called the idea “a cynical ploy to exploit the thirst of faithful customers,” the *Honolulu Star-Bulletin* labeled it “a lunk-headed idea” (Memminger

1999), and the *Miami Herald* referred to Coca-Cola management as a bunch of “soda jerks” (Hays 1999, 18).

Coke began to backpedal immediately. “We are not introducing vending machines that raise the price of soft drinks in hot weather,” said spokesperson Ben Deutsch (Reich 1999). Although Coca-Cola is rumored to have tested weather-sensitive machines in Japan, it has never employed them in the United States. And the temperature-sensitive vending machine gaffe is widely considered to be one of the factors contributing to Ivester’s losing his job.

Amazon flubs DVD pricing. One night in August 2000, a customer ordered from Amazon the DVD of Julie Taymor’s movie *Titus*, paying \$24.49. Browsing on Amazon a few nights later, he found that the price had jumped to \$26.24. As an experiment, he deleted the cookies on his computer that identified him to Amazon as a regular customer. Revisiting the Amazon website, he was quoted a price of \$22.74 for the same DVD (Streitfeld 2000).

This customer happened to be a regular purchaser of DVDs from Amazon, and he came to the reasonable conclusion that Amazon was charging higher prices to its most loyal customers. When he recounted his experience with Amazon on the website DVDTalk.com, the reaction was immediate and overwhelmingly negative. Comments posted on DVDTalk.com over the next weeks included:

- “Amazon is over in my book.”
- “I will never buy another thing from those guys!”
- “I am so offended by what they did that I’ll never buy another DVD from them again.”
- “Amazon is suck [*sic*].”

The tale spread via email and chat rooms, and *Computerworld* published a story about Amazon’s pricing policies. The national press picked up the story. Press coverage generally jumped to the conclusion that Amazon was experimenting with “dynamic pricing,” by which they would charge different customers different prices. As one reporter commented, “Dynamic pricing . . . involves putting customers into groups based on how price sensitive they appear to be. If someone accepts the \$10-higher price, say, three times in a row, maybe that’s all they will ever get in the future” (Coursey 2000).² While a few reporters noted that dynamic pricing was hardly unknown in the offline world, the response to Amazon’s action was almost universally negative.

In response to the negative publicity, Amazon apologized and claimed that it was merely running a price-sensitivity test: “It was done to determine consumer responses to different discount levels,” said Amazon spokesman Bill Curry. “This was a pure and simple price test. This was not dynamic pricing. We don’t do that and have no plans ever to do that” (Streitfeld 2000, A1).

What happened? In both cases, consumer reaction did not follow the standard model of rational economic decision making. After all, a rational consumer should care only about her price—not the prices charged to other customers or what the company might have charged under different conditions. In the cases of Amazon and Coke, apparently sensible

pricing schemes were abandoned as a result of consumer reaction. Clearly, the consumers quoted in these stories are not automatons comparing price to willingness to pay and dispassionately deciding whether or not to buy. Rather, they are human beings reacting emotionally to such seemingly irrelevant issues as the prices other people might be offered or how prices might change with the weather.

The topic of this chapter is how consumers react to prices in ways that deviate from the assumptions of rationality—as in the Amazon and Coca-Cola examples above—and the implications of these deviations for pricing and revenue optimization. Although the basic model of customer price response is quite robust, there are many types of behavior that it cannot represent. In particular, customers do not view buying and selling dispassionately as simple transactions among consenting adults. Rather, they often react to prices in ways that economists consider irrational.³ Customer response to price is not always based only on the price and product being offered. Instead, a number of other, seemingly irrelevant, factors can influence buying behavior, including these:

- How the price is presented and packaged
- How much profit the customer believes the merchant will realize
- How this price compares to the customer's expectation of what the price should be
- Prices the customer believes are being charged to other customers

All of these factors can influence consumer behavior, but none of them can be easily accommodated within the standard model in which willingness to pay and price alone govern whether or not a customer will purchase.

The study of deviations from rationality and their implications is called *behavioral economics*. Many behavioral studies have found effects of pricing that might be hard to use in the real world—for example, that customers are less sensitive to prices when the cents match their birthday (Coulter and Grewal 2014) or the final digits are associated with a favorite sports team, such as \$.49 for a fan of the San Francisco 49ers (Husemann-Kopetzky and Köcher 2017). While phenomena such as these may have little practical significance, three important and well-established categories of consumer irrationality are important for pricing and we examine them at length.

The first category consists of violations of the law of demand—the principle introduced in Chapter 3 that specifies that, all else being equal, an increase in price should lead to decreased demand, and a decrease in price should increase demand. However, there are situations in which the opposite effect might occur—raising price might lead to increased demand, and lowering price might decrease demand.

The second category consists of *presentation issues* in which customers react not only to a price but also to the way in which it is communicated. One well-known example of a price presentation issue is the *digit-ending* phenomenon—buyers in Europe and North America tend to overweight the leading (leftmost) digits when comparing prices so that, in immediate impression, \$19.99 seems to be a greater savings from \$20.00 than \$19.93 from \$19.99. Because of this phenomenon, certain penny-endings such as 9 and 5 are far more common than others.⁴

The third category of irrational price response involves the issue of *fairness*. The belief that one is entitled to fair treatment is deeply ingrained in most people. Perceptions of unfair treatment can trigger strong emotional responses such as those expressed in the DVDtalk.com postings about Amazon.

These three categories all influence consumer decisions, but they manifest themselves in different ways. A company that is doing a poor job at price presentation will simply sell less than it might otherwise, but it is unlikely to hear anything from its customers. After all, customers are unlikely to complain if a merchant's prices end in \$.83 rather than \$.99. Similarly, customers are unlikely to complain if they think that an item is priced too low for them. In contrast, customers are likely to complain—and loudly—if their sense of fairness is violated. Poor price presentation will probably not land you on the front page of the newspaper, but pricing considered unfair might—just ask Coca-Cola and Amazon.

14.1 VIOLATIONS OF THE LAW OF DEMAND

The law of demand specifies that demand for a good decreases as its price is increased and that demand increases as price is decreased. This corresponds not only to intuition but also to the fact that the price-response function is derived from the distribution of willingness to pay across a population. If this is the case, then raising prices should always lower demand and vice versa. Therefore, any violations of the law of demand must arise from situations in which the willingness-to-pay assumption does not hold. Three kinds of cases have been identified in which individual purchase behavior is not well described by a maximum willingness to pay, and therefore the law of demand might be violated.

14.1.1 Giffen Goods

Economic theory allows for the possibility of *Giffen goods*, whose demand rises as their price rises because of substitution effects. An example might be the case of a student on a strict budget of \$8.00 per week for dinner. When hamburger costs \$1.00 per serving and steak costs \$2.00, she eats hamburger six times a week and treats herself to steak once a week. If the price of hamburger rises to \$1.10, she stops buying steak and buys hamburger seven times a week to stay within her budget. In this case, a rise in the price of hamburger causes her consumption of hamburger to increase. While this behavior might conceivably occur at an individual level, a Giffen good requires that enough buyers act this way that they overwhelm other buyers who would buy less hamburger as the price rises. Giffen goods are rarely, if ever, encountered in reality—in fact, many economists doubt whether they have ever existed in the real world.⁵

14.1.2 Price as an Indicator of Quality

In some cases, price may be used by some consumers as an indicator of quality: higher prices signal higher quality. In this case, lowering the price for a product may lead consumers to believe that it is of lower quality, and demand could drop as a result. Typically, markets in which this is an issue have a large number of alternatives, and some buyers who do

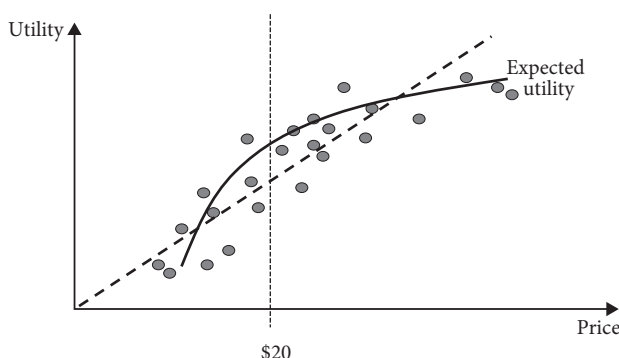


Figure 14.1 Price as signal of quality.

not have the time or resources to research the relative quality of all the alternatives instead use price as a proxy. Wine is a classic example: Faced with a daunting array of labels and varietals, many purchasers are likely to use a rule such as a \$10 bottle for dinner with the family, a \$15 bottle if the couple next door is dropping by, and a \$25 bottle if our wine-snob friends are joining us for dinner.

Using price as an indicator of quality is not necessarily an irrational strategy on the part of a consumer, particularly when quality is correlated with price. Consider a customer who is not highly knowledgeable about wine but has consumed a number of different California Chardonnays at different prices. Each dot in Figure 14.1 represents the utility that she enjoyed (denominated in dollars) from drinking a Chardonnay at that price point. Clearly her enjoyment of Chardonnay is correlated with its price, but the correlation is not perfect. In addition, her enjoyment demonstrates decreasing marginal utility—spending more at higher prices delivers less marginal utility than at lower prices. The solid curve shows her expected utility from purchasing a bottle as a function of price based on her experience. Assume that she is at a wine store shopping for a bottle and is unfamiliar with any of the brands on offer. With only price to guide her, it is perfectly rational for her to purchase at the price point that maximizes her surplus as defined by the difference between her expected utility and the price—which, in this case of the customer in Figure 14.1, occurs at about \$20. If enough customers share similar experiences, a Chardonnay might well sell more bottles at \$20 than at \$12.

The price-as-an-indicator-of-quality effect can be particularly important when a new product enters the market. A medical-product company developed a way of producing a home testing device at a cost 75% below the cost of the prevailing technology. It introduced the new product at a list price 60% lower than the list price of the leading competitors, expecting to dominate the market. When sales were slow, the company repackaged the product and sold it at a price only 20% lower than the leading competitor. This time, sales took off. The company's belief is that the initial rock-bottom price induced customers to believe its product was inferior and unreliable. The higher price was high enough not to raise quality concerns but low enough to drive high sales.

More recent studies have shown that not only may price be a signal of quality, but the price itself may influence customer enjoyment. In one study, the price of an energy drink

influenced the ability of purchasers to solve puzzles after consuming the drink (Shiv, Camon, and Ariely 2005). And in a widely publicized study, researchers performed a brain scan while subjects drank wine. They found that subjects reported enjoying a wine more if they thought that it was priced higher. Perhaps more surprisingly, the brain scan showed that they apparently *did* enjoy the same wine more if they thought it cost more—regions of the brain associated with pleasure were more highly activated when a subject consumed what they believed was a high-priced wine than when they consumed the same wine but believed it was low-priced (Plassmann et al. 2008).

14.1.3 Conspicuous Consumption

The sociologist Thorstein Veblen coined the term *conspicuous consumption* for the situation in which a consumer makes a purchase decision in order to advertise his ability to spend large amounts. As he put it, “Conspicuous consumption of valuable goods is a means of reputability to the gentleman of leisure” (Veblen 1899, 57). Thus, it is likely that a large segment of the purchasers of Bentley Continental GT automobiles (advertised as starting at \$202,500) and Louis Roederer Cristal Brut Champagne (about \$200–\$3,000 per bottle) are not necessarily purchasing entirely based on their willingness to pay for qualities of the car and the taste of the champagne, respectively, but are also making a statement about their ability to afford such luxuries. For a product that has been successfully marketed as an expensive luxury product, there may be those who will purchase it in part because it has a high price—and others know it. For such products, maintaining the air of exclusivity will influence marketing and pricing—for example, many luxury brands will not allow their products to be sold through the usual channels at a discount, even though it might be profitable to do so, because of the potential damage to the brand. Instead, they will sell through alternative channels, such as outlet stores, or ship to other countries to sell or even destroy product to maintain the image of high price and exclusivity.

14.2 PRICE PRESENTATION AND FRAMING

Consider the following three scenarios:

1. You walk up to the box office to buy two tickets to a movie. A sign in the window reads:

General Admission:	\$15.00
Senior Citizen (65 and over):	\$13.00
Youth (18 and under):	\$13.00
2. You walk up to the box office and the sign in the window reads:

General Admission:	\$13.00
Middle Aged (over 18 and under 65):	\$15.00
3. You walk up to the box office and the sign in the window reads:

General Admission:	\$13.00
Middle Aged (over 18 and under 65):	\$2.00 surcharge

Chances are that most people would consider the first scenario to be unremarkable. However, the second scenario would likely raise eyebrows. The effect is likely to be even stronger under the third scenario. In fact, you can almost hear the cries of “Why should I be punished because of my age? It’s not fair!” Yet, of course, all three price schemes are exactly the same, at least in the sense that everyone would pay the same amount for admission under each scheme. However, people perceive them and react quite differently. The most widely accepted reason for why this is so is provided by *prospect theory*.

14.2.1 Prospect Theory

Prospect theory was introduced by Daniel Kahneman and Amos Tversky in 1979 based on their observation that people do not evaluate opportunities based on a strict evaluation of expected costs and benefits. Rather, they use various mental shortcuts and rules of thumb to make decisions. This can lead to outcomes that are inconsistent with economic rationality. For example, prospect theory proposes that people tend to experience losses much more intensely than they value gains: this is called *asymmetrical treatment of gains and losses*. This idea is illustrated in Figure 14.2. Customers value a gain of Δ much less than they feel the pain of a loss of equal value. Since paying a higher price is viewed as a loss, this means that a surcharge (or price rise) of \$5.00 is viewed much more negatively than a discount (or price drop) of \$5.00 is viewed positively. In short, price rises are felt more intensely than price drops.

An important implication of prospect theory for price presentation is that *the presentation of prices can establish the baseline against which the actual price is compared*. As an example, consider the following scenario.

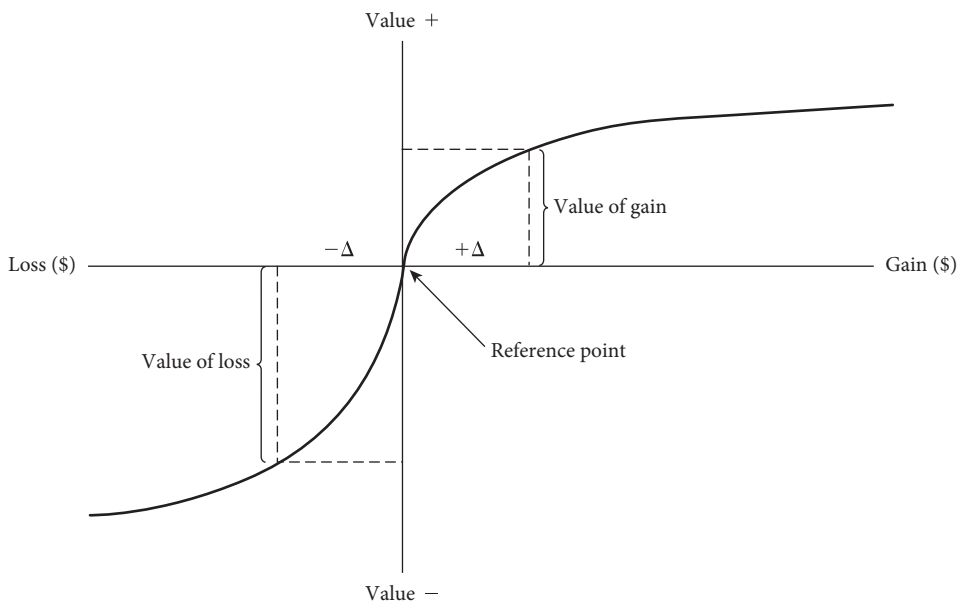


Figure 14.2 Asymmetrical treatment of gains and losses.

- *Station A* sells gasoline for \$2.60 per gallon but gives a discount of \$0.10 per gallon if you pay with cash.
- *Station B* sells gasoline for \$2.50 per gallon but charges a surcharge of \$0.10 per gallon if you pay with a credit card.

The two approaches are identical in terms of what any particular buyer will pay. Yet Station B's approach is never seen in the real world. Why? According to prospect theory, the reason is immediate from Figure 14.2. For each station, the list price establishes the baseline price in the buyer's mind. For Station A the \$0.10 discount ($+\Delta$) is viewed as a gain relative to the baseline price of \$2.60, while for Station B the \$0.10 surcharge ($-\Delta$) is viewed as a loss relative to the baseline price of \$2.50. Prospect theory predicts that a buyer using a credit card would be more likely to patronize Station A than accept the surcharge associated with Station B.

For each of the three movie-pricing scenarios, the general admission price establishes the baseline against which the price that a customer will actually be charged is compared. When the actual price includes a discount, it is viewed as a gain, which is viewed favorably. When the actual price includes a surcharge, it is viewed as a loss, which is felt negatively. Asymmetric treatment of gains and losses also explains why the middle-age-surcharge approach to movie pricing seems so jarring—and is never found in the real world.

Asymmetrical treatment of losses and gains may explain some of the negative reaction to Coca-Cola's announcement of temperature-dependent vending machine pricing. Note the wording of Douglas Ivester's announcement: "In a final summer championship, when people meet in a stadium to enjoy themselves, the utility of a chilled Coca-Cola is very high. *So it is fair that it should be more expensive.*" From a prospect theory point of view, he chose the worst price presentation possible—he framed the scheme in terms of a surcharge relative to a baseline price. This framed the higher price as a loss and triggered an immediate negative reaction. Coca-Cola would probably have fared better if Ivester had said something like: "On a cold day, the utility of a chilled Coca-Cola is not as high as on a hot day. So it is only fair that it should cost less—our customers deserve a price break in that situation." In fact, when faced with the negative reaction to the original suggestion, the company tried to reframe the description in just those terms by insisting that "its machines would be more likely to lower prices during off-peak hours in offices, for instance" (Martinson 1999).

Prospect theory provides an explanation for a nearly universal phenomenon in pricing—discounts are common, and surcharges are rare. As Chapter 12 notes, almost 80% of apparel sold in the United States in 2006 was sold at a discount from list price (Levy et al. 2007). It is safe to assume that only a tiny fraction (if any) of retail apparel sales took place at a surcharge from list price. Clearly, sellers believe that promotions and markdowns stimulate sales more than is explainable simply by the lower price. If the absolute price was all that mattered, sellers would be indifferent between running a sale and lowering list price. Rational consumers would simply compare their willingness to pay against the current price and decide whether to purchase. It would not matter to a rational consumer if the price she was paying was the list price or a sale price.

2-in-1 Take Apart Racing Car, DIY Toys 26 Pieces Stem Learning Assembly Car Toys with Electric Drill Tool, Lights and Sounds for Kids Boys Girls

by Beebeerun

★★★★☆ 15 customer reviews | 8 answered questions

List Price: ~~\$26.65~~

Price: **\$15.99** ✓prime

You Save: \$10.66 (40%)

VTech Pull and Sing Puppy

by VTech

★★★★☆ 618 customer reviews

| 28 answered questions

#1 Best Seller in Push & Pull Baby Toys

Was: ~~\$12.97~~

Price: **\$9.99** ✓prime

You Save: \$2.98 (23%)

Get \$70 off instantly: Pay \$0.00 upon approval for the Amazon Prime Rewards Visa Card. No annual fee.

Style: **Standard**

Figure 14.3 Two Amazon price listings showing discounts from a “List Price” (left) and from a “Was Price” (right). *Source:* Amazon.com.

But both brick-and-mortar retailers and online retailers like to present prices as discounts from higher standard prices. As an example, Figure 14.3 shows the prices of two different popular toys displayed on Amazon.com. The Pull and Sing Puppy on the right has a price of \$9.99 presented with a “Was Price” of \$12.97, while the Take Apart Racing Car on the left has a price of \$15.99 presented with a “List Price” of \$26.65. In both cases, the selling price is presented in combination with the List or Was price and the corresponding savings. The List and Was prices establish a higher benchmark price compared to which the actual price can be evaluated as a gain.

The practice has become sufficiently ubiquitous that some customers have become skeptical about the validity of the list price. A *New York Times* reporter wrote:

Le Creuset’s iron-handled skillet, 11 3/4 inches wide and cherry in color. Amazon said late last week that it would knock \$60 off the \$260 list price to sell the skillet for \$200. . . . The suggested price for the skillet at Williams-Sonoma.com is \$285 but customers can buy it for \$200. At AllModern.com, the list price is \$250 but its sales price is \$200. At Cutleryand-More.com, the list price is \$285 and the sales price is \$200. An additional 15 or so on-line retailers—some hosted by Amazon, others on Google Shopping—charge \$200. On Le Creuset’s own site, it sells the pan at \$200. (Streitfeld 2016)

Indeed, the power of perceived discounts is so strong that sellers are tempted to establish fictitious list prices against which they can pose their actual prices. A 1989 lawsuit held that the May Company “deceived customers by artificially inflating its so-called ‘original’ or ‘regular’ (reference) prices and then promoting discounts from these prices as bargains. This practice has become so rife in the U.S. retail environment that many consumers will only buy ‘on sale,’ and then are skeptical about whether they have really bought at a fair price” (Kaufmann, Ortmeier, and Smith 1991, 117). While displaying fictitious list prices is clearly deceptive (and likely illegal), whenever a seller is legitimately selling an item at a discount, it is extremely wise to make sure that customers are made aware of the fact.

14.2.2 Reference Prices

Kent Monroe (1973) introduced the idea that buyers shop with a reference price in mind that they consider fair. A price close to the reference price will be considered reasonable, and a consumer is likely to purchase if the selling price is close to her reference price. Prices much higher than the reference price may even trigger anger—“\$30 for a hamburger! That’s ridiculous!”

While it is good to be priced below a customer’s reference price, a price much lower than the reference price may cause suspicion that the item for sale is inferior or damaged. As an example, a start-up invented a process that could produce paper strips for use by diabetics to measure their blood sugar at a fraction of the cost of previous methods. Historically, these strips had sold for about \$2.00 each. With the new process, the company decided to sell its strips (which were of equal quality to those already on the market) for \$0.50 each. To its surprise, the strips did not sell well. However, when the company raised the price to \$1.75 each, sales skyrocketed. The historical selling price had become the reference price in consumers’ minds and the substantially lower price was interpreted as meaning that the strips were of lower quality—a major consideration to customers whose health depended on accuracy of measurement.

The reference price is not the same as willingness to pay. Willingness to pay is theoretically a property only of the buyer’s utility for the item under consideration, which should be independent of current or past prices. On the other hand, reference price is a function of observed prices, past and present. In the example of the gasoline station discussed earlier, we assumed that the reference price was the list price. The use of “list prices” and “was prices” by retailers is clearly an attempt to influence consumers’ reference prices.

This raises the issue of how reference prices are formed. As noted by Markus Husemann-Kopetzky, “People continuously learn new price information from past purchases, from current offerings of various competitors, from prices of comparable products, from regular or list prices, and so on. When consumers consider this information, they might adjust their internal reference price in either direction” (2018, 111). How different consumers process different types of information to update their reference prices is a topic of ongoing research. A useful survey lists a number of factors that could influence reference price, including price history, promotion history, store-visit history, brand, whether the purchase was planned or unplanned, store type (everyday-low-pricing [EDLP] stores like Walmart versus “hi-lo” stores that cycle between list prices and discount prices), advertised prices, frequency of purchases, durability, and associated services (Mazumdar, Raj, and Sinha 2005). These factors carry different weights for different types of customers. Given this complexity and the heterogeneity of consumers, it is impossible to perfectly engineer the reference price for any single consumer or chosen group.

While perfect reference price engineering may be impossible, there are several practical implications of reference pricing. If a customer becomes used to seeing an item at a low price, she will expect the price to be low in the future and will be less likely to purchase if it is high. This can create a dilemma for the seller. Dropping the price on an item now may be optimal from a tactical point of view (that is, it may maximize short-run contribution),

but it may have future repercussions if it lowers the reference prices of potential future customers. Several studies have shown that, if consumers utilize past prices in formulating their reference prices for future purchases, a cyclical hi-lo policy of periodic discounts can be optimal. It is also likely that if sellers ignore the effects of their pricing on reference price formation, then they will tend to underprice.⁶

14.2.3 Mental Budgeting

You are planning to go to a play by yourself one Sunday afternoon. Consider the following two scenarios.

- *Scenario 1:* You have already bought a \$20 ticket for the play. Just after leaving your apartment, you open your wallet, and the ticket slides out and slips down the storm drain, where it is lost.
- *Scenario 2:* You do not have a ticket but plan to buy one at the box office. Just after leaving your apartment, you open your wallet and a \$20 bill slips out of your wallet and is lost down the storm drain.

The question under both scenarios is: Would you go to the play and buy a \$20 ticket at the box office? (Assume in both cases that you know that tickets will still be available for \$20 when you arrive.) Kahneman and Tversky (1979) found that the percentage of people who would go to the box office and buy a ticket is much higher under the second scenario than the first. Of 250 college students presented with the second scenario, 187 said they would buy a ticket, while only 120 of 250 said they would in the first scenario. This is contrary to economic rationality. The wealth effect of losing a \$20 bill and a \$20 ticket are exactly the same. Why are people more willing to purchase a ticket in one case and not the other? The explanation, which has been supported by numerous experiments, is that people maintain mental budgets and evaluate expenditures against these budgets. Thus, someone might have a mental budget of \$20 (or \$30) for the day's entertainment. The initial purchase of a \$20 ticket is counted against this budget, so the purchase of a second ticket would exceed the mental budget. On the other hand, the loss of a \$20 bill is not counted against the day's entertainment budget (perhaps it is counted as a loss against general funds), so there is still sufficient entertainment budget to purchase a ticket.

Mental budgeting can have implications for how (and if) products and services should be bundled. If certain options, say, a top-of-the-line car stereo, are viewed as counting against a separate mental budget than the base product (the car itself), then it will generally be more effective to sell them separately.

14.2.4 Compromise Effects, Attraction Effects, and Decoys

Not only do customers use their prior price experiences in making current purchase decisions; they also use the prices of substitutes and variations in their purchase decisions, sometimes in surprising ways. A particularly striking example of this was described by Dan Ariely (2010). In a study, he offered students one of two different choice sets for subscribing to the *Economist*:

Choice Set 1	Choice Set 2
Internet subscription: \$59	Internet subscription: \$59
Print and internet subscription: \$125	Print-only subscription: \$125
	Print and internet subscription: \$125

A total of 68% of the students presented with Choice Set 1 chose the internet subscription, while 32% chose the print-and-internet subscription. However, when presented with Choice Set 2, 84% chose the print-and-internet option, while 16% chose internet only and 0% chose print only. The addition of an irrelevant third option substantially changed customer choice. In this case, it changed the average sales price per transaction from \$80.12 to \$114.44—a 43% increase resulting from nothing more than the addition of an option that was of no interest to any customer.

In the example above, the print-only subscription in Choice Set 2 is an example of a *decoy*: an irrelevant product that is added primarily to boost the sales of another product—in this case the print and internet subscription—by making it look like a bargain in comparison. Needless to say, the fact that decoys can work by boosting the sales of other alternatives appears irrational—if consumers were fully rational, the addition of an irrelevant alternative should not influence their choice between two relevant alternatives. Yet, as the *Economist* example shows, this is exactly what happens.

Decoys can be considered as particularly extreme examples of the *attraction effect*—the fact that introducing a dominated alternative can increase relative sales of a high-end product. The idea is illustrated in Figure 14.4. Here, we assume two products in the marketplace at two levels of quality and at different prices represented by points A and B. Corresponding to customer expectations, the lower-quality product A is priced less than the higher-quality product B—which means that neither product dominates the other. Note that any product in the shaded region is dominated because a better product is available at a lower price. Product C_1 is a decoy that is dominated by product A, which is the same quality but lower price. Product C_1 is unlikely to generate much demand on its own, but offering it will move some demand from product B to product A. Similarly, product C_2 is a decoy that is

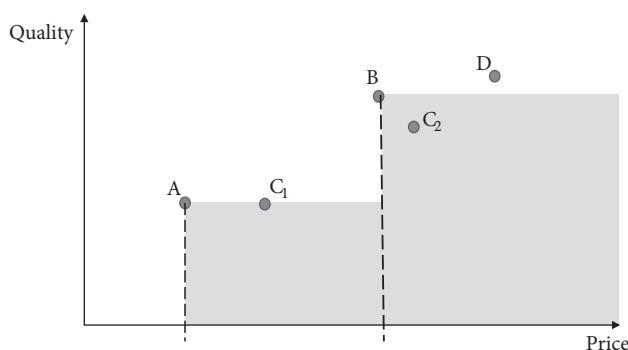


Figure 14.4 The attraction effect and the compromise effect.

dominated by product B. Again, product C_2 is unlikely to attract much demand itself, but it is likely to shift some demand from product A to product B.

A related phenomenon is the *compromise effect* in which adding a superior product to an existing product's market at a higher price will cause demand to shift to higher-price products. One example of the compromise effect is provided by Starbucks, which initially offered its drip coffee in three sizes: *small* (8 oz.), *tall* (12 oz.), and *grande* (16 oz.). In 1996, Starbucks introduced a fourth size, the 20-ounce *venti*. According to some sources, in addition to attracting sales itself, introduction of the venti led to a 10–15% increase in grande sales and a corresponding reduction in small and tall sales (Shartsis 2016). The increase in grande sales is an example of the compromise effect—the introduction of the superior alternative led to increased sales for the next best alternative. In Figure 14.4, the venti might be represented by point D—it is actually superior (i.e., larger) than the best existing alternative represented by point B. Because it is not dominated by any of the alternatives, it may attract some customers on its own merits, but it also will cause business to shift from point A to point B.

The compromise effect has been described as the tendency of customers to choose an intermediate option when they are uncertain about the relative quality of different options or their own strength of preference among them.⁷

14.3 FAIRNESS

The reaction to Amazon's DVD pricing illustrates the emotional response that differential pricing can evoke. Amazon learned the hard way that differential pricing can be the third rail of pricing. However, the Amazon story does not mean that differential pricing has to be entirely ruled out. On the one hand, a reputation for unfair pricing may alienate current and future customers. On the other hand, as Chapter 6 shows, the potential profitability increase from differentiated prices is great enough that companies that can find acceptable ways to segment their customers and charge differential prices have a strong incentive to do so.

This section addresses the issue of *fairness* as it applies to customer reactions to pricing. Fairness in the sense discussed here—and the way that most people mean it when they cry, “That’s not fair!”—was not part of neoclassical economics. However, behavioral economics has begun to research how consumers form their perceptions of fairness and how these perceptions influence the choices that they make. This is still an active area of research, and many aspects of perceived fairness in pricing are not fully understood. However, it has been shown that customers evaluate the fairness of a price in at least two ways. The first is relative to the perceived profit being earned by the seller. The second is relative to the prices the customer believes other buyers are paying. We consider these two in order.

14.3.1 Dual Entitlement

The first place that fairness can enter into pricing is via the buyer's perception of the seller's profit. Most people would agree that “greedy” companies should not generate “obscene”

profits by “gouging” customers with “unfairly high” prices. Most people could probably give examples of pricing they believe violated this principle—although they might be harder pressed to give an objective criterion of when profits become obscene or when a high price becomes gouging. Nonetheless, most customers believe that it is unfair for companies to charge too much for their goods and services—even if customers are willing to pay a high price for the item. This idea was formalized by Daniel Kahneman, Jack Knetsch, and Richard Thaler (1986b) as the concept of *dual entitlement*. Dual entitlement postulates that consumers believe they are entitled to a “reasonable” price and firms are entitled to a “reasonable” profit. Pricing that seems solely aimed to increase profit beyond what is viewed as reasonable is likely to be viewed as unfair. Specific implications of the principle of dual entitlement include the following.

- Raising prices to recoup increased costs is viewed as *fair* by most consumers.
- Raising prices simply to increase profits is likely to be viewed as *unfair*—even if people are willing to pay the higher price.
- Deviations from customary practices are often viewed with suspicion as hidden attempts to raise price.
- Providing information about the reason for increasing a price can often help acceptance.

One of the predictions of the theory of dual entitlement is that customers should be more accepting of price rises resulting from cost increases than they would be of price rises resulting from supply shortages. This does, in fact, appear to be the case—for example, charging more for vital supplies immediately before or after the arrival of a hurricane is viewed as very unfair by consumers. In this case, suppliers appear to be raising prices simply to grow profits unrelated to any change in costs in violation of dual entitlement. In the face of widespread consumer outrage after some suppliers hiked prices in the wake of Hurricane Andrew in 1992, Florida passed a law against price gouging. As of 2019, 35 states have passed laws against price gouging during periods of emergency (FindLaw 2020).

Further evidence of dual entitlement was gathered by Kahneman, Knetsch, and Thaler, who posed the following scenario to a sample of consumers: “A hardware store has been selling snow shovels for \$15. The morning after a large snowstorm, the store raises the price to \$20. Rate this action as: completely fair, acceptable, somewhat unfair, or very unfair” (1986b, 729). Of 107 respondents, 18% rated the pricing action as “acceptable or completely fair,” while 82% rated it as “somewhat unfair or very unfair.” In a similar vein, Kahneman, Knetsch, and Thaler asked what a store should do if it found it had exactly one item left in stock of that year’s most popular toy, assuming that all other stores were sold out of it (in their 1986 paper, it was a Cabbage Patch doll). A vast majority of those surveyed thought that it would be “very unfair” for the store simply to sell the toy to the highest bidder.

One problem created by dual entitlement is due to the fact that customers do not have a very good idea of sellers’ costs or profits. A 2001 survey of price and profit perceptions produced the results shown in Table 14.1. Each of the numbers in Table 14.1 is a gross overestimate of the actual margin. A typical discount store might realize a net margin of 4% and a typical grocery store 2%—not the 30% and 27.5%, respectively, that customers

TABLE 14.1
Responses to consumer survey questions regarding perceived profit

Question	Mean response
If a fancy department store pays \$40 to a manufacturer for a woman's blouse, what would be a fair price for the store to charge you?	\$58.16
If a fancy department store pays \$40 to a manufacturer for a woman's blouse, what would be a fair price for the store to charge you, keeping in mind the store must cover such costs as rent and payroll?	\$62.94
If a fancy department store pays \$40 to a manufacturer for a woman's blouse, how much do you think the store charges you for the blouse when it is not on sale?	\$76.58
For each \$100 a fancy department store makes in sales, how many dollars do you think are left over in pure profit after the store has covered all its costs?	\$33.09
... for a discount store?	\$30.24
... for a grocery store?	\$27.52

SOURCE: Bolton, Warlop, and Alba 2003.

apparently believe. Part of the discrepancy may be explained by reference pricing, since consumers tend to set their reference prices based on the lowest prices they see for an item and they assume that these prices are still profitable for the seller. Since merchants often mark down items below cost, consumers tend to assume that the original list price must have included a vast profit for the seller. This creates a problem. As described in Chapter 12, markdown pricing is often the optimal tactical policy for a merchant selling a stock of perishable goods. However, the short-term gain in profitability from the markdowns must be compared with the longer-term effect of creating the perception in consumers' minds that the cost of the item is low and that the merchant is reaping exorbitant profits at list price.

The principle of dual entitlement seems to be closely entwined with reference prices in the minds of customers. Consider the following two scenarios, posed by Kahneman, Knetsch, and Thaler (1986a).

- A shortage has developed for a popular model of automobile, and customers must now wait two months for delivery. A dealer has been selling these cars at list price. Now the dealer prices this model at \$200 above list price.
- A shortage has developed for a popular model of automobile, and customers must now wait two months for delivery. *A dealer has been selling these cars at a discount of \$200 below list price. Now the dealer sells this model only at list price.*

In the first case, 71% of those surveyed thought the dealer's action was unfair, while only 42% thought it was unfair in the second case. It appears that respondents are using the (unspecified) list price as the reference price at which the dealer is making a fair return. Pricing at this reference price is viewed as intrinsically more fair than pricing above it, independent of the absolute price being charged or the historical pricing practices of the seller. This is likely one reason why many customers view a dealer pricing a hot new model of car above the manufacturer's suggested retail price as unfair.

In view of this concern with fairness, music concerts and sporting events sometimes deliberately set prices below market based on the desire of the artist and/or promoter not to be perceived as gouging the fan base. As the late rock-star Tom Petty put it:

My top price is about \$65, and I turn a very healthy profit on that; I make millions on the road. I see no reason to bring the price up, even though I have heard many an anxious promoter say, “We could charge 150 bucks for this.” . . . It’s so wrong to say, “OK, we’ve got them on the ticket and we’ve got them on the beer and we’ve got them on everything else, let’s get them on the damn parking.” (Quoted in Wild 2002, 23)

Kid Rock decided to take a “pay cut” on his 2013 tour by selling tickets in every venue for \$20, even though the market-clearing price was much higher (Waddell 2013). Bruce Springsteen gave away an estimated \$3 million in potential revenue to his fans at a Philadelphia concert by selling tickets at a single price that was well below the price that would have cleared the market. Since the concert grossed about \$1.5 million, this is evidence of a serious desire to avoid perceptions of unfair pricing (Perry 1997).

Many economists would argue that these attempts to keep concerts affordable for the average fan are in vain because underpriced tickets are rapidly purchased by scalpers who either resell in person or use online marketplaces such as RazorGator to sell them at the much higher market-clearing price. Indeed, many of the artists and promoters who want to make tickets available more cheaply have resorted to complicated rationing methods to try to ensure that scalpers do not purchase all the cheap tickets to resell on the secondary market. The fact that some artists believe that such strenuous exercises are worthwhile is strong evidence that they want fans to believe that they are being fair in setting ticket prices and not merely chasing massive personal profits.

Dual entitlement also helps explain some of the negative reaction to Coke’s temperature-sensitive pricing scheme. Raising the price of a Coke on a hot day is viewed as similar to raising snow shovel prices after a snowstorm. This perception was reinforced by the fact that Coke’s announcement focused on the increased-price scenario, causing consumers to see the cold-weather price as the baseline price. It is likely that they believed Coke was making a reasonable profit at the cold-weather price and the hot-weather price was an attempt to gain an unreasonable profit, violating the principle of dual entitlement.

14.3.2 Interpersonal Fairness

As the Amazon case shows, price differentiation is an extremely sensitive issue. From the seller’s point of view, one of the most annoying aspects of interpersonal price comparison is that it can turn a happy customer into an unhappy customer for reasons totally unrelated to the quality of the underlying product or the responsiveness of customer service. The disgruntled Amazon customer who complained to DVDtalk.com about the price quoted for the DVD was not dissatisfied with his purchase or with Amazon’s service; he was unhappy because of the idea that someone may have been quoted a lower price. As one of the reporters covering the Amazon story wrote, “I don’t like the idea that someone is paying less than I am for the same product, *at the exact same instant*” (Coursey 2000). In other words, “I was happy with my purchase until I found out that someone else paid less.” This *ex post* effect will be familiar to anyone who has ever been responsible for setting salaries: employees who were perfectly happy with their raises become enraged upon hearing that someone considered less worthy received a larger raise. Similarly, learning that someone else received

(or was offered) a better price for the same good can instantly convert a happy customer to an unhappy one. This is irrational, at least in the narrow economic sense: my consumer's surplus should depend only on the price I paid, not on what others paid. Despite this, as the Amazon DVD pricing example shows, pricing that is perceived as unfair can generate a tremendous amount of negative emotion, ill will, and bad publicity.

Many cruise lines have faced this issue over the years. As discussed in Chapter 8, cruise lines have constrained, perishable capacity and most of the larger cruise lines actively practice revenue management to maximize the revenue for different sailings. This means that, if a sailing is not booking up as expected, the cruise line will lower the price for different cabin types to stimulate demand. Thus, different passengers can end up paying different prices for the same cabin type on the same sailing. It is likely that such price differences will not remain secret. People on a cruise dine together every evening. On the second or third night, the conversation will almost inevitably turn to prices. Mr. Smith mentions that his travel agent got a great deal for him and Mrs. Smith—an upper-deck, ocean-view stateroom for only \$3,200. Mr. Jones responds that his travel agent got him an upper-deck, ocean-view stateroom for \$2,800. Mr. Smith's face falls, and his cruise is suddenly ruined—at least for the moment. The cruise industry has tried various mechanisms to avoid this scenario. Currently the most popular appears to be a low-price guarantee or best-price guarantee in which a cruise line will provide a rebate—often in the form of an onboard credit—if a lower price is advertised within 48 hours of a purchase.

There is evidence that emotional reactions to unfair treatment are widespread across cultures and that the phenomenon has deep roots. Just how deep these roots might go was demonstrated by researchers at the Yerkes Primate Lab who performed a set of experiments in which capuchin monkeys could “buy” food using stone tokens. The food reward was either a cucumber slice or a grape. The monkeys preferred grapes, but those who received a cucumber slice in return for a token were happy until they saw that another monkey was getting a grape. When this happened, the behavior of the monkeys receiving cucumbers changed abruptly—they refused to eat the cucumber slices and threw them back at the researchers—reactions that never occurred when all of the monkeys received cucumber slices. The researchers concluded: “Capuchin monkeys . . . react negatively to previously acceptable awards if a partner gets a better deal” (Brosnan and de Waal 2003, 299).⁸

By this point, you might be forgiven for believing that any form of price differentiation will be considered unfair by consumers. However, this is not the case. We have seen that dynamic pricing is standard practice in many industries, including both online and brick-and-mortar retailers. Furthermore, price differentiation is widely exercised in many industries through such practices as couponing, sales promotions, markdowns, regional pricing, and peak-load pricing with little or no objection from consumers. In addition, some examples of direct, customer-based price differentiation that are widely practiced with little controversy include these:

- Senior citizen discounts
- Student discounts
- “Kids ride free”

- Family specials
- Lower tuition for in-state students than for out-of-state students at state universities
- College scholarships for low-income students, athletes, or those who score higher on tests

These examples are all pure price differentiation, or *group pricing* in the terminology of Chapter 6, in the sense that they are based entirely on characteristics of an individual customer and do not reflect any product or cost-of-service differences. Because of this, we would expect them to be controversial. Yet, in most cases, such policies are not particularly controversial, probably because they correspond to an ideal of giving discounts to “deserving” groups. What groups are deemed deserving will vary by country and culture. For many years, the French national railway (SNCF) maintained special discounts (“social tariffs”) for certain classes of workers, schoolchildren, apprentices, the unemployed, the disabled, the families of soldiers killed in World War I, and large families. The large-family (*Familles nombreuses*) tariff provided discounts of 30% for families with three children, 40% for families with four, 50% for families with five, and 75% for families with six or more children. This particular discount was adopted in 1921 to encourage repopulation of the country after World War I and remained in place until 2008 (De Boras et al. 2008; Chemin 2008).

Perhaps the most controversial form of group pricing that is still widely practiced is gender-based pricing. It is well established that women tend to be charged more than men for certain products and services. A study by the New York City Department of Consumer Affairs in 2015 found that across a sample of identical (or very similar) products marketed separately to men and women, the women’s version was priced higher 42% of the time, while the men’s version was priced higher only 18% of the time (Bessendorf 2015). A survey by researchers at the University of Central Florida showed that, on the average, women tended to be charged \$2.17 more than men for a haircut at a low-end salon, \$11.71 more at a midrange salon, and \$12.24 more at a high-end salon. Furthermore, women tended to be charged an average of \$3.95 for dry-cleaning a shirt compared to \$2.06 for men, and women’s deodorant averaged \$0.29 per ounce more than men’s deodorant. The prices of other types of dry-cleaning services, such as two-piece suits, blazers, and slacks, and other types of personal grooming products, such as razors, body spray, and shave gel, did not show statistically significant differences between women and men (Duesterhaus et al. 2011).

The surcharge that women pay over men for these products and services has been called the “pink tax.” There have been popular press articles on the pink tax (see Sebastian 2016; Ngabirano 2017), and a few jurisdictions, such as California, Massachusetts, Miami/Dade County, and New York City, have outlawed gender differences in pricing. On the other hand, anti-pink tax legislation has failed to pass in a number of locations in the face of opposition by business lobbyists. Even in jurisdictions with laws banning gender discrimination in pricing, enforcement has been sporadic and the practice persists. For example, in California, fewer than five suits were brought in the 20 years following the passage of the Gender Tax Repeal Act in 1995. And, although gender-based pricing for salons was (and still is) widely practiced in California, only five lawsuits were brought in the 10 years following the adoption of the legislation (Jacobsen 2018). This might suggest that much of the

public has learned to accept the practice—or is at least not sufficiently outraged by it to pursue legal remedies.⁹

You may realize by now that there is no universally reliable test to determine what types of pricing will be widely accepted by consumers and what will raise complaints. Cultural, social, and educational factors all seem to play a role—what is acceptable to one group (or one consumer) may not be to others. Policies that favor groups perceived as disadvantaged, such as the elderly, seem to be more acceptable than those that appear more arbitrary, despite the fact that senior discounts could often be taking advantage of the tendency of seniors to be more price sensitive than other groups. Surveys have shown that business and economics students tend to accept differentiated and dynamic pricing tactics more readily than the general public. For example, in the snow shovel scenario discussed earlier in the chapter, 76% of University of Chicago economics students said that raising the price of snow shovels after a snowstorm was acceptable, compared to 18% of the general public. This disparity would suggest that the prospect for future pricing debacles is good, since it is the business and economics students who are likely to be formulating the pricing schemes of tomorrow and, as a group, they may be oblivious to presentation and fairness issues.

14.4 IMPLICATIONS FOR PRICING AND REVENUE OPTIMIZATION

The topic of irrational consumer behavior and its influence on real-world pricing is relatively new and not perfectly understood—especially when it comes to judgments of fairness. As one researcher put it, “Fairness, it seems, is a complex and dynamic concept with many varied inputs” (Andreoni, Brown, and Vesterlund 2003, 22). Given this, you might conclude that the current state of knowledge is that consumers act rationally except when they do not. To some extent this is a fair judgment. Nonetheless, enough is known that it is possible to make specific recommendations to increase the effectiveness of pricing and revenue optimization and increase its chances for consumer acceptance. We start by looking at a summary of the factors that make tactical pricing schemes more acceptable. We then consider the implications of these for implementing pricing and revenue optimization. Finally, we return to the concept of fairness and examine two cases in which unfair pricing may have won out over fair pricing, despite consumer reactions.

Key elements that influence the acceptability of pricing and revenue optimization programs are shown in Table 14.2. The impact of these can be summarized as follows.

- *Product-based.* Product-based pricing differences are much more readily accepted than group pricing. As Chapter 6 shows, there is not a clear distinction between product-based differentiation and customer-based differentiation. To the extent that a pricing difference can be presented as product based rather than customer based, it is much more likely to be readily accepted.
- *Openness.* Customers certainly claim to prefer pricing schemes that are open. Some of the negative press that Amazon received for its DVD pricing dwelled on the idea that Amazon was doing something in secret. Furthermore, one of the lessons of the Amazon case is that any attempt to implement differential or dynamic pricing on

TABLE 14.2
Pricing acceptance factors

More acceptable	Less acceptable
Product based	Customer based
Open	Hidden
Cost-driven	Profit-driven
Discounts and promotions	Surcharges
Rewards	Penalties
Easy to understand	Hard to understand
Available to me	Unavailable to me
Familiar	Unfamiliar

the internet will be discovered and publicized. You should avoid any online pricing scheme that does not meet the *New York Times* test—that is, you would be averse to having it the topic of an article in the *New York Times*.

- *Reasonable.* Customers consider certain reasons for raising prices more reasonable than others. They are more accepting of a price raise due to increased costs than they are of one that is just about increasing profit. Customers also have implicit beliefs about a seller's costs and—via dual entitlement and reference pricing—some idea of what reasonable prices should be. Under the right circumstances, communicating the reasons behind discounts, but especially price increases, can help acceptance.
- *Discounts and promotions* are much more readily accepted than surcharges. Prospect theory predicts that the same price will be viewed more positively if it is the result of a discount rather than a surcharge. Furthermore, finding that another customer received a discount is much more likely to be received with equanimity than finding that one has been charged a surcharge—even when the net result is the same. As a result, list prices tend to be high, with discounts and promotions commonplace and surcharges rare.
- *Rewards.* In a similar vein, a reward for loyalty or volume purchases is much more likely to be acceptable than any kind of penalty. Airlines do much better offering frequent-flyer rewards than they would offering infrequent-flyer penalties.
- *Easy to understand.* Programs viewed as easy to understand are generally more accepted. Programs that are difficult to understand will cause customers to be concerned that they are not getting a good deal simply because they do not know how to play the game. Several studies have shown that, in some cases, customers may actually prefer suppliers who offer them fewer options—even if their resulting price is higher (Iyengar and Lepper 2000). Furthermore, customers often struggle to find the “best” option in cases in which computing the “real” price of an option is difficult—this is a particular issue in the case of financial products such as mortgages in which the total interest paid is a complex function of the interest rate, the term, the down payment, and other factors. In cases with many complex choices, many customers will revert to a simple heuristic such as “We’ll take the mortgage with the shortest term that has a monthly payment within our budget” even though that may not be the best choice.¹⁰

- *Available.* There seems to be a critical distinction in customers' minds between a deal that is potentially available to them and one that is not—a distinction closely related to the difference between product-based and customer-based segmentation. Customers are much more likely to be accepting of a discount that was (at least in theory) available to them than one that was not available to them under any circumstance. Thus, the acceptance of early-booking discounts in the airline industry was helped by the fact that any customer could take advantage of them, if only they had been able to make their travel plans earlier. In the same vein, the wide acceptance of discount coupons seems to stem in part from the fact that those who do not take advantage of the coupons realize that they could have done so by taking the time to clip them out of the newspaper. These *behavior-based* differentiation strategies seem to be much more broadly acceptable to customers than *identity-based* differentiation, which is predicated on factors such as race, age, and gender.
- *Familiar.* Customers are naturally conservative—they prefer traditional and customary ways of doing business and distrust innovations. Virtually any change in the way that prices are calculated, billed, or presented is likely to be viewed with suspicion as an attempt to raise prices. For example, there was widespread belief among consumers in the Eurozone that merchants took advantage of the conversion from the native currency to the Euro to raise prices.¹¹ As another example, ride-sharing companies such as Uber and Lyft traditionally based rates on a “surge multiplier” applied to the base fare (see Section 7.6.4 for details on how these multipliers are calculated and used). In 2019, Uber experimented with “additive surge,” in which the amount that drivers received would be a fixed surcharge—for example, \$2.10—to the base price of a ride rather than a multiplier (Garg and Nazerzadeh 2019). Many drivers assumed—largely without evidence—that the new practice was designed to reduce their earnings and enrich Uber.

You should anticipate negative reactions to virtually any deviation from traditional pricing practice. The least disruptive way to introduce pricing and revenue optimization in a market is within the context of existing and accepted tactical pricing practices, such as markdowns, promotions, and customized pricing.

One final note: While it is important to be sensitive to customer price perceptions, it is equally important not to allow anticipated customer reaction to squelch all possibility of pricing innovation. Many of the key findings on price perception, prospect theory, and fairness have been based on surveys, and, as any marketing professional knows, survey responses do not necessarily reflect how people will react in real life. For example, a majority of golfers stated that charging different prices depending on when they booked a tee time (e.g., two weeks in advance versus two days in advance) would be unacceptable (Kimes and Wirtz 2003). Nonetheless, several golf courses and online golf booking services have implemented time-of-booking pricing, with few or no customer complaints. While a pricing innovation may initially meet resistance and generate complaints, over time it may become the accepted norm. This is certainly what happened with airline fares—when revenue management was first initiated, customers complained about changing fares and that “the

person sitting next to me is paying half of what I paid.” However, over time, dynamic pricing and revenue management became accepted as the norm, and the complaints subsided.

The lesson is that customer complaints and negative press do not *necessarily* mean that a pricing program is a failure. Consider the case of the First National Bank of Chicago (usually called First Chicago). In the first quarter of 1992, the First Chicago realized that it had a problem with its retail banking operations. It calculated that 67% of its retail customers were not achieving adequate profitability. A primary reason: whereas many customers used tellers rarely if at all, a small number were so-called transaction hounds using tellers more than four times a month while maintaining less than \$2,500 in their accounts. Since teller salaries and benefits were a major part of the bank’s variable cost, it seemed only fair to charge customers for the service. As Jerry Jurgensen, First Chicago’s executive vice president in charge of branches, put it, “If nobody changed their behavior, 80% of my customers would never see the \$3.00 charge. For the other 20%, I want one of three things to happen: I need for them to change their behavior; I need for them to be willing to pay more, or I need for them to find another bank” (O’Sullivan 1997, 44).

First Chicago held a breakfast press conference to announce the new plan. It was not well received. The headline in the *Chicago Tribune* read, “First Chicago Loses Touch with Humans.” Competing banks leaped at the chance to tout their free checking services. The story hit the national papers and was even featured in a joke on *The Jay Leno Show*.

First Chicago was baffled by the reaction to its new pricing scheme. It had tested the plan for 18 months on different focus groups. It had gone out of its way (perhaps too far out of its way) to be open, announcing the new scheme at a press conference. And, after all, the \$3.00 teller fee was based on passing a real cost through to the customer, just the type of price change that the principle of dual entitlement predicted would be acceptable.

This would seem like a classic case of a pricing fiasco, and it is often presented as such: A clueless marketing department instituted a bonehead pricing program that ignored customer sensitivities and reaped a whirlwind of criticism. Another disaster, to be put alongside Amazon and Coca-Cola in the annals of pricing infamy. However, there is a twist to the story. From a financial point of view, the First Chicago program was a success. According to an article in the *American Banking Association Banking Journal*:

Banks are uniform in disassociating their profitability-probing behavior from First Chicago’s, which some consider to have been “PR suicide.” “They fired their marketing people the next day [after the teller-fee story broke],” said a source who preferred not to be named. However, it is widely conceded that First Chicago has had material success. It reportedly lost less than 1% of customers, not the 10% predicted—despite some other Chicago-area banks advertising against it—while branch staff decreased 30% and ATM activity “sky-rocketed” (ATM deposits grew 100% in three months). Speaking seven months after the restructuring, [former First Chicago senior vice president Marion Foote] said the bank was making an “adequate return” on 44% of customers, versus 33% beforehand, and customer satisfaction scores were higher than ever. (O’Sullivan 1997, 43)

Was teller surcharging a success or a failure at First Chicago? It depends on whom you ask. The marketing and public relations staff who lost their jobs would probably claim that

it was PR suicide. Anecdotal evidence seemed to indicate that some potential new customers were lost because of the negative publicity. But from the cold numbers it would appear that the teller fees were successful at achieving the announced goals: increasing customer profitability. It is suggestive that, despite the overwhelmingly negative publicity, First Chicago retained the fees for more than seven years, until it became part of BankOne. And since 1995, more and more banks have begun to charge teller fees (often as part of bundled pricing packages). For many banks, the purpose of the fee is to motivate customers to move from using human tellers (at an estimated cost of \$4 per transaction) to using video tellers (at an estimated cost of \$2 per transaction) (Lake 2018). However, the banks who have adopted teller fees since First Chicago have had the good sense not to call a press conference to announce that they were doing so.

Finally, competitive and market forces may not allow a company to simplify its pricing, even if it wants to. Consider the case of American Airlines.

By the spring of 1992, Robert Crandall, chairman and CEO of American Airlines, decided that airline fares had gotten too complex. While American's revenue management system had originally given it an advantage over its rivals, United and Continental had developed systems of their own and it was not clear that American retained any advantage. Furthermore, customers continued to complain about a system where business travelers paid much more money than leisure travelers for substantially the same service. Simplifying fares seemed to be a logical choice—it would allow all of the carriers to jettison the increasingly complex fare structures, eliminate much discounting, and improve customer satisfaction, all at the same time. American had led the majors into this swamp, and as the world's largest carrier, perhaps American could be the airline to lead them out.

On April 9, 1992, American Airlines called a press conference to announce its “value pricing” program. Crandall explained the rationale for change:

We have said for some time that our yields are too low, yet the conventional solution, higher and higher full fares and an ever-growing array of discount fares surrounded by an ever-changing plethora of restrictions, simply does not work. . . . In our unsuccessful pursuit of profits, we have made our pricing so complex that our customers neither understand it nor think it is fair. (Quoted in Silk and Michael 1993, 5)

Under value pricing, American sold only four fares in every market: first class, regular coach, 7-day advance purchase booking discount coach, and 21-day advance purchase booking discount coach. This simple structure was to replace the previous situation, in which American often offered 10 or even 20 different fares in a market, with different booking restrictions in different markets. Value pricing would reduce the total number of fares in the network from 500,000 to 70,000 (Silk and Michael 1993). As part of the plan, American also lowered its fares—first-class fares were reduced by 20–50% and regular coach fares were cut by about 40%. However, a myriad of other promotions and discount programs for groups and corporate customers were eliminated. Crandall's plan was that the new and simpler fares would induce enough additional traffic to outweigh the fare reductions.

American expected other carriers to follow its lead. After all, the complex fares associated with traditional revenue management were just as unpopular at competing carriers, and the

cost savings from fare simplification would be significant across the industry. American announced its intentions well ahead of time and did everything it could to signal to the other airlines that its new fare structure was about simplification, not starting a new fare war, and that simplification would be good for the entire industry. And the initial response was promising. Within a week, most major carriers, including United, Delta, Continental, and Northwest, adopted the value pricing fare structure, either in part or in whole. Initially, it looked like American had been able to lead the airlines successfully to a new, simpler structure without touching off a fare war.

Unfortunately, the hope was premature. TWA began a fare war in the second week of April. Initially, American stood by its new pricing scheme. However, in late May, Northwest issued a more serious challenge. It announced a special fare promotion called “Grown-Ups Fly Free” that offered free tickets to an adult when accompanied by a paying passenger between 2 and 17. The other airlines, including American, matched the promotions with 50% discounts of their own. These discounts were certainly effective in stimulating demand, and aircraft were flying with record loads—in many markets load factors approached 100%. However, the demand stimulated by the discounts did not outweigh the cannibalization, and many of the airlines suffered record losses along with their record load factors.

Increasingly, it became clear that value pricing was doomed. With an increasing variety of discounts and promotions being offered, things were reverting to the bad old days. American’s rivals were setting up fare structures that bracketed American’s three coach fares and then using their revenue management systems to feast on American, just as American had used its system to feast on PeopleExpress. By October 1992, it was evident that value pricing was finished. In an interview with the BBC, Crandall said that value pricing was “dying a slow death” and that the plan had “clearly failed” (Silk and Michael 1993, 7). By November, American had jettisoned value pricing completely and was back to setting and manipulating thousands of fares throughout its system.

Value pricing is often presented as a failed attempt at price leadership—which it was. Competitors found that by offering more complex fare structures, they could outcompete American in key markets, and the lure of doing so proved irresistible. Ultimately, complex fare structures backed up by sophisticated revenue management beat out the simplicity of value pricing.

But there is another lesson from the value pricing failure: Customers may prefer simple pricing to complex pricing, but they are not willing to pay for it. Customers praised the simplicity of value pricing, but when the time came to purchase a ticket, they bought from the airline that was offering the cheapest seat—not from the one offering the simplest fare structure. The customers who flocked to take advantage of “Grown-Ups Fly Free” tickets from Northwest did not care that they were destroying value pricing—they were simply looking for a bargain. This indicates that the success of *everyday low pricing* is primarily due to the fact that it is *low*, not to the fact that it is *every day*.

The most successful pricing and revenue optimization tends to be the most invisible. Systems that simply do a better job of setting prices within the context of an existing and accepted tactical pricing structure have the highest chance of success and acceptance. None-

theless, capturing the full benefits of pricing and revenue optimization may require new pricing practices. You can increase the chance of success by paying close attention to issues of price presentation and fairness in implementation. However, you should also expect that virtually any pricing innovation will be met with skepticism (at best). Part of the art of pricing and revenue optimization is knowing when to change in response to consumer resistance and when to persevere.

14.5 SUMMARY

- Customers do not always respond to prices in a way that is consistent with economic theory. Instead they are influenced by a variety of considerations, including the way the price is framed and the prices offered to others. When this occurs, it is said that customers behave *irrationally*. The study of the ways in which customers deviate from the predictions of economic rationality is called *behavioral economics*.
- In general, lower prices lead to higher demand and vice versa; this is the law of demand. However, there are three situations in which demand may actually increase with price:
 - With *Giffen goods*, an inferior good whose price is rising is substituted for a superior good.
 - *Price is considered a signal of quality* by consumers in some cases.
 - In *conspicuous consumption*, the price of a purchase signals to others the presumed wealth of the buyer.
- Customers are sensitive to the way that prices are presented—a price that is presented as a discount will typically attract more demand than the same price presented as list price. In addition, customers tend to be very averse to any price that is presented as a surcharge or a penalty. For this reason, sales promotions and discounts are common, while surcharges are rare. This phenomenon has been explained by *prospect theory*, which specifies that people are more averse to perceived losses than they are attracted to gains. A discount is perceived as a gain relative to a baseline price, which may be a reference price or a published list or “was” price, while a surcharge is perceived as a loss.
- Consumers believe it is fair that a company makes a reasonable profit and that they are offered a reasonable price—this is the principle of *dual entitlement*. Consumers are likely to react negatively if they believe that a company is violating this principle. This is an explanation for the fact that most consumers find price raises due to cost increases more acceptable than those due to scarcity.
- Consumers are also concerned with interpersonal fairness. In particular, they are likely to get upset if they learn that another customer paid (or was offered) a lower price for the same item at the same time. This makes many forms of group price differentiation controversial. Indeed, some forms of group price differentiation are illegal, depending on the jurisdiction. As a result, although individualized pricing on

the internet may be tempting, it raises the very real likelihood of strong consumer pushback.

- Consumers also tend to be conservative when it comes to pricing approaches. They are likely to believe that any change in pricing structure is an excuse for raising prices and violates the principle of dual entitlement.
- Pricing changes are more likely to be accepted if they are:
 - Product based rather than customer based
 - Open rather than hidden
 - Posed as a discount or reward rather than a surcharge or a penalty
 - Easy to understand rather than hard to understand
 - Available to all customers rather than restricted
 - Cost based rather than profit based
 - Familiar rather than unfamiliar
- Consumers' conservatism means that they are likely to object to any change in the way that goods are priced. But this does not mean all pricing innovation should be stifled. For example, customers objected to First Chicago's introduction of teller fees; however, it turned out to be reasonably successful. On the other hand, while consumers complained about the complexity and volatility of airline fares, American Airlines was unsuccessful in introducing a simplified fare structure because, at the end of the day, consumers preferred low fares to simple fares.

14.6 FURTHER READING

Behavioral economics has roots in the 1950s with the concept of bounded rationality developed by Herbert Simon. During the 1960s and 1970s, Daniel Kahneman and Amos Tversky published a series of papers that essentially established behavioral economics as a field. An entertaining description of the early development of behavioral economics through the collaboration between the very different personalities of Daniel Kahneman and Amos Tversky can be found in Lewis 2016.

The early part of this century saw an explosion not only of academic interest but also of books written for a wider audience, including *Nudge*, by Richard Thaler and Cass Sunstein (2008); *Predictably Irrational*, by Dan Ariely (2010); and *Thinking, Fast and Slow*, by Daniel Kahneman (2011). Behavioral economists study a wide range of topics, but pricing has been an important focus of research: surveys of research on the behavioral economics of pricing can be found in Özer and Zheng 2012, Hinterhuber 2015, and Husemann-Kopetzky 2018.

The use of brain scans to investigate the effect of prices and other marketing actions on cognition and behavior is relatively new but is rapidly expanding. A useful survey can be found in Hubert and Kenning 2008.

14.7 EXERCISES

1. *Giffen good*. A penurious graduate student has a food budget of \$100.00/week. To survive with sufficient energy to attend classes, he knows that he needs to consume 50 protein units per week. The only two foods he can stand to eat on a regular basis are beans and hamburger. He derives twice as much pleasure per protein unit from eating hamburger as he does from beans.
 - a. Assume that hamburger costs \$3.00 per protein unit and beans cost \$1.00 per protein unit. Formulate the student's diet problem as a linear program. (You can assume he wants to maximize his total utility from his diet and that he gets 1 utile from each protein unit of beans he consumes and 2 utiles from each protein unit of hamburger.) What is the optimal consumption of beans and hamburger in this case?
 - b. Plot the student's price-response function for beans as the price of beans goes from \$0.01 to \$2.00 per protein unit, assuming that everything else (including the price of hamburger) stays constant. Note that his individual price-response function is indeed upward sloping. Why? What happens when the price of beans exceeds \$2.00 per unit?
 - c. The student receives a scholarship that enables him to spend \$300 a week on food. What is his price-response function for beans now?
2. A particular market was served by two products offered by Acme and a competitor. The prices and quality ratings of the two products are as follows:

	Price	Quality rating
Acme brand	\$2.00	50
Competitor brand	\$2.50	70

The two brands each have a 50% share of the market. Now Acme introduces a new product with a price of \$2.20 and a quality rating of 30. What would you expect to happen in the market?¹²

- a. The new product will grab a dominant share of the market.
- b. The new product will split the market with the existing products into thirds.
- c. The new product will not gain much market share, but Acme will increase market share at the expense of the competitor.
- d. The new product will not gain much market share, but the competitor will gain market share at the expense of Acme.
- e. The new product will not gain much market share, and the relative shares of Acme and the competitor will be unchanged.

NOTES

1. Since it was an interview with a Brazilian magazine, Ivester was referring to a soccer championship match.

2. In this case, the term “dynamic pricing” was used as a misnomer for “price differentiation” or “price discrimination.”

3. The term “irrational” is not meant to be pejorative in this context—it simply refers to behavior that does not accord with the neoclassical economic model of rational consumer decision making.

4. The research on price-ending effects is quite extensive. For useful surveys, see Özer and Zheng 2012 and Husemann-Kopetzky 2018.

5. Although the idea of Giffen goods dates from the mid-nineteenth century, the first empirical evidence for a Giffen good in the real world—rice markets in Hunan Province in China—was not provided until the twenty-first century (Jensen and Miller 2008), although their claim is controversial. If Giffen goods do exist in the real world, they are quite rare and have little relevance to the practical problem of setting prices.

6. For the optimality of hi-lo pricing in the presence of reference price formation, see Greenleaf 1995 and Kopalle, Rao, and Assunção 1996. Popescu and Wu 2007 shows that cyclical pricing can be optimal and that ignoring reference price effects can lead to underpricing.

7. The idea of a decoy was introduced in Huber, Payne, and Puto 1982. For a discussion of attraction and compromise effects, see Özer and Zheng 2012 and Husemann-Kopetzky 2018.

8. A video showing the outraged reactions of capuchin monkeys to unequal pay can be found in a 2013 TED Talk by Frans de Waal, available at <http://bit.ly/1GO05tz>.

9. A counterexample to the pink tax is provided by bars that offer “ladies’ nights,” during which women are charged less than men for drinks (a practice that is illegal in California) or dance clubs that charge more for men than women in an attempt to attract more women and achieve gender balance (Lam 2014). In articles on both the pink tax and on ladies’ nights, the lower price is taken as the baseline price, and the higher price is seen as potentially unfair—a likely example of dual entitlement.

10. For a discussion of behavioral issues in mortgage selection and other consumer credit pricing situations, see Phillips 2018, chap. 8.

11. For a survey on consumer perception of price increases and actual increases following adoption of the Euro, see Pufnik 2017, which concludes that actual price increases were far lower than perceived increases for every country that joined the Eurozone up through 2016.

12. The results of this experiment are from Huber, Payne, and Puto 1982.

OPTIMIZATION

APPENDIX

A

Optimization refers to the process of finding the maximum or minimum of a function in some defined region, which could be either finite or infinite in extent. Here we look only at finding the maximum of a function, since that is the most relevant for pricing and revenue optimization.

A.1 CONTINUOUS OPTIMIZATION

In continuous optimization, we want to maximize a continuous, differentiable function $f(x)$ over a certain region—say, $x \geq 0$. Here, x may be either a single number—a *scalar*—or a vector $\mathbf{x} = (x_1, x_2, \dots, x_n)$. In either case, we seek the value of x , for which $f(x)$ is maximized. This general problem can be complex and subtle. In particular, a function may have many different local maxima within the region. If so, the problem of finding the global maximum among all the local maxima may be very difficult and may require the use of such approaches as genetic algorithms. Fortunately, the problems treated in this book typically have *unimodal* objective functions; that is, any local maximum will also be a global maximum. The problems we consider also generally have convex feasible regions. The combination of a unimodal objective function with a convex feasible region means that hill-climbing approaches typically work well. Starting at any point, we find an improving direction and move in that direction to a point with a higher objective function value. From that point we find a new improving direction and repeat the process until we get to a point where no further improvement is possible. This point is the global maximizer.

There are three possibilities.

1. The function may have no maximum. For example, Chapter 5 shows that a supplier facing a constant-elasticity price-response function with elasticity greater than 1 can always increase revenue by increasing his price. In this case, there is no finite price that maximizes revenue.

2. The function may attain its maximum value within the interior of the region.
3. The function may attain its maximum value on the boundary of the region

Given the restrictions we have put on the underlying function and the feasible region, an intelligent hill-climbing approach will find the optimal solution in the second and third cases. The first case will never occur in a well-posed problem—if we find that our pricing optimization problem has no finite solution, it generally means that the problem has been incorrectly specified or that one or more important constraints have been inadvertently excluded.

For an interior point $\mathbf{x}^*$ to be a maximizer of a function f , it must be the case that all of the derivatives are equal to zero at $\mathbf{x}^*$ —that is, $\partial f(\mathbf{x}^*)/\partial x_i^* = 0$ for all i . (If x is a scalar, this is equivalent to the requirement that $f'(x^*) = 0$.) If $\partial f(\mathbf{x})/\partial x_i > 0$ for some i , then $f(\mathbf{x})$ could be increased by increasing x_i^* by a small amount. If $\partial f(\mathbf{x})/\partial x_i < 0$ for some i , then $f(\mathbf{x})$ could be increased by decreasing x_i^* by a small amount. In either case, $\mathbf{x}^*$ is not a maximizer.

Thus, $\partial f(\mathbf{x}^*)/\partial x_i^* = 0$ for all i is a *necessary* condition for x to be an interior maximizer of f . It is not, however, a *sufficient* condition. There are two other possibilities: $\mathbf{x}^*$ could be a *minimizer* of $f(\mathbf{x})$. Or, more rarely, $\mathbf{x}^*$ could be an *inflection point* of $f(\mathbf{x})$. In the case when x is a scalar, the most common test to ensure that x^* is a maximizer of $f(x)$ is that the second derivative $f''(x) < 0$. If this condition is satisfied, then x^* will be a local maximizer of $f(x)$. And if $f(x)$ is unimodal, x^* must be a global maximizer as well.

Calculating the second derivative can be difficult. However, note that an x^* that satisfies the first-order conditions will also be a maximizer of $f(\mathbf{x})$ if perturbing any of its elements either up or down leads to a smaller value of $f(\mathbf{x})$.

Example A.1

Assume that we want to maximize the function $f(x) = -x^2 + 2x + 6$. The first derivative of $f(x)$ is $f'(x) = 2 - 2x$, which is equal to 0 only for $x^* = 1$ with $f(x^*) = 7$. The question is whether $x^* = 1$ is a maximizer or a minimizer of $f(x)$ (or possibly an inflection point). We note that $f''(x) = -2 < 0$ for all x , which indicates that x^* is a maximizer. Alternatively, we note that $f(.9) = 6.99 < f(x^*)$ and $f(1.1) = 6.99 < f(x^*)$, which is another way of showing that $x^* = 1$ is a maximizer.

This is a cursory treatment of a deep topic. It does not deal with the many exceptions, caveats, and subtleties to continuous optimization. However, for the purpose of basic pricing and revenue optimization, we usually do not need to deal with these subtleties; a simple approach will usually suffice. Most pricing and revenue optimization problems have a simple structure: Increasing a decision variable (price or capacity allotment) initially improves profit, but, at some point, further increases lead to lower profit. The maximum occurs at the point just before profit begins to degrade. With this structure, we can use simple approaches and not worry about some of the more subtle issues. This does not mean that these subtleties can always be ignored in practice. For example, in optimizing list prices, some products can be complements (like hot dogs and hot-dog buns), while others are substitutes (like Coke and Pepsi). In this situation, the problem of finding the

price that maximizes total expected contribution can have many local maxima, and a point that satisfies all of the first- and second-order conditions may not be the global maximum. This means that genetic algorithms or related approaches may be needed to find the global maximum.

A.2 LINEAR PROGRAMMING

Linear programming is a well-established approach to optimization in the case where a linear objective function is being maximized subject to linear constraints. The *simplex algorithm* for solving such problems was initially developed by George Dantzig in the 1960s and has been extended and refined many times since. The simplex algorithm, like all other commercially used algorithms for solving linear programs, is a hill-climbing approach, in the sense that it moves from point to point within the feasible region, increasing the objective function at every move, until no further improving direction within the feasible region can be found.

The standard form of a linear programming problem is

$$\max_{\mathbf{x}} \sum_{i=1}^n c_i x_i,$$

subject to

$$\sum_{i=1}^n a_{ij} x_i \leq b_j \quad \text{for } j = 1, 2, \dots, m,$$

$$x_i \geq 0 \quad \text{for } i = 1, 2, \dots, n.$$

The *objective function coefficients* c_i and the *constraint coefficients* a_{ij} can each be either positive, negative, or zero. The constraints have been specified as “less than or equal to,” but “greater than or equal to” constraints can also be incorporated by simply multiplying both sides by -1 . Equality constraints can be incorporated by including simultaneous “less than or equal to” and “greater than or equal to” constraints. The standard form has specified that each x_i is zero or greater, but it is also possible to constrain any or all of them to be zero or less or to take either positive or negative values. Thus, the standard form is quite flexible in representing any linear objective function subject to any set of linear constraints.

The standard form can also be written as

$$\max_{\mathbf{x}} \mathbf{c}^T \mathbf{x} \tag{A.1}$$

subject to

$$A\mathbf{x} \leq \mathbf{b},$$

$$\mathbf{x} \geq 0,$$

where $\mathbf{c} = (c_1, c_2, \dots, c_n)$, $\mathbf{x} = (x_1, x_2, \dots, x_n)$, $\mathbf{b} = (b_1, b_2, \dots, b_m)$, and the matrix A consists of the elements a_{ij} .

An optimization problem that can be specified in this form can be solved efficiently by a number of commercial software packages such as CPLEX. Microsoft Excel includes a linear programming capability within its SOLVER function.

A.3 DUALITY AND COMPLEMENTARY SLACKNESS

Every linear program of the form of Equation A.1 has an associated *dual* linear programming problem of the form

$$\max_{\lambda} \mathbf{b}^T \boldsymbol{\lambda}, \quad (\text{A.2})$$

subject to

$$\mathbf{A}^T \boldsymbol{\lambda} \geq \mathbf{c},$$

$$\boldsymbol{\lambda} \geq \mathbf{0},$$

where $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \dots, \lambda_m)$ is a vector of dual variables, one corresponding to each of the constraints in the original formulation in Equation A.1. The importance of the dual arises partly from the *complementary slackness* principle:

The incremental change in the objective function from an incremental change in constraint j in Formula A.1 will be (approximately) λ_j . If constraint j in Equation A.1 is not binding, then $\lambda_j = 0$ and relaxing the constraint will not change the value of the objective function. If $\lambda_j > 0$, then constraint j in Equation A.1 is binding.

In particular, if a constraint in a linear program is associated with a capacity or inventory limitation, the complementary slackness principle means that the corresponding dual variable will be the opportunity cost associated with that constraint.

A.4 DISCRETE OPTIMIZATION

Discrete optimization is in theory more applicable to pricing and revenue optimization than is continuous optimization, since most of the entities of interest (demand, bookings, no-shows, prices, etc.) are discrete rather than continuous. Unfortunately, discrete optimization is more computationally difficult, in general, than continuous optimization. In continuous optimization we are navigating a smooth surface, looking for the highest point. If the continuous function is differentiable (as we have assumed), either we are at a maximum or the local derivatives give us a clear direction to move. In contrast, discrete optimization is like hopping from island to island in an archipelago, looking for the island with the highest peak. When we are on a particular island, there may be no indicator to tell us which island to jump to next.

As a result, *integer linear programming*—finding the vector of *integers* $\mathbf{x}$ that solves Equation A.1—is much more difficult than standard linear programming, in which the answers do not need to be integers. In principle, the optimal integer solution to a linear program could be far from the optimal continuous solution found by standard linear programming

software. Since we are usually interested in integer solutions, this would seem to present a problem. Fortunately for most pricing and revenue optimization problems addressed in this book, rounding the continuous solution to a linear program to the nearest feasible integer solution usually turns out to give a good answer.

A particular case of interest in revenue management is the situation in which we need to find the integer $i \geq 0$ that maximizes $R(i)$, where $R(i)$ is known to be unimodal. This is the problem faced, for example, in setting a total booking limit for a flight or setting capacity limits with two booking classes. In this case, we can use the *principle of marginality*, which states that *the value i^* that maximizes $R(i)$ is the smallest value of i such that $R(i + 1) \leq R(i)$* . This means that we can start at $i = 0$, calculate $R(i)$ and $R(i + 1)$, and stop whenever $R(i + 1) \leq R(i)$. As long as we know in advance that $R(i)$ is unimodal, this approach will always terminate at the optimal value of i .

A.5 REINFORCEMENT LEARNING AND BANDIT APPROACHES

The methods that we have considered so far are based on maximizing (or minimizing) some function subject to a set of constraints. Note that optimization requires a stylized model that is always an imperfect representation of the real world. Optimization is a mathematical operation on this model that produces values of decision variables—in our case, primarily prices or booking limits—that can then be applied in the real world. For applications such as pricing and revenue optimization, this is often called a *forecast then optimize* approach—we build a mathematical model that forecasts what we believe would happen for different decisions we could make (often with a probability distribution) and then use an optimization model to calculate the values of the decision variables that maximize our objective function. This approach has two drawbacks: (1) it requires computational effort to estimate the parameters of the model and the forecast, such as the parameters of the price-response functions estimated in Chapter 4, and (2) the quality of the decisions is dependent on the quality of the assumptions underlying our model. If, for example, we assume demands by booking class are independent when, in fact, they are highly dependent, then our booking limits may be systematically biased.

The idea behind reinforcement learning is “learning by doing”—to systematically try different alternatives and do more of what works well and less of what does not work well. Reinforcement learning (RL) approaches were developed in part in emulation of the way that animals learn to negotiate their environment and meet such goals as obtaining food or shelter. RL approaches have posted some highly publicized successes—for example, the triumph of Google’s DeepMind AlphaGo over the world champion human Go player in 2018. Since about 2010, RL has become commonly applied to a wide variety of business decisions including pricing.

The basic idea behind reinforcement learning is shown in Figure A.1. At any time, an agent has a set of actions available to it and is in the current state. The agent chooses an action. On the basis of the current state and the action chosen, the environment will respond with a reward and a new state. Both the reward and the new state are likely to be

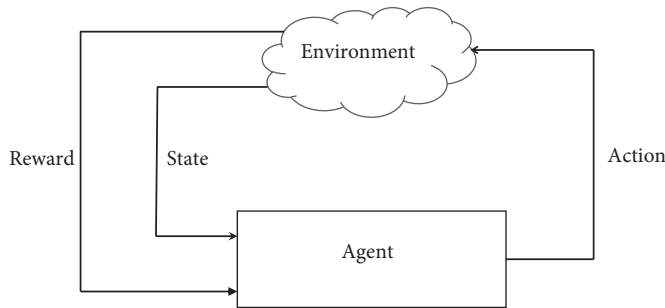


Figure A.1 Elements of reinforcement learning.

probabilistic, which is why the environment is represented as a cloud. The agent seeks to maximize the (possibly discounted) sum of the expected rewards that it will receive over time.

The structure in Figure A.1 is quite general. It could apply to such varied situations as these:

- A driver is trying to get from an origin to a destination. His state is his location; his available actions are the choices he could make at any time, such as turn right, turn left, or go straight; and his reward is the inverse of the time it takes him to get from the origin to the destination.
- A poker player is playing a hand of poker. The player's state is the cards she holds and the past actions of other players—how much they have bet in what order and which ones have folded. Her possible actions are all her possible bets, as well as “fold” or “stay,” and her reward is the payoff at the end of the hand.
- A chess player contemplates her next move. Her state is the position of the pieces on the board, her potential actions are her available moves, and the reward is either a win, a loss, or a draw at the end of the game.

As the examples might suggest, the problems attacked by reinforcement learning can be quite complex. In the examples above, the rewards do not occur primarily in response to each action but at the end when the driver reaches his destination or the chess player either wins, loses, or draws. Such applications can require deep and sophisticated approaches such as neural nets as well as lots of training to become effective.

Fortunately, the problems associated with pricing are much simpler—at least from a reinforcement learning point of view. The action available at each time is the price, and the reward is the immediate contribution—profit—received. There is no long-term goal such as winning a chess match or driving to a destination that requires thinking many steps ahead. And, at least as we consider it in Chapter 5, the pricing problem is stateless. This would not be the case if we were applying reinforcement learning to capacity control (Chapter 9), overbooking (Chapter 11), or markdown management (Chapter 12). In the case of these issues, the decision maker needs to keep track of the remaining unsold inventory, which would be a state variable.

Because of its relative simplicity, many pricing problems can be formulated as a *k*-armed bandit problem, which is one of the simplest forms of reinforcement learning. The name *k*-armed bandit arises from the metaphor of a gambler in a casino who has the option of playing one of *k* different slot machines (or one machine with *k* arms). The slot machines have different payoff probabilities. For simplicity, assume that they are \$1 slot machines and that, if you lose, you lose your dollar, but if you win, you get your original dollar back plus an additional \$1. The problem is that you do not know the payoff probabilities for the different machines but you want to maximize your expected winnings over some large number of plays. What should you do?

Let us start with the case of two machines—Machine 1 and Machine 2—and the following assumptions:

1. The probability of winning on either machine is independent of the probability of winning on the other.
2. The probabilities of winning on either machine do not change over time.
3. We have no prior information on the payoff probability of either machine.

For concreteness, let us assume that we bet \$1 each time and the payoff is \$2 if we win and \$0 if we lose. Let us consider the case where Machine 1 has a payoff probability of .7 and Machine 2 has a payoff probability of .3.

The most naïve strategy that we could play is to choose one of the machines at random and play it until we are tired of playing or run out of money—whichever comes first. Alternatively, we could alternate between the machines or flip a coin to determine which of the two machines to play on each turn. Under either strategy, the expected payoff per play is the same: On each play, there is a 50% chance that we are playing the .7 payoff machine with expected payoff $.7 \times \$1.00 - .3 \times \$1.00 = \$0.40$, and there is a 50% chance that we are playing the .3 payoff machine with expected payoff $.3 \times \$1.00 - .7 \times \$1.00 = -\$0.40$. Our expected payoff per turn for any of these naïve strategies is thus $.5 \times \$0.40 - .5 \times \$0.40 = 0$.

A more sophisticated strategy is to pick one of the two machines at random and play it. If we win, we play it again; if we lose, we move to the other machine. If we use this strategy, we will end up playing Machine 1 about 70% of the time and Machine 2 about 30% of the time on the average. Thus, using this strategy, our expected return is about $.7 \times \$0.40 - .3 \times \$0.40 = \$0.16$, which is not bad and far better than anything you will ever encounter in Las Vegas.

However, the “play again if we win, otherwise switch” strategy has an obvious drawback—it has no memory, so it cannot learn. Therefore, it can never realize that one machine is paying out more often on average than the other. The idea behind a bandit algorithm is to learn from experience. One particular approach—called an *epsilon-greedy algorithm*—is to do the following:

1. Set a small value of epsilon ϵ . For this example, we will choose $\epsilon = .1$.
2. We have just completed our *N*th play. For each machine, we record the number of times that we have played that machine, with n_{1N} being the number of times that we have played Machine 1 and n_{2N} the number of times that we have played Machine

2. Note that $n_{1N} + n_{2N} = N$. Also, we have recorded the number of wins that we have enjoyed so far on each machine, w_{1N} , w_{2N} .
3. Before play $N + 1$, calculate the win rates $r_{1N} = w_{1N}/n_{1N}$ and $r_{2N} = w_{2N}/n_{2N}$.
4. On play $N + 1$,
 - a. If $r_{1N} > r_{2N}$, play Machine 1 with probability $1 - \epsilon$ and Machine 2 with probability ϵ .
 - b. If $r_{2N} > r_{1N}$, play Machine 2 with probability $1 - \epsilon$ and Machine 1 with probability ϵ .
 - c. If $r_{1N} = r_{2N}$, play either machine with 50% probability.
5. For the machine that you played, i , update $n_{iN+1} = n_{iN} + 1$. If you won, set $w_{iN+1} = w_{iN} + 1$; if you lost, then $w_{iN+1} = w_{iN}$. For the machine not played—call it machine j —set $n_{jN+1} = n_{jN}$ and $w_{jN+1} = w_{jN}$. Increment $N \leftarrow N + 1$ and go to step 2.

We continue until we run out of money or get tired of playing.

How well does the ϵ -greedy multi-armed bandit algorithm perform? Over time, it will converge to the case in which Machine 1 is chosen 90% of the time. In this case, the average payoff will be $.9 \times \$0.40 - .1 \times \$0.40 = \$0.32$, double the average payoff from the “play again if we win, otherwise switch” approach. In simulating 100 plays, it picks Machine 1 about 85% of the time, so the average payoff is $.85 \times \$0.40 - .15 \times \$0.40 = \$0.28$. The average fraction of the time that Machine 1 is chosen in 100 plays is shown for 50 replications in the left-hand graph in Figure A.2. The fraction is .5 on play 2 because each machine is chosen once in the first two plays—from there the algorithm learns fairly rapidly to favor Machine 1.

The rapid learning displayed in the left-hand graph of Figure A.2 is partly a reflection of the fact that the payoffs from the two machines are so different—70% versus 30%. The algorithm learns much more slowly when the difference in payoff is smaller. The average learning trajectory in the case when the payoff from Machine 1 is 52% and that from Machine 2 is 48% is shown on the right-hand graph in Figure A.2. The learning is clearly slower and only an average of about 70% of the plays are on Machine 1 after 100

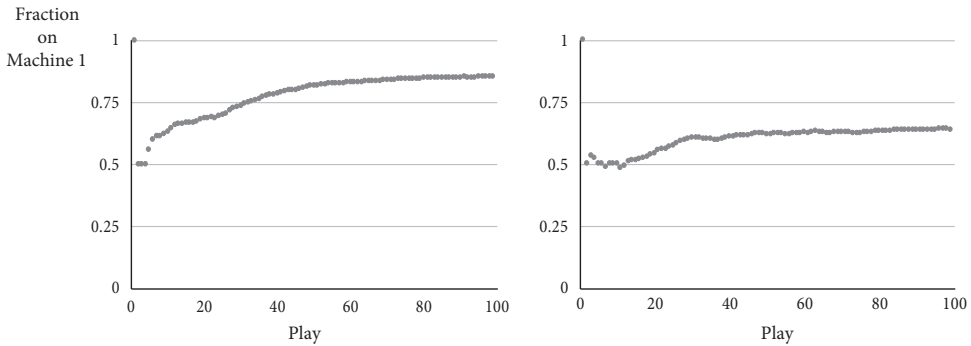


Figure A.2 Fraction of times that Machine 1 is played as a function of the number of plays averaged over 50 replications when the probabilities of winning are 70% and 30% (*left*) and 52% and 48% (*right*).

plays. On the other hand, when the difference between machines is smaller, learning is cheaper in the sense that there is less relative cost to spending time on Machine 2 rather than Machine 1.

While there are many refinements and extensions, the algorithm described above incorporates the fundamental features of k -armed bandit algorithms. The probability of picking a losing arm is critically important because, without occasional testing, there is a chance that the algorithm could pick the wrong arm and stick with it. In the case when the winning probabilities are 70% and 30%, there is a 9% chance that the first pull of Machine 1 will lose and the first pull of Machine 2 will win. In this case, a purely greedy algorithm (i.e., one with $\epsilon = 0$) would continue playing Machine 2 forever. By occasionally testing a seemingly inferior alternative, we eliminate this possibility.

The two-armed bandit in the example can be extended to multiple arms in a straightforward fashion. After trying each of the arms, we choose the arm with the best average performance a fraction of $1 - \epsilon$ of the time. We can choose among the other arms based on their relative performance—better-performing arms should have a higher probability of being chosen, but every arm should have some chance of being chosen.

The extension of the multi-armed bandit problem to the pricing optimization problem should be clear. For pricing, we have a discrete set of prices $p_1 < p_2 < \dots < p_n$ that we want to test. In period t , we decide on one of the n prices to charge. Assume that we have decided on price p_{i_t} . We observe the realized demand $\hat{d}(p_{i_t})$ and the corresponding contribution $\hat{d}(p_{i_t})(p_{i_t} - c)$. Let $I_i(t)$ be an indicator variable such that $I_i(t) = 1$ means that price i was the price chosen at time t and $I_i(t) = 0$ means that price i was *not* the price chosen, so that one and only one value of i has $I_i(t) = 1$ for any t . Then, at time T , the average contribution from price i is

$$\bar{\pi}_i(T) = \frac{\sum_{t=1}^T I_i(t) \hat{d}(p_{i_t})(p_{i_t} - c)}{\sum_{t=1}^T I_i(t)}.$$

A multi-armed bandit approach to pricing uses the values of $\bar{\pi}_i(T)$ to determine which price to charge in period $T + 1$.

While the multi-armed bandit approach is appealing in its simplicity and its lack of underlying assumptions, it has a number of potential drawbacks. First, it requires the ability to test frequently throughout the potential space of decisions. In some circumstances, testing may not be feasible or desirable. For example, an airline may not want to do extensive testing of high overbooking levels, which may lead to high levels of denied boardings. Second, the two-armed bandit example assumed that the payoff probabilities of each machine were constant over time and not influenced by the external environment. However, in pricing and revenue optimization, there are multiple external variables that might change demand—for example, day of the week, whether it is the Christmas holiday season, and known competitive prices. In this case, we are searching not just for a price but a *policy* that maps possible values of external variables to a best price. There are many different approaches to generating a policy using reinforcement learning—however, they all require significantly more testing than the simple examples we have seen.

A.6 FURTHER READING

There is a vast literature on classical optimization, both continuous and discrete. I am partial to *Linear and Nonlinear Programming* (Luenberger and Ye 2008) as a great single source for both continuous and discrete optimization. For those who want to go even deeper, *Convex Optimization* (Boyd and Vandenberghe 2004) is recommended for continuous optimization, and *Understanding and Using Linear Programming* (Matousek and Gärtner 2007) is recommended for discrete optimization.

For reinforcement learning in general and the bandit problem in particular, the best starting point is the excellent *Reinforcement Learning: An Introduction* (Sutton and Barto 2018); chapter 2 discusses the multi-armed bandit problem at length.

This appendix provides a brief review of the concepts and formulas from probability theory that are used in basic pricing and revenue optimization. A fuller treatment can be found in any text on the subject, such as Mood, Graybill, and Boes 1974; Jaynes 2003; or Bertsekas and Tsitsiklis 2008.

We use the notation $\Pr\{X\}$ to refer to the probability that event X will occur. X can be any event—for example, “Total demand during the week for a product will be greater than 100” or “No-shows will be exactly equal to 10 and shows will be equal to 95.” However, no matter what event X represents, its probability of occurrence will always be between 0 and 1, so $0 \leq \Pr\{X\} \leq 1$. The probability that event X does not occur is $1 - \Pr\{X\}$.

If X takes on numeric values—such as the number of heads when a fair coin is flipped 100 times or the number of sweaters that will be sold in a month—then X is termed a *random variable*. By convention, random variables are denoted with capital letters.

For any two events, X and Y , we use $\Pr\{X, Y\}$ (or sometimes $\Pr\{XY\}$) to refer to the probability that both X and Y occur. Thus, $\Pr\{\text{Shows are greater than 80, Demand is equal to 100}\}$ means “The probability that shows are greater than 80 *and* demand is equal to 100.” This should be contrasted with the notation $\Pr\{X|Y\}$, which denotes the *conditional probability* of event X *given* that event Y occurs. Thus, $\Pr\{\text{Shows are greater than 80}|\text{Demand is equal to 100}\}$ means “the probability that shows are greater than 80 *given* that demand is equal to 100.” The two concepts are linked by the formula

$$\Pr\{X|Y\} = \Pr\{XY\}/\Pr\{Y\}$$

for $\Pr\{Y\} > 0$. Note that since $\Pr\{Y\} \leq 1$, $\Pr\{X|Y\} \geq \Pr\{XY\}$.

Two events X and Y are called *independent* if $\Pr\{X|Y\} = \Pr\{X\}$. Intuitively, independence refers to the case in which the probability that event X will occur is unaffected by the value of Y . For example, if we say that weekly sales of product A are independent of product B, it means that our probability that sales of product B would exceed 10 units in that week would not change if we learned that sales of product A were 0, 10, or 100 units. Independence is always mutual; X independent of Y means that Y is independent of X , and vice versa. If X and Y are independent, then $\Pr\{XY\} = \Pr\{X\}\Pr\{Y\}$.

B.1 PROBABILITY DISTRIBUTIONS

A *probability distribution* is an assignment of probabilities to a set of events that are “mutually exclusive and collectively exhaustive.” *Mutually exclusive* means that, at most, one of the events will occur. *Collectively exhaustive* means that one of the events must occur. Together they guarantee that the sum of the probabilities of all events (or the integral of the probabilities if the events are continuous numbers) equals 1.

For a discrete distribution, $f(i)$ is the probability that event i will occur. In this book, we consider only discrete distributions defined on the nonnegative integers—that is, on $i = 0, 1, 2, \dots, \infty$.¹ By the *mutually exclusive and collectively exhaustive property*, we must have

$$\sum_{i=0}^{\infty} f(i) = 1.$$

The function $f(i)$ is known as a *probability density function*. Given a discrete probability density function, we define the *cumulative distribution function* by

$$F(i) = \sum_{j=0}^i f(j).$$

If f is the probability density function on demand in a particular week, then $F(i)$ is the probability that demand will be less than or equal to i . Note that $F(i)$ is increasing in i , $f(0) = F(0)$, and $\lim_{i \rightarrow \infty} F(i) = 1$.

Although discrete distributions are by far the most relevant for pricing and revenue optimization, we will use continuous distributions in some cases. For a continuous distribution, the cumulative distribution function $F(X)$ denotes the probability that the random variable x will take a value less than or equal to X . We assume that $F(X)$ is continuous and differentiable, in which case the corresponding probability density function can be computed as $f(x) = dF(x)/dx$. Then

$$F(x) = \int_{-\infty}^x f(y) dy.$$

For a continuous distribution, the probability that a random variable will be between a and b , with $a < b$, is $F(b) - F(a)$. The cumulative distribution function $F(x)$ is increasing in x and $\lim_{x \rightarrow \infty} F(x) = 1$.

For both continuous and discrete distributions, we define the *complementary cumulative distribution function* (CCDF) as $\bar{F}(x) = 1 - F(x)$. The CCDF $\bar{F}(x)$ is the probability that a random variable X with cumulative distribution function $F(x)$ will be greater than x . It is a nonincreasing function of x .

B.1.1 Expectation

Let the random variable X be distributed according to f . Then, the *expectation* or *mean* of X is defined by

$$\mu = E[X] = \sum_{i=0}^{\infty} i \times f(i)$$

when $f(\cdot)$ is a discrete distribution and by

$$\mu = E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

when $f(\cdot)$ is continuous. The notation $E[\cdot]$ denotes the expectation operator, meaning that it is the probability-weighted sum (for a discrete variable) or integral (for a continuous variable) of the function inside the brackets. The mean of the variable itself is denoted by μ .

The mean of a distribution should not be confused with the *mode* of the distribution, which is the most likely observation given a single sample. Nor should it be confused with the *median* of the distribution, which is the value such that the probability that a single observation will exceed that value is the same as the probability that it will be less than the value.

An alternative formula for calculating the mean of a discrete distribution defined over the nonnegative numbers $i = 0, 1, \dots, \infty$ is

$$\mu = \sum_{i=0}^{\infty} [1 - F(i)] = \sum_{i=0}^{\infty} \bar{F}(i). \quad (\text{B.1})$$

A similar result holds for a conditional probability distribution for a random variable X defined for $X \geq 0$ —namely,

$$\mu = \int_0^{\infty} \bar{F}(x)dx.$$

The concept of expectation can be extended to any function. Thus, if $g(i)$ is a function of i , the expectation of $g(i)$ is defined as

$$E[g(i)] = \sum_{i=0}^{\infty} g(i)f(i),$$

for a discrete variable and

$$E[g(x)] = \int_0^{\infty} g(x)f(x)dx$$

for a continuous variable. Note that, in general, $E[g(i)] \neq g[E(i)]$, with equality holding only if the function g is linear.

For any random variable X and any number a , $E[aX] = aE[X]$. Furthermore, for random variables, $X_1, X_2, \dots, X_n$; $E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i]$ and $E[X_i - X_j] = E[X_i] - E[X_j]$ for any i and j . This property does not apply to general functions of random variables—for example, $E[X/Y] \neq E[X]/E[Y]$ in general.

For purposes of pricing and revenue optimization, the expectation of the minimum of a random variable and a fixed capacity is often important. In other words, we want to calculate $E[\min(i, C)]$, where C is a fixed capacity and i follows some discrete distribution $f(i)$ on the nonnegative integers. The following formula is often useful in this case:

$$E[\min(i, C)] = \sum_{i=0}^{C-1} \bar{F}(i). \quad (\text{B.2})$$

We are also often interested in $E[(i - C)^+] \equiv E[\max(i - C, 0)]$. If $f(i)$ is a probability density function of total demand and C is capacity, then $E[(i - C)^+]$ is the expected number of customers in excess of capacity—that is, the expected number of customers turned away. Since the expected number of customers served plus the expected number turned away must equal expected demand, we can combine Equations B.1 and B.2 to derive

$$\begin{aligned} E[(i - C)^+] &= E[i] - E[\min(i, C)] \\ &= \sum_{i=0}^{\infty} i f(i) - \sum_{i=0}^{C-1} \bar{F}(i) \\ &= \sum_{i=C}^{\infty} \bar{F}(i). \end{aligned} \quad (\text{B.3})$$

B.1.2 Variance and Standard Deviation

Two other important properties of probability distributions are the *variance* and the *standard deviation*, both of which measure the *spread* of the distribution. The variance of a distribution is the expected square of the distance between a sample from the distribution and the mean. That is,

$$\text{var}[X] = E[(X - E[X])^2] = E[X^2] - (E[X])^2.$$

The variance of a distribution is always greater than or equal to 0; the higher the variance, the broader the spread of the distribution. For *independent* random variables, the variance of the sum is equal to the sum of the variances; that is,

$$\text{var}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \text{var}[X_i]. \quad (\text{B.4})$$

The *standard deviation*, denoted by σ , is equal to the positive square root of the variance: $\sigma = \sqrt{\text{var}[X]}$. Like the variance, $\sigma \geq 0$ for any distribution. Let $\sigma[X_i]$ be the standard deviation associated with random variable X_i . Then, from Equation B.4, we can derive the formula for the standard deviation of the sum of independent random variables:

$$\sigma\left[\sum_{i=1}^n X_i\right] = \sqrt{\sum_{i=1}^n (\sigma[X_i])^2}. \quad (\text{B.5})$$

Note that Equations B.4 and B.5 hold only when $X_1, X_2, \dots, X_n$ are independent.

The ratio of the standard deviation to the mean of a distribution is sometimes called the *coefficient of variation*, or CV, where $\text{CV} = \sigma/\mu$.

B.2 CONTINUOUS DISTRIBUTIONS

B.2.1 The Uniform Distribution

The uniform distribution is “the distribution of maximum ignorance.” It is used when we know nothing about the distribution of X except that it lies between bounds $a < b$, but we

have no reason to believe that it is more likely to lie at any one point within that interval than another. The uniform distribution has a probability density function,

$$f(x) = \begin{cases} 0 & \text{for } x < a, \\ \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{for } x > b. \end{cases}$$

The mean of the uniform distribution is $\mu = (a + b)/2$, and its standard deviation is $\sigma = (b - a) / \sqrt{12}$.

B.2.2 The Exponential Distribution

The exponential distribution has a single parameter, $\lambda > 0$, and a probability density function,

$$f(x) = \lambda e^{-\lambda x}$$

for $x \geq 0$. The mean of the exponential distribution is $\mu = 1/\lambda$, and the standard deviation is the same as the mean: $\sigma = 1/\lambda$.

B.2.3 The Normal Distribution

The normal (or Gaussian) distribution is the famous bell-shaped curve, the most widely used distribution in statistics and mathematical modeling. There are two reasons for its popularity. The first is that theory has shown that the normal distribution is, in many situations, a reasonable distribution to use when the actual underlying distribution is unknown or when the underlying distribution is the result of many different random influences. The second reason is that many practical techniques have been developed for making calculations with the normal distribution. Such techniques are easily accessible in most software and spreadsheet packages, such as Excel.

The probability density function for the normal distribution is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2},$$

where μ is the mean and σ is the standard deviation.

The normal distribution with mean 0 and standard deviation 1 is so important that it has its own notation:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-(x^2/2)},$$

where $\phi(x)$ denotes the normal distribution with mean 0 and standard deviation 1. The cumulative distribution function of the normal distribution with mean 0 and standard deviation 1 is denoted by $\Phi(x)$. Both $\phi(x)$ and $\Phi(x)$ are included in all statistical and spreadsheet packages. This is useful because they can be used to calculate $f(x)$ and $F(x)$ for any normal distribution, by means of the following transformations:

$$f(x) = \phi[(x - \mu) / \sigma] \quad F(x) = \Phi[(x - \mu) / \sigma], \quad (\text{B.6})$$

where $f(x)$ is the normal density function with mean μ and standard deviation σ , and $F(x)$ is the corresponding cumulative distribution function.

Example B.1

Monthly demand for a product has been observed to follow a normal distribution with mean of 10,000 and standard deviation of 8,000. The probability that demand will be less than 12,000 units is then given by $\Pr\{\text{Demand less than 12,000 units in a month}\} = \Phi[(12,000 - 10,000)/8,000] = \Phi(0.25) = 0.60$.

The normal distribution is widely used. However, a few words of caution are in order. First, the normal distribution is continuous, so care needs to be used in applying it to situations where outcomes are discrete. These issues are addressed in Section B.3.4 on the discrete normal distribution. Second, bear in mind that the normal distribution puts positive probabilities on all outcomes from $-\infty$ to ∞ . This can become a problem in situations where outcomes less than zero are not physically meaningful (e.g., demand, sales, no-shows, prices). Third, caution is particularly important when using the normal distribution in a situation where the standard deviation is large relative to the mean. For example, when the mean is equal to the standard deviation (e.g., the coefficient of variation is 1), the normal distribution places about a 16% probability on the outcome's being less than zero. In cases where the coefficient of variation is high, it may be necessary to use another distribution to obtain reasonable results.

B.3 DISCRETE DISTRIBUTIONS

B.3.1 The Bernoulli Distribution

The Bernoulli distribution is the simplest interesting discrete distribution. It is defined as

$$f(i) = \begin{cases} 1-p & \text{for } i=0, \\ p & \text{for } i=1, \\ 0 & \text{otherwise.} \end{cases}$$

The Bernoulli distribution applies to the situation in which a particular event has a probability p of occurring and, thus, a probability $1-p$ of not occurring. In this case, the outcome of the event's occurring can be arbitrarily labeled 1 and the outcome where the event does not occur can be labeled 0. The probability $1-p$ of the event's not occurring is often denoted by q .

Example B.2

A fair coin is to be flipped once. Let 0 denote heads and 1 denote tails. Then the outcome of the flip follows a Bernoulli distribution with $p = 0.5$.

The mean of the Bernoulli distribution is p and its variance is $p(1-p) = pq$, so its standard deviation is $\sigma = \sqrt{pq}$.

B.3.2 The Discrete Uniform Distribution

The discrete uniform distribution is appropriate whenever a number of mutually exclusive events all have equal probability of occurring. Then, if there are n events, the possibility that any one of them will occur is $1/n$. We assume that the events are indexed by consecutive integers $i = a, a + 1, a + 2, \dots, b$, where $a \geq 0$ and $b > a$. In this case, we can write the uniform probability density function as

$$f(i) = \begin{cases} 0 & \text{for } i < a, \\ \frac{1}{b-a+1} & \text{for } a \leq i \leq b, \\ 0 & \text{for } i > b. \end{cases}$$

The cumulative distribution function for the uniform distribution is

$$F(i) = \begin{cases} 0 & \text{for } i < a, \\ \frac{i-a+1}{b-a+1} & \text{for } a \leq i \leq b, \\ 1 & \text{for } i > b. \end{cases}$$

The cumulative distribution function for the uniform distribution function $\bar{F}(i) = 1 - F(i)$ is linear between a and b . This leads to the connection between the uniform willingness-to-pay distribution and the linear demand function discussed in Chapter 3.

The mean of the uniform distribution is $\mu = (a + b)/2$, and its standard deviation is $\sigma = \sqrt{(b-a)(b-a+2)} / \sqrt{12}$.

Example B.3

A seller believes that if he prices a new shirt at \$79.00, then total demand for the shirt will have equal probability of being any amount between 10,000 and 20,000 units. This corresponds to a uniform distribution with $a = 10,000$ and $b = 20,000$. This means the probability that demand will be less than or equal to 18,000 will be $F(18,000) = (18,000 - 10,000 + 1)/(20,000 - 10,000 + 1) = 0.8$. Mean demand is $(20,000 + 10,000)/2 = 15,000$ shirts, with a standard deviation of $\sqrt{10,000 \times 10,002} / \sqrt{12} = 2,887$.

B.3.3 The Binomial Distribution

The binomial distribution arises whenever each of a known number of people makes a decision between two mutually exclusive alternatives when each person has the same probability of choosing the first alternative and the decisions are all made independently (in the probabilistic sense). If the number of people choosing between the alternatives is n and the probability that any individual will choose the first alternative is p , then the total number choosing the first alternative will follow a binomial distribution with parameters n and p . In addition, the total number choosing alternative 2 will follow a binomial distribution with parameters n and q , where $q = 1 - p$.

Example B.4

One hundred passengers have booked the same flight. Each passenger has probability of 0.8 of showing, and passengers will make independent decisions whether or not to show. In this case, the number of shows will follow a binomial distribution with $p = 0.8$ and $n = 100$. Furthermore, no-shows will follow a binomial distribution with $p = 0.2$ and $n = 100$.

Example B.5

Ten thousand shoppers will log on to the website of a car loan provider. The supplier believes that if he offers an annual percentage rate of 6.9%, each shopper will have a 0.05 probability of filling out an application for a loan. The distribution of applications filled out would then be binomial with $p = 0.05$ and $n = 10,000$.

The classic example of a binomial distribution is drawing balls from a large urn with replacement. If an urn contains a large number of black and white balls such that p is the fraction of black balls, and n balls are drawn from the urn with the drawn balls replaced, then the number of black balls drawn will be binomial with parameters n and p . Similarly, if a fair coin is tossed n times, the number of heads tossed will be binomial with parameters n and $p = 0.5$.

The probability density function for the binomial distribution is

$$f(i) = \binom{n}{i} p^i (1-p)^{n-i} \quad \text{for } x = 0, 1, 2, \dots, n, \quad (\text{B.7})$$

where

$$\begin{aligned} \binom{n}{i} &= \frac{n!}{i!(n-i)!} \\ &= \frac{n \times (n-1) \times (n-2) \times \dots \times 1}{[i \times (i-1) \times (i-2) \times \dots \times 1] \times [(n-i) \times (n-i-1) \times (n-i-2) \times \dots \times 1]}. \end{aligned}$$

The mean of the binomial distribution is $\mu = pn$, the variance is $p(1-p)n$, and the standard deviation is $\sigma = \sqrt{p(1-p)n}$.

Example B.6

A bed-and-breakfast has 8 rooms and has accepted 10 bookings. On average, 85% of its bookings show, with the rest canceling. Assuming a show rate of .85, the expected number of guests who will show up is $.85 \times 10 = 8.5$, with a variance of $.85 \times .15 \times 10 = 1.275$ and a standard deviation of 1.13. The probability that exactly 7 guests will show up is

$$f(7) = \binom{10}{7} \times .85^7 \times .15^3 = \frac{10!}{7! \times 3!} \times .0011 = \frac{3,628,800}{5,040 \times 6} \times .0011 = .13.$$

Because of the terms of the form in Equation B.7, the binomial distribution can be very difficult to work with in practice. In particular, the term $n!$ grows very large very quickly.

Using Equation B.7 to calculate the number of shows given 100 bookings would require calculating $100!$, which is a 158-digit number. In addition, there is no closed-form solution for the cumulative distribution function $F(i)$ for a binomial distribution. Fortunately, for large n , the binomial distribution is well approximated by the normal distribution with mean $\mu = pn$ and standard deviation $\sigma = \sqrt{p(1-p)n}$.

Example B.7

An online tour broker believes that 30,000 people will visit his website in the next week, with each visitor having a 2% chance of purchasing a tour. The number of tours sold would then follow a binomial distribution with $p = 0.02$ and $n = 30,000$. His probability density function on demand can be approximated by a normal distribution with mean $0.02 \times 30,000 = 600$ and $\sigma = \sqrt{0.02 \times 0.98 \times 30,000} = 24.25$. The tour operator wants to know the probability that more than 580 customers will seek to purchase a tour. He can estimate this as $1 - \Phi[(580 - 600)/24.25] = 1 - \Phi(-0.8025) = 0.788$. Therefore, there is about a 79% chance that demand will be greater than 580.

B.3.4 The Discrete Normal Distribution

The normal distribution is a continuous distribution. However, in some cases we want to apply it to situations in which outcomes are discrete. For example, we might want to model demand for one-night arrivals at a hotel as “normal with mean 200 and standard deviation 100.” Furthermore, in most of these cases only nonnegative values will be meaningful—therefore, we need to restrict the distribution to the nonnegative numbers. We do this by defining

$$F(i) = \Phi[(i - \mu)/\sigma] \quad \text{for } i = 0, 1, 2, \dots, \infty$$

and

$$f(i) = \begin{cases} \Phi(-\mu/\sigma) & \text{for } i = 0 \\ \Phi[(i - \mu)/\sigma] - \Phi[(i - \mu - 1)/\sigma] & \text{for } i = 1, 2, 3, \dots, \infty. \end{cases}$$

Defining the normal distribution in this fashion makes it easy to use standard statistical functions in Excel, for example, for computations. However, a word of caution that this comes with a cost: The discrete normal distribution as defined here will now have a mean that is different (generally higher) than μ and a standard deviation that is somewhat lower than σ . Whether this is important or not depends on the application.

B.4 SAMPLE STATISTICS

So far, this appendix has discussed concepts such as the mean, standard deviation, and variance in terms of somewhat abstract probability distributions. In practice, we are often interested in calculating the best estimates of these statistics from data. Of course, all statistical software packages provide built-in functions for generating these estimates; thus, the

AVERAGE function in Excel estimates the statistical mean of a set of data, and the standard deviation is determined by the STDEV.P function.

Assume that we have N numerical data observations $x_1, x_2, \dots, x_N$. Then the mean of the observations is given by

$$\mu = \left(\frac{1}{N} \right) \sum_{i=1}^N x_i,$$

the variance is given by

$$\text{var} = \left(\frac{1}{N} \right) \sum_{i=1}^N (x_i - \mu)^2,$$

and the standard deviation, by $\delta = \sqrt{\text{var}}$.

These statistics are often measures of interest in themselves. For example, it may be of interest to know that the mean daily demand for a product over the last month was 140.3, with a standard deviation of 61.2. However, the mean and standard deviation can also be used as estimates of the properties of an underlying distribution; thus, we might model future demand for the product using a normal distribution with mean 140.3 and a standard deviation of 61.2. By the law of large numbers, the sample statistic estimates will approach the “true” values for a population as the number of observations N increases.

NOTE

1. Discrete distributions can also be defined with a range that includes negative integers. However, this book considers only distributions over nonnegative integers.

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